

MALTA AIR

Operations Manual Part A

Issue 21 Revision 2

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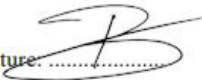
APPROVAL



Administration and Control of Operations Manual

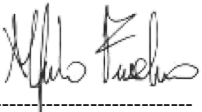
The operator's nominated person hereunder certifies that the list of effective pages / chapters and the contents of this volume of the Operations Manual **Part A (Issue 21)** satisfy the requirements as specified under 8.2 of Annex V to Regulation (EU) 2018/1139 and its Implementing Rules.

The elements of the Operations Manual that are specifically approved by the CAD are shown in the Operations Manual Approval Schedule.

On behalf of Malta Air		Transport Malta CAD Approval
Name: Roger Amaral Nominated Person Flight Operations Signature:  Date: 11/03/2025	Name: Alfredo Fuschino Compliance Monitoring Manager Signature:  Date: 19/03/2025	Name: Norman D'Amato Flight Operations Inspectorate  Signature: Date: 23/04/2025

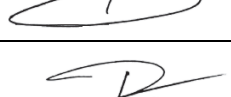
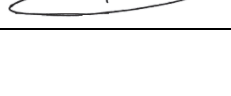


Administration and Control of Operations Manual

CHANGES NOT REQUIRING PRIOR APPROVAL		
Malta Air’s Nominated Person hereunder certifies that this revision, being a revision of content not requiring prior approval is made in accordance with the procedure stipulated in OMA, and that the list of effective pages / chapters and the contents of this volume of the Operations Manual satisfy the requirements as specified under 8.2 of Annex V to Regulation (EU) 2018/1139 and its implementing rules. TM-CAD have been notified of this revision.		
Name of Manual: Operations Manual Part A	Revision Number: Issue 21 Revision 2	Revision Date: 15 th May 2025
Name: Roger Amaral Nominated Person Flight Operations Signature: 	Name: Alfredo Fuschino Compliance Monitoring Manager Signature: 	

RECORD OF AMENDMENTS

<i>Amendment</i>		<i>TM CAD Approval</i>	<i>Amendment Implementation</i>		
<i>No</i>	<i>Date</i>	<i>Date</i>	<i>Date</i>	<i>Who</i>	<i>Signature</i>
Issue 1, Rev 0	06/06/2019	12/06/2019	12/06/2019	N D'Amato	
Issue 2, Rev 0	06/04/2020	27/04/2020	07/05/2020	R Amaral	
Issue 3, Rev 0	01/10/2020	16/12/2020	18/12/2020	R Amaral	
Issue 3, Rev 1	08/01/2021	N/A	14/01/2021	R Amaral	
Issue 3, Rev 2	11/02/2021	N/A	11/02/2021	R Amaral	
Issue 3, Rev 3	18/03/2021	N/A	18/03/2021	R Amaral	
Issue 4, Rev 0	08/04/2021	30/06/2021	08/07/2021	R Amaral	
Issue 4, Rev 1	12/08/2021	N/A	12/08/2021	R Amaral	
Issue 4, Rev 2	16/09/2021	N/A	23/09/2021	R Amaral	

<i>Amendment</i>		<i>TM CAD Approval</i>	<i>Amendment Implementation</i>		
<i>No</i>	<i>Date</i>	<i>Date</i>	<i>Date</i>	<i>Who</i>	<i>Signature</i>
Issue 4, Rev 3	23/12/2021	N/A	23/12/2021	R Amaral	
Issue 5, Rev 0	30/12/2021	N/A	23/12/2021	R Amaral	
Issue 5, Rev 1	10/03/2022	N/A	31/03/2022	R Amaral	
Issue 6, Rev 0	08/04/2022	21/09/2022	29/09/2022	R Amaral	
Issue 7, Rev 0	20/10/2022	28/10/2022	04/11/2022	R Amaral	
Issue 7, Rev 1	16/11/2022	N/A	16/11/2022	R Amaral	
Issue 8, Rev 0	23/11/2022	22/12/2022	22/12/2022	R Amaral	
Issue 9, Rev 0	24/11/2022	28/12/2022	29/12/2022	R Amaral	
Issue 9, Rev 1	12/01/2023	N/A	12/01/2023	R Amaral	
Issue 9, Rev 2	02/03/2023	N/A	02/03/2023	R Amaral	

<i>Amendment</i>		<i>TM CAD Approval</i>	<i>Amendment Implementation</i>		
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Issue 10, Rev 0	26/01/2023	16/03/2023	16/03/2023	R Amaral	
Issue 10, Rev 1	06/04/2023	N/A	06/04/2023	R Amaral	
Issue 10, Rev 2	27/04/2023	N/A	27/04/2023	R Amaral	
Issue 11, Rev 0	24/02/2023	04/05/2023	04/05/2023	R Amaral	
Issue 11, Rev 1	15/06/2023	N/A	15/06/2023	R Amaral	
Issue 11, Rev 2	29/06/2023	N/A	29/06/2023	R Amaral	
Issue 11, Rev 3	26/07/2023	N/A	26/07/2023	R Amaral	
Issue 11, Rev 4	31/08/2023	N/A	31/08/2023	R Amaral	
Issue 12, Rev 0	26/10/2023	27/10/2023	02/11/2023	R Amaral	
Issue 12, Rev 1	16/11/2023	N/A	16/11/2023	R Amaral	

Amendment		TM CAD Approval	Amendment Implementation		
No	Date	Date	Date	Who	Signature
Issue 13, Rev 0	07/12/2023	15/12/2023	21/12/2023	R. Amaral	
Issue 13, Rev 1	18/01/2024	N/A	18/01/2024	R. Amaral	
Issue 13, Rev 2	01/02/2024	N/A	01/02/2024	R. Amaral	
Issue 13, Rev 3	08/02/2024	N/A	08/02/2024	R. Amaral	
Issue 14, Rev 0	08/03/2024	14/03/2024	15/03/2024	R. Amaral	
Issue 15, Rev 0	27/03/2024	28/03/2024	01/04/2024	R. Amaral	
Issue 15, Rev 1	30/05/2024	N/A	30/05/2024	R. Amaral	
Issue 15, Rev 2	22/08/2024	N/A	22/08/2024	R. Amaral	
Issue 15, Rev 3	29/08/2024	N/A	29/08/2024	R. Amaral	
Issue 16, Rev 0	30/08/2024	09/09/2024	12/09/2024	R. Amaral	

<i>Amendment</i>		<i>TM CAD Approval</i>	<i>Amendment Implementation</i>		
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Issue 16, Rev 1	19/09/2024	N/A	19/09/2024	R. Amaral	
Issue 16, Rev 2	10/10/2024	N/A	10/10/2024	R. Amaral	
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Issue 19, Rev 0	05/12/2024	08/01/2025	09/01/2025	R. Amaral	
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Issue 21, Rev 1	28/04/2025	N/A	01/05/2025	R. Amaral	
Issue 21, Rev 2	15/05/2025	N/A	15/05/2025	R. Amaral	

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AMENDMENT HIGHLIGHTS

Since this is a new revision, sections of text have been replaced and indicated by revision bar. The following pages are a brief descriptions of significant changes from the previous Issue/Revision of OMA.

Note: The following may not have an entry in the Amendment Highlights:

- Spelling Corrections;
- Additional EASA and other regulatory references.

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AMENDMENT HIGHLIGHTS

Page	Description of Change
6–15	Diabetes Updated Text.
8–166	Anti-Collision Light Clarification.

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LIST OF EFFECTIVE PAGES

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R	APP-2	21	2	15/05/2025

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0 ADMINISTRATION AND CONTROL OF OPERATIONS MANUAL**0.1 The Operations Manual – General**

CAT.GEN.100, ORO.GEN.105 Competent Authority

Malta Air has its principal place of business in Malta and Transport Malta – Civil Aviation Directorate (TM CAD) is designated the competent authority in Malta.

The ‘Ryanair Group’ consists of Malta Air and other companies, including airlines.

‘A Ryanair Group Airline’ is either Lauda Europe, Ryanair DAC, Ryanair UK, Buzz or Malta Air.

0.1.1 Operations Manual Introductory Statement

ORO.MLR.100, ORO.GEN.005

Malta Air has established an operations manual (OM). Flight Operations must only be undertaken in accordance with Malta Air operations manual as specified under 8.2 of Annex V to Regulation (EU) 2018/1139.

The content of the Malta Air OM reflects the requirements set out in Annexes III (Part-ORO), IV (Part-CAT), V (Part-SPA), Annex VIII (Part-SPO) and Part SERA as applicable, and will not contravene the conditions contained in the operations specifications to the air operator certificate (AOC) or Operations Approvals as issued by the TMCAD.

The OM is issued in separate parts. All parts are electronic and are uncontrolled if printed.

All operations personnel have easy access to the portions of the OM that are relevant to their duties.

The OM shall be kept up to date by the relevant nominated persons. All personnel shall be made aware of the changes that are relevant to their duties. Each crew member is provided with electronic access to the OMA.

Issues and Revisions are used to Amend the Operations Manual(s)

When amendment are required to be notified to the TMCAD (in accordance with ORO.GEN.115(a) and ORO.GEN.130), Malta Air shall supply the TMCAD with such amendments in advance of the effective date, and any amendments to procedures associated with prior approval items must be approved by the TMCAD before the amendment becomes effective. Notwithstanding the above, when immediate amendments are required in the interest of safety, they may be published by the nominated person for flight operations and applied immediately, provided that any approval required has been applied for.

Malta Air ensures that information taken from approved documents, and any amendment thereof, is correctly reflected in the OM. Malta Air may publish more conservative data and procedures in the OM.

Malta Air ensures that all personnel are able to understand the English language to at least English language proficiency level 4. The OM is written in the English language and observes human factors principles by presenting the contents of the OM as simply as practicable.

Operations Control Centre, Dublin

The Operations Control Centre (OCC is under the control of the NPFO Ryanair DAC as delegated to the Head of Operations Control Centre). The Operations Control Centre Manual (OCCM) is a controlled document in Ryanair under the control of the NPFO and as delegated to the Head of Operations Control. The OCCM describes the OCC organisation, duties and responsibilities of OCC personnel.

The OCC also provides services to other AOCs within the Ryanair Group as detailed in relevant Service Level Agreements in force.

Management of a Safety Problem

ORO.GEN.155

Malta Air shall implement any relevant mandatory safety information issued by EASA, including airworthiness directives and any safety measures mandated by the TMCAD.

0.1.1.1 Operations Specification

AMC1 ORO.GEN.125

A certified copy of the Malta Air Operations Specification (OPS SPEC) can be found on each Pilot's EFB Cert File and carried on board at all time during flight operations. This provides information on types of operation carried out by Malta Air and any specific approvals.

0.1.2 Compliance with Operations Manual

ORO.GEN.110(b), 110(g), 210(d), CAT.GEN.MPA.105(b)

The procedures outlined in the Operations Manual (OM) and current regulations shall be adhered to at all times, including training, testing, delivery, demonstration and positioning flights by all crewmembers.

Note: When it is observed that a Company procedure, guidance or instruction violates a rule or regulation as established by the competent authority, the latter regulation shall be the governing one. Any suggestion for

improvement to any manual, errors and/or discrepancies shall be reported to the responsible manager via any of the reporting means indicated in OMA Chapter 11.

The commander shall, in an emergency situation that requires immediate decision and action, take any action he considers necessary under the circumstances in the interests of safety. In such cases he may deviate from rules, operational procedures and methods in the interest of safety.

0.1.3 Parts of the Operations Manual

ORO.MLR.101

OMA

This part comprises all non type-related operational policies, instructions and procedures and is compliant with all relevant regulations. It consists of:

Abbreviation	Designation
OMA	Operations Manual, Part A
SEP	Safety and Emergency Procedures Manual
GOM	Ground Operations Manual

OMB

This part comprises all type-related instructions and procedures, taking into account differences between types/classes, variants or individual aircraft used by the Company. It consists of:

Abbreviation	Designation
FCOM	Flight Crew Operating Manual
FCTM	Flight Crew Training Manual
OMB	Operations Manual Part B
MEL & CDL	Minimum Equipment List & Configuration Deviation List
QRH	Quick Reference Handbook

Note: The relevant parts of the AFM are included in the OM.

OMC

This part comprises route, area and aerodrome instructions and information. Airfield briefs (AFBs) are provided electronically on EFB. The OMC consists of information supplied by the company and Aeronautical Service Provider Route Manual.

Abbreviation	Designation
OMC	Operations Manual, Part C (Route Manual including any Jeppesen Publications (Pubs))

OMD

This part comprises of all training instructions for the delivery of training by training personnel.

Abbreviation	Designation
OMD	Operations Manual, Part D (Training Manual)

0.1.4 Explanations, Definition of Terms 1. Regulation (EU) No 965/2012 Annex 1

General

When used in the OMA, the following terms have the meaning explained below:

›**Shall**‹ or an action verb in the imperative sense, means that the application of a rule or procedure or provision is **mandatory**.

›**Should**‹ means that the application of a procedure or provision is **mandatory**.

›**May**‹ means that the application of a procedure or provision is **optional**.

Note: This manual is written in the third person singular, masculine, for convenience but ‘he’ also applies to ‘she’.

Abbreviations

This section is extracted from Annex 1 Definitions in the latest Air Ops Easy Access revision, Part SERA and company specific abbreviations are also included.

A	Aeroplane
a/c	Aircraft
A380	Airbus 380
AAC	Aeronautical Administrative Communications
AAL	Above Aerodrome Level
ABM	Abeam
ABN	Aerodrome beacon
AC	Alternating Current
AC	Advisory Circular

ACAS	Airborne Collision Avoidance System
ACC	Area Control Centre or Area Control
ACFT	Aircraft
AD	Aerodrome
ADF	Automatic Direction Finding
ADG	Air Driven Generator
ADIZ	Air Defence Identification Zone
ADR	Advisory Route
ADS	Automatic Dependent Surveillance
ADS-B	Automatic Dependent Surveillance – Broadcast
ADS-C	Automatic Dependent Surveillance – Contract
ADT	Autonomous Distress Tracking
AEO	All-Engines-Operative
AFB	Airfield Brief
AFFF	Aqueous Film Forming Foams
AFM	Aircraft Flight Manual
AFN	ATS Facilities Notification
AFN	Aircraft Flight Notification
AFTN	Aeronautical Fixed Telecommunications Network
AGL	Above Ground Level
AGNIS	Azimuth Guidance for Nose In Stand
AHRS	Attitude Heading Reference System
AIC	Aeronautical Information Circular
AIP	Aeronautical Information Publication
AIREP	Air Report
AIS	Aeronautical Information Service
AJG	Air to Ground
ALARP	As Low As Reasonably Practicable
ALSF	Approach Lighting System with Sequenced Flashing Lights
ALT	Altitude
AM	Accountable Manager
AMC	Acceptable Means of Compliance
AML	Aircraft Maintenance Licence
AMSL	Above Mean Sea Level

ANP	Actual Navigation Performance
AOC	Air Operator Certificate
AOC	Aeronautical Operational Control
AOM	Airfield Operating Minima
APCH	Approach
App	Approach control office or approach control
APR	April
APRX	Approximate or approximately
APT	Airport
APU	Auxiliary Power Unit
APV	Approach Procedure with Vertical Guidance
ARA	Authority Requirements for Aircrew
ARA	Airborne Radar Approach
ARO	Authority Requirements for Air Operations
ARP	Aerospace Recommended Practices
ARP	Aerodrome Reference Point
ASC	Air Safety Committee
ASDA	Accelerate-stop Distance Available
ASE	Altimeter System Error
ASR	Air Safety Report
AT	Aircraft Tracking
ASRI	Air Safety Report Investigation
ATA	Air Transport Association
ATA	Actual time of arrival
ATC	Air Traffic Control (in general)
ATIS	Automatic Terminal Information Service
ATO	Approved Training Organization
ATOW	Actual Take-off Weight
ATPL	Airline Transport Pilots Licence
ATQP	Alternative Training and Qualification Programme
ATS	Air Traffic Services
ATSC	Air Traffic Service Communication
ATZ	Aerodrome Traffic Zone
AUG	August

AUW	All Up Weight
AVASIS	Abbreviated Visual Approach Slope Indicator System
AVBL	Available
AVGAS	Aviation Gasoline
AVTAG	Aviation Turbine Gasoline (wide-cut fuel)
AWO	All Weather Operations
AWY	Airway
B37M	Boeing 737-7
B38M	Boeing 737-8/-8200
B39M	Boeing 737-9
B737	Boeing 737-700
B738	Boeing 737-800
B744	Boeing 747-400
B748	Boeing 747-800
B787	Boeing 787 family of aircraft
B737-NG	Boeing 737 (600/700/800/900)
BA	Braking Action
BALS	Basic Approach Lighting System
BCAR	British Civil Airworthiness Requirements
BC	Base Captain
BCN	Beacon (aeronautical ground light)
BCST	Broadcast
BDRY	Boundary
BHP	Brake Horse Power
BITD	Basic Instrument Training Device
BLDG	Building
BM	Base Manager
BRG	Bearing
BS	Base Supervisor
BTN	Between
BTRE	Base TRE
CAME	Continuing Airworthiness Management Exposition
C-PED	Controlled Portable Electronic Device
C/L	Centre-Line (RWY or TWY)

C/W	Cross wind
CAA	Civil Aviation Authority
CAP	Controller Access Parameters
CAT	Commercial Air Transport
CAT I/II/III	Category I/II/III
CBT	Computer-based Training
CC	Cabin Crew
CCM	Cabin Crew Manager
CDFA	Continuous Descent Final Approach
CDL	Configuration Deviation List
CG	Centre of Gravity
CH	Channel
CFIT	Controlled Flight into Terrain
CIS	Chief Instructor Safety
CLSD	Closed
CM	Context Management
CMV	Converted Meteorological Visibility
CNL	Cancel
CofA	Certificate of Airworthiness
COP	Code Of Practice
CoR	Certificate of Registration
CP	Chief Pilot NPFO
CPA	Closest Point of Approach
CPDLC	Controller Pilot Data Link Communication
CPL	Commercial Pilots Licence
CPT	Captain
CPTSR	Captain Special Report
CRE	Class Rating Examiner
CRI	Class Rating Instructor
CRM	Crew Resource Management
CRS	Course
CS	Certification Specifications
CSA	Cabin Services Agent
CSFF	Cold-soaked Fuel Frost

CSR	Cabin Crew Special Report
CSS	Cabin Services Supervisor
CTA	Control area
CTL	Control
CTR	Control zone
CU	Command Upgrade
CVR	Cockpit Voice Recorder
CWY	Clearway
D	Danger area (followed by identification)
D-METAR	Data link – Meteorological Aerodrome Report
D-ATIS	Digital Automatic Terminal Information Service
D-FIS	Data Link Flight Information Service
DA/H	Decision Altitude/Height
DALTA	Descent, Approach, Land, Taxi, Apron
DAP	Down linked Aircraft Parameters
DC	Direct Current
DCL	Departure Clearance
DCT	Direct
DDL	Deferred Defects List
DEC	December
DEG	Degrees
DENT	Identification
DEST	Destination
DEV	Deviation
DF	Direction Finding
DFMC	Dual Frequency Multi Constellation
DG	Dangerous Goods
DH	Decision Height
DI	Daily Inspection
DIFF	Deck Integrated Fire Fighting system
DIST	Distance
DLR	Data Link Recorder
DME	Distance Measuring Equipment
DPATO	Defined Point After Take-Off

DPBL	Defined Point Before Landing
DPCT	Deputy Pilot Crew Training
DPFO	Deputy Pilot Flight Operations
DR	Decision Range
DR	Dead Reckoning
DSTRK	Desired Track
DUB	Dublin
DVOR	Doppler VHF Omni-directional range
E	East or Eastern Longitude
EASA	European Union Aviation Safety Agency
EAT	Expected Approach Time
EC	European Community
ECAC	European Civil Aviation Conference
EDTO	Extended Diversion Time Operation
EFB	Electronic Flight Bag
EFIS	Electronic Flight Instrument System
EGNOS	European Geostationary Navigation Overlay Service
EGT	Exhaust Gas Temperature
ELEV	Elevation
ELT	Emergency Locator Transmitter
ELT (AD)	Emergency Locator Transmitter (Automatically Deployable)
ELT (AF)	Emergency Locator Transmitter (Automatic Fixed)
ELT (AP)	Emergency Locator Transmitter (Automatic Portable)
EM	Emission, Emergency
EPE	Estimated Position of Error
EPR	Engine Pressure Ratio
EPU	Estimated Position of Uncertainty
EQPT	Equipment
ERA	En-route Alternate (Aerodrome)
ERP	Emergency Response Plan
ETA	Estimated Time of Arrival
ETD	Estimated Time of Departure
ETOPS	Extended Range Operations with Two-engined Aeroplanes
ETP	Equal Time Point

EU	European Union
EUROCAE	European Organisation for Civil Aviation Equipment
EVS	Enhanced Vision System
EXC	Except
F	Fahrenheit, Flashing
FAA	Federal Aviation Administration
FAF	Final Approach Fix
FALS	Full Approach Lighting System
FANS	Future Air Navigation Systems
FAP	Final Approach Point
FAR	Federal Aviation Regulation
FAS	Final Approach Segment
FATO	Final Approach and Take-off
FC	Flight Crew
FCI	Flight Crew Instruction
FCL	Flight Crew Licensing
FCOM	Flight Crew Operations Manual
FDM	Flight Data Monitoring
FDO	Flying Display Operation
FDR	Flight Data Recorder
FEB	February
FFS	Full Flight Simulator
FGS	Flight control/Guidance System
FI	Flight Instructor
FIC	Flight Information Centre
FIR	Flight Information Region
FIS	Flight Information Service
FL	Flight Level
FLG	Flashing
FLIPCY	Flight Plan Consistency
FLT	Flight
FLTA	Forward-Looking Terrain Avoidance
FMECA	Failure Mode, Effects and Criticality Analysis
FMS	Flight Management System

FNPT	Flight and Navigation Procedures Trainer
FO	First Officer
FOD	Foreign Object Damage
FOPS	Flight Operations
FPL	Flight plan (ATC)
Fpm	Feet per minute
FREQ	Frequency
FRI	Friday
FSO	Flight Safety Officer
FSTD	Flight Simulation Training Device
FT	Feet (dimensional unit)
Ft	feet
FTD	Flight Training Device
FTE	Full Time Equivalent
FTL	Flight and Duty Time Limitations
FM	Fatigue Management
FRF	Final Reserve Fuel
G	Gravity (Aircraft Load Factor)
g	gram
G/S	Glide Slope
GA	General Aviation
GADSS	Global Aeronautical Distress and Safety System
GAGAN	GPS Aided Geo Augmented Navigation
GAL	Gallanis
GAST	GBAS Approach Service Type
GBAS	Ground-Based Augmentation System
GCA	Ground Controlled Approach
GCI	Ground Crew Instruction
GCAS	Ground Collision Avoidance System
GEN	General
GEO	Geographic or true
GIDS	Ground Ice Detection System
GLS	GBAS landing system
GM	Guidance Material

GMP	General Medical Practitioner
GMT	Greenwich Mean Time
GND	Ground
GNSS	Global Navigation Satellite System
GOPS	Ground Operations
GP	Glide Path
GPS	Global Positioning System
GPWS	Ground Proximity Warning System
GRF	Global Reporting Format
GS	Ground Speed
GS	Glide Slope
GWT	Gross Weight
H	Helicopter
H24	Continuous day and night service
HBN	Hazard beacon
HDG	Heading
HEL	Helicopter
HEMS	Helicopter Emergency Medical Service
HF	High Frequency (3000 to 30,000 kHz)
Hg	Mercury
HGT	Height or height above
HHO	Helicopter Hoist Operation
HI	High intensity
HIALS	High Intensity Approach Lighting System
HIGE	Hover In Ground Effect
HIRL	High Intensity Runway Light
HJ	Sunrise to sunset
HLL	Helideck Limitations List
HN	Sunset to sunrise
HOGE	Hover Out of Ground Effect
HOL	Holiday
HOT	Holdover Time
HoT	Hold-over Time
hPa	Hectopascal (Millibar) KHZ

HPL	Human Performance and Limitations
HR	Hours
HS	Service available during hours of scheduled operations
HSI	Horizontal Situation Indicator
HST	High Speed taxi-way Turn-off
HUD	Head-Up Display
HUDLS	Head-Up Guidance Landing System
HUMS	Health Usage Monitor System
HZ	Hertz (cycle per second)
IAF	Initial Approach Fix
IAL	Instrument Approach and Landing (chart)
IALS	Intermediate Approach Lighting System
IAS	Indicated Air Speed
IATA	International Air Transport Association
IBN	Identification Beacon
ICAO	International Civil Aviation Organization
IDE	Instruments, Data and Equipment
IF	Intermediate Fix
IFR	Instrument Flight Rules
IFSD	In-Flight Shutdown
IGE	In Ground Effect
ILS	Instrument Landing System
IM	Inner Marker
IMC	Instrument Meteorological Conditions
iMMR	Integrated Multi Mode Receiver
In	inches
INFO	Information
INOP	Inoperative
INS	Inertial Navigation System, Inoles
INT	Intersection
INTXN	Intersection
IP	Intermediate Point
IR	Instrument Rating
IR	Implementing Rule

IRS	Inertial Reference System
IRVR	Instrumented RYR
ISA	International Standard Atmosphere
ISO	International Organization for Standardization
IV	Intravenous
JAA	Joint Aviation Authorities
JAN	January
JAR	Joint Aviation Requirements
JUL	July
JUN	June
Kg	kilograms
km	kilometres
kt	knots
L	Left (runway identification)
LAT	Latitude
LB(s)	Pound(s) (weight)
LCC	Line Check Captain
LCG	Load Classification Group
LCN	Load Classification Number
LDA	Low Drag Approach
LDA	Landing Distance Available
LDG	Landing
LDP	Landing Decision Point
LDTA	Landing Distance at Time of Arrival
LED	Light-Emitting Diode
LF	Low Frequency
LGT	Light or Lighting
LGTD	Lighted
LH	Left Hand
LHS	Left Hand Seat
LI	Low Intensity
LIFUS	Line Flying Under Supervision
LITAS	Low Intensity Approach System
LLZ	Localizer

LM (M)	Locator, Middle (Marker)
LNAV	Lateral Navigation
LO	Locator, Outer
LoA	Letter of Acceptance
LOC	Locator
LOC	Localiser
LOCE	Line-Oriented Compliance Evaluation
LOE	Line-Oriented Evaluation
LOFT	Line-Oriented Flight Training
LONG	Longitude
LORAN	LORAN (Long Range Air Navigation System)
LOS	Limited Obstacle Surface
LPV	Localiser Performance with Vertical Guidance
LRCS	Long Range Communication System
LRNS	Long Range Navigation System
LT	Left, Local Time
LTC	Line Training Captain
LT(S)	Light(s)
LVL	Level (Vertical Distance)
LVO	Low Visibility Operation
LVP	Low Visibility Procedures
LVTO	Low Visibility Take-Off
M	Metres
MA	Monitored Approach
MAG	Magnetic
MAINT	Maintenance
MALS	Medium Intensity Approach Lighting System
MALSF	Medium Intensity Approach lighting system with Sequenced Flashing Lights
MALSR	Medium Intensity Approach lighting system with Runway Alignment Indicator Lights
MAPt	Missed Approach Point (also MAP)
MAR	March
MAX	Maximum
MAY	Malta Air

mb(s)	Millibar(s)
MCTOM	Maximum Certified Take-Off Mass
MCAS	Maneuvering Characteristics Augmentation System
MCF	Maintenance Check Flight
MCP	Mode Control Panel
MCP	Maintenance Check Pilot
MCPS	Maintenance Check Pilot Seattle
MDA	Minimum Descent Altitude
MDH	Minimum Descent Height
MEA	Minimum En-route Altitude
MED	Medical
MED L	Mid Exit Door Left
MED R	Mid Exit Door Right
MEL	Minimum Equipment List
MET	Meteorological and Meteorology
METAR	Meteorological Aerodrome Report
MF	Medium Frequency
MFRH	Minimum Flap Retraction Height for Take-off
MGA	Minimum Grid Altitude
MHA	Minimum Holding Altitude
MHz	megahertz
MHZ	Megahertz
MID	midpoint
MIL	Military
MIN	Minimum, Minute
MKD	Marked
MKR	Marker Radio Beacon
MLR	Manuals, Logs and Records
MLS	Microwave Landing System
MLW	Maximum Certified Landing Weight
MLX	millilux
MM	Multi-Mode
mm	millimetres
MM	Middle Marker

MMEL	Master Minimum Equipment List
MNR	Minimum Noise Routings
MOC	Minimum Obstacle Clearance
MOCA	Minimum Obstruction Clearance Altitude
MOCA	Minimum Obstacle Clearance Altitude
MON	Monday
MOPSC	Maximum Operational Passenger Seating Configuration
MORA	Minimum Off-Route Altitude
MPSC	Maximum Passenger Seating Capacity
MRA	Minimum Reception Altitude
MSA	Minimum Sector Altitude
MSA	Minimum Safe Altitude
MSAS	Multi-functional Satellite Augmentation System
MSL	Mean Sea Level
MTOW	Maximum Certified Take-off Weight
MZFW	Maximum Certified Zero Fuel Weight
N	North or Northern Latitude
NAA	National Aviation Authority
NADP	Noise Abatement Departure Procedure
NALS	No Approach Lighting System
NAT	North Atlantic
NAT HLA	North Atlantic High-Level Airspace
NAVAID	Navigational Aid
NCC	Non-Commercial operations with Complex motor-powered aircraft
NCO	Non-Commercial operations with Other-than-complex motor-powered aircraft
NDB	Non-directional Radio Beacon
NEI	Non-environmental Icing, also CSFF
NF	Free power turbine speed
NHP	Non Handling Pilot
NM	Nautical Miles
NM(S)	Nautical Mile(s)
NNC	Non Normal Conditions
No	Number
NOTAM	Notice To Airmen

NOTAM	A notice containing information concerning the establishment, condition or change in any aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential to personnel concerned with flight operations.
NOTECHS	Non-technical skills evaluation
NOTOC	Notification to captain
NOV	November
NP	Nominated Person
NPA	Notice of Proposed Amendment
NPA	Non-Precision Approach
NPCT	Nominated Person Crew Training
NPCA	Nominated Person Continuing Airworthiness Management/Chief Engineer
NPFO	Nominated Person Flight Operations
NPGO	Nominated Person Ground Operations
NR	Number
NVD	Night Vision Device
NVG	Night Vision Goggles
NVIS	Night Vision Imaging System
O/R	On request
OAC	Oceanic Area Control Centre
OAT	Outside Air Temperature
OBST	Obstruction
OCA	Oceanic Control Area
OCA	Obstacle Clearance Altitude
OCC	Operations Control Center
OCC	Operator Conversion Course
OCH	Obstacle Clearance Height
OCL	Oceanic Clearance
OCL	Obstacle Clearance Limit
OCT	October
ODALS	Omni-Directional Approach Lighting System
OEI	One-Engine-Inoperative
OFS	Obstacle-Free Surface
OFDM	Operational Flight Data Monitoring
OFP	Operational Flight Plan

OGE	Out of Ground Effect
OIP	Offset Initiation Point
OK	Serviceable
OM	Outer Marker
OM	Operations Manual
OMA	Operations Manual Part A
OMB	Operations Manual Part B
OMC	Operations Manual Part C
OMD	Operations Manual Part D
OML	Operational Multi-pilot Limitation
OMR	Operations Manual Revision
ONC	Operational Navigation Chart
OPS	Operations
ORO	Organisation Requirements for Air Operations
OTS CAT II	Other Than Standard Category II
P	Prohibited Area (followed by identification)
PA	Public Address, Pressure Altitude
PAPI	Precision Approach Path Indicator
PAQ	Post a Query
PAR	Precision Approach Radar
PBE	Protective Breathing Equipment
PBN	Performance-Based Navigation
PCDS	Personnel Carrying Device System
PCF	Passenger Cum Freight
PCN	Pavement Classification Number
PD	Performance Department
PDA	Premature Descent Alert
PDP	Pre-Determined Point
PED	Portable Electronic Device
PERM	Permanent
PF	Pilot Flying
PFC	Porous Friction Course
PFLR	Post Flight Localization and Recovery
PH	Public Holiday

PIC	Pilot-In-Command, aka commander
PIN	Personal Identification Number
PIREP	Pilot Report
PJE	Parachute Jumping Exercise
PLN	Flight Plan
PLOC	Prolonged Loss of Communication
PM	Pilot Monitoring
PN	Prior Notice Required
PNR	Point of No Return
POH	Pilot's Operating Handbook
POSN	Position
PPR	Prior Permission Required
PRM	Person with Reduced Mobility
PROC	Procedure
PSN	Position
PT	Procedure Turn
QAR	Quick Access Recorder
QDM	Magnetic heading (zero wind), Bearing to facility
QDR	Magnetic Bearing from Facility
QFE	Atmospheric pressure at aerodrome elevation (or at runway threshold)
QNE	Standard pressure setting 1013.2
QNH	atmospheric pressure at nautical height
QNH	Altimeter sub-scale setting to obtain elevation (when on the ground) Station setting or Area Setting Altitude above sea level
QTE	True Bearing
R	Right (runway identification)
R	Restricted Area (followed by identification)
RA	Resolution Advisory
RA	Radio Altimeter
RCAM	Runway Condition Assessment Matrix
RCC	Rescue Co-ordination Centre
RCF	Reduced Contingency Fuel
RCLL	Runway Centre Line Lights
RCR	Runway Condition Report

RCRS	Malta Air Confidential Reporting System
RDL	Radial
REF	Reference to or Refer to
REIL	Rwy End Ident Light
REQ	Request or Requested
RF	Radio Frequency
RFC	Route Facility Chart
RH	Revision Highlights
RI	Rectification Interval
RI	Ramp Inspection
RIBETS	Route Check, Instrument Crosscheck, Takeoff Briefing, Initial Emergency Briefing, Taxi Routing, SID Briefing
RIE	Rectification Interval Extension
RMA	Regional Monitoring Agency
RNAV	area Navigation
RNG	Radio Range
RNP	Required Navigation Performance
ROC	Rate of Climb
ROD	Rate of Descent
RP	Rotation Point
RP	Reporting Point
RPL	Repetitive Flight Plan
RSR	En-route Surveillance Radar
RT	Right, Radio Telephony
RTA	Required Time of Arrival
RTCA	Radio Technical Commission for Aeronautics
RTE	Route
RTF	Radio Telephone
RTF	Radius to Fix
RTOM	Reduced Take-Off Mass
RTOW	Regulated Take-Off Weight
RTZL	Runway Touchdown Zone Lights
RVR	Runway Visual Range
RVSM	Reduced Vertical Separation Minima

RWY	Runway
RWYCC	Runway Condition Code
RYR	Ryanair
S	South or Southern Latitude
SA	Safety Altitude
SAFA	Safety Assessment of Foreign Aircraft
SALS	Simple Approach Lighting System
SALSF	Simple Approach lighting system with Sequenced Flashing Lights
SAP	System Access Parameters
SAP	Stabilised Approach
SAP	Soon as Possible
SAR	Search and Rescue
SAS	Stability Augmentation System
SAT	Saturday, Static Air Temperature
SBAS	Satellite-Based Augmentation System
SCC	Senior Cabin Crew
SCCM	Senior Cabin Crew Member (No 1)
SCP	Special Category of Passenger
SD	Special Duties
SDCM	System of Differential Correction and Monitoring
SE	Safety Examiner
SEC	Second(s)
SEC	Security Manager
SEP	September
SEP	Safety and Emergency Procedures (Manual)
SEPE	Safety and Emergency Procedures Examiner
SEPI	Safety and Emergency Procedures Instructor
SERA	Standardised European Rules of the Air
SFC	Surface
SFE	Synthetic Flight Examiner
SFI	Synthetic Flight Instructor
SFL	Sequenced Flashing Lights
SH	Safety Height
SI	Safety Instruction

SI	Safety Instructor
SI	Statutory Instrument
SID	Standard Instrument Departure
SKED	Schedule or Scheduled
SM	Security Manager
SMM	Safety Management Manual
SMS	Safety Management System
SOP	Standard Operating Procedure
SPA	Operations requiring specific approvals
SPECI	aviation selected special weather report
SPI	Safety Procedure Instruction
SPO	specialised operations
SR	Sunrise
SRA	Surveillance Radar Approach
SRA	Safety Risk Assessment
SRE	Surveillance Radar Element of Precision Radar System
SS	Sunset
SSALF	Simplified Short Approach Lighting system with Sequenced Flashing Lights
SSALR	Simplified Short Approach Lighting system with Runway Alignment Indicator Lights
SSALS	Simplified Short Approach Lighting System
SSB	Single Sideband
SSEC	Static Source Error Correction
SSeFM	Safety, Security and Fatigue Management
SSR	Secondary Surveillance Radar
SST	Supersonic Transport
STAR	Standard Terminal Arrival Route
STC	Supplemental Type Certificate
STD	Scheduled Departure Time, Standard
STN	Station
STOL	Short Take-off and Landing
SUN	Sunday
SVC	Service
SVCBL	Serviceable

SWY	Stop way
T	True (bearings)
T–PED	Transmitting Portable Electronic Device
T/O	Take-Off
TA	Transition Altitude
TA	Traffic Advisory
TAC	Terminal Approach Chart
TACAN	UHFTactical Air Navigation Aid
TAR	Terminal Area Surveillance Radar
TAS	True Airspeed
TAT	Total Air Temperature
TAWS	Terrain Awareness Warning System
TC	Type Certificate
TC	Technical Crew
TCAS	Traffic Collision Avoidance System
TCCA	Transport Canada Civil Aviation
TCH	Type Certificate Holder
TDP	Take-off Decision Point
TDZ	Touchdown Zone
TEL	Telephone
TEM	Threat and Error Management
TEMP	Temperature
TFC	Traffic
THR	Threshold
THU (RS)	Thursday
TI	Technical Instructions
TIL	Until
TIT	Turbine Inlet Temperature
TKDF	Take-Off U Transition Level
TMA	Terminal Control Area
TM CAD	Transport Malta – Civil Aviation Directorate
TMG	Touring Motor Glider
TMM	Transmissometer (IRVR)
TOC	Top of Climb

TOD	Top of Descent
TODA	Take-off Distance Available (Aeroplanes)
TORA	Take-off Run Available
TR (K)	Track
TRE	Type Rating Examiner
TRI	Type Rating Instructor
TSE	Total System Error
TUE(S)	Tuesday
TVE	Total Vertical Error
TVOR	Terminal VOR
TWIP	Terminal Weather Information for Pilots
TWR	Aerodrome Control Tower or Aerodrome Control
TWY	Taxi-way
U/S	Unserviceable
UAC	Upper Area Control Centre
UAR	Upper Air Route
UDF	Ultra high frequency Direction-Finding Station
UFN	Until Further Notice
UH	Ultra High Frequency (300 to 3000 MHz)
UIC	Upper Information Centre
UIR	Upper Flight Information Centre
ULD	Underwater Locating Devices
UMS	Usage Monitoring System
UNL	Unlimited
UTA	Upper Control Area
UTC	Co-ordinated Universal Time
V2	Take-off safety speed
VAL	Visual Approach and Landing (Chart)
VAR	Variation
VASIS	Visual Approach Slope Indicator System
VAT	Indicated airspeed at threshold
VDB	VHF Data Broadcast
VDF	Very High Frequency Direction-Finding
VFR	Visual Flight Rules

VHF	Very High Frequency (30 to 300 MHz)
VIA	By Way Of
VIP	Very Important Person
VIS	Visibility
VMC	Visual Meteorological Conditions
VMO	Maximum operating speed
VNAV	Vertical navigation
VOLMET	Meteorological Information for Aircraft in Flight
VOR	VHF Omni-Directional Radio Range
VORTAC	VOR and TACAN Combination
VOT VOR	Receiver Test Facility
VSI	Vertical Speed Indicator
VTOL	Vertical Take-off and Landing
VV	Vertical Visibility
W	West or Western Longitude
W/V	Wind Velocity
WAAS	Wide Area Augmentation System
WAC	World Aeronautical Chart
WED	Wednesday
WEF	With Effect From or Effective From
WIE	With Immediate Effect
WIFI	Wireless fidelity
WIP	Work In Progress
WSR	Wet Skid Resistant
WX	Weather
XBAR	Crossbar (of Approach Lighting System)
ZFTT	Zero Flight-Time Training

Definitions

Definitions provided are taken from Regulation (EU) No 965/2012, Regulation (EU) No 379/2014 except where stated otherwise.

- ‘Accelerate-Stop Distance Available (ASDA)’ means the length of the take-off run available plus the length of stop way, if such stop way is declared available by the State of the aerodrome and is capable of bearing the mass of the aeroplane under the prevailing operating conditions;

- ‘Acceptable Means of Compliance (AMC)’ means non-binding standards adopted by the Agency to illustrate means to establish compliance with Regulation (EU) 2018/1139 and its Implementing Rules;
- ‘Acceptance checklist’ means a document used to assist in carrying out a check on the external appearance of packages of dangerous goods and their associated documents to determine that all appropriate requirements have been met with;
- ‘Adequate aerodrome’ means an aerodrome on which the aircraft can be operated, taking account of the applicable performance requirements and runway characteristics;
- ‘Aerodrome operating minima’ means the limits of usability of an aerodrome for:
 1. take-off, expressed in terms of runway visual range (RVR) and/or visibility and, if necessary, ceiling;
 2. landing in 2D instrument approach operations, expressed in terms of visibility and/or RVR, minimum descent altitude/height (MDA/H) and, if necessary, ceiling;
 3. landing in 3D instrument approach operations, expressed in terms of visibility and/or RVR and decision altitude/height (DA/H) as appropriate to the type and/or category of the operation;
- ‘Aeroplane’ means an engine-driven fixed-wing aircraft heavier than air that is supported in flight by the dynamic reaction of the air on its wings;
- ‘Airport moving map display (AMMD)’ means a software application that displays an airport map on a display device and uses data from a navigation source to depict the aircraft current position on this map while the aircraft is on the ground.
- For the purpose of passenger classification:
- ‘Adult’ means a person of an age of 12 years and above;
- ‘Child/children’ means persons who are of an age of two years and above but who are less than 12 years of age;
- ‘Infant’ means a person under the age of two years;
- ‘Aircraft’ means a machine that can derive support in the atmosphere from the reactions of the air other than the reactions of the air against the earth’s surface;
- ‘Aircraft Tracking’ means a ground based process that maintains and updates, at standardised intervals, a record of the four dimensional position of individual aircraft in flight;

- ‘Aircraft Tracking System’ means a system that relies on aircraft tracking in order to identify abnormal flight behaviour and provide alert;
- ‘Alternative means of compliance’ means those means that propose an alternative to an existing acceptable means of compliance or those that propose new means to establish compliance with Regulation (EU) 2018/1139 and its Implementing Rules for which no associated AMC have been adopted by the Agency;
- ‘Anti-icing’, in the case of ground procedures, means a procedure that provides protection against the formation of frost or ice and accumulation of snow on treated surfaces of the aircraft for a limited period of time (hold-over time);
- ‘Cabin crew member’ means an appropriately qualified crew member, other than a flight crew or technical crew member, who is assigned by an operator to perform duties related to the safety of passengers and flight during operations;
- ‘Calendar Year’ The calendar year in Ryanair FTL is 01 Jan to 31 Dec;
- ‘Ceiling’ means the height above the ground or water of the base of the lowest layer of cloud below 6000 m (20,000 ft) covering more than half the sky;
- ‘Certification Specifications’ (CS) means technical standards adopted by the Agency indicating means to show compliance with Regulation (EU) 2018/1139 and its Implementing Rules and which can be used by an organisation for the purpose of certification;
- ‘Circling’ means the visual phase of a circling approach operation;
- ‘Circling approach operation’ means a Type A instrument approach operation to bring an aircraft into position for landing on a runway that is not suitably located for a straight-in approach;
- ‘Clearway’ means a defined rectangular area on the ground or water under the control of the appropriate authority, selected or prepared as a suitable area over which an aeroplane may make a portion of its initial climb to a specified height;
- ‘Cloud base’ means the height of the base of the lowest observed or forecast cloud element in the vicinity of an aerodrome normally measured above aerodrome elevation;
- ‘Cloud of operational significance’ means a cloud with the height of cloud base below 1500 m (5000 ft) or below the highest minimum sector altitude, whichever is greater, or a cumulonimbus cloud or a towering cumulus cloud at any height;

- ‘Cockpit voice recorder (CVR)’ means a crash-protected flight recorder that uses a combination of microphones and other audio and digital inputs to collect and record the aural environment of the flight crew compartment and communications to, from and between the flight crew members;
- ‘Contaminated runway’ means a runway of which a significant portion of its surface area (whether in isolated areas or not) within the length and width being used is covered by one or more of the substances listed under the runway surface condition descriptors;
- ‘Contingency fuel/energy’ means the fuel required to compensate for unforeseen factors that could have an influence on the fuel consumption to the destination aerodrome;
- ‘Continuous Descent Final Approach (CDFA)’ means a technique, consistent with stabilised approach procedures, for flying the Final Approach Segment (FAS) of an instrument Non-precision Approach (NPA) procedure as a continuous descent, without level-off, from an altitude/height at or above the final approach fix altitude/height:
 1. for straight-in approach operations, to a point approximately 15 m (50 ft) above the landing runway threshold or the point where the flare manoeuvre begins; or
 2. for circling approach operations, until MDA/H or visual flight manoeuvre altitude/height is reached;
- ‘Controlled portable electronic device (C-PED)’ means a PED subject to administrative control by the operator that uses it. This includes, inter alia, tracking the allocation of the devices to specific aircraft or persons and ensuring that no unauthorized changes are made to the hardware, software, or databases. C-PEDs can be assigned to the category of non-intentional transmitters or T-PEDs.
- ‘Converted Meteorological Visibility (CMV)’ means a value, equivalent to an RVR, which is derived from the reported meteorological visibility;
- ‘Crew member’ means a person assigned by an operator to perform duties on board an aircraft;
- ‘Critical phases of flight’ in the case of aeroplanes means the take-off run, the take-off flight path, the final approach, the missed approach, the landing, including the landing roll, and any other phases of flight as determined by the pilot-in-command or commander;
- ‘Current fuel scheme’ means the approved fuel scheme that is currently used by the operator;
- ‘Damp runway’ means a runway where the surface is not dry, but when the moisture on it does not give it a shiny appearance;

- ‘Dangerous Goods (DG)’ means articles or substances which are capable of posing a risk to health, safety, property or the environment and which are shown in the list of dangerous goods in the Technical Instructions or which are classified according to those instructions;
- ‘Dangerous goods accident’ means an occurrence associated with and related to the transport of dangerous goods by air which results in fatal or serious injury to a person, or major property or environmental damage;
- ‘Dangerous goods incident’ means:
 1. an occurrence other than a dangerous goods accident associated with and related to the transport of dangerous goods by air, not necessarily occurring on board an aircraft, which results in injury to a person, property damage, fire, breakage, spillage, leakage of fluid or radiation or other evidence that the integrity of the packaging has not been maintained;
 2. any occurrence relating to the transport of dangerous goods which seriously jeopardises an aircraft or its occupants;
- ‘Decision Altitude (DA) or Decision Height (DH)’ means a specified altitude or height in a 3D instrument approach operation at which a missed approach procedure must be initiated if the required visual reference to continue the approach has not been established;
- ‘De-icing’, in the case of ground procedures, means a procedure by which frost, ice, snow or slush is removed from an aircraft in order to provide uncontaminated surfaces;
- ‘Demonstration flight’ means a flight to demonstrate the aircraft’s capabilities to persons on board;
- ‘Dry lease agreement’ means an agreement between undertakings pursuant to which the aircraft is operated under the air operator certificate (AOC) of the lessee;
- ‘Dry operating mass’ means the total mass of the aircraft ready for a specific type of operation, excluding usable fuel and traffic load;
- ‘EFB Administrator’ is responsible for the complete EFB system with appropriate authority within Ryanair's management structure;
- ‘EFB application’ means a software application installed on an EFB host platform that provides one or more specific operational functions which support flight operations.
- ‘EFB host platform’ means the hardware equipment in which the computing capabilities and basic software reside, including the operating system and the input/output software.

- ‘EFB installed resources’ means certified EFB hardware components external to the EFB host platform itself, such as input/output components (installed remote displays, keyboards, pointing devices, switches, etc.) or a docking station.
- ‘EFB mounting device’ means an aircraft certified part that secures a portable or installed EFB, or EFB system components.
- ‘EFB system supplier’ means the company responsible for developing, or for having developed, the EFB system or part of it.
- ‘EFB system’ means the hardware equipment (including any battery, connectivity provisions, input/output components) and software (including databases and the operating system) needed to support the intended EFB application(s).
- ‘Electronic Flight Bag (EFB)’ means an electronic information system, comprised of equipment and applications for flight crew, which allows for the storing, updating, displaying and processing of EFB functions to support flight operations or duties.
- ‘Emergency exit’ means an installed exit-type egress point from the aircraft that allows maximum opportunity for cabin and flight crew compartment evacuation within an appropriate time period and includes floor level door, window exit or any other type of exit, for instance hatch in the flight crew compartment and tail cone exit;
- ‘Final Approach Segment (FAS)’ means that segment of an Instrument Approach Procedure (IAP) in which alignment and descent for landing are accomplished;
- ‘Flight Data Monitoring (FDM)’ means the proactive and non-punitive use of digital flight data from routine operations to improve aviation safety;
- ‘Flight Data Recorder (FDR)’ means a crash-protected flight recorder that uses a combination of data sources to collect and record parameters that reflect the state and performance of the aircraft;
- ‘Flight following’ means the recording in real time of departure and arrival messages by operational personnel to ensure that a flight is operating and has arrived at the destination aerodrome or an alternate aerodrome;
- ‘Flight monitoring’ means, in addition to the requirements defined for flight following:
 1. operational monitoring of flights by suitably qualified operational control personnel from departure throughout all phases of the flight;
 2. communication of all available and relevant safety information between the operational control personnel on the ground and the flight crew; and

3. critical assistance to the flight crew in the event of an in-flight emergency or security issue, or at the request of the flight crew;
- ‘Flight recorder’ means any type of recorder that is installed on the aircraft for the purpose of facilitating accident or incident safety investigations;
 - ‘Flight Simulation Training Device (FSTD)’ means a training device which is: in the case of aeroplanes, a Full Flight Simulator (FFS), or a Flight Training Device (FTD), a Flight And Navigation Procedures Trainer (FNPT), or a Basic Instrument Training Device (BITD);
 - ‘Flight time’ means the total time from the moment an aeroplane first moves for the purpose of taking off until the moment the aeroplane finally comes to rest at the end of the flight;
 - ‘Flight watch’ means, in addition to all elements defined for 'flight monitoring', the active tracking of a flight by suitably qualified operational control personnel throughout all phases of the flight to ensure that the flight is following its prescribed route without unplanned deviations, diversions or delays;
 - ‘GBAS Approach Service Type (GAST)’ means a matched set of airborne and ground performance and functional requirements that are intended to be used in concert in order to provide approach guidance with quantifiable performance. Four types of approach service, GAST A, GAST B, GAST C and GAST D, are defined. GAST A, B and C correspond to NPA with LNAV minima, NPA with LNAV/VNAV minima and precision GLS with Category I minima. GAST D refers to CAT III operations;
 - ‘GBAS Landing System (GLS)’ means an approach landing system using ground based augmented global navigation satellite system (GNSS/GBAS) information to provide guidance to the aircraft based on its lateral and vertical GNSS position. It uses geometric altitude reference for its final approach slope;
 - ‘Global Reporting Format (GRF)’ is the standardised methodology for assessing and reporting runway surface conditions, enabling the harmonised assessment and reporting of runway surface conditions.
 - ‘Go-around’ means a transition from an approach operation to a stabilised climb. This includes manoeuvres conducted at or above the MDA/H or DA/H, or below the DA/H (balked landings);
 - ‘Grounding’ means the formal prohibition of an aircraft to take-off and the taking of such steps as are necessary to detain it;
 - ‘Head-Up Display Landing System (HUDLS)’ means the total airborne system which provides head-up guidance to the pilot to enable the pilot to either control the aircraft or to monitor the autopilot during take-off (if

applicable), approach and landing (and roll-out if applicable), or go-around. It includes all the sensors, computers, power supplies, indications and controls;

- ‘Hold-over Time (HoT)’ means the estimated time the anti-icing fluid will prevent the formation of ice and frost and the accumulation of snow on the protected (treated) surfaces of an aeroplane;
- ‘Human–Machine Interface (HMI)’ means a component of certain devices that is capable of handling human– machine interactions. The interface consists of hardware and software that allow user inputs to be interpreted and processed by machines or systems that, in turn, provide the required results to the user.
- ‘Installed EFB’ means an EFB host platform installed in an aircraft, capable of hosting type A and/or type B EFB applications. It may also host certified applications. It is an aircraft part, and, is therefore, covered by the aircraft airworthiness approval.
- ‘Landing Decision Point (LDP)’ means the point used in determining landing performance from which, an engine failure having been recognised at this point, the landing may be safely continued or a bailed landing initiated;
- ‘Landing Distance Available (LDA)’ means the length of the runway which is declared available by the State of the aerodrome and suitable for the ground run of an aeroplane landing;
- ‘Landing Distance at Time of Arrival (LDTA)’ means a landing distance that is achievable in normal operations based on landing performance data and associated procedures determined for the prevailing conditions at the time of landing;
- ‘Low Visibility Operations (LVOs)’ means an approach or take-off operations on a runway with a runway visual range less than 550 m or with a decision height less than 200 ft;
- ‘Low Visibility Take-off (LVTO)’ means a take-off with an RVR less than 550 m;
- ‘Maintenance check flight’ means a flight carried out to provide reassurance of the aircraft’s performance or to establish the correct functioning of a system that cannot be fully established during ground checks:
 1. as required by the Aircraft Maintenance Manual (AMM) or any other maintenance data issued by Boeing for the continuing airworthiness of the aircraft; or
 2. after maintenance, as required by Malta Air; or
 3. as required by Malta Air engineering for verification of a successful defect rectification; or

4. to assist with fault isolation or troubleshooting.

- ‘Maximum Operational Passenger Seating Configuration (MOPSC)’ means the maximum passenger seating capacity of an individual aircraft, excluding crew seats, established for operational purposes and specified in the operations manual. Taking as a baseline the maximum passenger seating configuration established during the certification process conducted for the Type Certificate (TC), Supplemental Type Certificate (STC) or change to the TC or STC as relevant to the individual aircraft, the MOPSC may establish an equal or lower number of seats, depending on the operational constraints;
- ‘Minimum Descent Altitude (MDA) or Minimum Descent Height (MDH)’ means a specified altitude or height in a 2D instrument approach or circling approach operation below which descent must not be made without the required visual reference;
- ‘Minor failure condition’ means a failure condition that would not significantly reduce aircraft safety, and which involves flight crew actions that are well within their capabilities;
- ‘Miscellaneous (non-EFB) software applications’ means non-EFB applications that support function(s) not directly related to the tasks performed by the flight crew in the aircraft.
- ‘Obstacle Clearance Altitude (OCA) or Obstacle Clearance Height (OCH)’ means the lowest altitude or the lowest height above the elevation of the relevant runway threshold or the aerodrome elevation, as applicable, used in establishing compliance with the appropriate obstacle clearance criteria;
- ‘Operational control’ means the responsibility for the initiation, continuation, termination or diversion of a flight in the interest of safety;
- ‘Performance class A aeroplanes’ means multi-engined aeroplanes powered by turbo-propeller engines with an MOPSC of more than nine or a maximum take-off mass exceeding 5700 kg, and all multi-engined turbo-jet powered aeroplanes;
- ‘Pilot-in-command’ means the pilot designated as being in command and charged with the safe conduct of the flight. For the purpose of commercial air transport operations, the ‘pilot-in-command’ shall be termed the ‘commander’;
- ‘Portable EFB’ means a portable EFB host platform, used on the flight deck, which is not part of the configuration of the certified aircraft.
- ‘Portable Electronic Device (PED)’ means any kind of electronic device, typically but not limited to consumer electronics, brought on board the aircraft by crew members, passengers, or as part of the cargo, that is not included in the configuration of the certified aircraft. It includes all equipment that is able to consume electrical energy. The electrical energy can be

provided from internal sources such as batteries (chargeable or non-rechargeable) or the devices may also be connected to specific aircraft power sources.

- ‘Principal place of business’ means the head office or registered office of the organisation within which the principal financial functions and operational control of the activities referred to in this Regulation are exercised;
- ‘Proficient’ means having demonstrated the necessary skills, knowledge and attitudes that are required to perform any defined tasks to the prescribed standard;
- ‘Ramp inspection’ means the inspection of aircraft, of flight and cabin crew qualifications and of flight documentation in order to verify the compliance with the applicable requirements;
- ‘Rectification interval’ means a limitation on the duration of operations with inoperative equipment;
- ‘Relocation flight’ means a flight with the purpose of changing the aircraft’s location.
- ‘Required navigation performance (RNP) specification’ means a navigation specification for PBN operations which includes a requirement for on-board navigation performance monitoring and alerting;
- ‘Rules of the air’ means the rules established in Commission Implementing Regulation (EU) No 923/2012;
- ‘Runway condition assessment matrix (RCAM)’ means a matrix that allows the assessment of the runway condition code (RWYCC), using associated procedures, from a set of observed runway surface condition(s) and pilot report of braking action.
- ‘Runway condition code (RWYCC)’ means a number, to be used in the runway condition report (RCR), that describes the effect of the runway surface condition on aeroplane deceleration performance and lateral control.
- ‘Runway Condition Report (RCR)’ means a comprehensive standardised report relating to runway surface conditions and its effect on the aeroplane landing and take-off performance.
- ‘Runway Visual Range (RVR)’ means the range over which the pilot of an aircraft on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identifying its centre line;
- ‘Safe forced landing’ means an unavoidable landing or ditching with a reasonable expectancy of no injuries to persons in the aircraft or on the surface;

- ‘Separate runways’ means runways at the same aerodrome that are separate landing surfaces. These runways may overlay or cross in such a way that if one of the runways is blocked, it will not prevent the planned type of operations on the other runway. Each runway shall have a separate approach procedure based on a separate navigation aid;
- ‘Special VFR flight’ means a VFR flight cleared by air traffic control to operate within a control zone in meteorological conditions below VMC;
- ‘Specially prepared winter runway’ means a runway with a dry frozen surface of compacted snow or ice which has been treated with sand or grit or has been mechanically treated to improve runway friction;
- ‘Stabilised Approach (SAp)’ means an approach that is flown in a controlled and appropriate manner in terms of configuration, energy and control of the flight path from a pre-determined point or altitude/height down to a point 50 ft above the threshold or the point where the flare manoeuvre is initiated if higher;
- ‘Sterile flight crew compartment’ means any period of time when the flight crew members are not disturbed, except for matters critical to the safe operation of the aircraft and/or the safety of the occupants;
- ‘Take-off Alternate Aerodrome’ means an alternate aerodrome at which an aircraft can land should this become necessary shortly after take-off and if it is not possible to use the aerodrome of departure;
- ‘Take-off Decision Point (TDP)’ means the point used in determining take-off performance from which, an engine failure having been recognised at this point, either a rejected take-off may be made or a take-off safely continued;
- ‘Take-Off Distance Available (TODA)’ in the case of aeroplanes means the length of the take-off run available plus the length of the clearway, if provided;
- ‘Take-off flight path’ means the vertical and horizontal path, with the critical engine inoperative, from a specified point in the take-off for aeroplanes to 1,500 ft above the surface;
- ‘Take-off mass’ means the mass including everything and everyone carried at the commencement of the take-off run for aeroplanes;
- ‘Take-Off Run Available (TORA)’ means the length of runway that is declared available by the State of the aerodrome and suitable for the ground run of an aeroplane taking off;
- ‘Technical Instructions (TI)’ means the latest effective edition of the ‘Technical instructions for the safe transport of dangerous goods by air’, including the supplement and any addenda, approved and published by the International Civil Aviation Organisation;

- ‘Traffic load’ means the total mass of passengers, baggage, cargo and carry-on specialist equipment, including any ballast;
- ‘Transmitting PED (T-PED)’ means a portable electronic device (PED) that has intentional radio frequency (RF) transmission capabilities.
- ‘Type A EFB application’ means an EFB application whose malfunction or misuse has no safety effect;
- ‘Type B EFB application’ means an EFB application:
 1. whose malfunction or misuse is classified as minor failure condition or below; and
 2. which neither replaces nor duplicates any system or functionality required by airworthiness regulations, airspace requirements, or operational rules.
- ‘ V_1 ’ means the maximum speed in the take-off at which the pilot must take the first action to stop the aeroplane within the accelerate-stop distance. V_1 also means the minimum speed in the take-off, following a failure of the critical engine at VEF, at which the pilot can continue the take-off and achieve the required height above the take-off surface within the take-off distance;
- ‘VEF’ means the speed at which the critical engine is assumed to fail during take-off;
- ‘Viewable stowage’ means a non-certified device that is attached to the flight crew member (e.g. with a kneeboard) or to an existing aircraft part (e.g. using suction cups), and is intended to hold charts or to hold low-mass portable electronic devices that are viewable by the flight crew members at their assigned duty stations.
- ‘Visual and non-visual aids and infrastructure’ refers to all equipment and facilities required for the procedure to be used for the intended instrument approach operation. This includes but is not limited to lights, markings, ground-or space-based radio aids, etc.
- ‘Visual approach operation’ means an approach operation by an IFR flight when either a part or all parts of an IAP is (are) not completed and the approach operation is executed with visual reference to terrain;
- ‘Weather permissible aerodrome’ means an adequate aerodrome where, for the anticipated time of use, meteorological reports, or forecasts, or any combination thereof, indicate that the meteorological conditions will be at or above the required aerodrome operating minima, and the runway surface condition reports indicate that a safe landing will be possible;
- ‘Wet lease agreement’ means an agreement between air carriers pursuant to which the aircraft is operated under the AOC of the lessor;

- ‘Wet runway’ means a runway whose surface is covered by any visible dampness or water up to and including 3 mm deep within the area intended to be used.

0.1.4.1 Rules of the Air and Other Definitions

Definitions

For the purpose of this Regulation the following definitions shall apply:

1. ‘ADS-B In’ means the receipt, processing, and display of other aircraft’s ADS-B transmissions;
2. ‘ADS-B Out’ means the provision of ADS-B surveillance data from an aircraft transmit perspective;
3. ‘Accuracy’ means a degree of conformance between the estimated or measured value and the true value;
4. ‘Advisory airspace’ means an airspace of defined dimensions, or designated route, within which air traffic advisory service is available;
5. ‘Advisory route’ means a designated route along which air traffic advisory service is available;
6. ‘Aerobatic flight’ means manoeuvres intentionally performed by an aircraft involving an abrupt change in its attitude, an abnormal attitude, or an abnormal variation in speed, not necessary for normal flight or for instruction for licenses or ratings other than aerobatic rating;
7. ‘Aerodrome’ means a defined area (including any buildings, installations and equipment) on land or water or on a fixed, fixed off-shore or floating structure intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft;
8. ‘Aerodrome control service’ means air traffic control service for aerodrome traffic;
9. ‘Aerodrome control tower’ means a unit established to provide air traffic control service to aerodrome traffic;
10. ‘Aerodrome traffic’ means all traffic on the manoeuvring area of an aerodrome and all aircraft flying in the vicinity of an aerodrome. An aircraft operating in the vicinity of an aerodrome includes but is not limited to aircraft entering or leaving an aerodrome traffic circuit;
11. ‘Aerodrome traffic circuit’ means the specified path to be flown by aircraft operating in the vicinity of an aerodrome;
12. ‘Aerodrome traffic zone’ means an airspace of defined dimensions established around an aerodrome for the protection of aerodrome traffic;

13. 'Aerial work' means an aircraft operation in which an aircraft is used for specialised services such as agriculture, construction, photography, surveying, observation and patrol, search and rescue, aerial advertisement, etc.;
14. 'Aeronautical Information Publication (AIP)' means a publication issued by or with the authority of a State and containing aeronautical information of a lasting character essential to air navigation;
15. 'Aeronautical mobile service' means a mobile service between aeronautical stations and aircraft stations, or between aircraft stations, in which survival craft stations may participate; emergency position-indicating radio beacon stations may also participate in this service on designated distress and emergency frequencies;
16. 'Aeronautical station' means a land station in the aeronautical mobile service. In certain instances, an aeronautical station may be located, for example, on boards hip or on a platform at sea;
17. 'Aeroplane' means a power-driven heavier-than-air aircraft, deriving its lift in flight chiefly from aerodynamic reactions on surfaces which remain fixed under given conditions of flight;
18. 'Airborne collision avoidance system (ACAS)' means an aircraft system based on secondary surveillance radar (SSR) transponder signals which operates independently of ground-based equipment to provide advice to the pilot on potential conflicting aircraft that are equipped with SSR transponders;
19. 'Aircraft' means any machine that can derive support in the atmosphere from the reactions of the air other than the reactions of the air against the earth's surface;
20. 'Aircraft address' means a unique combination of 24 bits available for assignment to an aircraft for the purpose of air-ground communications, navigation and surveillance;
21. 'Aircraft observation' means the evaluation of one or more meteorological elements made from an aircraft in flight;
22. 'AIRMET information' means information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en-route weather phenomena which may affect the safety of low-level aircraft operations and of the development of those phenomena in time and space, which was not already included in the forecast issued for low-level flights in the flight information region concerned or sub-area thereof;
23. 'Air-ground communication' means two-way communication between aircraft and stations or locations on the surface of the earth;

24. 'Air-ground control radio station' means an aeronautical telecommunication station having primary responsibility for handling communications pertaining to the operation and control of aircraft in a given area;
25. 'Air-report' means a report from an aircraft in flight prepared in conformity with requirements for position, and operational and/or meteorological reporting;
26. 'Air Taxiing' N/A MAY;
27. 'Air traffic' means all aircraft in flight or operating on the manoeuvring area of an aerodrome;
28. 'Air traffic advisory service' means a service provided within advisory airspace to ensure separation, in so far as practical, between aircraft which are operating on Instrument Flight Rules (IFR) flight plans;
29. 'Air Traffic Control (ATC) clearance' means authorisation for an aircraft to proceed under conditions specified by an air traffic control unit;
30. 'Air traffic control instruction' means directives issued by air traffic control for the purpose of requiring a pilot to take a specific action;
31. 'Air traffic control service' means a service provided for the purpose of:
 - a. Preventing collisions:
 - i. Between aircraft; and
 - ii. On the manoeuvring area between aircraft and obstructions; and
 - b. Expediting and maintaining an orderly flow of air traffic;
32. 'Air traffic control unit' means a generic term meaning variously, area control centre, approach control unit or aerodrome control tower;
33. 'Air traffic service (ATS)' means a generic term meaning variously, flight information service, alerting service, air traffic advisory service, air traffic control service (area control service, approach control service or aerodrome control service);
34. 'Air traffic services airspaces' mean airspaces of defined dimensions, alphabetically designated, within which specific types of flights may operate and for which air traffic services and rules of operation are specified;

35. 'Air Traffic Services (ATS) airspaces' mean airspaces of defined dimensions, alphabetically designated, within which specific types of flights may operate and for which air traffic services and rules of operation are specified;
 - a. 'Air Traffic Services (ATS) Reporting Office (ARO)' means a unit established for the purpose of receiving reports concerning air traffic services and flight plans submitted before departure.
36. 'Air Traffic Services (ATS) surveillance service' means a service provided directly by means of an ATS surveillance system;
37. 'Air Traffic Services (ATS) unit' means a generic term meaning, variously, air traffic control unit, flight information centre, aerodrome flight information service unit or air traffic services reporting office;
38. 'Alerting service' means a service provided to notify appropriate organisations regarding aircraft in need of search and rescue aid, and assist such organisations as required;
39. 'Airway' means a control area or portion thereof established in the form of a corridor;
40. 'Alternate aerodrome' means an adequate aerodrome to which an aircraft may proceed when it becomes either impossible or inadvisable to proceed to or land at the aerodrome of intended landing, where the necessary services and facilities are available, where aircraft performance requirements can be met, and which is operational at the expected time of use; 'alternate aerodrome' includes the following:
 - 'take-off alternate aerodrome': an alternate aerodrome at which an aircraft would be able to land if it becomes necessary shortly after take-off and it is not possible to use the aerodrome of departure;
 - 'En Route Alternate (ERA) aerodrome': an alternate aerodrome at which an aircraft would be able to land if a diversion becomes necessary while en route;
 - 'fuel En Route Alternate (fuel ERA) aerodrome' means an ERA aerodrome that is required at the planning stage for use in the calculation of fuel/energy;
 - 'destination alternate aerodrome': an alternate aerodrome at which an aircraft would be able to land if it becomes either impossible or inadvisable to land at the aerodrome of intended landing;
41. 'Altitude' means the vertical distance of a level, a point or an object considered as a point, measured from mean sea level (MSL);
42. 'Annex 3' means the Annex 3 "Meteorological Services" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;

43. 'Annex 6' means the Annex 6 "Operation of Aircraft" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
44. 'Annex 8' means the Annex 8 "Airworthiness of Aircraft" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
45. 'Annex 10' means the Annex 10 "Aeronautical Telecommunications" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
46. 'Annex 13' means the Annex 13 "Aircraft Accident and Incident Investigation" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
47. 'Annex 14' means the Annex 14 "Aerodromes" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
48. 'Annex 17' means the Annex 17 "Security" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
49. 'Annex 18' means the Annex 18 "Safe Transport of Dangerous Goods by Air" to the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
50. 'Approach control service' means air traffic control service for arriving or departing controlled flights;
51. 'Approach control unit' means a unit established to provide air traffic control service to controlled flights arriving at, or departing from, one or more aerodromes;
52. 'Apron' means a defined area, intended to accommodate aircraft for purposes of loading or unloading passengers, mail or cargo, fuelling, parking or maintenance;
53. 'Area control centre (ACC)' means a unit established to provide air traffic control service to controlled flights in control areas under its jurisdiction;
54. 'Area control service' means air traffic control service for controlled flights in control areas;
55. 'Area navigation (RNAV)' means a method of navigation which permits aircraft operation on any desired flight path within the coverage of ground- or space-based navigation aids or within the limits of the capability of self-contained aids, or a combination of these;
56. 'ATS route' means a specified route designed for channelling the flow of traffic as necessary for the provision of air traffic services;

57. ‘Automatic dependent surveillance – broadcast (ADS-B)’ means a means by which aircraft, aerodrome vehicles and other objects can automatically transmit and/or receive data such as identification, position and additional data, as appropriate, in a broadcast mode via a data link;
58. ‘Automatic dependent surveillance – contract (ADS-C)’ means a means by which the terms of an ADS-C agreement will be exchanged between the ground system and the aircraft, via a data link, specifying under what conditions ADS-C reports would be initiated, and what data would be contained in the reports;
- a. ‘Automatic Dependent Surveillance – Contract (ADS-C) agreement’ means a reporting plan which establishes the conditions of ADS-C data reporting (i.e. data required by the air traffic services unit and frequency of ADS-C reports which have to be agreed to, prior to using ADS-C in the provision of air traffic services).
59. ‘Automatic terminal information service (ATIS)’ means the automatic provision of current, routine information to arriving and departing aircraft throughout 24 hours or a specified portion thereof:
- a. ‘Data link-automatic terminal information service (D-ATIS)’ means the provision of ATIS via data link;
- b. ‘Voice-automatic terminal information service (Voice-ATIS)’ means the provision of ATIS by means of continuous and repetitive voice broadcasts;
60. ‘Ceiling’ means the height above the ground or water of the base of the lowest layer of cloud below 6000 m (20,000 ft) covering more than half the sky;
61. ‘Change-over point’ means the point at which an aircraft navigating on an ATS route segment defined by reference to very high frequency omnidirectional radio ranges is expected to transfer its primary navigational reference from the facility behind the aircraft to the next facility ahead of the aircraft;
62. ‘Chicago Convention’ means the Convention on International Civil Aviation, signed in Chicago on the 7th of December 1944, Chicago Convention;
63. ‘Clearance limit’ means the point to which an aircraft is granted an air traffic control clearance;
64. ‘Cloud of operational significance’ means a cloud with the height of cloud base below 1500 m (5000 ft) or below the highest minimum sector altitude, whichever is greater, or a cumulonimbus cloud or a towering cumulus cloud at any height;

65. 'Code (SSR)' means the number assigned to a particular multiple pulse reply signal transmitted by a transponder in Mode A or Mode C;
66. 'Communicable Disease' means an infectious disease caused by a contagious agent which is transmitted from person to person by direct contact with an infected individual or by indirect means such as exposure to a vector, animal, fomite, product or environment, or exchange of fluid, which is contaminated with the contagious agent;
67. 'Competent authority' means the TM CAD.
68. 'Control area' means a controlled airspace extending upwards from a specified limit above the earth;
69. 'Controlled aerodrome' means an aerodrome at which air traffic control service is provided to aerodrome traffic regardless whether or not a control zone exists;
70. 'Controlled airspace' means an airspace of defined dimensions within which air traffic control service is provided in accordance with the airspace classification;
71. 'Controlled flight' means any flight which is subject to an air traffic control clearance;
72. 'Critical area' means an area of defined dimensions extending around the ground equipment of a precision instrument approach within which the presence of vehicles or aircraft will cause unacceptable disturbance of the guidance signals;
73. 'Controller-pilot data link communications (CPDLC)' mean a means of communication between controller and pilot, using data link for ATC communications;
74. 'Control zone' means a controlled airspace extending upwards from the surface of the earth to a specified upper limit;
75. 'Cruise climb' means an aeroplane cruising technique resulting in a net increase in altitude as the aeroplane mass decreases;
76. 'Cruising level' means a level maintained during a significant portion of a flight;
77. 'Current flight plan (CPL)' means the flight plan, including changes, if any, brought about by subsequent clearances;
78. 'Danger area' means an airspace of defined dimensions within which activities dangerous to the flight of aircraft may exist at specified times;
79. 'Data link communications' mean a form of communication intended for the exchange of messages via a datalink;

- 80. 'Datum' means any quantity or set of quantities that may serve as a reference or basis for the calculation of other quantities;
- 81. 'Directive 2000/79/EC' means the Council Directive 2000/79/EC of 27 November 2000 concerning the European Agreement on the Organisation of Working Time of Mobile Workers in Civil Aviation concluded by the Association of European Airlines (AEA), the European Transport Workers' Federation (ETF), the European Cockpit Association (ECA), the European Regions Airline Association (ERA) and the International Air Carrier Association (IACA);
- 82. 'Downstream clearance' means a clearance issued to an aircraft by an air traffic control unit that is not the current controlling authority of that aircraft;
- 83. 'Estimated elapsed time' means the estimated time required to proceed from one significant point to another;
- 84. 'Estimated off-block time' means the estimated time at which the aircraft will commence movement associated with departure;
- 85. 'Estimated Time of Arrival (ETA)' means for IFR flights, the time at which it is estimated that the aircraft will arrive over that designated point, defined by reference to navigation aids, from which it is intended that an instrument approach procedure will be commenced, or, if no navigation aid is associated with the aerodrome, the time at which the aircraft will arrive over the aerodrome. For Visual Flight Rules (VFR) flights, the time at which it is estimated that the aircraft will arrive over the aerodrome;
- 86. 'Expected approach time' means the time at which ATC expects that an arriving aircraft, following a delay, will leave the holding fix to complete its approach for a landing. The actual time of leaving the holding fix will depend upon the approach clearance;
- 87. 'Filed flight plan (FPL)' means the flight plan as filed with an ATS unit by the pilot or a designated representative, without any subsequent changes;
- 88. 'Flight crew member' means a licensed crew member charged with duties essential to the operation of an aircraft during a flight duty period;
- 89. 'Flight information centre' means a unit established to provide flight information service and alerting service;
- 90. 'Flight information region' means an airspace of defined dimensions within which flight information service and alerting service are provided;
- 91. 'Flight information service' means a service provided for the purpose of giving advice and information useful for the safe and efficient conduct of flights;

92. 'Flight level (FL)' means a surface of constant atmospheric pressure which is related to a specific pressure datum, 1013,2 hectopascals (hPa), and is separated from other such surfaces by specific pressure intervals;
93. 'Flight plan' means specified information provided to air traffic services units, relative to an intended flight or portion of a flight of an aircraft;
94. 'Flight visibility' means the visibility forward from the cockpit of an aircraft in flight;
95. 'Forecast' means a statement of expected meteorological conditions for a specified time or period, and for a specified area or portion of airspace;
96. 'Ground visibility' means the visibility at an aerodrome, as reported by an accredited observer or by automatic systems;
97. 'Heading' means the direction in which the longitudinal axis of an aircraft is pointed, usually expressed in degrees from North (true, magnetic, compass or grid);
98. 'Height' means the vertical distance of a level, a point or an object considered as a point, measured from a specified datum;
99. 'Helicopter' N/A MAY;
100. 'High seas airspace' means airspace beyond land territory and territorial seas, as specified in the United Nations Convention on the Law of the Sea (Montego Bay, 1982);
101. 'Hotspot' means a location on an aerodrome movement area with a history or potential risk of collision or runway incursion, and where heightened attention by pilots or drivers is necessary;
102. 'IFR' means the symbol used to designate the instrument flight rules;
103. 'IFR flight' means a flight conducted in accordance with the instrument flight rules;
104. 'IMC' means the symbol used to designate instrument meteorological conditions;
105. 'Instrument approach operation' means an approach and landing using instruments for navigation guidance based on an Instrument Approach Procedure. There are two methods for executing instrument approach operations:
 - a. a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and
 - b. a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance;

106. 'Instrument Approach Procedure (IAP)' means a series of predetermined manoeuvres by reference to flight instruments with specified protection from obstacles from the initial approach fix or, where applicable, from the beginning of a defined arrival route to a point from which a landing can be completed and thereafter, if a landing is not completed, to a position at which holding or en route obstacle clearance criteria apply. IAPs are classified as follows:
- a. Non Precision Approach (NPA) procedure, which means an IAP designed for 2D instrument approach operations Type A;
 - b. Approach Procedure with Vertical Guidance (APV) means a Performance-Based Navigation (PBN) IAP designed for 3D instrument approach operations Type A;
 - c. Precision Approach (PA) procedure means an IAP based on navigation systems designed for 3D instrument approach operations Type A or B;
107. 'Instrument meteorological conditions (IMC)' mean meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, less than the minima specified for visual meteorological conditions;
108. 'Landing area' means that part of a movement area intended for the landing or take-off of aircraft;
109. 'Level' means a generic term relating to the vertical position of an aircraft in flight and meaning variously, height, altitude or flight level;
110. 'Line check' means a check conducted by the operator and completed by the pilot or the technical crew member to demonstrate competence in carrying out normal line operations described in the operations manual;
111. 'Manoeuvring area' means that part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons;
112. 'Meteorological Watch Office (MWO)' means an office monitoring meteorological conditions affecting flight operations and providing information concerning the occurrence or expected occurrence of specified en-route weather and other phenomena in the atmosphere which may affect the safety of aircraft operations within its specified area of responsibility;
113. 'Minimum fuel' is a term used to describe a situation in which an aircraft's fuel supply has reached a state where the flight is committed to land at a specific aerodrome and no additional delay can be accepted;

- 114. 'Misuse of substances' means the use of one or more psychoactive substances by flight crew, cabin crew members, and other safety-sensitive personnel in a way that:
 - a. Constitutes a direct hazard to the user or endangers the lives, health or welfare of others; and/or
 - b. Causes or worsens an occupational, social, mental or physical problem or disorder;
- 115. 'Mode (SSR)' means the conventional identifier related to specific functions of the interrogation signals transmitted by an SSR interrogator. There are four modes specified in ICAO Annex 10: A, C, S and intermode;
- 116. 'Model aircraft' means an unmanned aircraft, other than a toy aircraft, having an operating mass not exceeding limits prescribed by the competent authority, that is capable of sustained flight in the atmosphere and that is used exclusively for display or recreational activities;
- 117. 'Mountainous area' means an area of changing terrain profile where the changes of terrain elevation exceed 900 m (3000 ft) within a distance of 18,5 km (10,0 NM);
- 118. 'Movement area' means that part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s);
- 119. 'Night' means the period between the end of evening civil twilight and the beginning of morning civil twilight or such other period between sunset and sunrise as may be prescribed by the appropriate authority, as defined by the Member State;
- 120. 'Obstacle' means all fixed (whether temporary or permanent) and mobile objects, or parts thereof, that:
 - a. Are located on an area intended for the surface movement of aircraft; or
 - b. Extend above a defined surface intended to protect aircraft in flight; or
 - c. Stand outside those defined surfaces and that have been assessed as being a hazard to air navigation;
- 121. 'Operating site' means a site selected by the operator or pilot-in-command for landing, take-off and/or hoist operations;
- 122. 'Operational credit' means a credit for operations with an advanced aircraft enabling lower aerodrome operating minima than would normally be established by the operator for a basic aircraft, based upon the performance of advanced aircraft systems utilising the available external infrastructure.

Lower operating minima may include a lower decision height/altitude or minimum descent height/altitude, reduced visibility requirements or reduced ground facilities or a combination of these;

123. 'Operator proficiency check' means a check conducted by the operator and completed by the pilot or the technical crew member to demonstrate competence in carrying out normal, abnormal and emergency procedures;
124. 'Performance-Based Navigation (PBN)' means area navigation based on performance requirements for aircraft operating along an ATS route, on an instrument approach procedure or in a designated airspace;
125. 'Pilot-in-command' means the pilot designated by the operator, or in the case of general aviation, the owner, as being in command and charged with the safe conduct of a flight;
126. 'Pressure-altitude' means an atmospheric pressure expressed in terms of altitude which corresponds to that pressure in the Standard Atmosphere, as defined in Annex 8, Part 1 to the Chicago Convention;
127. 'Prohibited area' means an airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is prohibited;
128. 'Psychoactive substances' mean alcohol, opioids, cannabinoids, sedatives and hypnotics, cocaine, other psychostimulants, hallucinogens, and volatile solvents, whereas coffee and tobacco are excluded;
129. 'Public health' means all elements related to health, namely health status, including morbidity and disability, the determinants having an effect on that health status, health care needs, resources allocated to health care, the provision of, and universal access to, health care as well as health care expenditure and financing, and the causes of mortality;
130. 'Radar' means a radio detection device which provides information on range, azimuth and/or elevation of objects;
131. 'Radio mandatory zone (RMZ)' means an airspace of defined dimensions wherein the carriage and operation of radio equipment is mandatory;
132. 'Radio navigation service' means a service providing guidance information or position data for the efficient and safe operation of aircraft supported by one or more radio navigation aids;
133. 'Radiotelephony' means a form of radio communication primarily intended for the exchange of information in the form of speech;
134. 'Repetitive flight plan' means a flight plan related to a series of frequently recurring, regularly operated individual flights with identical basic features, submitted by an operator for retention and repetitive use by ATS units;

135. 'Reporting point' means a specified geographical location in relation to which the position of an aircraft can be reported;
136. 'Restricted area' means an airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is restricted in accordance with certain specified conditions;
137. 'Route segment' means a route or portion of route usually flown without an intermediate stop;
138. 'Runway' means a defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft;
139. 'Runway Condition Report (RCR)' means a comprehensive standardised report relating to the conditions of the runway surface and their effect on the aeroplane landing and take-off performance, described by means of runway conditions code;
140. 'Runway-holding position' means a designated position intended to protect a runway, an obstacle limitation surface, or an Instrument Landing System (ILS)/Microwave Landing System (MLS) critical/sensitive area at which taxiing aircraft and vehicles are to stop and hold, unless otherwise authorised by the aerodrome control tower;
141. 'Runway visual range (RVR)' means the range over which the pilot of an aircraft on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identifying its centre line;
142. 'Safe landing' means, in the context of the fuel policy or fuel schemes, a landing at an adequate aerodrome or operating site with no less than the final reserve fuel remaining and in compliance with the applicable operational procedures and aerodrome operating minima;
143. 'Safety-sensitive personnel' means persons who might endanger aviation safety if they perform their duties and functions improperly, including crew members, (flight crew members and cabin crew) aircraft maintenance personnel, and air traffic controllers;
144. 'Secondary surveillance radar (SSR)' means a surveillance radar system which uses transmitters/receivers (interrogators) and transponders;
145. 'Sensitive area' means an area extending beyond the critical area where the parking or movement, or both, of aircraft or vehicles will affect the guidance signal to the extent that it may be rendered as an unacceptable disturbance to aircraft using the signal;
146. 'SERA' Standardised European Rules of the Air;

- 147. 'SIGMET' means information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en-route weather and other phenomena in the atmosphere which may affect the safety of aircraft operations and of the development of those phenomena in time and space;
- 148. 'Signal area' means an area on an aerodrome used for the display of ground signals;
- 149. 'Significant point' means a specified geographical location used in defining an ATS route or the flight path of an aircraft and for other navigation and ATS purposes;
- 150. 'Special VFR flight' means a VFR flight cleared by air traffic control to operate within a control zone in meteorological conditions below VMC;
- 151. 'Strayed aircraft' means an aircraft which has deviated significantly from its intended track or which reports that it is lost;
- 152. 'Surveillance Data' means any data item, time stamped or not, within the surveillance system that pertains to:
 - a. Aircraft 2D position;
 - b. Aircraft vertical position;
 - c. Aircraft attitude;
 - d. Aircraft identity;
 - e. 24-bit ICAO aircraft address;
 - f. Aircraft intent;
 - g. Aircraft velocity;
 - h. Aircraft acceleration.
- 153. 'Surveillance radar' means radar equipment used to determine the position of an aircraft in range and azimuth;
- 154. 'Taxiing' means movement of an aircraft on the surface of an aerodrome or an operating site under its own power, excluding take-off and landing;
- 155. 'Taxiway' means a defined path on a land aerodrome established for the taxiing of aircraft and intended to provide a link between one part of the aerodrome and another, including:
 - a. Aircraft stand taxilane means a portion of an apron designated as a taxiway and intended to provide access to aircraft stands only.
 - b. Apron taxiway means a portion of a taxiway system located on an apron and intended to provide a through taxi route across the apron.

- c. Rapid exit taxiway means a taxiway connected to a runway at an acute angle and designed to allow landing aeroplanes to turn off at higher speeds than are achieved on other exit taxiways thereby minimising runway occupancy times;
- 156. 'Territory' means the land areas and territorial waters adjacent thereto under the sovereignty, suzerainty, protection or mandate of a State;
- 157. 'Threshold' means the beginning of that portion of the runway usable for landing;
- 158. 'Total estimated elapsed time' means:
 - a. For IFR flights, the estimated time required from take-off to arrive over that designated point, defined by reference to navigation aids, from which it is intended that an instrument approach procedure will be commenced, or, if no navigation aid is associated with the destination aerodrome, to arrive over the destination aerodrome;
 - b. For VFR flights, the estimated time required from take-off to arrive over the destination aerodrome;
- 159. 'Toy Aircraft' means an unmanned aircraft designed or intended for use, whether or not exclusively, in play by children under 14 years of age;
- 160. 'Track' means the projection on the earth's surface of the path of an aircraft, the direction of which path at any point is usually expressed in degrees from North (true, magnetic or grid);
- 161. 'Traffic avoidance advice' means an advice provided by an air traffic services unit specifying manoeuvres to assist a pilot to avoid a collision;
- 162. 'Traffic information' means information issued by an air traffic services unit to alert a pilot to other known or observed air traffic which may be in proximity to the position or intended route of flight and to help the pilot avoid a collision;
- 163. 'Training to proficiency' means training designed to achieve end-state performance objectives, providing sufficient assurance that the trained individual is capable of consistently carrying out specific tasks safely and effectively;
- 164. 'Transfer of control point' means a defined point located along the flight path of an aircraft, at which the responsibility for providing air traffic control service to the aircraft is transferred from one control unit or control position to the next;
- 165. 'Transition altitude' means the altitude at or below which the vertical position of an aircraft is controlled by reference to altitudes;
- 166. 'Transition level' means the lowest flight level available for use above the transition altitude;

- 167. 'Transponder mandatory zone (TMZ)' means an airspace of defined dimensions wherein the carriage and operation of pressure-altitude reporting transponders is mandatory;
- 168. 'Type A instrument approach operation' means an instrument approach operation with a MDH or a DH at or above 250 ft;
- 169. 'Type B instrument approach operation' means an operation with a DH below 250 ft. Type B instrument approach operations are categorised as:
 - a. Category I (CAT I): a DH not lower than 200 ft and with either a visibility not less than 800 m or an RVR not less than 550 m;
 - b. Category II (CAT II): a DH lower than 200 ft but not lower than 100 ft, and an RVR not less than 300 m;
 - c. Category III (CAT III): a DH lower than 100 ft or no DH, and an RVR less than 300 m or no RVR limitation;
- 170. 'Unidentified aircraft' means an aircraft which has been observed or reported to be operating in a given area but whose identity has not been established;
- 171. 'Unmanned Free Balloon' N/A MAY;
- 172. 'VFR' means the symbol used to designate the visual flight rules;
- 173. 'VFR flight' means a flight conducted in accordance with the visual flight rules;
- 174. 'Visibility (VIS)' means visibility for aeronautical purposes, which is the greater of:
 - a. the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognised when observed against a bright background; and
 - b. the greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background;
- 175. 'Visual meteorological conditions' mean meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, equal to or better than specified minima;
- 176. 'VMC' means the symbol used to designate visual meteorological conditions.

0.2 System of Amendment and Revision

ORO.MLR.100, (e)

The OM shall be kept up to date. All personnel shall be made aware of the changes that are relevant to their duties

In accordance with ORO.GEN.200(a)(5), the System of Amendment and Revision described in the Malta Air OMA is valid and the same for the other volumes of the OMs. To avoid text duplication, the System of Amendment and Revision's chapter of Malta Air OMs refers to OMA Section 0.2.

The Nominated Persons responsible for the OMs and other documents are specified in OMA 0.2.1.

Since the OMs have been prepared for use by Malta Air staff and contain detailed instructions and information on matters pertaining to the safety and conduct of flight operations, they must be kept up to date.

This is achieved by the System of Amendment and Revision. For this purpose, Malta Air defines in OM A 0.2.6 Issues and Revisions of a Manual.

Crew Members will be informed via electronic means that a new Issue or Revision of a Manual will be published, in accordance with OM A 0.2.6.

0.2.1 Responsibility

ORO.MLR.100, (i), ORO.AOC.150

The relevant Nominated Person is responsible for the overall control of the Operations Manual as outlined in the below table. He will authorise the content of amendments, new Issues of the Manuals and new Revisions of the Manuals in accordance with OMA 0.2.6. The EFB administrator is responsible for the administration of all electronic publications to all approved Malta Air electronic devices.

Malta Air shall incorporate all amendments required by TM CAD and relevant mandatory safety information as supplied by Air Ops ORO.GEN.155.

The responsibility of the following documents is assigned below:

Table 1

Ops Manual Part A	NPFO
Ops Manual Part B	NPFO
Ops Manual Part C, Route Manual	NPFO
Ops Manual Part D, Training (Crew)	NPCT
Ground Operations Manual	NPGO
Compliance Monitoring Manual	Compliance Monitoring Manager
Security Programme	Security Manager
MEL/CDL	NPFO
Safety Management Manual	Safety Manager

Note: Any suggestion for improvement to this manual, errors and/or discrepancies with the OMA or between the OMA and other regulations shall be submitted to the OMA section via Zendesk.

0.2.2 Amendment Record

ORO.MLR.100

Since the OM has been prepared for use by Company staff and contains detailed instructions and information on matters pertaining to the safety and conduct of flight operations, it must be kept up to date. Crew members will be informed via electronic means that a new issue or revision has been published.

Issues and revisions of the manuals will be numbered as described in OMA 0.2.6 and the method of incorporation will be as follows:

1. Amendments Record:

Table containing the issue/revision number, the date and a signature of the person filing the amendment.

Revision		TM CAD Approval		Revision Implementation		
Nr	Date	Nr	Date	Date	Who	Signature

2. Change Revision Summary:

Table containing the scope of the particular revision.

Page		Change
From	To	

0.2.3 Handwritten Amendments

Handwritten amendments and issues/revisions on paper versions are not permitted.

0.2.4 System for the Annotation of Pages

AMC3 ORO.MLR.100, ORO.MLR.100

For easy reference the manual is divided into chapters with consecutive numbering of pages. For a list of effective pages refer to the List of effective pages (LOEP) section at the beginning of this OM. The Air Operations numbering/lettering protocol is used.

0.2.5 Annotation of Changes

AMC3 ORO.MLR.100

Additions, changes or deletions are identified by text underlined with a dotted line, indicating where changes occurred in the amendment process.

0.2.6 Amendment Procedure

ORO.GEN.115, 130

The operations manual is subject to regular amendment. An amendment of the operations manual may lead to a new issue number or to a new revision number of the current Issue of a manual, depending on whether prior approval is required from TM CAD:

1. **Issue No of a Manual: an amendment of a Manual that requires prior approval from the TMCAD shall be published as new issue of the Manual.** The initial issue Manual is published as Issue 0, Revision 0, Month, Year. Whenever an amendment requiring prior approval is required, a new issue number of the Manual is published, with the revision number of the new issue re-set to 0 (e.g. Issue 1, Revision 3 becomes Issue 2, Revision 0). A major change of Regulation/Instrument/Order may require an issue of a manual. Subsequent issues of the manual are numbered sequentially.

Each issue of a manual has all changes described in a ‘Amendment Highlights’ section of the manual. Dotted underlines are included in each issue of the manual indicating the changes implemented.

2. **Revision of an issue No of a manual: an amendment to a manual that does not require prior approval from the TMCAD shall be published as a new revision** of the current Issue of the manual

Each revision to a current issue No of a manual has all changes described in a “Revision Highlights” section of the manual. Dotted underlines are included in each revision of a manual indicating the changes implemented.

In exceptional circumstances, the TM CAD may grant an exemption from meeting the requirements of EU Regulation (EU) 2018/1139 and its implementing rules, based on article 71 of that Regulation.

OM effective date will be specified on a dedicated Flight Operations Memo to inform crew of the new issue/revision No. of the manual and it is generally equivalent to the date on which the issue/revision is published.

0.2.6.1 Amendment Procedure – Changes Requiring Prior Approval (Issue No. of a Manual)

The NP is responsible for the preparation of the relevant OM amendment that is subject to prior approval from TM CAD, as outlined in Table 1. An amendment of the OM that requires prior approval will be referred to as ‘**Issue**’. After assessing the necessity for the issue, the NP sets out its scope and designates if necessary, a person to prepare it (Project Manager). All relevant departments concerned or potentially affected by the change shall be consulted as part of the management of change (MOC) process, and a Safety Risk Assessment shall be completed. The person preparing the issue is responsible for preparing it only on the basis of current source materials and regulations.

Once the issue is prepared, the NP responsible shall forward it to the CMM for compliance verification. Once this process is completed, all required documentation is submitted to TMCAD for approval.

The following preliminary set of documents shall be prepared and submitted to the TMCAD:

1. Operations Manual Approval Schedule (signature sheet);
2. Operations Manual Provisional Text, detailing the proposed changes to the manual;
3. Signed ‘Provisional Prior Approval of Change to Operations Manual’ Form
4. Signed ‘Administration and Control of Operations Manual’ Form; and
5. Any applications.

The project owner also retains copies of all submitted documents together with the TM CAD confirmation of receipt.

TMCAD comments on the proposed changes shall be responded to as required.

When TMCAD confirms that no further amendments are required to the proposed changes, Malta Air shall prepare and submit the following set of documents to the TMCAD:

1. Updated List of Effective Pages (LOEP);
2. Highlights/Summary of the amendments;
3. Operations Manual, incorporating the amendments as approved by TMCAD in their provisional approval;
4. Any applications.

After receiving the approval from the TM CAD, the NPFO creates a Flight Ops Memo informing crew of the new OM issue and its effective date. The Flight Ops Memo and the approved OM Issue are distributed accordingly.

The application for the amendment of an Air Operator Certificate is submitted at least 30 days before the date of the intended changes. In the case of a planned change of a Nominated Person, the Malta Air informs the TM CAD at least 20 days before the date of the proposed Issue. In cases of unforeseen changes the TM CAD is notified at the earliest opportunity, in order to enable the TM CAD to determine continued compliance with the applicable requirements and to amend, if necessary, the Air Operator Certificate and related terms of approval.

Below is a list of changes that may affect the Air Operator Certificate or the operations specifications or the operator's management system, that require prior TM CAD approval. The list includes items likely to be relevant to Malta Air's operations. When assessing the applicability of changes requiring prior approval, the latest requirements as published by TM CAD must be consulted.

1. The name of the Operator;
2. A change of legal entity;
3. The operator's principal place of business;
4. Additional locations of the operator;
5. The accountable manager;
6. Any of the persons referred to in ORO.GEN.210(a) and (b);
7. The operator's documentation, as required by regulation;
8. The facilities;
9. Procedures regarding items to be notified to the competent authority;
10. Use of on-board mass and balance computer systems;
11. Specific approvals in accordance with Annex V (Part -SPA);
12. Changes affecting scope of the certificate or the operations specifications of an operator, or any of the elements of the operator's management system as required in ORO.GEN.200(a)(1) AND (a)(2);
13. Procedure for management and notification of changes not requiring prior approval by TM CAD;
14. Management System & Safety Policy;
15. Procedure for carriage of weapons of war and munitions of war;
16. Use of isolated aerodromes;

17. Use of non-stabilised approaches for particular approaches to particular runways;
18. Use of non-CDFA technique to particular approaches to particular runways;
19. The method specified for establishing minimum flight altitudes;
20. Fuel schemes and special refuelling or defuelling of aeroplanes;
21. Application of a lower landing mass than the maximum certified landing mass for determining the indicated airspeed at threshold (VAT);
22. Approval of operations with increased bank angles;
23. Approval of steep approach operations;
24. Approval of short landing operations;
25. Use of revised standard mass values for passengers and baggage. (Use of a passenger and baggage mass survey plan);
26. Use of standard mass values for items other than passengers and baggage;
27. Use and implementation of alternative means of compliance;
28. Any leasing agreement (except wet-lease out);
29. Approval to provide cabin crew training and to issue cabin crew attestation;
30. Procedure for the use of aircraft included in an AOC by other operators for NCC, NCO and specialised operations, as required by ORO.GEN.310.
31. **Cabin crew:**
 - Conduct of the training, examination and checking required by Annex V (Part-CC) to Commission Regulation (EU) No 1178/2011 and issue of;
 - Cabin crew attestations;
 - Procedures for cabin crew to operate on four aircraft types;
 - Training programmes, including syllabi;
32. Evacuation procedures with a reduced numbers of required cabin crew;
33. Cabin crew operating on more than three types;
34. Training and checking programmes, including syllabi and use of individual flight simulation training devices (FSTDs);
35. Procedures for Flight Crew to operate on more than one type or variant;
36. Alternative Training and Qualification Programme (ATQPs);
37. Minimum Equipment List;

38. Rectification interval extension (RIE) procedures;
39. Outside the constraints of the MEL but within the constraints of the MMEL;
40. Transport of Dangerous Goods;
41. Dangerous Goods training programme;
42. Extended range operations with two-engined aeroplanes;
43. Low visibility take-offs;
44. Descent below Category I minima;
45. Approach operation utilizing enhanced vision systems (EVS);
46. The specified flight crew low visibility training and checking syllabus;
47. Operations in designated NAT HLA;
48. **Min flight altitudes:**
 - The method for establishing minimum flight altitudes
 - Descent procedures to fly below specified minimum altitudes;
49. **Performance:**
 - Approval of operations with increased bank angles;
 - Short landing operations (for performance class A and B aeroplanes);
 - Steep approach operations (for performance class A and B aeroplanes);
 - Reduced required landing distance operations (for performance class A and B aeroplanes);
50. Operations in Areas with Specified Navigation Performance Requirements;
51. Operations in defined airspace with reduced vertical separation minima;
52. Use of type B electronic flight bags (EFBs) applications;
53. The scheme specified for flight and duty time limitations and rest requirements.

Means of Compliance

ORO.GEN.120 (AltMoC)

Alternative means of compliance to those adopted by EASA may be used by Malta Air to establish compliance with Regulation (EU) 2018/1139 and its Implementing Rules.

Whenever Malta Air wishes to use an alternative means of compliance to the acceptable means of compliance (AMC) adopted by EASA, to establish compliance with Regulation (EU) 2018/1139 and its Implementing Rules, it shall, prior to implementing it, provide the TMCAD with a full description of the alternative means of compliance. The description shall include any revisions to manuals or procedures that may be relevant, as well as an assessment demonstrating that the Implementing Rules are met.

In order to demonstrate that the Implementing Rules are met, a Management of Change (MoC) process including risk assessment as appropriate shall be completed and documented. The MoC shall demonstrate that an equivalent level of safety to that established by the Acceptable Means of Compliance (AMC) adopted by EASA is reached.

Malta Air may implement these alternative means of compliance subject to prior approval by TMCAD and upon receipt of the notification as prescribed in ARO.GEN.120(d).

ARO.GEN.120(d). When the TMCAD finds that the alternative means of compliance are in accordance with the Implementing Rules, it shall without undue delay:

- Notify Malta Air that the alternative means of compliance may be implemented and, if applicable, amend the Malta Air AOC accordingly; and
- Notify EASA of their content, including copies of all relevant documents.

0.2.6.1.1 Section Not in Use

0.2.6.2 Amendment Procedure – Changes Not Requiring Prior Approval (Revision No. of the Current Issue of a Manual)

The NP is responsible for the preparation of the relevant OM revision not requiring prior approval, as outlined in Table 1. A revision of the OM that does not require prior approval will be referred as “Revision”. After assessing the necessity for the revision, the NP sets out its scope and designates a person to prepare it. All relevant departments concerned or potentially affected by the change, shall be consulted as part of the change process, and a safety risk assessment shall be completed unless it has been agreed through the consultation that one is not required.

The person preparing the revision is responsible for preparing it only on the basis of current source materials and regulations.

Once the Revision is prepared, all required documentation is submitted to the TMCAD for notification.

The FOPS SME will retain a copy of all submitted documents together with the TM CAD confirmation of receipt.

The documentation submitted to TM CAD includes:

1. Complete manual;
2. Amended Section/Pages;
3. Control Form for Changes Not Requiring Prior Approval; and
4. Revised List Of Effective Pages.

A revision of the OM not requiring prior approval must be submitted to the TMCAD as a notification by no later than the date of its intended publication.

If the TMCAD submits comments on the implemented revision, the NP responsible may consider withdrawing the revision. The NPFO then prepares the amendment taking into account the comments of the TMCAD and the procedure is repeated.

Changes not requiring prior approval from the TM CAD concern all changes other than those described under 0.2.6.1. If TM CAD states that the amendment requires approval, the issue procedure described in point 0.2.6.1 of this chapter applies. The TMCAD will acknowledge the notification receipt within 10 working days.

Before publishing the revision, the NPFO creates a Flight Ops Memo informing Crews of the new OM revision and its effective date. The Flight Ops Memo and the approved OM revision are distributed accordingly.

Note: Should a change made without prior approval and implemented by Malta Air subsequently be found to be not in compliance with either EASA Implementing rules or EU regulation, then the TMCAD may consider issuing a finding. This will be dealt with as prescribed in ORO.GEN.150.

0.2.6.3 Revision Process FCOM

Flight Crew Instructions (FCI's) are used to amend the FCOM.

FCI B (Boeing) – A 'Boeing' FCI amends the FCOM;

- FCI B737-NG amends the B737-800 FCOM and is applicable to all Boeing 737 – Next Generation aircraft
- FCI B737-8200 amends the B737-8200 FCOM and is applicable to the Boeing 737-8200 variant only

The FCI header will show the subject title and the departments for distribution and SRA status.

FCI's will remain in effect until incorporated in the manual or rescinded.

0.2.6.4 Amendment Process Other Manuals

Ground Operations Manual

Ground Crew Instructions (GCI) amend the GOPS manual. Any GCIs will normally be valid until the date of the update of the GOPS Manual. Any GCI's not included in this amendment will be either rescinded or reissued. When a GCI is published for a defined period of time prior to the next Ground Operations Manual revision this defined period of time shall be specified in the GCI. The GCI header will show the subject title, the departments for distribution and SRA status.

Operations Control Centre Manual

Operations Control Instructions (OCI) amend the Operations Control Centre Manual. OCI's will normally be valid until the date of the update of the OCCM. Any OCI's not included in this amendment will be either rescinded or reissued. When a OCI is issued for a defined period of time prior to the next OCCM revision this defined period of time shall be specified in the OCI. The OCI header will show the subject title, the departments for distribution and SRA status.

Safety Management System Manual

Safety Instructions (SI) amend the SMM (Safety Management System Manual). SI's will normally be valid until the date of the next amendment of the SMM (Safety Management System Manual). Any SI's not included in this amendment will be either rescinded or reissued. When a SI is published for a defined period of time prior to the next Safety Management System Manual amendment this defined period of time shall be specified in the SI. The SI header will show the subject title, the departments for distribution and SRA status.

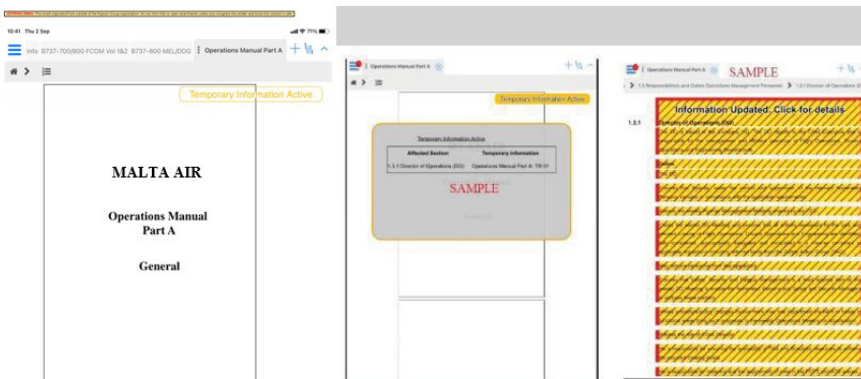
0.2.7 Temporary Revisions

ORO.MLR.100

Temporary revisions shall follow the same administrative flow as permanent revisions. The NP responsible however is authorised to publish any revision prior to TM CAD approval in the interest of safety provided it has been submitted to TM CAD.

Any temporary revisions will be published on approved electronic devices to enable operational personnel have the latest information available to them.

Temporary revisions in OMA, OMB, OMC, OMD and OMA-SEP will be published in a way that amended pages of the operations manual are clearly identified in yellow with Red Stripes on them. By clicking anywhere on the page, the amended text will be shown to the crew member.



0.2.8 Distribution System

ORO.MLR.100, (d)

The distribution system ensures all operational personnel have access to the most current manuals as relevant to the performance of their duties.

The NP is responsible for the proper operation of the distribution system of the OM he is responsible for, as outlined in Table 1.

The distribution system is as follows:

1. As outlined in OM A 0.2.6, the NP prepares the file to be uploaded into the electronic document platform. The file is then sent to the persons responsible for uploading it into the system.
2. The electronic document platform system automatically notify all users of the availability of new documents and the need to familiarize themselves with them.

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1 ORGANISATION AND RESPONSIBILITIES

1.0 Prime Responsibility

Malta Air shall only operate its' aircraft for the purpose of commercial air transportation (CAT) in accordance with Air Ops Regulations as specified in Annexes III, IV and V and the Basic Regulation (EU) 2018/1139.

Malta Air shall comply with the applicable airworthiness requirements for aeroplanes operated for the purpose of commercial air transportation.

Malta Air shall comply with relevant Requirements of Air Ops and Aircrew Regulations, the European Working Time Directive, and the Malta Air Navigation Order. Malta Air will conduct air transportation operations only when in possession of a valid Air Operators Certificate (AOC) issued by Transport Malta – Civil Aviation Directorate (TM CAD).

Malta Air has appointed Nominated Persons, as required by AIR OPS and FCL and ensures, through the processes defined in this chapter, that Nominated Persons have the capacity and resources to discharge their responsibilities, as identified in AIR OPS and Aircrew Regulations.

In Malta Air, the NPFO is responsible for the safe conduct of all flight operations conducted under the Malta Air AOC issued by TM CAD.

Malta Air Flight and Cabin Crew are subject to the limitations prescribed in the European Working Time Directive. No crew member shall exceed 2000 duty hours annually. Malta Air ensures that all crew members duty times are recorded and such records shall be made available to Regulatory Authorities when required.

1.0.1 Immediate Reaction to a Safety Problem

ORO.GEN.155

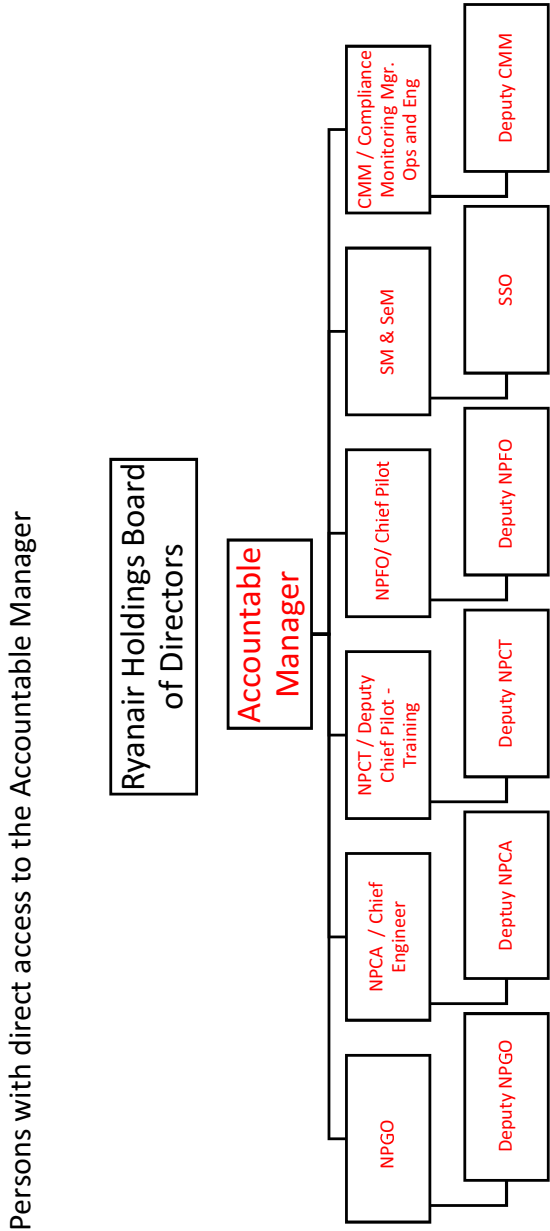
Malta Air shall implement:

1. Any safety measures mandated by TMCAD
2. Any relevant mandatory safety information issued by EASA

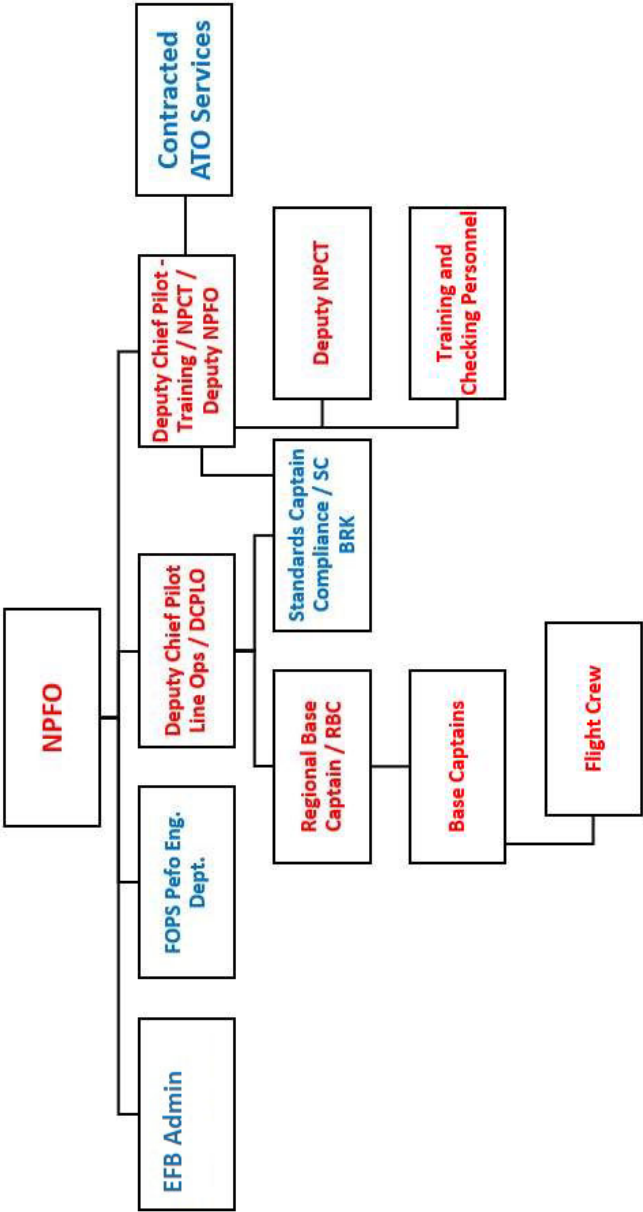
1.1 Management System: Organisational Structure

ORO.GEN.200

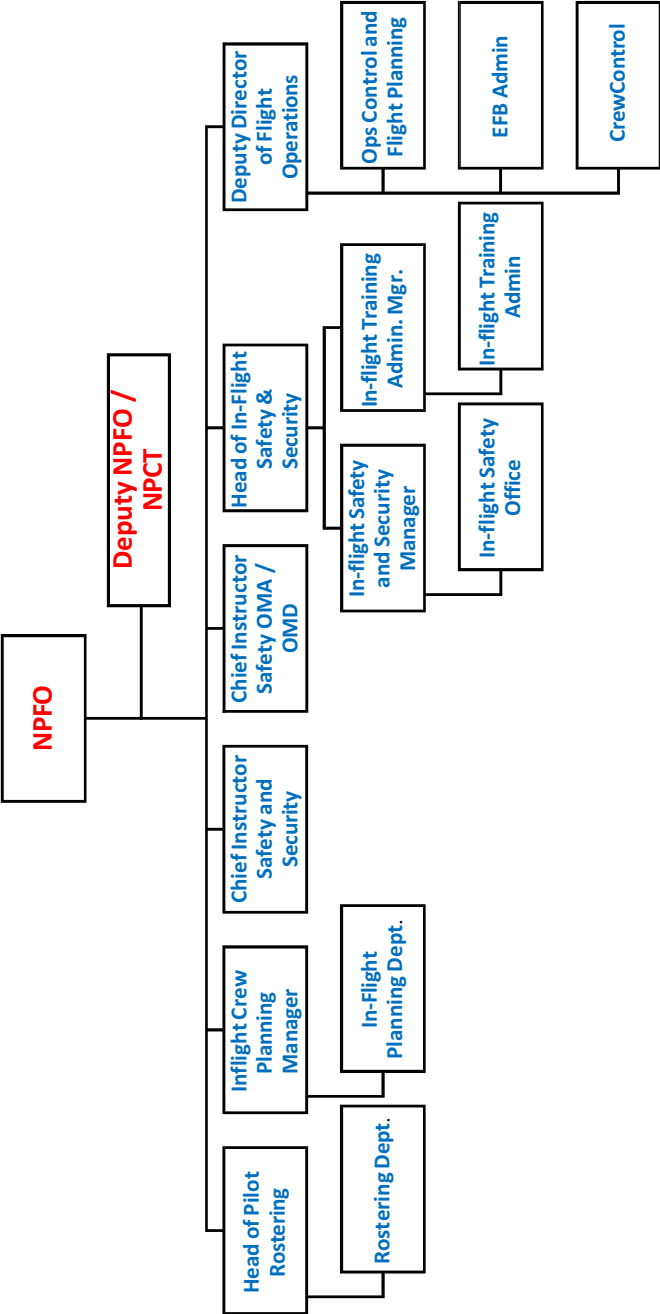
This section describes the Organisational structure in Malta Air.



Persons reporting to the NPFO:
Flight Operations and Training



Persons reporting to the NPFO
Inflight, Rostering and Operations Control



1.2 Accountable Manager and Nominated Persons

ORO.GEN.210, ORO.AOC.135

Detailed Terms of Reference, functions and responsibilities for Nominated Persons and Operations Management are maintained in the relevant sections of the Operations Manual.

1.2.1 Names of Accountable Manager, Nominated Persons and Deputy Titles

AMC1 ORO.AOC.135

Malta Air has an Accountable Manager and the following Nominated Persons and Managers. These persons are accountable to the AM for their nominated responsibilities.

Position	Name	Deputy
Accountable Manager (AM)	Richard Higgins	-----
Nominated Person Flight Operations (NPFO)	Roger Amaral	John Said
Nominated Person Continuing Airworthiness (NPCA)	Alessandro Sette	Josef Mizzi
Nominated Person Crew Training (NPCT)	John Said	Raphael Soler
Nominated Person Ground Operations (NPGO)	Thomas Kelly	David Lawrenson
Safety Manager	Carlo Peronti	Stephan Kronenberger
Security Manager	Carlo Peronti	Stephan Kronenberger
Compliance Monitoring Manager (Operations)	Alfredo Fuschino	Jody Anfossi

Note: Currently in Malta Air the Safety Manager and Security Manager roles are combined (Safety and Security Manager – SMM), as are the Compliance Monitoring Manager (Operations) and Compliance Monitoring Manager (Engineering). References to the Compliance Monitoring Manager shall be construed as referring to both operations and engineering compliance roles.

1.2.1.1 Competence of Nominated Persons (NP)

ORO.AOC.135

1. Each Nominated Person in Malta Air possesses at least the experience and licensing provisions that are listed below:
 - a. practical experience and expertise in the application of aviation safety standards and safe operating practices;

- b. comprehensive knowledge of:
 - i. the applicable EASA safety regulations and any associated requirements and procedures;
 - ii. Malta Air operations specifications; and
 - iii. the need for, and content of, the relevant parts of the Malta Air operations manual;
 - c. familiarity with Aviation Management Systems;
 - d. Management experience in Malta Air or in a comparable organisation; and
 - e. 5 years of relevant work experience including at least 2 years in the aeronautical industry in an appropriate position.
2. Flight operations. The nominated person shall have a valid flight crew licence and the associated ratings appropriate to the Malta Air operation. In case the nominated person's licence and ratings are not current, his deputy will hold a valid flight crew licence and the associated ratings.
 3. Crew training. The nominated person or his deputy will be a current type rating instructor on B737-NG. The nominated person will have a thorough knowledge of the Malta Air crew training concept for aircrew.
 4. Ground operations. The nominated person or his deputy will have a thorough knowledge of the Malta Air's ground operations concept.
 5. Continuing Airworthiness. The nominated person or his deputy will have the relevant knowledge and appropriate experience of the requirements related to aircraft continuing airworthiness as detailed in Part-M.

1.2.2 The Accountable Manager (AM)

ORO.AOC.135, ORO.GEN.210

The Accountable Manager has corporate authority for ensuring that all operations, and maintenance activities are financed and carried out to the standard required by the EASA, and that the Nominated Persons have the capacity and authority to discharge their responsibilities as identified in AIR OPS and Aircrew Regulations.

The accountable manager has direct safety accountability for all activities related to the AOC.ORO.GEN.200 (a) (1).

The Accountable Manager has overall responsibility for the Malta Air Compliance Monitoring System including the frequency, format and structure of the internal management evaluation process. ORO.GEN.200 (a) (6)

The Accountable Manager ensures that facilities allowing the performance and management of all planned tasks and activities in accordance with the applicable requirements remain available. ORO.GEN.215

The Accountable Manager has direct Fatigue Management (FM) accountability, and ensures that FM is embedded in all relevant Malta Air activities and is in the terms of reference of all management personnel described in OMA. The AM demonstrates his endorsement of FM by signing the Safety and Security Policy statement as amended to include FM. ORO.FTL.120

The Accountable Manager has direct safety accountability for contracted activities as described below: ORO.GEN.205

1. Contracted activities include all activities within the Malta Air scope of approval that are performed by another organisation either itself certified to carry out such activity or if not certified, working under Malta Air's approval. Malta Air shall ensure that when contracting or purchasing any part of its activity, the contracted or purchased service or product conforms to the applicable requirements.
2. When Malta Air contracts any part of its activity to an organisation that is not itself certified in accordance EASA Regulations to carry out such activity, the contracted organisation shall work under the approval of Malta Air. The contracting organisation shall ensure that the TM CAD is given access to the contracted organisation, to determine continued compliance with the applicable requirements. AM ensures that the providers of subcontracted activities are included in the Malta Air Safety Management System.

Duties of the AM

1. Ensures that Malta Air, under the control and supervision of the relevant Nominated Persons, remains in compliance with the applicable regulatory requirements.
2. Attends the weekly Malta Air Management Meeting.
3. Chairs the weekly AM Meeting and ensure that all actions necessary for the safe and efficient running of Flight Operations, Ground Operations and Engineering are identified, fully considered, appropriately delegated and completed in a manner compliant to applicable regulations.
4. Ensures that safety is a fixed agenda item at the weekly AM Meeting to enable the Nominated Persons to address safety matters.

AM is responsible for setting the Malta Air Safety Policy.

5. Ensures Fatigue Management (FM) is a fixed agenda item at the weekly AM Meeting to enable the Nominated Persons to address FM matters.

6. AM is the Crisis Director.

1.2.3 Nominated Person for Flight Operations (NPFO)

The NPFO has final authority on all matters pertaining to Safety, FM and Regulatory Compliance. The NPFO will ensure compliance with relevant International, European and National Requirements in the management of flight operations.

NPFO is the director of all the operational personnel.

The NPFO shall represent Malta Air at selected TM CAD and EASA activities.

The NPFO reports and is accountable to the Accountable Manager for safety and FM in Flight Operations.

The NPFO is a trustee of the OFDM programme.

The NPFO and all management personnel shall continually promote the Malta Air Safety and Security policy and FM policy to all personnel and demonstrate their commitment to it. AMC1 ORO.GEN.200 (a) (2) ORO.FTL.120.

Safety management functions within Flight Operations include:

- development, authorisation and control of operating standards and procedures,
- direction of crew training,
- direction and supervision of line operations, and
- review of safety performance,
- review of FM,
- assure NPFO representation at SAG and SRB Meeting,
- ensures that Regulatory Authorities are notified of incidents as required.

The Safety Manager has an administrative report to the NPFO and provides a weekly report to the NPFO summarising safety activities and identifying trends.

Records

The NPFO holds flight crew records, crew training records and other records as are statutorily required or as otherwise stated.

Responsibilities of NPFO

The NPFO has overall responsibility for the control and supervision of flight operations in Malta Air. He is responsible for compliance with international standards, statutory regulations and Air OPS and Aircrew Regulations in the management of Flight Operations. He will ensure that the supervision of Flight Operations is such as to assure this compliance.

Crew and passenger safety is the prime responsibility of the NPFO. Cabin crew are subject to NPFO control and supervision during revenue flights and any other flights on which they are assigned a crew function.

The NPFO prepares a weekly safety report for the Accountable Manager and reports quarterly to the Board. In addition the NPFO attends the weekly AM Meeting where he shall raise pertinent safety and FM matters with the individual department managers for follow up and closure.

The NPFO has the necessary authority to discharge his duties, specifically:

1. Changes to the Malta Air Operations Manual are made only with the written approval of the NPFO,
2. Flight Crew Instructions and Safety Procedure Instructions are issued only with the approval of the NPFO,
3. No Flight Crew member is permitted to operate or command a Malta Air aircraft without the approval of NPFO,
4. Changes to Malta Air OMs/Operating Procedures are made only with the approval of the NPFO,
5. Airports will be utilised for Malta Air operations only with the approval of the NPFO,
6. Publication of Crew Alert Bulletins and Company Notams.

1.2.3.1 Deputy Nominated Person Flight Operations (DNPFO)

The Deputy Nominated Person Flight Operations deputises the NPFO in his absence on all matters pertaining to Safety and Regulatory Compliance.

The Deputy NPFO will ensure compliance with relevant International, European and National Requirements in the management of flight operations as well as the safety and FM in his areas of responsibility.

The Deputy NPFO reports directly to the NPFO.

The Deputy NPFO shall represent Malta Air at selected TM CAD and EASA activities as required by NPFO.

The Deputy NPFO ensures that all Regulatory Authorities are notified of incidents as specified in the Safety Manual.

The Deputy NPFO attends the weekly, Safety, Security and Fatigue Management review meeting. He shall assure confidentiality, deidentification and anonymisation of personnel involved in safety reports in the minutes and follow-up actions arising to the greatest extent possible. A weekly summary of this meeting is provided to the NPFO.

Additionally, the Deputy NPFO has specific responsibility for:

- Regulatory compliance of OMA, SEP Manual, Operations Manual Part B – Performance, Operations Manual Part B – Mass and Balance and OMC.
- Responsible for the closure of Audit Findings in the Flight Operations functional areas.
- FLT OPs liaison with Boeing.
- OMC Jeppesen route Charts.
- Timely completion of Compliance Monitoring audit corrective actions.
- Management and implementation of the Boeing (Seattle) delivery programme.
- Management and implementation of the Maintenance Check Flight programme.

Designated Base Captains report functionally to the Deputy NPFO as assigned.

The Deputy NPFO will provide a weekly report on performance in his area of responsibility.

1.2.4 Nominated Person Crew Training (NPCT)

The NPCT actively promotes safety and FM in his areas of responsibility. The NPCT reports directly to the AM. He will be qualified in accordance with the relevant regulation and as a minimum a TRI/TRE on type.

The NPCT is also the Deputy Nominated Person Flight Operations.

The NPCT ensures that the training and checking requirements comply with the requirements of AIR OPS and Aircrew Regulations. He reports to the AM on all matters of regulatory compliance, training, safety and training standards. He also reports to the AM on all matters relating to resourcing, management and administration of the Malta Air Training Department.

NPCT has the authority and responsibility for implementation, control and supervision of crew training to ensure that the training and checking requirements and the contents of training comply with requirements of AIR OPS and Aircrew Regulations.

NPCT is responsible for ensuring the AOC required level of personnel training is achieved and maintained.

Additionally, the NPCT has specific responsibility for:

- OMD
- FCTM compliance
- EASA – Part FCL
- Contracted Services – Training
- Air Crew Standards management
- Ensures compliance with domestic training regulations
- FM Training programmes
- Management of Deputy Nominated Person Crew Training (DNPCT) and CI-LINE

NPCT will provide a weekly report on performance in his area of responsibility.

1.2.4.1 Deputy Nominated Person Crew Training (DNPCT)

The DNPCT will deputise the NPCT in his absence on all regulatory required matters pertaining to crew training.

The DNPCT actively promotes safety and FM in his areas of responsibility. The DNPCT reports directly to the NPCT.

The DNPCT ensures that the training and checking requirements comply with the requirements of AIR OPS and Aircrew Regulations.

Additionally, the DNPCT has specific responsibility for:

- All matters relating to management and administration of the training contracted services to Ryanair DAC, including the management and administration of Line Training
- Management of FSTD approvals
- Management of AOC facilities
- Management and tracking of training approvals and regulatory compliance

- Management and tracking of all training and checking for Non-Commercial operations

DNPCT will provide a weekly report on performance in his area of responsibility.

1.2.4.2 Section Not in Use

1.2.5 Nominated Person Continuing Airworthiness (NPCA)

Malta Air has appointed an NPCA with responsibility for the management and efficient operation of the CAMO and Engineering Department. The NPCA and has direct access to the Accountable Manager. The NPCA shall have the relevant knowledge, background and experience in accordance with Regulation (EU) No 1321/2014.

NPCA Duties:

1. Attends the weekly AM Meeting and ensure that all actions necessary for the safe and efficient running of Engineering are identified, fully considered, appropriately delegated and completed in a manner compliant to applicable regulations including referred items from Engineering Operations/Flight Operations/Ground Operations interface.
2. Ensures that all required corrective actions arising from the Malta Air SMS and FM are addressed and closed where possible prior to the end of the following quarter.
3. Produces a weekly report for the AM detailing Engineering activity and highlighting any SMS or FM items.
4. Ensures any new procedures/changes in policy are reviewed for risk analysis and agreed upon by engineering management group prior to implementation.
5. Where procedure/policy changes involve more than one department the NPCA brings such items to the weekly AM Meeting for information, review and risk assessment as applicable.

1.2.6 Nominated Person – Ground Operations (NPGO)

Malta Air has appointed a Nominated Person for Ground Operations (NPGO). The NPGO for Malta Air is the General Manager – Ground Operations. The NPGO has final authority on all matters pertaining to safety and regulatory compliance. The NPGO will ensure compliance with relevant International, European and National Requirements in the management of Ground Operations including subcontracted services.

The NPGO reports to the Accountable Manager on all matters relating to safety and regulatory affairs in Ground Operations. The NPGO shall have a thorough knowledge of Ground Operations. The NPGO shall continually promote the Malta Air Safety and Security Policy and Objectives to all personnel and demonstrate commitment to it.

Malta Air provides ACMI services to Ryanair Group airlines. The scheduling and commercial deployment of the Malta Air fleet assigned to ACMI operations is carried out by the operating AOC's Scheduling Department and daily operations are overseen by the respective OCC. The responsibility for control and supervision of all Ground Operations activities including those specified in SLA rests with the Malta Air Nominated Person Ground Operations.

1.2.7 Safety and Security Manager

AMC1 ORO.GEN.200, ORO.FTL.120

The Safety and Security Manager reports to the Accountable Manager. The Safety and Security Manager is the focal point for safety in Malta Air. He is responsible for the development, administration and maintenance of an effective safety management system. Detailed description of safety related duties and responsibilities are included in the Safety Management Manual and include facilitating hazard identification, risk analysis, monitoring the effectiveness of the Safety Action Plan, provision of reports on safety performance, maintenance of safety management documentation, availability of appropriate safety management training, ensure initiation and follow-up of internal occurrence/accident investigations and provide safety advice to the Nominated Persons and Accountable Manager.

The Safety and Security Manager is also the Fatigue Manager and is the focal point for FM in Malta Air. He is responsible for the development, administration and maintenance of an effective FM system.

Flight Data Monitoring (FDM) Programme

AMC1 ORO.AOC.130

The Safety and Security Manager is responsible for the identification and assessment of issues arising from the OFDM programme, and their transmission to the manager(s) responsible for the process(es) concerned. These managers are responsible for taking appropriate and practicable safety action within a

reasonable period of time that reflects the severity of the issue. He is responsible for the oversight and response to issues identified in the OFDM programme and shall take appropriate and practicable safety action within a reasonable period of time that reflects the severity of the issues identified.

The Safety and Security Manager is assisted by additional personnel for the performance of safety and FM related tasks.

Security

The Safety and Security Manager also performs assignments regarding the airline security program and coordination of security training. The detailed scope of the SSM responsibilities for security are described in the Malta Air Security Manual.

1.2.8 Section Not in Use

1.2.9 Compliance Monitoring Manager

The role of the Compliance Monitoring and Engineering Quality Manager is to ensure that the compliance monitoring programme is properly implemented, maintained, reviewed and improved. He develops the audit plans and ensures realisation. He administers the quality system documentation and ensures the corrective actions are completed and the adequate procedures implemented.

The Compliance Monitoring Manager will ensure that activities in flight operations, crew training, aircraft maintenance and ground operations are monitored for compliance with the applicable regulatory requirements and additional Malta Air requirements, and that such activities are being carried out properly under the supervision of the relevant Nominated Person.

The Compliance Monitoring Manager has direct access to the Accountable Manager and may communicate to him all information regarding Compliance matters.

A detailed description of the compliance monitoring responsibilities is included in the Malta Air Compliance Monitoring Manual.

1.2.10 Section Not in Use**1.3 Responsibilities and Duties Operations Management Personnel****1.3.1 Section Not in Use****1.3.2 Section Not in Use****1.3.3 Deputy Chief Pilot Line Operations (DCPLO)**

The DCPLO reports directly to the NPFO. The DCPLO responsibilities include:

- Ensuring ongoing operational compliance throughout the network and maximising efficiencies on all flight operations
- Flight Operations Liaison with Boeing
- FCOM Revisions
- Operational Fuel Management
- Winter Ops Documentation

The DCPLO shall represent Malta Air at selected external activities.

The DCPLO has administrative responsibility for line operations as assigned by the NPFO. Administrative responsibility is effected by routine RBC Meetings and review of the RBC weekly reports and production of a weekly summary of items to be raised by the NPFO at the management meetings, flight operations meeting and the NPFO meeting.

The DCPLO is responsible for safety, security and FM in his areas of responsibility.

The DCPLO provides a weekly report on performance in his area of responsibility.

1.3.4 Section Not in Use**1.3.5 Chief Instructor – Simulator (CI-SIM)**

The Chief Instructor – Simulator reports directly to the NPCT on all matters pertaining to simulator training and checking in the AOC. He will be qualified as a TRI/TRE and will act as the Chief Instructor and Chief Examiner for the AOC. He is responsible for all AOC SFIs, TRIs, SFEs and TREs.

The CI-SIM is also the Chief Instructor and Chief Examiner for the Ryanair DAC ATO and is responsible for all ATO SFIs, TRIs, SFEs and TREs.

The CI-SIM is responsible for the standards of all AOC and ATO Instructors and Examiners.

The CI-SIM has specific responsibilities for:

- Production and delivery of required AOC and ATO Flight Crew Training and Checking Programmes
- Management and supervision of Standards Instructors (Sis)
- Management and supervision of all Simulator Instructors and Examiners
- Management and supervision of all AOC and ATO students throughout training programmes
- Implementation of operational changes into relevant training programmes
- Management and supervision of Command Upgrade Programme (CU) in conjunction with CI-LINE

1.3.6 Chief Instructor – Line (CI-LINE)

The Chief Instructor – Line reports directly to the NPCT on all matter pertaining to line training and checking in the AOC. He will be qualified as a TRI/TRE and will act as the Chief Instructor for all Line Training and Line Checking activities.

He is responsible for the management and the standards of all AOC Base TREs (BTREs), Line Training Captains (LTCs), and Line Check Captains (LCCs).

The CI-LINE has specific responsibilities for:

- Production and delivery of required AOC Flight Crew Line Training and Checking programmes
- Management of BTREs for Line Training and Checking activities
- Implementation of operational changes into relevant training programmes
- Management and supervision of all AOC and ATO students throughout training programmes
- Management and supervision of Command Upgrade (CU) in conjunction with the Ryanair DAC ATO CI-SIM

The CI-LINE will provide a weekly report to the NPCT on performance in his area of responsibility.

1.3.7 Section Not in Use**1.3.8 Regional Base Captains**

RBCs are appointed by the NPFO. They report to the DCPLO. RBCs represent the NPFO office in all their activities. They will promote current Operational and Safety and FM issues and ensure compliance with the duties and responsibilities of the Base Captains assigned to him.

RBCs will perform regular scheduled and unscheduled base visits and liaise with the Base Captain, Base TRE, LASG, GOPS Regional Base Managers and FOPS Regional Base Managers regarding safety, FM, operational KPIs and training matters.

RBCs report administratively to the Deputy NPFO on areas of regulatory compliance, safety and FM.

Detailed terms of reference for RBCs are available from the DCPLO. RBCs are also qualified Flight Operations Auditors and perform FOPS safety audits, including when positioning to/from bases and complete the associated administration.

RBCs shall complete the Base Captains Training Course and will be familiar with the BC ToR and OPSDOC (Base Captain Terms of Reference and Operations Document).

1.3.9 Base Captains

Base Captains are appointed for those stations at which Malta Air Flight Crew are based. Base Captains report to the NPFO through the DPFO on all matters of regulatory compliance, aircraft operation, safety, FM and standards. Base Captains will collect and provide all necessary information to the NPFO to ensure Flight Operations from their bases are conducted in a safe, efficient, punctual and regulatory compliant manner according to approved Malta Air policies and procedures. Base Captains are trained ASR Investigators and conduct local safety investigations as directed by the Safety Office.

Base Captains are the NPFO representatives to locally based crew and liaise with other/external agencies as directed by the NPFO. Base Captains do not have the power to conduct business autonomously or to enter into contracts on behalf of the company with third parties.

Base Captains liaise with their Bases Manager for ground operational activities, Airport Facilities and administrative matters, supplies etc. Base Captains liaise with Base Supervisors for any local safety related matters.

Base Captains duties shall normally be conducted in the Base Office and crew facilities – providing BC visibility and/or availability to crew, in particular, being available in base for returning First Wave crew and departing Second Wave crew. Any variation shall be approved by the DCPLO.

Base Captains are trained ASR Investigators and conduct local safety investigations as directed by the Safety Office.

Base Captains will provide a weekly report on performance in their base.

Detailed Terms of Reference for BCs are contained in the DCP/RBC/BC ToR document available from the DCPLO.

1.3.9.1 Standards Captain (Contract Pilots and Compliance)

The Standards Captain Compliance is responsible for the supervision and control of all operations conducted by contract pilots at Malta Air as directed by the NPFO.

The Standards Captain Contract Pilots and Compliance reports to the Deputy NPFO and DCPLO and furnishes a weekly report on the conduct and compliance of contract pilots, with particular emphasis on FM and safety during line operations.

This report also contains matters of compliance monitoring and audit closure administrative allocated.

1.3.10 Section Not in Use

1.3.11 Section Not in Use

1.3.12 Section Not in Use

1.3.13 Flight Safety Officer (FSO)

The FSO reports to the Safety and Security Manager.

The FSO is a member of the Safety Action Group. His duties are detailed in the Safety Management Manual.

For the purpose of the SMS, the FSO is independent of the Nominated Persons and does not carry executive responsibility for safety but works through and with the co-operation of departmental managers.

The FSO needs to have access to all facets of the organisation in order to fulfil his role.

Duties of the FSO:

1. Provide the Safety Manager with a weekly report
2. Attends the weekly FOPS Safety, Security and FM Review Meeting
3. The FSO will undertake routine base visits and/or on-line meetings to:
 - Promote the Confidential Reporting System (CRS)
 - Ensure effective implementation of the Local Air Safety Groups (LASG)
 - Instruct and liaise with the Base Captains on the Base Investigation Process
 - Ensure OFDM facilitator training completion
4. The FSO is the Chief Incident and Accident Investigator

1.3.14 Chief Instructor Safety

1.3.14.1 Chief Instructor Safety – Safety, OMA and OMD

The CI-SFTY reports to the NPFO and NPCT. Reporting to the NPFO is on all matters of safety, operational standards and regulatory compliance relating to the maintenance of the SEP manual. Reporting to the NPCT is pertaining to the conduct of Cabin Crew training required by AIR OPS. They should be a current CCA holder and qualified as an SEPI/E.

The CI-SFTY (OMA and OMD) will have specific responsibility for the following:

1. Reports to NPFO on regulatory compliance matters of SEP manual.
2. Revises and upkeep the SEP Manual and B737 Quick Reference Guide.
3. Reports to NPCT on all matters of Safety and Emergency Procedures training.
4. Participates in recruitment process and approves new SEPIs and SEPES.
5. Conducts Instructor Proficiency Checks on selected SEPIs and SEPES.
6. Visits to main training bases in accordance with approved schedule.
7. Management of AOC Cabin Crew Training Facilities.
8. Production and delivery of all AOC Cabin Crew training courses.

1.3.14.2 Chief Instructor Safety – Safety and Security (CI-SFTY & SEC)

The CI-SFTY & SEC reports directly to the NPCT on all matters of safety, operational standards and regulatory compliance relating to the maintenance of the Cabin Crew Safety and Security training. They should be a current CCA holder, qualified as an SEPI/E and Security Instructor/Examiner.

The CI-SFTY & SEC will have specific responsibility for the following:

1. Production and delivery of all required Safety and Security training courses.
2. Participates in recruitment process and approves new SEPIs and SEPEs.
3. Conducts Instructor Proficiency Checks on selected SEPIs and SEPEs.
4. Visits to main training bases in accordance with approved schedule.

1.3.15 Duty Pilot

Flight Operations schedules a Duty Pilot who is available as a point of contact for line crews H24. The Duty Pilot is a qualified Captain acceptable to the NPFO.

The primary role of the Duty Pilot is to provide operational assistance to crews, OCC, Maintrol and Crew Control. Full details of the duties and responsibilities can be found in the Duty Pilot terms of reference.

1.3.16 Section Not in Use

1.3.17 Section Not in Use

1.3.18 The Malta Air Management Process

1.3.18.1 Malta Air Management Meeting (Z Meeting)

This section is to assist aircrew to understand the management process.

Malta Air business and operations are advanced through specific, minuted, regular meetings which Nominated Persons attend and contribute to. In this way they discharge their responsibilities by way of direct access to the Accountable Manager and regular interface with other Nominated Persons.

The Accountable Manager delegates the necessary authority to the Nominated Persons to discharge their responsibilities. The Malta Air management process supports the supervision of operations as required by EASA Regulations, including ORO.GEN.200.

The Malta Air meetings are as follows:

1.3.18.2 Section Not in Use**1.3.18.3 Section Not in Use****1.3.18.4 AM Meeting**

The AM Meeting is scheduled weekly and chaired by the AM or his nominee.

Attendees include:

Nominated Person for Flight Operations,

Nominated Person Ground Operations,

Nominated Person Continuing Airworthiness,

Safety and Security Manager,

and other invited personnel, as appropriate.

The purpose of the meeting is to provide attendees with the opportunity to raise all matters pertaining to operations, FM, safety and procedural changes. The meeting also provides the opportunity to review the operational efficiency and to follow up on corrective actions raised through the SMS and Compliance Monitoring System.

1.3.18.5 NPFO Management Meetings**1.3.18.5.1 NPFO Weekly Management Meeting**

This meeting is scheduled weekly and chaired by the NPFO or his nominee.

Attendees include the DCPs and other invited personnel.

The purpose of the meeting is to provide attendees with the opportunity to raise all matters pertaining to operations, safety, FM and procedural changes. This meeting reviews and actions all issues arising for the previous week and from other management meetings relating to the flight operations department safety, FM and performance. FOPS policy is continually reviewed at this meeting to ensure regulatory compliance and compliance with best practice.

1.3.18.5.2 Monthly OFDM Review Meeting

This minuted meeting is held monthly and is co-chaired by the NPFO and SM, and their nominee.

The meeting is attended by the NPCT, FSO, OFDM Trustee, Safety and Security Manager and other invited personnel.

The purpose of this meeting is to review OFDM performance, data capture rate and to review the Safety and Security Manager assessment of the OFDM data. OFDM data is assessed for its operational significance, identification of trends and recommendation of training and/or OPS initiatives as suggested by the data analysis.

Managers are responsible for taking appropriate and practicable safety action within a reasonable period of time that reflects the severity of the issue.

1.3.18.5.3 FOPS Safety, Security and FM Review Meeting

This minuted weekly meeting is chaired by the NPFO or his nominee. Attendees include: The Safety and Security Manager, FSO, NPCT (or his nominee), Inflight Safety Manager, GOPS Safety and Compliance Manager, NPCA representative and other invited personnel.

The purpose of the meeting is to have expertise from all relevant Nominated Persons and functional areas present to enable an operational review of:

- The consolidated report from the weekly ASR/FM Ops Review meetings over the previous week.
- Investigations arising from the weekly ASR/FM review meeting.
- Review draft investigations (BI, IBI, GBI, EBI) and FR.
- Progress any open investigation, formal review, ASR, CPTSR etc to closure including internal and external Safety Recommendations.
- Agree 'Top 3' ASR/FM threats for the week.

1.3.18.5.4 Daily ASR & FM Ops Review Meeting

This daily meeting takes place via Teams for the purpose of reviewing all ASRs filed in the previous 24 hrs and to agree the follow-up action required. The meeting is attended by representatives from Flight Operations, Engineering, GOPS, In-Flight Safety Office, Safety Services Office, Crew Training. Minutes are taken and agreed action items are circulated to all attendees and nominated persons after the meeting has concluded.

1.3.18.5.5 Base Captains Operational Standards Meetings

These meetings are scheduled at least twice per year. A meeting may be a face to face and/or web-conferencing meeting. A meeting is attended by the DCPs, RBCs, BCs and other invite personnel, as appropriate.

The purpose for the meeting is to review the policy implications of regulatory change, operational performance, new routes, new destinations, operational incidents, Safety FM policy, OFDM trends and exceedances.

1.3.18.5.6 Flight Safety Forum

This meeting is chaired by the Safety Manager or nominee. Delegates to this meeting are from the LASGs in Bases and the meeting enables the Safety Manager to receive their views and opinions as front-line personnel from FOPS, In-flight, Engineering and GOPS.

Full details are contained in the Ryanair Safety Management System Manual (SMM).

1.3.19 Regulatory Reporting Lines to the NPFO for Inflight**1.3.19.1 Head of Inflight Safety and Security**

AMC1 ORO.GEN.200(a)(2), ORO.FTL.120

The Head of Inflight Safety & Security has a direct regulatory report to the NPFO for all cabin operations relating to safety and security and all matters of ongoing regulatory compliance as approved by TM CAD. The Head of Inflight Safety and Security also reports to the NPCT for all matters related to cabin crew training.

1.3.19.2 Inflight Safety and Security Manager

The Inflight Safety and Security Manager has a direct regulatory report to the Safety and Security Manager on matters relating to the management, development, administration and maintenance of effective safety management processes in Inflight.

The Inflight Safety and Security Manager reports to the Head of Inflight Safety and Security for the functioning of the Inflight Safety Department and follow up on inflight related findings arising from internal Compliance Monitoring System audits and inspections.

The Inflight Safety and Security Manager shall hold or will have held a cabin crew attestation.

The Inflight Safety and Security Manager is a member of the Safety Action Group (SAG) and the Safety and Security Review Board (SSRB).

He/she deputises for the Head of Inflight Safety and Security.

The duties and responsibilities of the Inflight Safety and Security Manager include:

1. Manage the Inflight Safety office to ensure:
 - a. The Cabin Safety Reporting System is administered as required;
 - b. Relevant Cabin Safety Reports are brought to the Daily ASR Review Meeting for review and appropriate follow up action;

- c. Inflight Base Investigations and Inflight Formal Reviews are carried out as necessary.
2. Attendance at the weekly SSeFM meeting;
3. Attendance at the weekly Safety, Security and Compliance Meeting;
4. Submission of a weekly inflight report to the Safety Manager;
5. The Inflight Safety and Security Manager is the custodian of the Malta Air Inflight Confidential Reporting System;
6. Safety and Security Policy and Just Culture principles promotion within the Inflight Department;
7. In consultation with the Safety and Security Manager, the Inflight Safety and Security Manager will proactively monitor and identify inflight hazards, conduct risk assessments and confirm the effectiveness of actions taken to mitigate risks;
8. Ensure all Inflight related findings arising from internal Compliance Monitoring System Audits/Inspections are addressed on time for the quarterly MEP.

1.3.19.3 Inflight Training Administration Manager

The Inflight Training Administration Manager reports to the Head of Inflight Safety and Security for ensuring that all crew operating on Malta Air flights remain in check for all required training and that all Safety and Emergency Procedures Instructors and Examiners have rostered duties compliant with the Malta Air approved Individual Flight Time Specification Scheme (IFTSS) and in accordance with Fatigue Management principles.

1.3.19.4 Inflight Crew Planning Manager

The Inflight Crew Planning Manager has regulatory responsibilities to the NPFO in relation to Cabin Crew rostering and fatigue management of Cabin Crew personnel and is responsible for ensuring that all cabin crew operating on Malta Air flights are rostered in accordance with Malta Air approved Flight Time Limitation scheme.

The Inflight Crew Planning Manager ensures all personnel associated with Cabin Crew rostering are trained in accordance with the approved FTL Scheme.

A weekly report is submitted to the NPFO identifying any non-compliances with the approved FTL scheme.

1.3.19.5 Base Supervisor (BS)

Base Supervisor (BS) safety and security accountabilities are to the Head of Inflight Safety and Security. Each Malta Air Base shall have at least one Base Supervisor and sufficient deputy Base Supervisors to enable supervision of Cabin Crew personnel operating in that Base.

Base Supervisors duties and responsibilities include the monitoring of Cabin Crew incidents and the reporting of all safety related incidents to the NPFO through the Head of Inflight Safety and Security.

1.3.19.6 Section Not in Use**1.3.19.7 Head of Pilot Rostering**

Malta Air contracts Rostering to Ryanair DAC. The Head of Pilot Rostering has a direct regulatory report to the NPFO for the regulatory compliance of all rosters.

- Ensuring that rosters comply with the approved FTL Scheme and should be published at least 14 days in advance to enable operating crew to plan appropriate rest
- Fatigue Management principles are applied to the roster
- Ensuring that all personnel involved in pilot roosting are trained accordingly and have demonstrated competence in their roosting area
- Ensuring that all roosting personnel receive Fatigue Management training as outlined in SMM 6.7

A weekly report is submitted to the NPFO confirming published rosters comply with the approved FTL scheme.

1.3.20 Contracted Activities

AMC2 ORO.GEN.205

A Ryanair Group Airline is either Ryanair DAC or another airline within the Ryanair Group.

Each Ryanair Group Airline shall be subject to regulation by the competent National Authority for that airline. The listed below contracted activities will be outsourced to Ryanair DAC or other Ryanair Group organisations as required in accordance with the following regulations:

- ORO.GEN.205

List of contracted activities includes:

- Flight planning
- Crew planning

- Navigation services
- Crew training
- Continuing Airworthiness Management tasks
- Maintenance
- Operational base management
- Common usage of flight documentation (for example FCOM/QRH)
- Electronic Flight Bag's System

Ryanair DAC is an Irish registered company number 104547 with a registered address of:

Ryanair Dublin Office
Airside 3 Park Swords
Co Dublin
Ireland

It is an AOC holder with its principal place of business in Ireland and the Irish Aviation Authority (IAA) is the designated competent authority.

All activities contracted to Ryanair DAC will be conducted under Ryanair DAC's approval from the Irish Aviation Authority (IAA). Ryanair DAC have all of the necessary authorisations and approvals, along with the resources and competence to carry out contracted activities as directed.

Malta Air is responsible for ensuring that all contracted activities are subject to hazard identification and risk management, as required by ORO.GEN.200 (a)(3), and to compliance monitoring, as required by ORO.GEN.200 (a)(6). Where the Ryanair DAC approval is being utilised the Malta Air AM shall ensure that the approval covering the contracted activities is still valid.

A detailed list of contracted organizations, including the scope of cooperation, is listed in the Approved Suppliers and Subcontractors List, maintained by the Group.

Ryanair DAC and Malta Air shall use identical operating procedures. The Malta Air NPFO shall monitor differences between the Ryanair DAC and Malta Air documentation to ensure that the identical nature of the operating procedures and syllabi is maintained.

The Malta Air NPFO remains ultimately responsible for the product and delivery of any flight operational activity delivered by an external organisation.

1.3.20.1 Operations Control Center OCC

Operations Control Centre of Ryanair DAC ensures overall operations control on a 24/7 basis for Malta Air.

OCC provides:

- Operational Control and Supervising
- Maintenance control
- GOPS control
- Crew control
- Dispatch and flight watch
- Navigation and Performances calculation
- Collection and storage of Documentation and Flight Records
- Managing of Emergency situations
- Operational communication
- Meteorological information
- Cooperation with airport operations services, ATC
- Managing of allocated slots

1.3.20.2 Performance Department PD

The Performance Department is a contract service.

Performance Department ensures the regulatory compliance of all the performance calculations for Malta Air operations as delegated by the NPFO.

The Performance Department ensures:

- Regulatory compliance of the OMA, OMB and OMC related areas
- Control and support of all Electronic Flight Bag (EFB) PD related functions
- Airport/runway obstacle analysis to determine the optimum take-off/landing performance for all current & potential destinations/alternates including the calculation & provision of Engine Failure Turn Procedures, where required
- Evaluation of potential new airports on their operational viability to include Airport Categorisations and pay load range
- Conduct airport visits as required
- Daily monitoring and assessment of operational impact of published NOTAMs

- Monitoring of AIP, Aeronautical Charts and FMC data for action item affecting PD issued/published data
- Liaison with external agencies including aircraft manufacturer, regulatory authorities, contractors, suppliers, and airport authorities to ensure timely availability of accurate data meeting all requirements for operations

1.3.20.3 EFB Project Manager

The management and administration of the EFB system is contracted to Ryanair DAC. The Flight Operations EFB Project Manager has a safety, security and regulatory responsibility to the NPFO for the management and development of Electronic Flight Bag (EFB) applications.

The authority, duties and responsibilities of the EFB Project Manager are as follows:

1. Maintain regulatory compliance of EFB
2. Conduct aircraft and simulator trials/evaluations of new operational applications
3. Management of approval processes for new EFB Applications
4. Developing the EFB as required to increase operational efficiency and usability
5. Providing technical support and development of existing EFB applications
6. Developing Pilot Training requirements for all new operational applications
7. Liaison with required authorities to gain EFB regulatory approvals
8. Set up and support of organisations/partners relating to EFB systems

1.3.20.4 EFB Administrator

Administration of the Malta Air EFB system has been contracted to Ryanair DAC and ensures continuity of the management of the EFB system on a 24/7 basis.

EFB administration shall be subject to independent routine audits conducted by the Malta Air compliance monitoring programme.

Duties of EFB Administrator include but are not limited to:

- Ensure current/valid version of application software and data packages are installed.
- Ensure integrity of data packages used by applications.
- Provide support to EFB users on installed applications.

- Software configuration management.
- Ensure no unauthorised software is installed.
- Main contact between Malta Air and the EFB system and software suppliers.
- Management of returned flight data storage and backup.
- Co-ordinate the issue and return of EFB Devices.
- Planning of required Hardware to meet operational requirements.

1.3.20.5 Rostering

Rostering is contracted to Ryanair DAC. The Ryanair DAC Rostering Department ensures that all principles of Malta Air FTL's and FM are followed at all times as delegated by NPFO. Rostering activities shall be subject to independent routine audits conducted by the Malta Air compliance monitoring programme.

1.3.20.6 Approved Training Organization ATO

Refer to OMD.

1.3.20.7 Airworthiness Tasks

Refer to CAME Manual.

1.4 Authority, Duties and Responsibilities of the Commander

CAT.GEN.MPA.110

1.4.1 Basic Authority of the Commander

The Company designates one flight crew member to act as Commander of a flight or series of flights. In the case of non-commercial flights, as described in OMA 8.7, the Company will designate a 'Pilot-in-Command'. This difference in nomenclature reflects the different legal status of a non-commercial flight, since Commander is a term applicable to Commercial Air Transport flights.

The duties and responsibilities of a 'Pilot in Command' are the same as that of a 'Commander' and references in OMA to requirements associated with the 'Commander' are equally applicable to the 'Pilot-in-Command' of a Malta Air operated aircraft on a non-commercial flight.

The authority vested in the Commander is delegated from the NPFO.

The Commander is responsible for the operation and safety of the aircraft and for the safety of all persons on board as soon as the commander arrives on board the aircraft, until the commander leaves the aircraft at the end of the flight. For this purpose, he shall have final authority for the disposition of the aircraft during the time in which he is in command. He shall have authority to give all commands he

deems necessary for the purpose of securing the safety of the aeroplane and of persons or property carried therein. All persons carried in the aeroplane shall obey such commands.

Delegation of duties to other crew members in accordance with procedures laid down in the Operations Manual shall in no way affect his ultimate responsibility for the safe conduct of the flight.

Authority of the Commander

CAT.GEN.MPA.110

By direction of Malta Air, all persons carried in the aircraft shall obey all lawful commands given by the commander for the purpose of securing the safety of the aircraft and of persons or property carried therein.

The Commander is accountable to the NPFO for the safe, punctual and efficient execution of his duties and responsibilities.

Nothing in this manual shall be construed as limiting or derogating from this authority.

1.4.2 New Commander Restrictions

For the first 100 hrs following initial command check new commanders have the following restrictions:

1. Must not operate into Category B RYR Restricted airfields or Category C airfields.

Note 1: A destination, a take-off, enroute or destination alternate which is Category B RYR Restricted shall not be selected by a New Commander and ideally will not be presented in a flight plan for a New Commander. New Commanders shall check the Airfields (Aerodromes) Categorisation as published in the OMC Appendix A, Approved Aerodromes List on EFB for current B, RYR Restricted locations.

In an emergency situation this Category B RYR Restricted guidance does not apply.

Note 2: Refer to OMA 5.1.2.5 for Aerodrome categories.

2. Are required to confirm that the selected alternates used when planning flights are Category A or Category B airports.
3. Must not operate with an inexperienced co-pilot i.e. with a designation of NEW (OM-A 4.1.4). It is essential that new Commanders verify the status of the co-pilot at crew check-in.

4. Are not permitted to carry out visual approach manoeuvring unless no other approach procedure is available. Any visual approach must be conducted strictly in accordance with FCOM Vol 1, NP 21. Visual Approach.

1.4.2.1 Training Required for New Commanders to Operate from Specific Category B RYR Restricted Locations:

New commanders who have been assigned a Home Base that is Category B RYR Restricted, may operate from that base, including circling, provided they have completed the Training Module (Ref OM Part D 3.1.6.14). They are not permitted to operate to any other Category B RYR restricted or conduct circling approaches at any other airport until they have achieved 100 hours following initial line check.

1.4.3 Responsibilities of the Commander

CAT.GEN.MPA.105 Regulation (EU) 2019/1384

The commander, in addition to complying with CAT.GEN.MPA.100, shall:

1. Be responsible for the safety of all crew members, passengers and cargo on board, as soon as the commander arrives on board the aircraft, until the commander leaves the aircraft at the end of the flight;
2. Be responsible for the operation and safety of the aircraft, from the moment the aeroplane is first ready to move for the purpose of taxiing prior to take-off, until the moment it finally comes to rest at the end of the flight and the engines are shut down;
3. Have authority to give all commands and take any appropriate actions for the purpose of securing the safety of the aircraft and of persons and/or property carried therein in accordance with Regulation (EU) 2018/1139;
4. Have authority to disembark any person, or any part of the cargo, that may represent a potential hazard to the safety of the aircraft or its occupants;
5. Not allow a person to be carried in the aircraft who appears to be under the influence of alcohol or drugs to the extent that the safety of the aircraft or its occupants is likely to be endangered;
6. Have the right to refuse transportation of inadmissible passengers, deportees or persons in custody if their carriage increases the risk to the safety of the aircraft or its occupants;
7. Ensure that all passengers are briefed on the location of emergency exits and the location and use of relevant safety and emergency equipment;
8. Ensure that all operational procedures and checklists are complied with in accordance with the operations manual;

9. Not permit any crew member to perform any activity during critical phases of flight, except duties required for the safe operation of the aircraft;
10. Ensure that:
 - Flight recorders are not disabled or switched off during flight;
 - In the event of an occurrence other than an accident or a serious incident that shall be reported according to ORO.GEN.160(a), flight recorders' recordings are not intentionally erased; and
 - In the event of an accident or a serious incident, or if preservation of recordings of flight recorders is directed by the investigating authority:
 - a. Flight recorders' recordings are not intentionally erased;
 - b. Flight recorders are deactivated immediately after the flight is completed; and
 - c. Precautionary measures to preserve the recordings of flight recorders are taken before leaving the flight crew compartment;
11. Decide on acceptance of the aircraft with unserviceability's in accordance with the configuration deviation list (CDL) or the minimum equipment list (MEL);
12. Ensure that the pre-flight inspection has been carried out in accordance with the requirements of Annex I (Part-M) to Regulation (EU) No 1321/2014;
13. Be satisfied that relevant emergency equipment remains easily accessible for immediate use;
14. Record, at the termination of the flight, utilisation data and all known or suspected defects of the aircraft in the aircraft technical log or journey log of the aircraft to ensure continued flight safety.

The Commander signifies acceptance of his responsibilities for the flight by signing the paper LID (or Loadsheel), the paper Technical Log and by entering their 4 digit PIN in the OPT W&B e-signature panel. Signing the OPT W&B attaches your e-signature to the emailed statement (OFP, COMPULSORY READS, NOTAMS & WEATHER CHECKED/LICENCES VALID/AFB REVIEWED (WHERE PUBLISHED)/EFB VALIDITY CHECKED).

Note 1: The LID may be in electronic format (eLID). The Commander signifies acceptance of the eLID by entering their PIN code in the EFL App.

Note 2: A copy of the Technical Log must be retained on the ground. This may be in paper copy or electronically attached and submitted with the eLID.

- 1.4.3.1 **The commander to whom conduct of the flight has been delegated, shall, in an emergency situation that requires immediate decision and action, take any action he/she considers necessary under the circumstances in accordance with Regulation (EU) No 2018/1139. In such cases he/she may deviate from rules, operational procedures and methods in the interest of safety.**
- 1.4.3.2 Whenever an aircraft in flight has manoeuvred in response to a TCAS resolution advisory (RA), the commander shall submit a report to the competent authority*.
- 1.4.3.3 Bird hazards and strikes:
- Whenever a potential bird hazard is observed, the commander shall inform the air traffic service (ATS) unit as soon as flight crew workload allows.
- Whenever an aircraft for which the commander is responsible suffers a bird strike that results in significant damage to the aircraft or the loss or malfunction of any essential service, the commander shall submit a written bird strike report after landing to the competent authority*.
- 1.4.3.4 The commander shall, as soon as possible, report to the appropriate air traffic services (ATS) unit any hazardous weather or flight conditions encountered that are likely to affect the safety of other aircraft.

Note: *This responsibility is discharged by filing an ASR in the normal manner.

1.5 Duties and Responsibilities of Crew Members Other than the Commander

1.5.1 Co-pilot

The Co-pilot is responsible to the Commander during the preparation for and operation of his/her assigned flight, and to the Base Captain for duties other than a flight duty.

A co-pilot may be a Second Officer (≤ 500 hours CS 25) or a First Officer (> 500 hours CS 25).

The Co-pilot shall:

1. Observe the provisions of the Operations Manual in general, and such provisions of the Operations Manual as are necessary to fulfil his function and responsibilities as a Co-pilot;
2. Assist in the flight preparation;

3. Assist the Commander in the management of the flight and the manipulation of the aircraft controls in accordance with the directions of the Commander who will be guided by the modern principles of the CRM;
4. Participate in the execution of cockpit procedures, emergency procedures, check list procedures and instrument approach procedures as directed by the Commander and in accordance with the procedures and drills laid down in the Operations Manual;
5. Pay particular attention to monitoring the critical phases of flight. He has a duty to inform the Commander, in good time, if in his judgement there is any deviation from a safe standard of operation;
6. Observe Standard Operating Procedures. Draw attention of the Commander of any deviation from Standard Operating Procedures;
7. Carry out take-offs and landings only under the specific authority of the Commander. He shall carry out, in preparation for flight or during flight, such other duties as the Commander may delegate to him;
8. Assume the authority and responsibilities of the Commander only in the event of the latter's incapacitation; and
9. In the absence of competent ground handling staff be responsible to the Commander for supervising the loading of the aircraft and the preparation of the Load Sheet.

1.5.2 Cabin Services Supervisor (CSS/SCCM/No 1)

ORO.CC.200(d), AMC1 ORO.CC.200(d)

The primary statutory function of CSS/SCCM/No 1 is that of safety.

The senior cabin crew member shall be responsible to the commander for the conduct and co-ordination of normal and emergency procedures specified in OM(A) Safety and Emergency Procedures, including discontinuing non-safety related duties for safety or security purposes.

When the level of turbulence so requires, and in the absence of any instructions from the flight crew, the senior cabin crew member shall be entitled to discontinue non-safety related duties and advise the flight crew of the level of turbulence being experienced and the need for the fasten seat belt signs to be switched on. This shall be followed by the cabin crew securing the passenger cabin and other relevant areas.

The Cabin Services Supervisor, or No. 1, directs the activities of the other CCMs, and reports to the Commander.

The CSS is responsible for the organisation and completion of the following safety critical functions as and when required by the Safety and Emergency Procedures manual, or as directed by the commander:

1. Cabin preparation before flight including cabin safety equipment checks;
2. Passenger Briefing;
3. Securing the passenger cabin for take-off, and reporting to the Commander;
4. Passenger safety requirements observed during flight;
5. Securing the cabin for landing, and reporting to the Commander;
6. Ensures that all cabin crew are assigned meal breaks in compliance with 'Nutrition' in OMA Chapter 7.

1.5.3 Cabin Services Agent (CSA/CCM)

The primary statutory function of a CSA is that of safety.

The CSA/CCM is required to complete the following safety critical functions as and when directed by CSS/SCCM/No 1:

1. Cabin preparation before flight, including cabin safety equipment checks;
2. Passenger Briefing;
3. Securing the passenger cabin for take-off, and reporting to the CSS;
4. Passenger safety requirements observed during flight;
5. Securing the cabin for landing and reporting to the CSS.

Cabin Crew must observe the provisions in the Operations Manual in relation to their function.

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2 OPERATIONAL CONTROL AND SUPERVISION

2.0 Supervision of the Operation by Malta Air

ORO.GEN.110(c), ORO.MLR.115, AMC1 ORO.GEN.200(a)(4)

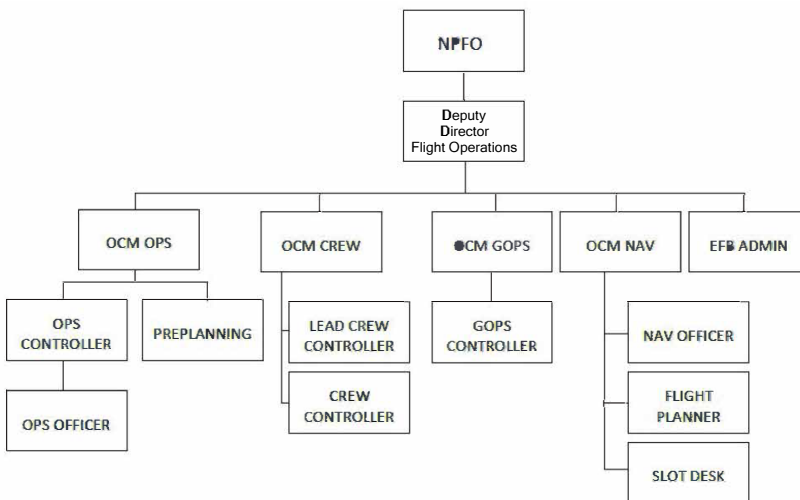
Definition – Operational Control

The exercise of authority over the initiation, continuation, diversion or termination of the flight in the interest of the safety of the aircraft and the regularity and efficiency of the flight. (ICAO Annex 6)

Malta Air has established a system for exercising operational control over any flight operated under the terms of the Malta Air AOC. This service is provided via Service Level Agreement (SLA) by the Ryanair DAC Operations Control Department, under the supervision of the MAY NPFO.

Ryanair DAC Operations Control is located in CHO in Dublin and is operational H24. Ryanair DAC Ops Control has access to the Eurocontrol Network Manager portals enabling control and supervision of flight operations. All Ryanair DAC Operations Control personnel are aware that operational and safety responsibility rests with the NPFO who will be contacted as required during flight operations in accordance with prescribed procedures.

Means of Communications with Ryanair DAC Operations Control is published on the OFP.



Operations Control is achieved by:

- Ensuring that all operations control personnel understand the Malta Air organigram as presented in Chapter 1.

- Ensuring that all operations control personnel have been properly instructed, have demonstrated their abilities in their particular duties and are aware of their responsibilities.
- On the job training and recurrent training to assure proficiency of operations control personnel in the operational procedures and instructions for all types of flight and ground operations.
- Ensuring that flight operations comply with the AOC and its associated Operations Specifications.
- Ensuring that planned operations are in compliance with the terms of the Certificates of Airworthiness and within the approved limitations contained in the Operations Manuals and the Aeroplane Flight Manuals.
- Checking, analysing and storing returned flight documentation for the statutory periods.

Information to monitor the operation comes from:

1. Electronic Flight Log (EFL);
2. Commander's Special Reports;
3. ASRs and other reports as detailed in Chapter 11;
4. Reports from the Safety and Security Manager;
5. The Technical Log; and
6. Verbal and confidential reports by Crew members to the Safety and Security Manager or Nominated Person Flight Operations.

Line Checks and Audit Reports as part of the Compliance Assurance Programme are an essential verification of compliance of EASA Regulations and Operations Manual Procedures.

- Commanders are expected to follow the operational direction of the controller. In practice where flights are subject to operational constraints the course of action is likely to be mutually agreed between the controller and the Commander.

2.0.1 Licence and Qualification Validity

Operations Control personnel are not licenced.

The qualification for appointment as Operations Controllers is the successful completion of the FOPS Training Procedure: Operations Personnel other than Crew Members, (OMD) including DG Training, Security Training, Risk Assessor training, FM training, on the job training and continuous assessment in the discharge of their responsibilities to assure maintenance of Operations Controller competencies and standards required.

Crewing personnel in the Operations Control department have received on the job training to satisfy the Ryanair Deputy Director Flight Operations that they have the required competencies and experience to discharge their responsibilities. They control and supervise all crew duties using the legality check function in the Crew Management System to ensure the following are valid:

- Qualifications and flight pairings;
- FTL checks;
- Checking that Crew Members designated to fly have valid medical, flight license, type rating and other company and FCL qualifications appropriate to the scheduled flights.

Each license/attestation entitles its holder to exercise its privileges as long as it remains valid. Whilst it is the holder's responsibility to perform the required checks and tests and medical examination to maintain validity, this will be monitored by crewing personnel in Operations Control.

Whenever a Crew Member discovers that the responsible department has overlooked an approaching date of licence or qualification expiry, he shall immediately notify the appropriate personnel and ensure that all necessary steps are taken to ensure the licence/qualification requirements are met.

2.0.2 Competence of Operations Personnel

The supervision of the competence of the operations personnel is achieved by:

- Ensuring that the personnel assigned to, or directly involved in, ground and flight operations are properly instructed and have demonstrated their abilities in their particular duties.
- Ensuring that the personnel can communicate in a common language (English) and that they are able to understand those parts of the Operations Manual which pertains to their duties and responsibilities.

The continued competence of operations control personnel is assured by ongoing supervision by the operation manager and successful participation in annual recurrent training and checking in their particular duties.

Supervision of training and assessment for operations control personnel is the responsibility of Ryanair Deputy Director Flight Operations.

2.0.3 Control, Analysis and Storage of Records, Flight Documents, Additional Information and Data

AMC1 ORO.GEN.220(b)

Malta Air maintains at a secure location a system for the storage and control of all relevant operational and technical information for each individual flight for the periods prescribed. The information/documentation to be stored and the storage period is prescribed in the tables below.

Attestation records shall be kept and stored by Malta Air as long as the cabin crew member is employed by Malta Air.

List of records which will be maintained by Malta Air ORO.MLR.115 ORO.GEN.220	
Operational flight plan	Three months
Aeroplane technical log	36 months after the date of the last entry in accordance with Annex I (Part-M)
Route specific NOTAM/AIS briefing documentation if edited by the operator	Three months
Mass and balance documentation	Three months
Notification of the special loads including written information to the commander about dangerous goods	Three months

Reports	
Journey log	Three months
Relevant Fuel Data (EFL) AMC1 CAT.OP.MPA.185(a)	Two years
Flight report(s) for recording details of any occurrence, or any event which the commander deems necessary to report/record (ORO.GEN.160). Refer to SMM Chapter 0.5 Definitions	Three months
Reports on exceedances of duty and/or reducing rest periods	24 months

Flight Crew Records	
Flight, duty and rest time	24 months
Licence	As long as the flight crew member is exercising the privileges of the licence for the operator
Conversion training and checking	Three years
Command course (including checking)	Three years
Recurrent training and checking	Three years
Training and checking to operate in either pilot's seat	Three years
Recent experience (FCL.060 (b) refers)	15 months
Recent and aerodrome competence (ORO.FC.105(b)(2);(c)refers)	Three years
Training and qualification for specific operations when required by Air-OPS (e.g. ETOPS CAT II/III operations)	Three years
Dangerous goods training as appropriate	Three years

Cabin Crew Records	
Flight, duty and rest time	24 months
Initial training, conversion and differences training (including checking)	As long as the cabin crew member is employed by the operator
Recurrent training and refresher (including checking)	Until 12 months after the cabin crew member has left the employ of the operator
Dangerous goods training as appropriate	Three years

Records for Other Operations Personnel	
Training/qualification records of the other personnel for whom an approved training programme is required by Air-OPS	Last two training records

Other Records	
Records on cosmic and solar radiation dosage	Until 12 months after the crew member has left the employ of the operator
SMM and CMS records (ORO.GEN.200)	Five years
Dangerous goods transport document	Three months after completion of the flight
Dangerous goods acceptance checklist	Three months after completion of the flight

Record-keeping: Paper and Electronic

Malta Air utilises both electronic and paper record keeping.

Electronic record keeping systems in Malta Air ensure that all records are accessible whenever needed within a reasonable time and the system is regularly tested and audited to assure compliance. These records are organised in a way that ensures traceability and retrievability throughout the required retention period.

Computer systems used in Malta Air record-keeping always have at least one backup system which is updated within 24 hours of any new entry. These systems include safeguards to assure that unauthorised personnel to cannot alter any data. All computer record-keeping data backup is stored in locations which comply with industry standards for such facilities and not in the same location as the working documentation.

The paper used is robust material which has demonstrated over time that it can withstand normal handling and filing, and that the records have remained legible throughout the required retention period. The retention period starts when the record has been created or last amended. Paper record-keeping storage is contracted out to storage organisations which have demonstrated competence and resilience in being compliant with applicable regulations¹.

Preservation of Documentation

ORO.MLR.115

Malta Air shall:

1. Preserve the information used for the preparation and execution of a flight and personnel training records, even if Malta Air ceases to be the operator of that aircraft or the employer of that crew member, provided this is within the timescales prescribed in the table above.

2. Where a crew member, in respect of whom an previous operator has kept flight duty, duty and rest period records, becomes a crew member for Malta Air that Flight Time record is made available to Malta Air.

Malta Air will enter in the Crew Management System the Flight Time records accrued since 1st January in the current FTL Year.

These records shall be valid on the first day of employment or provision of services to Malta Air.

3. Make a crew member's records available to a new operator when leaving Malta Air, provided this is within the prescribed timescales.

2.1 System and Responsibility for Promulgation of Additional Operational Instructions and Information

In addition to FCI/SPI/TI and NOTAMs Malta Air utilises the following means to provide additional operational instructions subject to the approval of the NPFO:

Ryanair Connect Web

The Ryanair Connect website connect.ryanair.com is an electronic communication and reporting portal.

There are various areas within Ryanair Connect which may be used to host operational information to crews depending on the nature of the information. The NPFO controls all information pertaining to flight operations on Ryanair Connect. Ryanair Connect is the equivalent portal for cabin crew with the same standards applied.

Ryanair Connect App

The Ryanair Connect App is an electronic communication and reporting system available on EFB and VPOS. The Ryanair Connect App contains the Documents App which is the company method of verifying that crew have read compulsory documents. All crew shall have read and confirmed all compulsory reads prior to operating on any given day. This procedure ensures all crew are up to date with operational documentation before flying.

MEMO's

Operational reminders, administrative reminders or non-operational information may be issued as a MEMO. Memos are not controlled documents however all current memos are listed in the MEMO master-list and are available electronically. MEMOs are usually listed as a compulsory read.

INFLIGHT SAFETY MEMOS

1. Safety related memos can only be issued by the Head of Inflight Safety and Security or the Inflight Safety and Security Manager.

2. Such memos will be titled “Inflight Safety Memo” and refer only to duties/procedures of Cabin Crew.

Inflight Safety Memos can only refer to currently approved procedures. Such memos are intended to serve as reminders. Each memo must refer to a relevant section of authority approved Malta Air Manuals.

COMPANY NOTAM

The NPFO authorizes all Company Notams in the standardized format with a validity period. Company NOTAMs will appear throughout the NOTAM briefing sections as and where they apply. They are specific to Malta Air. For example;

CO2/12 COMPANY NOTAM

LFOB/BVA BEAUVAIS

RWY 04/22 NOT SUITABLE FOR COMPANY OPS

CREW ALERT

The NPFO authorizes all crew alerts in the standardized format with a validity period. Crew Alerts will always appear at the end of the NOTAM brief. Crew Alerts are company wide alerts, and are messages of operational importance. For example;

‘Aircraft Library – Ships Documents update today. Ensure the necessary documents are updated prior to flight.’

CREW BULLETIN

The NPFO authorizes all crew bulletins in the standardized format with a validity period. Crew Bulletins are messages of operational importance specific to the flights in the briefing pack.

Airfield Briefs

The NPFO may promulgate significant information on an Airfield Brief including any validity period. Airfield Briefs are not intended to repeat information already promulgated by the aeronautical chart provider, but to identify threats and differences unique to the location.

The NPFO authorizes the Head of Flight Operations Performance Engineering to issue OPT or RTOW updates in response to any item affecting performance, including validity dates. Occasionally runway declared distances are changed at short notice and amended data will be published.

Ryanair Connect Bulletin Board/Safety Alert App

This is used to highlight any operational changes at short notice outside the normal production period to operating crews. The NPFO controls any published operational information. The Safety Alert App installed on EFB may be used to communicate safety information to crew.

Ryanair Connect Compulsory Read

This function is used to provide information to a crewmember and requires the crewmember to certify the material as read. All FCI/SPI/TI are compulsory read documents and at the discretion of the NPFO memos and safety bulletins may be included.

Similar processes apply on the cabin crew and engineering electronic portals.

The Safety and Security Manager has the ability to post Safety Instructions, Bulletins and Notices and at their discretion these may be compulsory reads.

2.2 Operational Control

ORO.GEN.110 (c), AMC1 ORO.GEN.110 (c)

Malta Air is required to exercise operational control and establish a method of supervision of flight operations approved by the TM CAD. Operations control is vested in the Nominated Person Flight Operations.

After the Malta Air commercial schedule is agreed it is implemented, supervised and controlled by the Ryanair Deputy Director Flight Operations under the authority vested in him by the NPFO and in turn the implementation, supervision and control of the daily flight schedule is delegated to the duty operations controller. Operational Control will always be provided whenever Malta Air flight operations are in progress.

The NPFO authorises any new routes and or aerodromes which Malta Air may use as originating aerodromes, destinations and alternates. The conditions under which flights are undertaken are contained in the Operations Manual which is issued on the authority of the NPFO. The LIDO flight planning system assures that all flight plans generated comply with OMA.

Operating crew is scheduled by the publication of crew rosters which are compliant with Flight and Duty Time Limitations and rest requirements. Any tactical changes by crew control are also subject to legality checks.

Airworthy aircraft for flight operations are assured by the Malta Air Engineering Department. The aircraft will be certified as airworthy in the Technical Log before acceptance by the commander. The MEL is jointly developed by the NPFO and the NPCA however the ultimate responsibility for this document rests with NPFO. The MEL must be approved by TM CAD.

Further oversight of the operation is maintained by the NPFO through the monitoring of post-flight documents and reports listed in OMA 2.0. The NPFO may assign the follow-up of any crew reports to other Malta Air personnel or Ryanair DAC as applicable.

2.2.1 The Functions of Operations Control Are Divided into Four Areas

1. Control of the operation;
2. Planning activities both short and long term;
3. Provision of services and documentation; and
4. Document control and maintenance of records as required EASA regulations.

2.2.2 Control Functions

The control functions include:

1. Initiation, diversion or termination of flights by Company aircraft – movement control;
2. Aircraft scheduling;
3. On-the-day crew deployment;
4. Flight watch;
5. Communications centre for the Malta Air route network;
6. Filing ATC flight plans; and
7. Obtaining ATC “slots”.

2.2.3 Planning Functions

The planning functions include:

1. Maintenance of AOC and obtaining other regulatory permits and authorisations etc.;
2. Consultation with the Commercial and Technical areas to agree seasonal schedules and charter programmes or leases on a long term basis to ensure feasibility, practicality and cost effectiveness;
3. Crew rostering – crew scheduling;
4. Charter airport and route assessment and obtaining Flight Operations approval from the NPFO as required;
5. Filing of ATC stored Flight Plans.

2.2.4 Provision of Services and Documentation

The provision of services and documentation include:

1. Supply of meteorological data and aeronautical briefing information for flight preparation;
2. Provision of computerised operational flight plans or manual flight planning data;
3. Provision of charter destination brief;
4. Supply of other operating documentation – Electronic Flight Log, paper voyage reports, data sheets, charts and other material carried on board;
5. Preparation of the Daily Movement Record;
6. Provision, maintenance and amendment of manuals included in the ships library on board the aircraft;
7. Distribution and filing of Electronic Flight Log data/paper voyage reports;
8. Threat assessment;
9. EFB Administration;
10. SAFA/SACA/CO-NOTAM production; and
11. Crew Control functions including crew management system legality checks.

2.2.5 Document Control and Records

1. Storage of Flight Ops Documentation
2. Retrieval of stored documentation as requested by Fops Managers.

2.3 Powers of the Authority

ORO.GEN.105

The TM CAD is the Authority with the power to issue an Air Operator Certificate (AOC) to operators whose principal place of business is Malta. The AOC may be varied, suspended or revoked if the TM CAD is no longer satisfied that the operation is safe.

2.3.1 TM CAD Inspector Access

ORO.GEN.140

The Company policy is that any person authorised by TM CAD is granted access to any facility, aircraft, documents, records, data, procedures and any other material relevant to its activity.

2.3.1.1 Access to Aircraft

CAT.GEN.MPA.190

The Company ensures that any person authorized by the TM CAD is permitted at any time to board and fly in any Malta Air aircraft and to enter and remain on the flight deck.

When occupying a flight deck jumpseat the TM CAD representative will be seated there prior to flight deck door locked and normally remain there for the duration of the flight. The SOP for vacating and re-entering the flight deck during flight applies to TM CAD representatives. However, at any time, the Commander may refuse access to the flight deck if, in his opinion, the safety of the aircraft would thereby be endangered.

The Commander shall produce documentation required to be carried on board when so requested by the TM CAD or ramp inspectors.

2.3.2 Ramp Inspection General

AMC1 ARO.RAMP.125, 125 (b), ARO.RAMP.140, ORO.GEN.140, 150

SAFA is Safety Assessment of Foreign Aircraft. SACA is Safety Assessment of Community Aircraft. SANA is Safety Assessment of National Aircraft. Most Ramp Inspections are conducted under one of the above categories, principally SAFA. Other inspections occur on the ramp including security inspections. The term Ramp Inspection includes any inspection conducted when the aircraft is on the ground by representatives of an Authority or State.

Ramp Inspections may be conducted as safety oversight by competent authorities in airports throughout the area of operation. Inspectors shall identify themselves to the Commander of the aircraft or, in his absence, to a member of the operating crew or to the most senior representative of Malta Air prior to commencing the on-board part of their Ramp Inspection.

On completion of a ramp check, the inspector will present the Commander or, in his absence, another member of the operating crew or a representative of Malta Air with a proof of Ramp Inspection form. In most cases, that is good for 48 to 72 hours, in satisfying a SAFA inspector at another location, unless that station has a reason to follow-up (bird or hail strike, violation or incident, etc.).

All SAFA inspection reports which are received by the Company are processed by the Compliance Department. All findings from a SAFA/SACA/SANA inspection are entered in the EASA's SAFA/SACA/SANA Database system where further investigation and possible follow-up are managed using Company procedures for corrective, preventive and verification actions by the relevant nominated person(s).

The following classification for findings is used:

- Level 1 findings: Information to the Commander.
- Level 2 findings: Information to the Authority and Malta Air.
- Level 3a findings: Restriction on aircraft flight operation.
- Level 3b findings: Corrective actions before flight.
- Level 3c findings: Aircraft grounded by inspecting Authority.
- Level 3d findings: Immediate operating ban.

In case of a level 3 finding the aircraft may be grounded by the inspector where it appears that the aircraft is intended or likely to be flown without completion of the appropriate corrective action by Malta Air. In such cases, the Commander or Malta Air shall be notified by the inspector.

Actions for Flight Crew Members

The inspector(s) must identify themselves by use of ID-card. A crewmember shall ask what type of ramp inspection will be conducted. The inspected crew will allow the inspector(s) admittance to the aircraft and provide the necessary documentation upon request. A Crew Member has to accompany the inspector during his stay on board. Assistance will be provided to inspectors to open hatches, doors and panels when required for the scope of the inspection. The list of current documentation is provided on the OFP for such inspections.

It is EASA policy that inspectors shall make all possible efforts to avoid delay to the aircraft inspected, except for safety reasons, including discontinuing an inspection when the turnaround is completed if there are no findings greater than Level 1 at that point.

The Commander shall file a report including:

- Names of inspectors.
- Name of the inspecting authority.
- Reason of the inspection.
- Specification of the laws and rules the inspection is based on.

2.4 Global Aeronautical Distress and Safety System (GADSS)

CAT.IDE.A.190, 280, 285, CAT.GEN.MPA.205, 210

Malta Air's Operations Control Manual and Emergency Response Plan incorporate GADSS requirements and procedures. These procedures include ATC coverage checks conducted by Ryanair Operations Control which occur, at a minimum, every 180 days.

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3 MANAGEMENT SYSTEM

3.1 Introduction

ORO.GEN.200

Under the EASA Implementing Rules, Malta Air has established, implements and maintains a management system which includes:

- Clearly defined lines of responsibility and accountability throughout the company, including a direct safety accountability of the Accountable Manager,
- a description of the overall philosophies and principles of the company with regard to safety and security, referred to as the Safety and Security Policy,
- the identification of aviation safety hazards entailed by the company's activities, their evaluation and the management of associated risks, including taking actions to mitigate risks and verifying the effectiveness of those actions,
- provision of personnel trained and competent to perform their tasks;
- documentation of all management system key processes, including a process for making personnel aware of their responsibilities and the procedure for amending this documentation,
- a function to monitor compliance with the relevant requirements. Compliance monitoring includes a feedback system of findings to the Accountable Manager to ensure effective implementation of corrective actions as necessary.

The Malta Air safety management system (SMS) is applicable to all departments, including, but not limited to, Flight Operations, In-flight, Customer Services, Engineering and Ground Operations. Malta Air shall ensure that Contracted services comply with Air Ops regulations and are compliant with the Malta Air SMS. The SMS has its own Manual called the Safety Management manual, (SMM). ORO.GEN.200(a)(5) (AMC2 ORO.GEN.200(a)(5)).

The responsibility for implementing the Malta Air SMS is delegated from the Accountable Manager to the Safety and Security Manager, and to Nominated Persons (Nominated Persons), who ensure that an integrated approach to all airworthiness matters and operating standards is achieved and that all necessary regulatory and legal requirements are satisfied. However, the overall responsibility for the establishment and maintenance of an effective management system rests with the Accountable Manager. ORO.GEN.210.

The Safety and Security Manager has direct access to the Accountable Manager for the purpose of compliance.

The Malta Air SMS is a pro-active, integrated approach to safety management, intended to be part of the company culture – ‘how do we do this safely’, and encompassing the management activities which contribute to a safe operation. The Safety Management System (SMS) is part of an overall management process which Malta Air has adopted in order to ensure that the safety objectives of the company are realised.

The purpose of this section is not to describe the Malta Air SMS in detail, but to provide an overview of its function for flight and cabin crew. Full details of the SMS are contained in the Safety Management System Manual (SMM).

The Malta Air safety and security policy includes a commitment to continuously improve to achieve the highest safety and security standards, comply with all the applicable legal requirements, consider best practices and provide appropriate resources.

3.1.1 Safety and Security Policy

Malta Air Operations Safety and Security Policy and Objectives

“Our primary operations objective is to conduct our air transport activities safely and securely”

ORO.GEN.200(a)(5)(b)(2), (a)(4)(b), AMC1 ORO.GEN.200(a)(2)(a), CAMO.A.200(a)(2)

Malta Air has established and shall maintain a Safety Management System, which includes a Security Management System, whose primary objective will be to reduce and control the risks involved in flight operations, ground operations, aircraft maintenance and engineering, and is committed to delivering the highest levels of safety and security.

Our policy is to promote safety and security awareness throughout the organisation. By monitoring our performance towards safety and security objectives and by conforming to the best operating practices of the commercial air transport industry, we will continually strive to improve the standard of safety and security in our operations.

To assist us in fulfilling our commitment to safety we shall:

1. Provide all necessary resources to implement this policy;
2. Require compliance with all relevant procedures, standards and regulations at all times;
3. Promote a positive safety and security culture and encourage personnel to identify and report hazards through the reporting system;
4. Communicate this policy, and any other relevant information throughout the organisation and provide a forum for staff communication of safety and security concerns and their resolution;

5. Periodically review this policy for continued relevance;
6. Ensure a Just Culture prevails by recognising that human error will occur and that resolution of such problems takes precedence over apportionment of blame or punitive action;
7. Ensure Fatigue Management forms an integral part of the SMS and includes a continuous process for the monitoring and management of fatigue;
8. Ensure the effective implementation of Human Factor Principles into operational practices.

Each person in Malta Air has responsibility for his or her own actions. We must all be aware of our safety and security responsibilities.

As each manager has responsibility for the safety and security performance of his or her own organisational area it shall be their overriding objective to focus on safety and security and, at all times strive to operate to the “how do we do this safely and securely” philosophy.

3.1.2 Safety and Security Policy Statement

The Board of Malta Air recognises the importance of effective safety management to ensure that our customers, staff, and all others affected by our activities are not exposed to unacceptable risks. Our aim is to minimise risks in all our company’s activities to a level that is as low as reasonably practicable.

The senior management of Malta Air is committed to ensure our safety processes are conducted in accordance with the best industry standards, thus ensuring the health and safety of our staff, customers, contractors and all others affected by our operation. Safety is a core value integrated into every activity carried out in the company and we commit to continually promoting, developing and improving our safety processes.

The safety and security policy is endorsed by the Accountable Manager. A framed hard copy of the Safety and Security Policy statement is displayed in every Malta Air Base.

3.2 Hazard Identification and Risk Management, SMM

The Safety Management Manual (SMM) defines Hazard Identification, Risk Assessment and Risk Management Policy in Malta Air.

3.2.1 Hazard Identification

Effective identification of hazards is achieved by:

- A review of accidents/incidents from both internal and external sources;

- A review of near-miss/dangerous occurrences from internal and external sources;
- Identifying deficiencies during safety and compliance audits;
- Identifying hazards using ‘brainstorming’ techniques (management & staff);
- Periodic review of hazards if there are significant changes to processes or equipment.

Line Managers are responsible for putting in place measures to remove, or mitigate, the risks of the identified hazards. The Safety and Security Manager provide support and guidance to line management.

3.2.2 Risk Assessment

Risk assessment is a subjective, qualitative, method of evaluating the likelihood and severity of loss arising from an identified hazard. Malta Air operates several systems to assist in the performance of hazard identification and risk management through the risk-assessment process, including ARMS methodology and ICAO Risk Tolerability Matrix.

3.2.3 The Malta Air Corporate Safety Strategy

The Corporate Safety Strategy is the highest-level definition of Malta Air Safety Management programme. The plan serves to focus safety efforts on those hazards considered to pose the greatest safety risks to the airline and/or the aviation industry in general. The Corporate Safety strategy is regenerated every three years to ensure that safety management responds to evidence based safety analysis. The Strategy also demonstrates Malta Airs’ commitment to develop processes, policies and procedures in order to improve safety. The current Strategy highlights safety issues and identifies risks which are classified into 5 Operational Focus Areas being Risk Assessment, Key Operational Risk Areas (KORAs), SMS Review, Security Management System (SeMS) and Dynamic Flight Planning. The Strategy provides Malta Air with a strategic risk-management process.

3.2.4 Key Operation Risk Areas’ (KORA)

These KORAs were identified as a result of

- Extensive operational experience,
- Feedback from the line through the Local Air Safety Group (LASG) and Flight Safety Forum (FSF).
- Flight Operations Safety Survey,
- Membership of various Industry Groups [including the EASA High Level Advisory Group, the EASA Safety Standards Consultative Committee (SSCC) and the SSCC Flight Standards Sub-committee],

- Assessment of published EU, TM CAD, UK Civil Aviation Authority (UK CAA) key operational risk areas.

KORA	RYP	EASA	IAA	ICAO	ACG	TM CAD	UKCAA
Aircraft Fire (F-NI)	X	X*	X*		NA	X	X
Bird Strike (BIRD)	X		X		NA	X	
Controlled Flight Into Terrain (CFIT)	X	X	X	X	NA	X	X
Ground Collision (GCOL)	X		X**		NA	X	
High Energy Approach (HEA)	X				NA	X	
Loss of Control In-Flight (LOC-I)	X	X	X		NA	X	X
Mid Air Collision (MAC)	X	X	X		NA	X	X
Runway Excursion (RE)	X	X	X	X	NA	X	X
Runway Incursion (RI)	X	X	X		NA	X	X
Safety of Ground Operations (RAMP, ICE)	X	X	X		NA	X	X***

* Fire, smoke and fumes

** Ground Handling

*** Included in 'Safety of Ground Operations'

Malta Air considers human factors to be an integral part of each of the KORAs and not an individual KORA in its own right.

Malta Air considers OFDM as an essential safety tool and not a risk area. Uniquely, and subject to strict protocols, four specified Malta Air safety personnel have direct access to de-identified OFDM data.

3.2.5 Flight Operations Department KORA Action Plan

The 'Bow-Tie' risk methodology will be applied to each KORA. Each 'Bow-Tie' will have a unique safety action plan detailed in the SMM 8.4.

3.3 Compliance Monitoring

AMC1 ORO.GEN.200(a)(6)

- The implementation and use of a compliance monitoring function enables Malta Air to monitor compliance with the relevant requirements of EASA Air Ops Regulations.
 - Malta Air has developed a compliance structure applicable to Malta Air operations training and maintenance activities and this is described in detail in the Compliance Monitoring manual.
 - The compliance monitoring function reflects the size and the complexity of Malta Air's Air Transport activities to be monitored.

2. Malta Air compliance monitoring procedures are designed to ensure safe activities and include auditing to establish compliance or otherwise in areas such as the terms of the Malta Air AOC, returned flight documentation and Technical Log completion, training and management system processes.

3. Organisational Set Up

The compliance monitoring manager is responsible for ensuring that the compliance monitoring programme is properly implemented, maintained and continually reviewed and improved.

The compliance monitoring manager has direct access to the accountable manager, can demonstrate relevant knowledge, background and appropriate experience related to the activities of Malta Air, including knowledge and experience in compliance.

The operation of Compliance Management in Malta Air is considered an independent function. Audits will not normally be carried out by personnel responsible for the function, procedure or product being audited.

All personnel subject to the Operations Manual shall render every assistance to the CMMs and their staff in the performance of their duties.

Detailed terms of reference are contained in the Malta Air Compliance Monitoring System Manual.

3.3.0.1 Compliance Monitoring Manager – Operations & Engineering (CMM)

The primary role of the CMM is to ensure that the compliance monitoring programme is properly implemented, maintained, reviewed and improved. The CMM will ensure that activities in flight operations, crew training, ground operations and engineering are monitored for compliance with the applicable regulatory requirements and additional Malta Air requirements, and that such activities are being carried out properly under the supervision of the relevant Nominated Person. The CMM has direct access to the Accountable Manager. Detailed terms of reference are contained in the Malta Air Compliance Monitoring System Manual.

3.3.1 Management Reviews

It is the responsibility of each Nominated Person to conduct regular reviews of those parts of the SMS which come within their areas of responsibility to ensure that the system is suitable and effective. Such reviews must be undertaken at least annually. Management reviews form an integral part of the overall safety-review process. Each review is documented and can include results of audits and other indicators, identification and correction of adverse trends, prevention of non-compliances, and assessment of areas for improvement (including, potentially, organisational structure, reporting lines, authority given to individuals, policies, processes and procedures).

Completed reviews are presented to applicable department heads and also presented to the Safety Review Board.

3.3.2 Compliance Monitoring Audits

Compliance Monitoring Audits consider operational safety processes, procedures and practices to monitor compliance with the requirements of EASA OPS, other EASA Rules and/or any National Aviation Authority regulations, as applicable.

3.3.3 Safety Audits

Safety auditing is a core safety management activity. Similar to financial audits, safety audits are designed to ensure that:

- The structure of the SMS is sound in terms of the provision of appropriate levels of staff; their compliance with approved procedures and instructions; and satisfactory levels of personal competence and training to operate equipment and facilities, and to maintain performance;
- Equipment performance is adequate for the safety of the service provided;
- Effective arrangements exist for promoting safety, monitoring safety performance and processing safety issues;
- Adequate arrangements exist to handle foreseeable emergencies.

3.4 Allocation of Duties and Responsibilities

3.4.1 Section Not in Use

3.4.2 Section Not in Use

3.4.3 Safety Manager

AMC1 ORO.GEN.200(a)

The Safety and Security Manager is the Safety Manager in Malta Air. He has the overall responsibility for ensuring that:

- he is the focal point and is responsible for the development, administration and maintenance of an effective safety management system;
- the Malta Air SMS is being implemented to ensure that the relevant requirements are being satisfied;
- he facilitates safety hazard identification, risk analysis and management;
- he monitors the implementation of actions taken to mitigate risks as listed in the safety action plan;
- he provides periodic reports on safety performance;
- he ensures the maintenance of safety management documentation;
- he ensures that safety management training is available, is provided and meets the industry standards;
- he provides advice on safety matters;
- he ensures initiation and follow up of internal occurrence/accident investigations;
- he attends Safety Review Board meetings;
- he has authority to communicate to the Accountable Manager all information necessary to allow decision making based on safety data.

The Safety Manager's authority is delegated from the Accountable Manager. The Safety Manager shall establish common procedures in Malta Air for the management and control of the SMS, to ensure that procedures are comprehensive and to provide a positive link between each part of the organisation where responsibilities may cross departmental boundaries.

3.4.4 ‘Safety Pyramid’



3.4.4.1 Local Air Safety Group (LASG)

Each base has a Local Air Safety Group (LASG). Ideally each LASG will consist of a minimum of the Base Captain (Facilitator), 1 Captain (Chairman), 1 First Officer, 1 Cabin Crew Member, the SEP/TSEP Instructor (if appointed) and the OFDM Facilitator. Engineering and GOPS are also represented.

There are scheduled and minuted LASG meetings each year in advance of the Flight Safety Forum (FSF) Meetings.

The LASG is intended to enable any personnel to raise safety issues with the LASG representatives who can raise these at the LASG meetings. Equally top-down issues can be communicated through the LASG meeting to front line personnel.

The LASG is the base of the Safety Pyramid as indicated in the Corporate Safety Strategy Publication. It is the primary meeting to engage front line personnel in the safety processes.

Personnel are nominated from the LASGs to attend the FSF meetings. Attendance at FSF is rostered.

3.4.4.2 Flight Safety Forum

The FSF meets at least twice per year and is chaired by the Safety Manager or his nominee. The Base Captains ensures at least one and not more than two members from their base are rostered to attend each meeting.

The FSF is minuted and the chairman provides a report with the FSF minutes to the Safety Action Group. Minuted items are distributed as required throughout the Company for action.

3.4.4.3 Safety and Security Action Group Meeting

A Safety and Security Action Group (SAG) Meeting occurs monthly and is attended by the Safety Manager, Flight Safety Officer and other departmental representatives as specified in the SMM. The Safety and Security Manager chairs the Safety and Security Action Group.

The terms of reference for the SAG are included in the SMM. The SAG reports to the Malta Air Safety and Security Review Board (SRB). In accordance with the provisions of ORO.GEN.200 (a) (1) the SAG shall assist the Safety Department with development, administration and maintenance of the Malta Air SMS.

3.4.4.4 Safety and Security Review Board Meeting

A Safety and Security Review Board (SRB) Meeting occurs 4 times per year and is chaired by the Accountable manager, Directors, all nominated persons, Safety Manager and other personnel invited.

The terms of reference for the SRB are included in the SMM. The SRB chairman makes a report to the Main Board Director-Safety who presents this to the main Board of Directors at their scheduled meetings.

3.5 Group Operations Management System

The development, delivery and maintenance of a safe operation is achieved through an effective Group Operations Management System. This system is described in Chapter 1 of this manual and the SMM.

3.5.1 Documentation of All Key Management System Processes

The documentation and procedures used within the SMS are fundamental elements in the management of risk. Procedures describe the standards required in order to meet regulatory and company, requirements. Compliance with those procedures cannot be assumed and all managers are responsible for ensuring and monitoring compliance with the requirements.

As a minimum, control of SMM including:

- Assignment of responsibility for the maintenance of contents to an individual or group.
- Identification of responsibility for authorising the manual.
- A list of effective pages (if appropriate).
- A record of changes made since the last issue/revision.
- Identification and withdrawal of obsolete documents.
- A title page which generally identifies the applicability and functionality.
- A table of contents.

- An introduction.
- A defined distribution method and identification of recipients.

Manuals are subject to regular review to retain the currency of the contained information. Procedures are established which enable relevant personnel to propose amendments to departmental or corporate procedures, including 'Zendesk'.

3.5.2 Manuals and Procedures Included in the Safety Management System (SMS)

Malta Air Operations Manual, Parts A to D.

Continuing Airworthiness Management Organisation Exposition. (CAME)

De-icing Manual.

Winter Operations Guide.

Ground Operations Manual (GOPS Manual).

Dangerous Goods Manual.

SMM.

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4 CREW COMPOSITION

ORO.FC.100

Malta Air operates a fleet of B737 aircraft, primarily engaged in commercial air transport. Non-revenue flights are described in OMA 8.7. The area of operations is defined in the Malta Air AOC.

This section describes the determination of crew composition in Malta Air.

4.1 Crew Members Duties and Responsibilities: Boeing 737 Flight Crew

4.1.1 Composition of Flight Crew

ORO.FC.100 (e), 200

The composition of the flight crew and the number of flight crew members at designated crew stations for all phases of flight shall be not less than the minimum specified in the aircraft flight manual or operating limitations prescribed for the aircraft. Therefore the minimum number of flight crew is two (2), one of whom shall be designated as the commander.

The flight crew shall include additional flight crew members when required by the type of operation as specified in the approved Malta Air FTL scheme and shall not be reduced below the number specified in the operations manual.

All flight crew members shall hold a licence and ratings issued or accepted in accordance with Commission Regulation (EU) No 1178/2011(1) and appropriate to the duties assigned to them.

When engaging the services of flight crew members who are working on a freelance or part-time basis, Malta Air verifies that all applicable requirements of the relevant elements of Annex I (Part-FCL) to Regulation (EU) No 1178/2011 are adhered to.

4.1.2 Operational Multi-Pilot Limitation (OML)

ORO.FC.100 (C)

Pilots with an OML on their medical certificate may only operate Malta Air aircraft in multi-pilot operations when the other pilot is fully qualified on the relevant type of aircraft, is not subject to an OML and has not attained the age of 60 years. During the pre-flight brief it is the responsibility of the flight crew member holding an OML to inform the other flight crew member of his OML.

4.1.3 Designation of the Commander

ORO.FC.105 (a), 105 (b), 105 (b)(2), 105 (c), 200, AMC 1 ORO.CC.200 (c), (d)

In accordance with 8.6 of Annex V to Regulation (EU) 2018/1139, a Commander is designated by the Company for every flight operated. In the case of non-commercial flights, as described in OMA 8.7, a 'pilot-in-command' is designated. The commander or pilot in command shall be qualified in accordance with Annex I (Part-FCL) to Regulation (EU) No 1178/2011, to be so designated.

Requirements, duties and responsibilities described in OMA as being applicable to the commander of a commercial air transport flight operated by Malta Air are equally applicable to a pilot-in-command designated for a non-commercial flight, except that in the case of certain non-commercial flights described in OMA 8.7, Malta Air may designate a pilot-in-command who has attained the age of 65 or greater.

Malta Air only designates commanders who have:

- The minimum level of experience as specified in OMA 5.2.
- Adequate knowledge of the route or area to be flown and of the aerodromes, including alternate aerodromes, facilities and procedures to be used.
- Completed a Malta Air/Ryanair Group command course if upgrading from co-pilot to commander as prescribed in OMD.
- Completed initial familiarisation training of the route or area to be flown and of the aerodromes, facilities and procedures to be used as prescribed in OMD.
- Achieved and maintains the route, area and/or aerodrome knowledge.

The Commander is designated by being the first name listed on the Electronic Flight Log/Paper Voyage Report who occupies a seat at the controls.

4.1.4 Crewing Inexperienced Flight Crew Members

ORO.FC.135, 200, ORO.FC.A.201

There shall not be any more than one inexperienced flight crew member in any flight crew.

The following terms are used in Malta Air:

New Captain Malta Air

A captain is 'new' until he has completed 100 flight hours and 10 sectors in 120 days or 150 hours and 20 sectors with no time limit since first successful line check. All new Captains are indicated in the Crew Management system by the word 'NEW' after the crewcode.

New Co-pilot Malta Air

A co-pilot is 'new' until he has completed 100 flight hours and 10 sectors in 120 days or 150 hours and 20 sectors with no time limit since first successful line check. All new co-pilots are indicated in the Crew Management system by the letter 'N' after the crewcode.

Crew restriction with 'new' aircrew member:

'N' Co-pilots are subject to the following operational restrictions:

1. A new co-pilot shall not operate with a new Captain;
2. It is essential that 'N' Co-pilots verify the status of the Captain at crew briefing.

Co-pilot Malta Air

Co-Pilots will fly from the right seat except where they have been specifically authorised to fly from the left seat during Command Upgrade training, and then only at the discretion of the Training Captain. Refer Malta Air Operations Manual Part D Section 2.1.6 and 3.1.6.

Inexperienced Co-pilot Malta Air

In Malta Air a co-pilot is inexperienced until he has completed 500 hours of flight time in Malta Air/Ryanair Group.

Inexperienced Co-pilot Operating Restrictions

Every inexperienced co-pilot shall so advise the Commander prior to operating and shall not perform the take-off or landing himself when the following conditions are experienced:

1. Crosswinds more than 2/3rds limiting value.
2. Runway is performance limited.
 - a. Takeoff: full rated thrust required.
 - b. Landing: AB MAX/MAX MANUAL required.
3. Take-off and landing RVR's less than 1000 m.
4. Ceiling is less than 100' above decision altitude for a precision approach and 200' above minima (MDA) for a non-precision approach aid.
5. The runway is contaminated.
6. Abnormal procedures are required due to system defects.
7. Inexperienced co-pilots shall not conduct the landing when the crosswind is in excess of 15 kt during normal operations. This restriction does not preclude the co-pilot from flying a monitored approach.

8. Visual approaches.
9. Forecast or reported windshear.

‘Check Captain’. A check captain is a Line Training Captain, TRI or TRE, LCC and who normally occupies an observer seat during the performance of check duties. The check captain is part of the operating crew. The check captain is not the commander but in the interests of safety may assume command and replace the PM or PF and reassign crew responsibilities.

Safety Pilot

A Safety Pilot (SP) assigned to a training flight is considered to be part of the operating crew and will normally occupy an observer seat.

The SP must be vigilant during all phases of the flight and be prepared to draw the Captain’s attention to any possible “threats” to the safe operation of the flight.

Increased vigilance is especially required during:

- Flight deck preparation.
- Receipt of any ATC clearance.
- Taxi out/taxi in.
- Level changes.
- Altimeter subscale changes.
- Preparation for final approach.

The safety pilot may perform the preflight or transit exterior inspection and generally assists the trainee pilot with paperwork before, during and after the rostered flight duties.

4.1.5 Recency of Experience

Refer to OMA 5.2.

4.1.6 Designation of the SCCM (Senior Cabin Crew Member)

In Malta Air the SCCM is also referred to as the CSS (Cabin Services Supervisor), and as the NO1.

4.1.6.1 Number and Composition of Cabin Crew

ORO.CC.005, 100, 200 (b), (c), AMC1 ORO.CC.100, 200 (d), 200 (e), ORO.GEN.100, AMC2 ORO.CC.200 (e)

The number and composition of cabin crew is determined in accordance with 7a of Annex V to Regulation (EU) 2018/1139, taking into account operational factors or circumstances of the particular flight to be operated.

The normal number of cabin crew on a Malta Air Boeing 737 is 4, one of whom shall be designated SCCM. Malta Air does not use augmented crew and in-flight relief of the SCCM or any other cabin crew does not arise in the context of FTL.

The SCCM must have at least one year's experience as an operating Cabin Crew member and have completed the Malta Air/Ryanair Group SCCM training. The SCCM reports to the Commander and is responsible for cabin safety and actioning safety and emergency procedures as specified in the Malta Air Safety and Emergency Procedures Handbook.

Senior Cabin Crew Member and Turbulence

Responsibility to the Commander

When the level of turbulence so requires, and in the absence of any instructions from the flight crew, the SCCM is entitled to discontinue non-safety related duties, advise the flight crew of the level of turbulence being experienced and the need for the fasten seat belt signs to be switched on. This shall be followed by the cabin crew securing the passenger cabin and other relevant areas.

4.1.7

Reduction of the Number of Cabin Crew During Ground Operations and in Unforeseen Circumstances

ORO.CC.205

Unforeseen circumstances in this context refer to incapacitation and unavailability of a senior cabin crew member or a cabin crew member as follows:

1. 'Incapacitation' means a sudden degradation of medical fitness that occurs during flight duty period either in-flight or during a flight transit of the same flight duty period away from operator's base and that precludes the senior cabin crew member or cabin crew member from performing his/her duties. Incapacitation prior to dispatch of the aircraft from a base of the operator does not substantiate a reduction of the cabin crew complement below the minimum required.
2. 'Unavailability' means circumstances at a stopover (layover) destination that preclude the senior cabin crew member or cabin crew member from reporting for the flight duty period, such as traffic jams that prevent the senior cabin crew member or cabin crew member from presenting himself/herself at the crew pick-up point in time, difficulties with local authorities, health problems, death, etc. Unavailability does not refer to insufficient number or absence of cabin crew members on standby, or absence from work due to pregnancy, maternity/paternity leave, parental leave, medical leave, sick leave, or any other absence from work.
 - a. Whenever any passengers are on board a Malta Air aircraft, the minimum No. of cabin crew as required in accordance with ORO.CC.100 shall be present in the passenger compartment (4 on B737-800 and 4 on B737-8200).

- b. Subject to the conditions specified in OMA (iii) (below), this number may be reduced:
 - i. During normal ground operations not involving refuelling/defuelling when the aircraft is at its parking station; or
 - ii. In unforeseen circumstances if the number of passengers carried on the flight is reduced. In this case, the Commander shall submit an MASR, and Malta Air shall submit a report to the TMCAD after completion of the flight.
- c. Conditions:
 - i. Procedures ensuring that an equivalent level of safety is achieved with the reduced number of cabin crew, in particular for evacuation of passengers, are established in the operations manual;
 - ii. The reduced cabin crew must include a SCCM;
 - iii. At least one cabin crew member is required for every 50, or fraction of 50, passengers present on the same deck of the aircraft;
 - iv. In the case of normal ground operations the number determined in accordance with (c) (iii) shall be increased to include one cabin crew member per pair of floor level emergency exits.

4.1.7.1 Reduction of the Number of Cabin Crew During Ground Operations and in Unforeseen Circumstances

AMC1 ORO.CC.205 (c) (1)

Procedures with reduced number of cabin crew:

1. During ground operations, if reducing the applicable minimum required number of cabin crew, the following procedure shall be applied:
 - Electrical power will be available on the aircraft;
 - A means of initiating an evacuation will be available to the SCCM or at least one member of the flight crew is in the flight crew compartment;
 - Cabin crew stations and associated duties are specified in the SEP Manual; and
 - Cabin crew remain aware of the position of servicing and loading vehicles at and near the exits.

Additionally, in the case of passengers embarkation:

- The SCCM has performed the pre-boarding safety briefing to the cabin crew; and
 - The pre-boarding cabin checks have been completed.
2. If, in unforeseen circumstances, the number of cabin crew members is reduced below the applicable minimum required number, for example in the event of incapacitation or unavailability of cabin crew, consider the following:
- Reduction of passenger numbers;
 - Reseating of passengers with due regard to doors/exits and other applicable limitations; and
 - Relocation of cabin crew taking into account the factors specified in OMA 4.1.6.1 and any change of procedures.
3. A report shall be submitted to the Authority after completion of the flight.

4.1.8 Single Cabin Crew Member Operations

ORO.CC.255

OMA 8.7 describes non revenue commercial ‘fly-arounds’ with one SCCM and 10 or less non revenue passengers.

4.2 Designation of the Commander

ORO.FC.105

OMA Section 4.1.3 and 5.2.

4.3 Flight Crew Incapacitation/Succession of Command

4.3.1 Incapacitation Procedures Succession of Command

In the event that the Commander can no longer act in that capacity the duties and responsibilities of Commander are assumed by the Co-Pilot. If another Malta Air Flight Crew member on board is qualified as a commander on type, is current and fit for duty he shall be asked to assume command.

4.3.2 SCCM Incapacitation

If the SCCM becomes incapacitated then the duties and responsibilities associated with this position delegates to the next senior CCM, normally CCM2, refer to SEP Manual and OMA 5.

- 4.3.3 The incapacitation drills and procedures are in the QRG on board and SEP Manual.

4.4 Operation on More than One Type or Variant

ORO.CC.255

Refer to OMA 5.2.

4.4.1 Scheduling of Crew

ORO.FC.140, ORO.CC.250

Aircrew scheduling shall be compliant with the Malta Air approved FTL scheme. The Crew Management System incorporates a legality check system to detect non compliances prior to assigning rostered duty. Malta Air has also an ECDR tracking system which tracks use of Captains Discretion and any 'duty busts'. Malta Air submits to TM CAD all Captains Discretions exceeding one hour and any duty busts.

A cabin crew member shall not be assigned to operate on more than three aircraft types and for cabin crew training and qualifications, Malta Air shall determine:

1. Each aircraft as a type or a variant taking into account, where available, the relevant data established in accordance with Regulation (EU) No 748/2012 for the relevant aircraft type or variant; and
2. Variants of an aircraft type are considered different types if they are not similar in the following aspects:
 - a. Emergency exit operation;
 - b. Location and type of portable safety and emergency equipment;
 - c. Type-specific emergency procedures.

4.5 Taxiing of Aircraft

CAT.GEN.MPA.125

Malta Air shall ensure that an aeroplane is only taxied on the movement area of an aerodrome when the person at the controls:

1. Is an appropriately qualified pilot; or engineer
2. Who has been designated by the Malta Air and:
 - a. Is trained to taxi the aircraft;
 - b. Is trained to use the radio telephone;
 - c. Has received instruction in respect of aerodrome layout, routes, signs, marking, lights, air traffic control (ATC) signals and instructions, phraseology and procedures;
 - d. Is able to conform to the operational standards required for safe aeroplane movement at the aerodrome.

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5 QUALIFICATION REQUIREMENTS

ORO.GEN.110 (b), 110 (c), 110 (d), 110 (e), 110 (g)

5.1 Licence and Qualification Requirements

ORO.FC.005

5.1.1 Licences

5.1.1.1 General Requirements

A personal identification document containing a photo shall be carried to identify the holder of the licence. This is normally in the form of a passport but Company and/or Airport IDs usually suffice.

All Crew Members shall be suitably qualified and competent to conduct the duties assigned to them (including Safety and Emergency Procedures, Dangerous Goods regulations and any other requirements contained therein).

5.1.1.2 Additional Requirements – Flight Crew

A valid licence with the respective type rating and a valid medical certificate shall be carried by the Flight Crew when exercising the privileges of the licence. Where a medical certificate limitation is specified, the crew must comply with this requirement at all times, e.g. if corrective lenses are required, a spare pair shall be carried at all times. Flight Crew Members are personally responsible for the timely extension or renewal of their licence and medical certificate.

5.1.1.3 Additional Requirements – Cabin Crew

ORO.CC.215 (b)

Cabin Crew must carry a valid Cabin Crew Attestation. Under all circumstances it must be readily accessible to present to an Authority. For Cabin Crew Attestation validity refer to OMA 5.1.7.

In addition, the list of Aircraft Type/Variant Qualifications provided by the Company must be carried by each Cabin Crew Member.

5.1.1.4 Requalification Requirements

Refer to Malta Air OMD.

5.1.2 Ratings

A rating is an entry in a licence stating special conditions, privileges or limitations pertaining to that licence. All Malta Air/Ryanair Group crew are type rated on the aircraft used during operation.

5.1.2.1 Instrument Rating

ORO.FC.200

A pilot shall not operate as Malta Air aircrew unless he has a current instrument rating (IR).

An IR is valid for one year. If an IR is to be re-validated the holder shall complete:

- A proficiency (LPC) check within the three months immediately preceding the expiry date of the rating.
- A line check during the period of validity of the rating.

The revalidation of an IR shall be combined with the type rating revalidation. If an IR has expired, refer to OMD.

5.1.2.2 Type Rating

For revalidation of type ratings, the applicant shall complete:

1. A proficiency check (OPC) in accordance with current requirements in the relevant type within the three months immediately preceding the expiry date of the rating; and
2. At least ten route sectors during the period of validity of the rating as pilot of the relevant type; or
3. One route sector as pilot of the relevant type or FFS (Full Flight Sim), flown with an examiner during the period of validity of the rating. This route sector may be flown during the proficiency check.

If a type rating has expired, refer to OMD.

5.1.2.3 Route, Area and Aerodrome Knowledge

Route/Area Competence Qualifications

ORO.FC.105, AMC1 ORO.FC.105 (b)(2), (c)

1. Initial route or area knowledge is established during line training to ensure that the Commander has knowledge of:
 - Terrain and minimum safe altitudes.
 - Seasonal meteorological conditions.
 - Meteorological, communication and air traffic facilities, services and procedures.
 - Search and rescue procedures where available.
 - Navigational Facilities associated with the area or route along which the flight is to take place. (i.e. alternative airports en-route).

- The most significant underlying risks and threats of a route or an area that could affect operations following ‘threat and error management model’.

Note: The Commander certifies adequate knowledge of the route by entering their 4 digit PIN in the OPT W&B e-signature panel (or by signing the paper OFP).

2. Route or area knowledge shall be maintained by operating or by being rostered as an observer for at least one sector in the area of operations as described in the AOC within a 36-month period. In addition, refresher training is required regarding route or area knowledge if not operating one sector in the area of operations as listed in the AOC for 12 months within the 36-month period.

5.1.2.4 Aerodrome Knowledge and Recency

ORO.FC.105, AMC1 ORO.FC.105 (b)(2), (c)

The validity of the aerodrome knowledge shall be maintained by operating at least once on the aerodrome within a 12 calendar months’ period or completing the briefing requirements detailed below as appropriate for the Aerodrome Category.

By entering their 4-digit PIN in the OPT W&B e-signature panel (or by signing the paper OFP), the Commander confirms that he has carried out the instructions below for the relevant Category of aerodrome.

Category A Aerodrome Knowledge

Before operating to a Category A aerodrome, the Commander shall obtain adequate knowledge of the aerodrome by means of the OMC, Aeronautical Charts and AFB. This includes knowledge of below (where applicable):

- Procedures applicable to routes through heavily populated areas and areas of high traffic.
- Noise Abatement requirements.
- Obstructions.
- Terrain and Emergency turns as appropriate.
- Physical lay-out.
- Aerodrome lighting.
- Approach aids.
- Arrival, departure, holding and instrument approach procedures.
- Applicable aerodrome operating minima.

- Ground movement considerations.

Category B Aerodrome Knowledge

Prior to operating to a Category B aerodrome (planned destination or required alternate), the Commander shall, in addition to the briefing requirements for Category A Aerodromes:

- Comply with any requirements stipulated by the competent authority responsible for the aerodrome; and
- Be briefed, or self-brief by means of by means of programmed instruction (including the OMC, Aeronautical Charts, AFB and Self Brief where published), about the additional considerations applicable to operations to that category B aerodrome. Additional restrictions may be imposed by the NPFO and will be detailed in the AFB or Self-Brief.

Category B (RJR Restricted) Aerodrome Knowledge

RJR Restricted is company specific terminology alerting operating crews to specific challenges within the Category B designation.

A “New” commander may not operate into a Category B (RJR Restricted) aerodrome. Further restrictions may be imposed by the NPFO and will be detailed in the AFB or Self Brief.

Category C Aerodrome Knowledge

ORO.FC.105, AMC1 ORO.FC.105 (c)

Prior to operating to a Category C aerodrome (planned destination or required alternate), the Commander shall, in addition to the briefing requirements for Category A Aerodromes:

- Comply with any requirements stipulated by the competent authority responsible for the aerodrome; and
- Be briefed or self-brief by means of programmed instruction, about the additional considerations applicable to operations to that Category C aerodrome; and
- Visit the aerodrome as an observer and/or undertake instruction in a suitable FSTD. The observer should occupy an observer’s seat where installed. If an observer’s seat is not available and cannot be installed, the pilot-in-command/ commander may occupy a pilot seat to conduct the aerodrome visit with a suitably qualified commander nominated by the Category C aerodrome operator.

The completion of the briefing, visit and/or instruction shall be recorded.

A “New” commander may not operate into a Category C aerodrome.

5.1.2.5 Definition for Aerodrome Categories

ORO.FC.105, AMC1 ORO.FC.105 (b)(2), (c)

All aerodromes, including alternate aerodromes, to which Malta Air operates, are classified as Category A, B or C.

Category A:

An aerodrome meets all the following conditions:

- A straight-in 3D instrument approach procedure with a glide path angle of not more than 3.5 degrees to each runway expected to be used for landing;
- At least one runway with no performance limited procedure for take-off and/or landing such as no requirement to follow a contingency procedure for obstacle clearance in the event of an engine failure on take-off from any runway expected to be used for departure, e.g. a Special EOSID; and
- Night operations capability.

Category B:

An aerodrome which does not meet the Category A conditions or which requires extra considerations due to:

- Non-standard approach aids and/or approach patterns, such as restrictions on the availability of straight-in instrument approach procedures;
- Unusual local weather conditions, such as environmental features that can give rise to turbulence, windshear or unusual wind conditions;
- Unusual characteristics such as unusual runway characteristics in length, width, slope, markings or lighting that present an atypical visual perspective on approach, or performance limitations, including Special EOSID;
- Any other relevant considerations including obstructions, physical layout, lighting; etc., such as restrictions on circling in certain sectors due to obstacles in the circling area;
- Training or flight crew experience requirements stipulated by the competent authority responsible for the aerodrome that do not include instruction in an FSTD or visiting the aerodrome.

Within the classification of Category B, Malta Air uses the term RYR Restricted to alert operating crews to specific challenges within Category B, as follows:

- Narrow Runway
- Runway LDA less than 1800 m
- Circling requirement

- Any other consideration identified by the NPFO and published in an AFB and/or Company NOTAM

Recognising these challenges, Malta Air has prohibited 'New' Commanders (OMA 1.4.2) from operating to Category B RYR Restricted, or Category C aerodrome.

Category C:

An aerodrome:

- That requires additional considerations to those of a Category B aerodrome, or
- For which flight crew experience or qualification requirements stipulated by the competent authority responsible for the aerodrome include instruction in an FSTD or visiting the aerodrome.

The NPFO shall specify the training required to operate into Category C aerodromes. A "New" commander may not operate into a Category C aerodrome.

5.1.3 Qualification for Low Visibility Operations (LVO)

SPA.LVO.120

Low Visibility Operations as dealt within the following paragraphs comprise:

- Low Visibility Take-off Operations (LVTO) in visibility conditions of less than 550 m RVR, and
- Category II and III precision instrument approach and landing operations.

Completion of low visibility training according to OMD is required before conducting Low Visibility Operations (LVO).

Note: Please refer to OMA 8.4, LVO operation.

5.1.3.1 Commander LVO Qualification

AMC1 SPA.LVO.120, (e)

A commander is authorized to conduct a take off in minimum RVR (125 m) when he has successfully completed the LVO training prescribed in the RST (Recurrent Simulator Training).

5.1.3.1.1 New Commanders LVO Qualification

A Commander is only authorized to conduct CAT II/III approaches after the following prerequisites have been accomplished:

100 m must be added to the applicable CAT II or CAT III RVR minima until a total of 100 hours or 40 sectors (which ever occurs first), as Commander on the type have been accumulated in line flying, including line training as part of the Command upgrade course.

5.1.3.2 First Officer LVO Qualification

Due to the high workload involved with LVP's during ground operations the F/O shall act as PM during take-off operations in visibility conditions of less than 550 m RVR. For approach and landing the FO is designated as PF until minima and will remain PF unless the commander decides to take control and land. i.e a go-around shall be flown by the FO. Company policy is the Commander will announce "CONTINUE" if he intends to take control.

5.1.4 Other Crew Qualifications

Qualifications described below are excerpts only. For a complete description of training syllabi refer to OMD.

5.1.4.1 Initial Operation to New Destinations and New Routes**New Destination:**

For the first 4 days operations into a new destination an LTC, LCC or Base Captain shall be rostered as Commander on each day.

Each Commander shall complete a New Destination Checklist/New Route Checklist for each of the first 4 days operation to the new destination (via Docunet/Forms).

New Route:

For the first day of operations of a new route, each Commander shall complete a New Destination Checklist/New Route Checklist (via Docunet/Forms).

5.1.4.2 Qualification to Operate in Airspace with Special Authorization/ Approvals

Prior to operating in airspaces where special authorization/operational approvals (e.g. NAT HLA or RVSM) are required the Flight Crew must be qualified in accordance with OMD and familiar by study of applicable procedures laid down in the FCOM Vol 1, OMC and, if available, by computer based training on the eLearning Platform.

5.1.4.3 Qualification to Operate in Either Pilot's Seat

ORO.FC.135

Captains whose duties also require them to operate in the right-hand seat and carry out the duties of co-pilot, or Commanders required to conduct training, or examining duties from the right-hand seat, shall complete additional training and checking according to OMD.

Two Flight Crew members ranked as Captains may make up the required Flight Crew complement.

All Captains are qualified as Co-Pilots during initial conversion, including LVO. Captains are not normally required to operate the aircraft from the right hand seat but may be rostered to do so as PM, and as PF for LVO approaches/Go Around. These qualifications are automatically renewed during statutory competency checks. (Refer to Chapter 5.)

Where two line Captains are rostered together the minimum RVR for take-off is 125 m (or as restricted by runway lighting).

5.1.4.4 Dangerous Goods Qualification (Flight Crew/Cabin Crew)

DG training and checking is included in the Company's emergency/recurrent training. The Cabin Crew and Flight Crew shall complete the mandatory recurrent dangerous goods training and is valid for 2 years. Details of training and checking requirements can be found in OMD.

5.1.4.5 Security Qualification (Flight Crew/Cabin Crew)

The Company ensures that all Crew Members have received an initial security training course. Requalification is provided, assuring that all training items have been covered in accordance with the National Civil Aviation Security Program. SEP Training is either provided on a yearly basis or as otherwise regulated by the TM CAD or AVSEC Malta. Refer to OMD.

5.1.5 Flight Crew Checking

ORO.FC.130, 230

5.1.5.1 Operator Proficiency Check (OPC)

ORO.FC.230 (b)

Each Flight Crew Member shall complete operator proficiency checks as part of the normal crew complement to demonstrate competence in carrying out normal, abnormal and Emergency Procedures. The validity period of the operator proficiency check is six calendar months. The validity period is counted from the end of the month when the check was taken.

5.1.5.2 Line Check

ORO.FC.230 (c)

Line Check and Proficiency Training and Checking

1. Line checks, route and aerodrome knowledge and recent experience requirements are intended to ensure the crew member's ability to operate efficiently under normal conditions, whereas other checks and emergency and safety equipment training are primarily intended to prepare the crew member for abnormal/emergency procedures.
2. The line check is considered a particularly important factor in the development, maintenance and refinement of high operating standards, and can provide Malta Air with a valuable indication of the usefulness of training policy and methods. Line checks are a test of a flight crew member's ability to perform a complete line operation, including pre-flight and post-flight procedures and use of the equipment provided, and an opportunity for an overall assessment of his/her ability to perform the duties required as specified in the operations manual. The line check is not intended to determine knowledge on any particular route.

The emphasis of a Malta Air LCK is to reinforce a safe and efficient operation. While SOPs are the bedrock of our operation, crews will not fail a LCK based purely on SOP non-compliance. A pilot will not pass a line check if a safety issue has not been dealt with properly or the LCC Line Check Captain has had to intervene in order to prevent an undesired aircraft state.

The Pass/Fail criteria of a Malta Air line check is as follows:

- Pass – No fail item and less than 5 SOP items as described in Operations Manual OMD 3.1.4.4.5. Validity period of the line check will be to the end of the current month plus 1 year.
- Restricted Pass – No fail items but more than 5 SOP items as described in OMD 3.1.4.4.5. The validity period of the line check will be to the end of the current month plus 3 months.
- Fail – a breach of any safety item as described in OMD 3.1.4.4.5 Crews will be taken off line until the line check has been reviewed and any remedial training decided.

Each New Entry OCC Pilot must now undergo two Line Checks following Initial Line Check (LCK 1) before they fall into the normal Annual Line Check system. Each type-rated new entry pilot must undergo one Line Check following Initial Line Check before they fall into the normal Annual Line Check system.

The Line Check Form

The LCC will give a copy of the Line Check Form to the Pilot who has just been checked. The Pilot must produce this Line Check Form to the LCC conducting the Pilot’s next check. This requirement of the Pilot applies to all Pilots, New Entry or existing Malta Air employees.

The LCC conducting the next Line Check will deduce from the previous Check Type box what type of check is now required e.g.

Refer to 5.1.5.3 for sequence of checks.

Proficiency Checks

Refer Malta Air Operations Manual Part D Section 2.1.4 and 3.1.4.

LPC revalidation is by a Malta Air/Ryanair Group TRE, a renewal is by the licensing authority only.

5.1.5.3 Recurrent Training (Flight Crew/Cabin Crew)

ORO.FC.130, 140, 230

The validity period for the following recurrent training is counted from the end of the month where the training was taken, annual recurrent validity periods being 12 calendar month. When the training is taken within the last three months of the validity period, the new validity period is counted from the end of the month.

Training Course	Line Check Type	Code	Period of Validity	Next LC Type
OCC	OCC Initial	OCC LCK 1	End of month + 3 months	OCC LCK 2
OCC	2 nd OCC Check	OCC LCK 2	End of month + 3 months	OCC LCK 3
OCC	3 rd OCC Check	OCC LCK 3	End of month + 1 Year	Annual
Type Rating	Type Initial	Type LCK 1	End of month + 3 months	Type LCK 2
Type Rating	2 nd Type Rating	Type LCK 2	End of Month + 1 Year	Annual
RST	Annual	Annual	End of month + 1 Year	Annual
RST	Annual	Annual Restricted Pass	End of month + 3 months	Annual
Command Upgrade	Command Initial	CU Initial	End of month + 3 months	CMD LCK 2

Training Course	Line Check Type	Code	Period of Validity	Next LC Type
Command Upgrade	2 nd Command Check	CMD LCK 2	End of month + 3 months	CMD LCK 3
Command Upgrade	3 rd Command Check	CMD LCK 3	End of month + 1 Year	Annual
Line Training Cpt.	Instructor Proficiency	IPC	End of month + 3 Years	IPC
Line Training Cpt.	Instructor Proficiency	IPC Grade 4/5	End of month + 3 months	IPC

5.1.6 Emergency and Safety Equipment Training

AMC1 ORO.FC.230

Each Crew Member receives training on the location and use of all emergency and Safety Equipment carried. This training is also part of the conversion course. Every three years crew members shall participate in actual fire fighting training. For more detailed information refer to OMD.

5.1.6.1 Crew Resource Management Training

ORO.FC.115, 215, 220, 230, AMC1 ORO.FC.115 and 215

The Company ensures that all Crew Members have received CRM training. Flight Crew Members shall have completed their initial CRM Training course before commencing unsupervised line flying.

CRM training is part of the aircraft type training, recurrent training as well as part of the command course and operator conversion training course (OCC). Further CRM training can be provided concurrently with the emergency training requalification. Additionally, crew members receive annual CRM training.

All major topics of CRM training shall be covered by distributing modular training sessions as evenly as possible over each three-year period.

5.1.6.2 First Aid Training

The Company ensures that all crew members have received a first aid training in accordance with OMD. When a Flight Crew Member has not previously completed an operator's conversion course, the Company will ensure that the Flight Crew Member undergoes general first aid training.

5.1.6.3 Dangerous Goods Training

AMC1 SPA.DG.105 (a)

The Company ensures that all Crew Members receive DG training in:

- Actions in the event of emergencies in flight occurring in the passenger compartment or in the cargo compartment, and

- The notification to ATC should an inflight emergency occur.

Crew Members other than Flight Crew Members receive training in:

- Dealing with incidents arising from Dangerous Goods carried by passengers, or
- Dealing with damaged or leaking packages in flight.

Training is conducted at intervals no longer than one year. For details of Dangerous Goods Training refer to OMD.

5.1.7 Cabin Crew Attestation Validity

ORO.CC.110, 210, CAT.GEN.175, CAT.GEN.MPA.170

Malta Air shall ensure that each Cabin Crew member:

- Is at least 18 years of age.
- Has been assessed, in accordance with the applicable requirements of Annex IV to Regulation (EU) No 1178/2011, and (EU) 2018/1042 as physically and mentally fit to perform their duties and discharge their responsibilities safely.
- Is tested for Psychoactive Substances upon employment for operations.
- Has successfully completed all applicable training and checking required by Subpart CC.
- Holds a valid attestation issued in accordance with Annex V (Part-CC) to Regulation (EU) No 290/2012.
- Is competent to perform his/her duties in accordance with procedures specified in the Operations Manual.

Malta Air shall ensure that all Cabin Crew members wear the Malta Air/Ryanair Group Cabin Crew uniform and are clearly identifiable to the passengers as a Cabin Crew member. CAT.GEN.MPA.115.

The uniform to be worn by operating cabin crew is designed so as not to impede the performance of their duties as required for the safety of passengers and flight during operations, and will allow passengers to identify the operating cabin crew including in an emergency situation.

5.1.8 Flight Crew Member

AMC1 ORO.FC.100 (c), EU Reg. 1178/2011 ANNEX IV - [PART-MED] SUBPART B - REQUIREMENTS FOR PILOT MEDICAL CERTIFICATES SECTION 1 - General, MED.B.001 - Limitations to medical certificates

Operating aircraft in multi-pilot operations includes the following restrictions:

1. Age 60-64.

Pilots aged between 60 and 64 may fly together in a Malta Air aircraft engaged in commercial air transport (CAT) as members of a multi-pilot crew.

2. Age 65 and Above.

The holder of a pilot licence who has attained the age of 65 years shall not act as a pilot of an aircraft engaged in commercial air transport operations.

3. Operational Multi-pilot Limitation (OML).

In accordance with OMA 4.1.2 pilots with an OML on their medical certificate must only operate aircraft in multi-pilot operations when the other pilot is fully qualified on the relevant type of aircraft, is not subject to an OML and has not attained the age of 60 years.

4. Permitted Age and/or OML Combinations

2 pilots 60–64	Approved for CAT
2 pilots with OML any age	Not Approved
1 pilot 60-64 + 1 pilot OML	Not Approved
1 pilot under 60 + 1 pilot with OML	Approved For CAT

5.2 Minimum Requirements

AMC1 ORO.FC.100(c), 105 (b)(2), (c), ORO.FC.105, 105 (b), 105 (c), 205, 235 (f), (g), ORO.GEN.200 (a)(3), AMC1 CAT.GEN.MPA.175(b), CAT.GEN.MPA.175

The requirements for operating crewmembers are listed below:

- 1. Licence:** have a current and appropriate licence for flight crew duties and a current attestation certificate for cabin crew duties.
- 2. Ratings;** have a current rating on the B737 600-900.
- 3. Qualification:** shall be qualified in accordance with current training programmes specified in the Malta Air OMD. All Malta Air personnel engaged in flight operations shall be at least 18 years of age, or at least 21 years of age to be a commander.
- 4. Competency:** shall have demonstrated competency as required by the Malta Air OMD training and checking programmes including as indicated in this chapter.

5. Experience: shall satisfy the experience requirements specified in the OMA, OMD manuals and in this chapter for Commander, co-pilot, CSS or CSA.
6. Checking; shall have completed specified training and checking in the Malta Air OMD manuals appropriate to duties assigned.
7. Recency: shall achieve recency requirements as follows:

Malta Air Requirement Recent Experience

Crew are required to record which crew member carried out each Take-off and Landing. The recording of Pilot Flying duties within EFL allows Malta Air to ensure that:

1. A pilot is not assigned to operate a B737 as part of the minimum certificated crew, either as pilot flying or pilot monitoring unless he has carried out three take-offs and three landings in the previous 90 days as pilot flying in a B737, or in a Malta Air approved flight B737 simulator.
2. The 90-day period above may be extended up to a maximum of 120 days by line flying under the supervision of a type rating instructor or examiner.
3. For periods beyond 120 days, the recency requirement is satisfied by a training flight or use of a flight simulator of the aeroplane type to be used.

Additional Malta Air Restrictions

Additional Malta Air restrictions apply as follows:

- Flight Crew members who have not flown in their capacity or in a type of aircraft for a period greater than 45 days shall complete a line flight as an operating crew member with a Type Rating Instructor (TRI) or Line Training Captain or a session of simulator before flying on the line. At the discretion of the NPFO these limits may be extended or modified but always subject to the Recent Experience requirements above.
- When a Flight Crew member has not flown an aircraft type for a period of six months or longer then he must complete an appropriate refresher reconversion course and carry out a Line check before flying on the line.
- This training will be complemented by line familiarization, line training and/or line checks, the extent depending on the period of time off the aircraft.

Aircraft Type: The aircraft type and number of aircraft operated by Malta Air are as specified on the Malta Air AOC, a current copy of which is always on board.

Malta Air operates Commercial Air Transport revenue flights and other non-revenue flights as indicated in OMA CH 8.7.

Crew Composition: This is described in OMA CH 4.

Designation as Pilot-in-command/Commander

1. In accordance with 8.e of Annex IV to Regulation (EU) 2018/1139, one pilot amongst the flight crew, qualified as pilot-in-command in accordance with Annex I (Part-FCL) to Regulation (EU) No 1178/2011, will be designated by Malta Air as commander.
 - a. Malta Air will only designate a commander if he has:
 - b. The minimum level of experience specified in the operations manual:
 - At least 21 years of age.
 - Current Class 1 medical reflecting having been assessed, in accordance with the applicable requirements of Annex IV (Part-MED) to Regulation (EU) No 1178/2011, as physically and mentally fit to perform their duties and discharge their responsibilities safely.
 - Unfrozen ATPL.
 - 3000 hrs total flight time.
 - Successfully completed the B737-NG Type Rating Course for a Captain.
 - 800 hrs in command, 500 hrs on type for direct entry Captains operation conversion course.
 - 800 hrs on type including one Winter season in Ryanair Group operations.
 - Maintained the required recency of experience.
 - c. Adequate knowledge of the route or area to be flown and of the aerodromes, including alternate aerodromes, facilities and procedures to be used;
 - d. Completed a Malta Air command course if upgrading from co-pilot to commander as prescribed in Operations Manual Part D.
2. Has completed initial familiarization training of the route or area to be flown and of the aerodromes, facilities and procedures to be used.

Co-pilot

Co-Pilots will fly from the right seat except where they have been specifically authorised to fly from the left seat during Command Upgrade training, and then only at the discretion of the Training Captain. Refer Malta Air Operations Manual Part D Section 2.1.6 and 3.1.6.

New Co-pilots – Operating Restrictions

In Malta Air a new Co-pilot is restricted until he has completed 100 hours flight time in Malta Air/Ryanair Group since first successful line check (LCK). All new Co-pilots are indicated in the Crew Management system by the letter 'N' after the crewcode. All new Co-pilots are subject to the following operational restrictions:

Must not operate with a new Captain, i.e. one with less than 100 hours flight time in Malta Air/Ryanair Group since initial appointment as a Commander. All new Captains are indicated in the Crew Management system by the word 'NEW' after the crewcode.

It is essential that new Co-pilots verify the status of the Captain at crew check-in.

Inexperienced Co-pilot and Operating Restrictions

In Malta Air a Co-pilot is inexperienced until he has completed 500 flight time hours in Malta Air/Ryanair Group. Every inexperienced Co-pilot shall so advise the Commander prior to operating and shall operate in accordance with the restrictions set;

1. Crosswinds more than 2/3rds limiting value.
2. Runway is performance limited.
3. Take-off and landing RVR's less than 1000 m.
4. Ceiling is less than 100' above decision altitude for a precision approach and 200' above minima (MDA) for a non-precision approach aid.
5. The runway is contaminated.
6. Abnormal procedures are required due to system defects.
7. Inexperienced co-pilots shall not conduct the landing when the crosswind is in excess of 15 kt during normal operations. This restriction does not preclude the co-pilot from flying a monitored approach.
8. Visual approaches.
9. Forecast or reported windshear.

5.2.0.1 Operation on More than One Type or Variant

AMC1 ORO.FC.240, AMC2 ORO.FC.240, ORO.FC.125, 140, 240

When a Flight Crew Member operates more than one type or variant within one or more licence endorsement as determined by the operational suitability data established in accordance with Commission Regulation (EU) No 748/2012 and associated procedures, the Company ensures that:

- The minimum Flight Crew complement specified in the operations manual is the same for each type or variant to be operated;

- The Flight Crew Member does not operate more than two types or variants for which a separate licence endorsement is required, unless credits related to the training, checking, and recent experience requirements are defined in data established in accordance with Regulation (EU) No 748/2012 for the relevant types or variants;
- Only aeroplanes within one licence endorsement are flown in any one flight duty period.

Before exercising the privileges of more than one licence endorsement and for complete differences and familiarization training requirements refer to OM D.

5.2.0.2 Minimum Requirements for New Entry Malta Air Flight Crew

The following tables detail the minimum entry requirements for all New Entry Malta Air Flight Crew and the applicable training course.

Common Requirements for all New Entry Flight Crew	
1	Date of Birth – not over 65.
2	Valid Licence: a. EASA Part FCL or; b. National Licence compliant with ICAO standards.
3	Valid Medical – Must hold a valid EU (Part-MED) issued Class 1 medical certificate.
4	English Language Proficiency – minimum required proficiency Level is Operational Level (Level 4). (Written proficiency is required at Operational level). (Reference: ICAO Doc 9835).
5	Psychological Assessment – Appropriate to the particularity, the complexity and the challenges of the operational environment that the flight crew are exposed to and undertaken within the past 24 months before commencing flying. The assessment shall be validated and either directly performed or overseen by a psychologist with appropriate knowledge and expertise.
6	Psychoactive Substance Test – upon employment for operations.

5.2.0.2.1 Operator Conversion Course – External Entry

1. Minimum Requirements – B737 300-900 Type Rated Direct Entry Captain

B737 300-900 Type Rated <u>Direct Entry Captain</u>	
1	Minimum 3,500 hours total flying time.
2	Minimum of 2,000 hours on a Multi-crew, Multi-engine aircraft weighing in excess of 30,000 kg MTOW with an established airline, cargo operator, private operator or military transport.
3	Minimum of 800 hours Pilot in Command (PIC) on a Multi-crew, Multi-engine jet aircraft weighing in excess of 30,000 kg. (No PIC hours on Turboprop aircraft shall be considered).
4	Minimum 500 hours PIC on the B737-300 to 900 series.

B737 300-900 Type Rated <u>Direct Entry Captain</u>	
5	Must have the B737 300-900 type rating listed on the Licence.
6	For EASA licence holders the application must be made within 36 months of the most recent B737 300-900 “Valid Until” date specified in Section XII “Certificate of Revalidation”. i.e. the B737 300-900 Rating must not have expired for longer than 3 years at the intended (OCC) course commencement date.
7	Must have operated on B737 300-900 as Commander within the 36 months prior to the course start date.
Completes Operator Conversion Course to B737-NG in accordance with Operations Manual Part D Section 2.1.1.1	

2. Minimum Requirements – B737 300-900 Type Rated Direct Entry First Officer

B737 300-900 Type Rated <u>Direct Entry First Officer</u>	
1	Minimum 1,200 hours total flying time.
2	Minimum of 1000 hours on CS25 Type Aircraft.
3	Minimum 800 hours on the B737-300 to 900 series.
4	Must have the B737 300-900 type rating listed on the Licence.
5	For EASA licence holders the application must be made within 36 months of the most recent B737 300-900 “Valid Until” date specified in Section XII “Certificate of Revalidation”. i.e. the B737 300-900 Rating must not have expired for longer than 3 years at the intended (OCC) course commencement date.
6	Must have operated on B737 300-900 within the 36 months prior to the course start date.
Completes Operator Conversion Course to B737-NG in accordance with Operations Manual Part D Section 2.1.1.1	

3. Minimum Requirements – B737 300-900 Type Rated Direct Entry Captain (EX Ryanair Group Captain)

B737 300-900 Type Rated <u>Direct Entry Captain (EX Ryanair Group)</u>	
1	Minimum 3,500 hours total flying time.
2	Minimum of 2,000 hours on a Multi-crew, Multi-engine aircraft weighing in excess of 30,000 kg MTOW with an established airline, cargo operator, private operator or military transport.
3	Minimum of 800 hours Pilot in Command (PIC) on a Multi-crew, Multi-engine jet aircraft weighing in excess of 30,000 kg. (No PIC hours on Turboprop aircraft shall be considered).
4	Minimum 500 hours PIC on the B737-300 to 900 series.
5	Must have <u>current</u> B737 300-900 type rating listed on the Licence.

B737 300-900 Type Rated <u>Direct Entry Captain (EX Ryanair Group)</u>	
6	Must be an <u>active</u> pilot of Boeing 737-NG Variants.
7	Must have previously operated for Ryanair Group as a Commander.
8	Must have operated on B737 300-900 as Commander within the 36 months prior to the course start date.
Completes Operator Conversion Course – EX RYR in accordance with Operations Manual Part D Section 2.1.1.1	

4. Minimum Requirements – B737 300-900 Type Rated Direct First Officer (EX Ryanair Group First Officer)

B737 300-900 Type Rated <u>Direct Entry First Officer (EX Ryanair Group)</u>	
1	Minimum 500 hours on the B737-300 to 900 series.
2	Must have <u>current</u> B737 300-900 type rating listed on the Licence.
3	Must be an <u>active</u> pilot of Boeing 737-NG Variants.
4	For EASA licence holders the application must be made within 36 months of the most recent B737 300-900 “Valid Until” date specified in Section XII “Certificate of Revalidation”, i.e. the B737 300-900 Rating must not have expired for longer than 3 years at the intended (OCC) course commencement date.
5	Must have previously operated for Ryanair Group as a First Officer.
Completes Operator Conversion Course to B737-NG in accordance with Operations Manual Part D Section 2.1.1.1	

5.2.0.2.2 Type Rating Courses – External Entry

1. Minimum Requirements – Direct Entry Non-Type Rated Captain

Direct Entry Non-Type Rated Captain	
1	Minimum 3,500 hours total flying time.
2	Minimum of 2,000 hours on a Multi-crew, Multi-engine aircraft weighing in excess of 30,000 kg MTOW with an established airline, cargo operator, private operator or military transport.
3	Minimum of 800 hours Pilot in Command (PIC) on a Multi-crew, Multi-engine jet aircraft weighing in excess of 30,000 kg. (No PIC hours on Turboprop aircraft shall be considered).
4	For EASA licence holders the application must be made within 36 months of the most recent relevant Type Rating (as above) “Valid Until” date specified in Section XII “Certificate of Revalidation”.
Completes Transition Type Rating Course (B737-800) (TTR) in accordance with Ryanair ATO Training Manual Section 2.2	

2. Minimum Requirements – Direct Entry Non-Type Rated First Officer

Direct Entry Non-Type Rated First Officer	
1	Minimum 1,500 hours total flying time.
2	Minimum of 1000 hours on CS25 Type Aircraft.
3	Minimum of 500 hours on CS25 Type Aircraft Jet > 20,000 kg.
4	Must be current on CS25 Type Aircraft.
5	For EASA licence holders the application must be made within 36 months of the most recent relevant Type Rating (as above) “Valid Until” date specified in Section XII “Certificate of Revalidation”.
Completes Transition Type Rating Course (B737-800) (TTR) in accordance with Ryanair ATO Training Manual Section 2.2	

3. Minimum Requirements – Cadet

Cadet	
1	Holds a minimum of CPL(A) with a valid multi-engine instrument rating.
2	Must have passed the ATPL(A) theoretical knowledge examinations in accordance with FCL.515.
3	Hold a certificate of satisfactory completion of multi-crew co-operation (MCC).
4	Must have a flight school report.
5	100 hours as pilot-in-command, or 70 hours as Pilot-in-Command if completed during a course of integrated flying as set out in Appendix 3 A to Part FCL and AMC1 to Appendix 3 A. Applicants who have conducted a modular course need to have 100 hours as Pilot-in-Command (PIC), or 70 hours as PIC if completed during a course of integrated flying training as set out in Appendix 3 A, C & D to Part-FCL and AMC1 to Appendix 3 A, C & D.
Completes Transition Type Rating Course (B737-800) (TTR) in accordance with Ryanair ATO Training Manual Section 2.1	

4. Minimum Requirements – APC Cadet

APC Cadet	
1	Holds a minimum of CPL(A) with a valid multi-engine instrument rating.

APC Cadet	
2	Must have passed the ATPL(A) theoretical knowledge examinations in accordance with FCL.515.
3	Must have a flight school report.
4	100 hours as pilot-in-command, or 70 hours as Pilot-in-Command if completed during a course of integrated flying as set out in Appendix 3 A to Part FCL and AMC1 to Appendix 3 A. Applicants who have conducted a modular course need to have 100 hours as Pilot-in-Command (PIC), or 70 hours as PIC if completed during a course of integrated flying training as set out in Appendix 3 A, C & D to Part-FCL and AMC1 to Appendix 3 A, C & D.
Completes APC Type Rating Course (APCTR) in accordance with Ryanair ATO Training Manual Section 2.3	

5.2.0.2.3 ICAO Licence Holder – Validation Applicant (OCC Only)

- ALL categories included must be met.
- Entry for ICAO Licence holders is ONLY permissible for valid Type Rated applicants.

ICAO Licence Holder – Validation Applicant (OCC Only)	
1	Applicant must meet the current Minimum Entry Criteria for the OCC (Operator Conversion Course) with the addition that they must hold a current B737 300-900 type rating and be current on B737 300-900 aircraft type in the role they have applied for.
2	Applicant must hold a valid ICAO Licence. • Note: This Licence must remain valid at all times that the validation is being exercised.
3	Applicant must hold a valid B737 300-900 Type Rating on the ICAO Licence. • Note: This Rating must remain valid at all times that the validation is being exercised.

ICAO Licence Holder – Validation Applicant (OCC Only)																							
4	Applicant must hold a valid ICAO Class 1 Medical. • Note: This ICAO Class 1 Medical must remain valid at all times that the validation is being exercised.																						
5	Applicants shall agree to update 2, 3 and 4 above prior to Malta Air OCC Course commencement to provide a maximum validity period of Licence, Rating and Medical. The minimum acceptable validity of each being 9 months at Course start date.																						
6	Applicant must meet the English Language Assessment criteria at the interview/Simulator assessment.																						
7	Applicants must meet the Part FCL minimum hour requirements for PIC or Co-Pilot positions as outlined in the table below i.e.: • Direct Entry Captain must have >1500 hrs as PIC on Multi Pilot aeroplanes. • Direct Entry Co-Pilot must have >1500 hrs as PIC or co-Pilot on Multi Pilot aeroplanes. <table><tr><th>Licence held</th><th>Total flying hours experience</th><th>Privileges</th><th></th></tr><tr><th>(1)</th><th>(2)</th><th>(3)</th><th></th></tr><tr><td>ATPL(A)</td><td>> 1 500 hours as PIC on multi- pilot aeroplanes</td><td>Commercial air transport in multi- pilot aeroplanes as PIC</td><td>(a)</td></tr><tr><td>ATPL(A) or CPL(A)/IR (*)</td><td>> 1 500 hours as PIC or co-pilot on multi-pilot aeroplanes according to operational requirements</td><td>Commercial air transport in multi- pilot aeroplanes as co-pilot</td><td>(b)</td></tr><tr><td>CPL(A)/IR</td><td>> 1 000 hours as PIC in commercial air transport since gaining an IR</td><td>Commercial air transport in single-pilot aeroplanes as PIC</td><td>(c)</td></tr></table>			Licence held	Total flying hours experience	Privileges		(1)	(2)	(3)		ATPL(A)	> 1 500 hours as PIC on multi- pilot aeroplanes	Commercial air transport in multi- pilot aeroplanes as PIC	(a)	ATPL(A) or CPL(A)/IR (*)	> 1 500 hours as PIC or co-pilot on multi-pilot aeroplanes according to operational requirements	Commercial air transport in multi- pilot aeroplanes as co-pilot	(b)	CPL(A)/IR	> 1 000 hours as PIC in commercial air transport since gaining an IR	Commercial air transport in single-pilot aeroplanes as PIC	(c)
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5.3 Cabin Crew

5.3.1 Conditions for Assignment to Duties

ORO.CC.100, (a), 110, 120, 125, 130, 200, 200 (e), 210, 215 (d), 250, AMC1 ORO.CC.135, 250(b)

As a minimum requirement a Cabin Crew Member must be:

Note: Refer to 5.1.7.

Before a Cabin Crew Member is allowed to carry out duty on an aeroplane he shall have successfully completed Cabin Crew Member training, as well as the associated checks as prescribed in OM D, in addition to qualifications according OMA 5.1.5.

CAT.GEN.MPA.115. The Company ensures that all Cabin Crew Members wear the Company’s Cabin Crew uniform and are clearly identifiable to the passengers as Cabin Crew Members. AMC1 CAT.GEN.MPA.115(a)

When Malta Air/Ryanair Group personnel or crewmembers other than operating cabin crew members carry out duties in a passenger compartment (e.g. positioning, auditing), they shall ensure that they do not perform tasks or wear a uniform in such a way that might lead passengers to identify them as members of the operating cabin crew.

Additional Cabin Crew Members During Familiarisation Flights

The minimum number of qualified CCMs to be carried is determined in accordance with ORO.CC.100, (a) see Chapter 4, Section 4.1. Additional CCMs may be carried at the discretion of the Company. TM CAD may under exceptional circumstances require Malta Air to include in the crew additional Cabin Crew members Trainee CCMs assigned for familiarisation flights as part of their qualification training or CCMs assigned to re-establish currency are not to be included in the minimum number of qualified CCMs required.

Operation on More than One Aircraft Type or Variant

A cabin crew member will not be assigned to operate on more than three aircraft types. For the purpose of cabin crew training and qualifications Malta Air will shall determine:

1. Each aircraft as a type or variant taking into account, where available, the relevant data established in accordance with Regulation (EU) No 748/2012 for the relevant aircraft type or variant; and
2. Variants of an aircraft to be different types if they are not similar in the following aspects;
 - Emergency exit operation;
 - Location and type of portable safety and emergency equipment;
 - Type-specific emergency procedures.

5.3.2 Upgrading to Senior Cabin Crew Member

AMC1 ORO.CC.200 (c)

To become a senior Cabin Crew Member the applicant shall have at least:

- One year experience as an operating Cabin Crew Member, and
- Successfully completed the appropriate course and the associated check (refer to OM D).

5.3.3 Minimum Experience Levels

AMC1 ORO.CC.100(b)

New Cabin Crew are classed as “inexperienced” until they have achieved three months or 125 Flight hours operating experience as a Cabin Crew member. A minimum of 2 experienced CCMs shall be part of the operating crew

complement. “Inexperienced” Cabin Crew members will be marked as ‘NEW’ in the crew management system, until they have achieved 125 Flight hours or operated for 3 months from their 1st Line Operations day.

Note: Cabin Crew must complete an eLearning module to ensure familiarity with company ethos, policies and practices in conjunction with achieving 125 flight hours prior to becoming experienced.

5.3.4 Selection of Most Appropriately Qualified Cabin Crew to Replace CSS

The correct chain of command for Cabin Crew during normal operation will dictate which person will act as No 1 during the scheduled flights. Operating position is established by reference to the Certificate of Competence and the length of service in the rank. In descending order, longest in the rank being senior, the chain of command is:

1. SEPE.
2. SEPI.
3. TSEPI.
4. CSS LC.
5. Permanent CSS.
6. Temporary PU.
7. JU CSS [JU (PU) or JU AD HOC] (except when specifically rostered as No 1).
8. JU.
9. JU (NEW).

5.3.5 Recent Experience/Refresher Training

ORO.CC.145

When a Cabin Crew Member, during the preceding six months within the validity period of the last relevant recurrent training and checking:

1. Has not performed any flying duties, he shall, before being reassigned to such duties, complete refresher training and checking for each aircraft type to be operated; or
2. Has not performed flying duties on one particular aircraft type, he shall, before being reassigned to duties, complete on that aircraft type:
 - Refresher training and checking; or
 - Two familiarisation flights.

The Company may elect to replace refresher training by recurrent training if the reinstatement of the Cabin Crew Member's flying duties commences within the validity period of the last recurrent training and checking. If that validity period has expired, refresher training may only be replaced by aircraft type specific and operator conversion training.

5.4 Training and Checking and Supervisory Personnel

ORO.FC.145

The training requirements for appointment to the following positions is described in the relevant sections of OMD.

Type Rating Examiner, TRE;

Type Rating Instructor, TRI;

Synthetic Flight Examiner, SFE;

Synthetic Flight Instructor (SFI);

Nominated Commander;

Line Training Captain (LTC);

Line Check Captain (LCC);

CRM Instructor (for Flight Crew Members);

CRM Instructor (for Cabin Crew Members);

First Aid Instructor;

SEP/TSEP Instructor;

DGR Instructor;

Ground Instructor;

Check CCM;

Service Instructor.

5.4.1 Training

AMC1 ORO.FC.145(b), 145(d)

Refer Malta Air OMD 3.1.0.

5.4.1.1 Flight Crew Training Conversion Training

ORO.FC.120 (a), AMC1 ORO.FC.220

Refer Malta Air OMD 2.1.1 and 3.1.1.

5.4.1.2 Differences Training, and Familiarisation Training

ORO.FC.125

Refer Malta Air OMD 2.1.3 and 3.1.3.

5.4.1.3 Recurrent Training and Checking

ORO.FC.130

Refer Malta Air OMD 2.1.4 and 3.1.4.

5.4.1.4 Command Training

ORO.FC.205

Refer Malta Air OMD 2.1.6 and 3.1.6.

5.4.1.5 Co-pilot Line Training

Refer Malta Air OMD 2.1.1 and 3.1.1 also Appendix A6 – Student Competency Checklist.

5.4.1.6 Route and Aerodrome Competence Qualification

The requirements and details are contained in Section 5.1.2.3.

5.4.1.7 Dangerous Goods Training

Refer Malta Air OMD 2.4 and 3.1.55.

5.4.1.8 Cockpit Resources Management Training

Refer Malta Air OMD 1.5.

5.4.1.9 Cabin Crew Training and Checking

AMC1 ORO.CC.115(c), 115(d), 115(e), 125, 125(b), 125(c), 125(d), ORO.CC.130, 130(c)

Refer Malta Air OMD 2.2.

5.4.1.10 Recurrent Training

AMC1 ORO.CC.140

Refer Malta Air OMD 2.2.7 and 3.1.44.

5.4.1.11 Refresher Training

AMC1 ORO.CC.145

Refer Malta Air OMD 2.2.10 and 3.1.47.

5.4.1.12 Checks

Refer Malta Air OMD 2.2 and 3.

Training Course	Line Check Type	Code	Period of Validity	Next LC Type
OCC	OCC Initial	OCC LCK 1	End of month + 3 months	OCC LCK 2
OCC	2 nd OCC Check	OCC LCK 2	End of month + 3 months	OCC LCK 3
OCC	3 rd OCC Check	OCC LCK 3	End of month + 1 Year	Annual
Type Rating	Type Initial	Type LCK 1	End of month + 3 months	Type LCK 2
Type Rating	2 nd Type Rating	Type LCK 2	End of month + 1 Year	Annual
RST	Annual	Annual	End of month + 1 Year	Annual
RST	Annual	Annual (Restricted Pass)	End of month + 3 months	Annual
Command Upgrade	Command Initial	CU Initial	End of month +3 months	CMD LCK 2
Command Upgrade	2 nd Command Check	CMD LCK 2	End of month + 3 months	CMD LCK 3
Command Upgrade	3 rd Command Check	CMD LCK 3	End of month + 1 Year	Annual
Line Training Captain	Instructor Proficiency	IPC	End of month + 3 Years	IPC
Line Training Captain	Instructor Proficiency	IPC Grade 4/5	End of month + 3 months	IPC

5.5 Operations Personnel Other than Aircrew and Engineering

This category includes tug drivers, deicing personnel and a range of contracted service personnel.

The Nominated Persons for FOPS, TRG, GOPS and ENGR ensure that any contracted service personnel in their area either have their own approvals or operate under the Malta Air approval. Prior to commencement of contracted activities the service provider is audited to ensure that the service providers are trained, speak english to at least level 4 and are competent to deliver the contracted service safely and to the standard contracted. (ORO.GEN.205)

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6 CREW HEALTH PRECAUTIONS

6.1 Preface

CAT.GEN.MPA.100, 170, Regulation (EU)2018/1042

All aircrew in Malta Air/Ryanair Group have industry leading stable rosters, scientifically evaluated to assure that all aircrew are rested prior to commencing a series of flight duties and have sufficient time off after a series of flight duties to recover.

During a rostered period of flight duties the Crew Management System assures compliance with Malta Air approved FTL scheme, including subsequent rest periods. Therefore all Malta Air/Ryanair Group and service provider aircrew are expected to report for rostered duties in a fit and rested state to complete these rostered duties.

It is the responsibility of every individual Crew Member to report for duty fully fit and rested and able to complete all duties and requirements of Malta Air during the subsequent period of duty. Chapter 7 of this manual details the Malta Air approved Scheme for Flight and Duty Time Limitations for both flight and cabin crew. The above also applies when attending mandatory SEP training and flight simulator training or assessment.

Malta Air shall take all reasonable measures to ensure that no person enters or is in an aircraft when under the influence of psychoactive substances to the extent that the safety of the aircraft or its occupants is likely to be endangered. Company approved policies and procedures facilitate the prevention and detection of psychoactive substance misuse by flight and cabin crew members under its direct control.

The flight and cabin crew member shall not perform duties on an aircraft:

1. When under the influence of psychoactive substances or when unfit due to injury, fatigue, medication, sickness or other similar causes;
2. Until a reasonable time period has elapsed after deep water diving or following blood donation;
3. If applicable medical requirements are not fulfilled;
4. If he is in any doubt of being able to accomplish his assigned duties; or
5. If he knows or suspects that he is suffering from fatigue as referred to in 7.f of Annex IV to Regulation (EU) 2018/1139 or feels otherwise unfit, to the extent that the flight may be endangered.

Note: The FTL Scheme in Section 7 fulfils the requirements of Para 7.f of Annex IV to Regulation (EU) 2018/1139.

6.1.0.1 Flight Crew

MED.A.020, CAT.GEN.MPA.215, Regulation (EU)2018/1042, No 376/2014

EASA Part-MED states:

1. Licence holders shall not exercise the privileges of their licence and related ratings or certificates at any time when they:
 - a. Are aware of any decrease in their medical fitness that might render them unable to safely exercise those privileges;
 - b. Take or use prescribed or non-prescribed medication that is likely to interfere with the safe exercise of the privileges of the applicable licence; or
 - c. Receive any medical, surgical or other treatment that is likely to interfere with flight safety.
2. In addition, licence holders shall, without undue delay, seek aeromedical advice when they:
 - a. Have undergone a surgical operation or invasive procedure;
 - b. Have commenced the regular use of any medication;
 - c. Have suffered any significant personal injury involving incapacity to function as a member of the flight crew;
 - d. Have been suffering from any significant illness involving incapacity to function as a member of the flight crew;
 - e. Are pregnant;
 - f. Have been admitted to hospital or medical clinic; or
 - g. First require correcting lenses.

In cases referred to in point (2):

Holders of class 1 medical certificates shall seek the aero-medical advice of an AeMC or AME. In that case, the AeMC or AME shall assess their medical fitness and decide whether they are fit to resume exercising the privileges of their license.

6.1.0.2 Cabin Crew

MED.A.020

EASA Part MED states:

1. Cabin crew members shall not perform duties on an aircraft and, where applicable, shall not exercise the privileges of their cabin crew attestation when they are aware of any decrease in their medical fitness, to the extent that this condition might render them unable to discharge their safety duties and responsibilities.
2. In addition, a cabin crew member shall, without undue delay, seek aeromedical advice when they:
 - a. Have undergone a surgical operation or invasive procedure;
 - b. Have commenced the regular use of any medication;
 - c. Have suffered any significant personal injury involving incapacity to function as a member of the cabin crew;
 - d. Have been suffering from any significant illness involving incapacity to function as a member of the cabin crew;
 - e. Are pregnant.

Aeromedical advice can be obtained from the Malta Air designated AME, any AME or the TM CAD. In that case, the AME or the AeMC shall assess the medical fitness of the cabin crew members and decide whether they are fit to resume their safety duties.

Medical Care Away from Base

Ryanair Operations will make arrangements to arrange and provide medical care for Company personnel away from base when required.

6.1.0.3 Medical Certificate**Medical Certificate (Flight Crew Members)**

Applicants for and holders of a commercial pilot licence (CPL), a multi-crew pilot licence (MPL), or an airline transport pilot licence (ATPL) shall hold a Class 1 medical certificate. (MED.A.030.)

Class 1 medical certificates shall be valid for a period of 12 months and 6 months for licence holders who have reached the age of 60. Examinations and/or assessments for the revalidation of a medical certificate may be undertaken up to 45 days prior to the expiry date of the medical certificate. (MED.A.045.)

Operational multi-pilot limitation (OML – Class 1 only): (MED.B.001.)

- When the holder of a CPL, ATPL or MPL does not fully meet the requirements for a Class 1 medical certificate and has been referred to the licensing authority, it shall be assessed whether the medical certificate may be issued with an OML ‘valid only as or with qualified co-pilot’. This assessment shall be performed by the licensing authority.
- The holder of a medical certificate with an OML shall only operate an aircraft in multi-pilot operations when the other pilot is fully qualified on the relevant type of aircraft, is not subject to an OML and has not attained the age of 60 years¹.
- The OML for Class 1 medical certificates may only be imposed and removed by the licensing authority.

6.1.0.4 Aero-Medical Assessments (Cabin Crew Members)

MED.C.005

Cabin Crew Members shall undergo aero-medical assessments to verify that they are free from any physical or mental illness which might lead to incapacitation or an inability to perform their assigned safety duties and responsibilities.

Each Cabin Crew Member shall undergo an aero-medical assessment before being first assigned to duties on an aircraft, and after that at maximum intervals of 60 months. Aero-medical assessments shall be conducted as specified in FCL MED. The Company ensures that each Crew Member has passed a medical examination or assessment at regular intervals as required by the Authority so as to ensure the medical fitness to discharge his duties in accordance with Regulation (EU) No 965/2012, ORO.CC.110 (a) 2.

6.1.0.5 Psychological Assessments

CAT.GEN.MPA.175, AMC1 CAT.GEN.MPA.175(b), ORO.GEN.200(a)(3), Regulation (EU)2018/1042, ED Decision2018/012/R

All flight crew shall undergo a psychological assessment within the past 24 months before operating in order to:

1. Identify psychological attributes and suitability of the flight crew in respect of the work environment;
2. Reduce the likelihood of negative interference with the safe operation of the aircraft.

The assessment will be validated and either directly performed or overseen by a psychologist with acquired knowledge in aviation relevant to the operation and with expertise in psychological assessment, and where possible, the psychological selection of aviation personnel.

6.1.1 Consumption of Psychoactive Substances

AMC1 CAT.GEN.MPA.100(c)(1), CAT.GEN.MPA.215, ED Decision2014/015/R

Flight and Cabin Crew members shall not to carry out duties on an aircraft when under the influence of psychoactive substances.

No person whose function is critical to the safety of aviation shall undertake that function while under the influence of alcohol, by reason of which human performance is impaired.

6.1.1.1 Alcoholic drinks must not be consumed by Crew Members during the eight hours before reporting for a rostered operating, positioning service, flight simulator or standby duty. There shall be no residual ingested alcohol in the bloodstream when reporting for such a duty. Blood alcohol levels should not exceed the lower of the National requirements or 0.2 per thousand at the start of a flight duty period.

6.1.1.2 Alcoholic drinks must not be consumed by Crew Members or any person whose function is critical to the safety of operations, whilst on duty. To comply with the above Crew Members will restrict themselves to light consumption of alcohol during the 24 hours preceding such duties. Positioning Crew Members must assume that there may be a requirement to undertake a flying duty until they have been checked-in for the positioning service.

6.1.1.3 Alcoholic drinks must not be consumed by aircrew wearing uniform when they are on Malta Air/Ryanair Group property or in public places. In this context public places include aircraft and crew transport.

6.1.1.4 Alcoholic drinks, or unsealed bottles or other unsealed containers of alcohol may not be taken onto the flight crew compartment at any time.

6.1.1.5 Psychoactive Substance Testing

ARO.RAMP.106, CAT.GEN.MPA.170, AMC1 CAT.GEN.MPA.170(b), AMC2 CAT.GEN.MPA.170(b)

Random psychoactive testing (including Alcohol) is required by regulation to be conducted by both the authority (Ramp Inspections) and the operator. A refusal by a crew member to cooperate during an alcohol test should be regarded as a refusal to grant access in accordance with ORO.GEN.140 in the case of an EU operator or in accordance with TCO.115 of Commission Regulation (EU) No 452/2014 in the case of a third-country operator.

6.1.1.5.1 Policy to Prevent Misuse of Psychoactive Substances

Malta Air will test for psychoactive substances in the following cases:

1. Upon employment.

2. With due cause in the following cases:
 - a. Following a reasonable suspicion, and following an assessment by appropriately trained personnel.
 - b. After a serious incidence or accident, that is confirmed to be subject to an ICAO Annex 13 investigation, provided that testing is possible with respect to the event's location.
 - c. By direction of the NPFO following a serious incident or occurrence which is not subject to a State Authority investigation.
3. In accordance with the Malta Air Psychoactive Substance Testing Programme.

Member States shall ensure that alcohol testing of flight and cabin crew members is carried out with regard to operators under their own oversight as well as with regard to operators under the oversight of another member state. Such testing will be performed by ramp inspectors within the framework of the ramp inspection programme.

By way of derogation, Member states may ensure alcohol testing of flight crew and cabin crew members to be carried out by other authorized officials and outside the framework of the ramp inspection programme.

Member States may carry out additional testing for psychoactive substances other than alcohol.

In that case, the Member State shall notify EASA and the European Commission.

6.1.1.5.2 Alcohol Testing

Alcohol testing may be conducted by the Competent Authority and/or authorized officials (as designated by the Member State) in the following cases:

1. Random testing (e.g. ramp inspection)
2. In cases of a reasonable cause or suspicion
3. After a serious incident or accident, that is confirmed to be subject to an ICAO Annex 13 investigation, provided that testing is possible with respect to the event's location

Testing will be carried out away from the public, in a discrete manner by accredited personnel, and in accordance with relevant EU and National legislation. A flight crew or cabin crew member who refuses to cooperate during test, or who has been identified to be under the influence of alcohol after a positive test, shall not be allowed to continue his or her duty. In case of a

confirmed positive alcohol test result or in the case of a refusal by a crew member to cooperate, the operator shall inform the crew member concerned and the competent authority responsible for the crew member concerned.

In the event of a positive test result Malta Air will remove the crew member from duty and all planned future duties, inform the competent Authority, and the Authority responsible for the personnel concerned, such as a medical assessor of the licensing Authority.

6.1.2 Narcotics See Section Below ‘Drugs’

6.1.3 Drugs

AMC3 ORO.MLR.100

It is a criminal offence in every State in the Malta Air area of operations to import, export, manufacture, supply or possess “controlled drugs”.

If, as a result of the effects of drugs, a Crew Member is incapable of discharging his duties, or his ability to carry them out is impaired, or has been convicted in any jurisdiction for a drugs offence, the Crew Member will be liable to disciplinary action. Furthermore, the Crew Member will also be liable to disciplinary action if involved in the import, export, manufacture, supply or possession of controlled drugs or if taking drugs, even if not convicted of a criminal offence (e.g. found in unlawful possession of drugs when going through Customs but not prosecuted).

For the purposes of this Policy, the whole period from reporting at base to completion of duty back at base is deemed to be duty time. Thus, any Crew Member taking or otherwise involved with illicit drugs during their free time on a trip will be subject to disciplinary action.

In all cases where there is an allegation of misuse of “controlled drugs”, the charge is one of gross misconduct. The sanction, if the case is upheld, is dismissal.

6.1.4 Sleeping Tablets

See Medication Below.

6.1.5 Anti-Depressants

See Medication Below.

6.1.6 Pharmaceutical Preparations

See Medication Below.

6.1.6.1 Medication

AMC3 ORO.MLR.100

This section addresses Sleeping Tablets (6.1.4.), Anti-depressants (6.1.5.) and pharmaceutical preparations (6.1.6.) under the title medication.

Many medications, including commonly used prescription, over-the-counter, or herbal/complementary medicines, may lower operational efficiency by causing drowsiness, increased reaction time or impaired judgement.

Flight crew must not fly whilst taking any medication unless it is known to be acceptable for use when flying. In particular, they must not fly whilst taking any medication that may cause impairment until at least 48 hours after such treatment has ended unless cleared by an AME. Advice may be sought from the Malta Air AME, any other AME or the TM CAD.

Cabin crew must not operate if using any medication that may cause impairment except where cleared to do so by the Malta Air AME, any other AME or the TM CAD.

Note: GPs and hospital specialists may not know if the medication they intend to prescribe is acceptable for use when flying.

6.1.6.1.1 Anaesthetics

Crew Members must not operate:

Within 48 hours following any general anaesthetic.

Within 12 hours following any local anaesthetic including dental.

6.1.6.1.2 Sleeping Medication

AMC3 ORO.MLR.100

Flight crew shall take advice from their AME, or the TM CAD before taking any medication for sleep whether prescribed or over the counter. Use of Melatonin is not acceptable for EASA Class 1 medical certification; therefore it may not be used by pilots.

Cabin crew: Cabin crew shall not take sleeping medication less than 12 hours before reporting and will only operate if free of any adverse effects of the medication. Malta Air advises against the use of Melatonin.

Note: A source of information on currently acceptable sleeping tablets medication can be found on the UK CAA website.

6.1.6.1.3 Anti-depressants

AMC3 ORO.MLR

Flight crew: Use of certain anti-depressant medication is acceptable in pilots but is restricted to a small number of drugs and subject to strict criteria and regular monitoring. Any pilot suffering from depression shall consult their GP and/or a consultant psychiatrist for appropriate treatment of their condition and seek advice from their AME, Malta Air AME or the TM CAD before starting any medication and before returning to flying.

Cabin crew suffering from or undergoing treatment for depression shall only operate if they are well enough to do so and not experiencing any adverse effects of medication which are likely to interfere with the safe performance of their safety duties and responsibilities. Crew members can obtain further advice from the Malta Air AME or the TM CAD.

6.1.7 Immunisation

AMC3 ORO.MLR.100

Medical advice is to be sought concerning the period to be observed before returning to flying duties following immunisation. Immunisations will be obtained at least 24 hours before operating due to possible adverse effects. Advice on requirements and recommendations on other immunisations can be obtained from the Malta Air AME.

Immunisation Requirements – Malta Air Area of Operations

There are no mandatory requirements for immunisation in the Malta Air Area of Operation.

6.1.7.1 Validity of Vaccinations

Yellow fever	10 days after vaccination, life long protection (all intercontinental Flight Crew shall have a valid vaccination).
Tetanus	After basic immunization or after booster 10 years (vaccination not required but strongly recommended).
Diphtheria	After basic immunization or after booster 10 years (vaccination not required but recommended).
Poliomyelitis	After basic immunization or after booster 10 years (vaccination not required but recommended).

Hepatitis A	Protection 10 days after the first vaccination up to twenty years if vaccination course is completed (vaccination not required but recommended).
Meningitis	7 days to 3 years after vaccination (vaccination required by certain countries, recommended for certain destinations).
Typhoid fever	8 days to 12 months after oral vaccination with Typhoral L, 8 days to 3 years after parenteral vaccination with Typhim Vi (vaccination not required but recommended in certain situations).
Cholera	7 days to 6 months after 2 doses of a currently by WHO recommended oral vaccination (not required by national health regulations, recommended only for defined high risk groups, not for Flight Crews).

6.1.8 Deep Sea Diving and Underwater (SCUBA) Swimming

AMC3 ORO.MLR

Crew Members are warned of the dangers of decompression sickness if they fly too soon after diving using under-water breathing equipment. Crew Members must not undertake:

- ‘No-decompression’ diving within 24 hours before a rostered duty.
- No Dives requiring decompression stops within 48 hours before a rostered duty.

Crew Members who have received treatment for decompression sickness must not fly within 72 hours of recovery and not until cleared by an AME.

6.1.9 Blood Donation and Bone Marrow/Stem Cell Donations

AMC3 ORO.MLR.100

Crew Members are permitted to donate blood only if there is a minimum time interval of 24 hours between a normal blood donation and the next flight duty.

After bone marrow donation a crew member is considered unfit until the time of a satisfactory medical re-examination. This usually includes a complete blood count evaluation to ensure a return to acceptable values. As individual response to this procedure varies a specialist medical opinion shall be sought. In addition to the above, aircrews donating stem cells are already considered unfit to fly from the commencement of pre donation hormone therapy. Both bone marrow and stem cell donations are considered surgical procedures and AME authorization is required prior to resumption of flight duty.

6.1.10 Meal Precautions Prior to and During Flight

AMC3 ORO.MLR.100

Crew Members must take all reasonable precautions to minimise the risk of contracting gastro-enteritis prior to and during duty periods.

Effects of Food Poisoning

Food poisoning may be mild and of little consequence. However, acute cases of food poisoning, some severe enough to cause incapacitation, continue to occur in flight and impinge on flight safety.

Storage of Food

Food which has been kept in relatively high ambient temperatures for several hours after preparation can be affected and shall be treated with suspicion as even severe contamination is rarely obvious. Also the re-heating process onboard is generally insufficient to destroy the organisms but more particularly the toxins which are produced.

The following foods can present a high risk of infection:

- Under-cooked or raw meats (bacterial and parasitic infection).
- Cold meat and fish dishes in warm climates (bacterial infection).
- Shellfish, molluscs and oysters (bacterial or Viral infection).
- Salads, including rice and pasta salads (hygiene and bacterial infection).
- Foods which contain milk, cream or ice-cream (possibly unpasteurised).
- Any bottled or canned drink delivered with a broken seal (bacterial or illicit drug contamination).

6.1.10.1 Choose Different Meals

It is strongly recommended that pilots choose different meals, thereby avoiding the possibility that food poisoning renders them unfit for duty simultaneously.

In order to eliminate, as far as possible, the risk of food poisoning, the operating crew (Commander and co-pilot) on any aircraft shall, as far as circumstances permit, partake of different food, prior to the commencement of any flight and during its subsequent operation.

Cases may arise where no choice is available for the vegetable or dessert courses, but each Flight Crew member must partake of a different fish or meat course. Where the choice is not limited, it is recommended that each Flight Crew member partake of completely different meals.

6.1.10.2 Receiving Fluids on the Flight Deck from CSS/CSA

To prevent spillages on the Control Stand and other locations *in the flight deck* the following precautions shall be observed:

- Only screw top containers are allowed in the flight deck, e.g. bottles, screw top cups.

- Containers must be covered to avoid spillage.
- Containers of any fluids shall not be passed across the Control Stand.
- Only installed cup holders shall be used for stowing cups while in use.
- All spillages on aircraft equipment, shall be reported to engineering and entered in the Technical Log.
- Do not place liquids or food on the oxygen mask container.

Oven Bags

Cabin crew shall not accept meals from flight deck for heating up in the aircraft ovens unless the meal is presented in a Malta Air approved oven bag. Oven bags are available in all bases.

6.1.11 Sleep and Rest

AMC3 ORO.MLR.100

It is the responsibility of every individual Crew Member to report for duty fully fit and rested and able to complete all duties and requirements of Malta Air during the subsequent period of duty.

OMA 7 describes the Malta Air approved FTL Scheme. Crew members will be mindful of lifestyle choices during OFF periods to minimise tiredness during subsequent duty periods.

6.1.11.1 Fatigue

General Policy

The Company's overriding policy is that Flight Crew members shall not commence a flight duty or continue a flight duty after an intermediate landing if they are aware that they are fatigued or will be fatigued before their next landing.

General Information and Symptoms

Fatigue affects essential faculties such as co-ordination, perception, judgement and critical thinking ability. Because of several of these features, the build-up of fatigue can be insidious and its recognition is dependent on understanding when it may be expected and the symptoms associated with it. It has been highlighted as a contributory factor in some accidents and incidents.

The degree of fatigue experienced by an individual can be variable and will depend on such factors as:

1. The difficulty and extent of the flight duty concerned;
2. The general fitness and mental state of the individual;
3. The quality of rest in the period proceeding a flight duty;

4. The phase of the circadian (day/night) cycle as well as other factors not as clearly identifiable.

The symptoms of tiredness and fatigue may include:

1. Eye discomfort;
2. Difficulty in focusing and reading instruments or text especially at night in reduced lighting;
3. Reduced awareness and monitoring functions;
4. Slower instrument scan and greater tendency towards fixation;
5. Bodily muscle and joint discomfort;
6. Irritability, as well as other symptoms.

The above identifies symptoms of fatigue and gives indications of how this matter can be managed by the Flight Crew member and his responsibility. In flight, the build-up of tiredness can be minimised by regular 3–4 hourly light meals and intake of proper fluids.

6.1.12 Surgical Operations, Anaesthetics

Refer to OMA 6.1.0.1/6.1.0.2 re Part Med A 020 requirements a summary of which is on each medical certificate.

Any condition rendering a crewmember unfit to fly shall be notified to Ryanair Operations.

6.1.13 Vision Correction, Wearing of Spectacles

Flight crew requiring spectacles as prescribed by an AME shall have two pairs of spectacles available when on duty.

6.1.14 Diabetic Pilot Procedures and OML

In Aircrew Regulation, Flight Crew members with diabetes mellitus are divided into two categories:

1. Flight Crew with diabetes mellitus that requires insulin; and
2. Flight Crew with diabetes mellitus with blood sugar control that can be achieved and stabilised without the required use of insulin.

Flight crew with diabetes mellitus requiring insulin, are prohibited from operating Malta Air aircraft. This also applies to flights operated under the provisions of the Group Operations. These flight crew will be identified by

having a State Specific Limitation (SSL) on their Medical Certificate, there will also be an aircraft registration(s) that the medical is valid for. E.g. “SSL EI register aircraft only”.

A pilot with diabetes mellitus not requiring insulin will have an OML restriction on his/her license. Only one OML crewmember is allowed in any flight crew complement. The management of the diabetic condition varies, and the diabetic pilot is responsible for full compliance with the OML conditions/restrictions.

Flight crew shall discuss any limitations imposed on their medical certificate with their AME to ensure that they are aware of the obligations and restrictions applicable to their certificate.

6.1.14.1 Crew Briefings

Diabetic pilots being treated with potentially hypoglycaemic antidiabetic medication should brief their colleague pilot(s) fully prior to the flight.

Possible symptoms of high or low blood glucose:

Low Blood Glucose (Hypoglycaemia)	High Blood Glucose (Hyperglycaemia)
Sweaty, pale skin	Thirst
Mood changes	Excess Urine Output
Poor concentration/distraction	Dehydration
Confusion	Mood Changes
	Excessive tiredness/sleepy
	Blurred Vision

Diabetic pilots should ensure that, if required, their colleague pilot cross checks their test result and should always say the reading aloud so that it is recorded on the Cockpit Voice Recorder (CVR).

Any crew intervention required to assist a pilot in controlling their blood glucose levels, by any means, is a reportable event and should be reported under the MOR scheme and the diabetic pilot should declare themselves unfit.

6.1.14.2 Blood Testing Protocol

Testing, if required, must always be undertaken whilst ensuring compliance with SOPs. The diabetic pilot shall not test his/her blood glucose during ground maneuvering in the vicinity of the runway holding areas or entering or crossing a runway, or in phases of flight associated with heavy workload including the take-off, approach and landing.

Any incapacitation of a diabetic pilot, whether sudden or subtle, shall be managed in the same way as any other pilot incapacitation.

6.1.14.3 Emergency Situations

If operational considerations prevent the pilot from undertaking a blood glucose test at the required time, 15 g of rapidly absorbable glucose/carbohydrate (eg, 3 jelly babies, 4 glucotabs) should be consumed immediately and blood glucose testing done as soon as possible.

In an event such as a rapid decompression, there would be no time to take precautionary carbohydrate and priority would be given to flying the aircraft. Carbohydrate should be taken once the emergency has stabilised. If a mask continues to be required, it could be quickly lifted, carbohydrate consumed, and the mask replaced within a couple of seconds. In any other emergency situation, 15 g carbohydrate should be taken as soon as practicable. If an operational emergency is prolonged, with no opportunity for blood glucose testing, this consumption of 15 g carbohydrate must be repeated every hour. Blood glucose testing should be undertaken hourly or more frequently if there was any concern about the pre-emergency glucose trend or if a lot of carbohydrate has been taken over the course of several hours without the possibility of testing.

6.1.14.4 Responsibilities of Other Pilot(s) (Whether Commander or Not)

Flight Crew shall respect the confidentiality of any medical information shared by the diabetic pilot.

6.2 Cosmic Radiation

The radiation we receive comes either from outer space or from the sun (intensity increasing with solar flare activity).

Natural protection from cosmic radiation is provided by the geomagnetic field and the attenuating effects of the earth's atmosphere. The level of cosmic radiation depends to some extent on the geographical position, but essentially on the altitude above the ground level. The maximum radiation level occurs at about 20,000 m. Polar Regions have greater radiation intensity and exposure is more important at higher altitudes.

The biological effect evidently also depends upon the length of time of exposure. The effect on biological tissue, or body cells depends not only on the total dose but also on the components of the radiation field.

The practical interest is in the biological effect of a radiation dose, and the dose equivalent is measured in Sieverts (mSv) per hour or millisieverts (mSv) per year (1 Sv = 1000 mSv = 10⁻³ J/kg).

It is worth noting that natural radiation occurs also at ground level and varies from location to location, e.g. in parts of Cornwall (UK) the natural radiation level is at about 6 mSv per year and in most of Finland is around 8 mSv per year.

6.2.1 Work Involved Health Risks

The probability of a radiation-caused health risk is related to the total amount of radiation accumulated by an individual. At very low exposures, however, the estimated increases in risk are very small. Council Directive 96/29/EURATOM.

The likelihood of developing a health risk because of occupational exposure as flight or cabin crew is a small addition to health risks experienced by the general population. Currently, it is not possible to establish that a health risk occurring in a particular individual resulted from exposure to cosmic radiation as a result of being aircrew considering the dosage received in-flight.

6.2.2 Company Obligations

The Company takes account of the in-flight exposure to cosmic radiation to all Crew Members while on duty (including positioning) and assesses all Crew Member's exposure through the use of an approved System. An individual record of exposure to cosmic radiation is kept for each Crew Member.

Such exposures are notified to the individual on request from the NPFO and on leaving the Company.

6.3 Temporary Relief from Flight Duties in the Interest of Flight Safety

There are a number of stressful events or situations which warrant Flight Crew members being relieved of their flying duties in the interests of maintaining the safety of the operation. Flight Crew may avail of temporary relief from flight duties via Operational channels (e.g. Base Captain, Deputy Chief Pilot, Chief Pilot), Human Resources (e.g., FOPS Base Manager or Personnel Officer), or the Peer Support Programme, as appropriate.

The stress implicit in a death within the immediate family has the potential to cause a degradation of operating performance and judgement. It is the individual pilot's responsibility to consider bringing such events to the attention of the Company through appropriate channels before engaging in any flight operations.

When requests or consideration for the temporary relief of flight duties is raised via Operational or Human Resources channels, the period of time for which a Flight Crew member is taken off flying duties is a matter for the discretion of the Chief Pilot. The period of time depends on the circumstances of each individual case and would usually involve consultation with the crew member. It is the duty of the Chief Pilot to take account of the potential risk and to default to the safest option.

6.4 Peer Support Programme (PSP)

CAT.GEN.MPA.215

Malta Air operates a Peer Support Programme (PSP), to ensure access to a confidential and non-punitive programme that assists and supports pilots in recognising, coping with and overcoming problems which might negatively affect their ability to safely exercise the privileges of their license.

The PSP is a proactive programme applying the principles of ‘Just Culture’ as defined in Regulation (EU) No 376/2014, whereby an Independent contracted PSP Management Organization (PSPMO), mental health professionals from a chosen Independent Health Care Organisation (IHO) and trained peers work together to enable self-declaration, referral, advice, counselling and/or treatment, where necessary, in case of a decrease in medical fitness, with support from Malta Air Flight Operations Management and Senior Management.

The PSP provides a facility for a pilot to contact a peer on a confidential basis for advice or assistance regarding a developing social, personal, health or professional standards issue. The protection of the confidentiality of data is a precondition for the PSP to encourage its use and ensures its integrity. It is intended that the PSP provides an adequate means of signalling a need for support at the earliest stages of a threat to a crew members physical or mental health, to provide education of pilots regarding self-awareness and facilitation of self-referral.

The PSP will be facilitated by an independent PSP Management Organization (PSPMO). The PSPMO ensures that Peers are available to all pilots within Malta Air. When required, the PSPMO may also coordinate access to Critical Incident Stress Management (CISM) for post-incident/accident crisis intervention services to pilots.

6.4.1 Peers

In the context of the PSP, a ‘peer’ is a pilot who shares common professional qualifications and backgrounds working in the Ryanair Group or a retired pilot from one of the Group AOCs.

Peers are nominated by the NPFO and receive initial and recurrent training facilitated by the contracted PSPMO and IHO. To ensure a high level of confidentiality for users of the service, Peer training includes emphasis on the Company Peer Code of Conduct and Data Protection Policies and Procedures.

6.4.2 Elements and Governance of PSP

a. Elements of PSP:

1. The education of pilots through CRM and SMS training regarding mental health in the aviation workplace, self-awareness and the facilitation of self-referral;
2. Critical Incident Stress Management;
3. Assistance provided by mental and psychological health professionals with relevant knowledge of the aviation environment;
4. Assistance provided by trained peers with relevant knowledge of a pilot’s working environment;
5. Monitoring to ensure the efficiency and effectiveness of the PSP;
6. Monitoring and support of the return to work process;
7. Management of risks resulting from fear of loss of license;
8. A referral system to an aero-medical examiner for defined cases which raise a serious safety concern;
9. Confidentiality and data protection procedures;
10. Selection and training of peers;
11. Oversight Committee.

b. Governance of PSP:

The PSP is linked to the Malta Air Management System for purposes of safety oversight and safety management. SMS Just Culture principles will prevail at all times.

An Oversight Committee will monitor the performance of the programme.

The Oversight Committee is comprised of:

1. NPFO (Chairperson)
2. Safety Manager

3. Representative from the PSPMO
4. Representative from HR

The Oversight Committee will meet quarterly to review anonymised and aggregated data provided by the PSPMO to support the Malta Air Safety Management System. Data reviewed by the Oversight Committee is anonymised and aggregated to ensure confidentiality and used solely for the purposes of safety management.

6.4.3 PSP Access and Processes

Access to the PSP is designed to:

1. Enable self-declaration or referral in case of a decrease in a pilot's medical fitness with an emphasis on prevention and early support; and
2. If appropriate, allow the pilot to receive temporary relief from flight duties and be referred to professional advice.
3. Be easy for pilots and provide adequate means of support at the earliest possible stage.

Pilots requesting support can contact the PSP via the confidential, dedicated email address available in the Peer Support Programme Handbook. PSPMO processes ensure daily monitoring and facilitation of the PSP contact portal. Pilots contacting the PSP will receive an initial response within 24 hours.

After the initial contact, the Peer liaises with the PSPMO to determine the appropriate level of support required/requested and coordinates accordingly. The Peer, PSPMO, and/or IHO will continue to facilitate assigned PSP processes until the pilot exits the programme.

6.4.4 Obligation to Seek Aero-Medical Advice in Case of a Decrease in Medical Fitness

Participation in the PSP does not remove a pilot's obligation to seek aero-medical advice in case of a decrease in medical fitness in accordance with MED.A.020 of Commission Regulation (EU) No 1178/2011.

6.4.5 Confidentiality and Protection of Data

- a. PSP and PSPMO processes ensure that personal data of pilots referred to the PSP will be handled in a confidential, non-stigmatising, and safe environment.

- b. A culture of mutual trust and cooperation is maintained so that the pilot is less likely to hide a condition and more likely to report and seek help. The PSP's effectiveness relies heavily on the establishment of trust between reporters and the recipients of reports. All reported concerns will be handled in a confidential manner.
- c. To avoid compromising voluntary participation in the PSP and/or flight safety, disclosure of data by the PSPMO to Malta Air may only occur in an anonymised manner (i.e., aggregated statistical data) and only for purposes of safety management. If required to facilitate PSP processes, pilots seeking support must give consent for their information to be referred, including when referral to a medical professional is indicated.
- d. Where the ongoing fitness of a pilot for medical certification is in doubt or there is a decrease in medical fitness, advice will be sought by the pilot from an aviation medical specialist (as mandated in MED.A.020 Decrease in Medical Fitness (EU. No. 1178/2011).
- e. Correspondence and records associated with the PSP and PSPMO are conducted confidentially and maintained securely, in line with GDPR. Medical confidentiality guidelines and data protection legislation are applied, including the Data Protection Act 1998, the EU Directive (95/46/EC) and General Medical Council guidance on Confidentiality 2009.
- f. Notwithstanding the above, an agreement with related procedures is in place between MaltaAir and the PSPMO on how to proceed in the case of a serious safety concern. A mechanism is in place to allow Peers who are concerned about information they have received to seek advice from the PSPMO and/or IHO (AME) if there is concern that the pilot may present an imminent risk to flight safety or pose a risk of harm to themselves or the public. In this circumstance, the associated aviation medical specialist must report this concern to the national authority which has issued the pilot's license without delay.

6.4.6 Promotion of the Peer Support Programme

Awareness and access to the Peer Support Programme are promoted via appropriate means (*Ryanair Connect*, SMS, LASG, ATO, EFB, Peers, colleagues, etc) and the 'Peer Support Programme Handbook'.

The 'Peer Support Programme Handbook', available on Ryanair Connect, contains insights on the programme's objectives, structure, processes, contacts, and wellbeing.

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7 FLIGHT TIME LIMITATION (FTL) SCHEME

7.0 Flight and Duty Time Limitations and Rest Requirements

General:

Malta Air has subcontracted all Rostering and Crewing functions to Ryanair DAC.

Malta Air conducts audits of Ryanair DAC roosting and crewing functions iaw ORO.GEN.205, ORO.GEN.200 (a)(6).

This Chapter follows the numbering, of the Air Ops Subpart FTL-Flight and Duty Time Limitations and Rest Requirements Regulation, All associated AMCs which are applicable to the Malta Air FTL, all CS applicable to the Malta Air FTL and selected Guidance Material applicable to the Malta Air FTL scheme is included.

Explanation of Air Ops References in this chapter:

Reference	Description
Article n	Article to regulation (EU) No 965/2012 (Air Ops Regulation)
ORO.XXX	Part-ORO: Organisation Requirements for Air operators, (Annex III to Regulation (EU) No 965/2012)
ORO.FTL.XXX	Subpart FTL (Flight and Duty Time Limitations and Rest Requirements)
ORO.FTL.1XX	Section 1 (General)
ORO.FTL.2XX	Section 2, Commercial Air Transport Operators (CAT)
AMCn ORO.FTL.XXX	Acceptable Means of Compliance to Part-ORO
GMn ORO.FTL.XXX	Guidance material to Part-ORO Subpart FTL
CS FTL.1.XXX	Certification Specifications: Commercial Air Transport by Aeroplane-Scheduled and Chartered Operations
GMn CS FTL.1.XXX	Guidance Material: Commercial Air Transport by Aeroplane-Scheduled and Chartered Operations

7.1 Malta Air Flight Time Limitation (FTL) Scheme

This Malta Air Flight Time Limitation (FTL) Scheme complies with:

- COMMISSION REGULATION (EU) No 83/2014 of 29 January 2014 (the Regulation) amending Regulation (EU) No 965/2012 laying down technical requirements and administrative procedures related to air operations pursuant to Regulation (EC) No 2018/1139 of the European Parliament and of the Council. This FTL scheme is effective from 18th February 2016 and remains in force until further notice.
- The Council Directive 2000/79/EC ‘Working Time Directive’ limit of 2000 working hours in a year applies to all personnel subject to this FTL.
- European Communities (Organisation of Working Time) (Mobile Staff in Civil Aviation) Regulations 2006.
- SI 507/2006, European Communities (Organisation of Working Time for Mobile Staff in Civil Aviation) Regulations 2006.

7.1.1 Scope

ORO.FTL.100

This Chapter establishes the requirements to be met by Malta Air and Malta Air crew members with regard to flight and duty time limitations and rest requirements. The personnel listed below engaged by Ryanair DAC ensure that Malta Air requirements are satisfied in accordance with the regulations and contractual obligations. These personnel are accountable to the NPFO Malta Air via their respective managers.

Owner	Document Title	Output
Deputy Director Flight Operations	Crew Management System Documentation	The control document for all rostering and crew control assignments, legality checks, and actual FTL records.
Head of Pilot Rostering	Flight Crew Planning Department: Roster Production Procedure	Pilot Rosters including legality checks and roster stability metrics
Pilot Training Delivery Manager	Training Department Flight Crew Planning: Roster Production Procedure	Rosters for pilots in Training, including legality checks and roster stability metrics
Crew Planning Manager	In-flight Crew Planning Department: Roster Production Procedure	Cabin Crew Rosters, including legality checks and roster stability metrics

Owner	Document Title	Output
In-flight Training admin Manager	In-flight Training: Roster Production Document	Cabin crew in Training Rosters, including legality checks and roster stability metrics
Operations Control Manager Operations Controllers Operations Officers	Ops Center Control Manual OCCM amended by OCI (Operations Control Instruction)	Tactical Management of the Commercial Schedule, including legality checks and Fatigue Management principles applied to tactical changes to the schedule
Operations Control Manager Crew Controllers	Ops Center Control Manual OCCM amended by OCI (Operations Control Instruction)	Tactical changes to rosters in the active week including legality checks and Fatigue Management principles applied to tactical changes to the roster
Operations Control Manager Flight Planners	Ops Center Control Manual OCCM amended by OCI (Operations Control Instruction)	Tactical Management of the Commercial Schedule including legality checks and Fatigue Management principles applied to tactical changes to the schedule

7.1.2 Applicability of Subpart FTL of ANNEX III

Article8

Commercial Air Transport Operations (CAT) is subject to Regulation EU 2018/1139.

7.2 Definitions

ORO.FTL.105

FTL definitions including those not applicable to the Malta Air FTL Scheme, are reproduced in this Chapter. Regulation (EU) No 83/2014.

7.2.1 “Acclimatised”

Acclimatised means a state in which a crew member’s circadian biological clock is synchronised to the time zone where the crew member is. A crew member is considered to be acclimatised to a 2-hour wide time zone surrounding the local time at the point of departure. When the local time at the place where a duty commences differs by more than 2 hours from the local time at the place where the next duty starts, the crew member, for the calculation of the maximum daily flight duty period, is considered to be acclimatised in accordance with the values in the following table:

Time difference (h) between reference time & local time where a crew member starts the next duty	Time elapsed since reporting at reference time				
	<48	48–71.59	72–95.59	96–119.59	≥120
<4	B	D	D	D	D
≤6	B	X	D	D	D
≤9	B	X	X	D	D
≤12	B	X	X	X	D

“B” means acclimatised to the local time of the departure time zone;
“D” means acclimatised to the local time where the crew member starts his/her next duty;
“X” means that a crew member is in an unknown state of acclimatisation.

Note 1: Acclimatised

ORO.FTL.105

A crew member remains acclimatised to the local time of his/her reference time during 47 hours 59 minutes after reporting no matter how many time zones he/she has crossed. The maximum daily FDP for acclimatised crew members is determined using the table provided with the reference time of the point of departure. As soon as **48 hours** have elapsed, the state of acclimatisation is derived from the time elapsed since reporting at reference time and the number of time zones crossed.

Note 2: Acclimatised ‘point of departure’

ORO.FTL.105

The point of departure refers to the reporting point for a flight duty period or positioning duty after a rest period.

Note 3: Acclimatised ‘time elapsed since reporting at reference time’

ORO.FTL.105

The time elapsed since reporting at reference time for operations applying Certification Specifications at home base refers to the time elapsed since reporting for the first time at home base for a rotation. (CS FTL.1.235(b)(3)(ii))

7.2.2 “Reference Time”

Reference time means the local time at the reporting point situated in a 2-hour wide time zone band around the local time where a crew member is acclimatised;

Note: Reference time

ORO.FTL.105

Example: A crew member is acclimatised to the local time in Rome and reports for duty in Dublin. The reference time is the local time in Dublin.

7.2.3 “Accommodation”

Accommodation means, for the purpose of standby and split duty, a quiet and comfortable place not open to the public with the ability to control light and temperature, equipped with adequate furniture that provides a crew member with the possibility to sleep, with enough capacity to accommodate all crew members present at the same time and with access to food and drink.

7.2.4 “Suitable Accommodation”

Suitable accommodation means, for the purpose of standby, split duty and rest, a separate room for each crew member located in a quiet environment and equipped with a bed, which is sufficiently ventilated, has a device for regulating temperature and light intensity, and access to food and drink.

7.2.5 “Augmented Flight Crew”

Augmented flight crew means a flight crew which comprises more than the minimum number required to operate the aircraft, allowing each flight crew member to leave the assigned post, for the purpose of in-flight rest, and to be replaced by another appropriately qualified flight crew member.

7.2.6 “Break”

Break means a period of time within a flight duty period, shorter than a rest period, counting as duty and during which a crew member is free of all tasks.

7.2.7 “Delayed Reporting”

Delayed reporting means the postponement of a scheduled FDP by Malta Air before a crew member has left the place of rest.

7.2.8 “Disruptive Schedule”

Disruptive schedule means a crew member’s roster which disrupts the sleep opportunity during the optimal sleep time window by comprising an FDP or a combination of FDPs which encroach, start or finish during any portion of the day or of the night where a crew member is acclimatised.

Early Type disruptive schedule applies to the Malta Air FTL.

Note: If a crew member is acclimatised to the local time at his/her home base, the local time at the home base should be used to consider an FDP as 'disruptive schedule'. This applies to operations within the 2-hour wide time zone surrounding the local time at the home base, if a crew member is acclimatised to the local time at his/her home base. ORO.FTL.105.

- **“Early type”** of disruptive schedule means:
 - for **“early start”** a duty period starting in the period between **0500** and **0559** in the time zone to which a crew member is acclimatised; and
 - for **“late finish”** a duty period finishing in the period between **2300** and **0159** in the time zone to which a crew member is acclimatised; and

7.2.9 **“Night Duty”**

Night duty means a duty period encroaching any portion of the period between **0200** and **0459** in the time zone to which the crew is acclimatised.

7.2.10 **“Duty”**

Duty means any task that a crew member performs for Malta Air, including flight duty, administrative work, giving or receiving training and checking, positioning, and some elements of standby.

7.2.11 **“Duty Period” (DP)**

Duty period means a period which starts when a crew member is required by Malta Air to report for or to commence a duty and ends when that person is free of all duties, including post-flight duty.

7.2.12 **“Flight Duty Period (‘FDP’)**

Flight Duty period means a period that commences when a crew member is required to report for duty, which includes a sector or a series of sectors, and finishes when the aircraft finally comes to rest and the engines are shut down, at the end of the last sector on which the crew member acts as an operating crew member.

7.2.13 **“Flight Time”**

Flight time means the time between an aircraft first moving from its parking place for the purpose of taking off until it comes to rest on the designated parking position and all engines or propellers are shut down.

7.2.14 **“Home Base”**

Home Base means the location, assigned by Malta Air to the crew member, from where the crew member normally starts and ends a duty period or a series of duty periods and where, under normal circumstances, Malta Air is not responsible for the accommodation of the crew member concerned.

7.2.15 “Local Day”

Local day means a 24-hour period commencing at **0000** local time.

7.2.16 “Local Night”

Local night means a period of 8 hours falling between **2200** and **0800** local time.

7.2.17 “Operating Crew Member”

Operating crew member means a crew member carrying out duties in an aircraft during a sector.

7.2.18 “Positioning”

Positioning means the transferring of a non-operating crew member from one place to another, at the behest of Malta Air, excluding:

- the time of travel from a private place of rest to the designated reporting place at home base and vice versa; and
- the time for local transfer from a place of rest to the commencement of duty and vice versa.

7.2.19 “Rest Facility”

Rest facility means a bunk or seat with leg and foot support suitable for crew members’ sleeping on board an aircraft.

7.2.20 “Reserve”

Reserve means a period of time during which a crew member is required by Malta Air to be available to receive an assignment for an FDP, positioning or other duty notified at least **10 hours** in advance.

7.2.21 “Rest Period”

Rest period means a continuous, uninterrupted and defined period of time, following duty or prior to duty, during which a crew member is free of all duties, standby and reserve.

7.2.22 “Rotation”

Rotation is a duty or a series of duties, including at least one flight duty, and rest periods out of home base, starting at home base and ending when returning to home base for a rest period where Malta Air is no longer responsible for the accommodation of the crew member.

7.2.23 “Single Day Free of Duty”

Single day free of duty means, for the purpose of complying with the provisions of Council Directive 2000/79/EC, a time free of all duties and standby consisting of one day and two local nights, which is notified in advance. A rest period may be included as part of the single day free of duty.

7.2.24 “Sector”

Sector means the segment of an FDP between an aircraft first moving for the purpose of taking off until it comes to rest after landing on the designated parking position.

7.2.25 “Standby”

Standby means a pre-notified and defined period of time during which a crew member is required by the operator to be available to receive an assignment for a flight, positioning or other duty without an intervening rest period.

7.2.26 “Airport Standby”

Airport standby means a standby performed at the airport.

Note: Airport standby counts in full as duty period ORO.FTL.105

7.2.27 “Other Standby”

Other standby means a standby either at home or in a suitable accommodation.

7.2.28 “Window of Circadian Low (‘WOCL’)”

WOCL means the period between **0200** and **0559** hours in the time zone to which a crew member is acclimatised.

7.2.29 Special Duties

“Special Duty” means a period of duty other than a flying or Standby duty.

7.3 Malta Air Responsibilities as the Operator

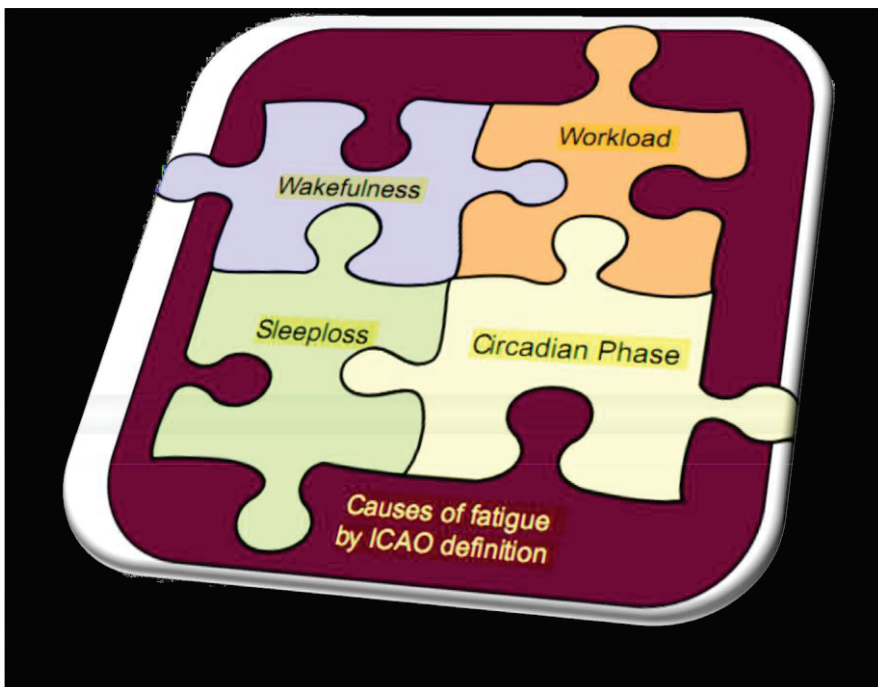
ORO.FTL.110

Malta Air provides policies and procedures, organisation and personnel, publicity, training, reporting and data analysis tools to enable the organisation and operational personnel to manage fatigue.

Crewmember fatigue ICAO Definition can be defined as:

A physiological state of reduced mental or physical performance capability resulting from sleep loss or extended wakefulness, circadian phase, or workload (mental and/or physical activity) that can impair a crew member's alertness and ability to safely operate an aircraft or perform safety related duties.

The ICAO definition is summarised in the picture below.

**Fatigue Hazard Identification**

Fatigue is identified as a hazard in the Malta Air Hazard Log and the causes which could promulgate fatigue are communicated and promoted through recurrent training and safety media. A Fatigue Safety Performance indicator has been created, this indicator includes time, place, extent and nature.

Investigations conducted by the Safety and Security Manager include fatigue as a possible hazard and each investigation includes crew's roster for the previous 28 days to the event.

In coordination with Ryanair DAC, Malta Air utilises the following aids to identity possible consequences of fatigue:

1. Risk assessments;

2. Safety reports filed through the safety reporting system;
3. External occurrence and accident reports;
4. Audits/non-compliance reports;
5. Internal reviews including the Daily Safety, Security and FM Review Meeting, Weekly Safety and Fatigue Management Review Meeting and SAG;
6. Zendesk, RCRS and;
7. Safety surveys.

In addition to the above Malta Air has appointed a designated Base Captain (eCDR BC) as the responsible individual for monitoring the use of Commanders Discretion in Malta Air. The eCDR BC reports directly to the NPFO for this purpose.

On a weekly basis, the eCDR BC reviews all Commanders Discretion Reports (ECDRs) to:

- Ensure that ECDRs submitted are complete and accurate.
- Report all exceedances in excess of 1 hour to TM CAD.
- Identify trends.
- Identify where FTL exceedances are caused by scheduling issues.

The eCDR BC provides a monthly eCDR Report to the DCPLO and NPFO. This report contains detailed information related to all eCDR filed during the preceding 28 day period including duration of discretion used, route, reason for discretion and details previous cases where discretion was used on the specific route. The eCDR reports are utilised to facilitate the seasonal schedule review. The NPFO will provide a monthly report to the Accountable Manager.

Fatigue as a risk is managed the same as any other risk under the Malta Air Safety Management System (SMS) and as described in the Safety Management Manual. ORO.GEN.200

In Malta Air fatigue management is a scientifically-based, data driven, flexible, ‘continuous process’ of monitoring and managing fatigue. Fatigue management is made up of the following components:

- Malta Air FTL Scheme.
- Occurrence Reporting.
- FTL Audit Schedule.

- Scientific Analysis.

The purpose of fatigue management is to:

- Ensure that the safety objectives of Regulation (EU) 2018/1139 are met.
- Prevent crew fatigue (transient and cumulative).
- Manage the operational risk(s) which may arise from crew member fatigue.
- Provide crew members with adequate rest opportunities. Crew members must use these opportunities effectively to ensure that they are sufficiently alert to operate to a satisfactory level of performance and safety.
- Provide analysis and support to the risk assessment of proposed changes to fatigue management.

Fatigue management is applicable in all departments. Heads of department have specific responsibility to actively manage fatigue risks, to continually assess fatigue risk measures and to continuously improve fatigue management strategies in place. Heads of Departments continually review all aspects of our operation including specific operational aspects where fatigue could generate an increased risk.

Every person in the organisation, responsible for preparation and maintenance of crew rosters and management personnel concerned receive initial and thereafter recurrent fatigue management training every three years to ensure continued compliance with best practices in the area of fatigue identification and fatigue management. This training is provided by and tracked by the Safety Manager.

Every person subject to this manual has the obligation to report issues relating to safety, which includes fatigue, using the Malta Air reporting systems available. All such reports are captured in the Safety Reporting System and are processed in accordance with the procedures detailed in OMA and SMM. Fatigue management is a fixed agenda item at the daily and weekly safety and fatigue management review meetings. ORO.GEN.200(A)(3)

This section details Malta Air's responsibilities to crewmembers in compliance with EU Regulations: ORO.FTL.110

- a. Duty rosters are published, normally 4 weeks in advance, but at least 14 days in advance to provide the opportunity for crew members to plan adequate rest. Rosters are published by 1700 hrs each Friday and are made available to individual crewmembers via the internal electronic portal appropriate to each functional area, AMC1 ORO.FTL.110(a).

- b. Malta Air crews are normally scheduled to start and finish a duty assignment at the same base. This has the distinct advantage that crews have the opportunity for sleep, usually at home base during the night period. We are ‘programmed’ to sleep during the night and this also minimises disruption to Circadian Rhythm. The same benefits apply during OFF days whereby aircrew can reduce any sleep deficit acquired during the previous duty period and because of the numbers of days off at a time, report for the next duty period with their ‘sleep reservoir’ replenished. Sleep is the only antidote to fatigue. This Malta Air model provides significant fatigue management and mitigation most of which is programmed into the Crew Management System FTL ruleset.

All Rosters are constructed in compliance with the Malta Air FTL or IFTSS (as applicable) including fatigue management principles therein. OFF days (opportunities for rest) are compliant with EU regulations and working time directive requirements thereby ensuring that crew members have sufficient rest opportunities. This enables fatigue to be managed to be as low as reasonably practicable and enables crew members to remain sufficiently free from fatigue so that they can operate to a satisfactory level of safety under all circumstances. Malta Air typically rosters more ‘OFF’ days than are required by EU regulations.

Crew controllers ensure that any roster changes on the day or in the crewing period, e.g. working day off, additional sectors etc are firstly legal by using the Crew Management System and thereafter managed as much as possible to avoid elevated levels of fatigue during the duty. Operating crewmembers normally have 4 or more earlies/lates in the 5 ON period. The Early Disruptive Schedule requires that there is a rest period of 60 consecutive hours after four or more early starts (pre 0600), late finishes (2300–0159) or night duties (0200–0459) and Malta Air complies with this. Working OFF days during OFF periods is subject to a Crew Management System legality check which would trigger a Disruptive Schedule notification if the 60 hrs is violated, thereby preventing this working OFF day assignment.

Additionally, each WOFF is subject to an assessment conducted in rostering or Operations Control, to ensure that at least an equivalent level of safety and fatigue management is achieved and maintained.

Malta Air has various metrics to assure that flights assigned are completed in the scheduled times and in particular within the max flight duty times. Punctuality is a clear measure that flights are being executed as planned. Any deviations, including eCDR (electronic Commanders Discretion Reports) are analysed and presented to the Safety Action Group Meeting (SAG) and when appropriate schedules and/or pairings are changed as a fatigue management mitigation strategy as indicated earlier in this chapter. Disruption due weather, medical emergencies, associated diversions and technical issues are frequent causes of roster disruption. Industrial action is

also a significant contributor. When this analysis is completed it will be apparent where the schedule is challenged by FTL and fatigue management principles and appropriate changes are made. The Safety Manager prepares this report. It is presented to the first SAG meeting following the summer or winter season.

- c. Reporting times in Malta Air allow sufficient time for ground duties. Malta Air expects that the commute from home/residence to the reporting point is achievable in 60 mins. Therefore, the report time excludes passing through staff channels and security in the airport and the commute to the reporting point. The report time for all locations*, crewroom, boarding gate or aircraft is STD-45 mins.

Note: *Reporting point will be identified in the Base Information document on EFB/VPOS.

This 45 minute report time in every airport has, over many years, proven resilient to achieve the completion of regulatory and non-regulatory tasks pre-flight. Malta Air has base management in each base lead by the Base Captain to monitor these activities. Better and faster delivery systems have enhanced pre-flight efficiencies. The commute to the aircraft is continuously reviewed to ensure it remains reasonable. The necessary preparations for departure including security checks, PDIs, boarding and fuelling, aircrew briefings and completion of pre-departure flows, checklists and performance calculations are continually assessed. The introduction of the Electronic Flight Bag has greatly assisted the Flight crew in this period. Where any element of the 45 minute report is challenged as reported by aircrew or detected from punctuality data, the 45 minute report and/or non-regulatory procedures will be changed to ensure continued compliance.

The frequency of operations in Malta Air ensures crew familiarity and practice to achieve compliance during the reporting period and enables expeditious completion of these tasks whilst maintaining high level of safety – as supported by SMS data. When there are widespread reporting delays and crews are rolled (reassigned) to other same wave flights, crews reporting may be instructed to report directly to the aircraft/crewroom/gate.

Crew members called off standby may be instructed to report to the crew room, the gate or directly to aircraft. If the crew has overnighted away from a Malta Air base or when returning to duty after the break on the ground during a split duty the most suitable reporting point may be the gate or aircraft. In all cases the standard pre-flight crew briefings must be completed prior to dispatch.

- d. The Malta Air model does not normally involve long duty periods combined with minimum rest periods. It is one of the tasks of the Ryanair DAC Rostering Manager to identify these and where possible make

appropriate roster amendments. In periods of disruption this combination may occur but each crewing or ops control response must be FTL compliant and every effort is made to minimise the fatigue impact. The duty may be completed providing it is FTL compliant (no legality issues flagged). Potential rest opportunities identified prior to the next duty period will be applied by crewing or ops control, where possible including changes to pairings.

Malta Air has significant experience in scheduling and rostering and it is Malta Air policy that our rosters deliver the required hours as evenly as possible across crews throughout the year. This fatigue management principle is included in the configuration of the roster optimiser function in the Crew Management System. The auditing schedule and crew reports in particular will confirm any roster allocation imbalance.

Minimum rest issues rarely arise and when they do it is invariably as a result of a widespread disruptive event such as strikes or weather. Minimum rest is always provided before commencing a rostered duty. Typically, disruptive periods impact over a few days and crew fatigue is managed by rest opportunities during the night period, which are never less than minimum rest. Any potential sleep deficit is contained within the '5 ON' roster and recovered during OFF days.

Very exceptionally, e.g. Volcanic Ash events, the period of disruption can continue whereby a re-rostering solution would be applied. All such rosters will be FTL compliant and every effort will be made to allocate pairings to keep fatigue as low as reasonably practical. Fatigue mitigation measures will continue after this period of disruption until rosters are stabilised and normal rostering service is resumed. During these periods crews will be reminded of appropriate sleep strategies in conjunction with other rostering tools such as split duty and delayed reporting as appropriate.

- e. Scheduling rules ensure that the rosters allocate duty patterns which avoid causing serious disruption of an established sleep/work pattern. Malta Air roster patterns are normally consecutive 'Early'-'Late'-'Early' patterns with a sleep opportunity during the night time when we are 'programmed' to sleep thereby minimising disruption to circadian rhythms. Transition from Late to Early roster periods is managed by sufficient OFF days to ensure adequate sleep opportunities and compliance with local night requirements. Other rosters as requested by individual crew members, eg all 'earlies' or all 'lates' are subject to the same fatigue management principles and FTL rules.

The Malta Air schedule does not have crew rest opportunities in locations of 4 or more time zones away from a base for scheduled CAT flights.

- f. Schedules and rosters comply with the provisions concerning early disruptive schedules by ensuring at least one local night prior to commencement of 'earlies' after 'lates'. This is one of the legality checks

built into the Crew Management System ruleset and the audit and crew reporting processes ensure any errors are trapped and rectified. All roster patterns in Malta Air comply with extended recovery rest periods and early disruptive schedule requirements: 36 hr including 2 Local Nights, 2 local days twice per month and 60 consecutive hours after 4 or more early starts, late finishes or night duties.

- g. Rest periods are provided to enable crew members to overcome the effects of the previous duties and to be rested by the start of the following flight duty period. Minimum rest periods are compliant with EU regulations regardless of geographical location of the individual or the location of the rest period in the 24 hr day, notwithstanding that rest periods in Malta Air are invariably during the night period. Any worked day off or calls off standby are managed to comply with fatigue management guidelines indicated in the Crewing documentation and as indicated in (b) above. Rest periods are always compliant with EU Regulations. Malta Air endeavours to maximise rest periods allocated to crew members including rest periods in excess of EU FTL requirements.

Non-commercial flights will be subject to bespoke scheduling and rostering, will always be FTL compliant and fatigue management principles applicable to CAT operations equally apply to non-commercial operations. Refer OMA 8.7 for categories of non-commercial operations.

The bespoke management of each ad hoc VIP Charter flight will apply fatigue management principles to the scheduling, rostering, control and supervision of these flights. Where crews are called off standby or complete a working day-off (WOFF) or other non-rostered activity, fatigue management principles are applied either to ensure that the crewmember has adequate rest but never less than minimum rest or this duty is changed to achieve an adequate rest opportunity.

- h. The OFF periods (Rest Periods) in any roster assures compliance with Recurrent Extended Recovery Rest Periods requirements which are not specifically notified to crew members in the published rosters.

Roster stability metrics described earlier combined with the audit schedule, crew reports and disruptive schedule legality check in the Crew Management System will trap any rostering errors of extended recovery rest periods and enable corrective actions to be taken to ensure compliance.

- i. Schedule guidelines enable aircrew to have through the night rest and all flight duties are scheduled to be completed within the allowable flight duty period taking into account the time necessary for pre-flight duties, the sectors and turnaround times.
- j. Malta Air will change a schedule and/or crew arrangements if the actual operation exceeds the maximum flight duty period on more than 33% of the flight duties in that schedule during a seasonal period. Additionally, the

performance metrics for roster stability/disruption will indicate any roster duties which frequently are outliers and Malta Air will take action to mitigate these utilising established Commander's Discretion Seasonal Review Management of Change processes.

7.3.1 Scheduling and Rostering

AMC1 ORO.FTL.110

The Malta Air schedule is prepared under the direction of the Accountable Manager in consultation with the NPFO. The procedures documentation in each rostering area demonstrates how fatigue risks are identified and assessed, monitored and controlled.

The commercial schedule is presented to the rostering managers who task the rostering departments to roster the commercial schedule. Consideration of fatigue is a fundamental element of the Scheduling and Rostering processes.

Malta Air rosters deliver the schedule in compliance with the following:

- a. Scheduling has an important impact on a crew member's ability to sleep and to maintain a proper level of alertness. Malta Air strikes a balance between the commercial needs and the capacity of individual crew members to work effectively when developing a workable roster to enable completion of the schedule and these rosters are constructed to distribute the amount of work as evenly among those that are involved over the calendar year.

Schedule disruption is managed by Operations Control and Crew Control. Fatigue management considerations are applied within schedule completion. Duty assignments to achieve this will always comply with the IFTSS scheme.

- b. The schedule construction guidelines require flights to be completed within the maximum permitted flight duty periods and Malta Air crew rosters take into account the time needed for pre-flight duties, taxiing, the flight and turnaround times to include the following:
 1. Malta Air schedules and rosters allocate work patterns which avoid undesirable practices. Malta Air scheduling normally avoids alternating day/night duties, alternating eastward-westward or westward-eastward time zone transitions, positioning of crew members so that a serious disruption of established sleep/work patterns occurs;
 2. Sufficient rest periods are always scheduled. The Malta Air commercial schedule does not have long flights crossing many time zones; and

3. Duty rosters are prepared and published sufficiently in advance with planning of recurrent extended recovery rest periods and notification of to the crew members well in advance to plan adequate pre-duty rest.

The Malta Air area of operations spans 5 time zones, UTC-1 to UTC+3 (5 hour time difference).

Furthermore Malta Air does not normally roster alternating late-early-late duties in the same roster period. During periods of considerable disruption such assignments may arise and will be managed by crew control, including the application of fatigue management to the greatest extent possible. Any extension of the 'ON' days will be carefully considered and have fatigue management principles applied to assure that completion of such additional duty is safe, FTL complaint and manages fatigue to the greatest extent possible.

Scheduling of VIP charter operations is ad-hoc but is subject to the same rostering guidelines as any other flight and is operated under CAT rules. Effectively there is bespoke crew control and operations control applied to charter operations including fatigue management principles and FTL compliance.

Where applicable the operating crews on (VIP) charter flights shall be supplied with the Safety Risk Assessment (SRA) for such flights to alert them to the different fatigue management issues including crossing time zones, night duties etc arising from the different nature and characteristics of charter flights.

7.3.2

Operational Robustness of Rosters

AMC1 ORO.FTL.110(j)

Fatigue Risks (Fatigue Hazard Identification, 7.3) are identified in advance, are assessed and monitored on a continuous basis and controlled as part of fatigue management by adherence to the IFTSS, corrective actions from data from the occurrence reporting scheme, compliance audits and scientific analysis confirming robustness of published rosters.

The Company seasonal schedule does not have east-west and west-east transitions, nor are rest periods in excess of 4 hours from home base required. East-west and west-east transitions are considered as amongst the most fatiguing elements of a schedule. The Company has various roster combinations ranging from the least fatiguing 5 ON × 4 OFF roster to 5 ON × 2 OFF roster. All roster combinations are from time to time subjected to scientific analysis by an independent external organisation recognised as a leading specialist organisation in Fatigue Analysis worldwide.

Duties are spread as evenly as possible across rostered crewmembers. The rolling average of flight time for each crewmember for the previous 12 calendar months is monitored and rostering personnel increase or decrease a crewmembers duty assignments accordingly.

The following Operational Roster Robustness performance indicators are established, monitored, managed, and reported monthly to the NPFO:

1. Operating crew average Flight Hours (year-to-date, previous 12 months)
2. Operating crew average Duty Hours (year-to-date, previous 12 months)
3. Use of Commander's Discretion (number of events, percentage of flight duties)
 - Increasing Flight Duty Period (≤ 1 hour, > 1 hour)
 - Increasing Rest Period
 - Reducing Flight Duty Period
4. Use of Commander's Discretion during Seasons Schedule Period:
 - $> 25\%$ but $\leq 33\%$
 - $> 33\%$
5. Fatigue Reports Received – Target N/A
6. Fatigue Reports attributed to non-compliances with the approved FTL Scheme. Target 0%

Roster robustness and fatigue management performance indicators are monitored and managed through established triggers and targets. When an alert trigger is exceeded, the NPFO will direct a comprehensive review to establish root cause and progress appropriate corrective/preventative actions, if applicable, and document via the appropriate management system meeting/report (e.g DO and/or NPFO weekly).

1. **Average Flight and Duty Hours:** Limits are established in the crew management system to comply with FTLs, provide operational roster robustness, and mitigate fatigue. Crew Flight and Duty hours are monitored daily by Crew Control and reported weekly by the Head of Pilot Rostering and Inflight Crew Planning Manager to the DDFOPS and NPFO for the purposes of monitoring and managing crewing processes, FTLs, and Fatigue. The crew management system tracks hours for crewmembers, and where appropriate, rostered duties are increased or decreased to restore the crewmember flight/duty hours to ensure completion of the planned schedule within the approved FTL Scheme.

2. Commander's Discretion:

Scheduled Route Pairing

> 25% – Monitor, adjust schedule and/or crew arrangements if appropriate

> 33% – Adjustments to the schedule and/or crew arrangements required

3. Fatigue Reports:

Fatigue reports filed: Target N/A

Fatigue reports filed as a result of FTL noncompliance: Target 0%

The NPFO will present Fatigue Management performance at the Safety and Security Action Group and Safety and Security Review Board where negative performance trends will be addressed accordingly for follow-up action. SMM 3.13 sections refer.

The NPFO provides the TMCAD with an annual FTL report. This report presents the actual Flight Time and Duty Time for each crewmember for the previous calendar year

7.4 Crew Member Responsibilities

ORO.FTL.115

The EU regulations below are specifically applicable to crew members who shall:

- a. comply with all flight and duty time limitations (FTL) and rest requirements applicable to their activities; CAT.GEN.MPA.100(b)
- b. make optimum use of the opportunities and facilities for rest provided and plan and use their rest periods properly. Crew member selection of accommodation should establish that it is suitable and conducive to providing uninterrupted sleep opportunities to enable the crewmember to be properly rested prior to the next duty and also that fatigue can be managed to as low as reasonably practicable.
- c. when undertaking duties for more than one operator crew members shall:
 - have written prior permission from Malta Air
 - maintain his/her individual records regarding flight and duty times and rest periods in Malta Air and the other operator FTL schemes
 - provide Malta Air and the other operator with the data needed to schedule and record activities in accordance with the Malta Air and other operator FTL requirements. CAT.GEN.MPA.100(b)
- d. crew members are responsible for completion of rostered assignments.

- e. crew members are obliged to report fatigue events using the fatigue report selectable in the ASR section of the Safety Reporting System.

Malta Air will provide additional communication of fatigue management principles via internal e-platforms. There will be a recurring modular educational programme including Circadian Rhythm, Workload and Sleep Science in particular reflecting that sleep is the only antidote to fatigue. Modules will include advice on sleep strategies, awake strategies and other fatigue mitigation strategies.

The Malta Air Occurrence Reporting scheme is rigorously presented to aircrew, rosterers and controllers as an essential tool in the management of fatigue. The robust reporting culture in Malta Air is encouraged, with fatigue selectable on reporting forms with an additional alertness questionnaire to enable Malta Air and crew members to better understand fatigue, fatigue mitigation and provide continuous review of the occurrence reporting data set.

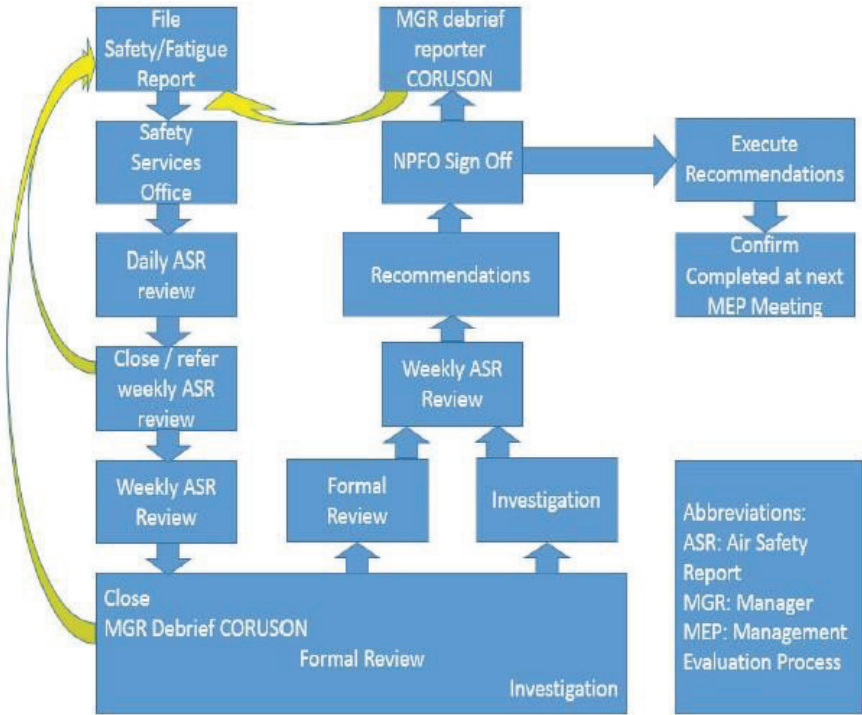
7.4.1 Fatigue Reporting Process

Malta Air provides all crew members with various reporting channels including ASRs, Captains Special Reports, Cabin Crew Safety Reports and Malta Air Confidential Reporting System.

Cabin Crew Fatigue Report – Should a Cabin Crew member call crewing with a fatigue claim, the crew member shall complete a Cabin Crew Fatigue Report as soon as possible on the company reporting system. The Cabin Crew Member should use their duty plan from the crewlink check in/out system to assist them in completing this form.

All 'fatigue' reports filed by cabin crew members are processed in the same manner as those filed by pilots. In the first instance the report is assessed at the Daily Ryanair DAC Safety and Fatigue Management Review Meeting. The cabin crew member who files the report will receive progress updates through the Coruson System.

All fatigue reports are forwarded to the TM CAD.



7.5 Fatigue Risk Management (FRM)

ORO.FTL.120

Fatigue is a risk to the safe conduct of Air Transport activities. All risks in Malta Air are managed by the Safety Management System which Malta Air has established, implements and maintains. For this reason there is not a stand-alone Malta Air Fatigue Risk Management System (FRMS), and the provisions of ORO.FTL.120, its associated AMC and GM are not applicable in this Manual.

7.6 Individual Flight Time Specification Scheme (IFTSS)

ORO.FTL.125

Malta Air has established, implemented and maintains this IFTSS, which is appropriate to its operation and will comply with EU Regulations. This IFTSS is approved by TM CAD and complies with EASA certification specifications. Malta Air currently has no deviations or derogations from EU regulations. Malta Air collects and analyses the data available in its IFTSS to continually review any effects on Fatigue. This process is described in greater depth in the SMM.

7.7 **Malta Air Commercial Air Transport Operations (CAT): Applicability**

CS.FTL.1.100

The EASA Certification Specifications are applicable to Malta Air as a Commercial Air Transport Operator for scheduled and charter operations.

7.8 **Home Base**

ORO.FTL.200

Malta Air assigns each crew member a home base.

CS FTL.1.200

1. The home base is a single airport location assigned with a high degree of permanence.
2. In the case of a change of home base, the first recurrent extended recovery rest period prior to starting duty at the new home base is increased to 72 hours, including 3 local nights. Travelling time between the former home base and the new home base is positioning.
3. Malta Air maintains a record of each home base as assigned to each crew member.

Crew members are required to make arrangements for temporary accommodation closer to their home base if the travelling time from their residence to the reporting point in the home base usually exceeds 60 minutes. CS FTL.1.200.

High Degree of Permanence:

The reality of the Aviation Business dictates that bases open, close, expand and contract in response to commercial necessities. Moreover the seasonality of our business may require varying the crewing levels in a Base for the duration of a particular season. All of the above must be managed within the confines of individual contractual agreements in place between Malta Air and its crew members.

Crew members may request a change of home base and Malta Air has no limit on the number of requests allowed during a crewmember's employment or service provision. Malta Air recognises that there are many reasons why a crew member would request a change of home base.

In the normal course base assignments are planned to be permanent until the crew member is assigned to another home base or the commercial/operational or contractual provisions alter this. In the normal course roster assignments are conducted in home Base, however Malta Air reserves the right to transfer or position a crew member to other locations for operational requirements.

Note: Home base: Travelling time

Crew members are required to make arrangements for temporary accommodation closer to their home base if the travelling time from their residence to the reporting point in the home base usually exceeds **60 minutes**. CS FTL.1.200

7.9 Flight Duty Period (FDP)

ORO.FTL.205

a. Flight Duty Period:

ORO.FTL.205 (a)

1. Reporting Times.

Malta Air has established the following reporting times that have been demonstrated over many years to include sufficient time for ground duties, pre-flight preparation, commute to aircraft and pre-departure activities. ORO.FTL.110(c).

The following reporting times also take into account the scheduled nature of the operation, occasional charter operations, the size and type of aircraft and the reporting airport conditions. ORO.FTL.205.

Report Time	
Home Base operations	45 mins
Out of base operations	45 mins

Malta Air continuously reviews pre-flight tasks which must be completed by both flight and cabin crew, including any changes arising from regulatory change or change of company policy. All Bases are reviewed to ensure that from reporting at the reporting point briefings, provision of documentation, commute to aircraft parking and completion of on-board pre departure activities are achievable within specified report times. Crew members shall report recurring instances which may indicate that reporting time is insufficient. The NPFO reviews reporting times and may shorten or lengthen.

2. Commanders Discretion procedure: refer to 7.9 (f).

b. Basic Maximum Daily FDP

ORO.FTL.205(b)

1. The maximum daily FDP (without the use of extensions) is in accordance with the following table for acclimatised crew members:

Table 1 Maximum Daily FDP – Acclimatised Crew Members

Start of FDP at reference time	1–2 sectors	3 sectors	4 sectors	5 sectors	6 sectors	7 sectors	8 sectors	9 sectors	10 sectors
0600–1329	13:00	12:30	12:00	11:30	11:00	10:30	10:00	09:30	09:00
1330–1359	12:45	12:15	11:45	11:15	10:45	10:15	09:45	09:15	09:00
1400–1429	12:30	12:00	11:30	11:00	10:30	10:00	09:30	09:00	09:00
1430–1459	12:15	11:45	11:15	10:45	10:15	09:45	09:15	09:00	09:00
1500–1529	12:00	11:30	11:00	10:30	10:00	09:30	09:00	09:00	09:00
1530–1559	11:45	11:15	10:45	10:15	09:45	09:15	09:00	09:00	09:00
1600–1629	11:30	11:00	10:30	10:00	09:30	09:00	09:00	09:00	09:00
1630–1659	11:15	10:45	10:15	09:45	09:15	09:00	09:00	09:00	09:00
1700–0459	11:00	10:30	10:00	09:30	09:00	09:00	09:00	09:00	09:00
0500–0514	12:00	11:30	11:00	10:30	10:00	09:30	09:00	09:00	09:00
0515–0529	12:15	11:45	11:15	10:45	10:15	09:45	09:15	09:00	09:00
0530–0544	12:30	12:00	11:30	11:00	10:30	10:00	09:30	09:00	09:00
0545–0559	12:45	12:15	11:45	11:15	10:45	10:15	09:45	09:15	09:00

Note: Flight Duty Period (FDP) Reference time:

The start time of the FDP in the table refers to the ‘reference time’. That means, to the local time of the point of departure, if this point of departure is within a 2-hour wide time zone band around the local time where a crew member is acclimatised. ORO.FTL.205.

2. The Malta Air commercial schedule would not normally result in a crew member being in an unknown state of acclimatisation. This can occur on delivery flights or ad hoc VIP charter flights and the table below addresses the max FDP when a crew member is in an unknown state of acclimatisation.

Table 2 Crew Members in an Unknown State of Acclimatisation

Maximum Daily FDP According to Sectors						
1–2	3	4	5	6	7	8
11:00	10:30	10:00	09:30	09:00	09:00	09:00

c. FDP with different reporting times for flight crew and cabin crew.

ORO.FTL.205 (c)

Malta Air flight and cabin crew have the same reporting times.

d. Max daily FDP for acclimatised crew members with the use of extensions without in-flight rest

ORO.FTL.205 (d)

1. The maximum daily FDP may be extended by up to 1 hour, not more than twice in any 7 consecutive days. In this case the minimum pre-flight and post-flight rest periods are increased by 2 hours or the post flight rest period is increased by 4 hours.
2. When extensions are used for consecutive FDPs the additional pre- and post-flight rest between the two extended FDPs required above shall be provided consecutively.
3. The use of the extension is planned in advance and is limited to a maximum of:
 - i. Five (5) sectors when the WOCL is not encroached, or
 - ii. Four (4) sectors when the WOCL is encroached by 2 hours or less, or
 - iii. Two (2) sectors, when the WOCL is encroached by more than 2 hours.
4. Extension of the maximum basic daily FDP without in-flight rest cannot be combined with extensions due to split duty in the same duty period.
5. The limits for extensions of the maximum basic daily FDP in accordance with the certification specifications applicable in this chapter, taking into account the number of sectors flown and WOCL encroachment are as detailed in 7.10 (c): CS FTL.1.205(b)

Table 3

Starting time of FDP	1–2 sectors (in hours)	3 sectors (in hours)	4 sectors (in hours)	5 sectors (in hours)
0600–0614	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0615–0629	13:15	12:45	12:15	11:45
0630–0644	13:30	13:00	12:30	12:00
0645–0659	13:45	13:15	12:45	12:15
0700–1329	14:00	13:30	13:00	12:30
1330–1359	13:45	13:15	12:45	Not Allowed
1400–1429	13:30	13:00	12:30	Not Allowed
1430–1459	13:15	12:45	12:15	Not Allowed
1500–1529	13:00	12:30	12:00	Not Allowed
1530–1559	12:45	Not Allowed	Not Allowed	Not Allowed
1600–1629	12:30	Not Allowed	Not Allowed	Not Allowed
1630–1659	12:15	Not Allowed	Not Allowed	Not Allowed
1700–1729	12:00	Not Allowed	Not Allowed	Not Allowed
1730–1759	11:45	Not Allowed	Not Allowed	Not Allowed
1800–1829	11:30	Not Allowed	Not Allowed	Not Allowed
1830–1859	11:15	Not Allowed	Not Allowed	Not Allowed
1900–0359	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0400–0414	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0415–0429	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0430–0444	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0445–0459	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0500–0514	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0515–0529	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0530–0544	Not Allowed	Not Allowed	Not Allowed	Not Allowed
0545–0559	Not Allowed	Not Allowed	Not Allowed	Not Allowed

Rostered extensions do not require submission of an eCDR when such assignments are completed in the max FDP using planned extensions.

e. In-flight rest: N/A to Malta Air

ORO.FTL.205(e)

f. Unforeseen Circumstances in Flight Operations – Commander’s Discretion.

ORO.FTL.205 (f)

It is Malta Air policy that the commander shall, in case of special circumstances which could lead to severe fatigue, and after consultation with the crew members concerned on their alertness levels, reduce the actual FDP and/or increase the rest period in order to eliminate any detrimental effect on flight safety.

1. **The Commander shall communicate his decision to Malta Air Operations enabling Operations to complete the alternative arrangements required.**
2. **The Commander shall file an ASR (Fatigue & ECDR) indicating the unforeseen circumstances leading to his decisions.**
3. **The Commander shall inform Malta Air Operations, currently subcontracted to Ryanair DAC, when the duty may be continued on the basis that the levels of alertness are acceptable to complete the duty unless the affected crew member(s) are stood down.**
4. **Crew members affected shall have sufficient rest after this duty to assure acceptable levels of alertness to complete the remaining roster period and sufficient OFF days (RERRP) thereafter to increase alertness levels to as high as reasonably practicable prior to the next duty period.**
5. **Ryanair DAC Operations shall make the necessary arrangements to change crew or if this is not practicable enable the levels of alertness to be increased to acceptable levels prior to any subsequent operations by the crew members affected and thereafter to achieve levels of alertness as high as reasonably practicable prior to the next rostered duty period.**

Commander’s discretion developed in the industry as a service to passengers to enable flights to be completed safely in unforeseen circumstances. The exercise or non-exercise of commander’s discretion is now a regulated procedure with rules determining how it may be used safely, including fatigue management, and to prevent any misuse.

Only the Commander shall make the decision to exercise Commander’s discretion. Malta Air accepts its shared responsibility in the safe exercise, or non-exercise, of commanders discretion with Flight crew and cabin crew. The exercise of Commander’s discretion is considered exceptional and shall be avoided at home base where stand-by or airport duty crewmembers are available.

Malta Air regularly reviews every series of pairings where Commander’s discretion has been exercised in order to be aware of possible inconsistencies in scheduling or rostering. Malta Air has developed and

implemented an eCDR application to host this function electronically. The reports generated from this application are made available internally including to the SAG Meeting and to the TM CAD.

The primary safety objective for Malta Air and the Commander is that the exercise of Commander's discretion is safe, having considered the factors below and the potential to avoid further disruption to our passengers. Exercise of Commander's discretion only arises in unforeseen circumstances occurring after reporting, especially in the case of an extended FDP or reduced rest, and the Commander shall take due consideration of additional factors that might decrease a crew member's alertness levels, such as: AMC1 ORO.FTL.205(f)

- WOCL encroachment;
- weather conditions;
- complexity of the operation and/or airport environment;
- aeroplane malfunctions or specifications;
- flight with training or supervisory duties;
- increased number of sectors;
- circadian disruption; and
- individual conditions of affected crew members (time since awake, sleep-related factor, workload, etc).

Rules for Exercise of Commander's Discretion

1. The conditions to modify the limits on flight duty, duty and rest periods by the Commander in the case of unforeseen circumstances in flight operations which start at or after the reporting are as follows:
 - i. the maximum daily FDP which results after applying Table 1 of this chapter or Split Duty may not be increased by more than **2 hours**.
 - ii. if on the final sector within an FDP the allowed increase is exceeded because of unforeseen circumstances after take-off, the flight may continue to the planned destination or alternate aerodrome, and
 - iii. the rest period following the FDP may be reduced but can never be less than 10 hours.
2. In case of unforeseen circumstances which could lead to severe fatigue, the Commander shall reduce the actual FDP and/or increase the rest period in order to eliminate any detrimental effect on flight safety.

3. The Commander shall consult all crew members on their alertness levels before deciding the modifications in subparagraphs 1 and 2 above.

As part of his shared responsibility and prior to the use of Commander's Discretion the Commander as a minimum shall establish:

1. If any crew member has a report time that would cause the planned use of discretion to exceed any legal limits.
2. If any crew member on that day has operated more sectors than the Captain.
3. If any crew member reported for duty with reduced rest (i.e. less than 12 hrs since the end of their last duty).
4. If any crew members are working a sixth consecutive day.
5. If there are there any other safety concerns to be considered prior to the decision to depart.

If the answer to all of the above questions is NO the alertness levels of the crew would indicate that the flight would dispatch. If the answer to any of the questions is YES the Commander should reassess what reduction in alertness, if any, is attributable to the use of Commander's discretion and what alternative measures are available.

4. The Commander shall submit a Commander's Discretion Report in accordance with established Malta Air procedures when an FDP is increased or a rest period is reduced at his or her discretion.
5. Where the increase of an FDP or reduction of a rest period exceeds 1 hour, a copy of the Commander's report, with Malta Air comments added, will be sent by Malta Air to the TM CAD, not later than 28 days after the event.
6. It is Malta Air policy that the Commander's discretion process is non-punitive when exercised in accordance with this section of the Operations Manual and the provisions of 'Just Culture' as described in OMA CH 11 and SMM, apply.
7. When the commander has elected to use Commanders Discretion and it is expected to exceed 1 hr and 30 mins the commander must confirm the latest airborne time with OCC to achieve arrival at the parking position and both engines shut down at the destination within the maximum cd available. Crew control will confirm that the use of this CD will not cause any crew member to exceed FT/FDT/

FTL.210 hard limits as found in Section 7.11. If the aircraft has not commenced the take-off roll by the latest airborne time the flight shall not depart.

Note: Flight Duty Period (FDP) Commander's Discretion

ORO.FTL.205

The Max Daily FDP for a duty planned with extensions is indicated in Table 3.

For a two pilot extended FDP operation, the use of commander's discretion is always based on the maximum daily FDP table ORO.FTL.205 (b) (1). (Table 1).

If Commanders discretion will be considered on a duty planned with extensions, the max discretion is 2 hours but added to the Max Daily FDP as calculated from Table 1 (not Table 3) for the same duty.

g. Unforeseen Circumstances in Flight Operations – Delayed Reporting

ORO.FTL.205(g)

Delayed reporting is used to notify a crewmember that the report time is delayed thereby avoiding a crewmember leaving their accommodation unnecessarily early. The restrictions below limit a delayed report being extended more than twice.

The delayed reporting procedures in the event of unforeseen circumstances are as follows: CS FTL.1. 205(d).

The maximum notification time for delayed reporting is 90 minutes prior to the report time.

The minimum notification time for delayed reporting is 60 minutes prior to the report time.

- The methods of contacting crew include the crewmember primary personal phone data in Crewlink, and/or email to EFB or personal email. Regardless of the communication used the crew member must confirm acknowledgement of the delayed report with Crewing including when electronic communication is used.

Communicating with crew during a rest period is as follows:

Electronic communications.

Notification can be sent at any time to a crew member:

- IDP*
- Email

- SMS

Note: *In situations of delayed reporting and in order to maximise the opportunity for further rest, crewmembers should check their IDP before leaving home or suitable accommodation when away from home base.

Phone communications

Malta Air protects an 8 hr sleep opportunity during rest periods by restricting phone contact to the first or last 90 minutes of a planned rest period. Malta Air can contact a crewmember within 90 mins of the end of previous Duty Period as the crewmember is assumed to be awake. Malta Air can contact a crewmember in the 90 mins prior to the report time of the next FDP as the crewmember is assumed to be awake.

Additionally, for crew operating night duties (any portion of the duty 02.00–04.59), it is not permitted to call a crewmember before 90 minutes before report time. This restriction does not prevent Crew Control using IDP communications as a method of advising roster change and assigning duties acknowledged through the IDP. (Ref OMA CH 7.1.1. The operating crew and operations control should plan a sleep opportunity as close as practicable to report time and Malta Air will not contact crewmembers during this time.)

All instances of delayed reporting procedures used are documented by Malta Air.

The FDP in the case of delayed reporting is calculated as follows:

CS FTL.1.205(d)

- one notification of delayed reporting results in the calculation of the MAX FDP according to (iii) or (iv) below
- if Malta Air amends the required reporting time for a second time, the FDP starts **1 hour** after the second notification or at the first delayed reporting time, if this is earlier.
- when the delay is less than **4 hours**, the MAX FDP is calculated based on the original reporting time and the FDP starts counting at the delayed reporting time
- when the delay is **4 hours** or more, the MAX FDP is calculated according to the more limiting of the original or delayed reporting time, but the FDP starts at the delayed reporting time
- As an exception to (i) and (ii) above, if Malta Air informs crew members of a delay of **10 hours** or more in reporting time and Malta Air does not further disturb the crew member, such delay of 10 hrs or more counts as a rest period.

7.10 Night Duties, Late Finishes and Extensions: Certification Specifications Applicable to Flight Duty Period (FDP)

CS FTL.1.205

a. Night Duties.

Malta Air does not normally roster night duties other than standby commencing before 05:00 local time. Charter operations or non-commercial flights as described in 8.7 may include night duties. Night duties and late finish duties under the provisions of ORO.FTL.205(b) and (d) comply with the following:

1. When establishing the maximum FDP for consecutive night duties, the number of sectors is limited to 4 sectors per duty.
2. Malta Air applies appropriate fatigue risk management (OMA 7.3, 7.4) to actively manage the fatiguing effect of night duties and late finish duties in relation to the surrounding duties and rest periods.
3. When planning and implementing appropriate FRM (OMA 7.3, 7.4) measures to reduce fatigue during night duties, Malta Air distinguishes between the following subtypes of night duties and ranks them based on the probability of occurrence of high levels of fatigue at Top of Descent (TOD):
 - i. FDPs with a start time between 02:00 and 04:59;
 - ii. FDPs with an end time between 02:00 and 05:59 and a start time at 01:59 or earlier; and
 - iii. FDPs with an end time at 06:00 or later and a start time at 01:59 or earlier.

Night Duties and Late Finish Duties – Appropriate Fatigue Risk Management (OMA 7.3, 7.4)

In the case of late finish duties (23:00-01:59), this is assigned as part of the normal Malta Air roster for scheduled operations and will be managed in accordance with the Malta Air fatigue management process outlined in OMA 7.3.

In the case of night duties described in 7.10 point 3, these subtypes are not normally assigned as part of the normal Malta Air roster for scheduled operations. If operational disruption results in a FDP infringing in these start or end times, then this will be managed in accordance with the Malta Air fatigue management process outlined in OMA 7.3.

In the case of standby commencing before 05:00 and flight duties arising during such standby periods, this is assigned as part of normal Malta Air roster for scheduled operations and will be managed in accordance with the Malta Air fatigue management process outlined in 7.3. Early standby periods are normally

assigned from 04:00 local time onwards and the fatigue management considerations are similar to that of a rostered flight duty that has a report time of 05:00–06:00.

To optimise alertness on night duties, both Malta Air and the operating aircrew are aware that sleep opportunities prior to reporting are desirable particularly after a period of wakefulness after a rest period.

- a. In the case of Ad Hoc flights that constitute night duties, these will always be subject bespoke rostering and fatigue assessment as part of the Malta Air management system. For flights under 8.7 for which the NPFO has directed an SRA, fatigue management shall form part of this assessment.

Ad hoc night duties shall comply with the following:

- The Ryanair DAC OCC or Rostering department (as appropriate), in coordination with the NPFO or his nominee, will conduct an assessment to ensure that an equivalent level of safety and fatigue management is achieved and maintained for the planned duty; and
- The operating crew and operations control should plan a sleep opportunity as close as practicable to report time and Malta Air or the Ryanair DAC Operations Control Centre will not contact crewmembers during this time. This is one strategy whereby practices leading to extended wakefulness before reporting for duties are avoided.

- b. **Night Duties In-Flight Rest: N/A Malta Air**

CS FTL.1.205

- c. **Extension of FDP without In-flight rest. Refer to 7.9 (d)**

CS FTL.1.205(b)

- d. **Extension of FDP due in-flight rest N/A Malta Air**

CS FTL.1.205(c)

7.11 Flight Times & Duty Periods

ORO.FTL.210

- a. The total **duty periods** to which Malta Air assigns a crew member does not exceed:
 1. **60 duty hours in any 7 consecutive days;**
 2. **110 duty hours in any 14 consecutive days; and**
 3. **190 duty hours in any 28 consecutive days**, spread as evenly as practicable throughout that period.

- b. The total **flight time** of the sectors on which an individual crew member is assigned as an operating crew member does not exceed:
 - 1. **100 hours** of flight time in any **28 consecutive days**;
 - 2. **900 hours** of flight time in **any calendar year**; and
 - 3. **1,000 hours** of flight time in any **12 consecutive calendar months**.
- c. Post-flight duty counts as duty period, but not as FDP.

The allocated period for post-flight duties is **30 minutes**. AMC1 ORO.FTL.210(c)

Malta Air has assigned 30 mins post flight duty time which has proven resilient considering our aircraft size, commuting to crewrooms (where applicable), airport conditions (eg Shengen) and electronic completion of Electronic Flight Log/Paper Voyage Report/Journey Log data.

When any crew member has post-flight duties in excess of 30 mins the end of duty time shall be communicated to crew control and this time is entered in the Crew Management System and is used to calculate the subsequent rest period. The reason for the delay is noted on the Electronic Flight Log/Paper Voyage Report/Journey Log e.g. 'awaiting SCP lift-off causing delay'.

7.12 Positioning

ORO.FTL.215

“Positioning”

Positioning means the transferring of a non-operating crew member from one place to another, at the behest of Malta Air, excluding:

- the time of travel from a private place of rest to the designated reporting place at home base and vice versa; and
- the time for local transfer from a place of rest to the commencement of duty and vice versa.

Each crewmember must know when to start the FDP/DP calculation. Each crewmember should always know the current MAX FDP (OMA CH 07 Table 1), including as circumstances change eg extra sector. Each crewmember should also calculate the length of the associated rest period when the current duty is completed. Unless the entire crew positions together there will be different start times for FDP/DP. The Commander must verify the start time of the FDP/DP for each crewmember to assure correct recording of FDP/DP, calculation of max FDP and subsequent rest periods. The Commander is responsible for actual times in EFL.

When positioning of crew is planned in the roster period it shall be indicated on the roster. If positioning is required in the crewing period the Crew Management System record will be amended accordingly. Positioning may be by any modes of transportation, including own transport.

The time required to commute from home base residence to home base is not DP, and the time to commute post DP from home base to home base residence is not DP.

7.12.1 Positioning Table

Rostering/Crew Control will assign 'Ground Transport' (GT) in Crew Management System in compliance with the positioning table below. The reporting time starts as indicated in the Reporting Time Outbound column in the table depending on the mode of transport (including flights) and is either duty time (positioning followed by rest period) or flight duty time (positioning followed by sector(s). Positioning is not a sector. The Reporting Time Return column indicates the time the duty ends.

Mode of Transport	Reporting point	Reporting Time Outbound: Rostered/ Notified	End Time of Duty Period
RYR Group Airline	Departure Gate	STD – 20	STA + 20 or actual time at terminal exit
Other airline	Terminal Entrance	STD – 1 hr	STA + 20 or actual time at terminal exit
Taxi	Pick up point	Rostered Pick up time	Drop off time as rostered or actual time if later
Own Car	Home	Rostered time of commencement of GT	Rostered or actual time (if later) of completion of GT
Car Rental	Car rental desk	Rostered time of commencement of GT	Rostered or actual time (if later) of completion of GT



The crewmember must communicate the time of the end of the FDP/DP with Operations Control/Crew Control to ensure correct DT/FDT records and rest period calculations unless these changes are already recorded in the duty plan (EFB).

7.12.2 ‘Long Haul’ Positioning

When positioning on flights outside the Malta Air Area of Operations (e.g. to Seattle) the reporting time is STD-60 or the time notified to Crew Control by the crewmember concerned. The DP starts at STD-60 or the time notified to Crew Control by the crewmember concerned and finishes at STA + 30 minutes or

ATA+30 if later. This accounts for longer time at the departure gate (larger aircraft) and customs/immigration clearance before/after arrival at your destination.

7.12.3 Calculation of Rest Period

When a duty is longer than planned, the actual end time of the duty period and the end time of the rest period should be communicated and agreed with Crew Control, unless these are already recorded on the duty plan on the EFB.

7.12.4 Travelling Other than as Planned by Malta Air

All positioning is planned in accordance with the IFTSS requirements. Crewmembers are expected to travel on the positioning flights or other modes of transport as planned by Malta Air. Where a crewmember, in his/her OFF time, chooses to travel by any means of transportation to or from their operating location (MAY flight, other airline, train, boat etc.) this activity is not recorded or tracked in Crew Management System.

Crewmembers are required to report for duty fit to fly and on time.

7.13 Split Duty

ORO.FTL.220, CS FTL.1.220

A Split duty is a regulated process to extend the basic maximum daily FDP due to a break on the ground. The period spent on the ground is Flight Duty period (FDP). Split duty cannot follow reduced rest. Additionally, any increase of limits on flight duty using split duty complies with the following.

- a. The break on the ground has a minimum duration of **3 consecutive hours**
- b. The break **excludes** the time allowed for post and pre-flight duties and travelling. The minimum total time for post and pre-flight duties and travelling is 30 minutes.
- c. The maximum FDP specified in Table 1 of this chapter may be increased by up to **50%** of the break.
- d. Suitable accommodation is provided either for a break of **6 hours** or more or for a break that encroaches the window of circadian low (WOCL).
- e. In all other cases
 1. accommodation is provided; and
 2. any time of the actual break exceeding 6 hours or any time of the break that encroaches the WOCL does not count for the extension of the FDP

7.13.1 Split Duty: Post, Pre-Flight Duty and Travelling Times

CS FTL.1.220

Where a split duty is required, OCC will book hotel accommodation. The commander will agree the time of the start of the break on the ground with Operations. The start of the break will be 30 mins after on blocks or as notified by the Captain.

After the break on the ground, 45 minutes report time is allocated after reporting at the designated reporting point to enable completion of the standard pre-flight preparations. The commander will agree the ‘on duty’ time after the break on the ground with Operations.

The minimum travelling time to the location of the break on the ground is 10 minutes. The minimum travelling time after the break on the ground is 15 minutes. When actual travelling times are greater, the actual times will be used as the DP/FDP record. These times are influenced by size of airport including security procedures, weather conditions, time of day, etc. The actual times must be communicated to Operations by the Commander to enable correct calculation of the max FDP.

7.14 Airport Standby, Home Standby

ORO.FTL.205(b), (d), 225, 235, CS FTL.1.225

Airport Standby (where facilities permit)	Home Standby
Rostered, with start and end times defined, ideally 14 days in advance.	Rostered, with start and end times defined, ideally 14 days in advance.
Duty period commences at time of reporting at the reporting point.	Home standby commences at rostered start time.
Counts in full as duty period if no FDP assigned.	25% of time on home standby counts as duty time if no FDP assigned.
Max DP is 12 hours if no FDP assigned.	Max duration of home standby is 16 hours.
If a FDP is assigned during airport standby the FDP starts at the start of the assigned FDP minus the time on standby in excess of 4 hours.	Home standby ceases when the crew member reports at the reporting point.
The maximum combined duration of airport standby and FDP is 16 hours.	If home standby ceases within the first 6 hours the Max FDP counts from reporting. (1)

Airport Standby (where facilities permit)	Home Standby
N/A	If home standby ceases after the first 6 hours the Max FDP is reduced by the amount of home standby time exceeding 6 hours. (2)
	If the FDP is extended due to Split-Duty the 6 hours above is extended to 8 hours. (3)
	If the home standby starts between 2300 and 0700 the time between 2300 and 0700 does NOT count towards the reduction of the (1), (2) and (3) FDP calculations above until the crew member is contacted by Malta Air.
If no FDP is assigned, minimum rest of 12 hours in home base applies after airport standby ceases.	If no FDP is assigned, minimum rest of 12 hours in home base applies after home standby ceases. If away from home base and in suitable accommodation minimum rest of 10 hours applies after home standby ceases.
If a FDP is assigned, the rest period required is the greater of the previous duty period or 12 hours in home base or the greater of the previous duty period or 10 hours away from home base.	N/A
If airport standby initially assigned is reduced by Malta Air during a standby that does not lead to a FDP being assigned minimum rest in home base of 12 hours applies.	If Home standby initially assigned is reduced by Malta Air during a standby that does not lead to a FDP being assigned minimum rest in home base of 12 hours applies.
N/A	If minimum rest in home base (or away from base if this is the case) is provided before reporting for the duty assigned during the standby, this time period does not count as standby.
	If a crew member receives an assignment during home standby the actual reporting time at the designated reporting point is used for the purpose of rest periods.
	It is a Crew control procedure that notification of assigned duties during home standby avoids interference with sleep patterns of home standby crew where possible.
Confirm no awake time of more than 18 hours.	Confirm the combination of home standby and FDP does not lead to more than 18 hours awake time***.
Accommodation provided.	Own accommodation provided.
N/A	Report at reporting point within 60 mins of notification.

*** 18 hours awake time: the following assumptions apply.

Start time of SBY	Mandatory end time of combined SBY/FDP
2300–0700 Local	18 Hours from 0700 <u>OR</u> Time of callout + 18 hrs if earlier
0700–2000 Local	18 Hours from 1000 <u>OR</u> Time of callout + 18 hrs if earlier
Night SBY (2000–0600)	18 Hours from 1900

Airport Duty

Airport Duty means a duty carried out at the airport where a crew member is carrying out tasks for Malta Air during the airport duty, and is available, should the need arise, for an immediate flight duty assignment. Crew are assigned airport duties in order to assist a base manager or a nominated deputy on local supporting activities.

Rostering

Start and end times defined, ideally 14 days in advance.

Duty Period

Commences at time of reporting at the reporting point. Counts in full as duty period if no FDP assigned.

Max Duty Period

12 hours if no FDP assigned.

Flight Duty Period

If assigned during airport duty the FDP starts at the time of commencement of the Duty Period.

Max Combined Airport Duty Period and Flight Duty Period

The maximum (post initial report time), that a combined Airport Duty Period and Flight Duty Period can assigned is calculated from Table 1 starting at the time of reporting at the initial reporting point.

Post Airport Duty Min Rest (no FDP)**

12 hours in home base applies after airport duty ceases.

Post Airport Duty Min Rest (with FDP)

The rest period required is the greater of the previous duty period or 12 hours in home base or the greater of the previous duty period or 10 hours away from home base.

Accommodation

Not Provided hence no awake time of more than 18 hours.

**If airport duty initially assigned is reduced by Malta Air during an airport duty that does not lead to an FDP being assigned minimum rest in home base of 12 hours applies.

The Max FDP may be impacted by the time spent on airport standby depending on at least the following: ORO.FTL.225 (f)

- rest facilities provided,
- need for immediate readiness,
- interference of airport standby with sleep,
- sufficient notification to protect a sleep opportunity between the call for duty and the assigned FDP.

If any of the above applies the crew member shall notify the commander and crew control. The max FDP may be reduced and/or a subsequent rest period may be increased by the Commander (discretion) or crew control to achieve a satisfactory level of fatigue for the assigned FDP or subsequent duty assignments.

Crew members may be rostered for either standby at the airport (**airport standby**) or other duties at the airport (**airport duty**) or standby at home or in suitable accommodation (**Home Standby**) and the procedures indicated in the table above apply. The term 'minimum rest' or 'min rest' as used in the table includes all the provisions of 7.16, Rest Periods. DP is duty period. FDP is flight duty period. During periods of disruption crew members may be assigned any of the standby duties in this section by Malta Air with sufficient notification to provide an opportunity to plan adequate rest. Crew members are considered to be on standby from the rostered start time to end time unless a FDP is assigned or Malta Air removes the crew member from standby.

Following a FDP If the subsequent duty period is home SBY the crewmember is contactable, including by phone, at the start of standby. If, as the result of a later than planned finish the revised Rest Period extends past the start time of the next rostered home SBY, the time at the end of the Rest Period becomes the start time of the home SBY. It is not necessary for the crew member to contact crew control to notify the new stby start time.

7.14.1 Special Duty (SD)

Special Duty means a period of duty other than a flying or Standby duty.

The standard crew management system duty time entry is 7 hours for each completed SD day.

The duty hours entered in the Crew Management System shall consist of the start time to the finish time.

7.15 Reserve

ORO.FTL.230, CS FTL.1.230

A crew member can be rostered for reserve duties, which means that the crew member is available to receive an assignment for a duty which could be a Flight Duty, Positioning or other Duty. Any such assignment is notified at least 10 hours in advance.

When Malta Air assigns crew members to reserve, the following requirements apply:

- Reserve assignments are rostered.
- The maximum period of a rostered reserve duty is 16 hours to protect an 8-hour sleep opportunity, the operator rosters a period of 8 hours, taking into account fatigue management principles, for each reserve day during which a crew member on reserve is not contacted by Malta Air. CS FTL.1.230(d)
- Not more than 7 consecutive reserve days may be assigned to a crew member in compliance with recurrent extended rest periods requirement. CS FTL.1.230(c)
- An assigned FDP counts from the reporting time.
- Reserve times do not count as duty period for the purpose of 7.11, Flight Times & Duty Periods and 7.16, Rest Periods.

Additional Requirements Applicable to Reserve:

ORO.FTL.230, CS FTL.1.230, 1.230

A reserve period that does not result in a duty period may not **retrospectively** be considered as part of a recurrent extended recovery rest period. The procedures for the notification of assigned duties during reserve avoid interference with

sleeping patterns if possible. The 10 hour notification of an assignment for any duty and reporting for that duty during reserve may include the 8 hour sleep opportunity during which a crew member on reserve is not contacted by Malta Air. Recurrent extended recovery rest period (7.16) applies to a crew member on reserve, i.e. there must be a recurrent extended recovery rest period after a maximum of 168 hrs (7 days).

7.16 Rest Periods

ORO.FTL.235

a. Minimum Rest Period at HOME Base

1. The minimum rest period provided before undertaking an FDP starting at home base is at least as long as the preceding duty period, or 12 hours, whichever is greater.
2. If Malta Air provides hotel accommodation to a crew member in home base this 12 hour period may be reduced to 10 hrs. This arrangement might arise during widespread inclement weather for example. Malta Air may apply the minimum rest period away from home base during a rotation which includes a rest period at a crew member's home base. This applies only if the crew member does not rest at his/her residence, or temporary accommodation, because Malta Air provides suitable accommodation. This type of roster is known as "back-to-back operation".
3. Malta Air policy is not to disturb crewmembers during a rest period. However, circumstances may arise whereby it is necessary to call crewmembers during a rest period such as a security event, or a health and safety event etc where there may be a risk to personnel or Company assets, e.g. a volcanic event whereby we may need to evacuate personnel and relocate assets.

Additionally as a last resort, and without prejudice to an 8 hour sleep opportunity, when a crewmember within 90 mins of report time has not responded to a duty change notified electronically, Malta Air may call crewmembers within 90 mins of a report time when a crewmember is considered to be awake preparing to report. A crew member is not obliged to answer their phone in this case, or even to have their phone available during a rest period.

b. Minimum Rest Period Away from HOME base

The minimum rest period provided before undertaking an FDP starting away from home base is at least as long as the preceding duty period, or 10 hours, whichever is greater. This period includes an 8-hour sleep opportunity in addition to the time for travelling and physiological needs.

The time allowed for physiological needs is 1 hour. Consequently, if the travelling time to the suitable accommodation is more than 30 minutes, Malta Air will increase the rest period by twice the amount of difference of travelling time above 30 minutes. AMC1 ORO.FTL.235(b)

c. Reduced Rest Derogation (N/A MAY, no FRMS)

CS FTL.1.235

d. Extended Recovery Rest Periods

Recurrent extended recovery rest periods (RERRP) compensate for cumulative fatigue. The minimum recurrent extended recovery rest period is **36 hours**, including **2 local nights** and the time between the end of one recurrent extended recovery rest period and the start of the next extended recovery rest period is not be more than **168 hours**. The recurrent extended recovery rest period is increased to **2 local days** twice every month.

If a crewmember has 4 or more early starts/late finishes or night duties as determined by Early Disruptive Schedule, the RERRP is extended to 60 consecutive hours.

e. Additional Rest Periods

Additional rest periods are provided in 7.16.1. to compensate for:

- the effects of time zone differences and extensions of the FDP;
- additional cumulative fatigue due to disruptive schedules; and
- a change of home base.

7.16.1 Early Disruptive Schedule and Time Zone Difference

CS CS FTL.1.235

a. Early Disruptive Schedules

1. If a transition from a **late finish to an early start** is planned at home base, the rest period between the 2 FDPs includes 1 local night.
2. If a crew member performs 4 or more night duties, early starts or late finishes between 2 extended recovery rest periods as defined in ORO.FTL. 235(d) the second extended recovery rest period is extended to 60 hours.

b. 'Rotation' and Time Zone Differences Crossing 4 or More Time Zones

Current Malta Air scheduling does not have flights across 4 or more time zones. This scenario may arise on ad hoc VIP charters. The text below remains in this section to assist with planning rotations including ad hoc VIP charters.

Malta Air will monitor rotations and combinations of rotations in terms of their effect on crew member fatigue and make changes to crewing arrangements reflecting fatigue management principles.

The current Malta Air schedule does not have any pairings which cross 4 time zones in any direction. The time zone differences section below is included to enable the rest calculations when applicable to ad hoc VIP charter operations which may be planned and which could cross 4 time zones. Time zone differences are compensated by additional rest, as follows:

1. **At home base**, if a rotation involves a 4 hour time difference or more, the minimum rest is as specified in the following table. (This is rest after completion of a rotation).

Minimum Local Nights of Rest at Home Base to Compensate for Time Zone Differences

Maximum Time Difference (h) Between Reference Time and Local Time Where a Crew Member Rests During a Rotation	Time Elapsed (h) Since Reporting for the First FDP in a Rotation Involving at Least 4-hour Time Difference to the Reference Time			
	<48	48–71:59	72–95:59	≥96
≤6	2	2	3	3
>6 and ≤9	2	3	3	4
>9 and ≤12	2	3	4	5

2. **Away from home base**, if an FDP involves a 4-hour time difference or more, the minimum rest following that FDP is at least as long as the preceding duty period, or 14 hours, whichever is greater.
3. In case of an Eastward-Westward or Westward-Eastward transition, at least 3 local nights of rest at home base are provided between alternating rotations.
4. The monitoring of combinations of rotations is conducted under the Malta Air SMS and an SRA is required prior to dispatch on any ad hoc VIP charter flight which crosses 4 time zones east or west. The SRA will inform Crew Control, Operations and operating crews of fatigue management principles and the significant differences in these operations.

7.16.2 Additional Provisions for Rest Periods

CS FTL.1.235(b)(3)

Rest Periods: Time Elapsed Since Reporting

The time elapsed since reporting for a rotation involving at least a 4-hour time difference to the reference time stops counting when the crew member returns to his/her home base for a rest period during which Malta Air is no longer responsible for the accommodation of the crew member.

Rest Periods – Rest Days

S.I. No 507 of 2006

Mobile staff in civil aviation shall be given days free of all duty and standby, which are notified in advance, consisting of at least 96 local days in each calendar year, which may include any rest periods required by law, and at least 7 local days in each calendar month, which may include rest periods required by law.

7.16.3 WOFF Out of Home Base

A crewmember may volunteer to operate a working day off (WOFF). Crew Management System legality check will establish if this WOFF is compliant with disruptive schedule and extended recovery rest requirements prior to assigning this WOFF. If the WOFF is operated at a location other than Home Base and no positioning occurs, positioning will not be indicated on the duty plan.

Example:

The crewmember Home Base is DUB and the crewmember is in Rome and volunteers to operate a WOFF from Rome. No positioning occurred and the WOFF happens in Rome. The Crew Management shall retain a file note of the circumstances for subsequent internal or external audit. (EASA FTL meeting 17 Jan 2018)

7.17 Nutrition

ORO.FTL.240, AMC1 ORO.FTL.240

Crew members are expected to report for duty well rested and nourished. Typically the first meal break will occur after 225 minutes of FDP as in the normal course people work 3 to 4 hours before a meal break.

Subsequent requested meal breaks are coordinated by the SCCM (in the cabin) and the commander (in the flight deck). These meal breaks should occur during normal meal times (Table 4) for the crew member and ideally be of 20 minutes duration to minimise the possibility of fatigue arising from nutritional needs of individual crew members. Meal breaks should ideally take place on the ground for all crew but not in the case of cabin crew when passengers are embarking or disembarking. When taken during flight, meal breaks must not occur during critical phases of flight or during the first on board service in the case of cabin crew.

It is Malta Air policy to facilitate crew with meal break(s) during the FDP. In the normal course meal breaks will be taken after 225 minutes of FDP and during the next normal meal times (base local times) as set out in the table below. Meal breaks are an important element in fatigue management strategies for crew members. Exceptionally (e.g. high density, short sectors less than 60 minutes) a meal break may be less than 20 minutes but won't be planned for less than 10 minutes duration. Typically there will be 20 minutes of meal break in the first 6 hours of FDP. To maintain hydration during FDP, crew members are encouraged to take regular breaks to consume liquids.

Cabin Crew:

Cabin crew requiring a meal break should request it from the CSS who will co-ordinate meal breaks at the earliest opportunity while ensuring that there are sufficient crew available to discharge their duties. Meal breaks will be planned in accordance with the following guidelines.

- Not normally within the first 225 mins of FDP (as when reporting for duty cabin crew should, in accordance with their own fatigue management strategy, be well rested and nourished for the first 225 minutes of FDP).
- Meal breaks taken on the ground are not permitted during embarkation/disembarkation.
- Meal breaks inflight cannot be taken during the critical phases of flight (meaning the take-off run, the take-off flight path, the final approach, the missed approach, the landing, including the landing roll, and any other phases of flight as determined by the commander) or during the first service.
- A meal break duration may be dependent on sector duration and will be planned for a minimum of 10 minutes and will be no longer than 20 minutes.
- A meal break taken in the cruise or on the ground should be at a time when the cabin crew member is less likely to be interrupted by the needs of customers.
- Once a crew member has completed their 20 minutes meal break, they should not in the normal course require another break until at least 225 minutes later.

Cabin crew should ensure they take liquids regularly throughout their FDP to guard against dehydration.

Flight Crew:

A flight crew member taking a meal break shall coordinate with the remaining crew member(s), including TEM principles. In particular VHF communications, PLOC prevention and DALTA brief must be accomplished as appropriate, and both crew members must be aware of and actively manage the increased threats during this process.

Short sectors have lower cruising levels reducing the time to climb and descend and usually resulting in a 20 minutes cruise segment on these sectors and meal breaks as required should be taken during this time.

Meal breaks inflight cannot be taken during the critical phases of flight meaning the take-off run, the take-off flight path, the final approach, the missed approach, the landing, including the landing roll, and any other phases of flight as determined by the commander.

Flight crew should ensure they take liquids regularly throughout their FDP to guard against dehydration.

Table 4

Meal Opportunity	Home Base Local Time
Breakfast	0700–1000
Lunch	1200–1500
Dinner	1800–2100

7.18 Records of Home Base, Flight Times, Duty & Rest Periods

ORO.FTL.245, 245

- a. For a period of 24 months Malta Air maintains
 - 1. individual records for each crew member including:
 - i. flight times;
 - ii. start, duration and end of each duty period and FDP;
 - iii. rest periods and days free of all duties; and
 - iv. assigned home base;
 - 2. reports on extended flight duty periods.
- b. Upon request, Malta Air will provide copies of individual records of flight times, duty periods and rest periods to:
 - 1. the crew member concerned; and

2. to another operator, in relation to a crew member who is or becomes a crew member of the operator concerned.
- c. Records for crew members who undertake duties for more than one operator will
 1. be provided to both operators to enable compliant rostering, and
 2. be kept for a period of 24 months.

7.18.1 Notification of Intention to Operate Outside the Malta Air FTL Scheme

Crew members are not permitted to engage in flying activities for hire or reward outside the Ryanair Group operation without written permission of the respective NPFO.

Any pilot intending to engage in recreational flying activities outside the respective Ryanair Group airline FTL Scheme must notify the associated NPFO using the form below.

Notification Form

Attention Nominated Person Flight Operations

Notification of intention to Operate Outside the IFTSS

I, _____ / of _____ Base, fly light aircraft for recreational purposes.

I am aware of the IFTSS restrictions and I undertake to abide by the duty and rest time restrictions detailed in OM(A) Chapter 7.

My external flying activities will not impact on my ability to legally perform my Malta Air rostered duties.

I will maintain a record of my external flying activities for inclusion in my personal flight records.

The completed form shall be returned to your base captain who will forward it to the NPFO office for approval.

Note: Permission is granted only when the form is countersigned by the NPFO and returned to the pilot.

7.19 Fatigue Management Training

ORO.FTL.250

Malta Air provides initial and recurrent fatigue management training every three years to crew members, personnel responsible for preparation and maintenance of crew rosters and management personnel concerned.

This training is in compliance with the Malta Air training programme as detailed in OM D. This training syllabus covers the possible causes and effects of fatigue and fatigue countermeasures, including:

- a. applicable regulatory requirements for flight, duty and rest;
- b. basics of fatigue including sleep fundamentals and the effects of disturbing the circadian rhythms;
- c. causes of fatigue, including medical conditions that may lead to fatigue;
- d. effect of fatigue on performance;
- e. fatigue countermeasures;
- f. influence of lifestyle, including nutrition, exercise, and family life, on fatigue;
- g. familiarity with sleep disorders and their possible treatments;
- h. where applicable, the effects of long range operations and heavy short range schedules on individuals;
- i. effect of operating through and within multiple time zones; and
- j. crew member responsibility for ensuring adequate rest and fitness for flight duty.

Malta Air will host information modules on e-platforms to include Sleep Science, Circadian Rhythms, Workload and guidance to crewmembers on sleep strategies before, during and after duty.

7.20 Training Assignments

Malta Air rosters training assignments. When training assignments are not in home base Malta Air plans positioning to/from such assignments including flights and Ground Transport (GT).

There is no regulatory requirement for minimum rest out of home base, or minimum rest in home base prior to a non-flight duty (training assignments/office duties etc). The fatigue management principles applied to training rostering include planning assignments such that

- Where the positioning logistics permit, at least a 12 hr rest period prior to the training assignment is planned in home base and at least 10 hours out of home base.
- A crewmember is planned to achieve at least an 8 hours sleep opportunity in suitable accommodation prior to a training assignment or positioning followed by a training assignment without an intervening rest period.
- Rostering will assess the training assignment, including positioning to/from the training location to ensure there is not more than 18 hours awake.

If the rostered positioning is disrupted and the guidelines above are not achieved the instructor/examiner shall decide whether or not to continue with the training assignment. If the decision is to continue it will be based on the confirmation by the crewmember that the reduced rest should not impact on the training objectives. If the decision is not to continue the crew member shall contact crew control at the earliest opportunity to enable alternative arrangements.

Instructors are expected to be properly rested prior to delivering instruction on any training assignment.

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8 OPERATING PROCEDURES**8.0 General Operating Procedures****8.0.1 Objectives**

8.0.1.1 Our primary operations objective is to conduct our air transport activity safely. It is also Malta Air policy that Flight Operations shall be conducted efficiently and punctually. This policy devolves on Flight Crew as their basic duty.

8.0.1.2 Commanders and Co-Pilots are required to adhere to the policies and directions contained in the Operations Manual.

8.0.1.3 Endangering Safety

CAT.GEN.MPA.175

Malta Air personnel and service providers shall take all reasonable measures to ensure that no person recklessly or negligently acts or omits to act so as to:

- endanger an aeroplane or person therein.
- cause or permit an aeroplane to endanger any person or property.

Malta Air approved procedures ensure that flight crew undergo a psychological assessment before commencing line flying in order to:

1. identify psychological attributes and suitability of the flight crew in respect of the work environment; and
2. reduce the likelihood of negative interference with the safe operation of the aircraft.

8.0.1.4 Unauthorised Carriage

Malta Air Commanders shall take all reasonable measures to ensure that no person secretes himself/herself or cargo on board a Malta Air aircraft. This is effected through security checks in the cabin, galleys and toilets, and by the pilot external walkaround checking holds and wheelwells. The handling agent checks that all holds are secured and closed prior to departure.

8.0.2 Standard Operating Procedures

8.0.2.1 Standard Operating Procedures (SOPs) fall into two categories, the first of which identifies a core of mandatory actions or procedures that must always be performed because the failure to do so has a direct adverse impact on the safety of operation.

These SOPs include formal items such as Standard Call-outs, altimeter setting and checking procedures, emergency drills, and non-exceedance of limitations.

The formal SOPs are also categorised by the imperatives of “shall” and “must” and include those procedures which require a singular means of compliance e.g. engine failure at take-off.

The operating philosophies are presented in the FCOM Vol 1, Normal procedures, QRH and FCTM (Flight Crew Training Manual).

- 8.0.2.2 The second category of standard operating procedure has the purpose of standardising procedures to make it possible to crew any two Flight Crew members together on a non-regular basis without lowering safety standards.

The consequential requirement is that personal methods or practices shall not be introduced. New or changed procedures may only be introduced or promulgated on the authority of the NPFO. The standardised operating procedures must always be used.

The operating philosophies are enunciated in the Vol 1 Normal procedures, QRH and FCTM.

- 8.0.2.3 A significant number of operating practices allow the discretion to the Commander to decide how the flight should be flown. These are characterised with the need to exercise sound judgement and prudent assessment and may be referred to as sound operating practises (included under the more general heading of airmanship during an earlier period).

There has been a trend within the industry to refer to all operating activity under the heading of SOPs or a desire to specify all procedures within such a concept. Such broadening of the scope of SOPs diminishes the merit and value of the concept and hence the need to highlight this differentiation.

8.0.3 Monitoring and Cross Checking

- 8.0.3.1 The primary task either Flight Crew member has to perform is to fly the aircraft. Monitoring and look-out are important secondary duties. Full freedom to monitor can only be achieved by making the most efficient use of flight deck resources and aircraft equipment.

- 8.0.3.2 The purpose of monitoring is to detect unintentional deviation from the planned flight path parameters and permit effective correction. FCOM Vol 1, Normal procedures directs the crew to the monitoring policy which creates the ideal of full awareness of both Flight Crew members as to aircraft position, situation and configuration.

Both Flight Crew members must be alert to the possibility of the other making errors in calculation and tabular selections, e.g. take-off or landing weights, engine power settings, speeds, descent range etc. Figures shall not be blindly accepted. There is a need to always check mentally for gross errors.

- 8.0.3.3 For monitoring in a Non-Normal or Emergency situation, refer to the Checklist introduction section in the QRH for normal and non normal checklist usage, and Chapter 8 in the FCTM. Section 8.0.6 also refers.

8.0.4 Briefing

- 8.0.4.1 The object of briefing is to ensure that both Flight Crew members are aware of and agree with a proposed plan of action.
- 8.0.4.2 Formal briefings are required before each take-off and prior to each descent and landing. The Take-off briefing is presented in FCOM Vol 1, Normal procedures. The timings of this and other briefings are presented in FCOM Vol 1, Normal procedures. Section 8.3 also refers.
- 8.0.4.3 The PF shall be aware of the need to let the PM know how he intends to conduct the operation. The term PM shall be applied in context as described in the FCOM Vol 1.
- 8.0.4.4 These briefing activities produce an environment in which the non-handling Flight Crew member can carry out his prime function of monitoring the flight more effectively.
- 8.0.4.5 Airfield Briefs contain important information for operations and when published, shall be used in the briefing processes including RIBETS and DALTA.

8.0.5 Checklists

8.0.5.1 General

One of the most important factors in good flight deck management is the proper utilisation of the checklist. Reliable normal operations and repeatable success in dealing with abnormal and emergency situations are clearly attributable to the checklist system. The system does, however, depend absolutely on the command being given for the correct checklist, and ensuring its completion.

Modern aircraft are increasingly complex. Crews “must know” items consist of recall items in QRH, limitations in FCOM Vol 1, and SOPs as presented in Vol 1. Crews shall complete appropriate checklists and time permitting refer to technical descriptions in FCOM Vol 2 and Boeing bulletins as appropriate. It is not best practise to perform actions other than QRH memory items from memory. Flight Crew members shall be familiar with the Checklist introduction section in the QRH for normal and non normal checklist usage, and Chapter 8 in the FCTM, Non Normal Operations. Guidance is provided on the balance required between “recall” technical knowledge and relying on written technical descriptions in FCOM Vol 2 (and Boeing bulletins) in this chapter – Basic Aerodynamics and System Knowledge.

The operating policies and procedures are enunciated in the FCOM Vol 1 Normal procedures, QRH and FCTM

8.0.5.2 Location of Manuals and Checklists

All normal, non normal checklists and Manuals required to complete assigned duties are contained in EFB/VPOS.

Normal and Non-Normal Checklists are contained in the ships library, in the Flight Crew Operations Manual (FCOM) Vol 1, the Quick Reference Handbook (QRH), and laminated checklists on the glareshield in the cockpit.

8.0.5.3 Normal Checklists

The operation and philosophy of checklist usage is described in FCOM Vol 1, Normal procedure <RYP>. Supplementary Procedure, and QRH, Checklist Introduction.

Where a checklist is interrupted, the check must be discontinued and when called for again, the challenge shall begin again at the last item previously called.

If any untoward situation should develop or discrepancy arise while a checklist is being carried out causing distraction then the complete checklist shall be reread.

8.0.5.4 Non-Normal and Emergency Procedures

Non-Normal and Emergency Procedures and the corresponding checklists require to be dealt with on a different basis.

This is described in the OMB, QRH, Non-Normal Checklist Operation and FCTM.

This does not mean that in all emergency/abnormal situations the Commander will take over the controls.

8.0.5.5 Overweight Landing

8.0.5.5.1 An Overweight landing is recommended in the following conditions:

- A serious malfunction that affects the airworthiness of the aircraft.
- A condition where an expeditious landing would prevent a non-normal situation developing into an unsafe situation (reduce the exposure to a degrading level of safety).
- A serious illness on board requiring prompt medical attention.
- If directed by the QRH to “land at the nearest suitable airfield”.

Overweight landings may be safely accomplished by using normal landing procedures and techniques. There are no adverse handling characteristics associated with overweight landings. Landing distance is normally less than take-off distances for flaps 30 or 40 landings at all gross weights.

If stopping distance is a concern, reduce landing weight as much as possible. At the Captain's discretion reduce weight by holding at low altitude with a high drag configuration (gear down) to achieve maximum fuel burn off.

Overweight autolands are not recommended. Autopilots on Boeing airplanes are not certified for autolands above maximum certified landing weight. At higher than normal speeds and weights, the performance of these systems may not be satisfactory and has not been thoroughly tested. An automatic approach may be attempted however the pilot should disengage the autopilot prior to flare height and accomplish a manual landing.

In an emergency, should the pilot determine that an overweight landing is the safest course of action, the approach and landing will be closely monitored by the pilot and the following factors considered:

- Touchdown may be beyond the normal touchdown zone, allow for additional landing distance.
- Touchdown at higher than normal sink rates may result in exceeding structural limits.
- Plan for a go-around or manual landing if autoland performance is unsatisfactory; automatic go arounds can be initiated until just prior to touchdown, and can be continued even if the airplane touches down after initiation of the go-around.

8.0.5.6 Suspected Hard Landing

Malta Air policy is that a pilot report is the only factor that consistently identifies a hard landing. If the aircraft commander believes that a hard landing may have occurred, a Technical Log entry must be completed and Maintrol must be informed. A hard landing inspection must be carried out by a suitably qualified engineer before the aircraft can be dispatched. The flight data g reading if available will only be used to determine the level of technical inspection required.

8.0.6 Monitoring in a Non-Normal or Emergency Situation

8.0.6.1 The importance of the monitoring of one Flight Crew member by the other during all non-normal and emergency procedures cannot be over-emphasised.

Particular attention is required to monitor the actions when shutting down an engine, and the flight instruments during the completion of the vital actions.

8.0.6.2 In the event of an engine failure during take-off the engine failure checklist will not be read until the clean enroute configuration is established, or the aircraft has stopped following an RTO, or, in the case of a return landing stabilised level flight is attained.

8.0.7 Standard Call-outs

8.0.7.1 Standard Call-outs are SOPs.

The standard call-outs are presented in FCOM Vol 1, Normal Procedures, and flight profiles are presented with configuration actions in FCOM Vol 1, Normal Procedures.

Non-normal flight profiles and configuration actions are presented in the QRH, Non-normal Manoeuvres.

8.0.7.2 Standard Calls, using unequivocal ENGLISH language as challenge and response, form the principal discipline to verify correct flight progress, highlight deviation and provide an alert system to detect crew incapacitation.

It is very important that the exact terminology is used precisely so that misunderstandings do not occur between crew members.

8.0.7.3 If a standard callout is not responded to, a possibility of Flight Crew member incapacitation exists and needs to be addressed with some further identification activity.

The QRH, Checklist Introduction describes Malta Air policy in this area. Operational Policies are described in FCTM, Chapter 1.

8.0.8 Communications

8.0.8.1 General

Effective communications is an important factor in securing safety in aviation and includes clear unambiguous transmission of messages, careful listening, and positive feedback where appropriate.

Communication refers not only to radio (RT) communication (OMC, Jeppesen Publications (Pubs)). It is ATC communication, inter-cockpit verbal exchange, Cabin Crew liaison and briefings, P.A. announcements and dealings with other Malta Air/Ryanair Group staff and agents.

8.0.8.1.1 Use of Headsets, Crew Member Interphone System and Flight Crew Interphone System

CAT.IDE.A.170 and AMC1.IDE.A.170, AMC1.CAT.IDE.A.325

Flight Crew Interphone System

Malta Air shall not operate an aeroplane on which a Flight Crew of more than one is required unless it is equipped with a Flight Crew interphone system, including headsets and microphones, not of a hand-held type, for use by all members of the Flight Crew.

Malta Air shall not operate an aeroplane with a maximum certificated take-off mass exceeding 15,000 kg or having a maximum approved passenger seating configuration of more than 19 unless it is equipped with a crew member interphone System.

The crew member interphone system required by this paragraph must:

1. Operate independently of the public address system except for handsets, headsets, microphones, selector switches and signalling devices;
2. Provide a means of two-way communication between the Flight Crew compartment and:
 - a. Each passenger compartment;
 - b. Each galley located other than on a passenger deck level; and
 - c. Each remote crew compartment that is not on the passenger deck and is not easily accessible from a passenger compartment.
3. Be readily accessible for use from each of the required Flight Crew stations in the Flight Crew compartment;
4. Be readily accessible for use at required cabin crew member stations close to each separate or pair of floor level emergency exits;
5. Have an alerting system incorporating aural or visual signals for use by Flight Crew members to alert the cabin crew and for use by cabin crew members to alert the Flight Crew;
6. Have a means for the recipient of a call to determine whether it is a normal call or an emergency call; and
7. Provide on the ground a means of two-way communication between ground personnel and at least two Flight Crew members.

Each Flight Crew member required to be on flight deck duty shall wear a headset with boom microphone (where available) or equivalent and use it as the primary device to listen to the voice communications with ATS:

- On the ground when receiving the ATC departure clearance via voice communications.
- On the ground when engine(s) are running.
- In flight except when in the Cruise.
- Whenever deemed necessary by the Commander.

The boom microphone or equivalent shall be in a position which permits its use for two-way radio communications.

Headsets shall be used whenever possible during pushback and start up for communication with ground crew. When it is not possible to use headsets, standard ICAO hand signals shall be used with ground crew.

On self-manoeuvring stands, unless local bylaws require use of headsets, the marshaller will provide standard hand signals (as per Standardised Rules of the Air) from the designated marshalling position for engine start and taxi guidance on/off the stand.

Malta Air headsets should be left hanging from the pilot grab handles located above the L1 and R1 windows during turnarounds when in use to avoid damage from operation of the L2 or R2 window.

Malta Air supplied pilot headsets, which are not being used, must be stored in the forward facing, storage compartment behind the Right-Hand Seat.

Two headsets are required for dispatch and can be either the Malta Air provided TELEX or a pilot's personal headset which meets regulatory requirements. MEL 23-14 refers. Missing or broken Malta Air provided headsets should be recorded in the aircraft's Technical Log.

8.0.8.2 Inter-cockpit Communication

Inter-cockpit communication includes the following formal items:

- Checklists
- Standard Call-outs
- Briefings
- Monitoring Cross Checks

8.0.8.3 Reading of Checklists

Reading of checklists requires the use of clear diction and must be audible. Challenge and response checklists will reflect the tone of challenge, and responses shall be equally positive.

Response shall reflect positive unambiguous feedback. This is made more effective by reading and speaking where specified, the standard terminology.

8.0.8.4 Briefings

Briefings ensure that both Flight Crew members are aware of the proposed plan of action.

8.0.8.5 Drawing Attention to Errors and Deviations

Monitoring and cross checking is only effective to the extent that the PM feels that he can draw attention to errors and deviations. It is important therefore, that the inter cockpit relationship encourages this to happen. This is a function not only of the PF but also of the PM.

8.0.8.6 Sterile Cockpit

AMC1 ORO.GEN.110 (f)

Flight Crew shall avoid conversation not directly related to the safe operation of the aircraft from commencement of push back to TOC, and from TOD to engine shutdown on stand. Headsets shall be worn and boom microphones used in preference to hand microphones for communication with ATC during these phases of flight. No unnecessary paperwork shall be accomplished during sterile cockpit operations.

Sterile Comms

Flight Crew shall avoid operating on or listening to any frequency not required for current operations from commencement of pushback to TOC and from TOD to engine shutdown on stand. The intention is that both pilots are giving full attention to live ATC frequency. All unnecessary calls to handling agents, maintenance, unnecessary weather, are prohibited.

8.0.8.7 Cabin Crew Liaison

Cabin Crew Members are called Customer Service Agents (CSAs) and the Cabin Services Supervisor is called the Customer Service Supervisor (CSS). Cabin Crew come under the command of the Commander from reporting point and remain under the command of the Commander until completion of the duty period. To be effective in their safety function the relationship between cabin crew and flight crew is reinforced by communications.

The operating crew should introduce themselves to one another at the reporting point. The Commander shall then follow the guidance in OMA 8.1.0.2 on where to conduct the briefing.

This will include as a minimum:

1. The aircraft registration and location (if applicable).
2. Establish a security code, review security procedures and security status for the day.
3. Discuss operational issues such as flight times, slots, special considerations, weather/icing conditions/turbulence.
4. Defects which will impact on the operation, e.g. A/C serviceability, air stairs inoperative, health and safety issues, any conflicts, etc.

If the crew complement is incomplete at this time, pertinent issues shall be addressed with all crew prior to dispatch.

The Commander, shall make a point of re-confirming that he would like to be informed of anything untoward which may occur in the cabin such as unusual noises and reports from passengers and shall confirm the cockpit security code once on board. Cabin Crew shall never take for granted that the Flight Crew are aware of serious malfunctions such as engine fires, fuel leaks or loose external structure.

Cabin Crew shall alert the flight deck using the cabin interphone or other prescribed mechanism. When time permits the flight deck shall acknowledge this communication and receive the Cabin Crew report.

Formal reports between cabin and cockpit crew are listed in checklists. Routine or non-operational liaison between cockpit and Cabin Crew shall not take place during climb or descent.

It is particularly important to advise Cabin Crew of the de-icing requirements anticipated throughout the day; brief the associated threats and remind Cabin Crew of their responsibility to report wing contamination to the flight deck. Cabin Crew shall be alerted to the possibility of momentary fumes in the cabin and also to close cabin doors as required. Cabin Crew have received training on the effects and appearance of contamination on wing surfaces and may be able to make a valuable contribution to safety by advising the Flight Crew of any contaminant on a wing before or after de-icing/anti-icing treatments.

The Cabin Crew will relay any passenger concern about wing contamination to the flight crew and a pre-take-off contamination check will be considered.

8.0.8.8 PA Announcements

The passenger PA has a very important safety role as well as being a useful commercial tool. The 'European Action Plan for the Prevention of Runway Incursions' recommends the taxi phase of operations be considered a 'Critical Phase of Flight'. This has implications for the use of the PA system, as detailed below.

The following bullet points outline the safety and operational procedures in relation to the use of the passenger PA system:

- The PA system must be checked for serviceability prior to departure on the first flight of the day. It is prudent this be done prior to passenger boarding. The checklist item 'PA' will be used to confirm that the PA is complete and the equipment is serviceable and confirmed by the No.1 to be audible in the cabin.
- The Captain must always make the initial welcoming PA. This establishes his/her voice, which is important in the event of a subsequent Non Normal or Emergency situation where a passenger PA is required. This PA must be done prior to pushback.
- Only essential PA's will be made during taxi out or taxi in. Stop the aircraft and set the parking brake when doing so. When advising the Cabin Crew to take their seats for departure keep the address short and concise (eg 'Cabin Crew Seats for departure'). Do not let the PA announcements infringe on runway safety.
- During the cruise the Captain, or at his/her discretion the First Officer, should make a PA advising of the expected time of arrival, weather and anything else of interest.
- Only essential or safety related PAs shall be made in the descent or holding. Standard calls "Cabin Crew 10 mins to landing" and "Cabin Crew seats for landing" shall be made at the appropriate time.
- Do not make a PA crossing or entering a runway except in an emergency.
- There is NO requirement to make a PA on taxi in except for essential safety PAs.

When the surface wind at the destination exceeds 20 kt the Captain will make the following PA when releasing Cabin Crew;

“Cabin Crew doors to manual and crosscheck – exercise caution due to high wind.”

A PA list is available on the EFB listing PA text for various circumstances for use by crew.

Note PA following a diversion. Malta Air Policy is that passengers are generally not to disembark at the diversion airport when it is intended that the flight will continue to the original destination for security. Therefore your PA should focus on the short delay to the continued flight to your intended destination.

8.0.8.9 Communication with Maintenance Personnel

It shall be considered best practice that in addition to reporting aircraft defects in the technical log, it is extremely helpful to the subsequent maintenance investigations if the Commander amplifies the problem experienced and the indications experienced by verbal debriefing of maintenance personnel. This may be achieved by utilising the base maintenance frequency, by telephone, or any other means.

This process helps to generate improved understanding of operating problems and leads to improved Engineering response.

This process may be utilised in flight, only during periods of reduced workload and consistent with the sterile cockpit policy. When a non normal condition is identified after take-off and the aircraft is not intended to climb to FL100 the above procedure shall be utilised when the requisite checklists are completed and the aircraft is established in a holding pattern or is under radar control and ATC have been advised that communications with engineering is required.

8.0.8.10 Crew and Passenger Use of Portable Electronic Devices (PED)

8.0.8.10.1 Crew Use of Portable Electronic Devices (PED)

The Malta Air EFB is classified as a C-PED (Controlled Portable Electronic Device). The only device authorised for the provision of any data associated with the operation of any flight is the company issued Electronic Flight Bag. Information obtained from the EFB may be used for the planning or operation of any Malta Air flight. Crew must follow company procedures to ensure up to date information is available on the EFB.

8.0.8.10.1.1 Restrictions on Cabin Crew Use of Personal Portable Electronic Devices (PED)

The use of personal PEDs is not permitted during critical stages of the flight or during the performance of any safety and security related duty e.g., pre-departure inspections, boarding and disembarking, security check/search, arming/disarming slides etc.

Cabin Crew members are only permitted to charge a PED on board the aircraft if the power source is kept out of baggage and monitored at all times.

The use of headphones by Cabin Crew is strictly prohibited onboard the aircraft during a flight duty period.

8.0.8.10.2 Passenger Use of Portable Electronic Devices (PED)

Mobile phones and tablet computers are permitted for use for the whole duration of the flight in Airplane or Flight mode. Devices that do not have Airplane or Flight mode must be switched off and remain switched off for the whole duration of the flight. Mobile phones and other transmitting PED's must be selected to Airplane or Flight mode from the time the doors are closed for departure until the doors are open on arrival.

8.0.8.10.3 Device Status with Aircraft Doors Closed

All devices that transmit or receive calls or text messages and do not have an Airplane or Flight mode must be completely turned off for take-off and landing and for the duration of the flight except:

Devices adhering to the Bluetooth standard (e.g. wireless headsets that use a Bluetooth connection to a PC) except telephones, may be used during flights but must be switched off during take-off and landing.

Devices that weigh one kilo or less and have an Airplane or Flight mode setting may be used in flight provided the Airplane or Flight mode is enabled after the main doors are closed. Laptops and larger PEDs shall be stowed during taxi, take-off and landing (any other time as requested by the Captain) to prevent possible injuries from projectiles and must not obstruct a passengers exit from a seat row in the event of an emergency.

The use of the wireless LAN (WLAN) is not permitted on Malta Air aircraft. Any device with this facility (i.e. laptops, electronic games, mobile phones, iPads, etc.) must have the facility switched off if they are to be used in flight. Devices in which it is not possible to deactivate WLAN/WiFi may not be used at any time during flight.

Larger PEDs such as disc players and similar entertainment equipment may be used during flight but must be switched off during take-off and landing.

8.0.8.10.4 Passenger Briefing on PED's

Cabin Crew must ensure that all passenger headphones are removed for the Safety Demo when completing 'cabin secure' checks. Cabin Crew must ensure that laptops and larger PEDs are properly stowed. Take the opportunity to remind passengers that are using allowed devices to ensure that they are in Airplane or Flight mode.

8.0.8.10.5 Usage of PED's in Case of Extended Delay

The commander may allow passengers to use their phones in transmitting mode on the ground before take-off with doors closed where a significant delay to the flight has occurred. Portable electronic devices such as tablets and mobile phones may be used on board in Airplane or Flight mode during all stages of the flight.

During push and hold, whether or not the engines are running, the Commander is authorised to permit the use of mobile phones providing an announcement is made to ensure all phones are switched off prior to take-off.

- 8.0.8.10.6 Malta Air Boeing 737 aircraft meet the requirements set out in EASA SIB No2013-21 2.a and are not susceptible to spurious electromagnetic interference from non-transmitting PEDs in the passenger cabin. However, crews shall be vigilant to the possibility of transmitting PEDs being active in the cabin. Possible indications of spurious electromagnetic interference include noise interference or distortion to the PA system. Possible flight deck indications include erratic cabin pressure control, VHF radio interference or radio altimeter indications. Excessive aileron and elevator activity during the approach with the autopilot engaged. Crews shall always follow Malta Air SOPs and approved checklists in any event. If spurious electromagnetic interference is suspected the aircraft Commander shall instruct the Cabin Crew to ensure that all portable electronic devices are switched off for the remainder of the flight. On landing a Technical Log entry shall be made and an ASR submitted.

8.0.8.10.7 Special Requests to Operate a PED or T-PED

The Commander may grant permission for special requests during any phase of the flight for specific reasons (e.g. for security measures) to operate a PED or TPED. This will be assessed on a case by case basis.

8.0.9 Aircraft Operating Limitations

“Each aeroplane shall be operated in compliance with the terms of its Certificate of Airworthiness and within the approved limitations contained in its Aeroplane Flight Manual” The Airplane Flight Manual limitations are expanded into the FCOM Vol 1, which shall be used.

Compliance with this regulation is ensured by adherence to the Malta Air normal and emergency checklists, Standard Operating Procedures and observation of the Aircraft Flight Manual Limitations.

Commanders are responsible for ensuring compliance with this regulation.

8.0.10 Co-pilot Flying

(ORO.FC.200)

It is the Company's policy that the Co-pilot will maintain flying proficiency. It is recognised in the industry that this activity is part of the important process of developing a safe airline environment. While elements of the process and mechanisms by which safety is achieved is considered elsewhere in the Operations Manual, some constraints in this area must be recognised.

Co-pilots generally have less experience than Commanders who will therefore exercise judgement in deciding which sectors to offer the Co-pilot particularly when the conditions are challenging at the destination airfield. This choice shall be a premeditated one by the Captain who will advise the Co-pilot to allow timely adjustment and re-arrangements of functions. The Commander obviously retains the prerogative to change his decision if circumstances change or deteriorate. The advice also seeks to pre-empt non-judgemental complacent routines such as “leg and leg about” from developing.

A Co-pilot who has successfully completed his/her Command Check may exercise the privilege of his/her licence by operating from the LHS, subject to the occupant of the RHS being an LTC, TRI or TRE and then only at his discretion, pending his formal letter of appointment as a Commander from the NPFO.

The Commander, when flying with inexperienced co-pilots or a new co-pilot, shall perform the take-off or landing himself when the following conditions are experienced:

1. crosswinds more than 2/3rds limiting value.
2. runway is performance limited.
3. take-off and landing RVR's less than 1,000 m.
4. ceiling is less than 100' above decision altitude for a precision approach and 200' above minima (MDA) for a non-precision approach aid.
5. the runway is contaminated.
6. abnormal procedures are required due to system defects.
7. inexperienced co-pilots shall not conduct the landing when the crosswind is in excess of 15 kt during normal operations. This restriction does not preclude the co-pilot from flying a monitored approach.
8. visual approaches.
9. forecast or reported windshear.

8.0.10.1 Handover of Control

The handover of control from the Captain to the First Officer following a Captain’s only take-off shall not take place until after reaching the top of climb. Line Training Captain’s may exercise discretion on the hand over when engaged in Line Training.

8.1 Flight Preparation and Operation Instructions

8.1.0 Planning and Preparation of Documentation, EFB, Ryanair Connect

ORO.AOC.140

The table below describes the locations of each section of information required prior to dispatch, and the default locations in order of preference. The general principle is that crews shall exhaust all avenues prior to approaching Base Ops or Dub Ops personnel, as these locations can be extremely busy at peak times. Personnel at these locations will however address crew enquiries expeditiously and courteously.

Content	Location 1	Location 2	Location 3
Airfield Briefs <i>(including Self-Briefs)</i>	EFB	FDF	–
Airfield Briiefs LOEP	EFB	FDF	–
Flight Plans	EFB	https://ryr.lido.aero/briefing	Base Ops/Handling Agent
Maintenance status/H.I.L.	Tech Log	EFB	Maintrol
Enroute Charts	EFB	FDF Trip Kit	–
Jeppesen Publications (Pubs)	EFB	FDF Trip Kit	–
Terminal Charts	EFB	FDF Trip Kit	–
NOTAMs	EFB	https://ryr.lido.aero/briefing	Base Ops/Handling Agent
OMB	EFB	FDF	–
OMC <i>(including Appendices)</i>	EFB	FDF	–
Payload	EFB	Base Ops/Handling Agent	–
RTOW Charts	FDF	–	–
RTOW LOEP	FDF	–	–
Security Manual	BC EFB	–	–

Content	Location 1	Location 2	Location 3
Weather	EFB	https://ryr.lido.aero/briefing	Base Ops/Handling Agent
Weight Certificate	EFB	FDF	—

Ryanair Connect App

All crew shall have read and confirmed all compulsory reads prior to operating on any given day. This procedure ensures all crew are up to date with operational documentation before flying.

8.1.0.1 Aircraft Pre-departure Checks and Inspections

The following manuals and publications shall be used for reference as appropriate:

- Airplane Flight Manual/CDL.
- Boeing Flight Planning and Performance Manual.
- FCOM Vol 1.2 and QRH and FCTM.
- OMB.
- OMC, Electronic Airways and Route Manual.
- Enroute Charts (EFB).
- OMC, Airfield Briefs.
- EFB containing the following approved applications (EFL, Docunet, Boeing OPT, Jeppesen FD Pro).

Other Web Based Providers Approved by Malta Air

LIDO Brief is an approved web based provider of Flight Plans. Additional features of the website include delivery of weather and NOTAMs. Crews will normally receive this documentation electronically. Jeppesen is the approved provider of OMC, airways and route manual information.

Security of Web Based providers: The Malta Air IT Department shall be responsible for the security of all approved electronic sources. The normal delivery of documentation and information required to complete assigned flight operations shall be electronic. Backup delivery systems shall be in place to ensure continuity of operations.

8.1.0.1.1 Onboard Performance Tool (OPT)

Onboard Performance Tool (OPT) is a Boeing software application which calculates a “real time” Performance and Mass and Balance data for the actual conditions existing at the time of departure. Variations in the aircraft configuration loading, ambient conditions, runway length and some common MEL/CDL downgrades can be accounted for.

Take-off analysis data in OPT reflects the AFM performance. The OPT software is administered by the Performance Department.

OPT software, software updates and a current airport database is installed on each portable EFB device. Normally operating crew will make all performance calculations using the Boeing OPT application. The Performance Department shall provide OPT solutions on request to operating crews where crew are unable to do so.

Crews shall complete the performance calculations in advance of each flight. When the OPT is used to calculate take-off performance, the result shall be emailed to the default address for archiving. There is no requirement to complete a manual RTOW calculation unless the OPT or email applications are unavailable. If the RTOW figures have been manually calculated, they must be attached to the EFL submission (or recorded on the paper OFP or RTOW calculation table).

Where EFB is not available, an OPT request may be completed verbally between the Commander or a pilot designated by the Commander, the duty pilot or the Performance Department.

If a hard copy of the OPT solution is delivered to the Commander prior to take-off it shall be retained in the returned flight envelope. If a soft (electronic) copy has been provided (i.e. via email), then the crew/designated pilot/duty pilot/Perf Dept must forward the calculation to the default e-mail address “takeoffperformance@ryanair.com”.

OPT is the primary method of calculating take-off (including Weight and Balance) and landing performance.

8.1.0.1.2 The Trip Kit System

If EFB is unserviceable, refer to OMA 8.9.1.6.1 and 8.9.1.6.2. If a trip kit is required, the trip kit shall be assembled with reference to the Operational Flight Plan (eFPL).

Required Document	Print from
Terminal Charts	Trip Kit
Enroute Charts	Trip Kit
AFB	Ryanair Connect
RTOW	Ryanair Connect
OFP	Lidobrief
Manuals	Ryanair Connect

Crew must dispose of any printed charts post flight. Refer to EFB User Guide for Trip Kit instructions.

Operational Flight Plan General

All Malta Air flights are operated

- in accordance with the regulations laid down by EASA,
- in compliance with the specifications in the Company's Air Operator's Certificate,
- in compliance the limitations in the Aircraft Flight Manual, and
- in compliance with any restriction imposed by the TM CAD.

Malta Air specifies flight planning procedures to provide for the safe conduct of the flight and a LIDO Operational Flight Plan (OFP) is completed for each flight based on aircraft performance, operating limitations, relevant expected conditions on the route to be followed and the aerodromes concerned. The OFP shall include at least one alternate.

The flight crew shall complete the OFP as the flight progresses.

The Commander shall not commence a flight unless he is satisfied that:

1. the aeroplane is airworthy;
2. the aeroplane is not operated contrary to the provision of the Configuration Deviation List (CDL);
3. the instruments and equipment required for the flight to be conducted are available;
4. the instruments and equipment are in operable condition except as provided in the MEL;
5. the aircraft has been subject to a security check/security search;

6. those parts of the Operations Manual that are required for conduct of the flight are available, OM-A and SEP, OM B and OM C;
7. the documents, additional information and forms required to be available, ref. section 8.1.12 are on board;
8. current maps, charts and associated documents or equivalent data are available to cover the intended operation of the aeroplane including any diversion which may reasonably be expected, OM C. Any conversion tables necessary to support operations where metric heights, altitudes and flight levels must be used and are available on board (operations in area U);
9. ground facilities and services required for the planned flight are available and adequate;
10. the provisions specified in the Operations Manual in respect of fuel, oil, oxygen, minimum safe altitudes, aerodrome operating minima and availability of alternate aerodromes, where required, can be complied with for the planned flight;
11. the load is properly distributed and secured;
12. the mass of the aeroplane, at the commencement of the take-off roll, is be such that the flight plan can be conducted in compliance with the performance and weight and balance limitations of the aeroplane;
13. any additional operational limitation can be complied with.

8.1.0.2 Reporting Directly to Aircraft/Gate

Report Directly to Aircraft/Gate Bases can be identified in the base information guide located in the VPOS and the EFB.

Briefings

Safety Briefings are a vital part of the pre-departure sequence and ensure that all members of the crew establish and maintain a shared ‘mental model’ for the days operation. The Commander, or the CSS in the absence of the Commander, are required to ensure that all applicable safety briefings are completed prior to embarkation of passengers, that they are completed in full and are not subject to undue interruption. Under normal circumstances the briefing will be conducted in the aircraft cabin, where privacy and control of the environment is assured. The yellow safety strap will remain in place until briefings are completed and the aircraft ready for boarding. If due to the delayed arrival of the aircraft the crew may elect to use another suitable alternate location for the briefing. A suitable alternate location must satisfy the following requirements:

1. It must be sufficiently far away from members of the public, that the crew cannot be overheard.

2. It must be in a place where the crew are not subject to interruption from members of the public or ground staff.
3. If the weather is inclement then it must be conducted inside.

Note: If the above conditions cannot be met then the crew will wait for the aircraft to arrive or proceed to the base facilities where the briefing can be conducted.

Flight Crew Briefing

As part of his introduction the commander shall confirm with the first officer that both EFB are up to date and at the minimum accepted charge as per OMA 8.9, and that the eFPL is downloaded in EFL.

The Commander will then discuss with all members of the flight crew as a minimum the following.

- Aircraft location, defect and status
- Weather conditions above the applicable minima using OMA guidance for destinations and chosen alternates
- NOTAMs/Co. NOTAMS that are applicable to the flight
- Performance issues affecting the days flying
- Fuel requirements
- Loading and required aircraft weight for the sectors
- De-icing requirement if applicable
- Review appropriate TEM aspects affecting the planned operation

Training Flights

In addition to the above the Commander will review the Students training file and brief any training objectives for the duty consistent with the guidance containing in the Line training techniques document (LTTD).

Flight Crew Briefing of Cabin Crew

As part of the introduction the commander shall confirm with all members of the crew that all updates on EFB have been successfully completed. He will also confirm the report time of all members of the crew and establish if any member has been called from STBY or has had their report delayed.

Note: If for reasons of poor mobile network signal or airport WIFI a crew member reports that this has not been possible then the crew member will return to the base facilities to use the Company WIFI provided.

The Commander shall conduct a crew briefing in accordance with OM-A 8.0.8.7, taking the opportunity to reinforce any current NPFO safety messages.

- Missing crew or crew changes must be notified to Crew Control by the commander prior to dispatch.
- Late crew changes require EFL to be updated prior to dispatch to ensure FTL compliance.

In the event of a delayed inbound aircraft or prolonged delay, crew should agree a time with Operations Control to return to the reporting point.

8.1.1 Minimum Flight Altitudes

CAT.OP.MPA.145, CAT.OP.MPA.270

General

Minimum flight altitudes are determined for all flights performed by the Company. These are indicated on the LIDO Operational Flight Plan. Corrections for wind, temperature and QNH are always the Commander's responsibility except when under radar vectoring. A clearance issued under Radar control assures terrain and obstacle clearance in all temperature and allowable pressure conditions, in most states. AMC1 CAT.OP.MPA.145 (a).

Note: UK controllers will not correct for temperature

Operations are prohibited for take off, enroute, landing and selection of alternates when the QNH is not selectable on the altimeter sub-scale setting.

Where minimum flight altitudes established by national Authority are higher than those established by the Company, the higher values shall apply.

Malta Air flight crew shall not fly below specified minimum altitudes except when necessary for take-off or landing or descending in accordance with procedures approved by the TM CAD.

Procedure to Determine Terrain Clearance and Minimum Usable Flight Level (MUFL)

The cruising flight level/altitude shall be equal to or higher than the true minimum safe en-route altitude. When selecting cruising levels other than those specified in the OFP crews shall take into account that pressure and/or temperature below ISA require an increase of safety margins and that wind speed correction may have to be applied in addition:

- QNH: each 1 hPa below ISA requires an increase of 30 ft.

- OAT: when below 0deg the MSA shall be temperature corrected. Each minimum altitude below the corrected MSA shall be temperature corrected. The glare shield 'handy dandy' is available to calculate temperature corrections.

Example:

Conditions:	MEA 12,000 ft FL160, actual OAT -30°C at FL160 Lowest local QNH 977 hPa
Step 1: Temperature correction	ISA Temperature FL160 is -17 °C Deviation -13°C 13°C corresponds to 5% 12,000 ft + 5% = 12 600 ft Please follow the Handy Dandy provided on the Glare shield for Deviation Values.
Step 2: QNH correction (1013-local QNH) hPa × 30 ft	1013 hPa-977 hPa = 36 hPa 12,600 ft + (36 × 30) ft = 13,680 ft
Step 3: MUFL	FL140

8.1.2 Criteria and Responsibilities for Determining the Adequacy of Aerodromes

8.1.2.1 Use of Aerodromes

CAT.OP.MPA.105

Only NPFO approved aerodromes adequate for the aircraft and the operations concerned are used. This is assured by the Performance Department aerodrome categorisation process which requires the sign-off of the NPFO for any approved aerodrome.

OMC A – Approved Aerodromes List details all approved aerodromes, with details of Aerodrome Category, RYR Restricted Status (if applicable), plus any pertinent additional information, such as Emergency Use Only.

Diversion to Emergency Use Only aerodromes may cause significant operational disruption, for example, due to the lack of ground/passenger handling, commercial contracts and runway/airport slots. Before diversion to an Emergency Use Only aerodrome, Flight Crew must consider such implications, making reference to the Airfield Brief, and where possible, contacting OCC (via ACARS).

Notwithstanding the above, in an emergency situation the Commander is authorised to use any aerodrome as required, but if time is available, he is expected to select an approved aerodrome.

Isolated Aerodrome

Isolated aerodrome operations not currently in use.

8.1.2.2 Adequate Aerodrome

CAT.OP.MPA.107

Malta Air considers an aerodrome as adequate if, at the expected time of use, the aerodrome is available and equipped with necessary ancillary services such as Air Traffic Services (ATS), sufficient lighting, communications, meteorological reports, navigation aids and emergency services.

1. Diversion Distance is one hour flying time at an OEI cruising speed according to the AFM in still air standard conditions based on the actual take-off mass, being 427 nm (B737-800) and 416 nm (B737-8200)*.
2. The Diversion Distance is at maximum planning range, calculated at M.79/330 Kts.
3. The Commander must select an appropriate speed for diversion to the 'Nearest Suitable Airport' which may require up to M.79/330 Kts.

Note: * In Accordance with CAT.IDE.A.285 for over water flights the Maximum Diversion Distance is limited to 400 NM unless Liferafts are carried.

8.1.2.3 Rescue and Fire Fighting Services (RFFS)

8.1.2.3.1 Aerodromes selected by Malta Air undergo airport categorisation process in compliance with Air Ops, OMC. The RFFS requirements shall comply with ICAO Annex 14, Section 9.2.

8.1.2.3.2 The following definitions apply for compliance:

RFFS Category:

Rescue and Fire Fighting Services as defined in ICAO Annex 14.

Aeroplane RFFS Category:

The category specified in ICAO Annex 14 Table 9-1 for a given aeroplane type.

Aerodrome RFFS Category:

The published RFFS category for a given aerodrome.

8.1.2.3.3 The Aeroplane RFFS category for B737-8200 series aircraft is Category 7.

The Aeroplane RFFS category for B737-800 series aircraft is Category 7.

- 8.1.2.3.4 At the planning stage, if the Aerodrome RFFS category is not equal to or better than the Aeroplane RFFS category, Malta Air shall ensure that the Aerodrome RFFS category at each aerodrome required to be specified in the OFP meets the criteria in Table 1 for the type of operation and the time of expected use.

Table 1 Minimum Aerodrome RFFS Category

Aerodromes (Required to be specified in the operational flight plan) Note: If an individual aerodrome serves more than one purpose, the highest required category corresponding to that purpose at the time of expected use applies.	Minimum Aerodrome RFFS Category (Based on published aerodrome RFFS category, including any modification by NOTAM)
Departure and Destination Aerodrome	RFFS category for each aerodrome should be equal to or better than the aeroplane RFFS category. Where a suitable risk assessment has been conducted by the operator (refer to below for details): • One category below the aeroplane RFFS category, or • Two categories below the aeroplane RFFS category, in the case of a temporary downgrade of 72 hours or less but not lower than RFFS Category 4.
Take-off Alternate and Destination Alternate Aerodromes⁽¹⁾	Where, a suitable risk assessment has been conducted by the operator (refer to below for details): • Two categories below the aeroplane RFFS category, or • Three categories below the aeroplane RFFS category, in the case of a temporary downgrade of 72 hours or less but not lower than RFFS Category 4.
En-route Alternate Aerodromes	If at least 30 minutes notice is given to the aerodrome operator prior to the arrival of the aeroplane, a minimum of RFFS Category 4. If less than 30 minutes notice can be given to the aerodrome operator prior to the arrival of the aeroplane: • Two categories below the aeroplane RFFS category, or • Three categories below the aeroplane RFFS category in the case of a temporary downgrade of 72 hours or less but not lower than RFFS Category 4.

For non-scheduled maintenance, cargo, positioning and ferry flights, irrespective of the number of movements the minimum RFFS CAT is 5. Only crew and essential staff shall be on board.

Approval for the Selection and specification in the OFP of an Aerodrome with an RFFS category below the aeroplane RFFS category, in accordance with Table 1, shall be published to Crew by Airfield Brief or company notam if the downgrade exceeds 72 hours.

In flight, the Commander may decide to land at an Aerodrome where the Aerodrome RFFS Category is lower than specified above, if in his judgment and after due consideration of all the prevailing circumstances, to do so would be safer than to divert.

8.1.2.4 Aerodrome Selection Policy

CAT.OP.MPA.182, AMC1 CAT.OP.MPA.182

1. At the planning stage, it must be ensured that once the flight has commenced, there is reasonable certainty that an aerodrome where a safe landing can be made will be available at the estimated time of use of that aerodrome.
2. At the planning stage, to allow for a safe landing in case of an abnormal or emergency situation after take-off, a take-off alternate aerodrome shall be selected and specified in the operational flight plan if either:
 - a. the meteorological conditions at the aerodrome of departure are below Malta Air's established aerodrome landing minima for that operation; or
 - b. it would be impossible to return to the aerodrome of departure for other reasons.
3. The take-off alternate aerodrome shall be located within a distance from the departure aerodrome that minimises the risk of exposure to potential abnormal or emergency situations. In selecting the take-off alternate aerodrome, at least the following shall be considered:
 - a. actual and forecast meteorological conditions;
 - b. availability and quality of the aerodrome infrastructure;
 - c. navigation and landing capabilities of the aircraft in abnormal or emergency conditions, taking into account the redundancy of critical systems; and
 - d. approvals held (e.g. Low Visibility Operation (LVO), etc.).

- e. where it is not possible to use the departure aerodrome as a take-off alternate aerodrome due to meteorological or performance reasons, Malta Air selects another adequate take-off alternate aerodrome that is no further from the departure aerodrome than maximum Diversion Distance in accordance with OMA 8.1.2.
- 4. At the planning stage, for each Instrument Flight Rules (IFR) flight, one or more aerodromes shall be selected and specify in the operational and Air Traffic Services (ATS) flight plans, so that two safe-landing options are available during normal operation when reaching the destination aerodrome.

Note: When planning the flight, two safe-landing options are expected to remain available until the flight reaches its destination, where a decision will be made to commit to land or divert. This will typically be a runway at the destination aerodrome itself and a runway at a destination alternate aerodrome.

The requirement may also be satisfied by two landing runways at the destination aerodrome, provided that the risk of a single event (such as an aircraft accident) or meteorological deterioration at that single aerodrome will not eliminate both options.

Malta Air is not approved for the use of an isolated aerodrome as destination aerodrome.

- 5. Appropriate safety margins to flight planning take into account a possible deterioration of the available forecast meteorological conditions at the estimated time of landing in accordance with OMA 8.1.5.4.
- 6. For each IFR flight, Ryanair shall ensure that sufficient means are available to navigate to and land at the destination aerodrome or at any destination alternate aerodrome in the event of loss of capability for the intended approach and landing operation.

8.1.2.4.1 Basic Fuel Scheme – Destination Aerodromes – PBN Operations

AMC1 CAT.OP.MPA.182 (f)

To comply with point 8.1.2.4(6) when intending to use PBN, an aerodrome shall be selected as a destination alternate aerodrome only if an instrument approach procedure that does not rely on a GNSS is available either at that aerodrome or at the destination aerodrome.

GM1 CAT.OP.MPA.182 (f)

OMA 8.1.2.4.1 applies only to destination alternate aerodromes in flights that require a destination alternate aerodrome. A take-off or an ERA aerodrome with instrument approach procedures that rely on a GNSS may be planned without

restrictions. A destination aerodrome with all instrument approach procedures that rely solely on a GNSS may be used without a destination alternate aerodrome if the conditions for a flight without a destination alternate aerodrome are met.

The term ‘sufficient means are available to navigate to and land at’ means that the procedure can be used in the planning stage and should comply with planning minima requirements.

8.1.2.4.2 Basic Fuel Scheme – Destination Aerodromes – Operational Credits

AMC1 CAT.OP.MPA.182 (f)

N/A Malta Air.

8.1.2.5 Basic Fuel Scheme – Take-off Alternate Aerodrome

AMC1 CAT.OP.MPA.182

The take-off alternate aerodrome should not be farther from the departure aerodrome than 1-hour flight time at a One-Engine-Inoperative (OEI) cruising speed according to the AFM in ISA and still-air conditions using the actual take-off mass in accordance with OMA 8.1.2.

8.1.2.6 Basic Fuel Scheme – Destination Alternate Aerodrome

AMC2 CAT.OP.MPA.182

1. For each IFR flight, at least one destination alternate aerodrome should be selected and specified in the operational and ATS flight plans
2. For each IFR flight, two destination alternate aerodromes should be selected and specified in the operational and ATS flight plans, when the safety margins for meteorological conditions of OMA 8.1.5.4 (AMC5 CAT.OP.MPA.182), and the planning minima of OMA 8.1.5.5 (AMC6 CAT.OP.MPA.182) cannot be met, or when no meteorological information is available for the selected destination aerodrome.
3. Malta Air may operate with no destination alternate aerodrome when specific Flight Operations approval has been received from the company and when the following two conditions are met:
 - a. the duration of the planned flight from take-off to landing does not exceed 6 hours or, in the event of in-flight re-planning, in accordance with point OMA 8.1.7.3(4) (CAT.OP.MPA.181(d)), the remaining flying time to destination does not exceed 4 hours; and
 - b. two separate runways are usable at the destination aerodrome and the appropriate weather reports and/or weather forecasts indicate that for the period from 1 hour before to 1 hour after the expected time of arrival, the ceiling is at least 2000 ft (600 m) or the circling height + 500 ft (150 m), whichever is greater, and ground visibility is at least 5 km.

8.1.3 Use of Aerodromes

CAT.OP.MPA.105, 110

Malta Air only uses aerodromes that are adequate for the 800/8200 and for Commercial Air transport operations. The NPFO may approve flights to other aerodromes which are non CAT operations, e.g. maintenance positioning flights.

Malta Air has established minimum values for vertical and horizontal visibility for planning, take-off and landing. Jeppesen is the aeronautical Navigation Services Provider. The methods for determination of these minimum values are described in the following sections:

8.1.3.1 Take-off Conditions

CAT.OP.MPA.265

Before commencing take-off, the commander shall be satisfied that:

1. The meteorological conditions at the aerodrome and the condition of the runway/FATO intended to be used will not prevent a safe take-off and departure; and
2. The selected aerodrome operating minima are consistent with all of the following:
 - a. the operative ground equipment;
 - b. the operative aircraft systems;
 - c. the aircraft performance;
 - d. flight crew qualifications.

Meteorological Conditions for Take-off-runways

AMC1 CAT.OP.MPA.265 (a)

- The commander should not commence take-off unless the weather conditions at the aerodrome of departure are equal to or better than the applicable minima for landing at that aerodrome unless a weather-permissible take-off alternate aerodrome is available.
- If the reported VIS is below the minimum specified for take-off and RVR is not reported, then take-off should only be commenced if the commander can determine that the visibility along the take-off runway is equal to or better than the required minimum.

8.1.3.2 Approach and Landing Conditions

CAT.OP.MPA.300

Before commencing an approach operation, the commander shall be satisfied that:

1. The meteorological conditions at the aerodrome and the condition of the runway/FATO intended to be used will not prevent a safe approach, landing or go-around, considering the performance information contained in the operations manual; and
2. The selected aerodrome operating minima are consistent with all of the following:
 - a. the operative ground equipment;
 - b. the operative aircraft systems;
 - c. the aircraft performance;
 - d. flight crew qualifications;

Flight crew shall complete a 'Landing Distance at Time of Arrival' (LDTA) assessment, where required, in accordance with 8.1.3.2.2.

8.1.3.2.1 Approach and Landing Conditions

AMC1 CAT.OP.MPA.300(a)

Landing Distance at Time of Arrival (LTDA) Assessment

1. The in-flight landing distance assessment should be based on the latest available weather report and runway condition report (RCR) or equivalent information based on the RCR.
2. The assessment should be initially carried out when the weather report and the RCR are obtained, usually around top of descent. If the planned duration of the flight does not allow the flight crew to carry out the assessment in non-critical phases of flight, the assessment should be carried out before departure.
3. When meteorological conditions may lead to a degradation of the runway surface condition, the assessment should include consideration of how much deterioration in runway surface friction characteristics may be tolerated, so that a quick decision can be made prior to landing.
4. The flight crew should monitor the evolution of the actual conditions during the approach, to ensure that they do not degrade below the condition that was previously determined to be the minimum acceptable.

8.1.3.2.2 In-flight Check of the Landing Distance at Time of Arrival

CAT.OP.MPA.303

No approach to land shall be continued unless the landing distance available (LDA) on the intended runway is at least 115 % of the landing distance at the estimated time of landing, determined in accordance with the performance information for the assessment of the landing distance at time of arrival (LDTA).

Performance information for the assessment of the LDTA shall be based on approved data contained in the AFM.

8.1.3.3 Aerodrome Operating Minima

CAT.OP.MPA.110 (a), (b)

Malta Air establishes Aerodrome operating minima for each departure, destination or alternate aerodrome. These are published in the OM C. They are the lowest permissible minima taking into account various factors, such as:

1. The type, performance, and handling characteristics of the aircraft;
2. The equipment available on the aircraft for the purpose of navigation, acquisition of visual references, and/or control of the flight path during take-off, approach, landing, and the missed approach;
3. Any conditions or limitations stated in the Aircraft Flight Manual (AFM);
4. The relevant operational experience of Malta Air;
5. The dimensions and characteristics of the runways/Final Approach And Take-off Areas (FATOs) that may be selected for use;
6. The adequacy and performance of the available visual and non-visual aids and infrastructure;
7. The Obstacle Clearance Altitude/Height (OCA/H) for the Instrument Approach Procedures (IAPs);
8. The obstacles in the climb-out areas and necessary clearance margins;
9. The composition of the flight crew, their competence and experience;
10. The IAP;
11. The aerodrome characteristics and the available Air Navigation Services (ANS);
12. Any minima that may be promulgated by the State of the aerodrome;
13. The conditions prescribed in the operations specifications including any specific approvals for Low Visibility Operations (LVOs);
14. Any non-standard characteristics of the aerodrome, the IAP or the environment.

Heights used for calculation of Circling minima are related to the aerodrome elevation. Circling altitude is temperature corrected then rounded up to the nearest 100 ft.

Heights used for calculation of non-precision procedures may be related to aerodrome or threshold elevation. Refer to OM C for respective reference datum applied.

Decision Height (DH) for precision approaches are related to the threshold elevation of the landing runway.

Minimum Descent Altitude (DA/MDA) is temperature corrected if required then an additional 40 ft is added.

CAT II Decision Heights are related to the threshold elevation.

CAT II Radio Altimeter setting heights are related to the highest terrain elevation underneath the approach path within a radius of 30 m around the point on centerline above which the nominal glide path height equals the DH.

CAT III Decision Height/Radio Altimeter setting heights are not calculated individually since the standard DH value is approved for the particular aeroplane type and is published in Table 8.14 and 8.15.

Dimensional Units

• Decision Height/Altitude • Minimum Descent Height/Altitude • Ceiling	Feet
Visibility/RVR	Meters

8.1.3.3.1 Classification of Aeroplanes

CAT.OP.MPA.320

The approach classification for Malta Air aircraft is Category C as derived from the table below:

Aeroplane Classification

Aircraft categories corresponding to VAT values	
A	VAT
A	Less than 91 kt

Aircraft categories corresponding to VAT values	
B	From 91 to 120 kt
C	From 121 to 140 kt
D	From 141 to 165 kt
E	From 166 to 210 kt

8.1.3.3.2 Determination of DH, DA, MDH, MDA and T/O Minima

CAT.OP.MPA.110 (a), (c), (d)

Jeppesen provides aeronautical charts as an external service provider. The values for DAs/DHs, MDAs/MDHs, RA setting heights, ceiling and RVR/VIS values published on Jeppesen charts, are determined by Jeppesen in accordance with prescribed calculation procedures taking into account:

1. The provisions of CAT.OP.MPA.110.
2. OCAs, OCHs, OCLs and/or official minima values published by state authorities.
3. Any other particular regulations prescribed by state authorities.

Landing minima may have to be modified on short term basis. State authorities will publish raised or lowered OCA and/or OCH values only.

If an OCA/OCH published in NOTAM is higher than DH/DA/MDH/MDA indicated on IAC this higher value shall be used.

8.1.3.4 Aerodrome Operating Minima

AMC1 CAT.OP.MPA.110

Take-off Operations – Aeroplanes

8.1.3.4.1 Take-off Minima

Take-off minima are expressed as Visibility (VIS) or Runway Visual Range (RVR) limits, taking into account all relevant factors for each runway planned to be used and aircraft characteristics and equipment. Where there is a specific need to see and avoid obstacles on departure and/or for a forced landing, additional conditions, e.g. ceiling, should be specified. This ceiling requirement will be annotated in the Airfield Brief when applicable.

8.1.3.4.2 Visual Reference

1. The take-off minima will be selected to ensure sufficient guidance to control the aircraft in the event of both a rejected take-off in adverse circumstances and a continued take-off after failure of the critical engine.
2. For night operations, the prescribed runway lights shall be in operation.

8.1.3.4.3 Required RVR/VIS

1. Take-off minima specified by Malta Air are expressed as RVR or VIS Values not lower than those specified in Table 1.

Table 1 – Take-off (without LVTO Approval) RVR or VIS

Minimum RVR* or VIS*	Facilities
500 m (day)	Nil**
400 m (day)	Centre line markings; or Runway edge lights; or Runway centre line lights
400 m (night)	Runway end lights***; and Runway edge lights; or Runway centre line lights

*: The reported RVR or VIS value representative of the initial part of the take-off run can be replaced by pilot assessment.

**: The pilot is able to continuously identify the take-off surface and maintain directional control.

***: Runway end lights may be substituted by colour coded runway edge lights or colour-coded runway centre line lights.

8.1.3.4.4 Aerodrome Operating Minima – Determination of DH/MDH for Instrument Approach Operations

AMC3 CAT.OP.MPA.110

1. The Decision Height (DH) to be used for a 3D approach operation or a 2D approach operation flown using the Continuous Descent Final Approach (CDFA) technique should not be lower than the highest of:
 - a. the Obstacle Clearance Height (OCH) for the category of aircraft;
 - b. the published approach procedure DH or Minimum Descent Height (MDH) where applicable;
 - c. the system minima specified in Table 4;
 - d. the minimum DH permitted for the runway specified in Table 5; or
 - e. the minimum DH specified in the Aircraft Flight Manual (AFM) or equivalent document, if stated.

2.
- MAY does **NOT** currently conduct non CDFA approaches (OMA 8.1.3.4.15).

Table 4 – System Minima

Facility	Lowest DH/MDH (ft)
ILS/MLS/GLS	200
GNSS/SBAS (LPV)	200*
Precision approach radar (PAR)	200
GNSS/SBAS (LP)	250
GNSS (LNAV)	250
GNSS/Baro-VNAV (LNAV/VNAV)	250
LOC with or without DME	250
SRA (terminating at ½ NM)	250
SRA (terminating at 1 NM)	300
SRA (terminating at 2 NM or more)	350
VOR	300
VOR/DME	250
NDB	350
NDB/DME	300
VDF	350

*: For Localiser Performance with Vertical guidance (LPV), a DH of 200 ft may be used only if the published FAS datablock sets a vertical alert limit not exceeding 35 m. Otherwise, the DH should not be lower than 250 ft.

Table 5 Runway Type Minima

Runway Type		Lowest DH/MDH (ft)
Instrument runway	Precision Approach (PA) runway, Category I	200
	NPA runway	250
Non-Instrument runway	Non-instrument runway	Circling minima as shown in Table 15

3.
- Where a barometric DA/H or MDA/H is used, this should be adjusted where the ambient temperature is significantly below International Standard Atmosphere (ISA). The OMB Quick Reference Guide provides a cold temperature correction table for adjustment of minimum promulgated heights/altitudes.

8.1.3.4.5 Determination of RVR or VIS for Instrument Approach Operations

AMC5 CAT.OP.MPA.110

1. The RVR or VIS for straight-in instrument approach operations should be not less than the greatest of:
 - a. the minimum RVR or VIS for the type of runway used according to Table 8;
 - b. the minimum RVR determined according to the MDH or DH and class of lighting facility according to Table 9; or
 - c. the minimum RVR according to the visual and non-visual aids and on-board equipment used according to Table 10.

If the value determined in (a) is a VIS, then the result is a minimum VIS. In all other cases, the result is a minimum RVR.

2. If the approach is flown with a level flight segment at or above the MDA/H, then 400 m should be added to the RVR calculated in accordance with (1) for Category C and D aeroplanes.
3. The visual aids should comprise standard runway day markings, runway edge lights, threshold lights, runway end lights and approach lights as defined in Table 11.

Table 8 Type of Runway Versus Minimum RVR or VIS

Type of Runway	Minimum RVR or VIS (m)
PA runway Category I	RVR 550
NPA runway	RVR 750
Non-instrument runway	VIS according to Table 15 (circling minima)

Table 9 – RVR Versus DH/MDH

DH or MDH (ft)			Class of Lighting Facility			
			FALS	IALS	BALS	NALS
			RVR (m)			
200	–	210	550	750	1000	1200
211	–	240	550	800	1000	1200
241	–	250	550	800	1000	1300
251	–	260	600	800	1100	1300
261	–	280	600	900	1100	1300
281	–	300	650	900	1200	1400
301	–	320	700	1000	1200	1400

DH or MDH (ft)			Class of Lighting Facility			
			FALS	IALS	BALS	NALS
			RVR (m)			
321	–	340	800	1100	1300	1500
341	–	360	900	1200	1400	1600
361	–	380	1000	1300	1500	1700
381	–	400	1100	1400	1600	1800
401	–	420	1200	1500	1700	1900
421	–	440	1300	1600	1800	2000
441	–	460	1400	1700	1900	2100
461	–	480	1500	1800	2000	2200
481	–	500	1500	1800	2100	2300
501	–	520	1600	1900	2100	2400
521	–	540	1700	2000	2200	2400
541	–	560	1800	2100	2300	2400
561	–	580	1900	2200	2400	2400
581	–	600	2000	2300	2400	2400
601	–	620	2100	2400	2400	2400
621	–	640	2200	2400	2400	2400
641		660	2300	2400	2400	2400
661	and above		2400	2400	2400	2400

Table 10 – Visual and Non-visual Aids and/or On-board Equipment Versus Minimum RVR

Type of Approach	Facilities	Lowest RVR	
		Multi-pilot Operations	Single-pilot Operations
3D operations Final approach track offset $\leq 5^\circ$ for Category C and D aeroplanes	Runway touchdown zone lights (RTZL) and runway centre line lights (RCLL)	No limitation	
	Without RTZL and RCLL but using HUDLS or equivalent system; autopilot or flight director to the DH	No limitation	600 m
	No RTZL and RCLL, not using HUDLS or equivalent system or autopilot to the DH	750 m	800 m
3D operations	Runway touchdown zone lights (RTZL) and runway centre line lights (RCLL) and Final approach track offset $> 5^\circ$ for Category C and D aeroplanes	800 m	1000 m
	without RTZL and RCLL but using HUDLS or equivalent system; autopilot or flight director to the DH and Final approach track offset $> 5^\circ$ for Category C and D aeroplanes	800 m	1000 m
2D operations	Final approach track offset $\leq 5^\circ$ for Category C and D aeroplanes	750 m	800 m
	Final approach track offset $> 5^\circ$ for Category C and D aeroplanes	1200 m	1200 m

Table 11 – Approach Lighting Systems

Class of Lighting Facility	Length, Configuration, and Intensity of Approach Lights
FALS	CAT I lighting system (HIALS ≥ 720 m) distance coded centre line, barrette centre line
IALS	Simple approach lighting system (HIALS 420–719 m) single source, barrette
BALS	Any other approach lighting system (HIALS, MALS or ALS 210–419 m)
NALS	Any other approach lighting system (HIALS, MALS or ALS < 210 m) or no approach lights

4. For night operations or for any operation where credit for visual aids is required, the lights should be on and serviceable except as provided for in Table 17.

5. Where any visual or non-visual aid specified for the approach and assumed to be available in the determination of operating minima is unavailable, revised operating minima will need to be determined.

8.1.3.4.6 Aerodrome Operating Minima

GM4 CAT.OP.MPA.110

Means to Determine the Required RVR Based on DH and Lighting Facilities

The values in Table 9 are derived from the formula below:

Minimum RVR (m) = $[(DH/MDH \text{ (ft)} \times 0.3048)/\tan\alpha] - \text{length of approach lights (m)}$ where α is the calculation angle, being a default value of 3.00° increasing in steps of 0.10° for each line in Table 9 up to 3.77° and then remaining constant. An upper RVR limit of 2400 m has been applied to the table.

Use of DH for NPAS Flown Using the CDFA Technique

GM5 CAT.OP.MPA.110

OMA 8.1.3.4.4 (AMC3 CAT.OP.MPA.110) provides that, in certain circumstances, a published MDH may be used as a DH for a 2D operation flown using the CDFA technique.

The safety of the use of MDH as DH in CDFA operations has been verified by at least two independent analyses concluding that the CDFA using MDH as DH without any add-on is safer than the traditional step-down and level-flight NPA operation. A comparison has been made between the safety level of using MDH as DH without an add-on with the well-established safety level resulting from the ILS collision risk model. The NPA used was the most demanding, i.e. most tightly designed NPA, which offers the least additional margins. It should be noted that the design limits of the ILS approach design, e.g. the maximum GP angle of 3.5° , must be observed for the CDFA in order to keep the validity of the comparison.

Notwithstanding the text above, Minimum Descent Altitude (MDA/H) is temperature corrected if required then an additional 40 ft is added as required by the TMCAD in accordance with OMA 8.1.3.3.

8.1.3.4.7 Aerodrome Operating Minima

AMC7 CAT.OP.MPA.110

Circling Operations**1. Circling minima**

The following standards should apply for establishing circling minima for operations:

- a. the MDH for circling operation should not be lower than the highest of:
 - i. the published circling OCH for the aeroplane category;
 - ii. the minimum circling height derived from Table 15; or
 - iii. the DH/MDH of the preceding Instrument Approach Procedure (IAP);
- b. the MDA for circling should be calculated by adding the published aerodrome elevation to the MDH, as determined by (1)(a); and
- c. the minimum VIS for circling should be the higher of:
 - i. the circling VIS for the aeroplane category, if published; or
 - ii. the minimum VIS derived from Table 15.

Table 15 – Circling**MDH and Minimum VIS versus Aeroplane Category**

	Aeroplane Category	
	C	D
MDH (ft)	600	700
Minimum VIS (m)	2400	3600

2. Conduct of flight

- a. the MDH and OCH included in the procedure are referenced to aerodrome elevation;
- b. the MDA is referenced to mean sea level;
- c. for these procedures, the applicable visibility is the VIS; and
- d. Table 15 provides tabular guidance of the relationship between height above threshold and the in-flight visibility required to obtain and sustain visual contact during the circling manoeuvre.

3. Instrument approach followed by visual manoeuvring (circling) without prescribed tracks
 - a. When the aeroplane is on the initial instrument approach, before visual reference is established, but not below the MDA/H, the aeroplane should follow the corresponding IAP until the appropriate instrument MAPt is reached.
 - b. At the beginning of the level flight phase at or above the MDA/H, the instrument approach track should be maintained until the pilot:
 - i. estimates that, in all probability, visual contact with the runway of intended landing or the runway environment will be maintained during the entire circling procedure;
 - ii. estimates that the aeroplane is within the circling area before commencing circling; and
 - iii. is able to determine the aeroplane's position in relation to the runway of intended landing with the aid of the appropriate visual references.
 - c. If the pilot cannot comply with the conditions in (3)(b) at the MAPt then a missed approach should be executed in accordance with the IAP.
 - d. After the aeroplane has left the track of the initial instrument approach, the flight phase outbound from the runway should be limited to an appropriate distance, which is required to align the aeroplane onto the final approach. Such manoeuvres should be conducted to enable the aeroplane to:
 - i. attain a controlled and stable descent path to the intended landing runway; and
 - ii. remain within the circling area and in such way that visual contact with the runway of intended landing or runway environment is maintained at all times.
 - e. Flight manoeuvres should be carried out at an altitude/height that is not less than the circling MDA/H.
 - f. Descent below the MDA/H should not be initiated until the threshold of the runway to be used has been appropriately identified. The aeroplane should be in a position to continue with a normal rate of descent and land within the Touchdown Zone (TDZ).

4. Instrument approach followed by a visual manoeuvring (circling) with prescribed track

Note: Exceptionally Malta Air performs a circling manoeuvre with prescribed tracks in accordance with the standards below, and only when the procedure is described on the accompanying Airfield Brief.

- a. The aeroplane should remain on the initial instrument approach procedure until one of the following is reached:
 - i. the prescribed divergence point to commence circling on the prescribed track; or
 - ii. the MAPt.
- b. The aeroplane should be established on the instrument approach track in level flight at or above the MDA/H at or by the circling manoeuvre divergence point.
- c. If the divergence point is reached before the required visual reference is acquired, a missed approach should be initiated not later than the MAPt and completed in accordance with the instrument approach procedure.
- d. When commencing the prescribed circling manoeuvre at the published divergence point, the subsequent manoeuvres should be conducted to comply with the published routing and published heights/altitudes.
- e. Unless otherwise specified, once the aeroplane is established on the prescribed track(s), the published visual reference does not need to be maintained unless:
 - i. required by the State of the aerodrome; or
 - ii. the circling MAPt (if published) is reached.
- f. If the prescribed circling manoeuvre has a published MAPt and the required visual reference has not been obtained by that point, a missed approach should be executed in accordance with (5)(b) and (5)(c).
- g. Subsequent further descent below MDA/H should only commence when the required visual reference has been obtained.
- h. Unless otherwise specified in the procedure, final descent should not be commenced from MDA/H until the threshold of the intended landing runway has been identified and the aeroplane is in a position to continue with a normal rate of descent to land within the touchdown zone.

5. Missed approach

- a. Missed approach during the instrument procedure prior to circling:
 - i. If the missed approach procedure is required to be flown when the aeroplane is positioned on the instrument approach track defined by radio-navigation aids RNAV, RNP, or ILS, MLS, and before commencing the circling manoeuvre, the published missed approach for the instrument approach should be followed; or
 - ii. If the IAP is carried out with the aid of an ILS, an MLS or a Stabilised Approach (SAP), the MAPt associated with an ILS, or an MLS procedure without Glide Path (GP-out procedure) or the SAP, where applicable, should be used.
- b. If a prescribed missed approach is published for the circling manoeuvre, this overrides the manoeuvres prescribed below.
- c. If visual reference is lost while circling to land after the aeroplane has departed from the initial instrument approach track, the missed approach specified for that particular instrument approach should be followed. It is expected that the pilot will make an initial climbing turn toward the intended landing runway to a position overhead the aerodrome where the pilot will establish the aeroplane in a climb on the instrument missed approach segment.
- d. The aeroplane should not leave the visual manoeuvring (circling) area, which is obstacle-protected, unless:
 - i. established on the appropriate missed approach procedure; or
 - ii. at Minimum Sector Altitude (MSA).
- e. All turns should be made in the same direction and the aeroplane should remain within the circling protected area while climbing either:
 - i. to the altitude assigned to any published circling missed approach manoeuvre if applicable;
 - ii. to the altitude assigned to the missed approach of the initial instrument approach;
 - iii. to the MSA; or

- iv. to the Minimum Holding Altitude (MHA) applicable to transition to a holding facility or fix, or continue to climb to an MSA;

or as directed by ATS.

When the missed approach procedure is commenced on the 'downwind' leg of the circling manoeuvre, an 'S' turn may be undertaken to align the aeroplane on the initial instrument approach missed approach path, provided the aeroplane remains within the protected circling area.

The commander should be responsible for ensuring adequate terrain clearance during the above-stipulated manoeuvres, particularly during the execution of a missed approach initiated by ATS.

- f. Because the circling manoeuvre may be accomplished in more than one direction, different patterns will be required to establish the aeroplane on the prescribed missed approach course depending on its position at the time visual reference is lost. In particular, all turns are to be in the prescribed direction if this is restricted, e.g. to the west/east (left or right hand) to remain within the protected circling area.
- g. If a missed approach procedure is published for a particular runway onto which the aeroplane is conducting a circling approach and the aeroplane has commenced a manoeuvre to align with the runway, the missed approach for this direction may be accomplished. The ATS unit should be informed of the intention to fly the published missed approach procedure for that particular runway.
- h. The commander should advise ATS when any missed approach procedure has been commenced, the height/altitude the aeroplane is climbing to and the position the aeroplane is proceeding towards and/or heading the aeroplane is established on.

6. Circling at Night

- a. At night, the maximum reported surface crosswind component for a circling approach is 20 kts. Gust values shall not exceed 40 kts.
- b. At night, the ceiling required before commencing an approach shall be Circling Minimums +300 ft.
- c. "New" Captains may not conduct Circling approaches at night.

8.1.3.4.8 Aerodrome Operating Minima

AMC9 CAT.OP.MPA.110

Visual Approach Operations

Malta Air will not use an RVR of less than 800 m for a visual approach.

8.1.3.4.9 Aerodrome Operating Minima – Conversion of Visibility to CMV

AMC 10 CAT.OP.MPA.110

The following conditions apply to the use of Converted Meteorological Visibility (CMV) instead of RVR:

- 1. If the reported RVR is not available, a CMV may be substituted for the RVR, except:
 - a. to satisfy the take-off minima; or
 - b. for the purpose of continuation of an approach in LVOs.
- 2. If the minimum RVR for an approach is more than the maximum value assessed by the aerodrome operator, then CMV should be used.
- 3. In order to determine CMV from visibility:
 - a. for flight planning purposes, a factor of 1.0 should be used;
 - b. for purposes other than flight planning, the conversion factors specified in Table 16 should be used.

Table 16 – Conversion of Reported VIS to RVR/CMV

Light Elements in Operation	RVR/CMV = Reported VIS ×	
	Day	Night
HI approach and runway lights	1.5	2.0
Any type of light installation other than above	1.0	1.5
No lights	1.0	not applicable

8.1.3.4.10 Effect on Landing Minima of Temporarily Failed or Downgraded Ground Equipment

AMC11 CAT.OP.MPA.110

Aerodrome Operating Minima

- 1. General

These instructions are intended for use both before and during flight. It is, however, not expected that the commander would consult such instructions after passing 1000 ft above the aerodrome. If failures of ground aids are announced at such a late stage, the approach could be continued at the

commander's discretion. If failures are announced before such a late stage in the approach, their effect on the approach should be considered as described in Table 17, and the approach may have to be abandoned.

2. Conditions applicable to Table 17:
 - a. multiple failures of runway/FATO lights other than those indicated in Table 17 should not be acceptable;
 - b. failures of approach and runway/FATO lights are acceptable at the same time, and the most demanding consequence should be applied; and
 - c. failures other than ILS, GLS, MLS affect the RVR only and not the DH.

Table 17 – Failed or Downgraded Equipment – Effect on Landing Minima Operations without a LVO Approval

Failed or Downgraded Equipment	Effect on Landing Minima	
	Type B	Type A
Navaid stand-by transmitter	No effect	
Outer marker (ILS only)	Not allowed except if the required height versus glide path can be checked using other means, e.g. DME fix	APV – not applicable
		NPA with Final Approach Fix (FAF): no effect unless used as FAF
		If the FAF cannot be identified (e.g. no method available for timing of descent), NPA cannot be conducted
Middle marker (ILS only)	No effect	No effect unless used as MAPt.
RVR Assessment Systems	No effect	
Approach lights	Minima as for NALS	
Approach lights except the last 210 m	Minima as for BALS	
Approach lights except the last 420 m	Minima as for IALS	
Standby power for approach lights	No effect	
Edge lights, threshold lights and runway end lights	Day: no effect; Night: not allowed	

Failed or Downgraded Equipment	Effect on Landing Minima	
	Type B	Type A
Centre line lights	No effect if Flight Director (F/D), HUDLS or autoland; otherwise RVR 750 m	No effect
Centreline lights spacing increased to 30 m	No effect	
TDZ lights	No effect if F/D, HUDLS or autoland; otherwise RVR 750 m	No effect
Taxiway lighting system	No effect	

8.1.3.4.11 Aerodrome Operating Minima

GM2 CAT.OP.MPA.110

Approach Lighting Systems – ICAO, FAA

The following table provides a comparison of ICAO and FAA specifications.

Table 19 – Approach Lighting Systems – ICAO and FAA Specifications

Class of Lighting Facility	Length, Configuration and Intensity of Approach Lights
FALS	ICAO: CAT I lighting system (HIALS ≥ 720 m) distance coded centre line, barrette centre line FAA: ALSF1, ALSF2, SSALR, MALSR, high or medium intensity and/or flashing lights, 720 m or more
IALS	ICAO: simple approach lighting system (HIALS 420–719 m) single source, barrette FAA: MALSF, MALS, SALS/SALSF, SSALF, SSALS, high or medium intensity and/or flashing lights, 420–719 m
BALS	Any other approach lighting system (HIALS, MALS or ALS 210–419 m) FAA: ODALS, high or medium intensity or flashing lights 210–419 m
NALS	Any other approach lighting system (HIALS, MALS or ALS <210 m) or no approach lights

8.1.3.4.12 Aerodrome Operating Minima

CAT.OP.MPA.115

Approach Flight Technique

All instrument procedures, including NPAs, shall be flown using the Stabilised Approach technique (SAp) and from the FAF to the MAP or MDA the continuous descent final approach (CDFA) technique.

8.1.3.4.13 Increments Specified by the Competent Authority

GM6 CAT.OP.MPA.110

Aerodrome Operating Minima

Additional increments to the published minima may be specified by the TMCAD to take into account certain operations.

8.1.3.4.13.1 Aerodrome Operating Minima

GM8 CAT.OP.MPA.110

Low Temperature Correction

When the aerodrome temperature is below 0 degrees celsius a correction should be applied to the DA/H. The OMB Quick Reference Guide may be used to determine the correction that should be applied. The calculations in the table are for a sea level aerodrome; they are therefore conservative when applied at higher level aerodromes.

8.1.3.4.14 Continuous Descent Final Approach (CDFA)

AMC1 CAT.OP.MPA.115

Approach Flight Technique

The following criteria apply to CDFA:

1. The CDFA technique ensures that an approach can be flown on the desired vertical path and track in a stabilised manner, without significant vertical path changes during the final segment descent to the runway. While level flight is permitted prior to the FAF (e.g. V/S approaches), level flight is **not** permitted in the final approach segment, which commences at the FAF. This technique applies to an approach with no vertical guidance and controls the descent path until the DA/DH. Malta Air provides information allowing the flight crew to determine the appropriate descent path for each NPA procedure to be used. This information is either:
 - a. a descent path depicted on the approach chart including check altitude/heights against range;
 - b. a descent path coded into the aircraft flight management system; or
 - c. a recommended descent rate based on estimated ground speed.
2. The information provided to the crew observes human factors principles.

3. The descent path should be calculated to pass at or above the minimum altitude specified at any step-down fix. Jeppesen charts depict the appropriate cross check altitudes/heights with the corresponding appropriate range information. Generally, the MAPt is published on the chart.
4. The optimum angle for the descent path is 3 degrees and should not exceed 4.5 degrees.
5. Malta Air has established procedures that require:
 - a. the pilot monitoring to verbalise deviations from the required descent path;
 - b. the pilot flying to make prompt corrections to deviation from the required descent path; and
 - c. a call-out to be made when the aircraft is approaching the DA/H.
6. A missed approach should be executed promptly at the DA/H or the MAPt, whichever is first, if the required visual references have not been established.
7. For approaches other than circling approaches, the lateral part of the missed approach should be flown via the MAPt unless otherwise stated on the approach chart.

8.1.3.4.15 Approach Flight Technique

AMC2 CAT.OP.MPA.115

NPA Operations Without Applying the CDFA Technique

Any Non CDFA approach to any runway end requires prior approval from the TM CAD.

MAY does **NOT** currently conduct non CDFA approaches.

Circling is not considered CDFA, however the instrument approach to the circling altitude is CDFA from the FAF.

8.1.3.4.15.1 Approaches Offset by More than 5 Degrees

When final approach track is offset by more than 5 degrees the approach shall:

1. be flown using Stabilised Approach techniques (SAp) and CDFA.
2. have the published RVR/CMV increased by 400 m (max 5000 m).

8.1.3.4.16 Operational Procedures and Instructions and Training

AMC3 CAT.OP.MPA.115

Approach Flight Technique

It is Malta Air policy that the SAP and CDFA approaches will be flown including RNP APCH Operations. Maximum use of the AFDS applies. All approved and trained approach techniques are presented in FCOM Vol I and Boeing FCTM, including the duties of each crew member during the conduct of the approach.

Malta Air has established procedures which prohibit level flight at MDA/H without the flight crew having obtained the required visual references. It is not the intention to prohibit level flight at MDA/H when conducting a circling approach, which does not come within the definition of the CDFA technique.

Initial and recurrent flight crew training required by ORO.FC includes the use of the CDFA technique.

8.1.3.4.17 Stabilised Approach Operations

CAT.OP.MPA.115, AMC1 CAT.OP.MPA.115(a)

Approach Flight Technique

All approaches shall be flown as stabilised approach operations unless otherwise approved by the Competent Authority for a particular approach to a particular runway.

The following criteria should be satisfied for all stabilised approach operations:

1. The flight management systems and approach aids should be correctly set, and any required radio aids identified before reaching a predetermined point or altitude/height on the approach.
2. The aeroplane should be flown according to the following criteria from a predetermined point or altitude/height on the approach:
 - a. the angle of bank should be less than 30 degrees; and
 - b. the target rate of descent should be that required to maintain the correct vertical path at the planned approach speed.
3. Variations in the rate of descent should normally not exceed 50% of the target rate of descent.
4. An aeroplane should be considered stabilised for landing when the following conditions are met (the below requirements are satisfied by complying with the LDA Stabilization Criteria for approaches, OMA 8.3.0.3.4):
 - a. the aeroplane is tracking within an acceptable tolerance of the required lateral path;

- b. the aeroplane is tracking within an acceptable tolerance of the required vertical path;
 - c. the vertical speed of the aeroplane is within an acceptable tolerance of the required rate of descent;
 - d. the airspeed of the aeroplane is within an acceptable tolerance of the intended landing speed;
 - e. the aeroplane is in the correct configuration for landing, unless operating procedures require a final configuration change for performance reasons after visual reference is acquired; and
 - f. the thrust/power and trim settings are appropriate.
- 5. The aeroplane should be stabilised for landing before reaching 500 ft above the landing runway threshold elevation. Refer to OMA 8.3.0.3.4.
- 6. For approach operations where the pilot does not have visual reference with the ground, the aeroplane should additionally be stabilised for landing before reaching 1000 ft above the landing runway threshold elevation. In the case of circling approaches flown after a CDFA, the aircraft will be stabilised in the circling configuration (Landing Gear Down, Flap 15, on speed) not later than passing 1000 ft above the runway elevation.
- 7. Malta Air specify the following in OMA 8.3.0.3.4:
 - a. the acceptable tolerances referred to in (4);
 - b. the means to identify the predetermined points referred to in (1) and (2) is described in OMA 8.3.0.3.9. This should normally be the FAF.
- 8. When an operator requests approval for an alternative to the stabilised approach criteria for a particular approach to a particular runway, the operator should demonstrate that the proposed alternative will ensure that an acceptable level of safety is achieved.

8.1.3.4.18 Noise Abatement Procedures NADP DESIGN

AMC1 CAT.OP.MPA.130

- 1. For each aeroplane type two departure procedures are defined, in accordance with ICAO Doc. 8168 (Procedures for Air Navigation Services, 'PANS-OPS'), Volume I:
 - a. noise abatement departure procedure one (NADP 1), is designed to meet the close-in noise abatement objective; and
 - b. noise abatement departure procedure two (NADP 2), is designed to meet the distant noise abatement objective.

2. For each type of NADP (1 and 2), a single climb profile is specified for use at all aerodromes, which is associated with a single sequence of actions.

The NADP procedures are described in detail in FCOM Vol I. Airfield Briefs will indicate which NADP profile is required but it is Malta Air policy to use NADP2 when possible. Where there is a discrepancy between Malta Air policy and a charted procedure Malta Air policy applies.

8.1.4 Enroute Operating Minima for VFR flights: N/A MAY

CAT.OP.MPA.135 (c)

8.1.5 Presentation and Application of Aerodrome and En-route Minima

CAT.OP.MPA.135

1. Malta Air operations are only conducted along routes, or within areas, for which:
 - a. Space-based facilities, ground facilities and services, including meteorological services, adequate for the planned operation are provided;
 - b. the performance of the aircraft is adequate to comply with minimum flight altitude requirements;
 - c. the equipment of the aircraft meets the minimum requirements for the planned operation; and
 - d. appropriate maps and charts are available;
2. Malta Air ensures that operations are conducted in accordance with any restriction on the routes or the areas of operation specified by the CAD
3. N/A Malta Air

8.1.5.1 Planning Minima for IFR Flights

CAT.OP.MPA.185

1. Malta Air has established procedures for in-flight fuel management that ensure:
 - a. continual validation of the assumptions made during the planning stage (pre-flight or in-flight re-planning, or both);
 - b. re-analysis and adjustment, if necessary;
 - c. that the amount of usable fuel remaining on board is protected and not less than the fuel that is required to proceed to an aerodrome where a safe landing can be made; and
 - d. relevant fuel data for the purpose of points (a), (b), and (c) shall be recorded.

2. Malta Air has procedures in place to require the commander to obtain delay information from a reliable source when unforeseen circumstances may result in landing at the destination aerodrome with less than the final reserve fuel plus any fuel to proceed to an alternate aerodrome, if required.
3. The commander shall advise Air Traffic Control (ATC) of a ‘minimum fuel’ state by declaring ‘MINIMUM FUEL’ in accordance with OMA when the commander has:
 - a. committed to land at a specific aerodrome; and
 - b. calculated that any change to the existing clearance to that aerodrome may result in landing with less than the planned final reserve fuel.
4. The commander shall declare a situation of ‘fuel emergency’ by broadcasting ‘MAYDAY, MAYDAY, MAYDAY FUEL’ in accordance with OMA, when the usable fuel that is calculated to be available upon landing at the nearest aerodrome where a safe landing can be made is less than the planned final reserve fuel.

8.1.5.2 Basic Fuel Scheme – Aerodrome Forecast Meteorological Conditions

AMC3 CAT.OP.MPA.182

Table 1 – Aerodrome Forecasts (TAFs) and Landing Forecasts (TRENDS) to be Used for Pre-flight Planning.

APPLICATION OF AERODROME FORECASTS (TAF AND TREND) TO PRE-FLIGHT PLANNING						
(a) APPLICATION OF INITIAL PART OF TAF						
(1) Application period: from the start of the TAF validity period to the time of applicability of the first subsequent 'FM...' or 'BECMG' or if no 'FM...' or 'BECMG' is given, to the end of the validity period of the TAF						
(2) Application of forecast: the forecast of the prevailing weather conditions in the initial part of the TAF should be fully applied, with the exception of mean wind and gusts that should be applied in accordance with the policy under column 'BECMG AT and FM...' in the table below. However, this may be temporarily superseded by a 'TEMPO' or 'PROB***', if applicable according to the table below.						
(b) APPLICATION OF FORECAST FOLLOWING CHANGE INDICATION IN THE TAF AND TREND						
TAF or TREND for AERODROME PLANNED AS:	FM... (alone) and BECMG AT:	BECMG (alone), BECMG FM, BECMG TL, BECMG FM... TL, in case of:		TEMPO (alone), TEMPO FM, TEMPO FM... TL, PROB 30/40 (alone)		PROB TEMPO
	Deterioration and improvement	Deterioration	Improvement	Deterioration		Improvement
				Transient / shower conditions in connection with short-lived weather phenomena, e.g. thunderstorms, showers	Persistent conditions in connection with e.g. haze, mist, fog, dust storm / sandstorm, continuous precipitations	In any case
DESTINATION at (ETA) ± 1 HR	Applicable from the start of change	Applicable from the start of change	Applicable from the end of change	Not applicable	Applicable	
TAKE-OFF ALTERNATE at ETA ± 1 HR	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind and gusts exceeding required limits may be disregarded	Mean wind should be within required limits	
DESTINATION ALTERNATE at ETA ± 1 HR	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied		Gusts exceeding crosswind limits should be fully applied	
FUEL ERA at ETA ± 1 HR						Should be disregarded
ETOPS ERA from earliest ETA to ETA + 1 HR	Applicable from the start of change	Applicable from the start of change	Applicable from the end of change	Applicable if below applicable landing minima	Applicable if below applicable landing minima	Improvement should be disregarded including mean wind and gusts
	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind should be within required limits	
	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	

* The space following 'FM' should always include a time group, e.g. FM1030.
 Note 1: 'required limits' are those contained in the OM.
 Note 2: If promulgated aerodrome forecasts do not comply with the provisions of ICAO Annex 3, operators should ensure that guidance on the application of these reports is provided.
 Note 3: For the definitions of the meteorological terms used in this table, see ICAO Annex 3.

8.1.5.3 Basic Fuel Scheme – Reaching the Destination Aerodrome

AMC4 CAT.OP.MPA.182

In the context of the basic fuel scheme and basic fuel scheme with variations, 'reaching the destination' means the point at which the aircraft has reached the applicable DA/H or MDA/H at the destination aerodrome.

8.1.5.4 Basic Fuel Scheme – Safety Margins for Meteorological Conditions

AMC5 CAT.OP.MPA.182

1. An aerodrome should only be selected as a:

- take-off alternate aerodrome; or
- destination aerodrome;

when the appropriate weather reports and/or forecasts indicate that during a period commencing 1 hour before and ending 1 hour after the estimated time of arrival at the aerodrome, the weather conditions will be at or above the applicable landing minima as follows:

- RVR or VIS specified in accordance with point CAT.OP.MPA.110 (OMA 8.1.3.4); and
- for a type A or a circling operation, ceiling at or above MDH.

2.

An aerodrome should only be selected as a:

a.

destination alternate aerodrome; or

b.

fuel ERA aerodrome;

when the appropriate weather reports and/or forecasts indicate that during a period commencing 1 hour before and ending 1 hour after the estimated time of arrival at the aerodrome, the weather conditions will be at or above the planning minima.
3.

For the take-off alternate aerodrome, any limitations related to OEI operations should be taken into account.

8.1.5.5 **Basic Fuel Scheme – Planning Minima**

AMC6 CAT.OP.MPA.182

An aerodrome should only be selected as a:

1.

Destination alternate aerodrome; or
2.

Fuel ERA aerodrome;

only when the appropriate weather reports and/or forecasts indicate that the weather conditions will be at or above the planning minima of Table 2 below (any limitations related to OEI operations are also taken into account):

Table 2 – Basic Fuel Scheme – Planning Minima Destination Alternate Aerodrome, Fuel ERA Aerodrome

Type of Approach Operation	Aerodrome Ceiling (Cloud Base or Vertical Visibility)	RVR/VIS
Type B instrument approach operations	DA/H +200 ft	RVR/VIS +800 m
Type A instrument approach operations	DA/H or MDA/H +400 ft	RVR/VIS +1500 m
Circling approach operations	MDA/H +400 ft	VIS +1500 m
Crosswind planning minima: see Table 1 of AMC3 CAT.OP.MPA182 (OMA 8.1.5.2)		
Wind limitations should be applied taking into account the runway condition (dry, wet, contaminated)		

8.1.5.6 **Meteorological Conditions Required**

CAT.OP.MPA.245

1.

On IFR flights, the commander shall only:

a.

commence the flight; or

- b. continue beyond the point from which a revised ATS flight plan applies in the event of in-flight re-planning,

when information is available indicating that the expected meteorological conditions, at the time of arrival, at the destination and/or required alternate aerodrome(s) are at or above the planning minima.
2. On IFR flights, the commander shall only continue towards the planned destination aerodrome when the latest information available indicates that, at the expected time of arrival, the meteorological conditions at the destination, or at least one destination alternate aerodrome, are at or above the applicable aerodrome operating minima.

8.1.6 Interpretation of Meteorological Information

The Jeppesen General Airway Manual (OMC) contains all necessary information and explanatory material on the decoding of MET forecasts and MET reports for each State in the area of operations. This includes interpretation of conditional expressions. This information is extracted by Jeppesen from ICAO documentation and the GEN sections of AIPs of States in the area of operations.

8.1.7 Determination of the Quantities of Fuel and Oil Carried

8.1.7.1 Use of Isolated Aerodromes

N/A Malta Air.

8.1.7.2 Malta Air Fuel Scheme

CAT.OP.MPA.180 (a)(3)(ii)

Malta Air is approved to utilise a Basic Fuel Scheme with Variations.

When applicable, Malta Air may decide to deviate from the Basic Fuel Scheme, satisfying the AMC for Basic Fuel Scheme with Variations applicable to the specific deviation. Whenever a Basic Fuel Scheme with Variation is applied, this is specified in the OMA section accordingly (e.g. “*Basic Fuel Scheme with Variation*” is indicated in the title).

Currently Malta Air is approved for Basic fuel scheme with variations – TAXI FUEL, and CONTINGENCY FUEL.

1. Malta Air has established, implemented, and maintains a fuel scheme that:
 - a. is appropriate for the type(s) of operation performed;
 - b. corresponds to the capability of Malta Air to support its implementation; and

- c. is either:
 - i. a basic fuel scheme, which shall form the basis for a basic fuel scheme with variations and an individual fuel scheme; the basic fuel scheme derives from a large-scale analysis of safety and operational data from previous performance and experience of the industry, applying scientific principles; the basic fuel scheme shall ensure, in this order, a safe, effective, and efficient operation of the aircraft; or
 - ii. a basic fuel scheme with variations, which is a basic fuel scheme where the analysis referred to in point (i) is used to establish a variation to the basic fuel scheme that ensures, in this order, a safe, effective, and efficient operation of the aircraft.
- 2. All fuel schemes shall comprise:
 - a. a fuel planning and in-flight re-planning policy;
 - b. an aerodrome selection policy; and
 - c. an in-flight fuel management policy.
- 3. The fuel scheme and any change to it shall require prior approval by the TMCAD.

8.1.7.3 Fuel Scheme – Fuel Planning and In-flight Re-planning Policy

CAT.OP.MPA.181

- 1. Malta Air:
 - a. has established a fuel planning and in-flight re-planning policy as part of the fuel scheme;
 - b. ensures that the aeroplane carries a sufficient amount of usable fuel to safely complete the planned flight and to allow for deviations from the planned operation;
 - c. has developed procedures for the fuel planning and in-flight re-planning policy that are contained in the operations manual.
 - d. ensures that the fuel planning of the flight is based on:
 - i. current aircraft-specific data derived from a fuel consumption monitoring system or, if not available;
 - ii. data provided by the aeroplane manufacturer.
- 2. Malta Air ensures that the planning of flights includes the operating conditions under which the flight is to be conducted; the operating conditions shall include at least:
 - a. aircraft fuel consumption data;

- b. anticipated masses;
 - c. anticipated meteorological conditions;
 - d. the effects of deferred maintenance items and/or of configuration deviations;
 - e. the expected departure and arrival routing and runways; and
 - f. anticipated delays.
3. Malta Air ensures that the pre-flight calculation of the usable fuel that is required for a flight includes:
- a. taxi fuel that shall not be less than the amount expected to be used prior to take-off;
 - b. trip fuel that shall be the amount of fuel that is required to enable the aeroplane to fly from take-off, or from the point of in-flight re-planning, to landing at the destination aerodrome;
 - c. contingency fuel that shall be the amount of fuel required to compensate for unforeseen factors;
 - d. destination alternate fuel:
 - i. when a flight is operated with at least one destination alternate aerodrome, it shall be the amount of fuel required to fly from the destination aerodrome to the destination alternate aerodrome; or
 - ii. when a flight is operated with no destination alternate aerodrome, it shall be the amount of fuel required to hold at the destination aerodrome, while enabling the aeroplane to perform a safe landing, and to allow for deviations from the planned operation; as a minimum, this amount shall be 15-minute fuel at holding speed at 1500 ft (450 m) above the aerodrome elevation in standard conditions, calculated according to the estimated aeroplane mass on arrival at the destination aerodrome;
 - e. final reserve fuel that shall be the amount of fuel that is calculated at holding speed at 1500 ft (450 m) above the aerodrome elevation in standard conditions according to the aeroplane estimated mass on arrival at the destination alternate aerodrome, or destination aerodrome when no destination alternate aerodrome is required, and shall not be less than (for turbine-engined aeroplanes) the fuel to fly for 30 minutes;
 - f. additional fuel, if required by the type of operation; it shall be the amount of fuel to enable the aeroplane to land at a fuel en route alternate aerodrome (fuel ERA aerodrome critical scenario) in the

event of an aircraft failure that significantly increases the fuel consumption at the most critical point along the route; this additional fuel is required only if the minimum amount of fuel that is calculated according to points (3)(b) to (3)(e) is not sufficient for such an event;

- g. extra fuel to take into account anticipated delays or specific operational constraints; and
 - h. discretionary fuel, if required by the commander.
4. Malta Air ensures that in-flight re-planning procedures for calculating the usable fuel that is required when a flight proceeds along a route or to a destination aerodrome other than the ones originally planned include points (3)(b) to (3)(g).

LIDO as the flight planning service provider complies fully with the Malta Air fuel scheme. No Malta Air revenue flight shall be planned to arrive at destination with less than 1800 kg of fuel. This means that the DEST Fuel in the PROGRESS page of the FMC shall indicate not less than 1800 kg at dispatch.

8.1.7.3.1 Basic Fuel Scheme – Pre-flight Calculation of Usable Fuel for Performance Class A Aeroplanes

AMC1 CAT.OP.MPA.181

For the pre-flight calculation of the usable fuel in accordance with OMA 8.1.7.3 (CAT.OP.MPA.181), Malta Air shall:

1. For taxi fuel, take into account the local conditions at the departure aerodrome and the APU consumption;

Note 1: LIDO Taxi Fuel includes fuel for taxi, APU usage, engine start and anti-ice. Lido plans taxi times based on available airport/runway taxi time data. If no data is available, then the default taxi time of 15 minutes is used.

Note 2: Malta Air may apply Basic Fuel Scheme with Variations – Taxi Fuel in accordance with OMA 8.1.7.3.1.2.

2. For trip fuel, include:
- a. fuel for take-off and climb from the aerodrome elevation to the initial cruising level/altitude, taking into account the expected departure routing;
 - b. fuel from the top of climb to the top of descent, including any step climb/descent;
 - c. fuel from the top of descent to the point where the approach procedure is initiated, taking into account the expected arrival routing; and

- d. fuel for making an approach and landing at the destination aerodrome;
- 3. For contingency fuel, calculate for unforeseen factors either:
 - a. 5% of the planned trip fuel or, in the event of in-flight re-planning, 5% of the trip fuel for the remainder of the flight; or
 - b. an amount to fly for 5 minutes at holding speed at 1500 ft (450 m) above the destination aerodrome in standard conditions,
whichever is the higher;

Note: Malta Air may apply Basic Fuel Scheme with Variations - Contingency Fuel in accordance with OMA 8.1.7.3.1.3.
- 4. For destination alternate fuel, include:
 - a. when the aircraft is operated with one destination alternate aerodrome:
 - i. fuel for a missed approach from the applicable DA/H or MDA/H at the destination aerodrome to the missed-approach altitude, taking into account the complete missed-approach procedure;
 - ii. fuel for climb from the missed-approach altitude to the cruising level/altitude, taking into account the expected departure routing;
 - iii. fuel for cruising from the top of climb to the top of descent, taking into account the expected routing;
 - iv. fuel for descent from the top of descent to the point where the approach is initiated, taking into account the expected arrival routing; and
 - v. fuel for making an approach and landing at the destination alternate aerodrome; and
 - b. when the aircraft is operated with two destination alternate aerodromes, the amount of fuel that is calculated in accordance with point (4)(a), based on the destination alternate aerodrome that requires the greater amount of fuel;
- 5. For FRF, comply with OMA 8.1.7.3(4) (CAT.OP.MPA.181(c));
- 6. For additional fuel, include an amount of fuel that allows the aeroplane to proceed, in the event of an engine failure or loss of pressurisation, from the most critical point along the route to a fuel en route alternate (fuel ERA)

aerodrome in the relevant aircraft configuration, hold there for 15 minutes at 1500 ft (450 m) above the aerodrome elevation in standard conditions, make an approach, and land;

7. For extra fuel, include anticipated delays or specific operational constraints that can be predicted; and
8. For discretionary fuel, include a quantity at the sole discretion of the commander.

8.1.7.3.1.1 Basic Fuel Scheme – Pre-flight Calculation of Usable Fuel

AMC4 CAT.OP.MPA.181

The additional fuel required by the type of operation in the event of an aircraft failure that significantly increases fuel consumption at the most critical point along the route should be calculated according to the engine failure or loss of pressurisation, whichever requires a greater amount of fuel. This additional fuel is described in OMA Section 8.3.2.4.2.5.1.

8.1.7.3.1.2 Basic Fuel with Variations – Taxi Fuel

AMC5 CAT.OP.MPA.181

To calculate taxi fuel for a basic fuel scheme with variations Malta Air may use statistical taxi fuel.

8.1.7.3.1.3 Basic Fuel with Variations – Contingency Fuel

AMC6 CAT.OP.MPA.181

1. Contingency fuel variations are methods of reducing the basic amount of contingency fuel based on established mitigating measures.
2. Malta Air has established and maintains a fuel consumption monitoring system for individual aeroplanes and uses valid data for fuel calculation based on such a system, and therefore may use any of the requirements in point (3) to calculate the contingency fuel.
3. The contingency fuel should be the fuel described in points (3)(a) or (3)(b) whichever is higher:
 - a. an amount of fuel that should be either:
 - i. not less than 3% of the planned trip fuel, or in the event of in-flight re-planning, 3% of the trip fuel for the remainder of the flight provided that a fuel en route alternate (fuel ERA) aerodrome is available; or
 - ii. an amount of fuel sufficient for 20-minute flying time based upon the planned trip fuel consumption;
 - b. an amount of fuel to fly for 5 minutes at holding speed at 1500 ft (450 m) above the destination aerodrome in standard conditions.

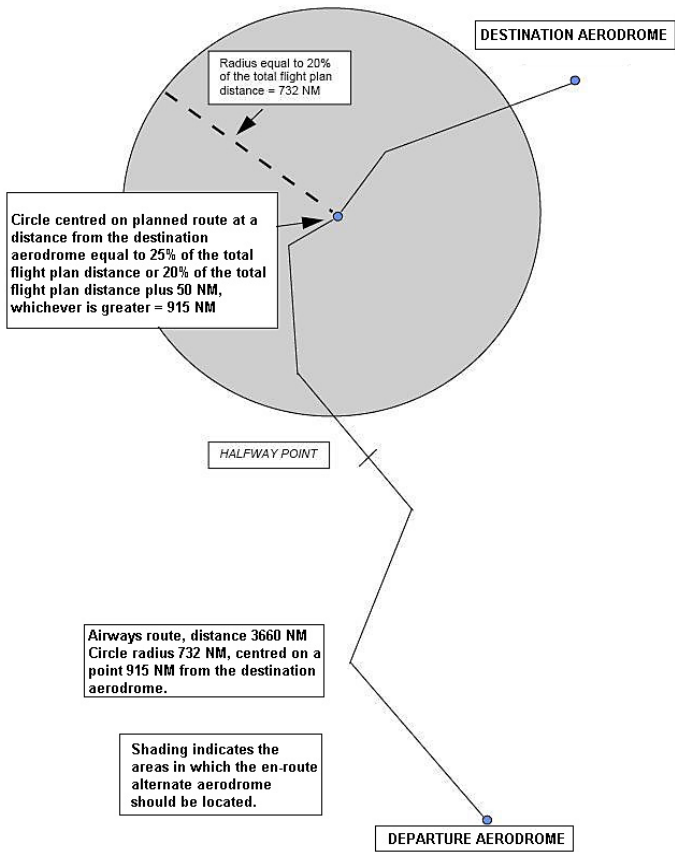
8.1.7.4 Basic Fuel with Variations – Location of the Fuel En Route Alternate Aerodrome to Reduce Contingency Fuel to 3%

AMC7 CAT.OP.MPA.181

The use of a fuel ERA allows contingency fuel to be reduced to 3% of the trip fuel.

The fuel en route alternate (fuel ERA) aerodrome should be located within a circle with a radius equal to 20% of the total flight plan distance; the centre of this circle lies on the planned route at a distance from the destination aerodrome equal to 25% of the total flight plan distance, or at least 20% of the total flight plan distance plus 50 NM, whichever is greater. All distances should be calculated in still-air conditions (see Figure 1). The fuel ERA aerodrome should be nominated in the operational flight plan.

Figure 1 Location of the Fuel ERA Aerodrome for the Purposes of Reducing Contingency Fuel to 3%



8.1.7.5 Basic Fuel with Variations – Statistical Contingency Fuel Method

GM2 CAT.OP.MPA.181

N/A Malta Air.

8.1.7.6 Contingency Fuel

Contingency fuel is the amount of fuel required to compensate for unforeseen factors.

Unforeseen factors are those that could have an influence on the fuel consumption to the destination aerodrome, such as deviations of an individual aeroplane from the expected fuel consumption data, deviations from forecast meteorological conditions, extended unexpected delays in flight, extended unexpected taxi times, and deviations from planned routings and/or cruising levels.

Malta Air identifies each flight on which contingency fuel is used and the frequency of this and adjusts the LIDO OFP accordingly until such flights are conducted with contingency fuel remaining in compliance with the fuel monitoring programme.

8.1.7.6.1 Departure Runway/SID OFP Discrepancy

Contingency fuel is carried to account for deviations from planned routings (including departures and arrivals). Prior to pushback/engine start for the purpose of take-off, if it is established that the expected departure route is significantly longer than the departure depicted on the OFP, the commander may accept the FMC arrival fuel prediction at destination on condition that the FMC has been updated with actual ZFW, TOC and average winds. Crews shall evaluate the effect, if any, of the pilot input FMC SID stop altitude as this can cause an artificially high fuel burn in certain circumstances. If required, remove the restriction, check arrival fuel prediction and reinstate the restriction. The commander shall review the operational impact of the change of routing in deciding whether to accept the OFP or to request a revised OFP depicting the actual RWY/SID in use. Following a Runway/SID change, prior to engine start for the purpose of take off the FMC predicted fuel at destination shall be at least the sum of $CONT + ALTN + FINRES + DEV + TAXI$.

8.1.7.6.2 If during taxi for the purpose of taking off, delays/departure routing/Rwy changes are encountered such that the FMC predicted fuel at destination is less than $ALTN + FINRES + DEV$ Fuel, the Commander, having updated the FMC with accurate TOC and average winds, shall return to stand unless the fuel required for alternate, final reserve and deviation fuel meet with flight plan requirements.

8.1.7.7 Fuel Policy Destination Alternate Aerodrome

The departure aerodrome may be selected as the destination alternate aerodrome. If an alternate is changed then the adjusted flight plan fuel required shall be on board.

8.1.7.8 Fuel and Oil Supply

CAT.OP.MPA.260

The commander shall only commence a flight or continue in the event of in-flight re-planning, when satisfied that the aircraft carries at least the planned amount of usable fuel and oil to safely complete the flight, taking into account the expected operating conditions.

8.1.7.9 Malta Air Fuel Policies

Malta Air's fuel policy is to carry the correct amount of fuel necessary to safely and efficiently complete the flight while meeting regulatory requirements and maintaining operational flexibility. Every flight shall carry a sufficient amount of fuel, to complete the planned flight safely and to allow for deviations from the planned operation.

It is important to note, that the safe conclusion of any flight depends on the accuracy and completeness of initial planning as well as the intelligent use of on-board resources. Flight planning activities must be complemented by practical in-flight fuel management policies and procedures. The final authority and responsibility for fuel loads to be carried and the fuel management rests with the Commander.

Having ascertained the ZFW or estimated ZFW the Commander will determine the final fuel in accordance with the Malta Air fuel policy of Section 8.1.7. The required final fuel will then be passed to the Ramp Agent and to the Refueller. Fuel uplift shall be ordered in kilogrammes (kg) of fuel.

Dispatch with over-fuelled aircraft.

Typically an over-fuel situation will arise from an aircraft swap whereby an aircraft fuelled for a long flight is swapped on to shorter flight which would result in arrival at destination heavier than the MLW.

It is prohibited to use high drag techniques for the purpose of increasing fuel burn during any phase of flight after take-off with the exception of Gear Down in a holding pattern having consulted PI-QRH, and having received an ATC clearance to hold or otherwise delay arrival.

In order to meet with Performance Dispatch requirements a new flight plan is required to reflect the swapped aircraft TOW. To ensure the flight plan reflects a landing at or below Maximum Landing Weight, Flight Planning will consider the following options:

- Reducing zero fuel weight (ZFW) by offloading cargo (consult with Ops).
- Extended taxi or 'Push and Hold' with engines running, (fuel burn attributable to this recorded as Planned Taxi Fuel Burn in OPT W&B or in the paper Loadsheets).

- Flight plan at a lower level.
- Flight plan with a longer routing.

Note: The new flight plan produced by Flight Planning will not plan holding with the gear extended in order to satisfy a landing at or below Maximum Landing Weight.

The landing must be completed at or below MLW unless a non-normal situation dictates otherwise.

The OFP will have the following company notam:

“This flight plan route/burn has been adjusted to take account of over-fuelled condition. Ensure MLW limitation is not exceeded at touchdown”.

8.1.7.9.1 PIC Discretionary Fuel

It is the Company's policy to achieve optimum fuel economy by means of fuel uplift management together with fuel saving techniques in operation.

In keeping with this policy:

Flight crew shall load PIC Discretionary Fuel only if, in the Commander's opinion, the extra fuel is required due to forecast meteorological or RCR requirements. PIC Discretionary Fuel should be considered in terms of endurance (time) and not quantity of fuel (for example 15 minutes is half final reserve fuel). PIC Discretionary Fuel is carried at the discretion of the Commander for operational reasons. This Discretionary fuel is additional to the required OFP fuel.

PIC Discretionary Fuel should not be carried unless there are sound operational or economic reasons for doing so. If potential disruption is not evident during pre-flight planning, then it is acceptable practice for the commander to dispatch with OFP Block Fuel or adjusted Block Fuel. To avoid first wave delays, GOPS will load OFP Block Fuel rounded up to the nearest 100 kgs + 100 kgs when pre-fueling aircraft.

Additional fuel will be included on the OFP where experience on a route has shown that flight level restrictions or destination holding are common this fuel will be added to the ATC or OPN fields of the OFP.

The penalty for carriage of extra fuel is approximately 2.5% of extra fuel per hour of flight. (i.e. on a 4 hour flight up to 10% of the extra fuel uplifted will be burned off due to the increased aircraft weight).

Even with the best planning information and practices, not all factors can be foreseen. If the flight is well planned and well executed, a diversion is not considered a failure. The occasional weather related diversion is preferable and less expensive than an overly conservative fuel carriage philosophy. Commanders shall anticipate flight disruption and arrival delays when the destination forecast indicates conditions approaching limiting values. When limiting weather conditions are forecast at airports crews will consider the implications of multiple flight disruptions when assessing the PIC Discretionary fuel or diversion fuel requirements, which may be more than 15 minutes.

When widespread disruption is forecasted (several airports in the same geographic region subjected to similar weather phenomena such as snow, gale force winds or fog) crews will consider selecting an 'out of area' alternate at the planning stage. As the flight progresses more accurate information can be gathered and a closer alternate can be selected. Selection of a closer alternate while enroute will allow the commander to consider using some of the extra fuel for the planned alternate for holding or other disruption at the planned destination before proceeding to the new closer alternate.

In addition to ensuring legal dispatch requirements are met crews must also give consideration to the primary objective of completing the planned flight. For example, an airport which is forecast PROB30 CB's or PROB40 CB's may only require 1 alternate provided the other required minima are met. In these circumstances, it may be prudent to take some holding fuel in order to increase the chances of completing the flight to the planned destination.

The following guidelines will be used in helping to determine when to load PIC Discretionary Fuel:

- Volcanic Ash Event in progress or operations in close proximity to areas known to contain increased levels of volcanic ash;
- Thunderstorms are forecast;
- Operational landing limits are forecast or reported to be 2/3 or less of the forecast RVR, for example landing RVR 600 m and forecast is 900 m RVR or less.
- An unscheduled runway closure, affecting the arrival rate, for example: Freezing precipitation (runway treatment) – Moderate to heavy intensity snowfall (snow removal);
- Crosswinds in excess of 2/3 of limiting value;
- Contaminated runway limitations (it may be desirable to reduce landing weight and not increase by loading extra fuel in these conditions);
- Forecast en-route adverse meteorological conditions.

When risk of diversion is high i.e. weather approaching limits, curfew etc. the Commander shall ensure that; weather and NOTAM permitting, the highest ranked commercial alternate (C1, C2, C3 etc.) is planned. In this situation if the commercial alternate is also listed on the OFP as a suitable alternate, the Commander may simply uplift the fuel required for the chosen commercial alternate.

If the commercial alternate is not listed as a suitable alternate the Commander shall contact Operations to request a new OFP with the agreed commercial alternate. The revised fuel figure shall be recorded as BLOCK FUEL in the Electronic Flight Log, or FPF FUEL on the paper Voyage Report. Where Operations are not informed of the change of alternate ATC will anticipate a diversion to the planned alternate unless advised otherwise. Commercial alternates are listed in order of preference as C1, C2 and C3, etc. If the commercial alternate is not listed as a suitable alternate on the OFP (failed suitability check at the planning stage etc.) the Commander may still choose to divert to the commercial alternate subject to:

1. A review of NOTAMs.
2. FMC predicted arrival fuel at the alternate is greater than final reserve fuel after entering the expected route, Long Range Cruise (LRC), expected cruise altitude, arrival and expected approach.
3. Negotiating ATC approval to divert to an alternate that may not be included in the OFP.

The final decision on the required fuel rests with the Commander.

When Total Fuel is greater than the Block Fuel rounded up to the nearest 100 kgs + 100 kgs, the Commander shall record the reason for the difference between the Total Fuel and the Block Fuel on the Electronic Flight Log (or paper Voyage Report). Such explanations will include reference to relevant supporting documentation, including but not limited to weather, NOTAMS etc.

8.1.7.9.2 Fuel Saving is Effected at the Planning Stage by:

Using the computerised plan which accurately accounts for enroute winds and estimated aircraft weight.

Using ECON speeds for the flight plan.

Using the alternate selected by LIDO subject to weather conditions.

8.1.7.10 Fuel Icing

The LIDO flight planning system will not plan a flight through areas of extremely low temperature. If an en-route segment has an average temperature of -70°C or below, the flight will either be rerouted or the cruising flight level will be limited to minimize the aircrafts exposure to extremely low temperatures.

Tankering and Environmental Icing

On Tanker sectors, if the crew considers that de-icing is expected at destination, then round-trip fuel may be carried subject to possibly restrictive landing performance, and the aircraft will be de/anti-iced as per Malta Air de/anti-icing procedures using holdover tables in the Technical Log.

Tankering Limitations

Tankering is not limited by variable MTOWs as per weight certificate. Placard weights shall be increased to accommodate tankering requirements. The commercial decision to tanker fuel will be made automatically by LIDO. Trip fuel is calculated to include the fuel used for carriage and will be included as TANKERING on the LIDO Flight Plan. To maximise the benefits of fuel tankering, captains are required to tanker the fuel recommended by LIDO unless there are sound operational reasons for not doing so. Flight crew shall adjust the LIDO tankering recommendation when take-off or landing is anticipated on a contaminated runway. Tankering in excess of fuel required on the subsequent sector is uneconomical, the carriage of any such extra fuel shall be explained on the Electronic Flight Log/Paper voyage report.

Tankering is subject to landing performance, and CSFF considerations.

LIDO calculates optimum tanker fuel. Do not uplift for multi-sector tankering unless specifically directed.

On tanker sectors the Next Leg fuel requirement is indicated under the Planned Fuel section on the OFP. This information is used to calculate the uplift when tankering subject to limitations described above.

When tankering is planned and included in the OFP, LIDO will consider the additional trip fuel required to carry this fuel. The flight plan trip fuel will be automatically adjusted to reflect this. Where tankering has not been planned or included in the OFP crews shall correct the trip fuel using the operational impacts section of the OFP or the formula below.

The following formula may be used to calculate the additional Trip Fuel burn when ATOW exceeds FPL TOW on non-tankering sectors:

2.5% of the additional weight (fuel in this example) carried in kg, multiplied by the flight time in hours. For the B737-NG, 2.5% of the additional fuel carried in kg, multiplied by the flight time in hours, e.g. 1500 kg additional weight (fuel) carried for a 2 hr flight = $0.025 \times 1500 \times 2.0 = 75$ kg.

8.1.8 Mass (Weight) and Center of Gravity

Malta Air has established a Mass & Balance (MAB) Programme derived from data from Boeing including the AFM. This policy is contained in the OMB, FCOM Vol 1, QRH and OMA. The Malta Air Mass & Balance programme is managed and maintained in regulatory compliance by the Head of Flight Operations Performance Engineering who reports to the NPFO. In this section the word weight and mass are interchangeable.

8.1.8.1 Mass and Balance, Loading

CAT.POL.MAB.100

1. Malta Air ensures that during any phase of operation, the loading, mass and Centre of Gravity (CG) of the aircraft complies with the limitations specified in the AFM, or the operations manual if more restrictive.
2. Malta Air establishes the mass and the CG of each aircraft by actual weighing prior to initial entry into service and thereafter at intervals of nine years as fleet masses are used. The accumulated effects of modifications and repairs on the mass and balance are accounted for and properly documented. Aircraft are reweighed if the effect of modifications on the mass and balance is not accurately known.
3. The weighing is accomplished by the manufacturer of the aircraft or by an approved maintenance organisation.
4. Malta Air determines the mass of all operating items and crew members included in the aircraft dry operating mass by using standard masses. The influence of their position on the aircraft's CG is also determined.
5. Malta Air establishes the mass of the traffic load in accordance with standard passenger and baggage masses.
6. In addition to standard masses for passengers and checked baggage, Malta Air can use standard masses for other load items, if it demonstrates to the TM CAD that these items have the same mass or that their masses are within specified tolerances.
7. Malta Air determines the mass of the fuel load by using the actual density or, if not known, a fuel density of .79.
8. Malta Air ensures that the loading of
 - a. its aircraft is performed under the supervision of qualified personnel; and

- b. traffic load is consistent with the data used for the calculation of the aircraft mass and balance.
- 9. Malta Air Mass and Balance Policy complies with additional structural limits such as the floor strength limitations, the maximum load per running metre, the maximum mass per cargo compartment and the maximum seating limit.
- 10. Malta Air specifies in OMB – Mass and Balance, the principles and methods involved in the loading and in the mass and balance system that meet the requirements contained in this section which covers all types of intended operations.

8.1.8.2

Mass and Balance Data and Documentation

CAT.POL.MAB.105

- 1. Malta Air establishes mass and balance data and produces mass and balance documentation (OPT W&B/Loadsheet) prior to each flight specifying the load and its distribution. The loadsheet may be in paper or electronic format. The OPT Weight and Balance App and/or the paper loadsheet enable the commander to determine that the load and its distribution is such that the mass and balance limits of the aircraft are not exceeded. The paper Loadsheet contains the following information:
 - a. Aircraft registration and type;
 - b. Flight identification, number and date;
 - c. Name of the commander;
 - d. Name of the person who prepared the document;
 - e. Dry operating mass and the corresponding CG of the aircraft;
 - f. Mass of the fuel at take-off and the mass of trip fuel;
 - g. Mass of consumables other than fuel, if applicable;
 - h. Load components including passengers, baggage, freight and ballast;
 - i. Take-off mass, landing mass and zero fuel mass;
 - j. Applicable aircraft CG positions; and
 - k. The limiting mass and CG values.
- 2. The person supervising the loading of the aircraft confirms by hand signature or equivalent that the load and its distribution are in accordance with the Load Instruction Document (LID) or the loadsheet given to the commander. The commander indicates his/her acceptance by hand signature or equivalent.

3. Malta Air has specified procedures for last minute changes to the paper loadsheet to ensure that:
 - a. any last minute change is entered on the loadsheet and entered in the FMC;
 - b. the maximum last minute change allowed is $\leq 500\text{kgs}$ whether passengers, cargo or any combination thereof; and
 - c. new mass and balance documentation is prepared if this maximum number is exceeded, i.e. for $> 501\text{ Kg}$.

4. Malta Air has specified procedures for last minute changes to the paper Load Instruction Document (LID) to ensure that:

last minute changes are easily recorded on the LID and entered in the OPT W&B App. There is no maximum limit to the electronic LMC weight adjustment as any change will trigger an entirely new calculation within the App.

Note: The LID may be in electronic format (eLID). The person supervising the loading of the aircraft confirms that the load and its distribution are in accordance with the eLID by entering their PIN code in the GOPs App. The Commander signifies acceptance of the eLID by entering their PIN code in the EFL App.

8.1.8.3 Mass and Balance, Loading Weighing of an Aircraft

AMC1 CAT.POL.MAB.100(b)

It is Malta Air policy to use standard crew, passenger and baggage weights as described Operations Manual part B, Mass and Balance, Section 3.

8.1.8.4 Mass and Balance, Loading Fleet Mass and CG Position

AMC2 CAT.POL.MAB.100(b)

The method for determining the applicable passenger, baggage and cargo mass is described in Operations Manual Part B, Mass and Balance, Section 3.

8.1.8.5 Mass and Balance, Loading Centre of Gravity Limits – Operational CG Envelope and In-flight CG

AMC1 CAT.POL.MAB.100(a)

In the Certificate Limitations section of the AFM, forward and aft CG limits are specified. These limits ensure that the certification stability and control criteria are met throughout the whole flight and allow the proper trim setting for take-off.

Malta Air ensures that these limits are respected by compliance with the approved procedures in OMB – Mass and Balance:

8.1.8.6 **Mass and Balance, Loading Dry Operating Mass**

AMC1 CAT.POL.MAB.100(d)

The dry operating mass includes:

- 1. Crew and crew baggage;
- 2. Catering and removable passenger service equipment; and
- 3. Tank water and lavatory chemicals.

8.1.8.7 **Mass and Balance, Loading Mass Values for Crew Members**

AMC2 CAT.POL.MAB.100(d)

- 1. Malta Air uses standard masses, including hand baggage, of 85 kg for flight crew/technical crew members and 75 kg for cabin crew members to determine the dry operating mass.
- 2. Malta Air corrects the dry operating mass to account for any additional baggage. The position of this additional baggage is accounted for when establishing the centre of gravity of the aeroplane.

8.1.8.8 **Mass and Balance, Loading Mass Values for Passengers and Baggage**

AMC1 CAT.POL.MAB.100(e)

- 1. When determining the mass of passengers by using standard mass values, the standard mass values in Tables 1, ‘30 and more’, ‘all adult’ mass, is used. The standard masses include hand baggage and the mass of any infant carried by an adult on one passenger seat. Infants occupying separate passenger seats are considered as children.

Table 2 Standard masses for passengers – aircraft with total number of passenger seats of 20 or more

Passenger Seats:	20 and More		30 and More
	Male	Female	All Adult
All flights except holiday charters	88 kg	70 kg	84 kg
Holiday charters*	83 kg	69 kg	76 kg
Children	35 kg	35 kg	35 kg

Holiday charter means a charter flight that is part of a holiday travel package. On such flights the entire passenger capacity is hired by one or more charterer(s) for the carriage of passengers who are travelling, all or in part by air, on a round-or circle-trip basis for holiday purposes. The holiday charter mass values apply provided that not more than 5% of passenger seats installed in the aircraft are used for the non-revenue carriage of certain categories of passengers. Categories of passengers such as

company personnel, tour operators' staff, representatives of the press, authority officials etc. can be included within the 5% without negating the use of holiday charter mass values.

Table 3 Standard masses for baggage – aeroplanes with a total number of passenger seats of 20 or more

Type of Flight	Baggage Standard Mass
Domestic	11 kg
Within the European region	13 kg
Intercontinental	15 kg
All other	13 kg

2. For the purpose of Table 3:
 - a. domestic flight means a flight with origin and destination within the borders of one State;
 - b. flights within the European region mean flights, other than domestic flights, whose origin and destination are within the area specified in (d)(5); and
 - c. intercontinental flight means flights beyond the European region with origin and destination in different continents.
3. Flights within the European region are flights conducted within the following area:
 - N7200 E04500
 - N4000 E04500
 - N3500 E03700
 - N3000 E03700
 - N3000 W00600
 - N2700 W00900
 - N2700 W03000
 - N6700 W03000
 - N7200 W01000
 - N7200 E04500

4. Figure 1, The European Region.

Figure 1: The European region



5. Other Standard masses.
- N/A MAY.
6. On any flight identified as carrying a significant number of passengers whose masses, including hand baggage, are expected to significantly deviate from the standard passenger mass, Malta Air will determine the actual mass of such passengers by weighing or by adding an adequate mass increment. This process is in OMB – Mass and Balance.
7. If standard mass values for checked baggage are used and a significant number of passengers checked baggage is expected to significantly deviate from the standard baggage mass, the Malta Air will determine the actual mass of such baggage by weighing or by adding an adequate mass increment. This process is in OMB – Mass and Balance.
8. Actual bag weights will only be provided at airports approved by Ground Operations and when an eLID is in use. This process is outlined in OMB – Mass and Balance. Actual bag weights are not available for use with paper loadsheets.

8.1.8.9 Mass and Balance, Loading Procedure for Establishing Revised Standard Mass Values for Passengers and Baggage

AMC2 CAT.POL.MAB.100(e)

See Operations Manual Part B – Mass and Balance.

8.1.8.10 Mass and Balance, Loading Adjustment of Standard Masses

See Operations Manual Part B – Mass and Balance.

8.1.8.11 Mass and Balance, Loading Statistical Evaluation of Passengers and Baggage Data

See Operations Manual Part B – Mass and Balance.

8.1.8.12 Mass and Balance, Loading Fuel Density

If the actual fuel density is not known, Malta Air uses standard fuel density values for determining the mass of the fuel load. Such standard values are based on current fuel density measurements for the airports or areas concerned. For JET A1 (Jet fuel JP 1) Malta Air uses .79.

8.1.8.13 Mass and Balance Data and Documentation Non Standard Load Calculation

AMC1 CAT.POL.MAB.105(a)

If the loadsheet is constructed using a non-standard method for determining the mass of the load, the commander shall be informed.

8.1.8.14 Mass and Balance Data and Documentation Integrity

AMC1 CAT.POL.MAB.105(b)

Malta Air verifies the integrity of mass and balance data and documentation generated by Malta Air computerised mass and balance system, at intervals not exceeding 6 months. The Malta Air computerised mass and balance system has a process to ensure that amendments of input data are properly incorporated and that the system operates correctly on a continuous basis.

8.1.8.15 Mass and Balance Data and Documentation Signature or Equivalent

AMC1 CAT.POL.MAB.105(c)

The Malta Air Load Information Document (LID) is the master loading document and must be signed by the Captain. The LID may be in electronic format (eLID). The Commander signifies acceptance of the eLID by entering their PIN code in the EFL App.

The Malta Air computerised mass and balance system uses an electronic PIN code on a tablet screen. This is considered a hand signature under this regulation. Any time this signature is required, the signee will also type in their crew code or equivalent identifier into the tablet device.

8.1.8.16 When the Commander considers that it maybe be necessary to leave bags behind the Commander shall:

- Establish the limiting RTOW.

- Choose the closest suitable destination alternate subject to weather conditions to minimise the fuel uplift. Advise Ryanair Dublin Operations and enter relevant remarks in EFL Captains Remarks.
- Establish the actual ZFW prior to fuel uplift.
- Personally check all OPT W&B/loadsheet entries for accuracy.
- Consider a tech stop in consultation with Ryanair DAC Operations.
- Do not uplift any fuel in excess of OFP Operation Flight Plan Fuel resulting in bags being left behind.
- Where fuel in excess of OFP has been loaded, it is preferable to extend taxi time and burn excess fuel in order to reduce the take-off weight to RTOW. The maximum taxi weight is published in the fleet weight schedule by fleet.
- “Push and Hold”: the aircraft pushes back, starts engines or powers off and proceeds to a designated hold area and remains there with engines running.

The weight reduction to be accomplished by fuel burn during “push and hold” shall be included in the taxi fuel on the loadsheet. The taxi limiting weight shall apply when the aircraft is taxied for take-off. It is preferable to burn 700 kg of fuel during an extended taxi/holding period, rather than leave 54 bags behind. ATC permission to accomplish this “push and hold” procedure is required.

8.1.8.17 Loadsheet Discrepancies and LMCs

Should a paper loadsheet discrepancy arise, or a LMC be required, the aircraft will be closed up as normal providing the paper loadsheet has been signed by the Commander. Any loadsheet discrepancy shall be resolved over the headset with the doors closed.

Procedure flow:

- Completed paper Load Sheet with the Captain.
- The Captain will sign the paper loadsheet and return a copy together with a copy of the Technical Log to the dispatcher.
- The Captain will sign the load sheet and return a copy together with a copy of the Technical Log to the dispatcher.
- The dispatcher will immediately leave the aircraft.
- The No 1 will select the flight deck access switch to NORM, lock the flight deck door and close the L1 door.
- The Cabin Crew will then assist the passengers to take their seats in the cabin.

- When the aircraft is on stand, entering “I” and “ENT” on the flight deck door access panel gives an audible tone twice in the flight deck this signifies all passengers are seated.
- In the event that a paper loadsheet amendment or LMC is required the paper loadsheet will be lowered to the ramp agent.
- Once the corrected paperwork is returned to the flight deck, the Before Start checklist is completed to the line and push and start is requested.

8.1.8.18 Verification of Mass and Balance Documentation Used

To determine the Allowed Traffic Load:

1. The maximum weight for take-off shall be the Regulated Take-off Weight (RTOW) for the conditions and runway in use and not a default maximum certified or structural take-off weight (MTOW), as used previously. This value shall be calculated for every take-off, regardless of the length of runway in use, and shall not exceed the placarded MTOW.
2. The maximum weight for landing under normal conditions shall be the Regulated Landing Weight for the expected runway in use and forecast or actual conditions as appropriate. This value shall not exceed maximum certified landing weight. Paper loadsheets must show the maximum weights as calculated and notified by you.

Newspapers

When newspapers are carried on board, refer to the dispatcher or Operations Manual Part B – Mass and Balance for instructions on Paper Loadsheets completion and storage of newspapers on board.

The Commander is required to examine the OPT W&B/paper loadsheet to ensure that the maximum structural weights and the regulated take-off weight for the expected flight are not exceeded and that the centre of gravity is within the aircraft limits.

Paper Loadsheets

In order to minimise the chance of single or cumulative errors, here is a short “checklist” of items to look out for:

- Have an up-to-date weight certificate to hand.
- Ensure any crew numbers above standard are included.
- Please ensure that the entered maximum TOW is the calculated RTOW for the departure and not simply the maximum certificated TOW.
- Check for gross errors.

- Cross-check ZFW, TOW and LW against calculated maximums at the top of the paper loadsheet and absolute limits. Check also that at least the relevant balance limits have been circled.
- Any shortfall in service levels by handling agents shall be noted in the loadsheet. Tick box if NOT supplied.

SCHEDULED		LOADING SCHEDULE 737-800 & 737-800W SERIES 189 SEATS				RYANAIR			
AIRCRAFT REG.	FLIGHT NUMBER	STATION	DESTINATION	CREW	DATE				
APS Weight		MAXIMUM WEIGHTS FOR		ZERO FUEL		TAKE-OFF		LANDING	
1st Observer (+ 90.1)		Take-off Fuel		a		b		c	
2nd Observer (+ 90.1)		ALLOWED WEIGHT FOR		TOW (limited if a, b or c)		Tip Fuel +			
SR Attendant (+ 79.9)		Operating Weight							
BR Attendant (+ 79.9)		LOWEST TRAFFIC LOAD							
Food Galley +		Take-off Fuel							
AK Galley +		Tus. Time (limited on time - (ZFW/WK by added - 20 mins)							
Dry Operating Weight =		GULPER		STEPS		HSET		ZFW	
Take-Off Fuel		Ramp		Taxi					
Operating Weight =									

8.1.8.19 Acceptance/Certification of OFP

The OFP No is automatically recorded when the Commander accepts the eLID by entering their PIN code in the EFL App. Where paper LID/Loadsheet are in use, the OFP request No. will be recorded by the Commander in the S.I. box on the paper loadsheet or the designated OFP number box on the LID.

Required if unable to sign/send OPT TOW CG CG	Security CHECK / SEARCH complete following flight from IATA @ TIME(Z)
Aircraft Loaded law RYR Instructions HANDLING AGENT SIGN HANDLING AGENT NAME (BLOCK CAPITALS)	Air-Ops & Reg (EC) 300/2008 compliant OFP # CAPTAIN SIGN CREWCODE (BLOCK CAPITALS)

FR 135/07 MAR/STN-LBC A/C: EIEKA						Page 1
FR135 07MAR2019 EGSS-EDHL B738 EIEKA RELEASE 0433 07MAR19 LONDON/STANSTED-LUXEBOCK BLANKENSEE WX PROG 0715 0718 0721 OBS 0700 0700 0700						
SCHEDULED LOADSHEET / 737-800W RYANAIR 189 PASSENGER SEATS/ All Weights in Kilograms						
AIRCRAFT REG EIEKA	FLIGHT NUMBER FR135	STATION STN	DESTINATION LBC	CREW 2 + 4	DATE 07 Mar 19	
APS Weight 1st Observer (+ 50.1) 2nd Observer (+ 90.1) 5th Attendant (+ 79.9) 6th Attendant (+ 79.9) Fwd Galley + Aft Galley + Dry Operating Weight = Take-Off Fuel Ramp 6500 - Taxi 100 Operating Weight =		MAXIMUM WEIGHTS Take-Off Fuel + ALLOWED WEIGHT FOR TOW (lowest of a, b or c) Operating Weight ALLOWED TRAFFIC LOAD = Tick if not supplied on time GPU/FEP STEPS H/SET C/2/WK SI: 15 FGB in hold 2				
4 2 6 0 5 6 4 2 6 0 5 6 6 4 1 9 8 4 9 0 2 5 4		ZERO FUEL 6 1 6 8 8 6 4 2 0 6 8 1 0 8 7 4 9 9 0 6 7 8 1 7 4 9 0 2 0 1 8 7 9 7				
4 2 6 0 5 6 6 4 1 9 8 4 9 0 2 5 4		OFF REQUEST NO. 1				

8.1.8.20 Load Information Document (LID)

The LID may be in paper or electronic format (eLID). The Load Information Document (LID) is prepared by the handling agent or the flight crew at gate closure. The policy of LID completion following gate closure is designed to minimise LMCs.

The flight crew will use the passenger and cargo distribution information contained in the LID/eLID to complete the electronic OPT W&B prior to flight.

8.1.8.20.1 LID Discrepancies and LMCs

In the event that a LID amendment or LMC is required after the aircraft doors have been closed, the eLID may be edited by the handling agent or the commander and electronically submitted. For a paper LID amendment, the document will be lowered to the ramp agent, amended by the ramp agent and returned to the flight deck, alternatively the flight crew may amend the paper LID when contact with the ramp agent is available on the headset.

Note: The handling agent must confirm that any amendment to the eLID is acknowledged by the commander.

8.1.9 ATS Flight Plan

Submission of the ATS Flight Plan

Malta Air Operation is responsible for ensuring that every Malta Air flight has a flight plan submitted and accepted by ATC prior to departure.

A flight plan may be submitted by Boeing or Malta Air Operations for delivery flights.

Contents of ATS flight plan can be found OMC, Jeppesen General Airway Manual. A detailed description of LIDO flight plan and associated terminology is presented in OMA 8.1.10.

A Malta Air flight plan is automatically cancelled in the ATS system when a delay of 15 minutes after filed departure time has occurred. Operating crews and dispatchers must manage the communication of revised departure times to Malta Air Operations to prevent or minimise Flight Ops delays for cancelled flight plans.

A flight plan may be refiled in flight with ATS.

8.1.9.1 In-Flight Changes to Flight Plans

Operations

Malta Air operations shall ensure that the in-flight operational instructions involving a change to the air traffic flight plan shall, when practicable, be coordinated with the appropriate Air Traffic Service unit before transmission to an aeroplane.

Flight Crew

When crew receive a rerouting, upon reaching the last cleared point crew should confirm next point with ATC. Do not clear route discontinuities without ATC confirmation.

8.1.9.2 Calculated Take-off Time (CTOT) Slot Allocation

If a flight is delayed or is expecting a delay beyond the STD/CTOT the crew shall advise handling agents to contact Dub Ops by the most expeditious means, indicating when they expect to be ready to depart, and Dub Ops shall secure a new CTOT. If it is not possible to contact Dub Ops then ATC shall be advised that you will miss your STD/CTOT.

8.1.9.3 Mode S

Flight Crews shall accurately operate their Mode S transponders. The flight callsign on the OFP shall be entered correctly in the data field provided on the RTE page and the format is **the three letter IATA designator for the aircraft operating agency, followed by the flight alphanumeric (e.g. MAY212 or MAY2G etc).**

8.1.10 Operational Flight Plan (OFP)

AMC1 CAT.OP.MPA.175, CAT.OP.MPA.180

8.1.10.1 Requirements

The Operational Flight Plan provides for the determination of the fuel to be carried in accordance with OMA Section 8.1.7.2, Malta Air Fuel Scheme (CAT.OP.MPA.180). The Lido plan can be calculated up to 10 hours before departure. At the time the plan is calculated, a suitability check will be performed on destination and alternate airports based on the latest weather forecast and NOTAM information. If the suitability check is disabled by operations, a remark will appear on the flight plan to advise crew.

The commander, must always review the most up to date forecast information for departure, destination and alternate aerodromes before deciding to accept or reject the OFP.

8.1.10.1.1 Operational Flight Plan (OFP)

AMC1 CAT.OP.MPA.175(a)

All entries on the operational flight plan will be made concurrently and be permanent in nature.

The flight plan requirements are specified below.

RYR	Page 1
[ATC Flight Plan FR 7148/09 MAY/DUB-VNO A/C: EIEMC]	

FF EUCHZMFP EUCBZMFP	
091430 EDDFDLHE	
AD UUUWZDZX ULLLDZDX EYKAYWYX	
(FPL-RYR5DX-IS	
-B738/M-SDGIRWY/SB1	
-EIDW1620	
-N0455F370 ROTEV1A ROTEV UP600 TRN UM89 SAB UL983 PETIL/N0455F390	
L983 VES L975 KOKAK DCT GISON/K0829F390 A945 BALIT N183 AKUMA L749	
NEDAM NEDAM2A	
-EYVIO235 EVRA	
-PEN/A1B1C1D1O1 COM/TCAS DOF/130509 REG/EIEMC EET/EGPX0009 EGT0015	
EGPX0019 EKDK0104 ESAA0138 UMKK0159 EYVL0205 OPR/RYANAIR RVR/175	
RMK/EMERGENCY CONTACT +353 1 8121544 TCAS)	

ATC FLIGHT PLAN FR 7148/9 MAY/ DUB-VNO	Flight Numbers/Date/From – To.
A/C: EIEMC	Aircraft Registration.
FF EUCHZMFP EUCBZMFP 091430 EDDFDLHE AD UUUWZDZX ULLLZDZX EYKAYWYX	AFTN addresses. Date/Time.
FPL – MAY5DX – IS	ATC Flight Plan/Callsign/I – IFR/S – Scheduled.
• B738/M – SDGIJPRWY/SB1 • B38M/M – SADFGHIJ1J7LM3RWXYZ/ SB1D1G1	Aircraft type B738/M – Boeing 737-800/M – Medium Wake B38M/M – Boeing 737-8200/M – Medium Wake /S – Standard equip (VHF RTF, VOR, ILS) /A – GBAS (GLS) /D – DME /D1 – ADS-C with FANS 1/A capabilities /E1 – FMC WPR ACARS (position reporting) /E2 – D-FIS ACARS Datalink Flight Information Services /E3 – PDC ACARS Pre-Departure Clearance /F – ADF /G – GNSS /G1 – ADS-C with ATN capabilities /H – HF RTF (when equipped) /I – Inertial Navigation /L – ILS /JX – FANS CPDLC Configuration • J1 CPDLC-ATN-B1 VDLM2 – Honeywell Mk II+ • J4 CPDLC-FANS VDL MODE 2 (VHF only) • J7 CPDLC-FANS VHF+ SATCOM (Iridium) CPDLC /M3 – ATC SATCOM VOICE Iridium. /P2 – indicates FANS RCP 240 NAT NLA compliant /R – PBN approved /W – RVSM approved /X – NAT HLA approved /Y – VHF with 8.33 khz capability

	/Z – Other equipment carried (whenever Z is being used in the equipment indicator of that flight plan, a specification shall be made by inserting the sub-field COM/and/or NAV/and/or DAT/of the flight plan to indicate the type of COM/and/or NAV/and/or DAT/used for this flight). /S – Transponder mode S, including both pressure-altitude and aircraft identification capability /B1 – ADS-B with dedicated 1090 MHz ADS-B ‘out’ capability /D1 ADS-C with FANS 1/A capabilities /G1 ADS-C with ATN capabilities
EGSS1620	Departure Airport ICAO code/STD.
N0455F370 ROTEV1A ROTEV UP600 TRN UM89 SAB UL983 PETIL/ N0455F390 L983 VES L975 KOKAK DCT GISON/K0829F390 A945 BALIT N183	Cruising speed/Flight Level/Routing with any speed and/or FL change.
EYVI0235 EVRA	Destination Airport ICAO code/EET to Destination IAF/Alternate Airport ICAO code.
PBN/A1B1C1D1L1O1S2	Performance Based Navigation (PBN) and/or the Required Navigation Performance (RNP) levels that can be met: A1 RNAV 10 (RNP 10) B1 RNAV 5 all permitted sensors B5 RNAV 5 INS or IRS C1 RNAV 2 all permitted sensors D1 RNAV 1 all permitted sensors D3 RNAV 1 DME/DME L1 RNP4 O1 Basic RNP 1 all permitted sensors S2 RNP APCH with BARO-VNAV
NAV/RNP2	Indicates significant data related to the navigation equipment of that aircraft, other than specified in PBN/
COM/TCAS	Communications/TCAS.
SUR/RSP180	SUR/RSP180 – indicates FANS ADS-C (NAT NLA compliant).
DOF/130509	Date of Flight/09 May 2013.
REG/EIEMC	Aircraft registration/EIEMC.

EET/EGPX0009 EGTT0015 EGPX0019 EKDK0104 ESAA0138 UMKK0159 EYVL0205	Estimated Elapsed Time to FIR boundaries.
SEL/AGHK; CODE/4CAC53	Aircraft unique SELCAL Code; Aircraft specific CPDLC Aircraft Address Code.
OPR/Malta Air	Aircraft Operator/Malta Air.
RVR/175	Lowest RVR for arrivals.
RMK/EMERGENCY CONTACT +353 19451990 TCAS	Remark – plain language remarks.

FR 7148/01 MAY/DUB-VNO A/C: EIDHG					Page 2
[OFP]					
FR7148	01MAY2014	EIDW-EYVI	B738 EIDHG	RELEASE 1021	01MAY14
OFF 1	DUBLIN	INTL-VILNIUS INTL			
		WX PROG	0115 0118 0121	OBS	0106 0106 0106
ATC C/S	RYR62EF	EIDW/DUB	EYVI/VNO	CRZ SYS	CI 6
01MAY2014	EIDHG	1550/1608	1854/1909	GND DIST	1195
737-800W /	CFM56-7B26 /		STA 1900	AIR DIST	1131
		CTOT:....		TOC WIND	317/032
				AVG WIND	288/028
MAXIMUM	TOW 75.0	LW 65.3	ZFW 61.7	AVG W/C	P024
ESTIMATED	TOW 62.6	LW 56.2	ZFW 53.3	TOC ISA	P005
				AVG FF KGS/HR	2294
				FUEL BIAS	P05.0
ALTN EVRA				TKOF ALTN	
FL STEPS	EIDW/0390/				
DISP RMKS	NIL				

FR7148	Malta Air ICAO code and flight number.
01MAY2014	Date.
EIDW-EYVI	Departure Airport ICAO code – Destination Airport ICAO code.
EIDHG	Aircraft Registration.
RELEASE 1021 01MAY14	Time and Date of OFP calculation.
OFP 1/X*/X**	OFP number/* Reclearance number/ **NAT HLA (ETOPS) number. (*/** only appears for a Reclearance/NAT HLA flight).
DUBLIN INTL-VILNIUS INTL	Departure Airport Name – Destination Airport Name.

WX PROG 0115 0118 0121	Date and Time of weather prognosis used.
OBS 0106 0106 0106	Date and Time of observations.
ATC C/S MAY 62EF	ATC Callsign MAY5DX.
EIDW/DUB	ICAO/IATA code of Departure Airport.
EYVI/VNO	ICAO/IATA code of Destination Airport.
CRZ SYS CI 6*	Cruise System used Cost Index 6. (*May show Cost Index/Mach Number/ LRC/or a combination of these).
01MAY2014	Date.
EIDHG	Aircraft Registration.
1550/1608	Estimated Off Block Time/ Estimated Airborne Time.
1854/1909	Estimated Landing Time/Estimated On Block Time.
GND DIST	Total Ground Distance.
737-800W CFM56-7B26	Aircraft Description.
STA 1900	Scheduled Time of Arrival.
AIR DIST	Total Air Distance.
CTOT...*	Calculated Take-Off Time (*may show twice).
TOC WIND	Top of climb wind for First FL planned.
AVG WIND	Average wind direction/speed for routing.
B737-800W: MAXIMUM TOW 75.0/ 78.0* B737-8200: MAXIMUM TOW 76.2	Planned Maximum Structural Take-Off Weight 74,990 kgs (B737-800W) or 76,200 kgs (B737-8200). (*78.0 may show for 737-800W aircraft certificated for 77,990 kgs MTOW).
LW 65.3	Planned Maximum Structural Landing Weight 65,317 kgs.
ZFW 61.7	Maximum Structural Zero Fuel Weight 61,688 kgs.
AVG W/C	Average Wind Component for routing (P- Tailwind/M-Headwind).
ESTIMATED TOW	Estimated Take-Off Weight.
LW	Estimated Landing Weight.
ZFW	Estimated Zero Fuel Weight.

TOC ISA	Top of Climb temperature deviation from ISA (P-Plus/M-Minus) for First FL planned.
AVG FF KGS/HR 2294	Calculated Average Fuel Flow for routing
FUEL BIAS P05.0	Considered Fuel Bias (performance correction) of the aircraft in percentage (1.0 equals 1%) (P for PLUS/M for MINUS). <i>The Airplane Flight Manual provides fuel burn information based on flight test data. Over time, the aircraft will begin to deviate away from the certified fuel burn figures. Malta Air ensures the fuel consumption monitoring system is used to determine an individual aeroplane's performance in comparison with its predicted one. This correction is reflected in the Operational Flight Plan as Fuel Bias.</i>
ALTN EVRA	ICAO code for First Planned Destination Alternate.
TKOF ALTN	ICAO code for selected Take-Off Alternate.
FL STEPS EIDW/390	List of all Planned Waypoint/Flight Level steps for the routing. List of all Planned Waypoint/Flight Level steps for the routing(TOC wind and TOC ISA relate to the first FL displayed here)
ETP FLIGHT	Will appear if NAT HLA (ETP) is considered for the flight.
DISP RMKS NIL	Dispatcher Remarks. The flight planning department can add specific remarks to an individual flight plan. If the suitability check is disabled by operations, a remark will appear here to advise crew. NIL shown if no remarks.

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-----
                PLANNED FUEL
-----
FUEL      ARPT    FUEL    TIME
-----
TRIP              VNO    6354    0246
CONT 5%              318    0008
ALTN              RIX    1492    0035
FINRES              1055    0030
ADD RES              0      0000
-----
PLANNED T/OFF FUEL    9219    0359
-----
OPN              99      0003
-----
T/OFF FUEL          9318    0402
TAXI              DUB     234    0018
-----
BLOCK FUEL          DUB    9552
PIC EXTRA              ....
TOTAL FUEL              ....
REASON FOR PIC EXTRA .....
-----
FMC INFO:
RESERVES              2547
TRIP+TAXI             6588
-----
NO TANKERING RECOMMENDED (P)
NEXT LEG DATA NOT AVAILABLE
-----

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PLANNED FUEL	Headline for Fuel section.
FUEL ARPT FUEL TIME	Headlines for Fuel sub sections.
TRIP VNO	Trip to Destination Airport IATA code.
CONT 5%	Contingency Fuel (may be expressed as 5%, 3%, 20 MINS, or MIN CONT)/ Planned Contingency Fuel/Planned Contingency Time. <i>MIN CONT displays when 5 mins contingency fuel is planned.</i> (* The En-Route Alternate IATA code will show if an ERA is planned).
ALTN RIX	1 st Alternate IATA code.
FINRES	Final Reserve Fuel at the 1 st Destination Alternate.
ADD RES	Additional Reserve Fuel.
ETP	Extra fuel required for a NAT HLA flight (*will only appear for a NAT HLA [ETP] flight).
PLANNED T/OFF FUEL	TRIP Fuel + CONT + ALTN + FINRES + ADDRES + NAT HLA (ETP).
DEV	Extra Fuel for Aircraft Deviations (*Will only appear when required).
ATC	Extra Fuel for ATC (*Will only appear when required).

WXX	Extra Fuel for Weather (*Will only appear when required).
OPN	Extra Fuel for Operational reasons (*Will only appear when required).
TANKERING	Planned Extra Fuel for Tankering (*Will only appear when required).
T/OFF FUEL	Take-Off Fuel.
TAXI STN	Planned Taxi Out Fuel/Departure Airport.
BLOCK FUEL STN	T/OFF Fuel + TAXI Fuel/Departure Airport.
PIC DISCR	Discretionary Fuel added manually at discretion of PIC.
TOTAL FUEL	Manually enter Total Fuel.
REASON FOR PIC DISCR	Enter a brief description for carriage of discretionary fuel.
FMC INFO: RESERVES	Automatic addition of ALTN + FIN RES + ADD RES.
TRIP + TAXI	Automatic addition of TRIP and TAXI Fuel.
NO TANKERING RECOMMENDED	Tankering recommendations for the flight.
(P)	(P) Reason for not tankering is PRICE. (*May show Price/Business/Handling/Volume).

TANKERING SECTOR (B): (FOR 2605) BASED ON ICING CONSTRAINT
FR6023 LFMU/BZR 27 - EGGD/BRS 27 ALTN EGGB/BHX 33
ZFW: 49.9 PLNTOF: 6659 TAXI: 195

TANKERING SECTOR	Indicates a Tankering sector.
FR6023 LFMU/BZR 27 – EGGD/BRS 27	Next leg flight number/Next leg Departure airport and runway – Destination airport and runway/Next leg Alternate and runway.
ZFW: 49.9	Next leg Zero Fuel Weight.
PLNTF: 6659	Next leg Planned Take Off fuel.
TAXI: 195	Next leg Taxi fuel.

OFP, COMPULSORY READS, NOTAMS & WEATHER CHECKED / LICENSES VALID / AFB REVIEWED (WHERE PUBLISHED) / EFB VALIDITY CHECKED		
DISPATCHER:	TEL:	
PIC CREWCODE:	F/O CREWCODE:	T/P CREWCODE:
PIC SIGN:	F/O SIGN:	T/P SIGN:

OFP, COMPULSORY READS, NOTAMS & WEATHER CHECKED/ LICENCES VALID/AFB REVIEWED (WHERE PUBLISHED)/EFB VALIDITY CHECKED	Flight Plan acceptance statement.
DISPATCHER: AOS TEL: 35 31 248 0557	Dispatcher Details: AOS/Contact Details (may show AOS – Airline Operations Support if automatically calculated or Dispatchers name if manually calculated).
PIC NAME	Manually enter Captains CREWCODE.
F/O NAME	Manually enter Co-Pilots CREWCODE.
T/P NAME	Manually enter the Third Pilots CREWCODE.
PIC SIGNATURE	To be signed by Captain.
F/O SIGNATURE	To be signed by Co-Pilot.
T/P SIGNATURE	To be signed by Third Pilot.

ALTERNATE ROUTE TO:	APT TRK DST			VIA	FL	MORA	WC	FINRES TIME	1094 FUEL	
C2	EGNM	045	78	POLSR POL DCT	050	29	P032	0019	841	
C3	EGNX	193	134	SANBA1R SANBA DCT DIPSO ROKUPLA	130	29	P000	0032	1376	
C4	EGBB	187	132	LIVSU1R LIVSU Y53 NANTI DCT CREWE CHASE1C	090	29	P000	0034	1402	
US	C1	EGGP	254	138	MONTY1R MONTY KEGUN1D	070	37	P002	0036	1493

Route to Alternate (RTA)

The Flight Level from DEST to ALTN displayed on the OFP indicates the highest FL as planned by Lido along the route to alternate. Fuel requirements for the route to alternate are calculated at the flight level displayed in the RTA section of the OFP.

The missed approach of the destination runway is calculated, followed by a DCT segment to the end point of the departure procedure at the destination aerodrome. Thereafter the already calculated route between departure and arrival procedure is used to proceed to the alternate aerodrome.

On short routings to alternate where the SID intersects the STAR, a low FL may be planned by Lido (see example above: EGCC to EGNM @ FL050). This does not prevent the crew from requesting a higher level from ATC if required.

Commercial Alternate

A commercial alternate is an aerodrome selected and approved by the company. It may not be the nearest airport to the destination, but it may be more desirable to use for commercial reasons. Commercial alternates are listed in the RTA section of the OFP. Commercial alternates are labelled in order of preference as C1, C2 and C3, etc. The sequence of alternates may change based on fuel requirements, suitability checks, or if manually sequenced by the flight planning department.

When risk of diversion is high, i.e. weather approaching limits, curfew etc. the Commander shall ensure that; weather and NOTAM permitting, the highest ranked commercial alternate (C1, C2, C3 etc.) is planned.

Fuel Alternate

A fuel alternate is an aerodrome selected and approved by the company. It is usually the nearest airport to the destination. It may not be desirable to use for commercial reasons (e.g. inadequate ground transport) but may be included for flight planning purposes. The letter **F** will appear alongside a fuel alternate to advise the crew. Fuel alternate shall only be planned when the risk of diversion is low.

When risk of diversion is low, i.e. weather is good, no curfew etc., the Commander may expect that; weather and NOTAM permitting, the fuel alternate is planned.

Unsuitable Alternate

The commercial alternates, in a small number of cases, are deemed 'US' (unsuitable) by Lido during the airport suitability check. This anomaly occurs when no weather forecast is available or the weather is deemed to be outside of limits for that airport at the time of flight plan calculation.

If a commercial alternate (C1 or C2) fails a weather suitability check, the alternate information (Track, Distance, Route, Flight Level, MORA, Wind Component, Time and Fuel) will still be displayed in the Route to Alternate section of the OFP.

The designator 'US' will appear beside the 'unsuitable' alternate to highlight the fact that it has failed a suitability check at the planning stage.

Unsuitable Alternate Procedure

If the commercial alternate is listed as ‘US’ on the OFP, the commander may elect to divert to this alternate following a review of NOTAMs, weather and fuel requirements.

Weather and NOTAMs for the ‘unsuitable’ alternate will be automatically included in the briefing package.

Expanded Route to Alternate (RTA)

The expanded route to alternate is displayed at the end of the flight plan. RTA will show the full routing from the destination to each alternate.

Alternate fuel calculations are based on the Lido Block Fuel. The fuel required for the flight planned commercial alternate is included in the OFP Block Fuel. Additional listed alternates may have higher fuel requirements.

Note: Alternates with higher fuel requirements may display an EFOB figure less than FINRES, or a negative EFOB figure in the expanded RTA section on the OFP. Crew must manage fuel requirements in accordance with sections 8.1.7.9.1 PIC Discretionary Fuel and 8.3.7.1 In-flight Fuel Management Policy when selecting and planning alternates.

FLIGHT LOG TO EGGP/LPL(C1)											
AWY				FL	IMT	MN	WIND	OAT	EFOB	PBRN	
POSITION	LAT	EET	ETO	MORA	ITT	TAS	COMP	TDV			
IDENT	LONG	TTLT	ATO	DIS	RDIS	GS	SHR	TRP	AFOB	ABRN	
FREQ											

ALTERNATE ROUTE TO	Headline for Alternate information section.
FIN RES 1046	Final Reserve Fuel at the 1 st Destination Alternate.
APT	Airport. Alternate Airport ICAO codes.
TRK	Track. Average Magnetic Track Distance for routing from Destination to Alternate.
DST	Distance. Ground Distance for routing from Destination to Alternate.
VIA	Via. ATS route description Distance for routing from Destination to Alternate.
FL	Flight Level. Highest planned Flight Level for routing from Destination to Alternate.
WC	Wind Component. Average track wind component from Destination to Alternate.
MORA	MINIMUM OFF ROUTE ALTITUDE
TIME	Time. Planned Trip Time for routing from Destination to Alternate.
FUEL	Fuel. Planned Trip Fuel for routing from Destination to Alternate.
<i>F</i>	<i>Fuel Alternate.</i>
<i>C 1</i>	<i>First preferred Commercial Alternate. (May show C 1 - C 99).</i>

MEL/CDL ITEMS	DESCRIPTION
XX-XX-XX-XX	

MEL/CDL ITEMS	Headline for Minimum Equipment List/ Configuration Deviation List Items.
DESCRIPTION	A Brief Description of the MEL Item for crew info.
XX-XX-XX-XX	MEL reference code.
	Brief description of operational restriction.

ROUTING:

ROUTE ID: DUBVNO
EIDW/10 ROTEV5E ROTEV UP600 TRN UM89 SAB UL983 PETIL L983 VES L975
KOKAK DCT GISON A945 BALIT N183 AKUMA L749 NEDAM NEDAM2A EYVI/02

ROUTING ROUTE ID:	Headline for Routing Description.
DUBVNO (E/MCT*/R)	Airport IATA codes will show here if routing is a predefined company route. Optimization Policy. E-ETOPS/MCT- Minimum Cost Track*/ R- Restrictions – Restrictions shows the route has been calculated taking airspace and notams into account. (*May show MCT/MFT/MDT/MTT).
EIDW/28 ROTEV1ANEDAM2A EYVI/02	Routing Summary. Includes planned runways, airways and waypoints but does not include flight levels.

OPERATIONAL IMPACTS

FL CHANGE2000 ABOVENOT AVAILABLE

FL CHANGE2000 BELOWTRIP P 0018 KGSTIME P 0003

WEIGHT CHANGEUP 1000TRIP P 0081 KGSTIME P 0000

OPERATIONAL IMPACTS*	Headline
FL CHANGE 2000 ABOVE	2000 FT ABOVE FLIGHT PLAN SUGGESTED FLIGHT LEVEL.
NOT AVAILABLE	Not Available will show if the associated FL is above the MAX FL. Info will not be displayed.
FL CHANGE 2000 BELOW	2000 FT BELOW FLIGHT PLAN SUGGESTED FLIGHT LEVEL.
TRIP P0018 KGS TIME P0003	Trip Fuel Difference is PLUS 18 kgs/Trip Time Difference is PLUS 3mins at that FL.
WEIGHT CHANGE UP 1000	INCREASE OF WEIGHT OF 1000 KG.

*The Operational Impacts section of the OFP provides crews with Fuel Burn/Time information for Flight Levels above and below the planned cruising level should this level be unavailable or deemed inappropriate by the commander.

Fuel Burn/Time information is also provided for change to the aircraft weight. Fuel burn adjustment may be applied by the crew when actual ZFW is up to 1000 kgs above the OFP eZFW. This correction, if applicable, shall be recorded on the Electronic Flight Log (Captains Remarks)/Paper voyage report. If the actual ZFW exceeds the OFP eZFW by more than 1000 kgs, then a new operational flight plan should be calculated.

RVSM (FL290 - FL410)	NO.1	STBY	NO.2
TOC			
TOC +1HR			
TOC +2HR			
TOC +3HR			

RVSM (FL290 – FL410) NO.1 STBY NO.2 TOC TOC +1HR TOC +2HR TOC +3HR TOC +4HR	If required, record the No 1, No 2 and STBY altimeter readings.
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All Operational flight Plans (OFP) contain a section to record the No 1, No 2 and SBY altimeter readings if required.

Altimeter readings will be checked in accordance with the FCOM Pre-departure Briefing ‘Instrument Crosscheck’ on the ground prior to departure. The two primary altimeters must be within the maximum value of acceptable altimeter differences which should not exceed 23 m (75 ft) if operations in RVSM airspace are planned. At intervals of approximately one hour, crosschecks between

primary altimeters will be made. Both altimeters need to agree within ± 200 feet. The hourly check of No 1 and No 2 altimeter during flight in RVSM airspace is not recorded. Refer to FCOM Vol 1 Limitations and SP for limiting differences.

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Page 4

DEPARTURE ATIS:

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-----RYP62EF-----

DEPARTURE ATC CLEARANCE:

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DEST ELEVATION: EYVI: 646FT

ALTN ELEVATION: EVRA: 36FT

DEPARTURE ATIS	RECORD DEPARTURE ATIS
DEPARTURE ATC CLEARENCE	RECORD DEPARTURE CLEARANCE
MAY62EF	CALLSIGN
DEST ELEVATION: EYVI 646FT	DESTINATION AIRPORT ELEVATION
ALTN ELEVATION: EVRA 36FT	ALTERNATE AIRPORT ELEVATION

	TIMES		
	ESTIMATED	SKED	ACTUAL
OFF BLOCKS	1550Z/1650L	1550Z/1650L	Z
AIRBORNE	1608Z/1708L		Z
LANDED	1854Z/2154L		Z
ON BLOCKS	1909Z/2209L	1900Z/2200L	Z
FLIGHT TIME	0246		
BLOCK TIME	0319	0310	

TIMES	Headline.
ESTIMATED/SKED/ACTUAL	Estimated/Scheduled/Actual Headline.
OFF BLOCKS Z/L	Off Blocks Time ZULU/LOCAL.
AIRBORNE	Airborne Time.
LANDED	Landing Time.
ON BLOCKS	On Blocks Time.
FLIGHT TIME	Total Flight Time.
BLOCK TIME	Total Block Time.

RUNWAY / INTERSECTION	ACTUAL TOW	
SURFACE CONDITION	CORRECTIONS	
FLAPS / THRUST	CORRECTED TOW	
	OAT	
QNH < 1013	RTOW LIMIT	
ANTI-ICE	BLEEDS OFF	Y N
MEL / CDL	BLEEDS OFF LIMIT	
TOTAL CORRECTIONS	ASSUMED TEMP	
	ASSUMED TEMP LIMIT	

Refer to Appendix 8.1.8.11 for RTOW Calculations.

RUNWAY/INTERSECTION	Runway Designator/Runway Intersection.
SURFACE CONDITION	Runway condition Dry/Wet/ Contaminated.
FLAPS/THRUST	T/O Flap section/Take Off Thrust setting.
QNH < 1013	QNH correction when below standard.
ANTI-ICE	ENGINE ANTI ICE ON for Take Off correction.
MEL/CDL	MEL/CDL correction for aircraft defects.

TOTAL CORRECTIONS	Total of QNH, ANTI ICE, MEL corrections.
ACTUAL TOW	Actual TOW from OPT W&B/ paper Loadsheet.
CORRECTIONS	Total corrections as calculated.
CORRECTED TOW	Corrected Take Off weight.
OAT	Outside Air Temperature.
RTOW LIMIT	Regulated Take Off Weight Limit from airport. RTOW chart.
BLEEDS OFF Y/N	Bleeds Off Yes/No.
BLEEDS OFF LIMIT	RTOW limit weight with Bleeds Off.
ASSUMED TEMP	Assumed temperature used for Take Off.
ASSUMED TEMP LIMIT	RTOW limit weight using assumed temperature.

FR 7148/01 MAY/DUB-VNO A/C: EIDHG

ARRIVAL ATIS / ALTERNATE ATIS	RECORD ARRIVAL AND ALTERNATE ATIS HERE.
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FLIGHT LOG										

MOST CRITICAL MORA 10900 FEET AT GCTS//MXSHR 04 AT TFN										

OCEANIC CLEARANCE:										
NAVIX 40N016W GONAN OMOKO GUNSO FL380 .77 XXXX										
MAX OCEANIC ENTRY FL380										

AWY		FL		IMT	MN	WIND	OAT	EFOB	PBRN	
POSITION	LAT	EET	ETO	MORA	ITT	TAS	COMP	TDV		
IDENT	LONG	TTLT	ATO	DIS	REDIS	GS	SHR	TRP	AFOB	ABRN

MOST CRITICAL MORA 10900 FEET	Most critical MORA for the entire route.
AT GCTS	Name of Waypoint where this occurs.
MXSHR	Max Shear Rate of the entire routing.
AT TFN	Name of Waypoint where the max shear occurs.
OCEANIC CLEARANCE	The oceanic part of the routing.
NAVIX 40N016W GONAN OMOKO GUNSO FL380.77 XXXX	Waypoint names in oceanic airspace. Flight Level for the first oceanic segment. Mach number for the first oceanic segment (XXXX is the aircraft SELCAL – <i>N/A MAY</i>).
MAX OCEANIC ENTRY	Max Flight Level at which the aircraft (considering its weight) could reach by the oceanic entry point.
AWY	Airway/SID/STAR/NAT Track.
POSITION*	Waypoint Name. (* May show TOC/TOD or FIR boundary name).
IDENT	The waypoint/navaid identifier.
LAT	Latitude.
LONG	Longitude.
EET	Estimated Elapsed Time FROM previous TO current routing position.
TTLT	Total Time from Take Off TO current position.
ETO	Estimated Time Over waypoint.
ATO	Actual Time Over waypoint.

FL	Flight Level AT current waypoint.
MORA	Minimum Off Route Altitude. Minimum safe altitude FROM current waypoint TO next waypoint.
DIST	Ground Distance FROM previous waypoint TO current waypoint.
IMT	Initial Magnetic Track FROM current waypoint TO NEXT waypoint.
ITT	Initial True Track FROM current waypoint to NEXT waypoint.
RDIS	Remaining Ground Distance.
MN	Mach Number AT the current waypoint.
TAS	True Air Speed FROM current waypoint TO the next waypoint.
GS	Average Ground Speed FROM current waypoint TO the next waypoint.
WIND	Wind Direction and Speed AT current waypoint.
COMP	Track Wind Component FROM current waypoint TO the next waypoint (P- tailwind/M- headwind).
SHR	Shear Rate for segment TO next waypoint. (Not shown during climb/descent.)
OAT	Outside Air Temperature for segment TO the next waypoint (M-minus).
TDV	Deviation between OAT and ISA Temperature (P-plus/M-minus).
TRP	Tropopause height at current waypoint.
EFOB	Estimated Fuel On Board at current waypoint.
AFOB	Actual Fuel On Board.
PBRN	Planned Fuel Burn (from Take Off to current position including planned taxi out fuel).
ABRN	Actual Fuel Burn

LONDON UIR					
-EGTT	N5439.6	0000	...		
	W00530.0	0015	...	5	1104

LONDON UIR	Airspace Name (FIR Boundary).
EGTT	Airspace ICAO code.
~ DP LOGSU ~	
DP LOGSU	Decision Point (for Reclearance Flights only). N/A MAY.

FR 6154/16 MAY/ACE-PIK A/C: EIEBX							Page 9
ETP CRITICAL POINT INFORMATION (ETP RULE TIME: 060MIN)							
CRITICAL POINT FOR FUEL REQUIREMENTS: N4722.2 W00900.0							
ETP INFORMATION							
ETP	SAP	ELTIME TIME (AAP)	DIST ISA	MORA W/C	ICE CPUEL	FOB	COND
ENTRY1	LPFS	0032 0041	203	028	416	3015	10556 DC
N3208.0	W01230.5	(GCRR)	P14	M010			
ETP1	LPFS/GMMX	0034 0042	206/225	028/038	421	3054	10465 DC
N3222.5	W01221.6		P14/P14	M011/P017			
ETP2	GMMX/LPPT	0100 0047	248/229	041/029	457	3315	9416 DC
N3508.4	W01034.9		P13/P08	P009/M011			
ETP3	LPPT/LEST	0150 0026	128/122	032/048	262	1902	7392 DC
N4053.3	W00846.2		P06/P08	P010/M009			
ETP4	LEST/EICK	0246 0049	270/269	034/020	513	3532	5216 DX
N4722.2	W00900.0		P09/P06	P003/P004			
EXIT1	EICK	0256 0037	200	020	404	2781	4852 DX
N4811.7	W00900.0	(EICK)	P05	P006			
ENRTR ALTMS (WEATHER SUITABILITY PERIOD)							
LPFS	10:32	12:44	WX MIN:	1220-3900	FCST WX:	9999-9999	
GMMX	10:38	13:11	WX MIN:	521-1600	FCST WX:	9999-8000	
LPPT	11:03	13:40	WX MIN:	400-1550	FCST WX:	9999-9999	
LEST	11:37	15:04	WX MIN:	217-950	FCST WX:	9999-9999	
EICK	12:55	15:04	WX MIN:	300-1100	FCST WX:	9999-9999	

**ETP CRITICAL POINT INFORMATION	Headline.
CRITICAL POINT FOR FUEL REQUIREMENTS	Most critical position for ETOPS/ETP calculation.
ETOPS INFORMATION	Headline.
ETOPS	Headline.
SAP	Suitable Airports.
ELTME	Elapsed Time from departure airport to current position.
TIME	Trip Time from current position to the Critical Suitable Airport.
DIST	Distance.
MORA	Minimum Off Route Altitude.
ICE	Icing Fuel (this is included in CFUEL).
CFUEL	Critical Fuel required from current position to most critical suitable airport.
FOB	Fuel On Board at PET (Point of Equal Time).
COND*	Condition for the respective ETOPS/ETP position. (*May show DX – 1 Eng Out with Decompression/DC – Decompression/ X – 1 Eng Out).
(AAP)	Adequate Airport.
ISA	Average temperature deviation for routing to respective suitable airport.
W/C	Average Wind Component for routing to respective suitable airport.
ENRTE ALTNS (WEATHER SUITABILITY PERIOD)	Headline.
LPPS 14:15 16:53	Suitable Airports suitability period (from – to).
WX MIN: 960 - 3900	Weather Minima for suitable airport. Vertical Visibility (DH;Ceiling) feet – Horizontal Visibility (RVR;VIS) meters.
FCST WX: 9999 - 9999	Weather Forecast for suitable airport. Vertical Visibility (DH;Ceiling) feet – Horizontal Visibility (RVR;VIS) meters.

ENTRY 1	Entry points relate to aircraft entering NAT HLA (ETOPS) rules. Entry points depicted on the OFP have no function during Malta Air NAT HLA flights.
ETP 1	Equal Time Point.
EXIT 1	Exit point relate to aircraft exiting NAT HLA (ETOPS) rules. Exit points depicted on the OFP have no function during Malta Air NAT HLA flights.

**N/A MAY. Malta Air is not authorised to conduct ETOPS however Malta Air is authorised to conduct NAT HLA operations. All LIDO flight plans are always planned to remain within 60 mins of a suitable airport. All NAT HLA flight plans will generate ETPs. CRITICAL FUEL is fuel required to divert to a suitable airport at the most critical point during a NAT HLA flight.

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WIND INFORMATION										

CLIMB	T O C			TRN	SAB					
350 258/020 -42	410	243/021	-44	410	241/020	-44	410	220/023	-43	
310 269/019 -42	390	248/020	-43	390	246/020	-43	390	221/023	-43	
200 295/033 -29	370	248/019	-43	370	247/018	-43	370	221/021	-43	
150 294/033 -18	350	253/018	-42	350	250/017	-43	350	219/021	-43	
100 294/029 -08	330	249/016	-43	330	246/015	-43	330	214/022	-44	

CLIMB			Position		
350	258/020	-42	Flight Level	Wind Direction/Speed	OAT

8.1.10.1.2 Shear Value on OFP

Theoretically there is no upper limit to the value but generally it will appear as a single digit. The wind shear value is calculated from wind speed and direction samples taken 2,000 ft above and 2,000 ft below the datum level. The lower the number the less the chance of turbulence.

Moderate turbulence can be encountered whenever the shear value is equal to or greater than 3. Severe turbulence can be anticipated whenever the shear value is equal to or greater than 5.

There is no exact correlation between the shear value and clear air turbulence (CAT). For example, a shear value of zero just means that the winds above and below the datum are the same; it gives no indication of the wind speed and direction at the datum level which could be a 100 kt jetstream (with high turbulence) or 20 kt light winds (with no turbulence).

Commanders shall review this section of the OFP at the pre-flight stage, compare it with forecast areas of turbulence and brief the SCCM on any anticipated areas of turbulence en route. Be aware that the predictions are based on forecast winds and should be used for guidance in conjunction with the information derived from Significant Weather charts and Sigmets.

8.1.10.1.3 Fuel ERA and 3% Contingency OFP

The use of a fuel ERA allows contingency fuel to be reduced to 3% of trip fuel, providing that 3% contingency fuel is greater than an amount of fuel to fly for 5 minutes at holding speed at 1500 ft (450 m) above the destination aerodrome in standard conditions.

The Fuel ERA location is automatically generated by LIDO in compliance with the formula in 8.1.7.4.

The same in-flight fuel checks, fuel management monitoring and re-planning in-flight considerations apply whether with 5% or 3% contingency as described in 8.3.7.1.

‘FUEL EN ROUTE ALTERNATE (ERA) AERODROME’

Fuel ERA aerodromes could be used in the following cases:

1. ‘Fuel ERA aerodrome critical scenario’: that aerodrome is used when additional fuel is required at the most critical point along the route to comply with point (3)(f) of OMA 8.1.7.3 (CAT.OP.MPA.181)
2. ‘Fuel ERA aerodrome 3%’: that aerodrome is used when Malta Air reduces the contingency fuel to 3%

LIDO Suitability Check 3% Contingency

Lido automatically performs suitability checks when selecting any alternate aerodrome. The Fuel ERA suitability check establishes that the Fuel ERA qualifies firstly as an alternate, and secondly that it is located in the geographical area required by the Fuel ERA formula. The suitability check includes:

- That the alternate is an adequate aerodrome on which the aircraft can be operated taking account of the applicable performance requirements and runway characteristics.
- That the weather forecast for +/-1 hour of expected time of use indicates minima enabling an instrument approach, or in the absence of an instrument approach a visual approach to be completed, and is therefore weather permissible.
- Check of NOTAMS to ensure that the aircraft can land safely at the FUEL ERA if required.

If an aerodrome fails the LIDO Fuel ERA suitability check it cannot be a Fuel ERA and LIDO will seek the next aerodrome which complies with Fuel ERA requirements. If LIDO cannot find a qualifying Fuel ERA in the area specified in the formula, LIDO will NOT generate a 3% Contingency OFP.

Identification of a 3% Contingency OFP

A 3% contingency OFP will be annotated in the Dispatch Remarks Text box on the OFP and have CONT 3% in the fuel section instead of CONT 5% and Fuel ERA IATA code on the OFP.

DISP RMK:

“THIS OFP IS PLANNED 3% CONTINGENCY FUEL. THE FUEL ERA HAS BEEN SELECTED IN COMPLIANCE WITH OMA 8.1.7.4. LIDO AUTOMATICALLY CONDUCTED A SUITABILITY CHECK WHICH INCLUDED WEATHER AND NOTAM CHECK AND VALID FOR +/- 1HR OF POTENTIAL TIME OF USE TO IDENTIFY THE FUEL ERA”

FUEL	ARPT	FUEL	TIME
TRIP	STN	11706	0444
CONT 3%	BOH	351	0009
ALTN	LTN	1111	0025
FINRES		1130	0030
ADD RES		0	0000
ETP		0	0000

3% Contingency Planning requirements

At flight planning, crews do not have to verify that the Fuel ERA automatically generated by LIDO complies with the formula in OMA 8.1.7.4. This is a required element of the LIDO certification.

Crews do have to establish that the briefing pack contains weather forecast and Notams for the Fuel ERA and verify that the Fuel ERA is weather permissible for the period of potential use, (OMA 8.1.5. refers), to enable crew acceptance of the 3% Contingency OFP.

3% Operational Impacts

When operating on a 3% OFP the standard in-flight fuel checks, in-flight re-planning considerations and continue/divert considerations apply. If a diversion is required en-route any en-route alternate may be used, including the Fuel ERA, as deemed appropriate by the aircraft commander.

8.1.10.2 Computerised OFP

The briefing pack may be paper or electronic format. All OFPs will be prepared by the OCC and presented on EFL/LIDOBRIEF. As required, it may be electronically transmitted to another station or station office for hard copy delivery to the Commander. OFPs should not normally be downloaded (or printed) prior to STD-45 mins on the first sector of the rostered duty to prevent an invalid OFP on the first sector, or subsequent sectors.

The OCC frequently change flight plans after they have been posted to EFL/LIDOBRIEF. If the briefing pack has been downloaded/printed before STD-45, it is imperative that the flight plan be rechecked at STD-45 mins to confirm its validity prior to departure.

- Electronically: For EFL, crew should open the EFL App and select 'Download' to check for the latest flight plan. If the EFL update is not available, then crew must contact Ryanair Ops or check the Lidobrief portal to verify validity of the OFP.
- Paper: For Lidobrief, check the printed OFP number against the onscreen OFP number. If the numbers do not match, then you must print the updated flight plan.

Browse Flights Flight Briefing WX/NOTAM									
My Flights All Flights									
AO	AL	FLT No.	OS	Reg.	DEP	DEST	From UTC Time	Crew Name	Search
							12Mar21 1441		
Print Flight List									
1 - 50 of 492									
AO	FLT	DEP	DEST	ETD	Reg.	Captain	Status	Confirmed on	Actions
FR	FR 3868 / 12	PFO	AMM	12/1450	EIDCH	-	FINAL in ORFZ OFP 1/100	-	
FR	FR 1782 / 12	STN	BOD	12/1450	EIEKN	-	CONFIRMED OFP 5/0/0	12/1441	
FR	FR 9677 / 12	CIA	VLC	12/1450	EIDLY	-	FINAL in ORFZ OFP 1/100	-	

FR 1782/12Mar21/STN-BOD Reg:EIEKN OFP:5/0/0									
OFP									
FR1782	12MAR2021	EGSS-LFBD	B738	EIEKN	RELEASE	0837	12MAR21		
OFP 5		LONDON/STANSTED-BORDEAUX/MERIGNAC							
			WX	PROG	1215	1218	OBS	1200	1200

Note: Each OFP is assigned a random OFP Number. Should your OFP be updated by the OCC, the latest assigned OFP number may be higher or lower than the original OFP Number.

When a flight plan change is made after STD-45 mins, OCC will contact the crew by phone/dispatcher/handling agent or any other appropriate means.

OFP Changes after STD-45

When there is a change to an OFP after reporting for duty, OCC will e-mail the OFP change notification to the Handling Agent. The OFP change notification e-mail is copied to the operating crews' EFBs solely to alert them of the OFP change notification, i.e. that a new OFP is required to enable completion of the flight preparation required for dispatch.

EFBs shall be checked for communications from Malta Air during all ground operations. Specifically the EFBs should be checked for an OFP change notification when commencing the RIBETS – Route Check. Prior to route entry, select 'Download' in EFL to check for the latest OFP.

Where a check of EFB emails or EFL App indicates an OFP update, then the latest flight plan must be downloaded for flight. If unable to download electronic flight plans, then the crew should contact the handling agent and request the hard copy plans at first available opportunity.

All electronic OFP change notifications on EFBs will be acknowledged by the flight crew member by replying to the OFP change notification email. There is no text required, simply open the e-mail and press REPLY – SEND as the acknowledgment.

The OCC will cancel the flight plan for a sector where an updated OFP has not been downloaded by the operating crew at STD-10 mins. In this case check with OPS for the new flight plan and new flight number/call sign.

8.1.10.3 OFP as a Navigation Log

The Operational Flight Plan as well as recording pertinent details of the Flight operated will be used as a navigation log for;

1. Recording fuel and time checks at the first convenient waypoint in the cruise, and thereafter at or abeam OFP waypoints at no more than 60 minute intervals. Fuel checks are made in accordance with OMA 8.3.7.1 – In-Flight Fuel Management Policy.
2. Enroute and destination diversions routings are to be logged.
3. All ATC clearances must be recorded accurately when the aircraft is on the ground and in flight in the cruise, usually by the PM.

Total Fuel figure must always be entered. The requirement to tanker is reflected in the calculated departure fuel, i.e. the ETOW of 69,990 would represent booked payload plus trip fuel plus tankered fuel to 69,990 kgs.

The corrected amount less taxi shall also be shown at the top of the EFOB column and subsequent figures altered accordingly. AFOB will be completed in tandem with ATOs.

The weather for the aerodrome of departure and landing shall be recorded on the OFP. Record alternate actuals and/or other en-route wx reports when required.

8.1.10.4 Flight Preparation Conversion Tables

Any conversion tables necessary to support operations where metric heights, altitudes and flight levels are required will be provided by Malta Air as a quick reference laminated checklist on the flight deck. These tables are in OMC Jeppesen General Airway Manual.

8.1.10.5 Ditching

CAT.GEN.MPA.150,, CAT.IDE.A.285

Malta Air aircraft will always be flight planned to remain within 400 nms from an adequate aerodrome unless Liferafts are carried.

8.1.10.6 Journey log

AMC1 ORO.MLR.110

General

1. The aircraft journey log, or voyage report will include the following items, where applicable:
 - a. Aircraft nationality and registration,
 - b. Date,
 - c. Name(s) of crew member(s),
 - d. Duty assignments of crew member(s),
 - e. Place of departure,
 - f. Place of arrival,
 - g. Time of departure,
 - h. Time of arrival,
 - i. Hours of flight,
 - j. Nature of flight (scheduled or non-scheduled),
 - k. Incidents, observations, if any,
 - l. Signature of person in charge.
2. The information, or parts thereof, may be recorded in a form other than on printed paper. This is the Electronic Flight Log or crewlink in Malta Air and accessibility, usability and reliability is assured by Malta Air IT.

3. 'Series of flights', means consecutive flights, which begin and end:
 - a. Within a 24 hour period;
 - b. At the same aerodrome or operating site or remain within a local area specified in the operations manual; and
 - c. With the same pilot-in-command/commander of the aircraft.

Paper Voyage Reports must be completed correctly and signed by the Commander on completion of the series of sectors which constitute his assigned flight duty. The Paper Voyage Report contains details of flight times, delay reports, fuel data, FDT and DT. The data fields on the VR shall be completed as follows:

- **Fuel Uplift LTS:** Exact fuel uplifts in litres as indicated on the fuel receipt(s) e.g. 7856 ltr.
- **FPL Fuel:** The exact Block Fuel on the OFP to the kg, e.g. 8456 kg.
- **Dep Fuel:** Fuel onboard to nearest 10 kg, as per fuel gauges and tech log, e.g. 7240 kg.
- **Arr Fuel:** Fuel onboard to nearest 10 kg, as per fuel gauges and tech log e.g. 2340 kg.
- **Burn Off:** Difference between Dep Fuel and Arr fuel to nearest 10 kg, e.g. 4270 kg.
- **PLOG BURN: APU + TRIP:** The total of APU + OFP TRIP + TAXI. (TRIP+ TAXI in LIDO as APU included in Taxi fuel)

Do not include contingency fuel in this calculation.

FLIGHT	DATE	TIME	FROM	TO	FUEL	REMARKS
P2FW 60878	11/03/2025	08:00	MLA	MLA	150KG	TAXI/APU
					3432KG	TRIP
					475KG	CONTINGENCY
					9642	
					1170KG	RESV/HOLD
					1110KG	GCFV
					2280	
					12397KG	FPL FUEL
					0KG	TANK
					0KG	EXTRA
					12397KG	BLOCK FUEL
					12500	FINAL FUEL

- **TOW:** Enter the Certificated MTOW for the sector.

Fuel Offload APU	FPL Fuel	Dep Fuel	Arr Fuel	Burn Off	PLG Burn APU+TRIP	Low	OffBlock Time	OnBlock Time	Total Block Time	Air Burn	Actual Pax	Landed Time	FRT	Fuel Extra
6202	12397	12500	2950	9550	9642	74990	12.20	16.32	04.12	12.27	182+2	16.26	0	-92
8150	13259	13390	3120	10270	10180	69990	17.10	21.30	04.20	17.16	180+8	21.26	0	+90

- **Off Block Time:** To the nearest minute.
- **On Block Time:** To the nearest minute.
- **Total Block Time:** Difference between ON and OFF times.
- **Airborne:** To the nearest minute.
- **Actal Pax:** TOB.
- **Landed:** To the nearest minute.
- **FRT:** Weight in kg.

Fuel extra: ± the difference between **OFF APU + TRIP** and **BURN OFF** (+ more than OFF, - less than OFF).

On the top RHS of the form there are four headings: CP remarks, delay 1, delay 2, and delay 3. Enter up to three constituent causes of the delay as

delay code 1/minutes,

delay code 2/minutes,

delay code 3/minutes.

Include a brief legible description in English of the cause of the delay on the voyage report.

On the bottom RHS are two boxes:

- Note 1:** Defuelling is not a normal flight operations activity. If required it will be conducted under engineering supervision.
- Note 2:** When revised VR is available it will be included with datafield explanations in the subsequent planned revision of OM A.

8.1.10.6.1 Electronic Flight Log (EFL)

The Electronic Flight Log must be completed correctly and verified by the Commander on completion of the series of sectors which constitute the assigned flight duty. The EFL enables crew to download and view the electronic briefing pack including the electronic OFP for their flight. The user may also view Crew and Flight Information, HIL, Weather data and eLID information. EFL allows crew to electronically submit Fuel, Payload, Times, Delays and GOPs data. ECDR reports are also submitted through the EFL App.

Expanded guidance for use and completion of EFL is provided in the EFB User Guide.

EFL Preflight Procedures

Crew must log-in to the EFL App at the start of each duty.

Crew must download the electronic briefing pack for each sector within EFL. If the EFL is unavailable or crew are unable to download the electronic briefing pack, then the full Lido briefing pack must be printed and used for flight.

EFL Normal Cockpit Procedures

Crew should enter the relevant information for the flight.

EFL Post Flight Procedure

The Electronic Flight Log data may be submitted by either the Captain or First Officer. Once all flight information has been submitted by the First Officer the data must be verified by the Captain.

Crew should ensure that they log out of the EFL App once all data has been successfully submitted.

8.1.10.6.2 Paper Voyage Report

Where Electronic Flight Log is not available, the paper voyage report must be completed correctly and signed by the Commander on completion of the series of sectors which constitute his assigned flight duty. The Voyage Report contains details of flight times, delay reports, fuel data, FDT and DT. The data fields on the VR should be completed as follows:

8.1.11 Malta Air Aircraft Technical Log (EU) No 1321/2014, M.A.306

8.1.11.1 General Requirements

The Technical Log for each aircraft is required to contain the following information for each aeroplane:

B737-800 Technical Log Contents

Page Number	Description	Form	Applicability	Issued	Inserted		Removed	
					Date	By	Date	By
1	Index page	MAY017	B737-800					
2.1	CRS Document.	RA803	B737-ALL					
2.5	Out of Phase Repeat Inspections (if applicable).	–	B737-800					
3.0	Flight Log & Workorder	(B737-800 – MAY001)	B737-800					
3.1	Notice To Crew.	MAY909	B737-ALL					
3.2	HOLD ITEM LIST [See end of Index Pages for active HOLD ITEM LIST sheet	MAY015	B737-ALL					
3.3	Auto Coupled Approach Report.	MAY906	B737-ALL					
3.4	Maintenance Support.	MAY908	B737-ALL					
3.4.1	Stations at which Commanders may NOT issue CRS	RA253	B737-800					
3.5	LIDO FMC (F-F FACTOR)	MAY907	B737 ALL					
4.1	Preflight. Transit/Preflight check	MAY802	B737-800					
4.2	Ramp-1 check form	MAY800	B737-800					
4.3	50FC Check	MAY858	B737-800					
5.1	Damage Record Forms	(B737-800 – MAY805)	B737-800					
5.2	Aircraft Flight Deck Window Damage Record	MAY827	B737 ALL					
5.3	Airstairs Manual Stowing Instructions	MAY845	B737 ALL					
5.4	Thrust reverser reset procedure guide	MAY403	B737-800					
5.5	Radio ahead codes	MAY808	B737 ALL					
5.6	Flight crew bird strike inspection procedure	RA210	B737-800					
5.7	Frost Measurement Tool	–	B737-800					
5.8	Return to Service Flight Deck Configuration	MAY1000	B737-800					
Note: Each Aircraft Model B737-800 Technical Log must contain the relevant sheet's, identified in this index.								

B737-8200 Technical Log Contents

Page Number	Description	Form	Applicability	Inserted		Removed	
				Date	Sign	Date	Sign
1	Index Pages	MAY017 M	B737-8200				
2.1	CRS Document	RA803	B737 ALL				
2.5	Out of Phase Repeat Inspections (if applicable)	–	B737-8200				
3	Flight Log & Workorder book	TL001	B737-8200				
3.1	Notice To Crew	MAY909	B737 ALL				
3.2	HOLD ITEM LIST [See end of Index Pages for active HOLD ITEM LIST sheet]	MAY015	B737 ALL				
3.3	Auto Coupled Approach Report	MAY906	B737 ALL				
3.4	Maintenance Support	MAY908	B737- ALL				
3.4.1	Stations at which Commanders may NOT issue CRS	RA253M	B737-8200				
3.5	LIDO FMC (F- FACTOR)	MAY907	B737 ALL				
4.1	Pre-flight Check	MAY970	B737-8200				
4.2	Ramp-1 Check	MAY971	B737-8200				
4.3	50FC Check	MAY972	B737-8200				
5.1	Damage Record Forms	MAY976	B737-8200				
5.2	Aircraft Flight Deck Window Damage Record	MAY827	B737 ALL				
5.3	Airstairs Manual Stowing Instructions	MAY845	B737 ALL				
5.4	Radio ahead codes	MAY808	B737 ALL				
5.5	Return To Service Flight Deck Configuration	MAY1000 M	B737-8200				
5.6	Flight Crew Bird Strike Inspection Procedure	RA210	B737 ALL				
Note:	Each B737-8200 Aircraft Technical Log must contain all relevant sheets identified in this index.						

8.1.11.2 Certification of Technical Log

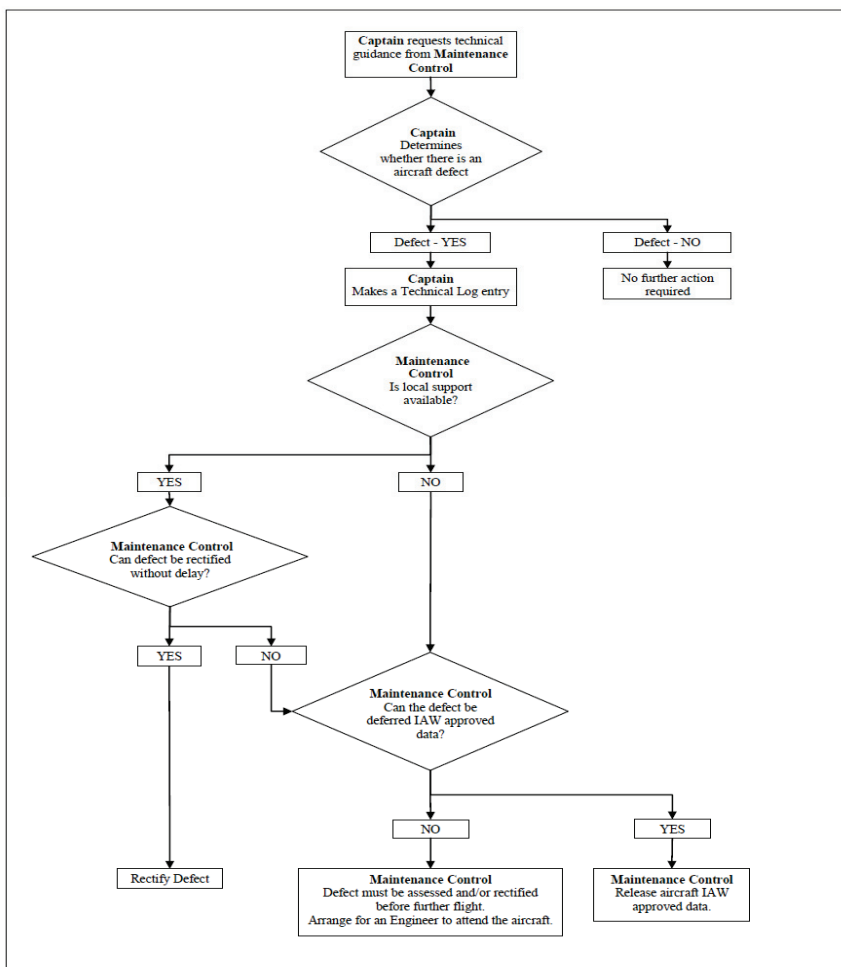
ORO.MLR.100

Commanders shall ensure that the following tasks are completed:

1. The recording, and rectification of defects, together with certification;
2. The recording and certification of servicing with fuel and oils and of de-icing;

3. Certification of completion of the Pre-flight/Transit inspection; and
4. Certification of acceptance of the aircraft for flight by the Commander;
5. If flight crew encounter a possible defect during flight operations the Commander shall, at the next shutdown use the MAINTROL/FLIGHT CREW COMMUNICATIONS FLOW CHART below which is also contained in the aircraft Technical Log to determine the appropriate action required.

Maintrol/Flight Crew Communications Flow Chart



All shaded fields require entry by the aircraft commander. Non-shaded fields require data where necessary.

If no data is required, data fields in the non-shaded section are not to be completed.

Whenever a ASR is filed to report a technical problem or failure, the corresponding tech log entry shall be made in the data field for entering the tech log page number in the electronic reporting ASR form.

If you have a technical issue on taxi out and need to return to stand the correct phraseology is “MAY ABXY requests a return to stand to contact our Company”. Once the matter has been resolved, inform ATC of your intentions. The handling agent/engineer will forward the necessary technical log documentation to the local NAA as required.

8.1.11.3 Aircraft Acceptance by Commander

CAT.GEN.MPA.105

The Commander is required to review and certify the aircraft Technical Log. By signing the Technical Log the Commander is certifying compliance with the requirements of Air OPS with respect to:

1. The aircraft is airworthy, and the Certificate of Release to Service is current, if such Certificate is required to be contained in the Technical Log (see below).
2. The aircraft configuration is in accordance with the Configuration Deviation List (CDL).
3. The instruments and equipment required for the flight to be conducted are available; Air OPS
4. The instruments and equipment are in operable condition except as provided in the MEL.
5. The manuals, documents, additional information and forms as required to be available and carried on board by CAT.GEN.MPA.180, AMC1 CAT.GEN.MPA.180 are on board;
6. The aircraft has been de-iced and inspected as required by CAT.OP.MPA.250;
7. The aircraft carries at least the planned amount of fuel and oil to complete the flight safely, taking into account the expected operating conditions.

For the purposes of compliance with respect to (1) above the Commander is deemed to have properly satisfied himself if the Technical Log has been duly completed and presented to him showing:

1. That defects from the previous flights have been rectified and signed off or deferred, and that any deferred defects and equipment found inoperative during the pre-departure inspection are acceptable as inoperative for flight in accordance with the MEL; and
2. That a daily check has been duly certified within the requisite period. The Commander is not required to check the running total of flying hours nor to verify the validity of the Certificate of Release to Service or to have knowledge of the next scheduled maintenance where this is controlled by other means acceptable to the TM CAD.

Flight Crew as a privilege of their licence, having been duly instructed, are entitled to conduct transit and pre-departure inspections, carry out or supervise the servicing with fuel, oils and hydraulic fluid or the de-icing of the aircraft they intend to operate, complete specified maintenance tasks as trained, and to certify the satisfactory completion of these activities in the Technical Log.

8.1.11.3.1 Pre-departure Inspections

Pre-flight/Transit Check

CAT.GEN.MPA.105

The Transit Check is the external check of the aircraft, or “walkaround”, completed by a qualified crew member prior to commencement of a flight. All Malta Air pilots are qualified to conduct a Transit Check. The conduct of the transit check is described in FCOM Vol 1. When the transit check has been completed the Captain will strike through “Pre-flight” and sign the Pre-Flight/Transit data field in the Technical Log.

The forward and rear hold shall always be opened by the Flight Crew member during the walkaround to preclude the possibility of inbound bags/freight not being offloaded.

He will pay particular attention to flight controls and gear and tyre condition; the removal of locks and covers, the high risk of damage by ground service vehicles and the possibility of fluid leaks from engine, hydraulic and fuel systems.

Malta Air Flight Crew members are qualified to perform the external transit checks. This check is certified completed by the Captain in the Technical Log.

8.1.11.4 Completion and Certification of Technical Log after Flight

The various sections of the Technical Log require the following to be completed by the Flight Crew and certified by the Commander after any flight:

8.1.11.4.1 Use of Aircraft Torches

Aircraft torches are emergency equipment and shall therefore be used in an emergency. They shall NOT be used for aircraft “walkaround”. Pilots are required to have their own serviceable torch. Use of aircraft torches shall be recorded in the Technical Log.

8.1.11.4.2 Radio Ahead Codes

When calling maintenance on VHF the crew should refer only to the code describing the affected system on the list of Radio Ahead Codes, e.g. “Station this is MAY 8TG, please advise Malta Air Engineering that we have a CODE 59”.

The Radio Ahead Code is in the Technical Log.

8.1.11.5 Technical Log Entries B737-800/-8200

	Ref No.	Recorded Data	Example																																		
<table><tr><td>FL Page / Sector</td><td colspan="3">1 632208 A</td></tr><tr><td>A/C Reg</td><td>9H-</td><td>2</td><td></td></tr><tr><td>Flt No / Date</td><td>3</td><td>Date</td><td>4</td></tr><tr><td rowspan="4">FLIGHT</td><td>To</td><td>5</td><td>Land</td></tr><tr><td>From</td><td>6</td><td>T/off</td></tr><tr><td>Cyc</td><td>7</td><td>hh mm</td></tr><tr><td>TG FS</td><td>10</td></tr><tr><td rowspan="3">Captain</td><td>Sign</td><td>11</td><td>Defect Raised / Nil</td></tr><tr><td></td><td>12</td><td>13</td></tr><tr><td>Print</td><td></td><td></td></tr></table>	FL Page / Sector	1 632208 A			A/C Reg	9H-	2		Flt No / Date	3	Date	4	FLIGHT	To	5	Land	From	6	T/off	Cyc	7	hh mm	TG FS	10	Captain	Sign	11	Defect Raised / Nil		12	13	Print			1	Technical Log page number	632208
	FL Page / Sector	1 632208 A																																			
	A/C Reg	9H-	2																																		
	Flt No / Date	3	Date	4																																	
	FLIGHT	To	5	Land																																	
		From	6	T/off																																	
		Cyc	7	hh mm																																	
		TG FS	10																																		
	Captain	Sign	11	Defect Raised / Nil																																	
			12	13																																	
		Print																																			
	2	Aircraft registration	ABC																																		
	3	Flight number	1234																																		
4	Date – DD/MM/YY	01/05/12																																			
5	To – 3 letter code	STN																																			
6	From – 3 letter code	DUB																																			
7	Circle and number of TG/FS as required	TG FS 1																																			
8	Land – landing time in UTC	08:30																																			
9	T/off – take-off time in UTC	07:30																																			
10	hh/mm – flight time this sector	1:00																																			
11	Sign – Captain’s signature	<i>J Bloggs</i>																																			
12	Print – Captain’s crew code	ABCDEF																																			
13	Circle as required	Defect Raised/ Nil																																			

					Ref No.	Recorded Data	Example
					14	Record arrival fuel	-----
					15	Record total arrival fuel	-----
					16	Prior Uplift (required if different to 14 above)	Leave blank if not required
					17	Total prior to uplift (sum of 16 above)	Leave blank if not required
					18	State SG (specific gravity) used to calculate uplift	Actual SG or default SG .79
					19	Enter expected uplift in Ltr/ Kg	-----
					20	Enter actual fuel uplift Ltr/ Kg	-----
					21	Record actual from gauges post uplift	-----
					22	Record total post uplift in kgs (sum of 21 above)	-----
					23	If adjustment to 22 is required, calculate adjustment in Ltrs and Kgs	Leave blank if not required
					24	Record actual from gauges (post adjustment)	Leave blank if not required
					25	Total of 24 above	Leave blank if not required
					26	Fluid type	75/25
					27	Fluid type	Type II
					28	Sign name	<i>J Bloggs</i>
					29	Record date	Leave blank if not required
					30	Record the time de-icing starts in UTC (for 2 nd step de-icing start timing at start of 2 nd step)	0600

FUEL Kgs		Fuel LH	Ctr	RH
Arrival	LH /Ctr / RH	14	14	14
	Total	Arrival Total 15 Kg		
Prior Uplift	LH /Ctr / RH	16	16	16
	Total	Prior Uplift Total 17 Kg		
Plan Uplift		sg 19	Ltr 19	Kg
Uplift		18	20	20 Kg
Totals	LH/Ctr/RH	21	21	21
	Total	Post Uplift Total 22 Kg		
Adjust		23	Ltr	23 Kg
Totals	LH/Ctr/RH	24	24	24
	Total	Adjusted Total 25 Kg		
Deice Mix/Type		Mix% 26	Type	27
Date / Start Time		Deice Sign 28	Start Time 30	
		Date 29		

					Ref No.	Recorded Data	Example																														
<table><tr><th>FLUIDS QTS</th><th>UPLIFT</th><th>TOTAL</th><th>UPLIFT</th><th>TOTAL</th></tr><tr><td>ENG</td><td>1 31</td><td>32</td><td>2 31</td><td>32</td></tr><tr><td>HYD</td><td>A 33</td><td>34 %</td><td>B 33</td><td>34 %</td></tr><tr><td>IDG</td><td>1 35</td><td>36</td><td>2 35</td><td>36</td></tr><tr><td>APU</td><td>37</td><td>38</td><td>Sign</td><td>39</td></tr><tr><td>Capt Accept</td><td colspan="4">For flight 40 check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight 41 Captain: Sign: 42 Date:</td></tr></table>					FLUIDS QTS	UPLIFT	TOTAL	UPLIFT	TOTAL	ENG	1 31	32	2 31	32	HYD	A 33	34 %	B 33	34 %	IDG	1 35	36	2 35	36	APU	37	38	Sign	39	Capt Accept	For flight 40 check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight 41 Captain: Sign: 42 Date:				31	State amount uplifted if applicable quarts). If no uplift leave blank	2
FLUIDS QTS	UPLIFT	TOTAL	UPLIFT	TOTAL																																	
ENG	1 31	32	2 31	32																																	
HYD	A 33	34 %	B 33	34 %																																	
IDG	1 35	36	2 35	36																																	
APU	37	38	Sign	39																																	
Capt Accept	For flight 40 check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight 41 Captain: Sign: 42 Date:																																				
					32	Record indicated amount (quarts)	18																														
					33	State amount uplifted if applicable (quarts). If no uplift leave blank	4																														
					34	Record indicated amount (%)	98																														
					35	Engineering use only. Leave blank	See <i>Note</i>																														
					36	Engineering use only. Leave blank	See <i>Note</i>																														
					37	Engineering use only. Leave blank	See <i>Note</i>																														
					38	Engineering use only. Leave blank	See <i>Note</i>																														
					39	Engineer's or Captain's signature as appropriate	<i>J Bloggs</i> See <i>Note</i>																														
					40	Flight number	1234																														
					41	Tick one to confirm Pre Flight OR Transit checks have been completed	√ See <i>Note</i>																														
					42	The Captain's crewcode, acceptance signature and date required for each sector operated	<i>J Bloggs</i> ABCDEF J Bloggs 01/05/12 See <i>Note</i>																														

Note: The aircraft Captain will sign for the Engine and Hydraulic fluid quantities for all sectors during his/her flight duty unless already signed for by an engineer.

IDG and APU quantities are for engineering use only.

Aircraft Captains are reminded that by signing data field **42** they are confirming the following has been checked prior to departure:

- That a Preflight has been carried out.
Tick box as appropriate.
- That a check of WO, HIL, APS, Fuel, Oil and Hydraulic quantities are complete.

	Ref No.	Recorded Data	Example																																																		
<table><tr><td colspan="2">Sign</td><td colspan="2">Sign</td><td></td></tr><tr><td>Cap't Account</td><td>For flight check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight</td><td>For flight check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight</td><td>For flight check of WO, HIL, AP Hydraulics and of Pre Flight</td><td></td></tr><tr><td>Captain: Sign: Date:</td><td></td><td>Captain: Sign: Date:</td><td>Captain: Sign: Date:</td><td></td></tr><tr><td colspan="5">***** NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST BE SET AT 77,990 KGS</td></tr><tr><td>MTOW ASD 1499</td><td>CRS Sign/Stamp</td><td>CRS Sign/Stamp</td><td></td><td></td></tr><tr><td>SB 737-11-1237</td><td>66,990 Kgs</td><td>66,990 Kgs</td><td>66,990 Kgs</td><td></td></tr><tr><td>SB 737-11-1236</td><td>69,990 Kgs</td><td>69,990 Kgs</td><td>69,990 Kgs</td><td></td></tr><tr><td>SB 737-11-1257</td><td>71,990 Kgs</td><td>71,990 Kgs</td><td>71,990 Kgs</td><td></td></tr><tr><td>SB 737-11-1235</td><td>74,990 Kgs</td><td>74,990 Kgs</td><td>74,990 Kgs</td><td></td></tr><tr><td>SB 737-11-1238</td><td>77,990 Kgs</td><td colspan="3">NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST BE SET AT 77,990 KGS</td></tr></table>	Sign		Sign			Cap't Account	For flight check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight	For flight check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight	For flight check of WO, HIL, AP Hydraulics and of Pre Flight		Captain: Sign: Date:		Captain: Sign: Date:	Captain: Sign: Date:		***** NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST BE SET AT 77,990 KGS					MTOW ASD 1499	CRS Sign/Stamp	CRS Sign/Stamp			SB 737-11-1237	66,990 Kgs	66,990 Kgs	66,990 Kgs		SB 737-11-1236	69,990 Kgs	69,990 Kgs	69,990 Kgs		SB 737-11-1257	71,990 Kgs	71,990 Kgs	71,990 Kgs		SB 737-11-1235	74,990 Kgs	74,990 Kgs	74,990 Kgs		SB 737-11-1238	77,990 Kgs	NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST BE SET AT 77,990 KGS			43	MTOW – tick box. Tick weight box for the current sector ONLY when weight change required	✓ See <i>Note</i>
Sign		Sign																																																			
Cap't Account	For flight check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight	For flight check of WO, HIL, APS, Fuel, Oil, Hydraulics and Completion of Pre Flight	For flight check of WO, HIL, AP Hydraulics and of Pre Flight																																																		
Captain: Sign: Date:		Captain: Sign: Date:	Captain: Sign: Date:																																																		
***** NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST BE SET AT 77,990 KGS																																																					
MTOW ASD 1499	CRS Sign/Stamp	CRS Sign/Stamp																																																			
SB 737-11-1237	66,990 Kgs	66,990 Kgs	66,990 Kgs																																																		
SB 737-11-1236	69,990 Kgs	69,990 Kgs	69,990 Kgs																																																		
SB 737-11-1257	71,990 Kgs	71,990 Kgs	71,990 Kgs																																																		
SB 737-11-1235	74,990 Kgs	74,990 Kgs	74,990 Kgs																																																		
SB 737-11-1238	77,990 Kgs	NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST BE SET AT 77,990 KGS																																																			
Note: During a weight change, by ticking the appropriate box, commanders are indicating the specific SB which authorises operations at that weight.	44	Company authorisation number, Captain's signature and license number <u>OR</u> Engineers stamp and company authorisation number. SB not required to be printed.	<i>J. Bloggs</i> IE.FCL.123456 IE.145.004.123 See <i>Note</i>																																																		

Note: Refer to OMA 8.1.11.9 and Aircraft Weight Certificate (EFB Cert File) regarding permissible weight changes as per approved aircraft variable weights programme.

	Recorded Data	Example
1	Ref number	12345
2	Aircraft Reg	9H-ABC
3	Engineering	-----
4	MM/DD/YY	-----
5	Tech Log Ref	12345
6	Sign	-----
7	Tick pilot box	-----
8	Tick relevant box	-----
9	ATA	-----
10	If required tick	-----
11	Detail defect	-----
12	Engineering	-----
13	Engineering	-----
14	Engineering action	Closure/HIL
15	Engineering codes	-----
16	Downgrade status	-----
17	Downgrade status	-----
18	Transfer/HIL WO	-----
19	MEL Category	-----
20	Station	-----
21	Reason	-----
22	Due Date	-----
23	Due Hours	-----
24	Due Cycles	-----
25	CRS stamp and Signature	-----

CRS NUMBER/WORK ORDER NUMBER (Engineering from This Page)													
1 Date		2 FLY PAGE REF		3 Issued By		4 PILOT/ENGINEER / SIGNATURE		5 DOWNGRADE / CODE		6 ATA		7 DISCLOSURE SIGN	
4		6		7		8		9		10		11	
General Description (Please print and enter as much detail as possible) Only One Defect Per Page:													12
Main Message: 13													
Closure/HIL Action (Please print): 14													
FM / AMM / ERM CODE				REV		MEL / DOWNGRADE CODE							
15				16		17							
Transfer / HIL WO : 18		Reason		Due Date		Due PM		Due TAD		Due Date		25	
19		20		21		22		23		24		26	

[illegible]



Technical Log
Boeing 737
Doc. No. MAY909
Rev 2 08th Feb 2021

9H-

NOTICE TO CREW

OPENED				CLOSED		
No.	Technical details	Date	Sign	Details	Date	Sign
1	2	3	4	5	6	7

	Record	Example
1	Technical Log number	1
2	Aircraft MTOW/ Door code	66990/1234
3	Date opened	
4	Engineer signature	
5	Closure of NOTOC	Weight change
6	Date closed	
7	Engineer signature	

	Technical Log Boeing 737 Doc. No. MAY906 Rev. 03 – 13/06/2023	Paper Auto Coupled Approach Report must be completed in addition to the Auto Coupled Approach Report Form in Docunet				AIRCRAFT REGISTRATION
3.3 – AUTO COUPLED APPROACH REPORT Successful or Unsuccessful Approaches						
DATE	FLIGHT NUMBER	CAPTAIN CREWCODE	AIRPORT ICAO CODE	SATISFACTORY APPROACH (Strikethrough as required)		ADDITIONAL INFORMATION
1	2	3	4	YES	NO	5
				YES	NO	
				YES	NO	

	Record	Example
1	Inbound Date	dd/mm/yy
2	Inbound Flight Number	FR1234
3	Captain CREWCODE	
4	Arrival airport	LMML
5	More info if relevant	More info if relevant

8.1.11.6 Recording of Defects

CAT.GEN.MPA.105 (a) (14)

The Commander shall record, at the termination of the flight, utilisation data and all known or suspected defects of the aircraft in the aircraft technical log of the aircraft to ensure continued flight safety.

The accurate recording of defects that have occurred during the flight is of paramount importance. Historically, fatal accidents have occurred as a result of an Aircraft Commanders failure to record the full details of a technical event allowing engineering to make an informed decision as to the airworthiness of the aircraft.

The following types of event shall be recorded in detail in the aircraft technical log.

- Any event that resulted in a PAN or Mayday as a result of technical issue.
- Any event where safety equipment was used.
- Any event where crew oxygen was used.
- Any stick shaker event that was not attributable to environmental conditions.
- Any exceedance as prescribed in the FCOM limitations section.
- Any fire warning.
- Any overheat warning.
- Any Failure of the pressurisation system (even if resettable).
- Any Event where the crew were required to reference the Boeing QRH for a technical reason.
- Any event where flight control issues were experienced.
- Any Severe turbulence encounter.

The above list is not exhaustive Commander's shall record any technical event deemed appropriate which has the potential to impact the safe operation of the aircraft due to a technical deficiency.

Commanders must ensure that all symptoms, indications and effects of the failure are recorded in the technical log and if necessary, communicated by phone to maintenance control.

Commanders should note it is not permissible to defer the recording of aircraft defects. The decision rest with maintenance control as to whether a HIL may be raised or the fault rectified before the next flight.

8.1.11.6.1 Dispatch with Defects at an Outstation

Malta Air Engineering has approval from the TM CAD (Airworthiness) for the Commander to release an aircraft for service at an outstation without a Malta Air engineer licensed on type subject to the conditions below:

- The captain shall always in the first instance advise MAINTROL in DUB of the defect.
- If the defect can be deferred with reference to the MEL, Maintrol will instruct the Captain how to complete the relevant work order and HIL item.
- Maintrol will record verbatim the HIL description electronically.
- The crew shall take no maintenance action other than to reset a circuit breaker, insert a cockpit placard*, or conduct a BITE reset in the cockpit (not the E&E bay) on the instructions of maintrol.
- The defect is to be reviewed by an appropriately approved engineer on return to base, and either rectified or formally raised as a deferred defect by the engineer.

* A placard shall take the form of an INOP sticker unless specified. Pilots are NOT authorised to alter the CAT I/II/III status

This procedure is only intended for use when maintrol cannot make available a suitable engineer and the defect meets all the requirements for dispatch according to the relevant MEL.

8.1.11.6.1.1 Technical Delays – Boarding

Crews shall embark the passengers on to the aircraft unless a technical defect renders it unsafe to board, reduces passenger comfort beyond acceptable levels, or is not possible due to fuelling and boarding regulations.

In general wheel changes, seat repairs, brake changes, birdstrike inspections, minor system failures, flight deck cautions etc are acceptable in terms of boarding unless engineering specifically directs that boarding is delayed.

All boarding delays shall be noted on the Electronic Flight Log/Paper Voyage report (Refer to Section 8.1.10.6).

8.1.11.7 Stowage of Technical Log

The CORRECT AIRCRAFT TECHNICAL LOG will be stowed behind the

Captain's seat, or in a bracket provided on the flight deck for take-off and landing in such a way that it cannot become a hazard in the flight deck.

8.1.11.8 Cabin Defects Log

To ensure airworthiness related items are transferred to the Aircraft Technical Log the Commander shall at the end of each flight, receive the Cabin defects Log from the SCCM. The Commander shall transfer any airworthiness or safety related items in the Aircraft Technical Log, (e.g. defective/missing safety equipment, cabin or emergency lighting deficiencies, missing/torn decals, damaged/malfunctioning seats, etc.). Engineering shall clear/defer defects in accordance with MEL/DDG.

8.1.11.9 Aircraft Variable Weights Programme**Boeing 737-800**

Boeing 737-800 are assigned to one of three different certified Maximum Take Off Weights (MTOWs):

1. 71,990 kg;
2. 74,990 kg; or
3. 77,990 kg.

In accordance with Boeing Service Bulletins SB 737 11 1237, SB 737 11 1236, SB 737-11-1257 and SB 737-11-1235, B737-800W aircraft listed in the Aircraft Weight Certificate as having a certified MTOW of either 74,990 kgs or 71,990 kgs may be operated at variable MTOWs of 69,990 kgs or 66,990 kgs.

B737-800W aircraft with a certified MTOW of 77,990 kgs are not part of the variable weights program and will operate with a fixed MTOW of 77,990 kgs and have:

1. An entry in the Notice to crew section of the Technical Log stating “MAXIMUM TAKEOFF WEIGHT IS 77,990 KGS”; and
2. A single fixed 77,990 kgs weight placard.

Boeing 737-8200

Boeing 737-8200 are assigned one Maximum Take Off Weight (MTOW) of 76,200 kg.

In accordance with Boeing Service Bulletins SB 737 11 1314, SB 737 11 1315, SB 737-11-1316, B737-8200 aircraft listed in the Aircraft Weight Certificate as having a certified MTOW of 76, 200 kgs may be operated at variable MTOWs of 72,600 kgs and 69,000 kgs.

A Captain of a Malta Air aircraft may certify a change to the Certificated Maximum Take-off Weight at an unsupported station under the authorisation issued by the quality system of Ryanair DAC Part 145 AMO.

In order for a Commander to be authorised to certify the release to service of an aircraft following a weight change at an unsupported station the Commander shall:

1. Hold a valid ATPL on aircraft type.
2. Be over 21 years old.
3. Have received adequate training approved by the Ryanair DAC EASA Part-145 Maintenance Organisation.
4. Have completed procedural training as specified in Part 145.
5. Have completed maintenance airworthiness regulation training.
6. Have their Personal Authorisation Certificate (PAC) endorsed by the Ryanair DAC ATO Head of Training.

The authorisation will be valid for 12 months subject to the satisfactory re-current training on the aircraft type. A list of Engineering supported stations (RA253 for 737-800 or RA253M for 737-8200) is contained in the Technical Log and the Electronic Flight Bag (EFB).

The eLearning Weight Change module and the PAC (Personal Authorisation Certificate) comply with the requirements as set out in 3, 4, 5 and 6 above.

8.1.11.10 Engineering Tasks Permitted to be Carried Out by Commanders

There are a limited number of tasks normally carried out by engineers that may be carried out by Commanders who have received adequate approved training. The tasks are carried out in accordance to Ryanair DAC MOE.

Flight Crew authorised in accordance with Ryanair DAC MOE 3.9.7 may perform the below set of tasks:

- Pre-flight Inspection iaw MAY CAME 1.11
- Birdstrike Inspection iaw DAC MOE L2.2.6
- Change of Maximum Take-off Weights iaw DAC MOE L2.2.5 and Malta Air CAME Appendix A LB-02-008 Multiple Operating Weights
- Deferral of Defects iaw Malta Air CAME 1.8.6

8.1.11.10.1 Weight Change by Commander

The commander may certify a change to the Maximum Take-off Weight (MTOW) IAW Ryanair DAC MOE 3.9.7.3 and Malta Air CAME Appendix A LB-02-008 Multiple Operating Weights or subsequent revision at an unsupported station only. A list of Engineering supported stations is included in the Technical Log. If a suitably qualified engineer is present at a station it is 'supported', and crews are not permitted to do weight changes. If a suitably qualified engineer is not present at the station (not rostered, not qualified, at another base etc.) it is 'unsupported' and commanders are permitted to carry out weight changes.

When a qualified engineer is present at station he/she will always complete the weight change, if there is no qualified engineer on station the qualified Commander will make the weight change.

STEP 1.

The Commander decides on the lowest MTOW required for each sector. When an adjustment and recertification is needed in the Technical Log, the weight placard must also reflect the dispatch MTOW. The Commander shall check the MTOW and Notice to Crew section of the Technical Log for prohibitions on changing the MTOW of the aircraft. In the MTOW adjustment section the Commander ticks the box beside the dispatch MTOW. The Commander will sign under the CRS Sign/Stamp heading and shall include the company authorisation number (IE.145.004) and ATPL license number (e.g. IE.145.004 IRL/AT/123456A/A).

***** NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST NOT BE ADJUSTED *****												
MTOW ADJ IAW			CRS Sign/Stamp			CRS Sign/Stamp			CRS Sign/Stamp			CRS Sign / Stamp
SB 737-11-1237	66,990 Kgs			66,990 Kgs			66,990 Kgs			66,990 Kgs		
SB 737-11-1236	69,990 Kgs			69,990 Kgs			69,990 Kgs			69,990 Kgs		
SB 737-11-1257	71,990 Kgs			71,990 Kgs			71,990 Kgs			71,990 Kgs		
SB 737-11-1235	74,990 Kgs			74,990 Kgs			74,990 Kgs			74,990 Kgs		
SB 737-11-1238	77,990 Kgs											

Form RA001 Issue 22, October 2015

The Commanders signature, company authorisation number and license number is required when a MTOW change is made.

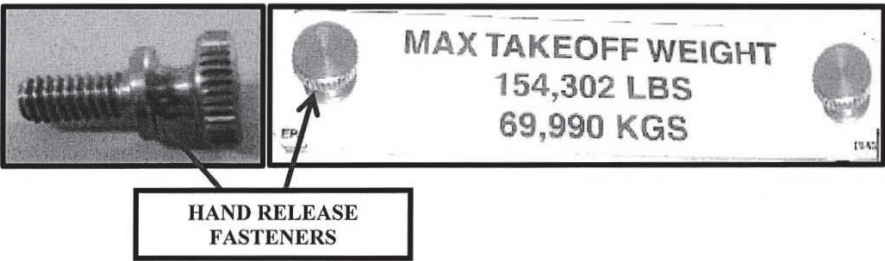
- Note:**
- Should an entry be made in error in this section of the Technical Log (including a weight change box ticked in error) the associated CRS certificate must be completed in full.
 - It is not necessary to make any entry in the weight change box section of the Technical Log unless a weight change is required.

STEP 2.

Update the weight placard to show the new MTOW (located on the right hand side of the flight deck entrance above the observer O₂ Mask).

Two screw fasteners hold the MTOW placard in place which are removed and replaced by hand.

Note: No tooling is required.



There are 2 plates with three placards. The alternate placard is stowed behind either of the plates currently displaying the MTOW.

B737-800:

MTOW placards display 66,990 kgs, 69,990 kgs, 71,990 kgs and 74,990 kgs. Select as applicable.

MAX TAKEOFF WEIGHT 147,688 LBS 66,990 KGS	MAX TAKEOFF WEIGHT 152,302 LBS 69,990 KGS	MAX TAKEOFF WEIGHT 158,711 LBS 71,990 KGS	MAX TAKEOFF WEIGHT 165,325 LBS 74,990 KGS
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B737-800 MTOW	Boeing Service Bulletin
66,990 kg	SB 737-11-1237
69,990 kg	SB 737-11-1236
71,990 kg	SB 737-11-1257
74,990 kg	SB 737-11-1235

B737-8200:

MTOW placards display 69,000 kgs, 72,600 kgs and 76,200 kgs. Select as applicable.

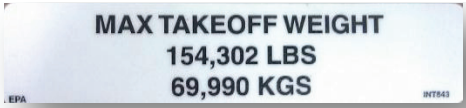
MAX TAKEOFF WEIGHT 152,119 LBS 69,000 KGS	MAX TAKEOFF WEIGHT 160,056 LBS 72,600 KGS	MAX TAKEOFF WEIGHT 167,993 LBS 76,200 KGS
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B737-8200 MTOW	Boeing Service Bulletin
69,000 kg	SB 737-11-1314
72,600 kg	SB 737-11-1315
76,200 kg	SB 737-11-1316

One placard displays 66,990 kgs. SB 737-11-1237.



One placard displays 69,990 kgs. SB 737-11-1236.



One placard displays 74,990 kgs. SB 737-11-1235 (on the reverse of any of the above plates).



8.1.11.10.2 Bird Strike Inspection

Training Requirements: Aircraft commanders have completed the bird strike inspection training as described in Ryanair DAC MOE 3.9.

Maintenance Control must be contacted to determine if the aircraft is at an unsupported station. Commander may complete a bird strike inspection of an aircraft in accordance with RA 210 if instructed to do so by engineering.

8.1.11.10.3 Raising of a HIL Malta Air CAME 1.8.6

In the event of an aircraft crew discovering that they have a defect at a station where there is no engineering support on site, the aircraft commander will contact Line Maintenance Control in Dublin and discuss the extent of the defect as evidenced by cockpit indications and/or BITE messages available to the crew.

***** NOTE IF AIRCRAFT MTOW IS SET AT 77,990 KGS THE WEIGHT MUST NOT BE ADJUSTED *****											
MTOW ADJ IAW			CRS Sign/Stamp			CRS Sign/Stamp			CRS Sign/Stamp		
SB 737-11-1237	66,990 Kgs			66,990 Kgs			66,990 Kgs			66,990 Kgs	
SB 737-11-1236	69,990 Kgs			69,990 Kgs			69,990 Kgs			69,990 Kgs	
SB 737-11-1235	71,990 Kgs			71,990 Kgs			71,990 Kgs			71,990 Kgs	
SB 737-11-1235	74,990 Kgs			74,990 Kgs			74,990 Kgs			74,990 Kgs	
SB 737-11-1238 / 737-11-1302		77,990 kgs		NOTE: IF AIRCRAFT MTOW IS SET AT 77,990KGS THE WEIGHT MUST NOT BE ADJUSTED							

8.1.11.10.4 Affixing a Placard (Inoperative Sticker)

Commanders may affix inoperative stickers to indicate a system is unserviceable upon the instruction of an engineer, they may not change Aircraft autoland category placards.

8.1.11.10.5 Resetting of a Circuit Breaker(s)

Commanders may upon the instructions of an Engineer pull/reset circuit breakers on the ground for diagnostic purposes, they may not collar or inhibit systems by pulling circuit breakers.

8.1.11.10.6 Issue a CRS

Commanders may issue a certificate of release to service for specific maintenance actions **ONLY** these are:

1. Bird-strike inspection
2. Weight Change

8.1.11.10.7 Ground Running of Engines for Diagnostic Purpose

Commanders may at the request of Engineering with an Engineer present run an engine at idle thrust to enable check to me made Engineer Procedure Notice 18.03 refers.

Note: This is not the same as the engine fuel leak procedure defined in FCOM VOL 1 NP 21.50. NP 21.50 is a flight crew procedure not an engineering procedure.

8.1.11.10.8 Exercising of Flight Control or System

The Commander may at the request of Engineering exercise any control or operate any switch on the flight deck for the purposes of validating whether or not a system is inoperative or to reproduce a fault with the aircraft on the ground.

Should a commander be unsure as to whether he is authorised to carry out a procedure requested by Engineering then he shall in the first instance contact the duty pilot. (+353 1248 0557 Option as per AOC then Option 9).

8.1.12 Documents, Manuals and Information to be Carried

CAT.GEN.MPA.180

1. The following documents, manuals and information shall be carried on each Malta Air flight, as originals or copies unless otherwise specified:
 - a. the Aircraft Flight Manual (AFM), or equivalent document(s);*
 - b. the original Certificate of Registration;*
 - c. the original Certificate of Airworthiness (CofA);*
 - d. Airworthiness Review Certificate;*
 - e. the original Noise Certificate*
 - f. a certified true copy of the Air Operator Certificate (AOC);*
 - g. the Operations Specifications relevant to the aircraft type, issued with the AOC;*
 - h. the original Aircraft (Radio) Station Licence, if applicable;*
 - i. the third party liability insurance certificate(s);*
 - j. the journey log for the aircraft;*
 - k. the aircraft technical log, in accordance with Annex I (Part-M) to Regulation (EU) No 1321/2014;
 - l. details of the filed ATS flight plan, contained in the LIDO OFP*;
 - m. current and suitable aeronautical charts for the route of the proposed flight and all routes along which it is reasonable to expect that the flight may be diverted (Jeppesen Charts);*
 - n. procedures and visual signals information for use by intercepting and intercepted aircraft (Jeppesen Publications (Pubs) > ICAO Rules of the Air – Annex 2);*
 - o. information concerning search and rescue services for the area of the intended flight, which shall be easily accessible in the flight crew compartment (Jeppesen Publications (Pubs) > Airway Manual);*
 - p. the current parts of the operations manual that are relevant to the duties of the crew members, which shall be easily accessible to the crew members;*
 - q. the MEL;*
 - r. appropriate notices to airmen (NOTAMs) and Aeronautical Information Service (AIS) briefing documentation;*
 - s. appropriate meteorological information;*
 - t. mass and balance documentation;*

- u. the operational flight plan;*
- v. notification of Special Categories of Passenger (SCPs) and special loads; and
- w. any other documentation that may be pertinent to the flight or is required by the States concerned with the flight.

Note: * Item so marked may be carried in an electronic format only.

2. **Documents, Manuals and Information to be Carried Electronic format:** AMC1 CAT.GEN.MPA.180

The documents, manuals and information may be available in a form other than on printed paper, accessibility, usability and reliability shall be assured.

3. Notwithstanding (1), in the case of loss or theft of documents specified in 1(b) to 1 (h), the operation may continue until the flight reaches its destination or a place where replacement documents can be provided.

The Specified Documentation is Located as Indicated in the Tables Below

Title	Location	Req'd for dispatch	Amended by
AFM (Airplane Flight Manual), incl CDL (Configuration Deviation List)	EFB	1	Engineering
FCOM Vol 1, 2	EFB	1 copy of each	Flt Ops
QRH	Flight Deck	1	Flt Ops
Operations Manual Part A	EFB	1	Flt Ops
Current FCIs	EFB	1 copy of all current FCIs	Flt Ops
SEP Manual	EFB/Flight Deck	1* (electronic or paper)	Flt Ops
Cert File	Flight deck	1*	Engineering Flt Ops

Title	Location	Req'd for dispatch	Amended by
Technical Log	Flight deck	1	Engineering
MEL (and approved extracts)	EFB – Docunet	1	Engineering
Laminated Normal Checklist	Flight Deck	1 (may be QRH checklist)	Flt Ops
Laminated Winter Ops Checklist	Flight Deck	0	Flt Ops

Note: * In case of loss or theft of documents * above, and Pilot Licences and ratings, the operation is allowed to continue until the flight reaches the base or a place where a replacement document can be provided.

Table 4 Cert File

Document	Req'd for Dispatch	Amended By	Electronic Location
Certificate of Registration*	1	Engineering	Docunet>Cert File
Certificate of Airworthiness*	1	Engineering	Docunet>Cert File
Airworthiness Review Certificate*	1	Engineering	Docunet>Cert File
Noise Certificate*	1	Engineering	Docunet>Cert File
Air Operators Certificate (AOC)*	1	Flt Ops	Docunet>Cert File
Mel Approval Certificate*	1	Flt Ops	Docunet>Cert File
Wireless telegraph (Aircraft Station Licence)*	1	Engineering	Docunet>Cert File
Certificate of Third party Insurance Liability*	1	Flt Ops	Docunet>Cert File
Aircraft Weight Certificate*	1	Flt Ops	Docunet>Cert File
AFM Status Sheet*	1	Engineering	Docunet>Cert File
Engineering Software Configuration Sheet	1	Engineering	n/a
Aircraft Radio Apparatus Status Sheet	1	Engineering	Docunet>Cert File

Note: * Item so marked may be carried in an electronic format only.

8.1.12.1 Information to be Retained on the Ground

CAT.GEN.MPA.185

1. Malta Air ensures that at least for the duration of each flight or series of flights:
 - a. information relevant to the flight and appropriate for the type of operation is preserved on the ground;
 - b. the information is retained until it has been duplicated at the place at which it will be stored;
2. The information referred to in (1) includes:
 - a. a copy of the operational flight plan;
 - b. copies of the relevant part(s) of the aircraft technical log;
 - c. route-specific NOTAM documentation if specifically edited Malta Air;
 - d. mass and balance documentation (or electronic equivalent); and
 - e. special loads notification.
3. A copy of the NOTOC shall be transmitted to:
 - a. The destination airport;
 - b. Flight Operations in Dublin;
 - c. A copy shall be returned to Flight Operations via EFL (or in the returned Flight Envelope);
 - d. The alternate airport in the event of a diversion and retained by the addressees for the duration of the flight. Information to Passengers and other Persons is provided for in the Ground Operations Manual and SEP Manual.

8.1.12.2 Documents, Manuals and Information to be Carried Electronic format

AMC1 CAT.GEN.MPA.180

The documents, manuals and information may be available in a form other than on printed paper, accessibility, usability and reliability shall be assured.

8.1.12.3 Documents, Manuals and Information to be Carried the Air Operator Certificate

Certified true copies are provided directly by the TM CAD.

8.1.12.4 Procedures and Visual Signals for use by Intercepting and Intercepted Aircraft

AMC1 CAT.GEN.MPA.180(a)(13)

The procedures and the visual signals for use by intercepting and intercepted aircraft reflect those contained in the International Civil Aviation Organisation (ICAO) Annex 2. This is part of the operations manual, OMC Jeppesen General Airway Manual.

8.1.12.5 Personal Documents and Information to be Carried

Flight Crew:

- Valid Pilots Licence with Type Rating.
- Instrument Ratings.
- R/T Licence (included in ATPL).
- Passport and Visas if required.
- Vaccination certs. if required.
- Current Medical Certificate.

Cabin Crew:

- Passport and visas if required.
- Vaccination certs. if required.
- ATQV or cert of competency or Licence.

8.1.12.5.1 Action in the Event of Lost/Stolen Documents – Flight Crew

Should a crew member's flight documentation be lost or stolen it is permitted for that crew member to operate for a maximum of 1 flight duty period in accordance with the procedure set out below.

- Inform the TM CAD or relevant aviation authority immediately to organise replacement documents and obtain a time frame for replacement.
- Lost documents shall be considered stolen, inform local police authority.
- Inform crew control of the loss/theft and provide a time frame for replacement.
- Print a copy of pilot licence and medical from Electronic record keeping system.
- The printed documents will include;
 1. Licence Parts I – XII.

2. Validity, Radiotelephony Privileges & Remarks.
 3. Certificate of Issue, Revalidation or Renewal.
 4. Medical Certificate.
 5. Validation (if applicable).
- The crew member shall inform crew control as soon as replacement documents have been received and ensure revised documents are uploaded to Electronic record keeping system.

8.2 Ground Handling Instructions

8.2.0 General

8.2.0.1 On-time Departure – Punctuality

Departures "on-time" is a key measure of the performance of a scheduled airline.

BLOCK TIME is defined as the time between an aeroplane first moving from its parking place for the purpose of taking off until it comes to rest on the designated parking position and all engines are stopped.

The Commander shall state 'OFF-BLOCKS hh.mm' as the time movement commences with pushback (or taxi for self-maneuvring operations) for the purpose of taking off and this shall be recorded on the Electronic Flight Log/Paper Voyage Report.

The Commander shall state 'ON-BLOCKS hh.mm' which shall be recorded on the Electronic Flight Log/Paper Voyage Report as the time when the last engine decelerates through 20% N2, (when pilots select ANTI-COLLISION Light Switch OFF). When Crewlink entries are made by the crew these shall be the times recorded on the Electronic Flight Log/Paper VR as stated by the Commander.

ARMD movement times are electronically recorded and uploaded on Crewlink/Crew Management System in the following hierarchy:

1. Crew entries;
2. Aircraft automatic recording;
3. Station SITA message.

From the above hierarchy crews can override ARMD entries. When the crew record is different than the ARMD record this shall be reported via Zendesk.

It is Malta Air policy to depart early when possible and departures ahead of schedule are permitted and should be anticipated. To achieve an on-time departure requires timely completion of a series of aircraft servicing and passenger handling activities. Commanders shall monitor these activities and use their authority to influence and facilitate the process.

Safety is our primary operational objective and crews shall resist any propensity to accelerate or rush the normal delivery of pre-flight/before start activities to avoid a delayed departure time. The integrity of checklist completion must never be compromised.

All components of a delayed departure shall be recorded on the Electronic Flight Log using the IATA Delay Codes.

8.2.0.2 Passenger Boarding Without Flight Crew on Board

Boarding without flight crew will **only** commence when the Cabin Crew have been instructed to do so by Operations Control and is only permitted if the following conditions can be met:

- Flight crew estimated arrival time is expected to be under 20 minutes.
- Fuelling is not taking place.
- Cabin crew have established the outside air temperature to be 25 degrees or less.
- The APU is off.
- Ground power is connected and running.
- Mobile steps must remain in place until at least one flight crew arrives.
- Cabin lights are illuminated.

Designated GHA Duties

- The 'Designated GHA' (DGHA) is identified to the CSS thereby confirming both are aware that this is a 'boarding without flight crew onboard' procedure.
- The DGHA shall have a serviceable radio and/or a mobile phone.
- The DGHA remains on station, within 20 m of the aircraft L1 door, in continuous line of sight of the CSS position and ready to act without delay until at least one flight crew arrives on board.
- The DGHA will alert the CSS to any event outside the aircraft which may affect the safety of passengers, crew or the aircraft.

- The DGHA will without delay, summon emergency or airport services required either by the developing situation at the aircraft or as instructed by the CSS for any situation on board the aircraft.

CSS Duties

- CSS is responsible for aircraft safety until arrival of the Flight Crew and shall have authority to give all commands he/she deems necessary for the purpose of securing the safety of the aircraft and of persons or property carried therein.
- All persons carried in the Aircraft shall obey such commands.
- The CSS will maintain the DGHA in line of sight during boarding and when boarding is completed until at least one flight crew member is on board.
- The CSS will without delay, alert and brief the DGHA when there is a situation on board requiring any intervention from services on the ground following the PAA Brief concept.

PAA

- Problem: State the problem to the DGHA.
- Action: (Stay onboard, disembarkation, rapid disembarkation, evacuation).
- Additional Information: Airport Emergency Services (Fire/Medical/Police etc.) required/not required.

The CSS Shall on Arriving at the Aircraft:

1. Shall select the Ground Service button to “on” in order to illuminate cabin lights.
2. Inform the Dispatcher or available Ground Crew member that Flight Crew is not present and that the boarding without flight crew procedure will be applied.

Embarkation

1. The Cabin crew will complete a security search, the CSS will record the time this is completed and inform the Commander when he arrives onboard the aircraft.
2. The CSS will brief the Cabin Crew on the alternative means of communication e.g. Loudhailers if an urgent situation develops as the Cabin Crew interphone will not be available.
3. The CSS will indicate assistance is required by calling or waving both arms, the DGHA goes to the vicinity of the L1 door to receive a briefing (PAA format).

Arrival of Flight Crew

1. The CSS will commence the boarding PA, this confirms that the flight crew are onboard and completes PDIs on the interphone.
2. The CSS shall confirm with the Commander that all PDIs have been completed and the time at which the security search was completed.
3. The Flight Crew complete the pre-flight briefing.
4. The CSS passes required information to the remaining Cabin Crew before the departure.
5. The CSS alerts the DGHA and confirms that the boarding without flight crew procedure is completed.

Note 1: Crew on Airport Duty may board passengers without Flight Crew. The Airport Duty crew must be from the same AOC as the Aircraft or qualified to operate on the AOC of the aircraft.

Note 2: In all cases, a full crew complement, including a CSS must be present to complete PDIs and Security Searches. The CSS will assign roles to the crew.

Note 3: PDIs and Security Searches may be completed by Airport Duty Crew.

Note 4: Crew on Airport Duty shall confirm with the operating CSS that all PDIs have been completed and the time at which the security search was completed.

Note 5: The operating Crew may take over once all PDIs and Security Search have been completed.

8.2.0.3 Operations on the Ground with APU Inoperative and High Outside Air Temperature

The MEL for the B737-800/8200 allows for dispatch with the APU inoperative. Dispatch with the APU – INOP presents additional consideration for engine starting and control of temperature in the cabin.

Good communication and briefing between flight crew, cabin crew and ground crew is essential when an aircraft is required to operate with an APU – INOP. Operations Control Centre will make best efforts to allocate the lines of flying to the most suitable destinations. Ideally APU – INOP aircraft will be allocated to two sector duties limiting the exposure to the number of turnarounds.

The temperature in the cabin can rise very quickly with passengers on board even in comparatively mild temperatures. There have been several significant disruptive passenger incidents caused by rising cabin temperatures that could have been avoided with proactive management, good communications and forward planning.

Handling agents will be notified in advance by OCC that the arriving aircrafts APU is INOP, however crews should call ahead and ensure the following are available:

- Ground Power ready
- ASU ready
- Ground source air-conditioning

The following restrictions apply to an aircraft with an inoperative APU:

- The Commander shall not depart to a destination CO-NOTAM'd as having "No ASU".
- If departing to a destination where the temperature is expected to be higher than 30 degrees centigrade then the commander shall contact OCC to confirm that ground source conditioned air is available.

Arrival

Taxing in, crews should take every opportunity to cool the cabin as much as possible, flight crew should ensure that all cabin temperature selectors are selected to maximize cooling – C. In the case of direct sunlight entering the cabin, the Commander should instruct cabin crew to lower window blinds on the effected side after disembarkation when completing the security checks.

On Stand

Commanders should request that ground source air conditioning where available be connected as soon as possible when ambient temperatures are above 25 degrees centigrade. If fixed base air-conditioning is not available, then a mobile unit should be requested. Crews should note that if an air-conditioning unit is not available then certain air start have the ability to be used for cabin air conditioning, GOPs will advise.

Boarding

Maximize the use of the ground sources air-conditioning. In ambient temperatures above 25 degrees ensure that air conditioning is switched on prior to boarding. Commanders should ensure that cabin temperature is maintained by reference to the forward/aft cabin temperature selector and good communications with the No 1.

If no ground source air-conditioning is available and ambient temperatures are above 25 degrees centigrade, good communications is required to manage the cabin temperature. Pre-boarding Commanders shall:

- Brief No1 that all cabin doors are to remain open until they are advised to close, this should be advised once start clearance has been received.
- Brief the GHA and confirm ASU serviceable connected and ready to start.
- Brief No1 any delay to start clearance such as tech ASU, missed slot etc will require cancel start clearance and reopen all cabin doors.

Commanders shall ensure that the No1 engine is started at the earliest opportunity to provide air-conditioning once boarding is complete and doors are closed. If extended delays are expected, crews should push, start both engines and Taxi out/hold remotely. Consideration to extra fuel burn during fuel uplift planning is advised. If remote holding is not available boarding should be delayed coordinating completion of boarding in accordance with CTOT.

Failure of APU During Turnaround

In the event of an APU becoming inoperative after boarding is completed, and where air-conditioning cannot be provided, if the cabin temperatures rises above 25 degrees centigrade the commander shall make all the necessary GOPS arrangements for all passengers to be disembarked.

The Commander should review options with the OCC and Engineering before making the decision when to re-board the passengers.

Failure of APU During Push Back and Engine Start

Following an APU failure during start and a return to stand, ensure that ground source air-conditioning is connected as a priority, if ground source air-conditioning ASU is not immediately available and the ambient temperatures are approaching 25 degrees centigrade then the Commander should ensure all cabin doors are opened and make the necessary GOPS arrangements for disembarkation of passengers into the terminal until an air start unit is available.

On Stand Delays with APU Available

When on stand with passengers boarded and doors closed cabin crew should be briefed to closely monitor cabin temperatures especially in the middle of the cabin and to report any evidence of cabin overheating to the captain at the first indication of passenger unrest.

8.2.1 Special Refuelling or Defuelling of the Aircraft

AMC CAT.OP.MPA.200

Malta Air has performed a risk assessment, developed procedures and has established a training programme for its personnel involved in Special Refuelling or Defuelling of the aircraft.

Special refuelling or defuelling applies to refuelling or defuelling with passengers embarking, on board, or disembarking.

Any special refuelling or defuelling procedures and any change to them shall require prior approval by the TMCAD.

8.2.1.1 Refuelling, Ground Equipment, Electrical Switching, Defueling Procedures and Precautions

The refuelling procedures in this manual are shared with Engineering and GOPS in their relevant manuals to the extent that Engineering and Ground OPS personnel are aware of refuelling procedures and in particular the designation of duties of the Refuelling Supervisor.

Malta Air aircraft may be fuelled with an APU running, and/or GPU/ASU or other ground equipment attached. Electrical switching is permitted but HF, when fitted, shall not be used. Refuelling or defueling is not authorised with engine(s) running.

Malta Air aircraft are not normally defueled. Consult Maintrol if defueling is a possibility.

Malta Air aircraft shall not be refueled with wide cut fuel.

Malta Air contracts with refuelling service provider should ensure that wide-cut fuel cannot be supplied to Malta Air aircraft.

If the commander suspects that wide cut fuel has been supplied he shall contact maintrol immediately. Flights can continue but Maintrol will supply special procedures to be applied and the NPFO shall authorise the flight or series of flights until it has been established that no more wide-cut fuel in in the fuel system.

8.2.1.2 Refuelling or Defueling with Passengers Embarking/On Board/ Disembarking

CAT.OP.MPA(b)(2)

Definitions to be Used in this Section

Qualified Person – A person on board the aircraft for refuelling operations who is capable of handling emergency procedures concerning fire protection and firefighting, handling communications, and initiating and directing an evacuation.

Refuelling Supervisor: A member of ground crew and/or air crew supervising the refuelling operation. A Refuelling Supervisor will alert the Qualified Person without delay if any hazardous situation occurs.

Refuelling supervision can be completed by:

1. Multiple persons tasked with specific functional roles (distributed supervision) during special refuelling operations (RFS-A).
2. A single person assigned by the commander (RFS-B).

Note: The Commander is accountable for ensuring that the refuelling procedures are followed during the refuelling operation.

Refuelling or Defuelling with Passengers Embarking, on Board or Disembarking

AMC5 CAT.OP.MPA.200

When refuelling or defueling with passengers embarking, on board, or disembarking, the necessary precautions should be taken, and the aircraft should be properly manned by qualified personnel that should be ready to initiate and direct an evacuation of the aircraft by the most practical and expeditious means available.

Operational Procedures with Passengers Embarking, on Board or Disembarking

ALT/24/0012

1. When refuelling/defuelling with passengers on board, ground servicing activities and work inside the aeroplane, such as catering and cleaning, should be conducted in such a manner that they do not create a hazard and allow emergency evacuation through those aisles and exits that are intended for emergency evacuation.
2. The deployment of integral aeroplane stairs or the opening of emergency exits are not necessarily a prerequisite to refuelling.

3. Operational procedures specify that at least the following precautions are taken:
 - a. one qualified person should remain at a specified location during refuelling/defuelling operations with passengers on board, and be capable of using emergency procedures for fire protection and firefighting, communications, as well as for initiating and directing an evacuation;
 - b. two-way communication should be established and remain available through suitable means, between the ground crew that supervises the refuelling and the qualified personnel on board the aeroplane;
 - c. crew, personnel, and passengers should be warned that refuelling/defuelling will take place;
 - d. the 'FASTEN SEAT BELT' signs should be off;
 - e. 'NO SMOKING' signs should be on, together with interior lighting to allow the identification of emergency exits;
 - f. passengers should be instructed to unfasten their seat belts and refrain from smoking;
 - g. the minimum required number of cabin crew should be on board and prepared for an immediate emergency evacuation;
 - h. if fuel vapour is detected inside the aeroplane, or any other hazard arises, refuelling/defuelling should be stopped immediately;
 - i. the ground area beneath the exits that are intended for emergency evacuation, as well as slide deployment areas, should be kept clear where stairs are not in position for use in the event of evacuation; and
 - j. provision is made for a safe and rapid evacuation.

8.2.1.2.1 Refuelling is permitted with passengers embarking, on board or disembarking.

Refuelling of aircraft is a routine procedure and precautions are always taken to preclude the possibility of fire, fuel spillage, etc. however, when passengers are involved, additional precautions must be taken to ensure that they can be disembarked/evacuated in the unlikely event of a hazard arising that affects safety.

Refuelling with passengers embarking, on board or disembarking is subject to the approval of the local Airport Authority and may not be available at certain airports.

8.2.1.2.2 Roles and Responsibilities**8.2.1.2.2.1 Flight Crew Responsibilities**

The Commander designates a Qualified Person on board the aircraft for refuelling operations with passengers on board, embarking or disembarking who:

- Is capable of handling emergency procedures concerning fire protection and firefighting, handling communications, and initiating and directing an evacuation.
- Is a Ryanair Group pilot qualified on type.
- Shall remain on the flight deck.
- May also be the Refuelling Supervisor.
- The Commander shall determine the refuelling supervision based on whether RFS-A, or RFS-B applies.

The Commander is responsible for ensuring that the refuelling procedures are followed during the refuelling operation and shall ensure the following:

- The applicable refuelling procedure is confirmed from the RFS approved airport list prior to commencement of refuelling.
- The applicable refuelling procedure is announced to the cabin crew.
- The aircraft is suitably immobilised by means of brakes and/or chocks.
- Passengers and Cabin Crew are informed so that Cabin Crew are appropriately positioned for emergency egress procedures.
- Interphone is serviceable between flight deck and cabin, or alternatively, and by prior arrangement, the qualified person on the flight deck briefs the SCCM to be in the forward galley to receive any instructions from the flight deck.
- The SCCM is advised of any problems outside the aircraft which might affect the Cabin Crews' decision on door usage in the event of an evacuation being required.
- Where a passenger boarding bridge is not in use, steps shall be in place at the doors (L1 or L2).
- The "No Smoking" sign shall be switched on and the "Seat Belt" sign shall be switched off.
- If advised that fuel vapour is detected within the cabin, fuelling must be ceased until it is safe to resume.
- Remain vigilant for fire or other warning of an unsafe condition via headset or visual communication and be prepared to initiate passenger evacuation if necessary.

- If a hazard arises, communicate with the refuelling operator to ensure that fuelling shall be stopped immediately until conditions permit resumption.
- Before and during refuelling, where applicable and in consultation with the GHA, ensure that Fire Service facilities are available.
- Every case of fuel spillage, irrespective of the amount, shall be reported immediately to the Airport Authorities. In this case, the flight crew member in the cockpit shall notify ATC that a fuel spillage has occurred and shall include details of the aircraft parking position.
- The commander shall ensure that, prior to the commencement of special refuelling, a suitable means to communicate with the qualified personnel on board the aircraft is available.

Note: When there are no passengers embarking, on board or disembarking the qualified person does not have to remain in the flight deck.

8.2.1.2.2.2 Refuelling Supervisor Responsibilities (RFS A)

The primary responsibility of a Refuelling Supervisor is to promptly alert the qualified person on board to any emerging hazard.

During refuelling with passengers embarking, on board, or disembarking, 'the Qualified Person' as designated by the commander, shall remain on the flight deck in a suitable position to maintain visual contact with the fuelling environment.

In order to ensure appropriate refuelling supervision during special refuelling operations, the Commander must:

- Announce 'RFS A' to the Cabin Crew.
- Advise the GHA that special refuelling will take place. The GHA will inform the ground staff team to be alert to any fuelling hazard that may require them to apply their respective procedures.
- Advise the Refuelling Operator as to when refuelling can commence.
- With both flight crew in the flight deck, each flight crew member supervises his respective area of the refuelling zone.
- Anytime there is only one flight crew member in the flight deck, he shall occupy the right-hand seat.
- During refuelling supervision crews can complete other cockpit duties, however, these duties must not detract from the overall supervision of refuelling operations.

During the refuelling process, the Fuelling Operator is viewable from the right-hand seat in the flight deck. In the case of any hazard or incident that may affect the safe conduct of refuelling operations, the following procedure shall apply to ensure that the fuelling stops immediately:

The Flight Crew shall:

- Visually attract the attention of the Refuelling Operator and/or use the 'GRD Call' pushbutton; or
- Call on the interphone system if the Refuelling Operator is using a headset; or
- Open the flight deck side window and use direct verbal or non-verbal communication including the use of hand signals; or,
- Dispatch the other flight crew member, or any other ground personnel to the Refuelling Operator.

The Refuelling Operator shall:

- Ensure that the fuelling stops immediately (prepare to move the bowser away from the aircraft if necessary).

Attract the attention of the Flight Crew by any means available, including:

- Banging on fuselage at flight deck position,
- Call on the interphone system if using a headset,
- Visual signals,
- Using 'PILOT CALL' pushbutton,
- Verbal communication via open side window,
- Entering aircraft to notify crew directly,
- Alert and dispatch any ground personnel in the vicinity to help.

In the event that two-way visual or audio communication becomes unavailable, or anytime supervision of the refuelling environment is impaired (e.g., poor apron lighting, inclement weather such as heavy rain/snow, or a ground handling vehicle blocking line-of-sight, etc.) refuelling shall be suspended until:

- Two-way visual or audio communication is re-established, or
- RFS Procedure B procedures below are commenced.

Note: The Qualified Person may also be a Refuelling Supervisor.

8.2.1.2.2.2.1 Refuelling Supervisor Responsibilities (RFS B)

Note: RFS B is applied where local by-laws do not allow the use of RFS-A.

The primary responsibility of the Refuelling Supervisor is to promptly alert the qualified person on board to any emerging hazard.

- Observe the safety of the Refuelling Operator,
- Verify that the refuelling operation remains safe,
- Maintain a suitable means to communicate with the qualified person,
- Ensure correct and prompt action is taken in response to any emerging emergency situation.

The Refuelling Supervisor shall:

- Position himself externally with access to the interphone connection point adjacent to the nose wheel, whereby communications are established between the Qualified Person and the Refuelling Supervisor using the aircraft intercommunications system and maintain the refuelling operator in sight during refuelling.
- Be trained in the Malta Air Aircraft refuelling procedures.
- Be trained in the operation of the aircraft communication systems.

The Refuelling Supervisor can complete other tasks (e.g., completing the walkaround) providing he can see the Refuelling Operator and remains within easy reach of a system of communication.

Note: Local by-laws may require the Refuelling Supervisor to remain on the headset at all times. (*Refer to the 'Refuelling Supervision Airport List' on EFB*).

8.2.1.2.2.3 Refuelling Operator Responsibilities

Refuelling Operators using own headsets have received own company training accepted by Malta Air to operate the refuelling panel and to establish communications using the aircraft communications system with the Qualified Person on the flight deck.

The Refuelling Operator shall:

- Ensure that the aircraft is bonded to the refuelling tanker/fuel pit.
- Establish visual communication with the Flight Crew.
- Ensure that refuelling does not commence without prior authorization from the flight crew or other person designated by the commander.

- Ensure that when hazards, unsafe acts or incidents related to refuelling operations are identified, stop the refuelling immediately.
- Be responsible for the safe positioning of the fuel truck, awareness/maintenance of the vehicle escape path and awareness to keep clear emergency evacuation paths from the aircraft.
- Notify the Qualified Person as soon as possible when hazards, unsafe acts or incidents in his area are identified.

8.2.1.2.2.4 Cabin Crew Special Refuelling Supervisory Responsibilities

As part of their responsibilities Cabin Crew are tasked to promptly alert the qualified person on board to any emerging hazard that they may identify, inside or outside of the aircraft. The SCCM must inform the Commander if there are any PRMs on board whose presence could affect refuelling procedures and ensure that SCPs are not seated in direct access seating and their embarkation or disembarkation does not block exits intended for emergency exit.

The SCCM shall inform all cabin crew that fuelling is about to take place.

- Cabin Crew shall be at their designated location.
- Cabin Crew shall make a PA announcement, advising passengers that fuelling is taking place, that their seatbelts must be unfastened.
- Cabin Crew shall immediately inform the flight crew of:
 - Any problems within the cabin during boarding and fuelling.
 - Passengers smoking.
 - Abnormal build-up of fumes.
 - Aisles/exits blocked or congested.
 - Any abnormality that increases risk.
- Additionally, during special refuelling (RFS A)
 - Cabin Crew assigned to the L1 and L2 doors must remain particularly vigilant for any hazard outside (including, but not limited to, passengers smoking, passengers walking under the wing, fuel leaking from the wing vent, etc.).

8.2.1.2.2.5 Ground Crew Special Refuelling Supervisory Responsibilities

As part of their responsibilities Ground Crew are tasked to promptly alert the qualified person on board to any emerging hazard that they may identify, inside or outside the aircraft.

- Ensuring that a clear path is maintained from the aircraft to allow for the quick removal of fuelling vehicles, if required. (Ramp/Fuellers).

- Ensuring that only one aircraft door may be accessed by a service vehicle (e.g., catering/cleaning truck). Passenger steps/Airbridges are deemed to be emergency exit paths and are not considered to be service vehicles. An Ambulift in position to embark/disembark PRMs is considered an emergency exit path.
- Ensuring that the ground area beneath aircraft exit doors not in use is kept clear of obstruction and that a safe and rapid evacuation is possible.
- In the unlikely event that the Fuelling Operator has become incapacitated the Ground Crew shall notify the Flight Crew and assist as required.
- When marshalling passengers, remain vigilant that passengers do not stray into the refuelling safety zone.
- When marshalling passengers who are embarking or disembarking during special refuelling operations, remain in the assigned position and remain vigilant for any fuel leaking from any wing vent.

8.2.1.2.2.6 Suitable Means of Communication During Special Refuelling Operations

The following methods of communications are permitted for use when refuelling with Passengers embarking, on board or disembarking:

- Use of the aircraft intercommunications system
- Use of the GRD CALL push button
- Use of the PILOT CALL push button
- Direct verbal/non-verbal communication
- Use of Hand Signals and or banging on the aircraft fuselage
- Entering/exiting the aircraft to communicate directly
- Dispatching a member of the crew to alert the Refuelling Operator

Any of these methods can be used by the refuelling supervisors, qualified personnel, and the refuelling operator provided that the method used is the most effective and expedient manner to relay information related to a developing hazard during special refuelling operations.

8.2.1.2.2.7 Fuel Operator Service Levels

The Refuelling Operator carries out the aircraft refuelling including making necessary nozzle connections to the fuel hydrant pit and the aircraft, operating the fuel vehicle and operating the deadman control. The Refuelling Operator opens, closes and operates the switching in the refuelling panel. When the Refuelling Operator does not operate the refuelling panel this is done by an engineer or one of the flight crew.

Note: The fuel bowser may be connected to the aircraft immediately on arrival on stand after engines are shut down and the anti-collision light is switched OFF but refuelling shall only be commenced with passengers embarking, on board, or disembarking, on the direction of the Commander or his designate.

8.2.1.2.2.8 Reporting – Special Refuelling Procedures

As a minimum, an Air Safety Report must contain the following details in respect of a fuel leak or spillage:

- Refuelling Procedure in force (A or B)
- The location of the leak/spillage
- The means of identification of the emerging/actual hazard
- Method of communication to the Qualified Person and the Refuelling Operator
- Any specific precautions taken (including precautionary disembarkation)
- Any breakdown in communication between the Qualified Person/Refuelling Supervisors/Refuelling Operator/Airport Services

8.2.1.2.2.9 On Completion of Refuelling

On completion of refuelling, Qualified Person shall ensure that:

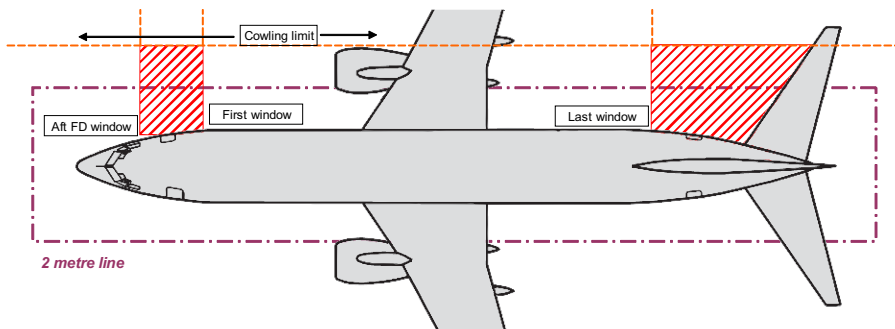
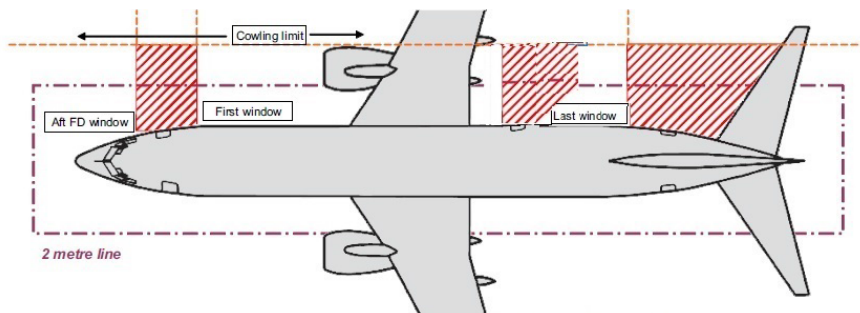
- The Cabin Crew are informed
- The “Seat Belt” sign is switched ON

On completion of refuelling, the Refuelling Operator will provide the flight crew with uplift details.

8.2.1.2.3 Additional Guidance During Refuelling Operations

When a pilot is required to complete switching on the refuelling panel during refuelling, he shall explain to the refueller that the refueller must not refuel the aircraft (pressurise the hose) when the pilot is operating the switches at the refuelling panel.

The diagram below shows the hatched areas that will be kept clear during refuelling. Also depicted is the “2 metre line”. This has been established by GOPS with all handling agencies that this is the closest that equipment not in use for embarking/disembarking passengers or loading/unloading bags may be left, e.g. GPU.

B737-800:**B737-8200:****Line Operations Safety Procedures During Refuelling**

1. Consider not commencing refuelling if brakes or engines show signs of being over-heated;
2. Ensure no HF transmissions;
3. Ensure no mobile phone usage within 3.5 m of a refuelling point (i.e. under the right wing);
4. Malta Air flights shall not be fuelled in a hangar.

8.2.1.3 Uplift Check

The Commander shall **always** perform an Uplift Check using the fuel record figures in the Technical Log after every refuelling to provide assurance of the fuel quantity system accuracy.

The Uplift Check consists of comparing the sum of the arrival fuel plus the uplift fuel with the total of the gauge readings after refuelling.

The tolerance is ± 200 kg for the Boeing 737 aircraft. If refuellers supply in excess of the fuel requested the Commander shall annotate the circumstances on the Electronic flight log/Paper Voyage Report/Journey Log and ideally on the fuel docket. In this case attach the fuel docket to the EFL submission (or return in the flight envelope).

If the Commander has any reason to doubt the integrity of the fuel quantity on board or the FQIS, then the drip sticks must be used to confirm the quantity of fuel in the tanks.

The acceptable tolerance between any tank gauge reading and the drip stick figure is ± 200 kg for the Boeing 737. Such a large figure is only acceptable due to the difficulty of reading the drip sticks and making the corrections for pitch and roll attitudes. Differences approaching this magnitude will be subject to further uplift checks on subsequent refuelling.

In such circumstances, Commanders shall refer to MEL and consider taking additional fuel in excess of flight plan fuel for the sector.

8.2.2 Passenger and Cargo Handling Procedures Related to Safety

CAT.OP.MPA.165

Passenger Seating

Malta Air has established procedures to ensure that passengers are seated where, in the event that an emergency evacuation is required, they are able to assist and not hinder evacuation of the aircraft.

Compliance with Operations Manual Part B – Mass and Balance assures that no unsafe loading condition arises during embarking and disembarking, loading or unloading.

Boeing anti tipping procedures are stated in the GOPS manual and SEP Manual, and are particularly relevant when disembarking via an airbridge/jetway.

It is essential that all crew are observant of passenger movement and distribution in the cabin during embarkation and disembarkation to prevent an out-of-balance condition which could lead to an aircraft tipping event. The crewmember/GHA performing the exterior inspection and/or refuelling supervision shall immediately contact the flight deck and ground crew should they observe excessive nose-gear strut extension, tipping, or abnormal off/loading or loading procedures. If required, the PIC/CSS should halt embarkation or disembarkation until the condition causing the tipping risk has been resolved.

Standard/Adjusted Airbridge Boarding

Adjusted Airbridge Boarding is NOT PERMITTED when Engineering Spares (AOG)/Cargo/Hold baggage are scheduled to be loaded or unloaded in aft holds.

Malta Air utilizes two styles of airbridge boarding – standard and adjusted airbridge boarding:

- Standard boarding (2 queues – Priority/Other Q).
- Adjusted Airbridge Boarding (3 queues – Priority/Seat Row 22-33 (B737-800) or 23-35 (B737-8200)/All other customers)).

“Adjusted Airbridge Boarding” applies after the dispatcher confirms:

- Holds 3 and 4 empty of Engineering Spares/Cargo.
- Engineering Spares/Cargo will not be loaded into Holds 3 or 4 during the boarding process.
- Bags will only be loaded in Hold 3 AFTER loading of Hold 2 is complete and only towards end of boarding.
- Prior to embarkation, dispatcher informs the Captain “Adjusted Airbridge Boarding Applies.”

The PIC must always confirm with the CSS prior to airbridge embarkation which boarding style applies: “standard airbridge boarding” or “adjusted airbridge boarding.” The CSS will then advise the cabin crew prior to embarkation.

When “Standard Airbridge Boarding” applies, anticipate the following sequence:

- Priority passengers board first.
- All other passengers.

When “Adjusted Airbridge Boarding” applies, anticipate the following sequence:

- Priority passengers board first.
- Passengers seated in Rows 22-33 (B737-800) or 23-35 (B737-8200).
- All other passengers.

Malta Air does not carry third party cargo. Cargo on Malta Air aircraft is Engineering Stores or other company mail. It is loaded in accordance with Operations Manual Part B – Mass and Balance.

8.2.2.1 Carriage of Special Categories of Passengers (SCPs)

CAT.OP.MPA.155

1. Persons requiring special conditions, assistance and/or devices when carried on a flight shall be considered as SCPs including at least:
 - a. persons with reduced mobility (PRMs) who, without prejudice to Regulation (EC) No 1107/2006, are understood to be any person whose mobility is reduced due to any physical disability, sensory or locomotory, permanent or temporary, intellectual disability or impairment, any other cause of disability, or age;
 - b. infants and unaccompanied children; and
 - c. deportees, inadmissible passengers or prisoners in custody.
2. The number of SCPs that may be carried on a given flight is specified in the SEP Manual
3. SCPs shall not be allocated, nor occupy, seats that permit direct access to emergency exits or where their presence could:
 - a. impede crew members in their duties;
 - b. obstruct access to emergency equipment; or
 - c. impede the emergency evacuation of the aircraft.
4. The commander shall be notified in advance when SCPs are to be carried on board.

A passenger who books WCHR assistance for convenience rather than necessity is not considered an SCP.

For the purpose of passenger classification: (Annex I definitions).

1. ‘adult’ means a person of an age of 12 years and above;
2. ‘child/children’ means persons who are of an age of two years and above but who are less than 12 years of age;
3. ‘infant’ means a person under the age of two years.

8.2.2.1.1 Travelling with an Accompanying Person (ABP)

Where a passenger is not self-reliant, Malta Air requires that they must travel with an able-bodied accompanying person aged 16 and over. Malta Air has adopted the guidelines as set out by the UK Department for Transport – Access to Air Travel for Disabled People: Code of Practice (July 2008). Walk on/walk off passengers travelling with an ABP are not included in the PRM figures.

8.2.2.1.2 Guidelines

If a passenger is not self-reliant in accordance with OMA SEP 2.17.1, then he or she must travel with an able-bodied accompanying person aged 16 and over who will be capable of providing the assistance required.

8.2.2.1.3 Seat Allocation for Disabled/Reduced Mobility Passengers

CAT.OP.MPA.165, SEP 5.4.5

The Malta Air computerised allocated seating booking programme should, and Malta Air cabin crew will ensure that access to emergency equipment and evacuation of the aircraft in an emergency will not be impeded before allocating seats.

The following categories may not be allocated direct access seating:

1. Passengers with reduced mobility, which may include but not be limited to someone with a mobility or sensory impairment.
2. Passengers whose physical size would prevent them from being able to move quickly.

The final decision regarding seat allocation in the cabin is made by the SCCM.

8.2.2.1.4 Blind or Significant Vision Impaired Passengers (BSVIP)

Blind or Significant Vision Impaired Passengers who meet all of the self-reliant requirements (OMA SEP 2.17.1) may travel without an accompanying person.

A Blind or Significant Vision Impaired Passenger who requires assistance to board or disembark, or to move around the cabin (e.g., assisted to their seat), regardless if travelling alone or accompanied by an ABP, is subject to Safety Limit of 4. Otherwise, Safety Limit of 50% of ABP's applies.

Travel with an assistance dog is permitted on certain routes (refer to OMA 8.2.2.1.6). Blind or Significant Vision Impaired Passenger travelling with a guide or assistance dog are always subject to a Safety Limit of 4.

8.2.2.1.5 Deaf or Significant Hearing Impaired Passengers (DSHIP)

Deaf or Significant Hearing Impaired passengers are not subject to SCP Safety Limit of 4, unless travelling with a guide dog. They are, however, subject to seating restrictions.

Deaf passengers who are able to fully understand an onboard safety demonstration and leaflet are not required to notify Ryanair of their condition.

Where a deaf passenger requires any individual assistance, they are required to indicate this in their booking or contact the Ryanair Special Assistance team.

Hard of Hearing Passengers

If a passenger is hard of hearing but uses a corrective device which enables the passenger to hear clearly, they are not considered an SCP.

8.2.2.1.6 Travel of Assistance/Guide Dogs

Malta Air may carry a maximum of four Assistance or Guide dogs in total on any one flight. The passenger responsible for the assistance/guide dog must be in possession of documentation which confirms that the dog has been trained to perform a specific function.

Malta Air Check-in staff or Ground Handling Agents confirm the necessary documentation has been checked in order for the animal to travel. Dogs must have a pet passport or, in countries where pet passports are not issued, they must have an official veterinary health certificate.

Malta Air Cabin Crew are only required to confirm that the assistance dog is listed on the boarding card.

8.2.2.1.7 Battery Operated Wheelchairs

Wet cell battery operated wheelchairs cannot be accepted for travel. Gel/Dry cell battery operated wheelchairs are permitted to be carried in the aircraft hold provided the battery is disconnected and the battery terminals are insulated by the user before the unit is handed over to loading personnel. Passenger Wheelchairs are not permitted to travel in the cabin.

8.2.2.1.8 Assistance In-flight

Water is available for purchase from the bar if required for the administration of oral medication. Cabin Crew may alternatively provide passengers with potable water once it has been boiled and allowed to cool. Cabin Crew are not allowed to provide medical services such as administering injections. For safety reasons Cabin Crew cannot lift passengers or assist with personal hygiene. Passengers requiring these types of assistance must travel with an accompanying passenger.

8.2.2.1.9 Essential Medical Equipment/Supplies

Passengers must bring the prescription with them to allow security clearance for the carriage of medication in hand luggage. Passengers who need to carry essential medical equipment/supplies in addition to their hand or checked baggage allowance are, subject to cabin safety restrictions, approved to carry these equipment/supplies.

8.2.2.1.10 Denying Travel

The following are considered grounds for denying a passenger travel:

1. If the attending physician recommends that the passenger not travel (or expresses doubt in this regard).

2. If the passenger suffers from any contagious/communicable disease.
3. If the passenger's condition is such that the health/safety/welfare of Malta Air staff or other persons may be at risk.
4. If the passenger's condition is such that it may cause distress or discomfort to other passengers.
5. If the special arrangements or level of assistance required would put an undue burden on Malta Air or its agents thereby degrading the level of service to other passengers.
6. If a stretcher is required on board.

8.2.2.1.11 Pregnancy

Malta Air reserves the right to refuse travel if a passenger over 28 weeks pregnant does not present a 'fit to fly' letter from their midwife/doctor.

8.2.2.1.12 Transportation of Inadmissible Passengers, Deportees or Persons in Custody

8.2.2.1.12.1 Definitions

Deportee

A person who is legally admitted to a State by its authorities or who enters a State illegally, and who at some time later, is formally ordered by the authorities to leave that State.

Inadmissible Person

Persons whose entry to a State is refused by the competent authorities and who are being transported back to their country of departure, or to any other country where the person is admissible.

Person in Lawful Custody (Escorted Prisoner)

Persons either under arrest or convicted by a court of law who have to be transported.

8.2.2.1.12.2 Policy

The decision to accept a deportee/inadmissible person/person in lawful custody on a flight ultimately rests with the Captain. If he/she considers the carriage of such persons might jeopardise the safety/security of the passengers, crew or aircraft, carriage will be refused.

Where an escort is required, a maximum of one deportee/inadmissible person/person in lawful custody is permitted for carriage on a flight.

Persons in lawful custody shall always be escorted.

8.2.2.1.12.3 Notification Requirements

The competent authority shall notify Malta Air in writing in advance (where possible 24 hours in advance) when they plan to embark a deportee, inadmissible person or person in lawful custody.

The written notification shall normally be in English and shall contain the following details:

1. Identity and gender of the person;
2. Reason for transportation;
3. Name and title of escort(s) (if provided);
4. Risk assessment by the competent authority including reasons to escort or not and details of any known medical condition(s);

Note: If the risk assessment indicates a ‘high’ level of risk an alternative means of transport should be used.

1. Prior seating arrangements (if required);
2. Nature of available travel documents.

Malta Air will make this information available to the Captain of the flight prior to passengers boarding the aircraft.

The procedures for refusal of carriage of potentially disruptive passengers are contained in the ground operations manual.

8.2.2.1.12.4 Escorts

Persons in lawful custody must always be escorted on a flight.

For deportees and inadmissible persons, the competent authority will, on the basis of their risk assessment, decide whether or not an escort is required.

If an escort is deemed necessary, a minimum of two escorts should be provided with additional escorts as necessary.

If an escort is deemed not necessary, the deportees or inadmissible person will be escorted to the aircraft.

The escort will not leave the aircraft until the doors are ready to be closed.

8.2.2.1.12.5 Guidance to Commanders

1. Limitations on the carriage of Inadmissible Persons or Deportees who do NOT require an escort;

Inadmissible Persons are considered Special Category of Passengers and are therefore limited to 50% of the available seats on board the aircraft, however, the number would normally be less than 2. **If presented with more than 2 Inadmissible Persons or Deportees, contact the duty pilot.**

2. Documentation to be carried on board;

The Commander must be notified of INAD passenger carriage approvals. In all cases, the Ryanair Group Immigration Support Team will notify the Commander prior dispatch via appropriate channels.

This will take the form of an email from immigrationsupport@ryanair.com which will contain details of the passenger and a risk assessment carried out by the Ryanair Group Immigration team.

All INAD pax must be included in the flight manifest.

In some states the Commander may be giving an envelope containing the passengers travel documents. This is not a requirement, however if you are presented with documentation, the documentation shall be retained on the flight deck.

If a passenger has been allowed to board in possession of their travel documentation crew shall not attempt to take it from them.

3. Boarding and during the flight
 - The Inadmissible Persons or Deportees and escort(s) are pre-boarded and will disembark last.
 - The Inadmissible Persons or Deportees and escort(s) if applicable are seated at the rear of the Aircraft.
 - Where possible handcuffs and other restraints are removed while onboard the aircraft.
 - The Inadmissible Persons or Deportees are not permitted to occupy emergency exit seats or aisle seats.
 - The Inadmissible Persons or Deportees shall have no access to alcohol.
 - The Inadmissible Persons or Deportees hold baggage is loaded last to avoid delays in the event that they do not travel.

4. Disembarking at destination

On arrival at the destination the passenger may or may not be met by immigration.

If no immigration officer is present at the stand, then request that the handling agent inform the immigration team of the passenger and ask if they intend to attend.

Once all other passengers have disembarked, and in the absence of instruction to the contrary, allow the passenger to disembark and give any travel documentation to the handling agent. Do NOT give the documentation to the passenger.

8.2.2.2 Permissible Size and Weight of Hand Baggage (Cabin Baggage)

(CAT.OP.MPA.160)

The cabin baggage specifications on the Malta Air website and Malta Air GOPS procedures in force assure that only hand baggage that can be adequately and securely stowed is taken into the passenger compartment. Passengers and crew are protected against items in overhead bins moving by the design of the bins and passenger announcements advising caution that such items may have moved. Cabin baggage placed underneath seats cannot move to obstruct aisles and exits under normal flight conditions.

8.2.2.3 Securing of Passenger Compartment and Galley(s)

(CAT.OP.MPA.230)

Before taxiing, take-off and landing all exits and escape paths will be unobstructed by complying with the procedures presented in the OMA-SEP.

8.2.2.4 Positioning of Ground Equipment

All ground equipment for the servicing of the aircraft whilst on a turnaround will be positioned on the starboard side of the aircraft. This is to ensure that passengers do not come in contact with any equipment while embarking or disembarking the aircraft. The ground equipment manual describes the position of ground equipment in the vicinity of the aircraft, including the 2 m rule, OMA 8.2.1.2.

8.2.2.5 Operation of Aircraft Doors

(CAT.OP.MPA.220)

Assisting means for emergency evacuation Malta Air has established procedures in OMA-SEP which ensure that before taxiing, take-off and landing and when safe and practicable to do so, all means of assistance for emergency evacuation that deploy automatically are armed.

Instructions for the operation of aircraft doors is contained in the FCOM and OMA-SEP.

Only personnel who have been given specific, hands-on training in the opening and closing of the L1 door while standing on the integral airstairs are permitted to do so. Operating L1 door from an airbridge positioned at the same level as the forward entry may be carried out by other authorised staff if unattended by flight or Cabin Crew.

8.2.2.5.1 Aircraft Operations in High Winds on the Ground

The chances of personal injury, aircraft or ground equipment damage increase as surface wind increases. Up to 20 kt the risk is relatively low. As the wind increases above 20 kt the risk increases exponentially.

All destination airport authorities and handling agents operate to the limits specified in this section unless they have demonstrated to Malta Air that their mobile steps are certified for use in higher windspeeds. The Malta Air acceptance of this is controlled by GOPS. If the handling agent in a destination presents steps in wind conditions in excess of 20 kt the operating crew can assume that the use of this equipment has been accepted by GOPS.

Wind Between 20 kt and 40 kt

Confirm all ground/ramp equipment has been secured (i.e. parking brakes set or chocked). Do not use mobile steps unless the handling agency has equipment accepted by GOPS for use in excess of 20 kt windspeed – see above.

The maximum windspeed for continuance of operations on the ground, including gusts, is an absolute value of 60 kts.

Windspeed is calculated as the sum of the steady wind plus $\frac{1}{2}$ the gust, e.g. wind 240/30 gusting 50 gives a total wind speed of 30 kt +10 kts ($\frac{1}{2}$ the gust of 20 kt); total wind = 40 kt.

Operations Between 41 kt and 60 kt

If winds are forecast to exceed 40 kts and aircraft are expected to be left unattended flight crew shall contact Maintrol for guidance on the protection of aircraft on the ground.

All unnecessary ramp equipment will be removed to a sheltered location.

The Stabilizer shall be set to Zero units of stab trim 'Airplane full Nose Down'.

The Parking Brake will be SET and wheels double chocked.

The Aircraft will be fueled to at least 7000 kg.

Aircraft must be parked tactically* to ensure door and air stairs are not operated when directly exposed to the wind (*ref: Maintenance Manual*).

Operation of selected doors with winds greater than 40 kt may be approved by the Commander where positioning of the aircraft relative to prevailing wind or proximity to local buildings or the terminal affords obvious protection.

The DO NOT DE-ICE placard shall be placed in the captains DV window. This procedure shall be applied even if there is no expectation that de-icing will be required.



Before de-icing or pre-flight return the aircraft to normal configuration for take-off. The DO NOT DE-ICE placard shall be stowed in the glare shield holder.

Note: If the aircraft has been de-iced with the stabilizer trim in the zero trim position Maintrol shall be contacted for guidance before dispatch.

Wind at or above 60 kt

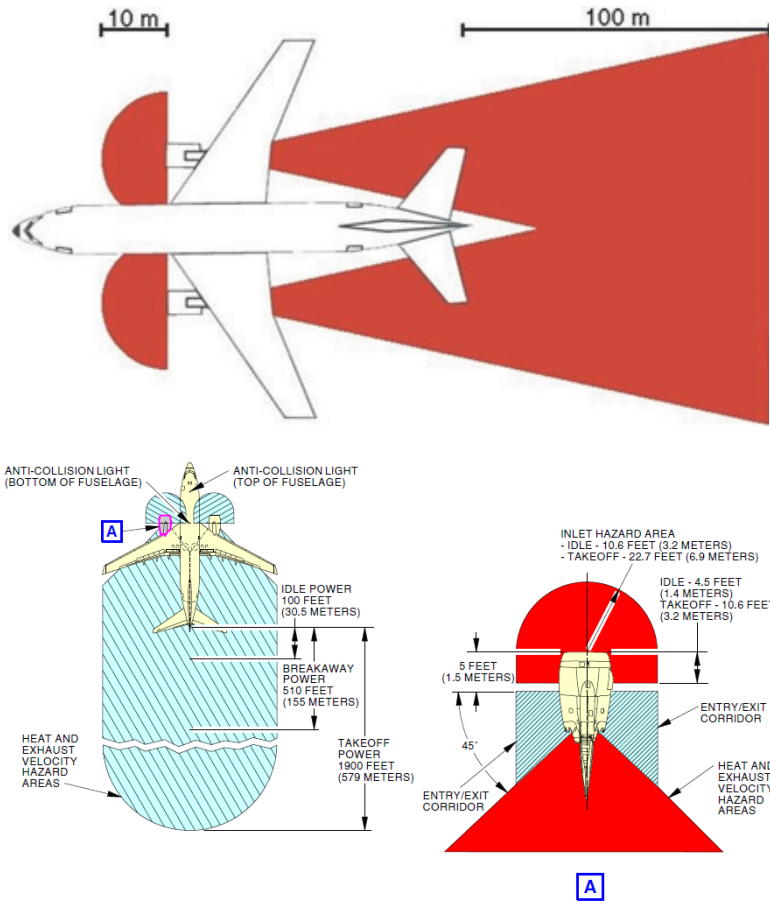
There is an absolute value of 60 kts (including gusts) at which all operations on the ground shall be suspended, and shall remain suspended until winds have abated to below 60 kts for a period of more than 30 minutes, e.g. wind 240/30 gusting 60 – all operations on the ground shall be suspended.

8.2.2.6 Safety on the Ramp – Fire Prevention, Blast and Suction Areas

Fire prevention on the ramp is managed by the Airport Authorities and procedures, instructions and guidance are promulgated to airport users via Directors Notices or equivalent instrument.

Blast and suction areas for engine start, various power applications and prior to engine shutdown are diagrammatically represented, including recommended safe distances in the Ground Operations Manual and by decals on both engines.

WARNING: The greatest danger exists when engines are running. Pay attention to anti-collision lights (flashing red light on the aircraft).



8.2.2.7 Start-up, Ramp Departure and Arrival Procedures

These procedures are contained in the FCOM and the GOPS Manual. Ground Handling Agents can access procedures, instructions and guidance from the GOPS Manual.

Use of Anti-collision Light

Switching on the anti-collision light is a universal signal in the industry that appropriate precautions to remain clear should be taken by personnel in the vicinity of an aircraft. Illumination of the anti-collision light is an indication that a hazard exists, engines are running (or about to start), the aircraft is being towed, or that the aircraft is moving under its own thrust.

Equally, switching off the anti-collision light is an indication that the engines no longer present a hazard on the aircraft.

Note: Use of the anti-collision light other than for designed, approved purposes is not permitted.

CAUTION: *If the ground power trips out and there is no other source of AC power selected, the anti-collision light does not function, though it is switched ON. The number one engine may still be running and GOPS personnel could assume it is safe to approach the aircraft as they would believe the anti-collision light was switched off.*

Push back and Towing – Aeroplanes

AMC1 CAT.OP.MPA.205

Barless Towing

Barless towing is based on the applicable SAE ARP (Aerospace Recommended Practices). Pre- or post-taxi positioning of the aeroplanes is only executed by barless towing when the aeroplane is protected by its own design from damage to the nose wheel steering system. Malta Air aircraft are protected by the bypass pin isolating the hydraulic system to the nosewheel steering and the crew procedure in FCOM Vol 1 NP.

On aircraft equipped with serviceable flight deck-to-ground intercom, this equipment will be used whenever the aircraft is on stand, and until clearance for hand signals has been given prior to departure.

All start-up, ramp departure and arrival procedures are applied as per aircraft type Standard Operating Procedures as laid down in the FCOM Vol I.

Whenever the use of hand signals is required, ground marshallers and Flight Crew Members shall use hand signals defined in ICAO Rules of the Air, Annex 2 (refer to OM A Chapter 12).

8.2.2.7.1 Start-up

After start-up clearance has been received from ATC, engines may be started only when an “all clear” signal or communication on the aircraft interphone system has been received from the Engineer/Mechanic or Handling Agent. Refer to OMA 12.6 for standard marshalling signals used during ground operations.

When interphone capability is not available for the pushback or start procedure:

- To indicate the ground crew’s readiness for push/start, the marshaller stands on the designated marshalling position (Ground Operations Manual refers), approximately 12 metres in front of the aircraft, or as per local apron design.
- Do not turn on the anti-collision light until the marshaller is on the designated marshalling position indicating that both handling pre-departure checks are complete and the ground crew is ready for start.

Start procedures and supplementary start procedures are contained in FCOM Vol 1, Normal and Supplementary procedures.

The PM shall monitor the engine start on instruments.

During engine start sequence the Captain shall verify that the marshaller is not indicating a start incident (e.g., cut engine(s), fire) using hand signals. The Captain shall immediately respond to hand signals from the marshaller.

8.2.2.7.2 Push-Back Procedures

General

Increasingly, aircraft are parked nose-in at airport terminal gates and have to be pushed back for departure.

The following are Malta Air procedures agreed with the Engineering Department. However, the general principles are good for, and should be applied at, other airports where ground personnel are provided by other organisations.

Although the Commander retains full command of the aircraft, it must be understood that the ground Crew Chief is in charge of the push-back manoeuvre and that the responsibility for the safety of the aircraft, the tug and any other person or objects in the immediate vicinity rests with him.

8.2.2.7.3 Ground/Flight Deck Communications

Although, in the following procedure, visual alternative signals are given for use when the ground intercom cannot be used, it must **always** be used if it is available or **can be made** available.

When using the ground intercom, it is essential that the words and phrases in FCOM Vol 1, Normal procedures are used exactly as given, since they have been chosen with care and are intended to avoid confusion. Communication from the operating crew is generally handled by the Commander. When calling, ground crew will be addressed as “GROUND” and the operating crew as “FLIGHT DECK”.

8.2.2.7.4 Interruption of Push-Back

Where the cockpit crew require the push-back to be stopped, the request is made on the service interphone, which is normally connected throughout the push-back.

If the tow bar becomes disconnected from the aircraft during the push-back, this will be immediately advised to the cockpit crew who will straight away apply the brakes with caution.

8.2.2.7.5 Re-establishment of Cockpit/Ground Communications

The flight crew will flash the aircraft landing lights to attract the ground personnel attention.

Particular care needs to be taken where it is necessary for ground crew to come back to the aircraft and re-establish headset communications after the engines have been started. The aircraft must remain stationary until the matter has been fully resolved and safety ensured.

The aircraft shall not taxi until a positive “Affirmative/all clear” or “thumbs-up” signal has been given to the Captain by the marshaller.

If the ground crew indicates a requirement to reconnect the headset and communicate with the cockpit, the aircraft will stop and remain stationary until the matter is fully resolved.

It is expected that a clear “thumbs-down” signal will be given, that the headset will be waved to the cockpit and that the aircraft will be approached from the front and in full view of the cockpit. However, the absence of any of these signals or indications must not in any circumstances be taken as clearance to proceed.

8.2.2.7.6 Power-Back, Reverse-Back Procedures

Reversing using Reverse thrust on Boeing 737 aircraft is prohibited for all Malta Air operations.

Where it is not possible to use forward thrust to manoeuvre out of a particular situation then the Commander shall first request the assistance of an aircraft tow-truck.

The Aircraft Marshaller/Ground Engineer must have two-way voice communication with the Commander, or hand signals may be used.

In addition, there must be one other competent person positioned who can confirm to the Marshaller that the area into which the aircraft is being manoeuvred is clear.

8.2.2.7.7 Pushback and Delayed Start Procedures

Pushbacks with delayed starts are used to free up stands and ground crew for other Malta Air aircraft to ensure punctuality. Expect to conduct push and delayed start operations at any Malta Air destination or base.

You may be towed to the hold position or you may taxi there.

If taxiing:

- Follow normal pushback and before taxi procedures;
- Start APU approaching the hold position;
- At the hold position, shut down the engines and complete the normal shut-down procedure;
- Start engines following normal engine start procedures to comply with your CTOT and depart as normal.

Note: Marshalling is not required.

Note 1: The time of the original push to hold is not block time and will be noted in the comments section of the Electronic Flight Log/Paper Voyage report. This is material information for Ground Operations statistics. After push and hold is completed, the time of start up for the purpose of take-off is the legal off block time for Company and pilot records.

Note 2: This procedure may be used during aircraft circuit training details to accomplish tire checks. In this instance, taxi the aircraft to the ATC designated hold position and:

- Start APU;
- At the hold position, shut down the engines and complete the normal shut down procedure;
- Complete the tyre check;
- Start engines following normal engine start procedures and depart as normal.

Note: Marshalling is not required.

8.2.2.7.8 Aircraft Towing to/from Remote Stands “Pilots Guarding Brakes”

When an aircraft will be towed a suitable qualified person will be required to remain in the flight deck to operate the brakes and respond to any non-normal situation.

Malta Air Captains are qualified to perform this task and only one pilot is required. Normal pre and post flight procedures apply.

The Captain shall receive a briefing from the Engineer in charge of the towing operation or Duty Pilot as to the procedures to be used, which are included in the Malta Air CAME.

8.2.2.8 Servicing of Aircraft

It is essential that ground clearance is received before initiating any action or operating any system which might be hazardous to personnel or equipment on the ground.

When crew approach an aircraft for the first time it is recommended that they make a mental note of work in progress around the aircraft, and establish from the Technical Log if any servicing or rectification of systems is in progress or due to take place before departure. FCOM Volume 1 specifies that the Technical Log shall be checked as the second item in the cockpit safety inspection.

Potable Water Supply

Malta Air ensures that the uplift of potable water occurs only at stations where water is bacteriologically and chemically safe and satisfactory in taste, and where water quality meets the specifications of the appropriate health authorities.

Ground equipment used in aircraft servicing shall be operated as described in the GOPS Manual and SEP Manual.

Potable Water – Minimum for Dispatch

Potable water is not mandatory for flight, is primarily a service issue and the following guidelines apply:

- Safety of operations takes precedence over commercial considerations.
- When established that potable water is empty and bars have NOT been loaded the aircraft will not normally depart until one of these is replenished.
- When it is established that potable water is 1/8 full and bars have NOT been loaded it is permissible to depart. On board service shall be managed to conserve remaining potable water for unexpected SEP issues.
- When potable water indication is > 1/8 full and bars ARE on the aircraft normal service resumes until potable water content reduces to 1/8 full and the restrictions above apply.
- 1/8 full is between E and ¼ mark.

Sensible application of these guidelines will ensure that our passengers travel safely and punctually with minimum disruption.

8.2.2.9 Document and Forms for Aircraft Handling

The documents and forms for aeroplane, passenger and cargo handling related to safety, but excluding Dangerous Goods paperwork, include:

- The Load and Balance Document (LID or Loadsheets)

- Notification Form for special categories of passengers

Note: The LID may be in paper or electronic format (eLID).

Mass and Balance Data and Documentation

CAT.POL.MAB.105

In Malta Air when weight is used it has the same meaning as mass in this section. If a nonstandard method is used to prepare the mass and balance documentation the commander shall be informed AMC1 CAT.POL.MAB.100 a.

Using standard masses (CAT.POL.MAB.100. d.), Malta Air has established mass and balance data and produces mass and balance documentation prior to each flight specifying the load and its distribution. The mass and balance document is the Load Instruction Document (LID) or the paper loadsheet which is prepared according to the procedures and instructions contained in the Operations Manual Part B – Mass and Balance. This Manual is prepared and maintained by the Head of Flight Operation Performance Engineering. The OPT Weight and Balance App/paper loadsheet enables the commander to determine that the load and its distribution is such that the mass and balance limits of the aircraft are not exceeded.

A completed paper Loadsheet contains the following information:

- aircraft registration and type;
- flight identification, number and date;
- name of the commander;
- name of the person who prepared the document;
- dry operating mass and the corresponding CG of the aircraft;
- mass of the fuel at take-off and the mass of trip fuel;
- load components including passengers, baggage, freight and ballast;
- take-off mass, landing mass and zero fuel mass;
- applicable aircraft CG positions; and
- the limiting mass and CG values.

The person supervising the loading of the aircraft confirms by signature or equivalent on the Load Information Document (LID)/paper loadsheet presented to the commander that the load and its distribution is compliant with the LCM. The commander indicates acceptance by writing his crewcode legibly and his signature on the paper LID/paper Loadsheet. The commander shall always ask the dispatcher to “confirm the status of the rear hold”.

Note: The LID may be in electronic format (eLID). The person supervising the loading of the aircraft confirms that the load and its distribution are in accordance with the eLID by entering their PIN code in the GOPs App. The Commander signifies acceptance of the eLID by entering their PIN code in the EFL App.

Malta Air has specified procedures for last minute changes (CAT.POL.MAB.105. d.) to the paper Loadsheel to ensure that:

1. any last minute change after the completion of the paper Loadsheel is brought to the attention of the commander and entered in the paper Loadsheel;
2. the maximum last minute change allowed in passenger numbers or hold load is specified as 500 kgs or less;
3. new mass and balance documentation is prepared if LMC of 501 kg or more (paper Loadsheel only).

Notification Form for special categories of passengers.

This information is provided by the dispatcher. When other categories of passengers are travelling these are notified in advance by the Safety and Security Manager.

NOTOC

NOTOC is an ICAO term, Notice to Captain. In Malta Air it indicates to the Captain that dangerous goods are being carried.

Shippers Declaration for Dangerous Goods

This documentation is prepared by the organisation shipping the DG. In Malta Air this is the Engineering Department moving spare parts and equipment related to aircraft servicing within the Malta Air network.

Consignment Note

This is prepared by the agency in an airport which processes the DG from the shipper to the aircraft.

8.2.2.10 Multiple Occupancy of Aircraft Seats

An infant may be carried by an adult in a single seat and secured for take-off and landing by a supplementary loop belt to the adults seat belt. They will not be allocated, nor occupy seats where they would obstruct access to emergency equipment, or impede the emergency evacuation of the aircraft. An adult may carry only one infant in this context.

Only one infant will be carried by an adult in any seat row unit (triple), as the number of drop down oxygen masks in any PSU is limited. There will be a minimum of 4 masks, sometimes more depending on the rows.

The different specifications of infant/child car seats and harnesses is specified on the Malta Air website and procedures for cabin crew to assure correct use is included in the OMA-SEP.

8.2.3 Refusal of Embarkation

CAT.GEN.MPA.170

Malta Air personnel and service providers shall take all reasonable measures to ensure that no person under the influence of psychoactive substances, to the extent that the safety of the aircraft or its occupants is likely to be endangered, is permitted to board, to enter or to be on board its aircraft.

The Commander shall not allow a person to be carried in the aircraft who appears to be under the influence of alcohol or drugs to the extent that the safety of the aircraft or its occupants is likely to be endangered.

The CSS shall inform the Commander if a passenger appears to be under the influence of alcohol or drugs and may cause a safety problem. If the safety of the aircraft or its occupants is likely to be endangered the Commander shall off load the passenger.

8.2.4 De-icing and Anti-icing on the Ground

CAT.OP.MPA.250

A Commander shall not take-off unless the external surfaces are clear of any deposit which might adversely affect the performance and or controllability of the aircraft except as permitted in the Aircraft Flight Manual and set out in the FCOM Volume 1, Supplementary Procedures, Adverse Weather Operation (SP 16). This is also known as a clean aircraft operating policy.

If contamination is expected or observed then the aircraft shall be de-iced and/or anti-iced in accordance with Malta Air procedures.

It is Malta Air policy to delay flap selection until reaching the holding point in Freezing Conditions as defined in FCOM Vol 1 SP 16 or following aircraft de icing. This policy assures the longest hold over times available.

If, for whatever reason, the flaps are extended prior to the commencement of taxi in Freezing Conditions or following anti-icing and remain in that position, the holdover time is no longer valid or applicable.

8.2.4.1 Proprietary or Commercial Names

The De-Icing Manual maintained by Ryanair DAC, in consultation with the NPCA Malta Air. It contains the proprietary or commercial names of all de-icing or anti-icing fluids approved for use on Malta Air aircraft. The holdover tables approved and the names of fluids are also in the Technical Log and available to operating crews.

8.2.4.2 Characteristics of De-icing/Anti-icing Fluids

The basic function of De/Anti-icing fluids is to lower the freezing point of water. This will delay the accumulation of frost, slush, snow or ice on critical surfaces or and/or will melt contamination already present.

Fluids used for de-icing and/or anti-icing shall be acceptable to Ryanair and the aircraft manufacturer. These fluids will conform to specifications such as SAE AMS1424 (Type I), SAE AMS1428 (Types II, III and IV). Use of non-conforming fluids is not permitted due to their characteristics being unknown. The anti-icing and aerodynamic properties of thickened fluids may be seriously degraded by, for example, inappropriate storage, treatment, application, application equipment, age and in case they are applied on top of non-chemically compatible de-icing fluids.

Fluids

1. ISO Type I (unthickened)

Type I fluid: Due to its properties, Type I fluid forms a thin, liquid-wetting film on surfaces to which it is applied which, under certain weather conditions, gives a very limited HoT. For anti-icing purposes the fluid/water mixture should have a freezing point of at least 10° C below OAT; increasing the concentration of fluid in the fluid/water mix does not provide any extension in HOT.

2. ISO Type II and IV (thickened)

Type II and Type IV fluids contain thickeners which enable the fluid to form a thicker liquid-wetting film on surfaces to which it is applied. Generally, this fluid provides a longer HoT than Type I fluids in similar conditions.

3. ISO Type III (thickened)

Type III fluid is a thickened fluid especially intended for use on aircraft with low rotation speeds.

Early Fluid Failure on Extended Slats and Flaps

Winter 2009-2010 anti-icing fluid testing confirmed that fluid failure occurs much more rapidly on the highly sloped critical surfaces of the airplane such as the flaps and slats when extended in the take-off position than it does on the less severely sloped other critical surfaces. The degree of degradation of the holdover and allowance times on these surfaces is affected by many elements; the amount of slope of the surface, the type of fluid, the airplane skin and ambient temperature, the type of precipitation, the amount of fluid applied, and the effective wind. Under conditions typical of normal operation in ground icing condition and flap angles fluids were observed to fail in as much as half of the published holdover time for the fluid and precipitation conditions present.

Malta Air has approved the risk mitigation strategy below:

- Delay flap and slat extension until just prior to take-off, minimizing the time that the slat and flap surfaces are in the highly sloped configuration.

8.2.4.3 Effects of Icing and Other Contamination on Aircraft Performance**8.2.4.3.1 Aerodynamics and the Contaminated Wing**

Any type of ice accretion is an obstacle to smooth airflow and will slow the airflow down causing flow separation, introduce turbulence thereby degrading lifting performance of the wing. Furthermore, freezing deposits may cause moving parts, such as elevators, ailerons, flap actuating mechanism, etc., to jam and create a potentially hazardous condition.

8.2.4.3.2 Weather-related Conditions

Icing conditions on the ground can be expected when air temperatures fall below freezing and when moisture or ice occurs in the form of either precipitation or condensation. Precipitation may be rain, sleet or snow. Frost can occur due to the condensation of fog or mist.

Cold Soaked Fuel Frost (CSFF)

The CSFF (Cold soaked Fuel Frost) relief as described in the AFM applies to Malta Air operations when the ambient temperature is +4°C or above. The presence of cold soaked fuel frost on the upper surface of the wing, within the confines of the defined CSFF area, and on the lower surface of the wing within the equivalent defined CSFF area is acceptable for departure in accordance with FCOM procedures.

FCOM Vol 1, SP.3 'Cold Soaked Fuel Frost' shall be used to enable dispatch with CSFF within the limits specified.

Dispatch with CSFF within the limits and using the procedure described in these documents is approved by the TM CAD.

Note 1: Remind the CSS that any passenger reports of wing contamination must be reported to the Captain. If CSFF relief is being applied, the Captain shall inform the CSS before departure.

Note 2: A Flight Crew Member must verify that the frost on the upper surface of the wing is within the confines of the defined CSFF area and there is no evidence of clear ice formation due to refreezing of melted CSFF.

Clear Ice Phenomenon

Under certain conditions, a Clear Ice layer or frost can form on the wing upper surfaces when the aircraft is on the ground. In most cases this is accompanied by frost on the underwing surface.

Severe conditions occur with precipitation when sub-zero fuel is in contact with the wing upper surface skin panels. During walk-around Clear Ice accumulations are very difficult to detect from ahead of the wing or behind it, especially in poor lighting and when the wing is wet. The leading edge may not feel particularly cold. The Clear Ice may not be detected from the cabin either because wing surface details show through.

The following factors contribute to the formation intensity and the final thickness of the Clear Ice layer:

- low temperature of fuel that was added to the aircraft during the previous ground stop and/or the long airborne time of the previous flight resulting in a situation that the remaining fuel in the wing tanks is below 0°C.
- abnormally large amount of remaining cold fuel in wing tanks causing the fuel level to be in contact with the wing upper surface panels as well as the lower surface, especially in the wing tank area.
- temperature of fuel added to the aircraft during the current ground stop, adding (relatively) warm fuel can melt dry, falling snow with the possibility of re-freezing. Drizzle/rain and ambient temperatures around 0°C on the ground is very critical. Heavy freezing has been reported during drizzle/rain even at temperatures of 8 to 14°C. The use of thermal leading edge Anti-icing may melt falling dry snow that re-freezes later.

The areas most vulnerable to freezing are:

- the wing root area between the front and rear spars.
- any part of the wing that will contain unused fuel after flight, the areas where different structures of the wing are concentrated (a lot of cold metal), such as areas above the spars and the main landing gear doubler plate.

Aircraft-related Conditions

The concept of icing is commonly associated only with exposure to inclement weather. However, even if the OAT is above freezing point, ice or frost can form if the aircraft structure is below 0° C and moisture or relatively high humidity is present.

With rain or drizzle falling on sub-zero structure, a clear Ice layer can form on the wing upper surfaces when the aircraft is on the ground. In most cases this is accompanied by frost on the underwing surface.

8.2.4.3.3 Icing on the Ground

Types of Ice

Frost and Hoar Frost – Ice crystal deposits forming directly on cold surfaces with a temperature below freezing when the dew point is also below zero.

Rime Ice – White or opaque deposits of crystalline and granular ice formed by supercooled water droplets impinging and freezing on impact with the exposed surfaces of the aircraft. Typical of freezing fog with the deposits generally thicker than frost.

Clear Ice – Essentially transparent and adhering closely with almost 100% bonding to the aircraft surfaces, and formed by freezing of water caused by condensation or precipitation including freezing rain onto the surface, or by larger supercooled water droplets which do not freeze directly on impact. Also called glaze ice when it occurs in flight.

Dry Snow – Small flakes of snow with no liquid moisture content at temperatures below freezing. As a consequence does not tend to stick to surfaces which are also below zero and can be removed blowing or brushing off.

Wet Snow – Snow close to freezing point and composed of generally larger flakes with a higher water content than dry snow, some of which is in liquid form. Generally sticks to aircraft surfaces, engines and exposed system components and if the surfaces are below zero or temperatures drop below freezing point can freeze onto the surface greatly increasing the bonding to the surface.

Snow Pellets – The snow column for all fluid types has been updated to include snow pellets. Recent testing has indicated that snow holdover times are appropriate for snow pellet conditions.

Ice Pellet Allowance Times – During the winter of 2009-2010 wind tunnel research conducted with a newer generation type airfoil showed that propylene glycol (PG) and Ethylene Glycol (EG) fluids behave differently under certain temperature and ice pellet conditions. Specifically, higher aircraft rotation speeds are required to effectively remove PG fluid contaminated with light or moderate

ice pellets at temperatures less than -10°C. The range of Boeing 737 rotation speeds is above the minimum worst-case rotation speed to remove fluid with Ice Pellets.

Sleet – A mixture of rain and snow or hail which can adhere to aircraft surfaces as with wet snow.

8.2.4.3.4 Ice Accretion on the Ground

Aircraft on the ground, while parked or during ground operations, are susceptible to accumulation of ice accretions in conditions peculiar to ground operations in addition to many of the conditions that can be encountered in flight. Precautions are necessary to avoid re-contamination of the aircraft structure and systems after anti-icing/de-icing. These precautions are given in FCOM Vol 1 Supplementary Procedures, Adverse Weather. The conditions giving rise to ice accretion on the ground include:

1. All forms of precipitation – snow, sleet, hail or rain adhering to and freezing onto aircraft structure because of a fall in ambient temperature or due to structural cold soak;
2. Conditions of high relative humidity that may produce frost accretions on aircraft surfaces having a temperature at or below the frost point. Frost accumulations are common during overnight ground parking and after landing where aircraft surface temperatures remain cold following descent from higher altitudes. This is a common occurrence on lower wing surfaces in the vicinity of fuel cells. Frost accretions can also occur on upper wing surfaces in contact with cold fuel;
3. Freezing fog;
4. Operation on ramps, taxiways, and runways containing moisture, slush, or snow;
5. Blown snow from snow drifts, other aircraft, buildings, or other ground structures;
6. Snow blown by winds, other aircraft or ground support equipment; and
7. Snow, slush and ice re-circulated by engine, propeller, or rotor wash.
8. Operation of jet engines in reverse thrust is a common cause of such contamination and ingestion damage.

Airframe icing may occur on the ground with ambient temperatures (OAT) above freezing point, if the precipitation temperature is below 0°C (supercooled droplets) and/or the aircraft wing is cold soaked (either due low temperatures for an extended period on the ground or in flight).

Ensure in such situations you De/Anti ice or establish conclusively that the wing is clean (Contamination Check).

If in doubt – De-Ice.

8.2.4.4 Hold over Times (HoT)

The responsibility regarding Hold-over Time and subsequent re-application of de-ice/anti-ice (if applicable) rests with the Commander from the time he has received the final report about the treatment from the Anti/De-icing person in charge.

The Hold-over Time starts at the beginning of the last treatment. ‘Holdover time (HOT)’ means the period of time during which an anti-icing fluid provides protection against frozen contamination to the treated aircraft surfaces. It depends among other variables, on the type and intensity of the precipitation, OAT, wind, the particular fluid (or fluid Type) and aircraft design and aircraft configuration during the treatment.

HoT will in effect have run out when frozen deposits starts to form/accumulate on the treated surfaces irrespective of the tabulated Hold-over Times, or at the commencement of the take-off roll due to aerodynamic shedding of fluid.

Hold-over Times are only guidelines because a variety of factors can change the effectiveness of the fluid. These factors include high winds, jet blasts, wet snow heavy precipitation, aircraft skin temperature lower than outside air temperature, direct sunlight, etc. Monitoring of the environmental conditions is therefore essential.

Holdover tables indicating the hold over times are published annually by FCI before the commencement of the Winter period. These constitute the hold over tables approved by DENG for use in Malta Air operations.

Lowest Operational Use Temperature (LOUT) Table

Lowest Operational Use Temperature (LOUT) information for Type I, II, III and IV fluids has been added to the holdover time guidelines. This information has been derived by the FAA based on data provided by the fluid manufacturers. The Lowest Operational Use Temperature, or LOUT is the lowest temperature at which a fluid has been tested and certified as acceptable in accordance with the appropriate aerodynamic acceptance test whilst still maintaining a freezing point buffer of not less than:

1. 10°C for a Type I fluid; or
2. 7°C for Type II, III or IV fluids.

There are two aerodynamic fluid flow off test protocols for fluids; the low speed test is for aircraft with rotation speeds less than 100 knots and the high speed test for aircraft with rotation speeds greater than 100 knots. Type II, and Type IV fluids generally do not pass the low speed test.

In the case of Types II, III, and IV fluids there can be multiple LOUTs to account for the neat (100/0) and the 50/50 and 75/25 dilutions. The LOUTs for these fluids are accounted for in the temperature bands which are an integral part of their holdover tables. If a fluid has a LOUT which is warmer than the lowest temperature in a particular band, this information will be noted on the appropriate table.

The LOUTs for Type I fluids are provided in Tables as well as the manufacturer specified fluid/ water concentration used to establish the LOUT for each fluid. This concentration shall not be exceeded.

The LOUTs for undiluted Type II, Type III, and Type IV fluid (100/0) are provided in Tables. The LOUTs for the 50/50, and 75/25 dilutions are shown only in the brand name holdover tables if the temperature is warmer than the lowest temperature in a given band.

8.2.4.4.1 Account is taken of the wing skin temperature versus OAT, as this may affect:

- a. The need to carry out aeroplane de-icing and/or anti-icing; and
- b. The performance of the de-icing/anti-icing fluids.

Wing skin temperatures may be lower than OAT. A stronger mix may need to be used to ensure a sufficient freeze point buffer. The de-ice service provider shall be informed about the wing skin temperature (as indicated by the fuel temperature) and fluid concentration will be based on the lower of OAT and wing skin temp.

8.2.4.5 De-icing and Anti-icing Procedures and Precautions During use

8.2.4.5.1 General

Depending on the facilities available at the aerodrome, and on the aeroplane type, removal may be achieved by brushing, the application of fluids, infra-red heat or forced air, or a combination of these methods. Flight crews shall familiarise themselves with the methods locally available.

8.2.4.5.2 Commanders Initiation of Procedure

On arrival at the aeroplane the Commander will decide whether or not the aeroplane is aerodynamically clean (including the flight controls).

The Commander is responsible for selecting an effective De/Anti-icing treatment and to conform to FCOM Vol I procedures, techniques and legal requirements. The Commander's request for such treatment and the fluid mixtures used, will always take precedence over locally recommended procedures.

A Commander's personal check for the correct treatment may not be practicable and, in this case, the person in charge representing the Anti/De-icing Company is responsible for verification of the Anti/De-icing results, (post treatment check).

The result of the Anti/De-icing procedure will be communicated to the Commander including the time for start of treatment, type of fluid(s) and mixture ratio.

8.2.4.5.2.1 At all Malta Air bases Engineers carry out early morning pre-flight/daily checks. These include a check as to whether the aircraft requires de-icing or not. They supervise the process, ensure the necessary fluid application time details are recorded in the Technical Log and carry out tactile checks as required. The de-icing crew do not need supervision but they must provide the fluid application details.

At other airfields, contracted ground handling agents complete de-icing. The Flight Crew shall complete the 'Exterior Inspection' as normal. The Captain shall request de-icing when required. It is imperative he/she does so as soon as possible. The de-icing crew do not need supervision but they must provide the fluid application details.

The Technical Log must be filled out and signed. This provides a written record to support any subsequent holdover time calculation.

Hoar Frost (overnight): Engineering/Ground Handling agents use a company checklist to determine if early morning de-icing due frost is likely. They may then "early Anti ice/De-ice" the aircraft with approved fluid to avoid a delay to the first departure. They are required to furnish the relevant Anti-icing details. Ensure you refer to the Hold over time (listed in the Technical Log) before dispatch.

Anti-icing against Hoar Frost has a long Hold Over time. Doing so overnight generally avoids the requirement to De/Anti ice prior to departure.

8.2.4.5.2.2 De-icing Admin: DAR Form

DAR is “De-icing/Anti-icing Request form”. This is a GOPS form. Spare copies may be carried in the aircraft library locker.

DAR form is divided into the following sections:

- **Flight Details:** normally completed by the dispatcher/handling agent before the aircraft arrives on stand.
- **Contamination Check:** this section establishes which parts of the aircraft need de-icing/anti icing. Normally completed by the Captain or Engineer.
- **De-icing/Anti-icing Procedure (Anti-icing code):** Outlines details of the required treatment including number of treatments/steps used, fluid type, mix and quantity. Normally completed by the service provider.
- **Post Treatment Check:** Normally completed by the Service provider. This certifies that the above procedure has been completed to the required standard and that the areas outlined in the Contamination Check are now clear of all contamination. The service provider will only certify in the post treatment check areas which have been requested in the contamination check section of this form. It remains the responsibility of the Commander to ensure the contamination check highlights all areas requiring treatment. The Commander shall sign the form and insert the relevant Technical Log page reference.
- **Bottom Copy:** The bottom copy is returned to the handling agent/service provider.
- **Top Copy:** The top copy is retained by the Commander who will clip this to the left hand control column where it will serve as a reminder when calling for the “Before Take-off Checks ‘Below the Line’” that (1) the holdover time is still valid, and (2) have the conditions changed. Having reviewed these items, the top copy is removed from the Control Column PRIOR to Take-off roll.

8.2.4.5.2.3 De-icing with Engines Running

- Ensure the Flight Details and Contamination Check sections of the form are completed. Sign the form and insert the Technical Log page reference.
- Provide a copy of the Technical Log to the handling agent/service provider PRIOR to starting engines.
- Start engines, taxi and de/anti-ice the aircraft. When this procedure is complete confirm over the headset the Anti-icing Code (section 2) and the name of the person responsible for the procedure and post Treatment Check. Write the name(s) on the form (top blue copy).
- Sign the form and clip to the control column as described above.

- Complete the aircraft Technical Log as normal, including the de-icing section.
- When the de-icing with engines running is complete and before taxi, action the Winter Ops post de-icing checklist. When taxi commences call for the “Before take-off Checklist” which shall be carried out in full.

8.2.4.5.2.4 Diagram of DAR Form

A diagram of the De-icing/Anti-icing Request form is available in the De-Icing and Anti-Icing Procedures Manual.

8.2.4.5.2.5 Service Provider Declines to Use DAR Form

Pending standardisation of this issue by EASA many airlines and service providers have designed their own forms. Some providers insist on not using the Malta Air DAR form, using their own documentation instead. This is acceptable, and a copy of the service providers form shall be attached to the EFL submission (or returned in the flight envelope).

8.2.4.5.2.6 Alternative Methods of Removal of Dry Snow

As an alternative to the application of fluid, service providers may use long handled soft brooms and squeegees have been approved for snow clearance from wings and stabilisers. Soft cotton rope and fabric hoses may be used for clearing the upper fuselage surfaces.

Under certain conditions (e.g. temperature below +3°C light snow) it may be advisable not to spray the aircraft surfaces with de-icing fluid if the structure is dry. Wetting the skin will cause the new snow to stick and turn to slush, with the possibility of these deposits re-freezing during take-off.

8.2.4.5.2.7 First Aid Measures

- **Ingestion:** Give large quantities of water to drink. Consult medical personnel.
- **Skin contact:** Wash off in flowing water. Launder contaminated clothing before reuse.
- **Eye contact:** Irrigate for five minutes. Obtain medical assistance if irritation persists.
- **Inhalation:** Remove to fresh air if feeling unwell. Consult medical personnel if symptoms persist.

8.2.4.5.2.8 Special Considerations Due to Residues of Dried Fluids**Dried fluid residues:**

Dried fluid residue could occur when surfaces have been treated but the aircraft has not subsequently been flown and not been subject to precipitation. The fluid may then have dried on the surfaces.

Re-hydrated fluid residues:

Repetitive application of thickened de-icing/anti-icing fluids may lead to the subsequent formation/build up of a dried residue in aerodynamically quiet areas, such as cavities and gaps. This residue may re-hydrate if exposed to high humidity conditions, precipitation, washing, etc., and increase to many times its original size/volume. This residue will freeze if exposed to conditions at or below 0° C. This may cause moving parts such as elevators, ailerons, and flap actuating mechanisms to stiffen or jam in flight.

Re-hydrated residues may also form on exterior surfaces, which can reduce lift, increase drag and stall speed. Re-hydrated residues may also collect inside control surface structures and cause clogging of drain holes or imbalances to flight controls. Residues may also collect in hidden areas: around flight control hinges, pulleys, grommets, on cables and in gaps refer to Section 3 of the De-ice/Anti-icing Manual.

8.2.4.5.3 Step Anti/De-icing

Anti/De-icing can be performed in one step or in two steps:

The “One Step” procedure means that De-icing and Anti-icing are carried out at the same time, using a mixture of De-icing/Anti-icing fluid and water. This assumes that the applied fluid will provide an adequate protection period (HoT), in the prevailing meteorological conditions, against re-freezing.

In the “Two-Step” procedure the aeroplane is first de-iced to ensure that it is clean of all frozen contaminants, followed immediately by the application of anti-ice fluid. The second step will be applied before the first step fluid freezes, typically within three minutes and, if necessary, area by area.

- With a “one-step” Anti/De-icing procedure the HoT begins at the start of the Procedure.
- With a “two-step” procedure the HoT begins at the start of the second (Anti/De-icing) step.

8.2.4.5.3.1 Removal of Local Area Contamination

Local area de-icing is the process of removing of frost and/or ice which is limited to localised areas on the surfaces of the aeroplane. This type of contamination will generally be found on the wing and/or Horizontal stabilizer leading edges or in patches on the wing and/or stabilizer upper surfaces for example CSFF.

The following conditions are applied:

- Freezing conditions do not exist or are forecasted before departure
- Application of the treatment must be symmetrical
- The same type of fluid, heated with a fluid/water mixture suitable for a One-Step Procedure

If crews are unable to determine the extent of contamination (i.e. on horizontal stabilizer during hours of darkness), the required action is to revert to standard de-icing procedures.

Note: Both sides of the aeroplane must be treated identically (same areas, same amount and type of fluid, same mixture strength), even if the contamination is only present on one side.

After this check has confirmed that the treated areas are clean, the following statement shall be given to the Commander:

“Local Area De-icing only. Holdover times do not apply”

The extract below from the DAR form indicates to the crew that Local area de-icing has taken place.

1(a). Local Area Contamination/
Cold Soaked Fuel Frost

**HOLD OVER TIMES DO
NOT APPLY**



Indicate areas to be treated on the diagram to the right
With this type of contamination only affected areas need to be treated. De-ice fluids must be applied symmetrically on both sides of the aircraft.



Crews shall use the TEM techniques to ensure they are aware of the type of de-icing that has taken place.

Crew shall be aware that should the environmental conditions change then they will consider that the aircraft is not de-iced and return to stand for a de-icing/anti-icing treatment.

8.2.4.5.4 External Pre Take-off Contamination Check

The Commander shall continually monitor the environmental situation (e.g. precipitation, wind and OAT) after the performed Anti/De-icing treatment. Prior to take-off the Commander shall assess whether the applied HoT is still appropriate.

When the HoT has been exceeded or if doubt exists regarding the continued effectiveness of the applied Anti-icing treatment, the Commander shall arrange for a pre take-off contamination check to be performed externally by a qualified person* in order to verify that the aircraft's surfaces are free of contamination. When required, advise ATC/Dispatch that “an external pre-take-off contamination check is required” and they will coordinate to facilitate the local procedure. If this external check cannot be performed just before take-off, the aircraft shall return to a place where re-treatment shall be applied.

Note: *A qualified person is one who has received training in aircraft de-icing techniques normally de-icing staff or an engineer.

8.2.4.5.5 General Precautions

Aeroplane surfaces treated with Anti/De-icing fluids should appear glossy, smooth and wet. An indication that protection is deteriorating is the loss of gloss or dulling of the surfaces as the fluid gradually turns to slush. If, at any time prior to take-off, the Commander requires confirmation of the ice-free state of the aeroplane, he must have a visual inspection carried out or return to the ramp.

Unless the aeroplane is being anti/de-iced by an approved anti/de-ice contractor, the Flight Crew or other authorised personnel shall supervise the Anti/De-icing operation being carried out by, or on behalf of, a handling agency.

In this case, care will be taken to ensure that whether removed by broom, or the application of fluid spray, deposits are swept away from control surface hinge areas and system intakes, and that the sprays themselves are not directed to these areas. Since the melting deposits, which the Anti/De-icing is supposed to remove, may further dilute the Anti/De-icing fluid, re-freezing may occur if the solution runs onto other parts of the aeroplane, and close attention shall be paid to this possibility.

When Anti/De-icing operations have been completed, ideally as close to the scheduled departure time as possible, a careful inspection of the aeroplane is to be completed in order to confirm that wing, tailplane and control surfaces have been cleared of deposits, and that intake and drain holes are free of any obstruction. Control surfaces shall be moved over their full range, and jet engine compressors rotated by hand to ensure that they have not become frozen in position.

Note: Before reconfiguring or moving the aircraft, the Flight Crew shall receive a confirmation from the personnel involved in the operation that all De-icing and/or Anti-icing operations are complete and that all personnel and equipment are clear of the aircraft.

CAUTION: *Under no circumstances may an aircraft that has been anti-iced receive a further coating of Anti/De-icing fluid directly on top of the contaminated film. If an additional treatment is required before flight, a complete Anti/De-icing/Anti/De-icing process must be performed to ensure that any residues from the previous treatment have been removed.*

The aircraft must always be treated symmetrically – the left hand and right hand sides must receive the same and complete treatment.

8.2.4.5.6 Information to the Commander

The person carrying out or monitoring the correct Anti/De-icing of the aeroplane at the Anti/De-icing platform is responsible for the visual check on completion even though the final responsibility rests with the Commander. The Flight Crew may perform the visual check (if possible).

The code is then transmitted to the Flight Crew to confirm satisfactory treatment and the Commander will decide whether the HoT is adequate for taxiing and take-off.

The code defines the treatment (Anti/De-icing) applied to the aeroplane. It includes the fluid classification, concentration and time of starting the treatment. Protection period (HoT) is measured from the time of start of the last treatment.

Example 1: TYPE I 75/25 0550 (local time)

Example 2: TYPE II 100/0 0600 (local time)

Note 1: The mix ratio is by volume with the Anti/De-icing fluid quoted first, e.g. a 75/25 mix contains 75% Anti/De-icing fluid and 25% water.

Note 2: When the mix used differs from those tabulated, use the next lower concentration code, e.g. an 80/20 mix is coded as 75/25.

8.2.4.5.7 Taxi-through De/Anti-icing

Anti/De-icing with engines running may be performed in taxi-through facilities at particular airports. The OM B FCOM SP 16 prescribes procedures and checklists for use under these circumstances.

Pre-briefing must be conducted between the Flight Crew and the ground personnel performing the Anti/De-icing to ensure that they are aware of any specific requirements called for by the Flight Crew. VHF contact must be maintained with the Anti/De-icing supervisor throughout the operation. The Anti/De-icing supervisor is responsible for confirming that the aeroplane is completely free of ice and snow before releasing it.

The lowest possible power setting shall be used, flaps retracted or extended, (according to OM B FCOM SP16 procedure), and the air-conditioning switched OFF to prevent fluid from entering the cabin.

Before takeoff, perform a flight control check.

As soon as the aeroplane is clear of the facility, the air-conditioning may be switched ON again. Ensure that taxi and take-off checklists are completed. Prior to releasing brakes for take-off, applicable engine De-icing procedures will be applied.

8.2.4.5.8 Technical Log

The Commander is to confirm that, whenever Anti/De-icing has taken place, an appropriate entry has been made and signed in the technical log. In particular, the fluid type/mix and start/completion times shall be recorded.

If there is any subsequent departure delay, or further deterioration in the weather conditions, the Commander shall use this information, together with the HoT, to form a realistic idea of whether further Anti/De-icing may be required.

Note: *Times passed to the flight crew by the deicing team will be in local time, Commanders will need to convert this to UTC for entry into the Technical Log.

8.2.4.5.9 Take-off Prohibition

Take-off is prohibited if any of the following conditions exist:

- Snow, ice or frost deposits are adhering to the wings tail plane, control surfaces or engines.
- Heavy fall of wet snow, slush and rain with ambient temperature around freezing point unless adequately protected by anti-ice treatment.
- Freezing rain or drizzle unless adequately protected by anti-ice treatment.

CAUTION: Under certain meteorological conditions, de-icing and/or anti-icing procedures may be ineffective in providing sufficient protection for continued operations. Examples of these conditions are freezing rain, ice pellets and hail snow exceeding certain intensities, high wind velocity, and fast-dropping OAT. No HOT guidelines exist for these conditions. Where HOT guidelines do not exist for these conditions, or if in doubt, take-of shall not be commenced.

8.2.4.6 **Icing**

CAT.OP.MPA.255, AMC1 CAT.OP.MPA.255

Malta Air aeroplanes are certified for flights in icing conditions. However, if actual icing exceeds the intensity of icing for which the aircraft is certified, the Commander shall exit the icing conditions without delay, by a change of level and/or route, and if necessary by declaring an emergency to ATC. The intensity of aeroplane icing is defined in accordance with definitions provided in the Table below:

Table Definition of Aeroplane Icing

Light:	Ice accumulation which does not necessitate a change of heading or altitude
Moderate:	Ice accumulation which makes a change of heading and/or altitude desirable
Severe:	The rate of accumulation is such that De/Anti-icing equipment fails to control or reduce the hazard

For procedures in flight refer to FCOM Vol 1 NP and SP 16.

8.2.5 **Crew Inspections**

8.2.5.1 **Pre-flight**

CAT.GEN.MPA.105(a)(12)

The Pre-flight Check is the external check of the aircraft, or “walkaround”, completed by a qualified crew member prior to commencement of a flight in compliance with Annex I (Part-M) to Regulation (EU) 1321/2014; pre-flight check. The Pre-flight check is described in FCOM Vol 1. All Malta Air Flight Crew members are qualified to perform the Pre-flight checks.

The forward and rear hold shall always be opened by the Flight Crew member during the walk around to preclude the possibility of inbound bags/freight not being offloaded. Where bags or cargo are found in the hold they shall be brought to the attention of the ramp agent.

The Flight Crew will pay particular attention to flight controls, gear and tyre condition; the removal of gear pins and covers (if used), the high risk of damage by ground service vehicles and the possibility of fluid leaks from engine, hydraulic and fuel systems.

The captain's signature in the aircraft Technical Log is certification that all aspects of Pre-flight, including flight deck door access system (required for the first sector of each flight duty) have been completed as per approved procedures.

8.3 Flight Procedures

8.3.0 Flight Procedures – Malta Air Guidance for Various Flight Phases

All Malta Air revenue flights shall be conducted under Instrument Flight Rules, IFR.

Instrument Departure and Approach Procedures

1. Malta Air shall ensure that instrument departure and approach procedures established by the State in which the aerodrome is located are used.
2. Notwithstanding sub-paragraph above, a Commander may accept an ATC clearance to deviate from a published departure or arrival route, provided obstacle clearance criteria are observed and full account is taken of the operating conditions. The final approach must be flown visually or in accordance with the established instrument approach procedure.
3. Different procedures to those required to be used in accordance with sub-paragraph (1) above may only be implemented by an operator provided they have been approved by the State in which the aerodrome is located, if required, and accepted by the Authority.

8.3.0.1 Take-off and Departure

8.3.0.1.1 Take-off Briefing

The Malta Air policy for, and content of briefings is presented in FCOM Vol 1, Normal procedures. Refer section 8.0.4.

If the departure circumstances change then a new briefing or update must be given prior to take-off at or before the checklist item "Take-off brief....Review".

The object of briefing is to ensure that both Flight Crew members are aware of and agree with a proposed plan of action. Briefings will be as short as possible but give a clear understanding of the intentions. It is normally unnecessary to reiterate Standard Operating Procedures, but it is necessary to cover any special requirements.

If the recipient of a briefing is unsure of the intentions to any degree he must make sure the points are clarified.

The statement “Standard Calls and Procedures” for take-off (also applicable for landing) will apply only when both Flight Crew members are familiar with the procedures for the airfield concerned and when further elaboration is clearly not required. “Standard Calls and Procedures” cannot be applied to Non Precision or Visual approaches, both of which require a comprehensive briefing.

The take-off briefing shall be initiated by the pilot flying (PF) and shall always be the full briefing for the first flight of the day.

8.3.0.1.2 Taxiing

Malta Air policy for taxiing is presented in FCOM Vol 1, Normal procedures. If a technical problem manifests itself crews having completed the applicable Non Normal checklist, shall check circuit breakers to establish if this is the primary cause.

If at any time the clearance between the wing tip and stationary or moving objects cannot be assured then the Commander shall STOP the aircraft and ensure that there is a Marshaller guiding the wing tip before proceeding further.

Additionally crews shall follow taxi markings meticulously during the power off procedure to ensure that no part of the aircraft presents a hazard to other aircraft parked on the ramp.

1. The Taxi phase of operations shall be considered a ‘Critical Phase of Flight’. A sterile cockpit is mandatory and only essential safety PA’s are allowed.
2. Expected taxi routings will be briefed for prior to push back for taxi out and prior to descent for taxi in, for airports both crew members are not intimately familiar with. Always brief temporary taxi routings/restrictions.
3. Write down all taxi clearances. The RHS pilot must ensure the Captain has acknowledged the clearance before reading back.
4. The ‘Sterile Cockpit’ and ‘Sterile Comms’ applies to all taxi phases and climb and descent. This forbids non-operational chatter.
5. Maximum taxi speeds must never be exceeded.
 - a. 50 kts on a runway when backtracking to vacate or take-off
 - b. 30 kts on a taxiway
 - c. 15 kts on the apron
 - d. 10 kts when executing a turn of 45 or greater
 - e. 5 kts on contaminated runways or taxiways

If unsure of a taxi routing stop the aircraft and ask ATC for clarification. If unsure of your position when taxiing, stop and immediately notify ATC (including last known position). If you inadvertently enter a runway immediately notify ATC (an ASR for runway incursion is required). If visibility enables you to safely vacate a runway, vacate the runway as expeditiously as possible, unless otherwise instructed by the air traffic services unit and then once vacated, stop the aircraft. If in LVPs and it is not safe to vacate a runway include this in your notification to ATC. When requested give the taxi clearance you believed you were following.

6. Strobe Lights will be turned ON when entering or crossing an active or non active runway.
7. Prior to crossing or entering a runway, either active or non active, crews must make a 'Two Phase Call' (inter cockpit). FCOM Vol 1 Normal Procedures detail this call.

This 'Two Phase Call' requires the Commander to establish with the RHS pilot that:

- a. That a clearance has been received to cross or enter the runway.
- b. The approach is checked clear as much as is possible.

This must be responded to by the RHS pilot, acknowledging that the approach on their side (right) has been checked as per the Captain.

8. Taxi and Braking on Slippery Surface

The FCTM advises using reverse thrust as required to taxi speed – This is 5 kts on slippery surfaces. The FCTM cautions against using prolonged use of reverse thrust at taxi speeds due to possible FOD and engine surge, but this must be balanced against the risk of a runway/taxiway excursion. The only braking available when the aircraft is skidding is reverse thrust, up to and including max reverse.

Promptly select reverse thrust at any stage that the aircraft is, or appears to be compromised.

Exercise extreme caution:

- On runways promulgated as wet when the OAT is 0°C or colder.
- At runway ends where it is difficult to perform friction tests.
- When crossing painted runway or taxiway markings.
- On clearways or stopways which may not be de-iced as not actual runway.

The Malta Air policy for external aircraft lighting configurations is presented in Ops Manual Part A, FCOM Vol 1, Normal procedures available on DocuNet. These policies are incorporated in the Checklist and Standard Operating Procedures, FCOM Vol 1.

An announcement is made on the P.A. before each start and before each take-off, the contents of which are promulgated in the PA checklist on the flight deck.

1. The take-off shall not take place until the Cabin Crew have made the cabin secure report, nor shall the crew accept a line-up clearance until this report is received, and the aircraft is actually ready to effect a take-off clearance.
2. When making PA's the PF when airborne, shall hand over control of the operational frequency to the PM, and instruct the PM to maintain a listening watch on this frequency, and to immediately inform the PF of any circumstances requiring his immediate attention. He shall minimise the time off the operational frequency. When the announcement is complete he must positively establish from the PM any instructions received from ATC.

8.3.0.1.3 Logging of ATC Instructions and Clearances

The airways clearance and other instructions will always be logged by the FO on the ground or the PM when airborne and in the cruise.

All ATC instructions must be repeated back in full.

Standard phraseology as presented in OMC, Jeppesen Publications (Pubs) > Air Traffic Management (Doc 4444) > Section 12 Phraseologies shall be used.

All communications, except greetings in local language, shall be in the English language.

Ideally both the Commander and co-pilot are required to monitor ATC clearances and have a responsibility to query any clearance that is not understood, incorrectly received, or incomplete. Any uncertainty must be resolved by asking ATC to repeat the clearance. It is recognised that certain events preclude both pilots being in the flight deck simultaneously.

On stand, ATC departure clearances may be requested/confirmed via datalink or voice. When utilising voice, both pilots must be present in the flight deck and on headset. When conducted via datalink, the PM may request and accept the clearance at the appropriate time during the preflight procedure. During the RIBETS briefing, both the Captain and FO must confirm the clearance via reference to the ATC LOG in the FMC and the OFP.

Note: As per AFM, when obtaining departure clearance via ACARS, pilots shall manually verify (compare) the filed flight plan versus the digital pre-departure clearance and shall initiate voice contact with Air Traffic Control if any question/confusion exists between the filed flight plan and the digital pre-departure clearance.

Particular attention is needed to ensure that the SID corresponds to that for which the take-off briefing was given. It is a typical human factor mistake to believe that the clearance received corresponds with the clearance filed and expected.

On the ground, the altitude alert shall be pre-set up to an expected altitude clearance plus 100 ft, and selected to the actual altitude cleared on receipt of the clearance.

If you receive either a different departure clearance to that shown on the OFP or a change of SID after receipt you shall:

- Ask for confirmation of the clearance given
- If confirmed, ask for the flight plan route
- Exit Airplane mode on EFB. Open the Summary tab within the EFL App and select 'Download' to check for the latest flight plan. If the EFL flight plan update is not available, then crew must contact Ryanair Ops
- If taxiing ask for clearance to stop and hold clear of the runway and re-brief the new SID and route
- Re-programme the FMC and conduct a route check
- Be aware that the new route may be significantly longer than the original and an additional fuel uplift may be necessary.

In circumstances where the OFP does not agree with the flight plan held by ATC an ASR must be filed at the earliest opportunity to ensure that any systemic flaw in operational communications is corrected immediately.

8.3.0.1.4 Radio Aids

The Malta Air policy for navigation and tuning and identification of Nav Aids is presented in FCOM Vol 1, Normal procedures.

Nav Aids shall be selected to enable the departure/SID, arrival/STAR and instrument approach to be flown in raw data. Enroute Nav aid tuning may be accomplished for situational awareness, but is subservient to RNP 5.

Loss of RNP 5 or RNP 1 shall be reported to ATC immediately to enable resumption of conventional navigation.

8.3.0.1.5 Take-off

Aircraft management procedures specific to aircraft type are given in the FCOM Vol 1 and/or the QRH for all take-off variations.

The take-off shall be abandoned for a take-off configuration warning. If this results in loss of line up allowance the aircraft shall reposition to assure line up allowance integrity.

Before commencing take-off, a Commander must satisfy himself that, according to the information available to him, the weather at the aerodrome and the condition of the runway intended to be used will not prevent a safe take-off and departure. This information is available in the Operations Manual Part B – Performance.

Take-off Warning Report Faults – Fault Isolation

When a Take-off Config warning is triggered and the aircraft is returning to stand for maintenance investigation do not advance the thrust levers beyond 40% N1 during taxi to stand.

This procedure ensures that the cause of the TOC warning is preserved in the PSEU BITE control.

Advancing the thrust levers beyond 40% risks a thrust lever angle which cancels the fault that caused the TOC Warning.

Similarly observe 40% N1 when taxiing after a RTO caused by a TOC warning. Loose LE Slat connector and stab trim connectors disturbed by vibration on take-off roll typically cause high speed TOC warning. You are free to use reverse thrust as required to stop the aircraft as it is forward thrust lever angle which cancels the cause of the warning.

Weather Radar, Thunderstorms

Aircraft shall not take-off or follow a path which takes the aircraft into a thunderstorm cell. The weather radar shall be switched on for take-off and the Captain shall monitor the take-off path for cell activity. Thunderstorm activity and wind shear are covered under adverse weather in Part A, Guide to AWO.

Wake Turbulence Categories

Wake turbulence categories are presented in OMC, Jeppesen Publications (Pubs) > Air Traffic Management (Doc 4444) > ATC Section and OMA 8.3.9.

Bird Hazard

Crews shall carefully consider initiating take-off if flocks of birds are seen on the runway or close to the runway edge.

Take-off Flaps

Flap 5 is the normal departure flap setting, flaps other than Flap 5 shall be used when operationally necessary. If crosswind component is in excess of 10 kt fixed derate is permitted, however assumed temperature thrust reduction is not permitted.

All Flap 1 departures shall be flown by the Captain as PF.

Intersection Take-off

Intersection take-offs are encouraged and policy summary is as follows:

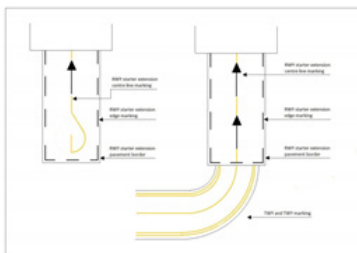
1. Performance plan for intersection take-off when appropriate.

Appropriate means when the intersection is en route to the threshold, use intersection numbers. If the threshold occurs before an intersection then you use the threshold numbers. It is not intended that a pilot prepares for Q3 at STN when 99% of the time departures are initiated from the threshold of RWY 22. It is however intended that pilots plan L1 when departing RWY 04 in STN as the outbound taxi route passes L1 on the way to the RWY 04 threshold. This issue is a combination of expediency, efficiency and airmanship.

2. Subject to performance limitations it is preferable to utilise intersection take-offs when available, permissible and safe to do so, rather than consume extra taxi fuel and time taken to depart from the full length. Up to maximum thrust may be considered to achieve this policy.
3. If you must use full length so advise ATC on start up.
4. On transfer to tower frequency reconfirm that you can take intersection take-off.

Starter Extension Take-off

Starter Extensions are an area of the runway prior to the threshold used for increasing TORA/TODA.



Dashed Marking Layout



Brussels International

In most European airports starter extensions are more commonly marked with solid white lines the same as runway edges, other airports use dashed lines to define the starter extension area.

Crew should remain vigilant to the possibility of confusing dashed edge marking with the centerline marking at airports that conform to the dashed marking layout, e.g. BGY.

When lining up on a runway using a starter extension confirm the aircraft is aligned with the centerline not the runway edge.

In Low Visibility conditions ensure that the ILS is tuned to the departing runway where available.

It is permissible to follow center line markings from the taxiway leading on to the runway but crews must give consideration to line up distances on performance limiting runways.

8.3.0.1.6 Rejected Take-off

The decision to reject a take-off will be made by the Commander, unless incapacitated. The F/O shall indicate to the Commander if he disagrees with a course of action during any flight phase, e.g. a continued take-off with a CONFIG WARNING.

1. The rejected take-off procedures are presented in FCOM Vol 1, Normal procedures.
2. Additional guidance material and best operating practise is provided in the Boeing Flight Crew Training Manual for Rejected take-off.
3. RTO > 80 kt and runway vacated:
 - Ask ATC for parking on taxiway/return to stand
 - Call Maintrol and report a high energy RTO
 - Do not take-off again until released by Maintrol
 - Calculate the turnaround time required from the brake cooling schedule
 - File a ASR on completion of duty.

8.3.0.1.7 Initial Climb-out

The after take-off and climb procedures are presented in FCOM Vol 1, and additional guidance and best operating practise is provided in the Boeing Flight Crew Training Manual.

Normal profiles are presented in the FCOM Vol 1, and non normal profiles in the QRH.

8.3.0.1.7.1 Minimum Turn Altitude

A turning altitude is specified in a SID if this altitude is less than 398 ft. Crews shall comply with these turn instructions. Where no turning altitude is specified turns shall not commence below 400 AAL.

Departures with an initial track up to 15° offset from the runway track are not considered to be turning departures. The published departure track shall be followed and requires selection of LNAV or an appropriate HDG SEL prior to 400 ft.

No turns may commence below 59 ft above aerodrome level.

No turns may commence before the Departure End of Runway unless following a published SID or Omnidirectional Departure Procedure.

8.3.0.1.7.2 Track Deviation: Prevention

Flight Crew members shall follow departure routings accurately. The primary causes of track deviations are non selection of LNAV, slow turning speed for initial turns not entered in FMC to contain the turn radius until the turn is completed, and inaccurate practice hand flying.

Standard ICAO speeds shall be flown until advised by ATC that no speed limit exists, and the aircraft is operating in Class A, B or C airspace, and is in the climb. Speed should be constrained to not more than 250 kt climb speed until the aircraft is heading in the direction of the on course routing to avoid flying extra mileage. This speed is comfortably within the max speed for initial turns in a SID of 265 kt for B737 800/8200. When LNAV is not engaged the lesser of rate one turns for the speed or 25° of bank shall be flown until established enroute, and the aircraft shall be operated in a manner that enables the crew to establish that the intended flight path is clear of traffic/obstacles. In the descent, 250 kt is recommended < FL 100 regardless of airspace classification, and crews shall decline offers from ATC of higher speeds. If ATC instructs a higher speed it shall be flown.

8.3.0.1.8 Noise Abatement

NADP2: Bug UP at 1,000 ft AAL, N1 at 1,500 ft AAL, retract flaps on schedule, maintain UP speed to 3,000 ft AAL, then accelerate.

When this Noise Abatement procedure does not provide adequate obstacle clearance an amended procedure shall be indicated in the RTOW charts in the same manner as EOSIDs, and used by the crew.

Some destinations may still make reference to NADP A, or NADP B etc. The term NADP A or B no longer exists. The Noise Abatement reference literature currently used at many MAY destinations predate NPA 53. Therefore in the absences of a specific instruction to fly NADP 1, Malta Air B737 aircraft will fly NADP 2. All other noise references are immaterial.

NADP1: N1 @ 1,500 feet AAL, Bug up @ 3,000 AAL, flap retraction on schedule

The THR RED Alt in the FMC may be preprogrammed to 3,000 ft AAL or may be manually programmed to 3,000 ft AAL, or as required.

A turning altitude is specified in a SID if this altitude is less than 398 ft. Crews shall comply with these turn instructions. Where no turning altitude is specified turns shall not commence below 400 AAL. The minimum altitude limitation for commencement of any turn is specified in FCOM Vol 1, limitations.

Noise abatement procedures do **not** take precedence over safe operation and emergency procedures when required, and may be cancelled during certain meteorological conditions.

Standard noise abatement departure procedures (FCOM Vol 1), shall be carried out for all normal departures. Crews shall follow the departure instructions presented on the OFP/AFB if different from the Standard MAY procedure. Speed in excess of that promulgated in the procedure will significantly degrade the effectiveness of noise abatement.

Strict adherence to routing and other procedures specified by the Airport Authority is important. These are specified in the Jeppesen Publications (Pubs).

There are noise measuring terminals points at many of the larger airports and noise violations will be filed by the airport authority, usually with a fine when noise limits are exceeded.

8.3.0.1.9 Systems Operation

The Malta Air standard operating procedures for systems operation, and area of operation concepts in the cockpit are presented in the FCOM Vol 1, Normal Operating Procedures, Supplementary Procedures, and QRH for non normal procedures.

8.3.0.1.10 Look Out

TCAS is fitted to all Malta Air aircraft. Crews shall respond to all TCAS RA instructions at all times. TCAS is not Secondary surveillance radar and shall not be used in this way. Standard Operating Procedure for TCAS system operation is presented in the QRH, and system description is in Vol 2. TCAS is not a substitute for “Lookout”.

Flight operations are IFR but the principle of "see and be seen" is deemed to apply. Crews must exercise increased levels of awareness when operating in the vicinity of airports and during climb and descent, particularly in destinations located in Class E, F and G airspace, or destinations requiring transit through Class E, F and G airspace.

In general, one Flight Crew member will always be in a position to maintain a look-out. Flight Crew members will be aware that while relative motion helps to detect another aircraft, an aircraft on a collision course will remain in the same position within the window frame and is consequently more difficult to detect. This also applies when airports are conducting simultaneous take-offs from intersecting runways.

Be particularly alert for GA aircraft and gliders in particular, many of which do not have transponders.

MAC (Mid Air Collision) has been identified in the MAY SMS as a key operational risk area (KORA) in the current Malta Air safety strategy.

8.3.0.1.11 ATC Communications**Stuck Mike Protection**

The Malta Air fleet has a stuck mike modification programme completed. This modification means that after 35 seconds of continuous transmission the VHF comm. transceiver will revert to the receive mode. Malta Air aircraft radio comply with the requirements of CAT.IDE.A.330, CAT.IDE.A.335 and AMC1 CAT.IDE.A.325.

ATC communication is covered in OMC, Jeppesen Publications (Pubs) > Air Traffic Management (Doc 4444). Both the PF and the PM have a responsibility to monitor ATC transmissions and to query any transmission that is not received fully and/or understood. Any uncertainty must be resolved by confirmation with ATC.

Both pilots shall react independently and simultaneously to an ATC instruction. Both pilots have defined actions to do on receipt of a clearance and there is a cross check built into the process. The reason for independent simultaneous action is to ensure that both pilots have heard the instruction, have actioned the instruction

and the crosscheck will identify any error by an individual pilot. If the response is not independent and simultaneous one pilot tends to follow the other and there is no opportunity to capture an error if one is made at this point.

All ATC communications shall be conducted in the English language.

Both Flight Crew members shall normally maintain a listening watch on the appropriate ATC frequency, including on the ground during pre-flight. If a Flight Crew member goes off frequency he must inform the other Flight Crew member.

Flight Crew members shall maintain good R/T discipline and make every effort to reduce speech traffic, particularly on congested frequencies. The following general rules apply:

1. Always listen out before transmitting.
2. The full call sign is to be used preceding the message. Enunciate clearly and avoid clipped responses or replying without giving a call sign.
3. On first contact after changing frequency during the departure, give call sign, SID, passing altitude/FL, cleared altitude/FL. Thereafter passing FL for cleared FL or current FL as appropriate, e.g." MAY 3015 FL 380".
4. Read back all clearances followed by your call sign. The word "CLEARED" shall only be used by ATC for an airways clearance, a take-off and a landing clearance.
5. A Take-off clearance shall be issued separately from any other clearance message. The expression "TAKE-OFF" shall only be used in radiotelephony when an aircraft is cleared for take-off or when cancelling a take-off clearance.
6. When requested to advise that you have vacated altitude or flight level, do not say you have vacated it until you have actually have.
7. NATS UK has produced a booklet addressing the most common ATC issues.

Change of Radio Frequency

It is essential to preset and crosscheck the new frequency before changing and contacting ATC. The following procedure shall apply to frequency changes:

CRZ	On receiving a frequency change the PF will enter the new frequency in the standby window on the RTP and the PM will respond to ATC and note the new frequency on the OFP. The PM will then crosscheck the standby frequency with the OFP before changing and contacting ATC.
CLB, DESC, or AP disengaged	The PM will be responsible for the frequency change. The PM will enter the new frequency in the standby window on the RTP while responding to ATC. The priority is to change frequency correctly before writing down a record of the new frequency or ATC instructions.

Awareness of FIR Boundaries

It is important to note the FIR boundaries on the OFP as a point where ATC handover can be expected:

- The EETs for all the FIR boundaries are given in the ATC flight plan section of the briefing package.
- FIR boundaries are also displayed in the appropriate positions throughout the OFP. Using this information, it is a simple task to identify the waypoints corresponding to the FIR boundaries. The time given to each boundary is the elapsed time from take-off. An example is shown below on a HHN-LPA OFP where MADRID (**LECM0133**) FIR boundary is expected 1hr 33m after take-off.

```
-N0367F220 DIK1Y DIK UN857 TOLVU/N0370F230 UN857 RAPOR/N0436F330
UN857 GIMER/N0445F350 UZ912 PON/N0450F370 UN872 ERIGA UZ218 BATA
DCT BABEX DCT DIGAL DCT MAGUM DCT ATECA DCT BAROK UN873
SAMAR
-GCLP0420 GCFV
-PBN/A1B1CID1Q1 COM/TCAS DOF/130326 REG/EIEMA EET/EBUR0011
LFFF0017 LECM0133 LPPC0200 GMMM0256 GCCC0352 OPR/RVANAIR RVR/200
```

- These boundaries will be identified and underlined on the OFP in anticipation of an ATC frequency change.
- For any direct routings the only abeam points required are those relating to these FIR boundaries. In the absence of OFP waypoints during an extended DCT routing an appropriate FIR boundary waypoint shall be created in the FMC. The lat/long coordinates for the boundary can be found on the OFP. A user waypoint and abeam point where applicable shall be created in the FMC fix page using this lat/log. Such waypoints should be named FIR1, FIR2 etc.
- Crews shall anticipate a frequency change prior to reaching an FIR boundary. If none is forthcoming then a radio check with ATC shall be initiated. For extended routings between FIR boundaries crews shall (in the absence of any other RT instructions) ensure that radio contact is maintained by performing

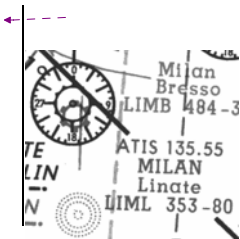
a radio check at intervals not exceeding 25 minutes and corresponding to a point identified along the route and highlighted on the ND from the FMC FIX page.

In summary, establish where the boundaries are, enter them sequentially as a FIX in the FMC, exercise R/T and CPDLC vigilance when approaching, and monitor 121.50 at all times in the cruise.

In addition to the MORA/altitude information contained on the flight plan, the low level Navigation Chart and/or airport area chart giving the minimum safe altitude shall be readily available for consultation in case of re- routing or if further amplification on safety heights is needed.

Checking Enroute Weather

During periods of general poor weather, enroute weather should be recorded during the cruise to satisfy the requirement to land at the nearest suitable airport in accordance with stated QRH procedure. Preference will be to get weather for enroute Malta Air/Ryanair Group bases as these are the most likely enroute diversion airports. You may use the ND to pick the airports that you want and then check the ATIS frequency on the E(LO) chart (above the airport name). Spending a lot of time listening to a VOLMET is not recommended. Never try and get weather when monitoring the ATC frequency. Always hand over the radio.



Monitoring 121.5

121.5 shall be monitored above FL100/10,000 ft. This frequency should be active on VHF 2, except when VHF 2 is used for other purposes. After 30 mins over water listening watch on 121.5 MHz is mandatory. Once you have passed FL100/10,000 ft climbing confirm 121.5 selected on VHF 2, with the volume adjusted to hear transmissions on this frequency.

R/T Discipline

Correct ATC terminology is essential. Accurate readback of all clearances, especially when using the terms “Heading” and “Altitude”, will ensure that there is no confusion with ATC and therefore reduce the amount of time spent on the frequency.

Before leaving a frequency it is important to pause in order to allow ATC the chance to correct your frequency readback. On selecting a new frequency it is important to pause before speaking in order not to block ATC or another aircraft transmission.

Remember how we sound on air will greatly influence others impression of our professionalism.

It is not necessary to maintain a listening watch while operating on HF provided the aircraft is Selcal equipped and this facility has been checked out on the ground or on initial contact with the radio station.

Position reports may be omitted under radar control if it is clearly understood from ATC that such reports are not required.

8.3.0.1.12 ATC Airspace and Speed Policy

FL 100 and below	Class A, B, C	Class D, E, F, G
Climb	250, faster if released by ATC	250 max
Descent	250 max, decline ATC offers of faster	250 max, no ATC discretion for faster

In Class A, B and C airspace, ATC have the discretion to cancel speed control. Where ATC advise “no speed restriction” or “high speed approved” in the climb the aircraft may, at the Captain’s discretion, be operated at the ECON climb speed.

Speed control instructions remain in effect unless explicitly cancelled or amended by the air traffic controller. Crew should comply with published SID and STAR speed restrictions unless the restrictions are explicitly cancelled or amended by the air traffic controller. Cancellation of any speed control instruction does not relieve the flight crew of compliance with the speed limitations associated with airspace classifications.

Some SID and STAR speed restrictions ensure containment within RNAV departure or arrival procedure (e.g., maximum speed associated with a constant radius arc to a fix (RF) leg).

When descending in Class A, B and C airspace, crews are restricted to 250 kt below FL100 unless otherwise instructed. Thus, crews shall decline ATC offers of discretionary high speed below FL 100, and request standard speeds. If a higher speed than 250 kt below FL100 is instructed by ATC whilst in A, B, or C airspace, crews shall fly this speed instructed until reaching FL60 or 6,000' AAL, whichever occurs first, where crews shall advise ATC, and reduce to standard SERA speeds.

The mandatory SERA speed restriction of 250 kt when operating in Class D, E, F and G airspace below FL100 remains unchanged. This restriction is to provide an acceptable level of safety when operating in airspace that is shared with VFR traffic. Unless instructed otherwise by ATC the 250 kt limit must be strictly observed. Consider reducing to 220Kts or minimum manoeuvre speed to improve ability to "See & Avoid" in airspace with significant presence of VFR traffic.

It is Malta Air policy to accept the highest standard radar service available while, regardless of airspace classification.

As OMC enroute charts do not always depict airspace classification/structure, refer to Malta Air Airfield Briefings for associated airspace and TEM guidance for airports where VFR traffic prevails.

Most importantly maintain lookout particularly in Class E, F and G airspace. At airports with dedicated glider sectors, crews must be alert to VFR activity which can take place inside or outside of charted areas. Utilize TEM to anticipate VFR traffic probability, presence, and intensity (i.e., consider Daylight VMC, higher cloud bases, airstrips in the vicinity, time of the day, etc.).

When VFR traffic conflicts with published or ATC directed routings, proactive TEM and coordination with ATC is essential. If necessary, consider delaying takeoff if uncontrolled VFR traffic impedes or interferes with departure routings transiting E, F, G airspace. When airborne, proactively request avoidance vectors if available, take avoiding action and advice advise ATC or enter a published holding pattern until well-clear of VFR traffic.

Visual approaches or departure/arrival runway changes are prohibited when glider/VFR activity is promulgated by ATIS/ATC unless doing so clearly reduces exposure to VFR traffic (e.g., enable aircraft to operate with a higher standard of radar service, airspace classification, etc.) or is otherwise explicitly specified in the Airfield Briefing.

It is good airmanship to take avoiding action (heading change) whenever radar provides conflicting traffic information, thereby ensuring that the conflict does not develop to a point where a TCAS warning is received.

Malta Air aircraft are fitted with Enhanced Mode S transponders and ATC can view:

- the flight number,
- current heading, altitude and speed selected on the MCP,
- the current V/S

It is considered best practice to request "standard speeds" from ATC when higher speeds are offered. These speeds are:

- 220 kt at IAF
- 180 kt on base
- 180 kt on Final approach track
- 160 kt usually from 8 nm to 4 nm

8.3.0.2 Climb, Cruise and Descent

8.3.0.2.1 Terrain Clearance

Particular care must be exercised when operating out of or into airports, close to mountains and other significant terrain:

1. Take-off Briefing and Descent and Approach Briefing shall identify terrain hazards and the minimum safe altitude.
2. After take-off when any emergency or compulsory turn has been completed the best gradient climb speed shall be flown until the safety altitude is achieved;
3. Direct To clearances shall not be accepted until at or above the current MSA.
4. In the CLIMB, only an unconditional "DIR TO" clearance issued by ATC may be accepted as ATC is responsible for terrain/obstacle clearance. When other conditions are attached to the "DIR TO", eg "with terrain in sight", pilots shall reject these clearances until at or above the MSA during climb. These issues do not arise in the descent.
5. "Radar Vectors" during departure, at or below the MSA, in compliance with 8.3.0.2 may be accepted. A commander may accept an ATC clearance to deviate from a published departure or arrival route, provided obstacle clearance criteria are observed and full account is taken of the operating conditions.

In addition to the safety height/altitude information contained on the flight plan, the low level Navigation Chart and/or airport area chart giving the minimum safe altitude shall be readily available for consultation in case of re-routing or if further amplification on safety heights is needed.

All altitudes require cold weather compensation correction when not under radar control, when self positioning, or when levelling off after an emergency descent.

Crews shall respond to EGPWS as specified in the QRH. Flight Crew members must not descend below minimum safe altitudes for the route or the sector at any time until they have positively identified and cross checked their position.

8.3.0.2.2 **Climb Phase**

The climb speed shall be limited to 250 kt below FL 100 as sound operating practice in class D, E, F and G airspace. ECON climb speeds are approved below FL 100 with ATC authorisation in A, B and C airspace.

Flight at ECON speed is Malta Air policy above FL 100. It is recognised that variations from ECON will be required and these are approved.

Changes to airspeed required by turbulence shall be notified to ATC immediately.

8.3.0.2.3 **BAT**

ATC has the altitude readout from mode S. ATC in some locations has the ability to identify incorrect altimeter setting (QNH) from this data. When this is detected ATC will reissue the QNH and ask crews to confirm given QNH is actually set.

What is BAT?

- BAT = Barometric Pressure Setting (BPS) Advisory Tool.
- Tool to identify significant QNH setting errors.

What does it do?

- Provide an advisory warning on the radar to London Terminal Control Approach controllers where there is a significant difference (greater than 5 mb) from the London QNH.
- It only supports arrival aircraft for aircraft at or below the transition level.

What does it not do?

- Alert on departure aircraft.
- Alert above the transition level (e.g. aircraft entering a terminal hold).

Is there any new phraseology?

- Where controllers choose to query the discrepancy, the phraseology to be used is:

“(Callsign), Check Altimeter Setting, QNH xxx” – where xxx is the local QNH being utilised.

- Where sufficient time exists, the use of the following phraseology may be used to query the Altimeter Setting used by the aircrew:

“(Callsign) Report QNH”

- This will enable flight crew to check selected pressure setting, irrespective of any BAT Advisory or QNH data displayed in the Mode S Data Window.

8.3.0.2.4 Altitude Awareness

Altitude awareness is achieved by the active attention of the Flight Crew. This awareness with associated continuous monitoring is the primary means of ensuring that cleared altitudes are adhered to and clearances are not infringed.

The standard operating procedure mandates altitude checks which confirm altitude awareness, and mandate vertical speeds to be flown to reduce closure rates. These policies and standard calls are presented in FCOM Vol 1, Normal procedures.

Climbing:

It is mandatory to select VNAV after all MCP Altitude/Flight Level changes during Climb. Below FL300 is prohibited to select V/S unless the ROC approaching the cleared Altitude/Flight Level exceeds the following limits.

- At 3000 ft to level off if ROC exceeds 3000 ft per min, select V/S and reduce to a maximum of 3000 ft per min.
- At 2000 ft to level off if ROC exceeds 2000 ft per min, select V/S and reduce to a maximum of 2000 ft per min.
- At 1000 ft to level off if ROC exceeds 1000 ft per min, select V/S and reduce to a maximum of 1000 ft per min.

Only select V/S when VNAV is providing a sustained climb rate in excess of 3000 ft, 2000 ft or 1000 ft per minute.

Above FL300 crew should remain in VNAV to avail of the system protections that include stall protection, minimum/maximum operating speed limits, thrust limits, and manoeuvre margins at high altitudes.

Above FL300 if sustained rates of climb in excess of 1500 ft per minute are encountered when approaching 1000' to the cleared level, both PF and PM must actively monitor aircraft performance and traffic display to prevent avoidable TCAS RAs. Where traffic is a consideration, adjust V/S as required and revert to VNAV as soon as traffic is no longer a potential conflict.

Aircraft performance must be closely monitored when using V/S, especially at high altitudes. Minimal and sudden environmental changes such as temperature and/or wind can severely deteriorate aircraft performance. Requests/Restrictions from ATC shall where possible be managed through VNAV.

Descending:

The preferred descent mode is VNAV. Other descent modes may be utilized to accommodate CDA's, level restrictions, ATC restrictions and requests. The ROD must be reduced unless a restriction is set by ATC, whenever the ROD approaching the cleared Flight Level/Altitude exceeds the following limits:

- At 3000 ft to level off if ROD exceeds 3000 ft per min, reduce to a maximum of 3000 ft per min.
- At 2000 ft to level off if ROD exceeds 2000 ft per min, reduce to a maximum of 2000 ft per min.
- At 1000 ft to level off if ROD exceeds 1000 ft per min, reduce to a maximum of 1000 ft per min.

Step descents are often a reflection of potential traffic below. ATC stepped-descents present a threat of avoidable TCAS RAs if the ROD is not effectively managed. If ATC instruct a stepped-descent (altitude changes of 2000 ft or less), crew should initially select V/S to manage these requirements.

- At 2000 ft to level off, select a maximum of 2000 ft per min.
- At 1000 ft to level off, select a maximum of 1000 ft per min.

Both PF and PM must actively monitor aircraft performance when utilising V/S, particularly at high altitudes. Do not engage VNAV or LVL CH when performing stepped-descents.

The altitude alert system should only be considered as a back-up aid. Its operation must not be permitted to replace the requirement to make the standard altitude calls.

The Malta Air Standard Operating Procedure in this area is presented in FCOM Vol 1, Normal procedures.

Incorrect setting of the QNH in the descent is an insidious mistake frequently made. An altitude crosscheck on final approach path is an essential crew action to detect this.

8.3.0.2.5 Systems Monitoring

Flight and navigation instruments and engine indications will be monitored on a routine on-going basis, but additionally the proper functioning of aircraft systems will be monitored regularly throughout the flight.

Whereas the aircraft checklists will provide a reminder for setting up the various systems, the orientation checks should be used as a trigger to periodically scan those panels and systems outside the regular forward instrument panel scan and not tied in to a central warning system. Typical items requiring re-selection or adjustment might include landing lights at FL 100, seat belt sign, anti-icing systems, and air conditioning systems etc.

8.3.0.2.6 Cruise

ECON cruise speed will be as filed on the operational flight plan.

Malta Air company policy is to operate at cost index or fixed speed/Mach as indicated on the operational flight plan. Crews shall operate at ECON speeds generated by the FMC. The only exception to this is for diversions where the route to alternate is always calculated at, and shall always be flown using LRC.

The flight levels on the OFP are optimum flight levels as determined by the LIDO flight planning optimization tool. These flight levels shall be flown as follows:

- When the flight plan cruise altitude is lower than the FMC OPT altitude crews shall fly the flight plan cruise altitude (See Note below).

- When the flight plan cruise altitude is higher than the FMC OPT altitude crews shall fly the FMC OPT altitude.

Note: Some flights may be planned at a lower level due to route restrictions. If the operational Impacts section of the OFP indicates a fuel saving at a higher level, a climb should be considered if deemed appropriate by the commander.

OPERATIONAL IMPACTS					
FL CHANGE	2000 ABOVE	TRIP	M 0031 KGS	TIME	M 0001
FL CHANGE	2000 BELOW	TRIP	P 0067 KGS	TIME	P 0002
WEIGHT CHANGE	UP 1.0	TRIP	P 0028 KGS	TIME	M 0000

Crew should reference the block wind information at the end of the OFP to determine if it is appropriate to climb to levels not displayed in the Operational Impacts section.

WIND INFORMATION							
CLIMB		T O C		LNO		PODAT	
350 035/011	-59	230 057/013	-27	230 060/013	-27	230 056/012	-27
310 044/009	-49	210 064/014	-23	210 066/014	-23	210 062/013	-24
200 068/015	-21	190 072/015	-19	190 072/014	-19	190 071/013	-20
150 084/015	-10	170 079/015	-15	170 079/015	-15	170 079/014	-15
100 085/017	-00	150 084/015	-10	150 086/015	-10	150 088/015	-10

However, the aircraft may be operated safely up to and including maximum altitude indicated in the FMC. Crews shall carefully consider the following prior to electing to operate above the FMC optimum altitude:

- Identify enroute areas of moderate turbulence and carefully consider reducing operating altitudes accordingly.
- Review the shear indications on the OFP.
- Avoid flight in severe turbulence (greater than 1.5 G).
- Not select wing anti-ice on with the SAT less than -40 degrees C (including when ice is visible on wipers and L1 and R1 windows).
- Select 10 degree bank angle on the MCP passing FL300 climbing (to provide over-bank protection in HDG SEL) and 25 degrees passing FL300 descending.
- For operations other than LNAV, fly at least 10 knots above the minimum manoeuvre speed (lower amber band).
- If speed drops below the lower amber band, immediately increase speed by reducing bank and/or increase thrust to max continuous (N1 page FMC) and/or descend.

Operations above the FMC MAX ALT are not permitted. The flight plan flight levels are calculated using 5% MAC. This is the most restrictive C of G position for generating cruising levels. Crews shall use a combination of the Operational Impacts information on the OFP and FMC information if a cruising FL above FMC MAX ALT is generated by the flight plan and a lower level is required.

Step climbs and descents shall be requested in accordance with the Operational Flight Plan. For the avoidance of doubt, it is only permitted to fly at speeds greater than ECON CLIMB, CRZ OR DESC in order to operate a Fixed speed/Mach flight plan, for profile adjustment, to comply with an ATC speed request or if directed by MEL, QRH or supplementary procedure for example, turbulence penetration.

The acceptance of any cruise altitude either enroute or during diversion indicates that the crew is aware that obstacle clearance is achieved. Accurate navigation and judicious requests for direct clearances can reduce the nautical airmiles (nam) flown and so reduce flight time and cost. Possible areas of clear air turbulence will be identified during flight preparation to provide prior warning to flight deck and Cabin Crew:

1. Severe CAT if identifiable shall be avoided.
2. When turbulence is experienced or anticipated the passenger seat belt sign shall be switched on. Best practice is that crews advise ATC of hazardous meteorological conditions enroute, which has the effect of notifying other traffic. It may be advisable also to limit cabin service.
3. Speed shall be adjusted at the Commanders discretion, having regard to the expected severity of the turbulence. It is not expected to reduce to the max severe turbulence penetration speed for light turbulence, it is expected to reduce to this speed for severe turbulence. Any speed reduction shall be notified to ATC, and Commanders shall advise ATC as to the continuing efficacy of RVSM in these conditions.
4. Crews shall follow FCOM Vol 1, SP 16 guidance, in-flight, in severe turbulence and mountain wave activity to maintain control of the aircraft.

Weather radar scanning shall be used to identify and avoid Cb build-up and thunderstorm activity. The Commander shall have WXR selected for take-off, climb, cruise and descent/approach in IMC and at night. This does not preclude the First Officer (when PF) from using the weather radar during the cruise to identify and avoid Cumulonimbus build-up and thunderstorm activity. Permission should be requested from ATC to detour around thunderstorm activity, preferably to upwind rather than downwind of the build-up. If the thunderstorm activity appears to be exceptionally severe and there is no apparent

path through or around the storm area/line then consideration should be given to diverting the flight. Crew's primary objective is to conduct our air transport activities safely.

Destination and primary alternate weather

Best practise is to obtain what weather you need to complete the flight, minimising the time the PM is off the operational ATC frequency. This is particularly important when the destination and alternates are just south of the alps. This weather is only available from VHF on line of sight transmission, which is too frequently coincident with ATC descent and arrival clearances. Proper planning should avoid this conflict of both pilots being on different frequencies.

Ideally crews shall avoid getting weather in the descent. It is recognised that occasionally actual destination weather is required in the descent/approach, and this is allowed. This shall be accomplished expeditiously, preferably by listening to ATIS, or if below 5,000 ft and within 25 miles, from the destination tower frequency. The PF shall be particularly vigilant on the live ATC frequency when the PM is getting this weather, and ALL ATC instructions received by the PF shall be briefed to the PM on his return to the ATC frequency.

Recording Weather on the OFP

The Wx pack on report includes up to 9 hr forecasts which complies with the requirements for destination and alternate aerodromes. The weather supplied on turnaround also supports these requirements.

When the crew decides to record weather during flight, this shall be recorded on the OFP ideally in the cruise. Best practise is to obtain what additional weather you need to complete the flight, **minimising the time the PM is off the operational ATC frequency.**

Use Wx information locations, e.g. Zurich, Vienna, to avoid the conflict of receiving weather reports coincident with descent planning or descent activities. Proper planning will minimise the threat of both pilots being on different frequencies.

The weather for the aerodrome of departure and landing shall be recorded on the OFP.

8.3.0.2.6.1 Management of ATC Clearances/MCP with One Pilot on Operational VHF Freq.

Where the PM is off the radio to get weather or operational information the PF will monitor the Operational VHF frequency. Where the PF gets an ATC instruction to change ALT or HDG, the PF will make the MCP change but will not execute the change until the PM is back on frequency and the change of HDG or ALT has been verified with the appropriate ATC centre.

Where ATC requires an immediate change of ALT or HDG the PF will interrupt the PM and point out the change to the MCP. The PM will immediately return to the operational frequency and ask for confirmation of the instruction. "XXX Control, confirm the HDG/ATL for MAY 123".

Where the PF gets an ATC instruction to change ALT or HDG via CPDLC, the PF will Respond WILCO/UNABLE or STANDBY. When the PM is back on frequency PF will brief all new CLOSED instructions on ATC log page.

8.3.0.2.7 Pre-descent Considerations

The top-of-descent point shall be determined taking into account the standard descent distance adjusted for wind component, anticipated ATC routing and possible holding, icing, safety heights, and runway in use. This is computed by the FMC based on routings and constraints entered.

The descent will be conducted in such a way as to achieve fuel economy. This is best achieved by VNAV and ECON speed.

FCOM Vol 1, Normal procedures describes the descent phase, including timings and content of descent and approach briefing. Flight Crew members shall be conscious of terrain under the descent path and aware of safety heights/altitudes along the arrival route and for the approach.

1. Flight Crew members must not descend below the MEA (minimum enroute altitude) or MORA (minimum off route altitude) or the minimum sector altitude (MSA) for the approach facility until they have positively identified and cross checked their position.
2. Flight Crew members shall not descend below the MSA to the minimum procedural altitude until the aircraft has passed over the approach fix or unless specific descent instructions are given in the approach chart except for paragraph (3) below. The extended ILS glidepath is not approved as an en-route descent aid and all aids shall be used only within their certified limits.
3. Where the aircraft is being vectored by radar, the radar controller is responsible for ensuring proper terrain clearance, including in cold weather/temperature compensation conditions. In these circumstances

clearance and descent to the cleared altitude below the MSA is authorised. When clearance for the approach is issued, the aircraft is expected to maintain the last assigned level until intercepting the specified, or nominal glide path of the approach procedure. If radar contact is lost the aircraft shall climb to the MSA without further delay.

4. STARs take the aircraft safely from the enroute phase to the initial approach fix and assure terrain/obstacle clearance. STARs are referenced to ground based nav aids.
5. RNAV stars are not necessarily referenced to ground based nav aids. RNAV descents below the MSA require a RNP1 approval. Malta Air has a PRNAV approval w.e.f. Jun 2007. (Note that self positioning on RNP1 approval must not take the aircraft outside the areas used to design the approach to ensure proper sequencing of the RNP during the descent approach (see OM-C RNP1, ATC Section) Self positioning relieves the radar controller of responsibility for terrain clearance.
6. Additionally accepting “Direct to” may take the aircraft over higher terrain/obstacles than the SH on the OFP indicates. Pilots shall establish that the minimum altitude in these circumstances is higher than the highest altitude in the current MORA and all MORAs enroute to the cleared waypoint/fix.

The Descent and Approach Briefing shall be completed as SOP prior to commencement of the descent, see FCOM Vol 1 for content.

MLW Consideration:

Crews shall establish that the aircraft shall not be above the Max Landing Weight on landing, particularly if it is close to the MLW at descent briefing. It is considered good airmanship to verify MLW will not be exceeded on landing prior to commencement of final approach.

8.3.0.2.8 Descent and Approach Briefing

The descent briefing is presented in FCOM Vol 1, Normal procedures, and the Flight Crew Training Manual provides additional operational guidance. The statement “Standard Calls and Procedures” will apply only when both Flight Crew members are familiar with the procedures for the airfield concerned and when further elaboration is not required.

The Airfield brief if published shall be consulted and briefed prior to top of descent. The airfield brief for an alternate if published shall be reviewed if a diversion is likely prior to top of descent, or as soon as practicable once a diversion is initiated. Airfield briefs are located within Docunet and FDF.

If a diversion is contemplated, the Commander will ensure that the co-pilot is aware of:

1. The nominated alternate airfield, and the expected routing to this alternate;
2. The fuel requirements for diversion bearing in mind air distance, weather and anticipated ATC delay;
3. The descent and approach procedures at the alternate airfield; and
4. If a definite decision has been made to divert after a particular approach, then a specific request will be made to ATC stating that in the event of a missed approach a clearance to the alternate airfield is required. The prior indication given should eliminate any ATC delays in setting course for the alternate.

8.3.0.2.9 Correcting Flight from above the Descent/Approach profile

The only lateral manoeuvring that is acceptable to regain the vertical approach profile is as follows:

- Request entry to a published intermediate holding fix in order to lose altitude in a clearly defined and controlled manner to achieve the platform or step altitude at the correct speed and aircraft configuration.
- Request an "orbit" that will be executed as a racetrack pattern for stability which is subject to the following:
 1. Not less than 10nms remaining to touchdown.
 2. Descent to not below the MSA or 3,000 ft whichever is higher.
 3. Max Flap 10, with speedbrake if operationally necessary.
 4. Met conditions VMC.
- If a visual approach has been requested and approved by ATC that is not straight in then consider extending the downwind or widening the base leg, subject to other traffic.
- If required conduct a full published go-around to another approach.
- It is not acceptable to join a visual circuit from an incomplete go-around.

If there is any doubt about the successful outcome of the approach a go-around must be the prime consideration and is preferred over any other method.

8.3.0.2.10 Orbits

It is only permitted to perform an orbit inside 10 nm from touchdown subject to the following restrictions:

- Requested by ATC for traffic separation

- Under Radar Control
- Not below 1500 ft AGL
- Max Flap 15
- Day VMC

Any other condition requires a go-around.

Malta Air does not permit orbits inside 10 nm from touchdown for the purpose of losing altitude or speed.

8.3.0.3 Approach and Landing

8.3.0.3.1 Landing Conditions

The destination meteorological information shall be obtained and evaluated before commencing an approach. This information together with the runway surface conditions is likely to determine whether an approach can or should be made, the type of approach to be made and the runway in use.

In all cases one should always plan and coordinate to utilize the lowest available aerodrome approach minima in order to avoid any unnecessary go arounds due to deteriorating or marginal weather conditions. E.g. an ILS CAT II/III should be requested when marginal CAT I conditions exists provided it is readily available.

Before commencing an approach to land, the commander shall adhere to the requirements of Section 8.1.3.2.

Crosswind component and limits shall be taken into account as it is noted that cross winds approaching limiting values can present significant handling challenges. In these circumstances it is prudent for the aircraft commander to act as pilot flying.

The weather minima to be used for the different types of approaches are displayed on the Approach Chart. The effect of downgraded equipment and meteorological terminology with regard to the determination and application of Airfield Operating Minima is presented in this manual.

In adverse weather conditions, and for landing on contaminated runways; see FCOM Vol 1, and QRH for procedures. The Flight Crew Training Manual provides additional guidance for descent, approach and landing techniques, including explanation of the consequences of errors.

An aircraft shall avoid making an approach if a thunderstorm is active over the airfield or there are cells on the final approach centre line within 3 nautical miles of the field. Crews shall be particularly aware of windshear conditions, and shall review the windshear escape manoeuvre if such conditions exist.

8.3.0.3.1 Landing Lights and Night Time Operations

CAT I ILS/NPA – (Manual Landing) Landing Lights – ON

CAT II/III – Landing Lights – Commander's Discretion

8.3.0.3.2 Spurious Predictive Windshear (PWS) Alerts

The Honeywell PWS alert system as fitted to the Boeing 737 fleet is prone to spurious Predictive Windshear alerts (PWS) in certain atmospheric conditions. PWS alerts are most likely to occur in calm conditions where the approach path is over an area of open water. Certain airports in the route network are known to suffer from this phenomenon. When so briefed, an approach may be continued at the discretion of the Commander after a spurious Predictive Windshear alert. The conditions required to make this judgement are no weather conditions existing or forecast that are associated with windshear.

Predictive Windshear Alerts are identified by the following Aural Alerts “WINDSHEAR AHEAD, GO AROUND WINDSHEAR AHEAD or MONITOR RADAR DISPLAY” and should not be confused by the Two-tone siren followed by “WINDSHEAR” which is indicative of a Reactive Windshear Warning.

A ASR shall be filed in all cases of activation of PWS, spurious or otherwise.

8.3.0.3.3 Conventional Approach (Non CAT II or III) Monitored Approach (MA) Policy

A conventional approach implies that either trained Flight Crew member may perform any type of approach and land at the Commander's discretion. Profiles for such approaches are presented in FCOM Vol 1, Normal procedures, and profiles for non normal approaches are presented in the QRH. Further limitations are specified for inexperienced co-pilots. When available all descents and approaches shall be conducted in LNAV/VNAV.

Monitored Approach (MA)

Malta Air procedure for MA is presented in FCOM Vol 1, QRH and is applicable to Precision Approaches and Non Precision Approaches (NPA) including LNAV/VNAV when the visibility is reported as less than specified minima. A Monitored Approach must be carried out when the weather conditions are below the following limits:

Approach	Minima: vis/ceiling	Exceptions
Type B (3D DH < 250 ft, >= 200 ft)	<= 1,000 m, 300 ft AAL	Nil
Type A (2D or 3D DH >= 250 ft)	<= 3,000 m, 1,000 ft AAL	Circling approach Visual approach (no published procedure)

Nothing prevents a crew conducting a Monitored Approach in better weather conditions.

Inexperienced co-pilots may fly a Monitored Approach (including when the crosswind is in excess of 15 kt) as the landing will be flown by the Captain. The Captain shall call ‘continue’ at or before minima when he has the required visual references for landing.

Competence – Conduct of Approaches

The conduct of approaches and the areas of responsibility for each crew member, the standard operating procedure, and the approach profiles are presented in FCOM Vol 1, Normal procedures.

Go-Around

The conduct of the GO-Around and the areas of responsibility for each crew member, the standard operating procedure, and the Go-Around profiles are presented in FCOM Vol 1, Normal procedures, and QRH for non normal Go-around procedures. Additional guidance is provided in the Boeing Flight Crew Training Manual.

8.3.0.3.4 Stabilised Approach

Stabilised Approach Procedure

The objective of a stabilised approach is to cross the runway threshold at a safe altitude, aligned with the runway centerline, with the aircraft in trim and thrust at appropriate approach settings, to arrive at the flare point at the required airspeed, and to touchdown as close as possible to the aiming point (1,000 feet for jet aircraft). To achieve this with consistency in various conditions of wind and weather requires:

- 1. Timely deceleration to final approach speed, adjusted for wind, matched with configuration change so that the landing configuration is achieved at 500 ft above airfield level (AAL) in VMC and 1,000 ft AAL in IMC.
- 2. Adherence to a constant angle of descent/flight path angle, e.g. the ILS glide slope, a visually assessed path, VNAV path on non-ILS/GLS approaches with the current V/S proportional to the aircraft current ground speed.

Crews are reminded that any aircraft that proceeds beyond the 500 ft VMC Landing Gate or 1000 ft IMC Landing Gate without being stabilised (parameters below) is required to submit an ASR detailing the circumstances behind the decision to continue the approach.

Stabilisation Approach Criteria

The following stabilised approach criteria must be achieved by the respective landing gate, and maintained until touchdown.

1. Target Approach speed VREF+5 to VREF+15 (minimum speed VREF, maximum speed VREF+20).
2. On the correct vertical and lateral path, where only small changes in heading and pitch are required to maintain the correct flight path. FCOM NP refer.

Note 1: Lateral manoeuvring to align with the runway extended centreline must be completed by 300 ft AAL during all Circling or Side-Step Approaches.

Note 2: Exceptions to this policy, where lateral manoeuvring below 300 ft AAL is permitted, shall be detailed in the relevant OMC AFB (CAT C aerodromes).

3. Appropriate thrust set.
4. Vertical speed proportional to the approach design and ground speed, but not more than 1000 fpm, unless briefed prior to the approach.
5. Landing Checklist complete (except landing lights).

8.3.0.3.4.1 CDA/LDA

Continuous descent approach (CDA) and Lo Drag approach (LDA) is considered best practise for CAT I ILS approach.

Additional guidance is provided in the Boeing Flight Crew Training Manual.

Coupled with the control of the aircraft descent profile on final approach is the need to maintain aircraft tracking along the aircraft centreline to:

1. permit easier recognition and handling of windshear;
2. assess the crosswind;
3. avoid large lateral deviation corrections necessitating excessive bank angles close to the runway; and

4. facilitate touchdown on the runway centreline and ensure that the aircraft is tracking the runway centreline after touchdown.

These criteria apply to both instrument and visual approaches even though the cues used for assessment and judgement differ. The stabilised approach facilitates the transfer from instrument to visual conditions, in which:

1. care is required to avoid “duck-under” after breakout into visual conditions at low altitude following an instrument approach;
2. care is also required to maintain the proper descent profile in conditions of decreasing visibility, and with the decreasing visibility of shallow fog, it is particularly important that the visual approach is fully stabilised before entering the fog layer;
3. Approach profile and path shall always be monitored by reference to all the nav aids, VNAV path and visual aids available. In addition, the PM must specifically direct attention to speed, rate-of-descent and glidepath throughout the final descent.

Utilising the Stabilised Approach concept with adherence to electronic and visual glide slope indicators, 1,000 ft visual runway aiming points, constant angle non-precision approaches, and “duck-under” avoidance constitute the essential operational procedures designed to ensure that an aircraft crosses the threshold by a safe margin, with the aircraft in the landing configuration and attitude. FCOM Vol 1 refers.

8.3.0.3.5 Visual Approaches

1. A visual approach may be carried out or continued only if it is clearly determined that the meteorological conditions are adequate for the approach, the aircraft will remain in visual contact with the ground and that the aircraft can be positioned for a landing by visual reference to the runway, and most importantly after prior double briefing of the procedure.
2. Verification of aircraft position before commencing a visual approach is particularly important and shall be cross checked by all available means. Radio aids will be used for this purpose and will in any case be selected, tuned and identified as for an instrument approach onto the landing runway and/or to facilitate monitoring of the approach when available.
3. A visual approach is a non-precision approach (NPA) by Malta Air standards and is therefore subject to the Double Brief. A crew may not elect to fly a visual approach without completing the Double Brief. The briefing must include the position of all configuration changes and autopilot/FD handling for the approach and the complete go-around manoeuvre. The crew must be alert for airfields, disused runways or roads that may be mistaken for the runway. Use the map display to back up the visual picture.

A visual approach may only be flown in conditions where constant contact with the runway and its environment can be maintained throughout the manoeuvre

4. Aircraft shall not fly below glidepath, or follow a descent path lower than the approach path normally indicated by the glidepath/PAPI.
5. The latest position to select Flap 1 on a visual approach is 10 nm from touchdown. Additional configuration changes must be made to allow the aircraft to be fully stable (as defined in FCOM Volume 1) with the landing checklist complete to the landing clearance by 500 ft AAL. Minimums therefore for Visual Approaches are 500 ft AAL.
6. Autopilot limitations (Boeing FCOM) must be considered when flying a visual approach without a LOC or GS. Crews should avoid the temptation to immediately disconnect all automatics in order to fly the approach and passenger comfort must not be compromised in the execution of the manoeuvre.
7. Visual approaches are required to utilise the stabilised approach technique and follow a constant angle electronic or VASI/PAPI glide path.
8. When manoeuvring for a visual approach, care must be taken to avoid and remain well clear of all terrain and obstruction.
9. This means that pilots shall not conduct self manoeuvring/positioning below the MSA at night. Pilots can conduct a visual landing at night only from a published approach procedure.

Where a night time circling approach is necessary the initial approach shall be conducted as an instrument approach with a break at MDA to a circling approach. Night landings from a circling approach require a PAPI/VASI

10. Night landings from a NPA to a runway less than 2000 m LDA without PAPI/VASI require NPFO approval.
11. ATC must be informed and have approved the visual manoeuvre prior to executing it.
12. The go-around is potentially the most difficult manoeuvre to execute from a visual approach, however, a well briefed and thought out manoeuvre that includes touch drilling the go-around is essential. FD's shall be used when available and the PF must push TOGA to activate GA mode if a go-around is required below the GA Alt.
13. In the event that a go-around is required the published missed approach procedure must be flown either to completion, or to level flight under radar vectors and after take-off checks completed.
14. Malta Air shall not use an RVR of less than 800 m for a visual approach.

15. Visual approaches shall be conducted by Commanders and experienced co-pilots (>500 hrs on type in Ryanair Group and 1500 flight hours on jet aircraft with a MTOW of greater than 20T).

8.3.0.3.5.1 Continuous Descent Approach (CDA)

The purpose of the Continuous Descent Approach is to reduce the environmental impact of the aircraft while improving passenger comfort and reducing overall fuel burn.

Crews should avoid any level segment in the flight path prior to Final Approach Point intercept (i.e. no ALT HOLD). Vertical speed should be adjusted to intercept the final approach path as early as possible during the initial approach phase.

When CDA procedure is published crews may use any V/S to achieve it, including V/S <500 fpm.

8.3.0.3.5.2 Low Drag Approach (LDA)

The purpose of the Low Drag Approach is to reduce the environmental impact of the aircraft while improving passenger comfort and reducing overall fuel burn.

Subject to ATC, crews will plan for a LDA.

Plan to fly the aircraft to intercept the final approach path at 180 kt, and flaps as required. Thereafter the aircraft is flown in the gear-up configuration to 4 nm (Precision approach) and 5/4 nm (NPA) from touchdown. Passing 4 nm or (5/4 nm NPA) but not later than 3.5 nm the PF should call for gear down, F15 and landing checklist and commence re-configuration for landing. The aircraft must be fully configured and stabilised by 500 ft AAL VMC or 1,000 ft AAL IMC.

8.3.0.3.5.3 Low drag Approach (Non-precision Approach)

Low drag non-precision approaches are not normally applicable to V/S NPA.

8.3.0.3.5.4 Low Drag Approach (Visual Approach)

Straight in arrivals with the runway in sight are approved for low drag technique.

8.3.0.3.5.5 Sidestep Approach

The criteria set out shall be observed unless associated AFB specifies otherwise. Collision avoidance, terrain and obstacle clearance is the responsibility of the aircrew upon commencement of the sidestep.

A sidestep manoeuvre can be commenced providing:

1. The sidestep has been requested by ATC and Missed Approach Instructions are confirmed,

2. The landing runway threshold is equal to or further away than the threshold of the instrument approach runway being flown,
3. Daylight VMC,
4. PAPI in sight,
5. Visual with the landing runway,
6. The manoeuvre must be commenced prior to 1000 feet above landing altitude indicated (Landing Altitude Reference Bar),
7. F30/40 selected prior to commencement of the manoeuvre,
8. The stabilised approach criteria can be met,
9. Landing performance must be calculated in advance for the most limiting runway.

8.3.0.3.6 Visual Company Coded RNV V

The availability of a VISUAL COMPANY CODED CIRCLING RNV V, at suitable Airports/Runways, will be detailed in the Airfield Brief (AFB).

The Performance Department will generate Circling (visual) profiles and waypoints for VISUAL COMPANY CODED CIRCLING RNV V to ensure compliance with the following standards in accordance with standard Company Circling SOPs:

1. The VISUAL COMPANY CODED CIRCLING RNV V will remain within the PANS-OPS 4.2 nm radius protected area.
2. All waypoints shall be based on the WGS84 standard and accurate to 0.01 seconds.
3. All tracks shall be Great Circle tracks.
4. Conventional IAPs only will be used to CIRCLING MDA.
5. Company generated waypoints are uniquely designated as XXXXZ1 through to XXXXZn, where XXX comprises the IATA aerodrome designator.
6. Waypoints defining the procedure are published on the associated VISUAL COMPANY CODED CIRCLING RNV V RWY XX chart.

Each procedure design achieves the following:

- The first waypoint in the procedure is at the start of the downwind leg,
- An intermediate waypoint is provided along the downwind leg,
- The base leg is coded as an RF turn terminating to coincide with the Runway Extended Centerline and PAPI angle,

- RF base turns are designed at 20° AOB and 168 kt,
- Descent will commence from a point representing the intersection of the final approach descent angle and the circling MDA which may be on the base turn RF leg to assure containment within 4.2 nm protected area,
- Descent gradient will be identical to the PAPI angle,
- Initial MAP will be coded straight ahead to CIRCLING MDA,
- MAP is a straight climb to CIRCLING MDA. Crew will follow the original published procedure in conventional navigation.

VISUAL COMPANY CODED CIRCLING RNV V will be applied only to suitable Airports/Runways where a circling MDA is published. Coded circling procedures are flown visually, with AFDS engaged.

Procedure Simulator Check

VISUAL COMPANY CODED CIRCLING RNV V has been extensively tested in a company level D simulator to ensure efficacy and safety of the procedure.

(AMC7 CAT.OP.MPA.110.)

‘Circling’ means the visual phase of an instrument approach to bring an aircraft into position for landing on a runway that is not suitably located for a straight-in approach.

Instrument Approach Minima

The minima for the instrument approach preceding the VISUAL COMPANY CODED CIRCLING RNV V is the published circling altitude (height) and the meteorological visibility. The associated visibility required for any instrument approach procedure is determined from Table 9 in OMA 8.1.3.4.5 and dependent on the approach lighting available to the landing runway.

FCOM

FCOM Vol 1, Normal procedures presents the flight profile and duties of each crew member during circling approaches. The QRH presents the flight profile and duties of each crew member for normal and non-normal circling procedures.

Landing Gate

The landing gate for a circling approach/sidestep is 500 ft AAL. Lateral manoeuvring to align with the runway extended centreline must be completed by 300 ft AAL during all Circling or Side-Step Approaches. The aircraft must be configured in the landing configuration by 500 ft AAL.

Exceptions to this policy, where lateral manoeuvring below 300 ft AAL is permitted, shall be detailed in the relevant OMC AFB (CAT C aerodromes).

Night Circling

A VISUAL COMPANY CODED CIRCLING RNV V will not be flown at night unless VASI/PAPI or electronic glideslope is available to the landing runway.

Approach Chart and FMC descriptors VISUAL COMPANY CODED CIRCLING RNV V Approach Procedure.

VISUAL COMPANY CODED CIRCLING RNV V charts are provided for approved aerodromes.

VISUAL COMPANY CODED CIRCLING RNV V

R is the nomenclature required by ARINC 424 for coding procedures in FMCs regardless of FMC manufacturer and represents the letter for Area Navigation (RNAV) approach. As the B737 fleet is certified for RNP 0.3 operations the navigation specifications for RNP 0.3 are applied for all visual coded circling procedures. These procedures are listed as RNV in the FMC.

Visual Company Coded Circling RNV V

The VISUAL COMPANY CODED CIRCLING indicates that this is a Visual, Company proprietary, (Flight Operations Performance Engineering Department) coded circling procedure produced by Jeppesen.

VISUAL COMPANY CODED CIRCLING RNV V RWY XX

XX is the runway identifier.

Navigation Specification and Temperature Limits

The navigation specification required is indicated on the chart and company aircraft are certified to Navigation Specification RNP 0.3. Each coded circling procedure has a base turn which is a Radius-to-fix (RF) leg, also relying on RNP 0.3 navigation specification. Upper and lower temperature limits as applicable are indicated on the chart available in the EFB.

RNAV Approach Limitations

Company Coded Circling procedures are subject to the Double Brief limitations in QRH NC.12.

AFDS

The AFDS is used during the visual coded circling procedure. LNAV provides lateral navigation to RNP 0.3 navigation specification. RF Legs are also to RNP 0.3 and executed visually. VNAV provides vertical guidance to within 75 ft above or below the FMC computed descent path and may remain engaged in vertical mode to MUH during the execution of the procedure visually.

Waypoint Integrity

Waypoints published on VISUAL COMPANY CODED CIRCLING RNV V charts have the equivalent integrity as waypoints created in any procedure design and comply with ANNEX VII PART-DAT (DAT.OR.100) thereby providing an equivalent level of safety. When cleared for the circling approach by ATC and a VISUAL COMPANY CODED CIRCLING RNV V is published and detailed in the AFB, crews are expected to execute the VISUAL COMPANY CODED CIRCLING RNV V procedure visually with AFDS engaged until continuing to land or executing a MAP.

Air Traffic Service (ATS)

ATS will have been briefed by the company on the design and advantages of these procedures in locations where these procedures exist including the tracks and corresponding location of the aircraft during these procedures.

VISUAL COMPANY CODED CIRCLING RNV V procedures reduce crew workload, increase AFDS usage, promote stabilised approaches and are the preferred method over conventional circling where only 2D AFDS guidance is available. These approaches are utilised when the aircraft cannot be flown on a straight course from the approach aid to the runway in use and visual manoeuvring is required.

The VISUAL COMPANY CODED CIRCLING RNV V commences after the IFR approach procedure where the final approach track intercepts the circling MDA. This is the point on the instrument procedure where the pilot must decide if the meteorological conditions are suitable to continue visually following the VISUAL COMPANY CODED CIRCLING RNV V procedure with sufficient visual references, otherwise the missed approach of the instrument procedure will be flown.

Visual contact with the runway environment is required during the VISUAL COMPANY CODED CIRCLING RNV V and the more stringent requirement of maintaining visibility with the landing runway threshold applies to permit descent below circling altitude. Once required visual contact has been established to continue circling, the VISUAL COMPANY CODED CIRCLING RNV V should be activated. Downwind waypoints are provided followed by a Radius to Fix (RF) leg to a point on final approach to the landing runway coincident with the descent angle. The VISUAL COMPANY CODED CIRCLING RNV V is flown visually with AFDS engaged in LNAV/VNAV. Descent point from MDA is calculated by the FMC and is coded to the landing runway threshold.

It is not permitted to route direct to a RF Leg waypoint. It is permitted to route direct to any waypoint preceding the first RF Leg waypoint. Aircraft configuration must be GEAR DOWN, FLAPS 15 at this waypoint.

Missed Approach Procedure

Standard Circling missed approach procedures apply.

This may require Gear Down, Flap 15 with associated gear warning.

The missed approach procedure is potentially a high workload, critical phase of the procedure, and must be fully briefed.

The Missed Approach Procedure can be split into three categories:

1. Missed approach on final to instrument approach runway.
 - a. Ensure MAA is set on the MCP before pressing TOGA.
 - b. LNAV is available and the FDs can be followed.
 - c. Follow the published missed approach procedure.

Once on crosswind with LNAV, VNAV PTH, SPD INT engaged the PM will select DEP/ARR, and re-select the original instrument approach procedure. This enables the utilisation of LNAV for the missed approach. Do not execute. This is a contingency plan for in the event of a missed approach.
2. Missed approach on crosswind or downwind with LNAV, VNAV PTH, SPD INT engaged:
 - a. Turn towards the runway in HDG SEL.
 - b. Intercept the missed approach procedure of the initial IAP in HDG SEL.
 - c. When time permits the PM will select the appropriate missed approach waypoint to L1 in LEGS and execute. Care must be taken that the turn direction is correct when executing 'direct-to' waypoints. LNAV is now available to provide guidance for the missed approach procedure of the initial IAP.
3. If a missed approach is required from final:
 - a. Climb ahead to circling MDA in HDG SEL.
 - b. Commence a turn on the circling side leaving flaps at 15.
 - c. Intercept the standard circling missed approach in HDG SEL.

- d. When time permits the PM will select the appropriate missed approach waypoint to L1 in LEGS and execute. Care must be taken that the turn direction is correct when executing 'direct-to' waypoints. LNAV is now available to provide guidance for the missed approach procedure of the initial IAP.

The missed approach procedure is potentially a high workload, critical phase of the procedure, and must be fully briefed.

Data Capture

The Commander shall indicate on the EFL when a VISUAL COMPANY CODED CIRCLING RNV V procedure has been used, indicating the success or failure of the Visual approach, providing relevant feedback and comments and including full details of failure causes.

The Performance Departments shall collate and assess the EFL data.

8.3.0.3.7 Circling Approach

After an IMC-letdown, a circling approach must be made when the aircraft cannot be flown on a straight course from the approach aid to the runway- in-use and the Flight Crew member must use visual guidance to position the aircraft on final approach. Such approaches require an accurate approach pattern and utmost attention:

1. At the circling minimum altitude a terrain clearance of at least 300 ft above the highest spot elevation within 4 nm of the runway system of the aerodrome is provided;
2. The PF shall, after having contact with the runway, proceed at an altitude not lower than the applicable circling minimum. This altitude must be maintained, keeping visual contact with the runway, until the final approach path is being intercepted on base leg or final. During descent, a normal safe final approach path shall be followed taking into account the spot-elevations in the approach path; and
3. A left downwind/base leg shall be flown unless cleared otherwise or when stipulated in the Terminal chart.

4. A circling approach will not be flown at night unless VASI/PAPI or electronic glideslope is available on the landing runway.

FCOM Vol 1, Normal procedures presents the flight profile and duties of each crew member during circling approaches. Additional guidance is provided in the Boeing Flight Crew Training Manual.

The QRH presents the flight profile and duties of each crew member for non normal circle to land procedure.

The landing gate for a circling/sidestep approach is 500 ft AAL. Lateral manoeuvring to align with the runway extended centreline must be completed by 300 ft AAL during all Circling or Side-Step Approaches. Exceptions to this policy, where lateral manoeuvring below 300 ft AAL is permitted, shall be detailed in the relevant OMC AFB (CAT C aerodromes).

8.3.0.3.8 NPA and Visual Landing in Marginal Conditions

- 8.3.0.3.8.1 Before electing to conduct a straight in approach on an NPA runway (rather than extend the track miles to the ILS runway) crews will carefully consider the current weather conditions.

As a general rule when the ceiling is < 1,000 ft and/or the visibility is < 3,000 m it is preferable to opt for the precision approach where the landing can be assured.

- 8.3.0.3.8.2 Extreme caution must be exercised when landing in reduced runway visibility just above the ground such as shallow fog, smoke, blowing snow or sand.

8.3.0.3.9 Instrument Approaches

The procedures to be used for an instrument approach are those published in the Jeppesen Manual, or procedures specifically approved for use by Malta Air.

1. An instrument approach procedure is defined as a series of predetermined manoeuvres by reference to flight instruments with specified protection from obstacles from the initial approach fix, or where applicable from the beginning of a defined arrival route to a point from which a landing can be completed and thereafter if a landing is not completed, to a position at which holding or en-route obstacle clearance criteria apply.
2. Initial and intermediate approach procedures are used to identify segments of the approach prior to the commencement of the final approach.
3. In the case of NPA, the final approach is that part of the procedure commencing at the final approach fix or when no FAF is published, when the aircraft is established on the final approach course, and ends at a point in the vicinity of the airfield from which a landing can be made, or a missed approach initiated. The Final Approach Fix (FAF) for an NPA published in the FMC database, with flight path angle displayed in FMC, and when no

Final Approach Fix is specified in the procedure, shall be 5/4 nm from the threshold when this approach is conducted in VNAV. When it is not practicable to identify a 5/4 nm point the FAF in VNAV shall be 2,000 ft AAL.

4. In some countries the FAF is a position identified by Radar controller, from which a constant angle approach is conducted. In some countries the distance from the threshold and the altitude corresponding to this point is published in the AIP for that country, e.g. Sweden.

Note: Where approaches are approved from fan markers, descents to the relevant altitudes specified are only authorised when both aural and visual marker signals have been received and identified.

Airports in some countries may be officially closed under certain conditions which obviously preclude the use of discretion by the Commander to make an approach.

Following a NPA the Missed Approach shall be executed when reaching the decision altitude or MAP whichever occurs first. The intention is that there shall be no level flight prior to the MDA or MAP, whichever occurs first. The MAY Missed Approach Procedure which requires ALT HOLD to be achieved before the MCP is set to the Missed Approach altitude complies with this requirement.

The Missed Approach/Circling shall commence at the circling minimum unless the required visual references exist to conduct the circling approach. If the required visual references are lost thereafter, a Missed Approach shall be executed.

Once the go-around has been initiated, resulting in de-stabilization of the approach, **no attempt** to reverse the decision and land shall be made even if the visual cues suddenly improve.

On **non-precision approaches** (NDB, SRA, VOR or other approaches without glide path information) it is vitally important that published procedures are rigidly adhered to. The correct descent profile must be followed in order to meet both safety and noise abatement requirements; consequently:

1. Final descent must not be commenced until “distance to go” information, locator or marker beacon, VOR cross-cut, DME or radar information, etc., associated with the approach has been correctly established, or until passing the final descent point in VNAV;
2. Normal descent procedures, correct rate of descent (R.O.D.), selection, shall ensure that the aircraft does not fly below the published approach descent profile and ensure that the aircraft does not reach its MDA at an

undesirable distance from the runway or in a hazardous area. A decision to go-around or continue an adjusted descent shall be made on reaching MDA.

3. The preferred LNAV/VNAV NPA procedure ensures lateral and vertical integrity during the approach, deviation scales and various alerts are presented to the crew to indicate system malfunction.
4. **Step-down non-precision technique is prohibited after the FAF.**
5. Some non-precision approaches may be timed from the final approach descent point. If sufficient visual reference to permit a safe landing has not been established before the MDA then the missed approach procedure will be immediately initiated.
6. The standard operating procedure, duties of each crew member and flight profile for NPAs is presented in FCOM Vol 1, Normal procedures, and additional guidance is provided in the Boeing Flight Crew training manual. The equivalent Non Normal procedures are presented in the QRH.

8.3.0.3.9.1 Commencement and Continuation of Approach “Approach Ban”

CAT.OP.MPA.305

1. If the reported visibility (VIS) or controlling RVR for the runway to be used for landing is less than the applicable minimum, then an instrument approach operation shall not be continued:
 - a. past a point at which the aeroplane is 1000 ft above the aerodrome elevation; or
 - b. into the Final Approach Segment (FAS) if the DH or MDH is higher than 1000 ft.
2. If the required visual reference is not established (OMA 8.3.0.3.11), then a missed approach shall be executed at or before the DA/H or the MDA/H.
3. If the required visual reference is not maintained after DA/H or MDA/H, then a go-around shall be executed promptly.
4. Notwithstanding point (1), in the case where no RVR is reported, and the reported VIS is less than the applicable minimum, but the Converted Meteorological Visibility (CMV) is equal or greater than the applicable minimum, then the instrument approach can be continued to the DA/H or MDA/H.

Application of RVR or VIS Reports

GM1 CAT.OP.MPA.305

1. There is no prohibition on the commencement of an approach based on the reported RVR or VIS. The restriction in OMA 8.3.0.3.9.1(1) applies only if the RVR or VIS is reported and applies to the continuation of the approach past a point where the aircraft is 1000 ft above the aerodrome elevation or in the FAS, as applicable.
2. If a deterioration in the RVR or VIS is reported once the aircraft is below 1000 ft or in the FAS, as applicable, then there is no requirement for the approach to be discontinued. In this situation, the normal visual reference requirements would apply at the DA/H.
3. Where additional RVR information is provided (e.g. midpoint and stop end), this is advisory; such information may be useful to the pilot in order to determine whether there will be sufficient visual reference to control the aircraft during roll-out and taxi. For operations where the aircraft is controlled manually during roll-out, OMA 8.4.1 Table 2 (AMC1 SPA.LVO.100(a)) provides an indication of the RVR that may be required to allow manual lateral control of the aircraft on the runway.

The touchdown zone RVR shall always be controlling. The minimum RVR value for the midpoint shall be 125 m, and 75 m for the stop end.

Minimum RVR for Continuation of Approach

AMC1 CAT.OP.MPA.305 (a)

1. The touchdown RVR should be the controlling RVR.
2. If the touchdown RVR is not reported, then the midpoint RVR should be the controlling RVR.
3. Where the RVR is not available, CMV should be used except for the purpose of continuation of an approach in LVO in accordance with OMA 8.1.3.4.9 Table 16.

8.3.0.3.10 Go-Around (GA)

A go-around shall be initiated if the required visual reference for landing has not been established on reaching the MDA or DA. Go-around policy is to fly the published MAP (missed approach procedure) to the specified altitude, and the go-around is NOT complete until the “after take-off checklist” is completed. Improvised/abbreviated go-arounds, e.g GA from one approach abbreviated into downwind to the opposite runway) are not recommended. Thereafter further options may be considered:

A Go-Around shall also be made:

1. If any of the Stabilised approach criteria will not be met by the landing gate refer to OMA 8.3.0.3.4.

2. Engine Failure on final approach when IMC and below 1,000 ft AAL.
3. If below 1,000 ft AAL, any significant departure from the normal approach path occurs and corrective action is not immediately effective.
4. If the approach aid in use or the respective aircraft installation is suspected to be malfunctioning, unless VMC.
5. Upon instruction of ATC; or
6. At the discretion of the Captain.
7. Flap load relief activation after the applicable landing gate – 1,000 ft AAL IMC, 500 ft AAL VMC.
8. TOO LOW GEAR, TOO LOW FLAP in normal operations.
9. Following Non-Routine RAAS/ORW Aural alerts requiring a Go-Around:
 - a. 737-NG RAAS
 - i. “UNSTABLE, UNSTABLE”
 - ii. “CAUTION TAXIWAY, CAUTION TAXIWAY”
 - iii. “LONG LANDING, LONG LANDING”
 - A. When operating to a short* or performance limited** runway, a go-around is mandatory. Refer to OMC AFB for short runway operational guidance.
 - B. When alert observed during landing after reverse thrust initiation, the pilot flying shall utilise increased levels of reverse thrust and braking, as appropriate.
 - *Short runway: runway LDA < 1800 m.
 - **Performance limited: calculated enroute landing performance requires MAX AUTO or MAX MANUAL braking selections to achieve $LDR \leq LDA$.
 - b. 737-8200 RAAS/ORW
 - i. “OVERRUN, GO AROUND”

All go-arounds shall be executed according to the procedures laid down for the aircraft. Once the decision has been made to go-around during final approach **no decision to abandon this go-around shall be taken.**

An ASR shall be filed for all go-arounds initiated below 1,000 ft AAL. It is not necessary to file ASR for a go-around above 1,000 ft AAL unless there are additional reportable circumstances.

Weather Related Missed Approaches

If a missed approach has been executed for meteorological reasons, another approach shall only be commenced if the Commander has reason to believe that a second approach will lead to a successful landing.

In the event that two weather related missed approaches have been carried out, The Commander shall divert to his alternate unless he receives weather information which indicates that an improvement to double the previously given weather has occurred. In this context, visibility would need to be increased by a factor of 2 and ceiling to twice the previously reported height.

The Company is aware that weather conditions are seldom stable and that a particular flight may encounter below limits weather where other flights approaching at the same time are able to land. Any economic penalty involved in complying this instruction is acceptable to the Company.

Our primary operations objective is to conduct our air operations activity safely.

8.3.0.3.10.1 It is not permitted to conduct a third approach following two abandoned approaches due to windshear.

8.3.0.3.11 Visual References for Instrument Approach Operations

AMC1 CAT.OP.MPA.305(c)

The change-over from flying on instruments to visual flying is the most difficult phase of the approach and needs good co-ordination and liaison between both Flight Crew members.

For instrument approach operations Type A and CAT I instrument approach operations Type B, at least one of the visual references specified below should be distinctly visible and identifiable to the pilot at the MDA/H or the DA/H:

1. Elements of the approach lighting system;
2. The threshold;
3. The threshold markings;
4. The threshold lights;
5. The threshold identification lights;
6. The visual glide path indicator;
7. The TDZ or TDZ markings;
8. The TDZ lights;
9. The runway edge lights; or

10. Other visual references specified in the operations manual.

Note: *OMA 8.3.0.3.5,

Night landings from a NPA to a runway less than 2000 m LDA without PAPI/VASI require NPFO approval.

Night landings from a circling approach require a PAPI/VASI.

This visual reference is required to allow Flight Crew member assessment of aircraft position and rate of change of position in relation to the desired flight path. An approach may not be continued below the DA/MDA unless the minimum visual reference required has been established as sufficient to continue visually, i.e. as at least equal to or better than the specified visibility minimum for the procedure being used.

8.3.0.3.11.1 Visual Descent After IMC Final Approach (CAT I, NPA)

The Flight Crew procedures and work distribution for the IMC approach are outlined in FCOM Vol 1, Normal procedures.

Care is needed to stay in the proper approach path by maintaining the stabilised aircraft condition and continuously monitoring the flight progress on flight instruments. Specifically:

1. Having directed his attention outside the cockpit during the last 100 feet above DA/MDA in order to pick-up and identify the approach lights or other visual reference, the PM must then redirect his scan on the flight instruments – in particular on deviations from glide slope and localiser, on aircraft pitch and roll, and on rate of descent;
2. The most common fault after transferring to visual flight in poor visibility is to lower the aircraft nose/pitch down and drop below the glide path (“duck under”). The PM must be expected to identify this deviation from the glide slope and “call” immediately; and
3. The illusion of pitch-up in rapidly decreasing visibility exacerbates this “duck under” tendency and must be monitored and guarded against by reference back to the flight instruments.

If visual reference is lost, **an immediate go-around is mandatory** as a matter of urgency:

1. In such circumstances, the instrument scan maintained by the PM becomes a vital monitoring function, with intervention if a critical condition should develop, until the PF's flight instrument scan has been re-established; and

2. The landing gear must not be retracted until a positive rate of climb is established. The go-around shall be continued even when an inadvertent touchdown on the runway occurs.

Under low visibility conditions near the minimum, dimming of the approach/runway lights shall not be requested as it results in reduced visual range for the lights:

1. Flight Crew members shall be aware of the intensity at which the approach/runway lights are operating, and to what intensity they will be reduced to if dimming is requested. This is particularly important if the approach and runway lights are on separate dimming circuits at certain airports, and Flight Crew members shall be familiar with the procedure for requesting individual dimming so as to avoid losing runway visibility if only the approach lights need to be dimmed.
2. Though very unusual in Europe some approach/runway lighting systems (Pilot Controlled Lighting – PCL, e.g. Dinard) may be dimmed from the cockpit by microphone “keying”.

The utmost caution is necessary in shallow fog conditions where the lighting is clearly visible from higher altitudes. However, in the top of the ground fog the visibility may suddenly drop to a very low value which then increases rather slowly on descending further.

There are a range of similar cases of such conditions in which the height of the layer of maximum fog density can vary from a few feet above the runway up to DA or higher which need to be carefully assessed in deciding how they should be handled.

Sequence flashing lights (SFL's) incorporated into some approach lighting systems are good for establishing visual contact with the full approach light system. However, by themselves, SFL's do not provide sufficient visual reference to assess aircraft position nor to allow correction of a lateral centre line correction as the roll reference is entirely missing.

Crews shall be familiar with the Malta Air Low Visibility operations document, which is part of the Malta Air SOP for low Visibility Operations.

De-tuning ILS

Crews shall not de-tune the ILS frequency during instrument approaches, including after becoming visual, unless required by the discontinued approach procedure or an ATC requested orbit.

8.3.0.3.12 Landing**8.3.0.3.12.1 “Land After” Clearances**

"Land After" clearances are authorised for Malta Air flight operations. Usually ATC will advise you prior to "MINIMUMS" to expect a late landing clearance. This is your cue to expect the actual landing clearance after "MINIMUMS". Execution of this approved clearance requires completion of the "LANDING LIGHTS" item on the Landing Checklist after the "MINIMUMS" call. The issuance of a LAND AFTER clearance authorises the aircraft to “cross the runway threshold” (OM C, ATC 7.10)

Crews shall be aware that Go-Around may be required after acceptance of the "land after" clearance up to a baulked landing.

8.3.0.3.12.1.1 Loss of Communications on Approach

1. If communications are lost with ATC on approach and clearance to land has not been received, the crew shall;
 - a. Check frequency selection and volume controls.
 - b. If frequency and volume are correctly set and no reply received, revert to previous frequency.
 - c. If still unable to establish communications attempt to establish communications on 121.5.
 - d. If no radio contact established or no signal received from the tower to land a go around shall be accomplished.

8.3.0.3.12.2 Threshold Crossing Altitude/Speed

CAT.OP.MPA.310

Malta Air has established operational procedures designed to ensure that an aeroplane conducting 3D instrument approach operations crosses the threshold of the runway by a safe margin, with the aeroplane in the landing configuration and attitude.

The standard prerequisites for a safe landing require the landing threshold to be crossed at the correct height and airspeed, and to land on the centreline within the prescribed touchdown area.

A stabilised approach is the means by which this effected.

The tower reported wind (10 m AGL) is the information on which Flight Crew member decisions are predicated.

Excess threshold height, speed and touchdown float will significantly increase landing/stopping distance.

Specifically:

1. The target threshold height for all aircraft is 50 feet;
2. The maximum allowable target airspeed over the threshold is VREF +15 kt for the Boeing 737; and
3. The fly speed which contains increments for wind speed and gust shall be flown accurately on the final approach and the additional speed increments reduced as the threshold is crossed so as to land at VREF plus gust increment. For a tail wind the fly speed shall be VREF +5 kt to touchdown at VREF.
4. With a 3° approach glide path and a target runway threshold height of 50 ft the touchdown aiming point can be identified from the standard runway markings.

Electronic and visual glide slopes may intersect the runway surface beyond the 1,000 foot point and some runways may have a displaced threshold. These shall be taken into account when determining the target touchdown point.

A shallower approach path of less than 3° and a down slope on the runway make an accurate touchdown progressively more difficult.

If for any reason the approach path is not maintained and it is likely that the touchdown will occur too short or too far beyond the touchdown area, or the threshold speed is excessive then a **go-around must be initiated**.

The Boeing Flight Crew Training Manual provides detailed guidance for the landing and roll out phase of flight.

1. The calculation of crosswind components is presented in the FCOM Vol 1, Normal procedures.
2. The recommended Boeing procedures for crosswind landings is presented in the FCTM where four options are given.

8.3.0.3.12.3 Runway Turn-off

Runway turn-offs are built to different designs. Flight Crew members will be prudent about aircraft turn-off speed until they are familiar with the turn-offs on specific runways:

1. Turn-offs at high speed can generate high side loads on the landing gear and increase tyre wear.
2. Turning forces on the aircraft are higher than experienced on the runway and special care needs to be taken where there is a possibility of ice or frost on the turn-offs and taxiways even though none exists on the runway.

3. Where ice or frost is present on runways, turn-offs or taxiways the aircraft shall be brought to a virtual standstill on the runway before turning-off to allow careful assessment of the surface conditions.

8.3.0.3.12.4 Reverse Thrust Policy

a. All Landings, a Serviceable Aircraft, No Applicable NNCs

The PF, with the agreement of the Commander shall decide on the Landing Flap, Autobrake and Reverse Thrust settings having consulted the QRH, PI section. When a OFP note or Airfield Brief specifies a particular landing configuration and/or technique these configurations/settings shall be used.

A minimum of Idle Reverse Thrust shall be used for all landings along with Autobrake 0-1-2-3-Max as appropriate. Where the use of higher Autobrake brake settings causes turnaround brake cooling issues, Reverse Thrust settings greater than idle shall be used.

Idle Reverse Thrust therefore shall be used in normal flight operations with RWYCC 5 (GOOD) or better as follows:

- On last flight of the day at all Malta Air/Ryanair Group bases.
- On the last flight of a series of flights when there is at least a 60 minute turnaround.
- Last landing prior to aircraft going out of service.
- Any landing when the turnaround schedule will not be adversely affected by the sole use of Autobrake.

The Commander is encouraged to use a combination of Flap/REV THR/AUTO BRK in order to exit the runway at a particular exit to minimise the taxi time to stand.

Note: The operational objective is to minimise runway occupancy times consistent with safety of operations.

b. Safety

Our primary operations objective is to conduct our air transport activities safely. Use of idle reverse thrust as described in this section is not considered detrimental to safety.

When required, up to and including MAX REV THR may be used.

8.3.0.3.13 Taxi-in

Taxiing comprises a relatively large proportion of engine running time for a short haul operation. While the Boeing 737 is normally taxied on both engines, the shutting down of an engine for taxi to the ramp/gate is acceptable and will generate a considerable fuel saving.

Care must be taken to avoid excessive jet blast when taxiing on one engine, particularly if the aircraft has had to stop or when turning at low speed.

It is considered best practise and crews shall have the airport map available for all operations on the ground at all times.

At many airports “Follow Me” vehicles are provided. Otherwise follow ATC instructions or ask for instructions if any doubt exists.

Arrival on Stand (Flight Deck)

Some stands require either electronic signalling or marshalling. It is not permitted to self position onto stands for which either electronic guidance or marshalling is required.

Some stands require neither. It is permitted to self position onto stands which do not require electronic guidance or marshalling.

Acknowledge all hand signals. Be aware that the “cut-throat” signal to shut down engines may be an urgent instruction rather than the last signal in a routine sequence of marshalling. If this is the case, do so even if you will be temporarily without AC power.

Crews shall not turn off the anti-collision light until N2 has fallen below 20% on both engines.

Parking

Stand Alignment and Stopping Guidance Systems

Stand alignment and stopping guidance systems using visual reference geometry are based on the Commanders/left seat eye position. When parking on a stand using such a system, the Flight Crew member operating from the left seat is required to taxi the aircraft, as only he can judge the correct positioning of the aircraft using these systems.

When clearances are judged not to be adequate marshalling assistance must be requested.

On non-guidance system stands a marshalling service is normally provided. However, a Flight Crew member may manoeuvre onto a self-parking stand at the Commanders discretion.

In either case, if there is any doubt about safety and clearance from an obstruction the Commander must stop short until the obstruction is removed, a marshaller is available, or the aircraft is towed onto the stand.

When the aircraft has reached its final parking position and the parking brake has been confirmed set, the flight crew shall provide the brakes set signal, which will be acknowledged by the ground crew.

Use of Parking Brake

The pilot monitoring (PM) will accomplish the exterior inspection at the earliest opportunity following completion of the shutdown procedure/checks (full or transit). The PM will connect the RFS headset and confirm with the pilot flying (PF) that the chocks are in-place/properly fitted and that all ground servicing equipment is properly in position.

The PF will then release the parking brake with caution and advise the PM when the parking brake is fully released. The PF and PM must monitor aircraft movement and pay particular attention during the walk around to ensure servicing equipment and aircraft (including air-stairs) has not been displaced and/or damaged. The confirmation of the proper placement of chocks/servicing equipment and release of the parking brake must be accomplished at the earliest opportunity to ensure optimum brake cooling. Report any abnormalities and damaged observed to the pilot flying, base engineers, and/or MAINTROL as appropriate.

Prior to securing the aircraft, notify ENG/MAINTROL if the aircraft will be left in a non-standard configuration (e.g., flaps not fully retracted).

8.3.0.3.14 Use of Ground Power

It is Malta Air policy to maximise the use of Ground Power when available for all turnarounds. Ideally the APU shall be started just prior to closing the L1 door. Without prejudice to Airport bylaws/Directors Notices APU is not normally required between temperatures of 5°C and 25°C and may be used for air-conditioning outside these temperature ranges during turnarounds.

8.3.0.3.15 Aircraft Changes, Crew Changes

The aircraft you are leaving shall be secured as described in the base airfield brief. This ensures regulations in that particular country/base are complied with.

The F/O on the inbound aircraft shall complete the shutdown checklist and immediately proceed to the outbound aircraft and commence pre-flight preparations.

The Captain shall complete the Technical Log, secure the aircraft, and proceed to the outbound aircraft.

When the outbound crew awaits your aircraft it is sufficient for the off going crew to vacate their aircraft having completed the shutdown checks and the Technical Log. The on coming crew shall complete a full pre-flight preparation.

Aircraft Changes

Planned aircraft changes are identified to operating crew in the EFL App.

The Electronic Flight Log/Paper Voyage report/Journey Log will have both registrations indicated where an aircraft change is planned.

Exercise caution with aircraft changes. The registration of the aircraft must be correctly reflected on the OFP. In the case of an unscheduled aircraft change the flight may not be commenced until a new OFP has been generated.

Boarding of Passengers During Aircraft or Crew Changes

Boarding of passengers on an aircraft that will have a crew change post-boarding is permitted subject to the following:

- The off-going crew shall remain on the aircraft and preserve the integrity of the Security Check/Search until the relieving crew arrives.
- The off-going Commander or SCCM shall brief the ongoing Commander or SCCM on the aircraft that passengers have been boarded and Security Check/Search has been completed on the aircraft in accordance with Malta Air procedures.

and:

For Security Checks:

- Either the off-going Commander or the ongoing Commander shall sign on the prescribed form that the Security Checks have been completed.

For Security Search:

- The off-going Commander shall sign on the prescribed form that the Security Search has been completed.

8.3.0.4 SAFA/SACA/SANA Inspections (Safety Assessment of Foreign/Community/National/Aircraft)

Commanders shall request a copy of the SAFA inspector's report if available, regardless of findings. This is to maintain the Ryanair database of reports and if findings are made to enable Ryanair to appeal such findings it considers unfair or inaccurate.

The Captain must inquire with the auditor(s) to determine and action the following:

1. What type of inspection is this (SAFA, SACA, etc.)?
2. What is your name or auditor identification number?
3. Were there any findings?
4. For all technical findings, call MAINTROL before departure.
5. If you disagree with findings, call the Duty Pilot for guidance prior to dispatch.
6. Ask for copy of report & take a picture with EFB.
7. Email picture of external agency inspection report to fileareport@maltaair.com.
8. File ASR indicating type of inspection, upload picture into ASR, submit ASR during post flightduties.

8.3.0.5 Post Flight Duties

A "Transit Checklist" or full Shut-down Checklist shall be completed as appropriate after each sector.

Upon leaving the aircraft at the end of a duty, at a base/airport where there is no engineer present, the Aircraft commander will perform a walk around inspection.

This inspection is to check for evidence of any defect that will affect the serviceability of the aircraft. In particular the aircraft commander will be checking for evidence of Bird strike and/or tyre damage.

Should the commander discover any defect or evidence of bird strike this will be reported to Maintrol on the telephone numbers listed in the Technical Log, but shall not leave the aircraft until maintrol has been contacted and the defect entered in the Technical Log.

The aircraft will be secured in accordance with current MAY security policies. Guidance for cold weather operations are provided in FCOM Vol 1.

Prior to securing the aircraft, notify ENG/MAINTROL if the aircraft will be left in a non-standard configuration (e.g., flaps not fully retracted).

The Commander will ensure that the crew adhere to all Customs clearance obligations.

The Electronic Flight Log/Paper Voyage Report/Journey Log, Electronic Voyage Report/Journey Log (Crewlink) and associated electronic documentation, Captain's Special Report, ASR and any other necessary reports, (see Chapter 11) shall be Completed, normally as part of 30 min post flight duty.

These reports together with the LID/Loadsheel and other specified flight documentation must be filed at the Base of Operations in accordance with Malta Air procedures below and in Chapter 11:

Electronic Flight Log: Submitted and Verified post flight.

Flight Envelope: File in base operations (if applicable).

8.3.0.5.1 Ryanair Connect App/Crewlink

The Ryanair Connect App is the primary source for roster information. The Roster App within Ryanair Connect provides 'live' duty details for the active week and displays rostered duty details for 3 weeks following the active week.

All operating crew are required to check-in via EFL at commencement of duty.

Crewlink reverts to back up as an electronic communication and reporting system. It is hosted electronically and accessible on base computers at workstations via Ryanair Connect "CHECKIN/CHECKOUT". If EFL is unavailable, all operating crew are required to sign in and log out on Crewlink at commencement and completion of duty. This option exists for 7 days post flight.

Where crew are unable to submit Electronic Flight Log data, the Commander is responsible to ensure that the data fields presented in Crewlink data input (effectively electronic voyage report) are completed correctly as part of post flight duties. If this is not possible this option remains for seven days to enable completion of data input. The same accuracy criteria as the Electronic Flight Log/Paper Voyage report applies. If the data input is not completed by this time, it causes disruption of Company account reconciliations, a non-compliance report is generated, and non-compliance file notes will be notified to Commanders.

Crew rosters are accessible via the Roster App. It is compulsory for crew members, on Friday, to check their roster for the subsequent week. This is because this is final notification of your actual duty plan for the following week. If there are changes after the Friday you will receive a notification from crew control via the Roster App, SMS, Email, Crewlink or phone.

There are ancillary functions available which are not flight operations functions in the context of this manual.

8.3.1 VFR/IFR Policy

CAT.OP.MPA.100

All flights shall be operated in accordance with Instrument Flight Rules. A flight plan shall be submitted prior to operating. Air Traffic Services (ATS) appropriate to the airspace and the applicable rules of the air shall be used for all flights whenever available. In-flight operational instructions involving a change to the ATS flight plan, when practicable, shall be coordinated with the appropriate ATS unit before transmission to an aircraft.

Commanders shall not cancel an IFR flight plan, or proceed VFR at any stage of a flight.

Aircraft operating under IFR may be cleared to conduct part of a flight in 'VMC', e.g. a visual approach. Such a clearance does not cancel the IFR plan, but for the duration of such a clearance, collision avoidance, terrain and obstacle clearance is the responsibility of the aircrew. The aircrew must be in visual contact with the runway to conduct a visual approach.

The NPFO may approve a portion of a flight under VFR. For CAT operations this is typically a departure from a location requiring the initial part of the departure to be flown VFR before resuming IFR. Occasionally a VFR departure may be authorized because the infrastructure to support a SID is temporarily unserviceable.

Clearances to maintain 'own separation in VMC' may be accepted by the Commander for limited portions of climb, descent or approach under the following conditions:

- Relevant traffic can be clearly identified and kept in sight.
- It is assured that sufficient separation will exist at all times.
- VMC can be maintained.
- ATC has coordinated the clearance with both aircraft.

When flights are conducted within class F and G airspace crews shall refer to the 'Air Traffic Services Outside Controlled Airspace' in the Jeppesen General Airway Manual > *ICAO Rules of the Air* > *Rules Applicable to IFR Flights Outside Controlled Airspace* and request the highest service available. Additional information may be available on the *Airway Manual* sections of the country overflown.

8.3.1.1 Operations Under IFR or at Night – Flight and Navigational Instruments and Associated Equipment

CAT.IDE.A.130 inc AMC

Malta Air aircraft are equipped to conduct flights at night or IFR flights.

8.3.2 Navigation Procedures

8.3.2.1 ‘Conventional Navigation’

All operating crews are trained to navigate using ‘conventional navigation’. Any navaid being used shall be identified (IDENT) and proper indication verified. ‘Conventional navigation’ is rarely used en-route as flight plans are RNAV5. The convention for being unable to remain RNAV5 is radar vectors not least because any ground based NAVAID probably has no relevance to your current filed en-route routing and any deviation towards a ground based NAVAID could compromise separation without an ATC clearance.

Procedures for Aircraft Equipped with Flight Management Systems (FMS)

All Malta Air Boeing 737 are equipped with dual FMS. The procedures for its operation and navigation, including independent cross check of keyboard entries utilising the FMS are detailed in FCOM Vol 1, Normal procedures, Supplemental procedures and system description in Vol 2.

Additional procedures for operations in NAT HLA are described in the ICAO NAT Doc 007 (North Atlantic Operations and Airspace) and the ‘Oceanic Guide and Checklist’. All charted procedures used shall be verified in the FMC against the Jeppesen charts and backed up with ground based navigation aids when required. No charted procedure shall be flown without the requisite chart being available on the flight deck.

All FMC navigation alerts shall be assessed for their operational significance depending on the navigation procedure in use and the considerations indicated in FCOM Vol 2 shall be applied.

The Area of Operations for Malta Air is indicated on the Operations Specifications Certificate which is available on the EFB.

8.3.2.2 Performance Based Navigation (PBN) excl RNP(AR)

CAT.OP.MPA.126

8.3.2.2.1 PBN Operations

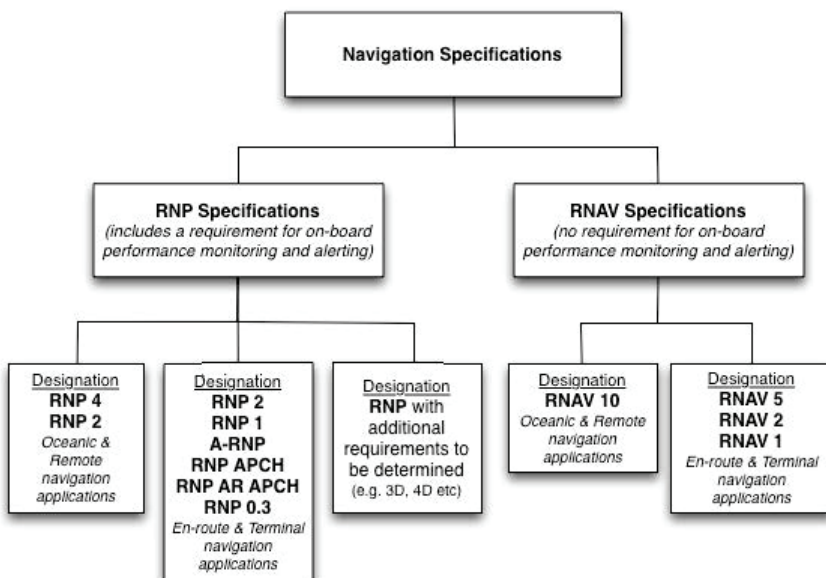
AMC1 CAT.OP.MPA.126, GM1 CAT.OP.MPA.126

Malta Air PBN navigation specifications are stated in the AFM and all PBN operations are conducted in accordance with these navigation specifications.

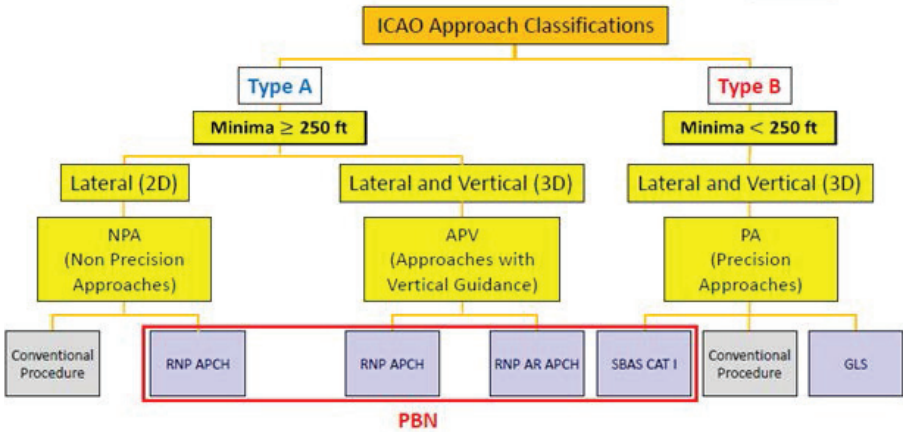
For both, RNP X and RNAV X designations, the 'X' (where stated) refers to the lateral navigation accuracy (total system error) in NM, which is expected to be achieved at least 95 % of the flight time by the population of aircraft operating within the airspace, route or procedure. For RNP APCH and A-RNP, the lateral navigation accuracy depends on the segment.

The operating crew must identify that PBN may be required on notified routes, for notified procedures and in notified airspace eg RNAV 10. However, it should be noted that many routes still use the designation 'RNP 10' instead of 'RNAV 10'. The terms 'RNP 10' and 'RNAV 10' should be considered equivalent.

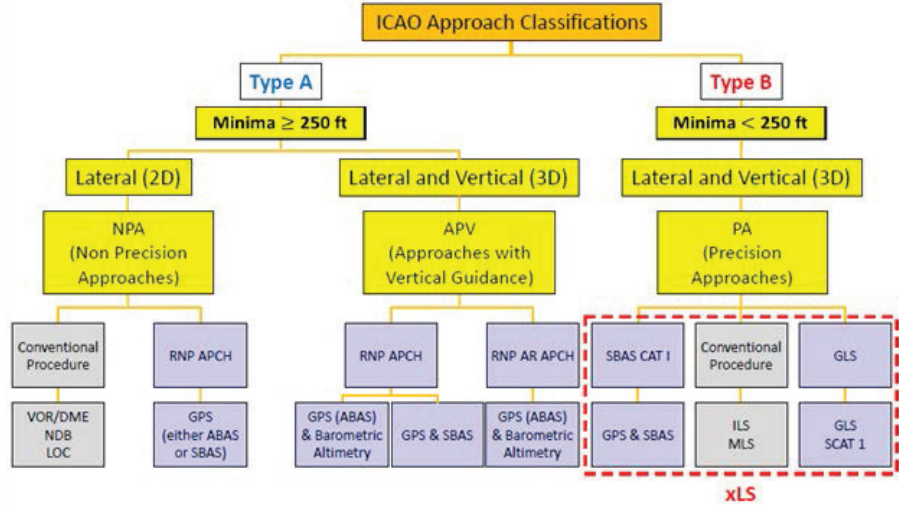
Navigation Specific Designations



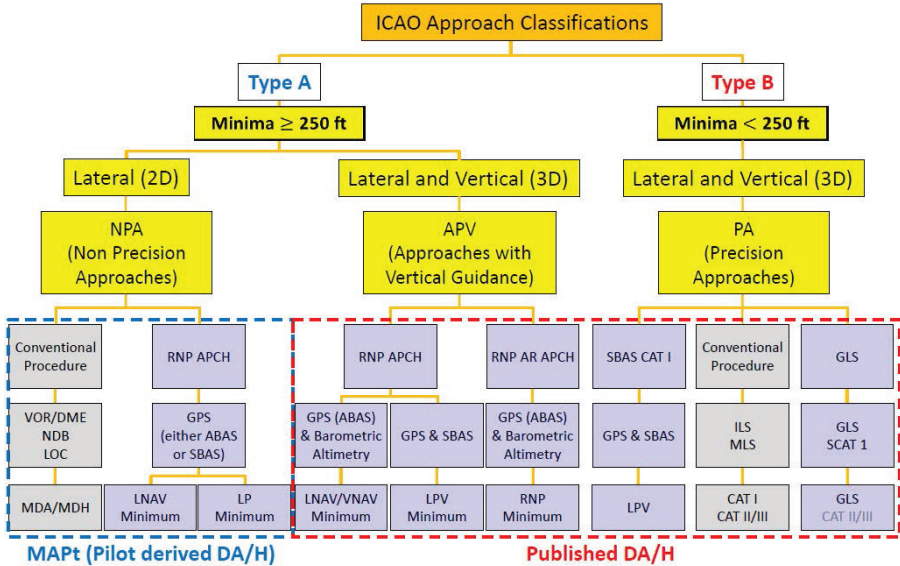
PBN Approach Operations



Approach Guidance XLS (ILS or GLS or MLS)



Decision Altitudes/Heights



Overview of PBN specifications								
	Flight Phase							
	En-route		Arrival	Approach				Departure
	Oceanic	Continental		Initial	Intermediate	Final	Missed	
RNAV 10	10							
RNAV 5		5	5					
RNAV 2		N/A MAY	N/A MAY					N/A MAY
RNAV 1		1	1	1	1		1	1
RNP 4	N/A MAY							
RNP 2	N/A MAY	N/A MAY						
RNP 1			1	1	1		1	1
A-RNP	2	2 or 1	1 - 0.3	1 - 0.3	1 - 0.3	0.3	1 - 0.3	1 - 0.3
RNP APCH (LNAV)				1	1	0.3	1	
RNP APCH (LNAV/VNAV)				1	1	0.3	1	
RNP APCH (LP)				N/A MAY	N/A MAY		N/A MAY	
RNP APCH (LPV)				N/A MAY	N/A MAY		N/A MAY	
RNP AR APCH				N/A MAY	N/A MAY	N/A MAY	N/A MAY	
Numbers specify the accuracy level								
	No specific approval required							
	Specific approval required							

For PBN operations using the navigation specification indicated in the table above, Malta Air:

1. Has established operating procedures specifying:
 - a. normal, abnormal and contingency procedures;
 - b. electronic navigation database management; and
 - c. relevant entries in the minimum equipment list (MEL)
 - d. monitoring and incident reporting procedures for PBN operations
2. Has specified the flight crew qualification and proficiency constraints and has ensured that the training programmes are consistent with the PBN operations indicated above; and
3. Ensures continued airworthiness of the area navigation system.

8.3.2.2.2 Monitoring and Verification

AMC2 CAT.OP.MPA.126

1. Pre-flight and general considerations
 - a. At navigation system initialisation, the flight crew confirms that the navigation database is current and verifies that the aircraft position has been entered correctly.
 - b. The active flight plan is checked including by comparing and confirming the charted procedures to be flown with the FMC and ND. Verify the departing runway and the waypoint sequence. Correct waypoint sequencing and the resultant 'picture' on the ND identifies reasonableness of track angles and distances. Verify any altitude or speed constraints and note which waypoints are flyby and which are fly-over. Where relevant, the RF leg arc radii should be confirmed.
 - c. Check that the navigation aids critical to the operation of the intended PBN procedure are available, if any.
 - d. Identify and confirm the navigation aids that should be excluded from the operation, if any.
 - e. An arrival, approach or departure procedure should not be used if the validity of the procedure in the navigation database has expired.
 - f. Verify that the navigation systems required for the intended operation are operational.
 - g. It is a requirement of RNP APCH operations that prior to dispatch, a prediction of the number of satellites available within the constellation must be carried out. The RAIM check is applicable to 15 min before Estimated Time of Arrival (ETA) until 15 min after

ETA. In the event of a predicted outage of more than five (5) minutes, the flight planning should be revised (e.g. delaying the departure or planning a different approach procedure).

- h. When intending to use PBN, the pilot-in-command should select an aerodrome only as a destination alternate aerodrome if an instrument approach procedure that does not rely on GNSS is available either at that aerodrome or at the destination aerodrome.

Note: It is required to conduct a RAIM check for destination and alternate airports at the planning stage, however, if as a result of emergency or en-route diversion, an RNP approach may be carried out at an airport that was not considered at the planning stage as any loss of RNP will be highlighted by the Aircraft systems. MEL restrictions for RNP APCH apply.

Flight Planning

For departures from an aerodrome which is served only with an APV BARO-LNAV/VNAV approach, a take -off alternate with NPA or CAT I minima is required. Crews intending to conduct operations on RNP APCH procedures shall ensure the aircraft is equipped with a dual FMC. Crews shall verify appropriate flight plan suffixes. The flight plan suffix R in item 10a S1 or S2 in item 18 indicates RNAV/RNP APCH capability.

The FMC navigation database must be current and must include the appropriate procedures. For dispatch with an expired navigation database Flight Ops must be contacted in advance to make required changes to the operational flight plan.

The instrument approach chart will clearly identify the RNP APCH operation as RNAV(GNSS), RNAV(GPS) or equivalent (e.g.: RNAV(GNSS) RWY 27,...). The RNP APCH procedures which may be used for the intended flight (including alternates aerodromes) must be selectable from a valid FMC navigation data base.

Malta Air Ops and flight crews must take account of any NOTAMs or briefing material that could adversely affect the aircraft system operation, or the availability or suitability of the procedures at the airport of landing, or any alternate airport.

If the missed approach procedures are based on conventional means (VOR, NDB), the appropriate airborne equipment required to fly this procedure must be installed in the aircraft and must be operational. The associated ground based nav aids must also be operational.

If the missed approach procedure is based on RNAV (no conventional or dead reckoning missed approach available), the appropriate airborne equipment required to fly this procedure must be available and serviceable

on board the aircraft. The contingency procedure shall be briefed in advance of conducting the approach to address possible failure of RNAV system components.

2. Departure

- a. Prior to commencing a take-off on a PBN procedure, check that the indicated aircraft position is consistent with the actual aircraft position at the start of the take-off roll, including verifying the RWY designator on the ND or RAAS.
- b. Where GNSS is used, the signal must be acquired before the take-off roll commences.
- c. Malta Air aircraft have automatic position updating including when TOGA is selected.

3. Arrival and Approach Preparation

Use DALTA and the double brief to check the procedure as loaded in the FMC. Prior to commencing the approach operation (before the IAF), the flight crew must verify the correctness of the loaded procedure by comparison with the appropriate approach charts.

This check includes:

- a. the waypoint sequence;
- b. reasonableness of the tracks and distances of the approach legs and the accuracy of the inbound course; and
- c. the vertical path angle, if applicable.
- d. any published altitude and speed constraints will be observed.

Verify aircraft is dual FMC equipped. Select or verify Approach RNP 0.3 in the FMC.

The crew must verify during the approach that a GNSS sensor is used for position computation. This is achieved operationally by verifying that the 'UNABLE REQ'D NAV PERF –RNP' message is not displayed.

4. Altimetry settings for RNP APCH operations using Baro VNAV

- a. Barometric settings
 - i. Set and confirm the correct altimeter setting and check that the two altimeters provide altitude values that do not differ more than 100 ft at or before the final approach fix (FAF).

- ii. Fly the procedure with:
 - A. a current local altimeter setting source available — a remote or regional altimeter setting source must not be used; and
 - B. the QNH set on the aircraft's altimeters.
- b. Temperature compensation
 - i. For RNP APCH operations to LNAV/VNAV minima using Baro VNAV:
 - A. Do not commence the approach when the aerodrome temperature is outside the promulgated aerodrome temperature limits for the procedure;
 - B. when the temperature is within promulgated limits, do not make compensation to the altitude at the FAF;
 - C. since only the final approach segment is protected by the promulgated aerodrome temperature limits, make cold weather altitude corrections as appropriate in other phases of flight.
 - ii. For RNP APCH operations to LNAV minima, make cold weather temperature corrections to all published altitudes and minima at or under the temperature corrected MSA, in particular on any step-down fix.
- 5. Sensor and lateral navigation accuracy selection
 - a. Verify, prior to approach, that the GNSS sensor is used for position computation.
 - b. Confirm that the indicated RNP value is appropriate for the PBN operation.

8.3.2.2.3 Management of the Navigation Database

AMC3 CAT.OP.MPA.126

1. For RNAV 1, RNAV 2, RNP 1, RNP 2, and RNP APCH, the flight crew should neither insert nor modify waypoints by manual entry into a procedure (departure, arrival or approach) that has been retrieved from the database. User-defined data may be entered and used for waypoint altitude/speed constraints on a procedure where said constraints are not included in the navigation database coding.

Note: It is permissible to edit altitude restrictions on arrivals XXXX replaced with XXXA in order to maintain a SAp/CDA approach prior to the FAF.

2. For RNP 4 operations, the flight crew shall not modify waypoints that have been retrieved from the database. User-defined data (e.g. for flex-track routes) may be entered and used.
3. The lateral and vertical definition of the flight path between the FAF and the missed approach point (MAPt) retrieved from the database shall not be revised by the flight crew.

8.3.2.2.3.1 Electronic Navigation Data Management

CAT.IDE.A.355

The Company only uses electronic navigation data products meeting standards of integrity that are adequate for the intended use of the data and has demonstrated this to the TM CAD. Malta Air continuously monitors the integrity of both the process and the products, either directly or by monitoring the compliance of third party providers. Malta Air Engineering ensures the timely distribution and upload of current, unaltered electronic navigation data to all aircraft.

8.3.2.2.3.2 Electronic Navigation Data Products

AMC1 CAT.IDE.A.355

1. When Malta Air uses a navigation database that supports an airborne navigation application as a primary means of navigation, the navigation database supplier must hold a Type 2 Letter of Acceptance (LoA), or equivalent.
2. If this airborne navigation application is needed for an operation requiring a specific approval in accordance with Annex V (Part-SPA), Malta Air procedures shall be based upon the Type 2 LoA acceptance process.

8.3.2.2.3.3 Letters of Acceptance (LoA) and Standards for Electronic Navigation Data Products

AMC1 CAT.IDE.A.355

A Type 2 LoA is issued by EASA. The FAA Type 2 LoA, and the Transport Canada Civil Aviation (TCCA) “Acknowledgement Letter of an Aeronautical Data Process”, is equivalent to an EASA Type 2 LoA. The navigation data products used by Malta Air are compliant with these requirements and copies of the relevant LoAs are maintained and available for inspection in Malta Air.

1. For RNAV 1, RNP 1 and RNP APCH, neither insert nor modify waypoints by manual entry into a procedure (departure, arrival or approach) that has been retrieved from the database. User-defined data may be entered and used for waypoint altitude/speed constraints on a procedure where said constraints are not included in the navigation database coding.
2. Do not revise or alter the lateral and vertical definition of the flight path between the FAF and the missed approach point (MAPt) as retrieved from the database.

8.3.2.2.4 Displays and Automation

AMC4 CAT.OP.MPA.126

Equipment to be Used:

Aircraft must be equipped with dual FMC, at least 1 flight director must be available and used. Use of autopilot when available is recommended.

VNAV is the recommended vertical mode for the initial and intermediate approach segments. Where “LNAV” minima is published the crew may use VNAV (Ref; Instrument Approaches using VNAV, FCOM Vol 1 and QRH MAN) or V/S (Ref; Instrument approaches using V/S, QRH MAN) on the final approach segment. VNAV is the preferred vertical AFDS mode. Where “LNAV/VNAV” minima are published the crew must use LNAV and VNAV on the final approach segment. (Ref; Instrument approaches using VNAV, FCOM Vol 1 and QRH MAN.)

1. For RNAV 1, RNP 1, and RNP APCH operations, use Navigation Performance Scale (as installed) or Progress page for lateral deviation indication, and where available, flight director and/or autopilot in lateral navigation mode.
2. The appropriate displays should be selected so that the following information can be monitored:
 - a. the computed desired path;
 - b. aircraft position relative to the lateral path (cross-track deviation) for FTE monitoring;
 - c. aircraft position relative to the vertical path (for a 3D operation).
3. Lateral deviation indicator scaling when available must be suitable for the navigation accuracy associated with the various segments of the procedure.
4. Maintain procedure centrelines unless authorised to deviate by air traffic control (ATC) or demanded by emergency conditions.
5. Cross-track error/deviation (the difference between the area-navigation-system-computed path and the aircraft-computed position) is limited to $\pm .5 \times \text{RNAV/RNP value}$ associated with the procedure. Brief deviations from this standard (e.g. overshoots or undershoots during and immediately after turns) up to a maximum of $1 \times \text{RNAV/RNP value}$ are allowable.
6. For a 3D (LNAV/VNAV) approach operation, a vertical deviation indicator must be used and where required by AFM limitations, a flight director or autopilot in vertical navigation mode.
7. Deviations below the vertical path should not exceed 75 ft at any time, (or half-scale deflection where angular deviation is indicated), and not more than 75 ft above the vertical profile, (or half-scale deflection where angular

deviation is indicated), at or below 1000 ft above aerodrome level. Execute a missed approach if the vertical deviation exceeds this criterion, unless the visual references required to continue the approach are in sight.

8.3.2.2.5 Vectoring and Positioning

AMC5 CAT.OP.MPA.126

The manual entry of coordinates into the FMC by the flight crew for operation within the terminal area is not permitted.

ATC tactical interventions in the terminal area may include radar headings, 'direct to' clearances which bypass the initial legs of an approach procedure, interceptions of an initial or intermediate segments of an approach procedure or the insertion of additional waypoints loaded from the database. In complying with ATC instructions, the flight crew must be aware of the implications for the navigation system.

'Direct to' clearances may be accepted to the IF provided that it is clear to the flight crew that the aircraft will be established on the final approach track at least 2 NM before the FAF.

'Direct to' clearance to the FAF is not acceptable. Modifying the procedure to intercept the final approach track prior to the FAF is acceptable for radar-vectorred arrivals or otherwise only with ATC approval. e.g. extending the centerline from the FAF/FAP, or a waypoint further out, to intercept the final approach course prior to the FAF/FAP for radar vectored arrivals.

The final approach trajectory must be intercepted no later than the FAF in order for the aircraft to be correctly established on the final approach track before starting the descent (to ensure terrain and obstacle clearance).

The lateral and vertical (for APV BARO-LNAV/VNAV operation) definition of the flight path between the FAF and the Missed Approach Point (MAPt) must not be revised by the flight crew under any circumstances.

1. 'Direct to' clearances to a fix that immediately precede an RF leg are not permitted.
2. For parallel offset operations en-route, transitions to and from the offset track should maintain an intercept angle of no more than 45° unless specified otherwise by ATC.

8.3.2.2.5.1 Alerting and Abort

AMC6 CAT.OP.MPA.126

1. Unless the flight crew has sufficient visual reference to continue the approach operation to a safe landing, an RNP APCH operation must be discontinued if:
 - a. navigation system failure is enunciated (e.g. warning flag);
 - b. lateral or vertical deviations exceed the tolerances;
 - c. loss of the on-board monitoring and alerting system.
2. Discontinuing the approach operation may not be necessary for a multi-sensor navigation system that includes demonstrated RNP capability without GNSS in accordance with the AFM. In this case the ANP/RNP continues to be indicated to the crew after the loss of GNSS.
3. Where vertical guidance is lost while the aircraft is still above 1000 ft AGL, the flight crew may decide to continue the approach to LNAV minima, when supported by the navigation system.

8.3.2.2.6 Contingency Procedures

AMC7 CAT.OP.MPA.126

1. The flight crew will make the necessary preparation to revert to a conventional arrival procedure where appropriate. If there is no conventional procedure expect radar vectors on notifying ATC of the loss of GNSS capability. The following conditions are considered in the presentation of minimum equipment required for conduct of PBN approaches, (LNAV and LNAV/VNAV):
 - a. failure of the navigation system components including navigation sensors, and a failure affecting flight technical error (e.g. failures of the flight director or autopilot);
 - b. multiple system failures affecting aircraft performance;
 - c. coasting on inertial sensors beyond a specified time limit; and
 - d. RAIM (or equivalent) alert or loss of integrity function.
2. In the event of loss of PBN capability, apply the contingency procedures indicated and navigate using an alternative means of navigation.
3. Notify ATC of any problem affecting PBN capability.
4. In the event of communication failure, the flight crew should continue with the operation in accordance with published lost communication procedures.

RNP APCH Operations, Contingency Procedures

The flight crew must notify ATC of any loss of the RNAV and/or RNP APCH capability, together with the proposed course of action as soon as possible. The loss of RNAV and/or RNP APCH capability includes any failure or event causing the aircraft to no longer satisfy the RNAV and/or RNP APCH requirements of the procedure. A clearance for a non-RNP approach must be requested. If no other procedure is available and acceptable and if below MSA/MOCA climb to a safe altitude along the prescribed track or be radar vectored.

In the event of communications failure, the flight crew shall continue with the procedure in accordance with lost communication procedures.

The procedure must be discontinued and a Go-around/missed approach is required if the 'UNABLE REQ'D NAV PERF-RNP, FMC DISAGREE', 'GPS INVALID' FMC Amber alert or VERIFY POSITION alert is displayed unless suitable visual reference is established and maintained.

The missed approach must be flown in accordance with the published procedure.

Use of the FMC during the missed approach is acceptable provided:

1. The FMC is operational.
2. The whole procedure (including the missed approach) is loaded from the navigation data base.

In the event of failure of one FMC during a procedure where two systems are necessary, the crew shall abort the procedure if the failure occurs before the FAF but may continue the approach if the failure occurs after FAF.

8.3.2.3 GLS (Ground Based Augmentation System) Landing System

GLS is a Type B, 3D approach and is a Category 1 (CAT I) precision approach

The GLS consists of three major elements:

- The satellite sub-system (space based constellation), which provides both the aircraft GNSS receiver and the ground GBAS station with ranging information;
- The ground station sub-system, which monitors the satellite signals and calculates/broadcasts pseudo-range corrections, integrity parameters, various locally relevant data such as atmospheric data and Final Approach Segments (FAS) defining the path in space to enable Precision Approach (PA) operations;

- The aircraft sub-system, which receives both the satellite signals and the GBAS signals, supplying navigation output/guidance to the flight instruments and to the autopilot.

GLS equipped aircraft are certified to conduct GLS operations using GNSS (GPS) signals for approach guidance.

8.3.2.3.1 Augmentation

Augmentation refers to the corrections to satellite signals made by the GBAS ground station, called pseudo-range corrections. The GBAS calculates pseudo-range corrections for all the satellites that meet the specified in-view criteria for both the aircraft and GBAS ground station. These corrections are applied to the GLS approach which is broadcast to the aircraft.

8.3.2.3.2 Operations Using GLS GBAS Ground Station

Each runway approach direction requires the definition of a final approach segment (FAS) to establish the desired reference path for an approach, landing, and rollout. The FAS data for each approach is determined by the GBAS service provider. Each approach is given a unique five-digit channel number and four-character identifier for a particular FAS, glideslope, and missed approach combination. FAS data for approaches supported by the particular GBAS facility are transmitted to the airplane through the VDB data link.

GBAS performs the following functions:

1. Provides locally relevant pseudo-range corrections;
2. Provides GBAS related data;
3. Provides final approach segment data when supporting precision approach;
4. Provides predicted ranging source availability data; and
5. Provides integrity monitoring for GNSS ranging sources.

iMMR (Integrated Multi Mode Receiver)

The iMMR functionality is described in the FCOM and training module.

The iMMRs process the pseudo-range correction and FAS data to produce an ILS-like deviation indication from the final approach path. These deviations are then displayed on the Primary Flight Display [PFD] and are used by airplane systems such as the flight guidance system (e.g. autopilot and flight director) for landing guidance.

The iMMR tunes the VHF receiver to a frequency based on the selected GLS channel. When the Ground Based Augmentation System (GBAS) VHF signal strength is sufficient, the iMMR is capable of decoding digital data broadcast by the GBAS. When the GBAS broadcast is received, the approach data block shows data as received from the GBAS, i.e. Ident/front course; DME to RWY threshold; Navigation Source Reference (GLS.) The iMMR can show distance to threshold when greater than the service volume established for GLS.

Threat:

The iMMR continues to show selected GLS identifier, runway ID, and distance to threshold after a GBAS failure. This means a continuous display of the GLS identifier does not necessarily mean a reliable GLS signal. The primary indication of a reliable GLS signal is the presence of localizer and glideslope pointers on the deviation scale (NPS).

iMMR Failure

In the event of an iMMR failure, localizer or glideslope pointers are not shown or are removed, and the amber LOC or G/S failure flags are shown. The absence of pointers prevents approach capture, or if LOC and G/S have already been captured, flight directors are removed from view and the autopilot disengages.

GLS Failure

Flight crew should not rely on the presence of an amber LOC failure flag, G/S failure flag, or GLS identifier to determine reliability of the GLS signal. In the event of a GLS ground station failure, localizer or glideslope pointers are not shown or are removed. The absence of pointers prevents approach capture, or if LOC or G/S have already been captured, flight directors are removed from view and the autopilot disengages.

If either of these two events occur execute a Go-Around and Missed Approach Procedure unless suitable visual reference is established and maintained to enable a safe landing.

The AFM contains GLS approvals. GLS systems and procedures are described in the FCOM and equipment failures are addressed in the MEL/DDG.

8.3.2.3.3 GLS Training and e-learning

The ILS-like implementation of the GLS was selected to support common flight deck and airplane systems integration for both safety and economic reasons. This implementation provides an optimal pilot and system interface with FMAs and deviation indications identical to those utilized during ILS approaches.

8.3.2.3.4 GLS Security

The accuracy, integrity, and continuity of service for the GLS have been specified to be the 'same as or better' than ICAO standards for an ILS ground station supporting Category I approaches.

8.3.2.3.5 Reportable Events of GLS Operations

A reportable event is an event that adversely affects the safety of the operation and may be caused by actions or events external to the functioning of the aircraft navigation system.

Technical defects and the exceedance of technical limitations are reportable events, including:

1. Significant navigation errors attributed to incorrect data or a database coding error;
2. Unexpected deviations in lateral/vertical flight path not caused by flight crew input or erroneous operation of equipment;
3. Significant misleading information without a failure warning;
4. Total loss or multiple navigation equipment failure;
5. Loss of integrity, e.g. RAIM function, whereas integrity was predicted to be available during preflight planning;
6. Any GLS system failure (refer FCOM).

An ASR shall be used to report all GLS anomalies. Coruson records ASRs GLS filed.

8.3.2.4 SPA

SPA.GEN.100, 105, 110, 115, 120, AMC1 SPA.GEN.105(a)

TM CAD is the responsible authority for issuance of specific approval(s) as listed in the Malta Air AOC. Malta Air SPAs are compliant with ORO.GEN.105.

Malta Air will document the use of each Specific Approval including all non-compliances observed and make such documentation available to the TM CAD on request. The operation of each approval shall be in compliance with the OMA and Part-ORO (Operator Requirements) of Annex III.

Malta Air will only carry out operations with those specific approvals which have been authorised by the authority and listed in the OP Spec. Specific approvals will remain in force until either amended or revoked by the authority

8.3.2.4.1 RNP(AR) Authorisation Required

Malta Air does not currently have an RNP(AR) approval.

8.3.2.4.2 NAT HLA

CAT.IDE.A.345, SPA.MNPS.100, 105, AMC1 SPA.GEN.105(a)

Malta Air has a restricted NAT HLA Approval issued by the TMCAD for T9 & T290 routes only in NAT HLA within the extent of the Malta Air Area of Operations as indicated in the current Operations Specifications Certificate and within the applicable diversion distance for the aircraft, 427 NM (B737-800)* or 416 NM (B737-8200)* of an adequate aerodrome at all times.

Note: *In accordance with CAT.IDE.A.285 for over water flights the Maximum Diversion Distance is limited to 400 NM unless Liferrafts are carried.

Pilots MUST NOT fly within NAT HLA unless they are qualified to do so and are in possession of the appropriate Approval(s). The Malta Air NAT HLA approval is indicated in the Operations Specifications Certificate.

The following documents, available on EFB, are applicable to NAT HLA operations:

- Oceanic Guide and Checklist
- HF Management Guidance Material (NAT Doc 003)
- Jeppesen Publications (Pubs) > Airway Manuals
- The North Atlantic Operations and Airspace Manual (NAT Doc 007).

NAT Doc 007 – North Atlantic Operations and Airspace Manual

The North Atlantic Operations and Airspace Manual (NAT Doc 007) has been produced with the approval and on behalf of the North Atlantic Systems Planning Group (NAT SPG) which is the North Atlantic regional planning body established under the auspices of the International Civil Aviation Organisation (ICAO). This Group is responsible for developing the required operational procedures, specifying the necessary services and facilities, and defining the aircraft and operator approval standards employed in the NAT Region.

Regulatory material relating to North Atlantic aircraft operations is contained in relevant ICAO Annexes, PANS/ATM (Doc.4444), Regional Supplementary Procedures (Doc.7030), State AIPs and current NOTAMs, which should be read in conjunction with the material contained in this Manual.

Jeppesen Publications (Pubs) > Airway Manuals

The Jeppesen Publications (Pubs) (e.g., General Airway Manual, Africa, Atlantic, Europe/Asia, North America, Pacific, etc.) extracts procedures, information and guidance from the North Atlantic Operations and Airspace Manual, PANS OPS 4444 and the Regional Supplementary Procedures, AIPs and other relevant documentation.

Oceanic Guide and Checklist

The Oceanic Guide and Checklist extracts the most significant information from both of the above and includes Malta Air SOP to be a more user friendly document for immediate access as required electronically for use in CAT operations in NAT HLA.

Malta Air HF Guidance Material

This document explains in detail the principles of High Frequency radio propagation, how to use and contingency procedures.

Flight Crew Training and Qualification

The flight crew training requirements are specified in OMD and include associated eLearning modules. This training in conjunction with this section of OMA and current NOTAMs ensures familiarity with the regulatory requirements for operations in NAT HLA.

Operations Controller Training and Qualification

OCC personnel engaged in NAT HLA operations are trained in compliance with Operations Controller Training, including NAT HLA, as stated in OMD. This training ensures such personnel understand the NAT HLA regulatory material relating to North Atlantic aircraft operations contained in relevant ICAO Annexes, PANS/ATM (Doc.4444), Regional Supplementary Procedures (Doc.7030), State AIPs and current NOTAMs.

Normal and Contingency Procedures

Normal and contingency operating procedures for operations in NAT HLA are described in great detail in the North Atlantic Operations and Airspace Manual (NAT Doc 007). Procedures are tailored for operational use in the 'Oceanic Guide and Checklist'. These include, but are not limited to, carrying out independent cross-checks of keyboard entries, RCL procedures, turn back and weather avoidance contingencies etc. reflecting the TEM concept of operations in NAT HLA.

Records of Operations in NAT HLA

Records of operations in NAT HLA are retained as applicable in compliance with OMA 2.0.3 and training records in an electronic record keeping system.

Safety Reporting

Any crew errors or omissions which did or could have compromised safety in NAT HLA shall be reported in accordance with the requirements of OMA Chapter 11.

Monitoring of Operations in NAT HLA

Malta Air monitors operations in NAT HLA to measure compliance and to have insight into failures and non-compliances. Non compliances, crew errors and equipment failures shall be reported in compliance with normal operating procedures and as appropriate in the technical log records. The ATS also monitors operations in NAT HLA and non-compliances are reported to TM CAD thence to Malta Air.

8.3.2.4.2.1 NAT HLA Equipment

CAT.IDE.A.345, AMC4 CAT.IDE.A.345, AMC1 SPA.MNPS.105

Aircraft authorised to operate within oceanic airspace or NAT HLA shall comply with Part SPA and AIP requirements.

Refer to MEL Annex 1 for NAT HLA Dispatch Requirements

8.3.2.4.2.1.1 Long Range Communications

HF

Aircraft operating in NAT HLA require at least one long range communication System. This is High Frequency (HF) radio on Malta Air aircraft, and is in addition to two independent radio communications systems, VHF1 & 2. This enables communication with an appropriate ground station from any point on the route, including diversions.

Whilst some Malta Air aircraft have one HF, the Malta Air NAT HLA approval is restricted to specific routes due to the relatively short duration of these flights outside VHF coverage. Contingency for HF blackout or loss of HF is presented in detail in the NAT Doc 007, Jeppesen Publications (Pubs) > Atlantic and the Oceanic Guide.

SATCOM

Currently not applicable to Malta Air Operations.

SELCAL

Currently not applicable to Malta Air Operations.

8.3.2.4.2.1.2 Long Range Navigation System (LRNS)

The Long Range Navigation System (LRNS) on board Malta Air aircraft is an FMC with GPS and IRS inputs. Dual FMC aircraft have two LRNS capable of filed oceanic route specification. It is Malta Air policy to assign dual FMC aircraft to operations in NAT HLA. However, a Malta Air aircraft may operate on T9 or T290 with one LRNS only (HF required).

8.3.2.4.2.1.3 RVSM

All Malta Air aircraft are RVSM compliant and all flight crew are RVSM qualified.

8.3.2.4.2.1.4 TCAS

All Malta Air aircraft are TCAS compliant and all flight crew are trained accordingly.

8.3.2.4.2.1.5 PBCS

Currently not applicable to Malta Air Operations.

8.3.2.4.2.1.6 Datalink

Malta Air aircraft planned on NAT HLA routings comply with route datalink requirements.

8.3.2.4.2.1.7 Underwater Locator Devices

Malta Air aircraft planned on NAT HLA routings comply with ULD requirements.

8.3.2.4.2.1.8 Liferafts

For over water flights, the Maximum Diversion Distance is limited to 400 NM unless Liferafts are carried.

8.3.2.4.2.2 NAT HLA Crew Qualification

To operate flights in NAT HLA, both pilots shall be NAT HLA qualified by completing the training specified in OMD, and so recorded on Electronic record keeping system.

A crew pairing of a 'new captain and a 'new' co-pilot is prohibited.

8.3.2.4.2.3 NAT HLA Organised Track System (OTS)

Malta Air flights are not planned on the OTS but the location of the OTS is useful for crew on any NAT HLA flight to have insight on the location of other NAT HLA traffic in case of contingencies or diversions. Therefore the organised track message and relevant amendments to it are provided in the NAT HLA preflight briefing pack from LIDO.

The OTS is described in detail in ICAO NAT Doc 007 and the 'Oceanic Guide and Checklist'. Malta Air operations in NAT HLA are not filed on the OTS, but on occasions may cross a track(s) on the OTS. Level-capping and fuelling considerations in these instances are explained in the 'Oceanic Guide and Checklist' on DocuNet.

8.3.2.4.2.4 Other Routes and Route Structures Within or Adjacent to NAT HLA Shannon Oceanic Transition Area (SOTA)

Part of the Shanwick OCA is designated as the Shannon Oceanic Transition Area (SOTA) and since 2009 this is not designated NAT HLA. SOTA has the same vertical extent as the Shanwick OCA and is bounded by lines joining successively the following points:

N5100 W01500 – N5100 W00800 – N4830 W00800 – N4900 W01500 – N5100 W01500

Air Traffic Service is provided by Shannon ACC using the call sign SHANNON CONTROL. Full details of the service provided and the procedures used are contained in OMC Jeppesen Publications (Pubs) > *Atlantic* > *North Atlantic Crossing Clearance Procedures and Frequencies* and Europe Airway Manual > *Air Traffic Control Data* > *North Atlantic (NAT) High Level Airspace (HLA)*.

Brest Oceanic Transition Area (BOTA)

Part of the Shanwick OCA is designated as the Brest Oceanic Transition Area (BOTA). NAT HLA requirements are still applicable from FL285 to FL420. BOTA has the same vertical extent as the Shanwick OCA and is bounded by lines joining successively the following points:

N4834 W00845 – N4830 W00800 – N4500 W00800 – N4500 W00845 – N4834 W00845

Air Traffic service is provided by the Brest ACC, call sign BREST CONTROL.

8.3.2.4.2.5 NAT HLA Flight Plan Requirements Random Routes

A route segment in NAT HLA that does not follow a published OTS track, in its entirety, is known as a “Random Route”. All flights which generally route in an eastbound or westbound direction should normally be flight planned so that specified ten degrees of longitude (20°W, 30°W, 40°W, etc.) are crossed at whole degrees of latitude. All generally northbound or southbound flights should normally be flight planned so that specified parallels of latitude spaced at five degree intervals (65°N, 60°N, 55°N etc.) are crossed at whole degrees of longitude.

All flights should plan to operate on great circle tracks joining successive significant waypoints.

OCC personnel planning a random route, should normally plan flight level(s) appropriate to the direction of flight. OCC flight planners are aware that the AIPs specify some exceptions to the use of “Appropriate Direction Levels” both during the OTS time periods and outside them. The current usage allocation of flight levels in the NAT HLA is published in the UK and Canadian AIPs as the NAT

Flight Level Allocation Scheme (FLAS). Therefore, OCC planners shall always consult the current AIPs and any supporting NOTAMs when flight planning random routes through NAT HLA.

Plotting:

In NAT HLA a plot of the filed routing and any re-clearances en-route is maintained on a suitable plotting chart. The Jeppesen FD Pro enroute map may be used as a plotting chart. Routes planned always remain within the diversion distance of one hour flying time at an OEI cruising speed of an adequate aerodrome. Any re-clearance en-route shall be declined if any part of the re-cleared route segment is outside the of the applicable diversion distance range ring when plotted.

When required to plot, it is good practice to plot your present position regularly on your chart because the present position output cannot normally be interfered with and its calculation is independent of the waypoint data. It is the one output which can be relied upon to detect gross tracking errors. **A position should be checked and preferably plotted approximately 10 minutes after passing each waypoint, and, if circumstances dictate, midway between waypoints. e.g. if one system has failed.**

However the following exceptions apply:

On T9, 213 & 16 – Do not plot.

These are airways programmed into the FMC so a plot is not required. However tracks and distances must be checked against the OFP in case of a programmed error by the service provider. If however you are routed ‘off-airways’ you must PLOT. Examples of this are:

- From one airway to another (LASNO on T9 to GONAN on T16) – plot required
- OMOKO on T16 to ‘47N11W’ then GONAN, back to T16 (i.e. a ‘dogleg’) – plot required
- A random LAT/LONG, which may not be on any of these three airways – plot required.

To/from Azores via Santa Maria OCA and Lisbon FIR – Do not plot.

The Portuguese Authority has confirmed that plotting is not required on these routes. However tracks and distances must be checked against the OFP in case of FMC insertion errors.

Note 1: All company flights are planned to be within a limit of an adequate Airport at any point during the flight. SOP's ensure that crew operating in NAT HLA enter the nearest enroute alternate on FIX page and circle with the applicable range ring for situational awareness. ETPs are computed for contingencies.

Note 2: *In Accordance with CAT.IDE.A.285 for over water flights the Maximum Diversion Distance is limited to 400 NM unless Liferafts are carried.

8.3.2.4.2.5.1 NAT HLA (ETP) Critical Fuel

There is a Critical Fuel calculation on NAT HLA flights. CRITICAL FUEL is fuel required for a scenario to divert to a suitable airport at the most critical point. Three Critical Fuel scenarios are considered.

1. A rapid decompression at the most critical point followed by descent to a 10,000 ft or a higher altitude if sufficient oxygen is provided in accordance with the applicable operational requirements.
2. Flight at the approved one-engine-inoperative cruise speed assuming a rapid decompression and a simultaneous engine failure at the most critical point followed by descent to a 10,000 ft or a higher altitude if sufficient oxygen is provided in accordance with the applicable operational requirements.
3. Flight at the approved one-engine-inoperative cruise speed assuming an engine failure at the most critical point followed by descent to the one-engine-inoperative cruise altitude. Upon reaching the alternate, fuel to hold at 1,500 ft above field elevation for 15 minutes and then conduct an instrument approach and landing is included.

Note: Critical Fuel includes ICE FUEL.

8.3.2.4.2.5.2 NAT HLA (ETOPS/ETP) Ice Fuel

There is an Ice Fuel calculation on NAT HLA flights. Ice Fuel is the expected fuel usage by anti-icing systems during a critical fuel situation. ICE FUEL is included in Critical Fuel.

Critical Fuel is corrected to include the greater of:

1. The effect of airframe icing during 10% of the time during which icing is forecast (including ice accumulation on unprotected surfaces, and the fuel used by engine and wing anti-ice during this period), or
2. Fuel for engine anti-ice, and if appropriate wing anti-ice for the entire time during which icing is forecast.

8.3.2.4.2.6 NAT HLA Request for Clearance (RCL)

General

There is a requirement to send a Request for Clearance (RCL) prior to Oceanic entry. An RCL is a voice or data link message via ACARS used to provide ETA at Oceanic Entry Point (OEP), requested flight level, and speed.

Flight Crew procedures for RCL are addressed in the 'Oceanic Guide and Checklist'. Contingencies including the Flight Level contingency rules, are extensively detailed in the North Atlantic Operations and Airspace Manual (NAT Doc 007) and Jeppesen Publications (Pubs) (e.g. Atlantic Reference and Airway Manuals). Enroute aircraft shall enter Oceanic Airspace in accordance with their current flight plan or as amended by ATC. Any change to the flight plan routing will be transmitted as part of the RCL.

8.3.2.4.2.7 LIDO NAT HLA Planning

Malta Air is not authorised to conduct ETOPS operations. LIDO will generate ETPs and will identify adequate ERA's when a NAT HLA flight plan is calculated. ***ETP FLIGHT*** will appear on the OFP. LIDO generated NAT HLA routes are always planned to remain within 60 mins of an ERA.

Planning Minima – NAT HLA

Malta Air will select an aerodrome as a NAT HLA fuel ERA only when the appropriate weather reports and/or forecasts indicate that the weather conditions will be at or above the planning minima of OMA 8.1.5.5 Table 2.

8.3.2.4.3 Reduced Vertical Separation Minima (RVSM)

SPA.RVSM.105, 100, 110, AMC1 SPA.RVSM.110(a), AMC2 SPA.RVSM.105

Company Aircraft may be operated in designated airspace where a reduced vertical separation minimum of 300 m (1000 ft) applies between FL290 and FL410 inclusive on the basis of the RVSM approval granted to Malta Air Ltd. by the TMCAD in the Operations Specifications in the AOC.

RVSM Operational Approval

SPA.RVSM.105, AMC1 SPA.RVSM.105

Malta Air Ltd. has provided to the TMCAD the items listed below to achieve an RVSM approval:

1. The RVSM airworthiness approval for each aircraft from the OEM;
2. Procedures for monitoring and reporting height-keeping errors have been established;
3. A training programme for the flight crew members involved in these operations has been established;

4. Operating procedures have been established specifying:
 - a. The equipment to be carried, including its operating limitations and appropriate entries in the MEL;
 - b. Flight crew composition and experience requirements;
 - c. Flight planning;
 - d. Pre-flight procedures;
 - e. Procedures prior to RVSM airspace entry;
 - f. In-flight procedures;
 - g. Post-flight procedures;
 - h. Incident reporting;
 - i. Specific regional operating procedures.

Malta Air has a RVSM Approval as stated in the Operations Specifications in the AOC. RVSM is represented by “W” on the filed flight plan in item 10. The operating procedures for operation in RVSM airspace, including the processes whereby an aircraft ceases to be RVSM compliant are presented in Operations Manual Part A and Part B including the MEL.

MEL RVSM Equipment Requirements (MEL Annex I)

SPA.RVSM.110

All Malta Air aircraft operating in RVSM airspace shall be equipped with:

1. Two independent altitude measurement systems;
2. An altitude alerting system;
3. An automatic altitude control system; and
4. A Secondary Surveillance Radar (SSR) transponder with altitude reporting system that can be connected to the altitude measurement system in use for altitude keeping.

RVSM Equipment Requirements

AMC1 SPA.RVSM.110(a)

Two independent altitude measurement systems:

All Malta Air aircraft comply with the following equipment requirements and each system has the following components:

- a. Cross-coupled static source/system, with ice protection if located in areas subject to ice accretion;

- b. Equipment for measuring static pressure sensed by the static source, converting it to pressure altitude and displaying the pressure altitude to the flight crew;
- c. Equipment for providing a digitally encoded signal corresponding to the displayed pressure altitude, for automatic altitude reporting purposes;
- d. Static Source Error Correction (SSEC), if needed to meet the performance criteria for RVSM flight envelopes; and
- e. Signals referenced to a flight crew selected altitude for automatic control and alerting. These signals are derived from aircraft altitude measurement systems meeting the performance criteria for RVSM flight envelopes.

Regional Monitoring Agency (RMA) Verification/Monitoring Programmes

Malta Air participates in Monitoring Programmes in the Area of Operations operated by Regional Monitoring Agencies. The data for all company aircraft is collated and available to the company. An altitude violation by a company aircraft is communicated by the RMA to the OCC. This violation is notified by the Manager OCC to the Safety Manager, NPFO, Maintrol and NPCA.

When height-keeping errors are recorded or received, Malta Air Ltd. shall take immediate action to rectify the conditions that caused the errors. Once rectified, this aircraft achieves RVSM capability, and the RVSM airworthiness approval is restored. RVSM violation notification shall be notified to TM CAD by the Safety Manager within 72 hrs from the moment the company becomes aware of the occurrence. Malta Air has a Maintenance Programme and Continuing Airworthiness Procedures in support of RVSM operations.

Note: The ADS-B capability is extensively used to compute aircraft altitude compliance in addition to Height Monitoring terminals. This captures a greater altitude monitoring sample more frequently by the Regional Monitoring Agencies.

8.3.2.4.3.1 RVSM Altitude Error Tolerance

SPA.RVSM.115

Occurrences of height-keeping errors caused by malfunction of aircraft equipment or of operational nature, equal to or greater than:

- A Total Vertical Error (TVE) of ± 90 m (± 300 ft);
- An Altimetry System Error (ASE) of ± 75 m (± 245 ft); and
- An Assigned Altitude Deviation (AAD) of ± 90 m (± 300 ft).

shall be reported by the Flight Crew so that the occurrence can be forwarded to TM CAD by the Safety Manager within 72 hrs from the moment the company becomes aware of the occurrence.

8.3.2.4.3.2 RVSM Airspace Procedures

AMC2.SPA.RVSM.105, CAT.OP.MPA.101, 105

Operating Procedures

1. Flight planning

During flight planning the flight crew shall pay particular attention to conditions that may affect operation in RVSM airspace. These include reported and forecast weather on the route of flight noting areas of turbulence in particular, minimum equipment requirements pertaining to height-keeping and alerting systems and any airframe or operating restriction related to RVSM operations.

2. Pre-flight procedures

The following actions shall be accomplished during the pre-flight procedure:

- a. Review the Technical Log to determine the condition of equipment required for flight in the RVSM airspace. Ensure that maintenance action has been taken to correct defects to required equipment.
- b. During the external inspection of aircraft, particular attention will be paid to the condition of static sources and the condition of the fuselage skin near each static source and any other component that affects altimetry system accuracy.
- c. Before take-off, the aircraft altimeters shall be set to the QNH of the airfield. The two primary altimeters must be within the maximum value of acceptable altimeter differences which should not exceed 23 m (75 ft) if operations in RVSM airspace are planned. The altimeters shall be crosschecked prior to departure.

Refer to FCOM Vol 1 Limitations Section.

- d. Before take-off, equipment required for flight in RVSM airspace shall be operative and any indications of malfunction will be resolved.

3. Prior to RVSM airspace entry

- a. The following equipment must be operating normally at entry into RVSM airspace:
 - i. Two primary altitude measurement systems.
 - A cross-check between the two primary altimeters must be made and must agree within ± 60 m (± 200 ft). Failure to meet this condition will require that the altimetry system is reported as defective and Air Traffic Control (ATC) notified.

The altimetry checks performed during the climb comply with this requirement.
 - ii. One automatic altitude-control system;
 - iii. One altitude-alerting device; and
 - iv. Operating transponder.
- b. Should any of the required equipment fail prior to the aircraft entering RVSM airspace, request a new clearance to avoid entering RVSM airspace.

4. In-flight procedures

- a. The following practices are included in Crew Training and Standard Operating Procedures for RVSM:
 - i. Flight crew shall comply with any aircraft operating restrictions given in the RVSM airworthiness approval.
 - ii. The sub-scale on all primary and standby altimeters must be set to 1013/29.92 in Hg when passing the transition altitude and rechecked for proper altimeter setting when reaching the initial cleared flight level.

The maximum in-flight difference between the Captain and First Officer altitude displays for RVSM operations is 200 ft which is also the trigger for the ALT DISAGREE alert.
 - iii. In level cruise it is essential that the aircraft is flown at the cleared flight level. This requires that particular care is taken to ensure that ATC clearances are fully understood and followed. The aircraft shall not intentionally depart from cleared flight level without a positive clearance from ATC unless the crew are conducting contingency or emergency manoeuvres.

- iv. When changing levels, the aircraft shall not be allowed to overshoot or undershoot the cleared flight level by more than 45 m (150 ft). Level off shall be accomplished using the altitude capture feature of the automatic altitude-control system, i.e. with an autopilot engaged.
- v. An automatic altitude-control system should be operative and engaged during level cruise. In any event, adherence to cruise altitude will be done by reference to one of the two primary altimeters. Following loss of the automatic height-keeping function, any consequential restrictions will need to be observed.
- vi. Ensure that the altitude-alerting system is operative.
- vii. At intervals of approximately 1 hour, cross-checks between the primary altimeters will be made. Both altimeters must agree within ± 60 m (± 200 ft). Failure to meet this condition will require that the altimetry system be reported as defective and ATC notified, or contingency procedures applied. Monitoring of the Autopilot Flight Director System and flight deck instruments suffices for altimeter cross-checking with one automatic altitude control system, one altitude alerting device and an operating transponder serviceable. The hourly check of No 1 and No 2 altimeter during flight in RVSM airspace is not recorded.
- viii. In normal operations, the altimetry system being used to control the aircraft will be selected for the input to the altitude reporting transponder transmitting information to ATC.
- ix. If the pilot is notified by ATC of a deviation from an assigned altitude exceeding ± 90 m (± 300 ft) then the pilot shall take action to return to the cleared flight level as quickly as possible.
- b. Contingency procedures after entering RVSM airspace are as follows:
 - i. Notify ATC of contingencies (equipment failures, weather) that affect the ability to maintain the cleared flight level and coordinate a plan of action appropriate to the airspace concerned as per OM(C) Chapter 13.
 - ii. Examples of equipment failures that shall be notified to ATC are:
 - Failure of all automatic altitude control systems aboard the aircraft;
 - Loss of redundancy of altimetry systems;

- Loss of thrust on an engine necessitating descent; or
 - Any other equipment failure affecting the ability to maintain cleared flight level.
- iii. Notify ATC when encountering greater than moderate turbulence.
 - iv. If unable to notify ATC and obtain an ATC clearance prior to deviating from the cleared flight level, follow any established contingency procedures for the region of operation and obtain ATC clearance as soon as possible.

5. Post-flight procedures

In making technical log entries against malfunctions in height-keeping systems, the commander will provide sufficient detail to enable maintenance to effectively troubleshoot and repair the system. Include the following information when reporting the defect:

- a. Primary and standby altimeter readings;
- b. Altitude selector setting;
- c. Subscale setting on altimeter;
- d. Autopilot used to control the aircraft and any differences when an alternative autopilot system was selected;
- e. Differences in altimeter readings;
- f. Use of air data computer selector for fault diagnosis procedure; and
- g. The transponder selected to provide altitude information to ATC and any difference noted when an alternative transponder was selected.

6. Reporting RVSM Altimetry error after Tech Log entry

If a Tech Log entry has been made regarding altimeter performance outside RVSM tolerances, an ASR must be filed on completion of duty. This is to enable Malta Air to report the RVSM altimetry non-compliance to TM CAD within 72 hrs from the moment the company becomes aware of the occurrence. Reports shall include an initial analysis of causal factors and measures taken to prevent repeat occurrences. When height-keeping errors are recorded or received, the operator shall take immediate action to rectify the conditions that caused the errors and provide follow-up reports, if requested by the competent authority.

8.3.2.4.4 Low Visibility Operations (LVO)

SPA.LVO.100

Malta Air shall only conduct the following operations when approved by the TMCAD:

1. Take-off operations with visibility conditions of less than 400 m RVR; and
2. Instrument approach operations in low visibility conditions.

8.3.2.4.5 EDTO/ETOPS

Malta Air does not have ETOPS Approval.

8.3.2.4.6 DG

Malta Air has a full Dangerous goods approval detailed in OM(A) Chapter 9.

8.3.2.4.7 Polar Navigation

Malta Air does not have an approval to conduct Polar Navigation, nor does this requirement arise in the current Malta Air area of operations.

8.3.2.5 Section Not in Use

8.3.2.6 Navigation in Other Designated Areas

Automatic Dependant Surveillance – Broadcast ADS-B

ADS-B used in Non Radar Airspace (NRA) is a surveillance technology that supports radar-like separation standards. The system brings significant safety and efficiency benefits, offering properly-equipped and certified aircraft more flexible, fuel-saving routes through airspace previously managed using only procedural air traffic control. Aircraft with ADS-B automatically transmit accurate position reports with integrity every second to ATC. As a result, ADS-B will reduce separation minima for equipped aircraft and allow more aircraft to follow the most efficient flight trajectory.

ADS-B is a surveillance system that uses satellites, aircraft avionics and ground infrastructure to relay a range of aircraft parameters such as identification, position and altitude to ATC. It is Automatic since no external stimulus is required for operation, and *Dependant since it relies on aircraft avionics to provide Surveillance through Broadcast messages.*

Airspace

ADS-B-NRA is designed for use in airspace classes A to G. These airspaces may include “en-route”, arrival and departure phases which are not covered by radar surveillance.

Note: The ground infrastructure will include ADS-B ground sensors so as to be able to provide surveillance to the ADS-B equipped aircraft within their range. However ADS-B coverage of the entire sector may not be complete.

Operational

The following operating principles apply:

- There are no fundamental changes in the roles and responsibilities of the flight crew or controllers. The controller remains responsible for the management of the airspace, maintaining separation and providing information to flight crews, whilst the flight crews are responsible for navigation, for acting upon ATC instructions and providing information to ATC.
- Procedural separation minima will be applied between ADS-B and non-ADS-B equipped aircraft, as well as between non-ADS-B equipped aircraft.
- When flying in an airspace providing ADS-B only surveillance, flight crews shall be aware if specific local procedures are in force.
- Airspace class dimensions and associated local rules (where applicable) are available in OM-C Route Manual, AIP, AIC and NOTAM.

Within the area where ADS-B only surveillance is implemented:

- Two-way radio (R/T) will be available as a primary means of communication between aircraft and ATC.

Note: CPDLC may co-exist as a means of communication.

- The flight crew will contact ATC when entering NRA in accordance with existing radio procedures.

Malta Air aircraft are ADS-B equipped.

8.3.2.7 Aircraft Data Communication Systems

ORO.MLR.105

8.3.2.7.1 ADS-B

Malta Air aircraft are ADS-B equipped.

Refer to Jeppesen Publications (Pubs) > *Atlantic* > *Operational Approval and Aircraft System Requirements for Flights in the NAT High Level Airspace (HLA)* regarding additional information and instructions associated with ADS-B airspace requirements. Refer to OMA 8.3.2.6 *Navigation in Other Designated Areas* for ADS-B operations in Non-Radar Airspace.

8.3.2.7.1.1 ADS-B In

Malta Air aircraft are not ADS-B 'In' equipped.

8.3.2.7.1.2 ADS-B Out

Malta Air aircraft are ADS-B 'Out' equipped.

There are no flight crew procedural changes associated with ADS-B 'Out' operations.

8.3.2.7.1.3 Section Not in Use

8.3.2.7.1.3.1 Section Not in Use

8.3.2.7.1.3.2 Section Not in Use

8.3.2.7.2 CPDLC

ORO.MLR.105

Malta Air aircraft are CPDLC equipped and shall conduct CPDLC operations in designated airspace.

8.3.2.7.3 DLM Airspace Contingency Procedures

N/A to Malta Air.

8.3.2.7.4 Aircraft Communications, Addressing and Reporting System (ACARS)

When installed, Technical Log Notice to Crew and the Operational Flight Plan, Item 10a identifies ACARS configuration.

On ACARS operational aircraft, system operation is restricted to obtaining departure clearance (DLC), pushback clearance, and Airline Operational Control (AOC) transmissions associated with safety/security alerts, diversions, and MAY808 Radio Ahead Codes.

ACARS AOC messages to/from OCC shall comply with standardised formats and content. Refer to the ACARS Supplementary Guide in the EFB for guidance.

When noted prior to dispatch, the PIC should advise Operations Control of any defective ACARS related equipment and revise OFP Item 10a accordingly, e.g.:

- E3 – PDC ACARS: ATC Pre-Departure Clearance received via ACARS.

8.3.3 Altimeter Setting Procedures

AMC1 CAT.POL.A.225, CAT.OP.MPA.101

The following altimeter setting procedures apply for flight operations:

On Ground

All Altimeters shall be set to QNH.

During Climb

When cleared to a flight level and above 3000AGL both pilots shall set primary altimeters to standard.

Cruise Level

When passing the transition altitude and above the MSA the standby altimeter shall be set to standard remain set to standard during cruise and descent until cleared to an altitude in the descent.

During Descent

When cleared to an altitude both pilots shall set primary altimeters to QNH as well as the standby altimeter.

8.3.3.1 Checking of Barometric Altimeters

After each setting of altimeters their readings shall be compared. This shall include the metric and standby altimeters when they are used (e.g. Area U).

The altimeter indications shall be cross-checked against the elevation of the aerodrome during preflight instrument check.

8.3.3.1.1 Temperature Error

Pressure altimeters are calibrated to indicate true altitude under International Standard Atmosphere (ISA) conditions. Any deviation from ISA will therefore result in an erroneous reading on the altimeter. The altimeter error may be significant under conditions of extremely cold temperature and appropriate corrections shall be applied.

8.3.3.1.2 For procedures on how to correct for temperature error refer to FCOM vol 1.

8.3.3.1.3 Minimum Flight Altitude Correction

When converting minimum flight altitude to minimum flight level, corrections shall be applied when actual QNH differs from the standard value. For each 1 hPa below 1013 hPa subtract 28 ft from your flight level to establish the true altitude of your flight level.

E.g. FL = 160

MEA = 15,000 ft

Local QNH 975 hPa = Minus 1,080 ft

True altitude = FL 16000 – 1,080 ft = 14,920 ft

MEA is 15,000 ft and true altitude of aircraft is 14,920 ft, therefore the minimum required obstacle clearance is not assured.

8.3.3.2 Types of Altimeter Settings

With regard to the ICAO standard atmosphere, each type of altimeter setting indicates a vertical distance above the corresponding reference datum shown in the table:

Type of Altimeter Setting	Reference Datum	Altimeter Indication
Standard	1 013.2 hPa/29,92 in Hg Level	Flight Level
QNH	QNH pressure level (approx. Mean Sea Level).	Altitude
QFE	QFE reference datum (aerodrome or threshold elevation).	Height above QFE reference datum.

Note: QFE settings are not allowed.

8.3.3.3 Metric Altimetry

Use of metric altimetry settings is not normally required for Malta Air operations. Metric Altimetry shall only be used as prescribed in the relevant Airfield Brief. Jeppesen charts provide a conversion table on the approach chart which cross references feet and meters, and QNH/QFE altitudes for all the charted altitudes of the procedure depicted. In addition, conversion tables are published on the Jeppesen Flight Level Transition Procedures section of the relevant region’s Airway Manual, where metric altimetry is required.

8.3.3.4 QFE Operating Procedure

N/A MAY OPS

8.3.4 Policy for Altitude Alert

CAT.IDE.A.140

The aircraft altitude alert system shall be considered as a back-up aid. Its operation must not be permitted to replace the requirement to make the standard altitude calls as presented in FCOM Vol 1, Normal procedures. Incorrect setting of the QNH in the descent is an insidious mistake frequently made. An altitude crosscheck on final approach path is an essential crew action to detect this. VSD when available also identifies blunder error.

8.3.5 Policy for Ground Proximity Detection

CAT.OP.MPA.290, CAT.IDE.A.150

When undue proximity to the ground is detected by a Flight Crew member or by a ground proximity warning system (GWPS/EGWPS), the pilot flying shall take corrective action immediately in accordance with procedures listed in the OM B to establish safe flight conditions.

- 8.3.5.1 The GPWS may not be deactivated/by pulling the circuit breakers or use of the flap or gear inhibit switch, except for approved checklist procedures where use of flaps at less than normal landing flap positions or landing gear up is specified.

8.3.5.2 Enhanced Ground Proximity Warning System (EGPWS)

EGPWS is a mandatory system and offers two advanced features, Terrain Awareness and Warning System (TAWS) based on an integral database, and Windshear Warning System.

Although there are few areas of known degradation of GPS signal, the consequences of interrupted GPS signal to some aircraft is a “TERRAIN, TERRAIN, PULL UP, PULL UP” command, EGPWS, or other look ahead terrain alerts. This will occur any time the signal is interrupted, regardless of the phase of flight of the aircraft, e.g., in the cruise. Ryanair flight planning requires assured terrain clearance, and by definition, a cruise altitude/flight level clears all obstacles enroute by the prescribed amounts.

Crews shall only inhibit EGPWS functionality when instructed to do so by a Boeing Checklist or Company procedure.

Unless specifically permitted by QRH or company procedure, and briefed in advance, crew reaction to an EGPWS ‘PULL UP’ warning shall be instantaneous. Operations into destinations not contained in WGS-84 database will cause GPWS warnings as the system does not recognise the destination and believes that the aircraft is too low. The warning is triggered as a result of system pre-programmed parameters relating aircraft altitude and distance from the nearest airport the GPWS system has in its database as distinct from the current non WGS-84 destination.

8.3.5.2.1 Management of the Enhanced Ground Proximity Warning System (EGPWS) Terrain Database (TDB)

CAT.IDE.A.355(B)

The Engineering Department ensures the timely distribution and upload of Terrain Data Bases (TDB) on all aircraft within eight weeks of the TDB release date. Each TDB will remain valid for a maximum of sixteen weeks.

The Performance Department reviews all TDB releases, cross-referencing the database updates with the specific Area of Operation (as per AOC Ops Spec). A Notice to Crew is raised in the Technical Logbook by Engineering, referring to the TDB update process. Safety or operationally critical TDB issues will be promulgated to Flight Crew by Company NOTAM and/or AFB. This structured process ensures that the Flight Crew operates with accurate and relevant information regarding obstacles and terrain in the intended Area of Operation.

The TDB version can be displayed on the Navigation Display by pressing the Ground proximity SYS TEST switch, which might be requested during SAFA/SACA Ramp Inspections.

8.3.6 Policy for Use of TCAS

CAT.OP.MPA.295, CAT.IDE.A.155

Malta Air aircraft are fitted with TCAS as required by current regulation. TCAS equipage is documented in FCOM/QRH and/or Aircraft Technical Log Notice to Crew/HIL as appropriate. The operating policy, procedures and call-outs are presented in the FCOM and QRH non-normal manoeuvres.

If an ATC clearance or instruction contradictory to the TCAS RA is received, the Flight Crew shall follow the RA and inform ATC directly: “[Callsign], Unable, TCAS RA”.

8.3.6.1 Traffic Advisory (TA)

The response to a TA is to immediately attempt to establish visual contact with the intruder. Pilots shall be aware that an aeroplane sighted may not necessarily be the intruder or the sole intruder TCAS is responding to.

On receipt of a TA, pilots shall use all available information to prepare for appropriate action if an RA occurs. While the PF shall continue to fly the aeroplane, the PM shall provide updates on the traffic location shown on the TCAS traffic display and use this information to help visually acquire the intruder.

Note: TCAS is not able to provide RAs for intruders without altitude reporting capability. Therefore, TAs without relative altitude data on the display shall be closely monitored.

8.3.6.2 Resolution Advisory (RA)

The PF must respond immediately to RA, according to the procedures outlined in the QRH Chapter MAN. The maneuver must be limited to the minimum required to satisfy the RA. The PF must strictly follow the TCAS RA even if:

- Conflicting instructions have been issued by ATC.
- The RA response capability is limited by aircraft performance.
- A traffic information has been given by ATC.
- There is a visual contact to other traffic. Pilots must be aware that an aeroplane sighted may not necessarily be the intruder or the sole intruder TCAS is responding to.

The PM shall monitor the response to the RA and shall provide updates on the traffic location by checking the traffic display.

When an RA occurs, the Flight Crew must notify the appropriate ATC unit as soon as possible, as permitted by Flight Crew workload, in the following manner:

“[Callsign], TCAS RA”

If the pilot is unable to comply with an ATC clearance or instruction because there is an RA, the appropriate response is:

“[Callsign] Unable, TCAS RA”

The ATC response in both cases should be:

“Roger”.

This relieves the controller of the responsibility to provide separation and he will not try to modify the aircraft flight path until the pilot reports “CLEAR OF CONFLICT”.

When TCAS announces “CLEAR OF CONFLICT” the PF must return the aeroplane promptly but smoothly to the ATC clearance. Announce the TCAS “CLEAR OF CONFLICT” message when the conflict is over using the phraseology:

“[Callsign], Clear of Conflict, Returning to (assigned clearance)”, or

“[Callsign], Clear of Conflict, (assigned clearance) Resumed”.

8.3.6.2.1 TCAS “TA only” shall be used in accordance with any QRH procedure, and/or MEL procedure. This includes, but is not limited to, EMERGENCY DESCENT and ENGINE FAILURE. TCAS will not track aircraft with vertical rates in excess of 10,000 fpm. Therefore, in an emergency descent, TCAS TA will be tracked by other TCAS II. As you are in TCAS TA mode you will not receive a RA, while an RA will be issued to an intruder, thereby not jeopardising collision avoidance.

TCAS climb can be generated at MAX ALT and shall be followed as the aircraft will have residual rate of climb capability, and if unable to climb (do not descend in this case) the system will generate another resolution in response.

Flying with an inoperative TCAS is permitted, including in RVSM airspace provided it is done in accordance with the applicable MEL. An aircraft that has a TCAS failure during flight can continue to its destination, and thereafter IAW MEL. There is no requirement to declare TCAS failure to ATC, and there is no requirement to annotate in the flight plan that TCAS II is unserviceable.

8.3.7 Fuel and Oil Supply

CAT.OP.MPA.260

The commander shall only commence a flight or continue in the event of in-flight re-planning, when satisfied that the aircraft carries at least the planned amount of usable fuel and oil to safely complete the flight, taking into account the expected operating conditions.

8.3.7.1 In-Flight Fuel Management Policy

AMC1 CAT.OP.MPA.185 (a)

8.3.7.1.1 Basic Fuel Scheme – Procedures for In-flight Fuel Management

1. In-flight fuel checks

- a. Malta Air have established a procedure to ensure that in-flight fuel checks are carried out at regular intervals or at specified points indicated in the operational flight plan (one check at least every 60 minutes) in accordance with OMA.
- b. the remaining usable fuel should be evaluated to:
 - i. compare the actual consumption with the planned consumption;
 - ii. check that the remaining usable fuel is sufficient to complete the flight, in accordance with point (2); and
 - iii. determine the usable fuel that is expected to remain upon landing at the destination aerodrome.

- c. in relation to the recording of relevant data, Malta Air should:
 - i. agree with the TMCAD on what constitutes relevant data for the purpose of recording;
 - ii. use the relevant data as Safety Performance Indicators (SPIs) of the current fuel scheme; and
 - iii. ensure that the recorded data is stored for at least 2 years.

Malta Air have established a procedure for the data to be de-identified to a level that ensures the implementation of a 'just culture'.

2. In-flight fuel management

- a. the flight should be conducted to ensure that the usable fuel expected to remain upon landing at the destination aerodrome is not less than:
 - i. the required alternate fuel plus the FRF; or
 - ii. the FRF if no alternate aerodrome is required.
- b. if an in-flight fuel check shows that the usable fuel expected to remain upon landing at the destination aerodrome is less than:
 - i. the required alternate fuel plus the FRF, the commander should request delay information from the ATC, and take into account the prevailing traffic and operational conditions at the destination aerodrome, at the destination alternate aerodrome, and at any other adequate aerodrome, to decide whether to proceed to the destination aerodrome or to divert in order to perform a safe landing with not less than the FRF; or
 - ii. the FRF, if no destination alternate aerodrome is required, the commander should take appropriate action and proceed to an aerodrome where a safe landing can be made with not less than the FRF.

The use of fuel after flight commencement for objectives other than the ones originally intended during pre-flight planning should require reanalysis and, if applicable, adjustment of the planned operation.

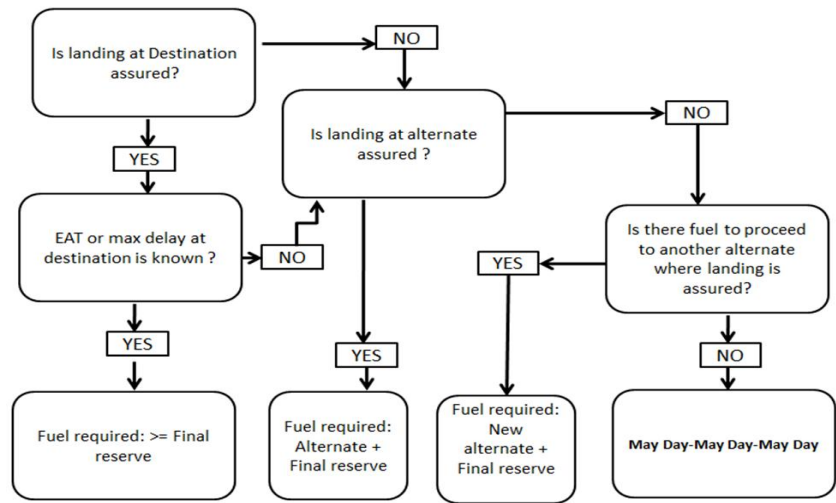
8.3.7.2 Re-planning In-flight

Re-planning in-flight may be conducted when planned operating conditions have changed or other reasons make further adherence to the original flight plan unacceptable or impractical, for example:

- Poor weather conditions or runway conditions at the planned destination and alternate.

- Fuel penalties due to ATC constraints or unfavourable wind.
- Degraded aircraft performance.

8.3.7.3 In-flight Fuel Chart – Options/Actions



Note: To consider a landing to be “assured”, the Commander shall take account of all operational considerations including any forecast deterioration in the weather and plausible single failures of ground and/or airborne facilities e.g. CAT II/III to CAT I.

Where the commander decides to divert, in order to preserve remaining fuel the commander shall advise ATC at the first available opportunity as follows:

“XXX Control, be advised that MAYXXX will be diverting to XXX for operational reasons if approach has not commenced by XX.XX hrs”.

8.3.7.4 MAYDAY – Fuel Emergency

The Commander shall declare a situation of fuel emergency by broadcasting “MAYDAY, MAYDAY, MAYDAY, FUEL, callsign”, when the calculated usable fuel on landing, at the nearest adequate aerodrome where a safe landing can be performed, is less than final reserve fuel.

8.3.7.5 Recording Inflight Fuel Checks

AMC1 CAT.OP.MPA.185 (a)

Actual fuel remaining shall be recorded:

- At or abeam the first convenient waypoint in the cruise.

- At or abeam OFP waypoints at not more than 60 minute intervals.

8.3.7.6 Minimum Fuel

When an impending fuel condition is identified for the current planned flight to intended aerodrome of landing the Commander shall take the following action.

The Commander shall request delay information from ATC when unanticipated circumstances may result in landing at the destination aerodrome with less than the final reserve fuel plus any fuel required to proceed to an alternate aerodrome. “ATC, request EAT, MAY 123”.

‘Minimum Fuel’ Declaration

When committed to land at a specific aerodrome, the commander should take into account any operational factor that may cause a delay to landing, and thus determine whether the aircraft will land with less than the planned FRF, even after receiving clearance from ATC. A change that may cause a delay to landing could be other than ATC, e.g. a change of weather conditions, etc. If any such factor is likely to result in landing with less than the planned FRF, the commander should declare ‘MINIMUM FUEL’ to ATC.

Note 1: The ‘MINIMUM FUEL’ declaration informs ATC that all planned aerodrome options have been reduced to a specific aerodrome of intended landing. It also informs the ATC that any change to the existing clearance may result in landing with less than the planned FRF. This is not an emergency situation but an indication that an emergency situation is possible, should any additional delay occur.

Note 2: The pilot should not expect any form of priority handling as a result of a ‘MINIMUM FUEL’ declaration. However, ATC should advise the flight crew of any additional expected delays, as well as coordinate with other ATC units when transferring the control of the aeroplane, to ensure that the other ATC units are aware of the flight’s fuel state.

It is important to note that a common element in every scenario is that each time MINIMUM FUEL is declared, the Commander has already committed to land at a specific aerodrome and is concerned that a landing may occur with less than final reserve fuel in the tanks. It is equally important to note that, although the coordinated escalation process (with ATC) related to the protection of final reserve typically occurs in 3 steps, each situation is different, and may be resolved at any stage in the process. The 3 steps in the escalation process are:

Step 1 Request delay information when required.	Seek information from ATC concerning any expected delays. “ATC, request EAT, MAY 123”.
Step 2 Declare MINIMUM FUEL when committed to land at a specific aerodrome and any change in the existing clearance may result in a landing with less than planned final reserve fuel.	Declare “ATC, MINIMUM FUEL, MAY 123” which should represent the last lines of defence in a multilayered strategy designed to ensure the protection of final reserve fuel and safe flight completion. Practically speaking, the Commander should declare “MINIMUM FUEL” when, based on the current ATC clearance, the anticipated amount of fuel remaining upon landing at the aerodrome to which the aeroplane is committed is approaching the planned Final Reserve Fuel quantity.
Step 3 The commander shall declare an emergency when calculated usable fuel on landing, at the nearest adequate aerodrome where a safe landing can be performed, is less than final reserve fuel.	Declare Fuel Emergency using “MAYDAY, MAYDAY, MAYDAY, FUEL, MAY 123”. The last in a series of procedural steps to ensure the safe completion of a flight is the declaration of an emergency. The “MAYDAY” declaration is used when all opportunities to protect final reserve fuel have been exploited and in the judgment of the Commander the flight will now land with less than final reserve fuel remaining in the tanks. The word FUEL is used as part of the declaration simply to convey the nature of the emergency to ATC. It is also important to note an emergency declaration not only opens all options for pilots (e.g. available closed runways, military fields, etc.) but it also allows ATC added flexibility in handling an aeroplane.

Note: Minimum Fuel declaration is required by the Standardised Rules of the Air SERA.11012 Minimum Fuel and Fuel Emergency. Crew must submit an ASR following a ‘Minimum Fuel’ call.

Ensuring a Safe Landing – Final Reserve Fuel Protection

The objective of the FRF protection is to ensure that a safe landing is made at any aerodrome when unforeseen circumstances may not allow to safely complete the flight, as originally planned.

The commander should always consider first planning a safe landing option and estimating whether this landing can be performed with more than the FRF. When this estimation indicates that the FRF can no longer be protected, then a fuel emergency should be declared and any landing option explored (e.g. aerodromes not assessed by Malta Air, military aerodromes, closed runways), including deviating from rules, operational procedures, and methods in the interest of safety (in accordance with OMA 1.4.3.1).

8.3.8 Adverse and Potentially Hazardous Atmospheric Conditions

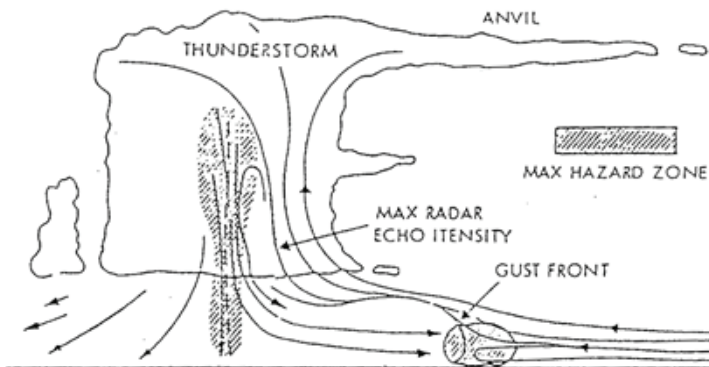
The following definitions are listed:

- Thunderstorms
- Icing Conditions
- Turbulence
- Windshear
- Jet Stream
- Volcanic Ash Clouds
- Heavy precipitation
- Sand Storms
- Mountrain Waves
- Significant temperature Inversions

8.3.8.1 Thunderstorms

1. In the case of detection of a significant cell or storm, the flight crew should observe the following recommendations:
 - To avoid a large storm, the flight crew should make decisions while still 40 nm from it;
 - Deviate upwind instead of downwind of a cell (less probability of turbulence or hail);
 - For storm-avoidance planning, consider the height of the storm:
 - Avoid all yellow, red, or magenta weather radar returns by at least 20 nm
 - Avoid all green, yellow, red, and magenta weather radar returns of cells taller than 28,000 ft by at least 20 nm
 - Cells exceeding 35,000 ft should be considered extremely hazardous and further separation (in addition to the 20 nm) should be used
 - If the top of cell is at or above 25,000 ft, do not overfly it to minimise the possibility of encountering turbulence;
 - Do not attempt to penetrate a cell or clear its top by less than 5000 ft vertically, otherwise the aircraft may encounter severe turbulence;

- Avoid flying under a thunderstorm because of possible windshear, microbursts, severe turbulence, or hail.



If the Commander finds himself committed to flying through an area of thunderstorm activity, the following procedures are recommended.

2. Approaching the Thunderstorm Area

- Ensure that Crew Members' and passengers' safety belts or harnesses are firmly fastened and any loose articles are secured.
- One pilot shall control the aeroplane and the other monitor the flight instruments and electrical supplies continuously.
- Agree with ATC an altitude for penetration whilst ensuring adequate terrain clearance.
- Check the operation of all Anti-icing and De-icing equipment and operate all these systems in accordance with Boeing and/or instructions.
- Follow Boeing and/or Malta Air recommendations on the use of the flight director and autopilot.
- Continue monitoring the weather radar in order to select the safest track for penetration.
- Avoid flying in close proximity to a thunderstorm whenever possible.

3. Air Traffic Control Considerations

A pilot intending to detour around observed weather should obtain clearance from or notify ATC so that separation from other aircraft can be maintained. If for any reason the pilot is unable to contact ATC to inform the controller of his intended action, any manoeuvre shall be limited to the extent necessary to avoid immediate danger and ATC must be informed as soon as possible.

4. Within the Storm Area

Maintain control of the aeroplane whilst concentrating on maintaining a constant pitch attitude appropriate to climb, cruise or descent, by reference to the attitude indicators and avoid harsh or excessive control movements. Do not be misled by conflicting indications on other instruments. Do not allow large attitude excursions in the rolling plane to persist. Attempt to maintain the original heading.

Do not correct for altitude gained or lost through up and down draughts unless absolutely necessary.

Maintain the trim settings and avoid changing the power setting except when necessary to restore margins from stall warning or high speed buffet.

If trim variations due to the autopilot (auto-trim) are large, the autopilot shall be disengaged. Movement of the mach trim, where it occurs, is however necessary and desirable. Check that the yaw-damper remains engaged.

8.3.8.2 Icing Conditions

8.3.8.2.1 General

Boeing Aeroplanes are certified to all applicable airworthiness regulations regarding flight in icing conditions.

The definitive holdover times for approved fluids are those published from the Manual of Aircraft De-icing, (Engineering), which are presented in the Technical Log.

A Commander shall not commence a take-off unless the external surfaces are clear of any deposit which might adversely affect the performance and/or controllability of the aircraft except as permitted in the AFM. A Commander shall not commence a flight under known or expected icing conditions unless the aircraft is certified and equipped to cope with such conditions.

8.3.8.2.2 Icing Conditions

Icing conditions exist when the OAT on the ground or TAT in-flight is 10 degrees C (50 degrees F) or below and visible moisture in any form is present (such as clouds, fog with visibility of one mile or less, rain, snow, sleet, ice crystals), or standing water, slush, ice or snow is present on the ramps, taxiways or runways.

Aircraft in flight can encounter a variety of atmospheric conditions which will individually or in combination produce ice accretions on various components of the aircraft. These conditions include:

1. Supercooled Clouds

Clouds containing water droplets (below 0°C) that have remained in the liquid state. Supercooled water droplets will freeze upon impact with another object. Water droplets can remain in the liquid state at ambient temperatures as low as -40°C. The rate of ice accretion on an aircraft component is dependent upon many factors such as droplet size, cloud liquid water content, ambient temperature, and component size, shape, and velocity;

2. Ice Crystal Clouds

Clouds existing usually at very cold temperatures where moisture has frozen to the solid or crystal state;

3. Mixed Conditions

Clouds at ambient temperatures below 0°C containing a mixture of ice crystals and supercooled water droplets;

4. Freezing Rain and Drizzle

Precipitation existing within clouds or below clouds at ambient temperatures below 0°C where rain droplets remain in the supercooled liquid state.

5. Freezing Fog (FZFG)

Fog (horizontal visibility reduced to less than 1,000 m) consisting mainly of super-cooled droplets that usually deposit rime or glaze on objects or surfaces with below freezing temperatures. The Outside Air Temperature range for freezing fog is -0. 1°C to -30.0°C.

In freezing fog, reported as FZFG in a METAR or ATIS, or when the horizontal visibility is less than 1,000 m with an OAT below 0°C, the aircraft must be anti-iced (or de-iced if such conditions exist) prior to departure.

A cold soaked aircraft with a wing skin temperature below 0°C may in reduced visibility be exposed to similar conditions as in FZFG. The Commander must carefully consider the requirement for de-icing/anti-

icing to ensure that the external surfaces are clear and remain clear of any deposit which might adversely affect the performance and/or controllability of the aircraft.

6. Severe Icing Associated with High Ground

Some incidents of very severe aircraft icing have been experienced in proximity to high ground (the Alps, and Pyrenees); Vigorous developed frontal systems particularly warm fronts (high water vapour concentration) which become slow moving or stationary due to the influence of high ground, shall be treated as capable of generating icing. Fronts that stagnate in mountainous areas may be subject to orographic lifting which develop areas of freezing rain, embedded cumulo-nimbus clouds, intense icing and severe turbulence;

Such icing may be difficult to forecast and when encountered can build-up to several inches extremely rapidly. It may be encountered at low temperatures and higher altitudes where icing of such severity would not be expected;

Heavy icing has also been experienced in mountain wave conditions.

Thrust settings need to be increased during descent in order to ensure adequate protection by maintaining sufficiently high temperature in the engine and wing anti-icing systems.

Types of ice deposits are detailed in 8.2.4.

Aircraft on the ground, while parked or during ground operations, are susceptible to accumulation of ice accretions in conditions peculiar to ground operations in addition to many of the conditions that can be encountered in flight (8.2.4). It is necessary to avoid re-contamination of the aircraft structure and systems after anti-icing/de-icing.

During the pre-flight inspection, the Commander is responsible for ensuring that the aircraft structure is clear of all snow, ice accretion and frost which could affect the safety of the aircraft in flight. The Pre-flight Inspection for contamination by frost, snow or ice prior to or after the de-icing/anti-icing process is set out in Section 8.2.4.

8.3.8.2.3 Procedures in Icing Conditions

The applicable procedures are presented detail in FCOM Vol 1.

8.3.8.2.4 Effects of Ice/Frost Accretion on Aircraft Performance and Flight Characteristics

General

During flight operations ice will form on leading edges of various components, within forward-facing air intakes (e.g. engine inlets) and frontal areas of the airframe. During ground storage or operations, ice will form on other portions of aircraft components such as upper surfaces of wings, fuselages, engine nacelles, horizontal stabiliser surfaces, and control surfaces. The effects that in-flight or ground ice accretions will have on aircraft performance and flight characteristics are many, varied and highly dependent upon aircraft design, ice surface roughness, ice shape, and areas covered. These effects will generally be reflected in the form of decreased thrust, decreased lift, increased drag, increased stall speed, trim changes, and altered stall characteristics and handling qualities. Slight weight increases will also occur, however, the effect of weight increase (with the exception of heavy snow and freezing rain deposits) is usually insignificant in comparison to aerodynamic degradation.

Wind tunnel and flight testing conducted in the past under research, development, and certification efforts, as well as operational experience, has shown that ice accretions on various aircraft components can have very significant and sometimes devastating effects on aircraft equipment operation, aircraft performance, and flight characteristics.

The effects of ice accretions on some of these components and the contribution to degradation of aircraft performance and flight characteristics are itemised on the following list:

1. Slight surface roughness can have significant effects on stall speed, stall behaviour and power required to achieve or to sustain flight;
2. Increased surface roughness, due to ice accretion, on wing leading and trailing edges will produce additional drag and further reduce lift. Leading edge surface roughness is more significant on most aerofoils.

This effect of surface roughness on the leading edge tends to be under estimated generally and is very great on modern aerofoil sections;

3. Stall characteristics will change and, depending upon aircraft design, the nature of ice accretions can cause either violent stall or a slower progression of stall. In some aircraft, pitch-up tendencies may be greater and roll-off tendencies can be exaggerated.
4. Controllability may be reduced requiring more stick deflection for manoeuvres or stall recovery.
5. Power available may be reduced due to ice accretions on propellers or jet engine inlets.

6. Ice or excessive quantities of FPD fluid have been known to cause control surface flutter.
7. Trim effectiveness can deteriorate with the accumulation of ice on unprotected surfaces.
8. Power failures may occur due to intake duct icing, carburettor icing, ingestion of ice particles into jet engines or clogging of fuel tank vents and fuel caps.
9. Severe vibrations may occur due to asymmetric shedding of ice from propellers or rotor blades. Helicopter auto-rotation capability can be significantly changed or lost.
10. Control surfaces such as ailerons, elevators, and wing spoilers can freeze in place if water deposits, snow, and FPD fluids are not properly cleaned or drained from critical areas. Movement of flying and trim control cables, rods or jacks can be impaired by freezing moisture in unpressurised areas, making control surfaces immovable. Attention to sealing and the use of anti-freeze lubricants can ensure protection.
11. Wing flaps may be damaged if retracted with ice accretions adhering to critical areas.
12. Landing gear mechanisms may be damaged or frozen in place if not properly cleared of ice accretions.
13. Forward visibility may be lost or significantly reduced if windscreen anti-icing systems are not available or are not properly utilised.
14. Radio, radar, and other communications and navigation antennae may be damaged or efficiency reduced due to ice accretions.
15. Ventilation, air conditioning, and other air inlets can be blocked or flow restricted.
16. Ice dislodged from fuselage or wing sections, antennae and other components forward of engine inlets and other critical components can produce damage.
17. Ice accretions, under certain conditions, may not have noticeable effects on aircraft performance and flight characteristics, however, these effects may become quite apparent in the event of engine failure or other emergencies.
18. Flight, engine, and other instruments are subject to error if ice accretions exist on external probes, in pressure lines, or on areas forward of or adjacent to external probes. Operational experience indicates that typical sources of error are icing of pitot-static probes used for airspeed, altitude, and engine pressure ratio measurements.

19. Automatic systems that utilise signal sources from external sensors such as Automatic Flight Control Systems, autothrottle speed command systems, or stability augmentation systems may be adversely affected by ice accretions on or in the vicinity of the sensors.

Residual moisture on door and cargo hatch seals may freeze under certain conditions causing leaks or seal damage, and the inability to open doors in an emergency.

8.3.8.2.5 Extended Operations in Icing Conditions

Although the process of certifying jet transport aircraft for operation in icing conditions involves many conservative practices, these practices have never intended to validate unlimited operations in severe icing conditions.

The safest course of action is to avoid prolonged operations in moderate to severe icing conditions.

8.3.8.2.6 CSFF

This is a condition whereby a thin deposit of hoar frost forms on surfaces which have been cold-soaked. In particular this hoar frost coating forms on the lower wing surface and on the upper wing surface above the fuel tanks, which contain very cold fuel (cold-soaked) as a result of extended operation in very cold temperatures.

It is a major safety issue that crews are aware of the effects of any precipitation falling on surfaces already cold-soaked, and the de/anti icing procedures contained in the FCOM Vol 1, in conjunction with the published holdover tables SHALL be applied.

8.3.8.2.7 Operation from Contaminated Runways and Taxiways

8.3.8.2.7.1 General

This section covers the general precautions and operating procedures that shall be taken and adhered to when operating from contaminated airport surfaces. Specific limitations, performance reductions, and operating procedures specific to aircraft type are contained the respective Flight Manual, and FCOM Vol 1, and QRH. Definitions of Contaminated Runways, Runway Surface Conditions including contaminant types and guidance material for Take-Off and Landing Performance Assessments are contained in Operations Manual Part B – Performance. The operational issues arising are detailed in the Ryanair Winter Operations Manual.

8.3.8.2.7.2 SNOTAMs

Information on runway and taxiway state is provided by a SNOTAM message transmitted by AIS.

Details of the SNOTAM message format and the codes used are contained in the Winter Operations Guide.

8.3.8.2.7.3 Braking Action

A subjective description of airplane stopping capability used by flight crew to characterise the deceleration associated with the wheel braking effort and directional controllability of the aircraft.

Maximum manoeuvring speed on the ground with RWYCC 3 (MEDIUM) or below shall be 5 kt.

8.3.8.3 Turbulence

No flights shall be planned through areas with severe turbulence. For procedures refer to FCOM Vol 1 and SEP.

Light:

- **Light Turbulence:** Slight changes in attitude or altitude or changes in IAS of 5-15 kts.
- **Light Chop:** Slight bumpiness without changes in attitude, altitude or IAS. Occupants may feel a slight strain against seat belts though no difficulty is encountered in walking.

Moderate:

- **Moderate Turbulence:** Turbulence that is similar to light turbulence but of greater intensity, changes in attitude and or altitude occur, IAS fluctuates 16-25 kts but the aircraft remains in positive control at all times.
- **Moderate Chop:** Turbulence that is similar to light chop but of greater intensity causing rapid bumps or jolts without appreciable changes in attitude or altitude. Occupants feel definite strain against seat belts or shoulder harness. Unsecured objects are dislodged.

Food service and walking are difficult.

Severe:

- **Severe Turbulence:** Turbulence that causes large, abrupt changes in attitude or altitude. Aircraft may be momentarily out of control. IAS fluctuates by more than 25 kts. Occupants are forced against seat belts or shoulder harness. Unsecured objects are tossed about. Food service and walking are impossible.

The continuity of turbulence will be described as:

- Occasional – less than 1/3 of the time.
- Intermittent – 1/3 to 2/3 of the time.
- Continuous – more than 2/3 of the time.

Severe turbulence must be reported to ATC giving position, time (GMT), FL/Altitude, A/C type, intensity, in or near cloud and duration. If severe turbulence is encountered an ASR must be filed and a Technical Log entry made. The Technical Log page reference shall be included in the ASR to facilitate Engineering admin.

8.3.8.4 Windshear

Wind shear is a variation in wind velocity (direction and/or speed) with distance in the vertical or horizontal plane. It can affect airspeed whenever an aircraft transverse from one wind velocity condition to another in less time than it takes for the aircraft to become adjusted in speed to the new environment.

Windshear

Windshear encountered on take-off or approach must be reported to ATC immediately by R/T if considered a hazard, giving details of height and severity. Severe windshear is considered to be uncontrollable changes from normal steady flight conditions below 1000 AGL, in excess of the following:

- 15 kts indicated airspeed.
- 500 fpm vertical speed.
- 5° pitch attitude
- 1 dot displacement from the glideslope/glidepath.
- Unusual thrust levels for a significant period of time.

If severe windshear is reported or forecast – **DO NOT TAKEOFF.**

Full thrust must be used if the nature or severity of the windshear cannot be determined.

Reduced thrust may be used if any windshear expected during or after take-off is caused by strong winds not associated with CBs or frontal activity.

Windshear Escape Manoeuvre

On aircraft fitted with aural windshear warning systems, if a GPWS “Windshear” warning is received, the windshear escape manoeuvre must be carried out, QRH MAN section, Windshear.

Exceptionally, in conditions of extreme turbulence associated with strong winds and not Cb or frontal activity, or areas of known spurious predictive windshear activity as identified in Boeing Bulletin or Malta Air/Ryanair Airfield Brief, a predictive windshear warning may be treated as spurious provided that this warning has been anticipated and briefed for in the Approach Briefing and appropriate allowance made.

If a subsequent approach is considered prudent, then the conditions shall be carefully evaluated and appropriate compensation made. If, on the subsequent approach, another warning is received and believed to be genuine, then the escape manoeuvre shall be made and a diversion considered. For restrictions on the number of approach attempts refer to OMA 8.3.0.3.10.1.

8.3.8.5 Jetstream

Jet streams are fast flowing, narrow air currents found in the Earth’s atmosphere. The main jet streams are located near the altitude of the tropopause, the transition between the troposphere and the stratosphere.

Jetstreams are indicated on SIG WX charts. Avoid flying along the edge of jetstreams due to the possibility of associated turbulence. Pilots will be aware of the effect of increased fuel consumption due to unexpected significant head wind components that can be experienced. It may be possible to avoid jetstreams by changing route and/or altitude.

8.3.8.6 Volcanic Ash Cloud

ORO.GEN.200

Volcanic Ash

Flight in visible/discernible ash must be avoided. When operating in areas where volcanic activity may occur pilots shall be vigilant and aware of their responsibilities to report any activity immediately. Flights planned near or through any areas of forecast contamination will comply with the Malta Air Safety Risk assessment accepted by the TM CAD. The operating crew will be provided with the latest information. The responsibility to avoid flight in visible/discernible ash remains with the operating Commander.

8.3.8.6.1 Operations in the Context of Volcanic Ash; Overflight and Underflight of Malta Air NFZs

Malta Air will not plan or intentionally conduct operations in areas known to contain visible or discernible Volcanic Ash (VA).

Threat and Error Management (TEM)

The generic threat is unintentional operations in airspace or aerodromes with known or predicted contamination by a volcanic ash cloud with characteristics harmful to the airworthiness and operation of the B737.

Arising from this generic hazard are specific risks to Malta Air's operation. These occur in maintenance, flight planning/dispatch and flight operations. A list of assessed hazards is located in the Hazard/Risk Identification & Assessment Register maintained by the Safety and Security Manager.

The Malta Air approach to managing its operations during a VA event is based on the TEM concepts of **Anticipation**, **Recognition** and **Recovery**. Anticipating potential hazards and risks associated with a future VA event builds vigilance at all levels in the company. Recognition of the existence of a threat allows FOPS and flight crew to deploy procedures and utilize their training to avoid the error of unintentional flight in levels of VA deemed to be harmful to the aircraft and its occupants.

The following paragraphs are grouped so that the activity undertaken by Flight Operations (FOPS) and Flight Crew are stated in each of the above categories.

In all cases, the fundamental principle underlying operations in the context of VA is to avoid flight in visible or discernible volcanic ash. Discernible ash is ash that can be sensed by human sight or smell or other senses.

8.3.8.6.2 Anticipation

8.3.8.6.2.1 Malta Air No Fly Zones

From the commencement of the Eruptive Stage of a VA event Malta Air will impose the following limits to its area of operations:

- A Malta Air No Fly Zone will be applied where the dimensions of a VAA/VAG overlap those of the WSI VA SIGMET. This combined graphic will be promulgated on a Malta Air No Fly Zone Chart. See below.
- Where a WSI VA SIGMET is not overlapped by the VAG Malta Air will regard this as a NFZ until the following has been established following discussion with a WSI forecaster:
 - The basis for the existence of the area not coinciding with the VAG,
 - When the forecast was made,

- Whether any new information has been considered (PIREPS, etc.) since the forecast,
- The level of confidence that WSI have in the accuracy of their forecast,
- An assessment of the risk of operating into that area.
- Following this analysis with WSI, the NPFO will conduct a risk assessment of operations onto this area of the WSI SIGMET. If the risk assessment results in a Tolerable or Acceptable category, then the operation may be planned and conducted. Otherwise this area of the WSI VA SIGMET will be regarded as a Malta Air No Fly Zone.
- The area in the VAG that does not coincide with the WSI SIGMET will be available to Malta Air operations for the planning and execution of flights. However, given that the VAAC has forecasted that some unspecified concentration of ash might be present in this area the following extra measures will be taken:
 - Malta Air will use Safety Reporting System to analyse Malta Air ASRs for reports of ash throughout its network.
 - Malta Air will send an SMS message and e mail to all Malta Air Base Captains that will require reports of any VA activity (or the absence of any VA activity) in their base to be submitted as a PIREP.
 - Industry reported PIREPs and other actual sightings of VA will be reviewed. Reports of No ash in areas where it had been forecast will be reviewed.
 - The Volcanic Ash Advisory Group daily web conference will be used to inform flight operations of the conditions within this area.
 - Flight crew will be provided with the following information.
 - ◆ The Malta Air No-Fly Zone as depicted on the Malta Air VA NOTAM.
 - ◆ Volcanic Ash Advisories and Graphics.
 - ◆ Intervened modelled ash concentration charts (if available, and for supplementary reference only).

Malta Air is prohibited from operating in any Initial Danger Area as defined by an Area Control Centre (ACC) in accordance with ICAO Doc 019 – Volcanic Ash Contingency Plan EUR and NAT Regions, Section 1 Alerting Phase, Paragraph 1.2.1.

8.3.8.6.2.2 Accepted Safety Risk Assessments (SRA)

The TM CAD has accepted SRA proposals from Malta Air which permit:

- Malta Air to operate in areas forecast to be contaminated with VA contamination indicated on the applicable VAA/VAG that do not overlap with a WSI VA SIGMET.
- Malta Air to fly under and to overfly a NFZ when required, e.g. the vertical extent of the applicable zone may be from FL200 to FL350, and operations below FL200 and above FL350 are approved. The requirements of OMA Section 8.1.1 – Minimum Flight Altitudes apply to all flights conducted under the NFZ.

8.3.8.6.2.3 Training

OMD Section 3.1.4.3 contains a requirement for Malta Air flight crew to be trained in the emergency Volcanic Ash procedure as detailed in the B737 QRH during the recurrent three year simulator training cycle. The training can be either theoretical or simulator based at the discretion of the HT in consultation with the NPFO.

FOPS controllers are trained in the use of CFMU, the associated WSI FUSION airspace management tool and procedures specifically designed to manage operations in the context of no-fly zones during a VA event.

8.3.8.6.2.4 FOPS Control

A detailed VA Event Response Procedure has been prepared that commences with the ICAO defined Pre-Eruption stage. See FOPS Volcanic Ash Event Response Procedure. The NPFO has overall responsibility for ensuring the safety of flight operations. For the duration of a VA event, a Suitably Qualified (SQ) Captain will be available to FOPS planning to assist in data interpretation and dissemination.

8.3.8.6.2.5 Pre Flight

Volcanic Ash Advisories and Graphs are those issued by Met Office VAAC which are published for various vertical blocks of airspace. Volcanic Ash information for pilots are available through Volcanic ASHTAMs and advisories as well as NOTAMs. These and the VA SIGMET are available through the LIDO website or company publication available to pilots.

A Malta Air VA NOTAM and RYR No Fly Zone Chart will be published by FOPS on Ryanair Connect within hours of a significant eruption. All charts must be downloaded, studied and included in flight briefing material carried on board.

8.3.8.6.2.6 Escape Routes

Malta Air policy is to circumnavigate any VASH NFZ. Flight Operations Planning will implement this policy during any VASH event. Overflight may be planned on a case-by-case basis, OCC flight planners shall ensure 2000 ft vertical clearance above the NFZ in the event of driftdown. Where avoidance of NFZ requires delayed climb or early descent, this will normally be reflected in the OFP. In all cases verify that the revised FL assures clearance above MSA.

Malta Air will endeavour to minimize flight over NFZ zones and flight plan accordingly. Ideally crews shall avoid or minimize flight operations over a NFZ where possible. To minimize time under/over a NFZ in an emergency or atmospheric event an ETP shall be calculated and displayed on the FMC and ND. This process shall be accomplished as described below for each transit of a NFZ en-route.

Airports in the NFZ will not be considered as suitable destinations or alternates at the planning stage with this in mind all flights shall be planned to be the diversion distance of one hour flying time at an OEI cruising speed of suitable Airports at any point during the flight.

If weather avoidance is anticipated in the area below a NFZ do not plan to overfly weather to the extent that the NFZ would be penetrated. Consider 180 turn, reroute or divert. Do not climb into a NFZ.

Where turbulence Shear Value 5 or greater is predicted in the section of flight above the NFZ and performance requirements cannot be assured, plan to re file clear of the area of turbulence. In the event of an inadvertent CAT encounter, follow manufacturer's procedures including descent if required. Where possible, avoid flight in NFZ unless no other option is available.

Crew shall refer to 'Drift-down' tables from Operation Manual Part B – Performance Section 2.6.1 which are provided for all engine and airframe configurations used by Malta Air. These tables provide guidance on the 'Distance to Drift-down in Nautical Miles' and 'Maximum Ash Cloud with Nautical Miles'.

8.3.8.6.2.7 Flight Crew Threat and Error Management (TEM)

Crews must have heightened vigilance and apply good TEM techniques, especially Anticipation and Recognition to ensure a safe operation. Procedures designed to mitigate the effect of predicted concentrations of VA can themselves become a hazard with associated risk. An example of this occurs when aircraft may have been sealed to prevent penetration of VA while parked.

The engineering department may have sealed all ports and orifices of aircraft that have been parked in areas where VA may have fallen to the ground. These seals will be removed by engineers using an approved procedure. When conducting an

Exterior Inspection, Flight Crew must adhere to the requirements of FCOM VOL 1 NP Exterior Inspection to ensure that their responsibilities are fulfilled and that any omissions by an engineer will be trapped and rectified.

8.3.8.6.2.8 Hazard Location

While crews can be confident that their route and flight level are safe and free from hazardous levels of VA, VA levels may be higher than normal. Accordingly, crews must apply TEM and be vigilant complying with related checklists and be fully briefed on the advisory information provided by FOPS. Airborne weather radar systems used on commercial aircraft are not designed to detect volcanic dust and in general will not and cannot be relied on to do so.

8.3.8.6.2.9 Avoidance

Crews are reminded to be vigilant against the possibility of an inadvertent encounter with volcanic ash. Maintain a close listening watch for PIREPs or reports from ATC. Flight operations in a RYR NFZ or in known areas of VA are prohibited except in non-normal situations.

8.3.8.6.2.10 Reporting

- Crew members are required to advise ATC of any volcanic ash encounter (PIREP).
- Operations in areas of predicted volcanic ash must be reported by ASR-Volcanic Ash – detailing the extent of the time spent in the relevant zone.
- Operations which are the subject of an actual volcanic ash encounter must be reported by ASR-Volcanic Ash. The encounter must be recorded in the aircraft Technical Log and MAINTROL advised.
- Operations outside a RYR NFZ but within an area of predicted VA by the VAAC where no volcanic ash encounter has occurred shall be reported as set out above.

Note: It is important to establish where ash has been predicted but has not been encountered.

- Even if an aircraft enters a NFZ and no volcanic ash encounter has been established, a Technical Log entry shall be made to initiate post flight maintenance action.

8.3.8.6.2.11 Legalities

Flight crew must exercise increased diligence with regards to the status of the aircraft's Pre Flight Inspection Status and the Hold Item List in the Aircraft Technical Log. If there is any doubt regarding the status of the aircraft's airworthiness or compliance with routine maintenance procedures then Maintrol must be contacted immediately.

If crews are disrupted and/or are located away from base for any period of time they must exercise increased diligence with regards to the status of their Checks as required by regulation. These include, but are not limited to Medical, Line Check, LPC, OPC and recurrent Safety Training. If there is any doubt regarding the status of your Checks you must contact Crew Control immediately prior to operating.

8.3.8.6.2.12 Review

The review process will operate as follows:

- At least twice a day the NPFO and the Deputy Director Flight Operations (or nominated deputies) will review the following prior to deciding on the extent of schedule change to the next "wave":
 - The Volcanic Ash Advisories and Graphs
 - The WSI VA SIGMET
 - The plot of the RYR NFZs
 - ASRs and VA Reports
- PIREPS from other operators.
- Input from the TM CAD, EACCC conference calls, VAAG Webex conferences and other such bodies.
- Malta Air will conduct further risk assessments as warranted by the situation or a change to the circumstances underpinning the current approval.

8.3.8.6.3 Recognition**8.3.8.6.3.1 Visible or Discernible Ash**

Flight in visible or discernible volcanic ash must be avoided. At night and in IMC the Inboard Landing Lights must be selected ON in all areas predicted to contain VA.

Discernible ash is ash that can be sensed by human sight or smell or other senses. Ash may not be visible during darkness or daytime IMC. However, ash can be discerned by odour, haze, changing engine conditions, airspeed, pressurization and static discharges.

Table X: The following table provides detailed guidance regarding visible or discernible ash in all conditions.

Table X Visible or discernible volcanic ash	Detectable signs of ash		
	Day	IMC	Night
Odor similar to electrical smoke or sulfur	X	X	X
Smoke/dust in cockpit (usually coming from the A/C system)	X	X	X
Ash clouds	X		X ¹
St. Elmo’s fire		X	X
Static discharges all around the windshield		X	X
Orange glow in the engine inlets			X
Landing lights cast sharp, distinct shadows on the volcanic ash cloud as opposed to the normally fuzzy, indistinct shadows cast on water/ice clouds			X
Strobe light reflections			X
“Frosting” of windows	X	X	X
¹ May be visible during night VMC with landing lights on but at a shorter distance.			

8.3.8.6.3.2 Low and Medium Levels of VA

- Static discharges. A phenomenon similar to St. Elmo’s fire or glow can occur. In these instances, blue-colored sparks can appear to flow up the outside of the windshield or a white or orange glow can appear at the leading edges of the wings or at the front of the engine inlets.
- Smoke/dust/haze. Most flight crews, as well as cabin crew or passengers, see a haze develop within the airplane. Dust can settle on surfaces.
- Odor. When encountering a volcanic ash cloud, flight crews usually notice a smoky or acrid odor that can smell like electrical smoke, burned dust, or sulphor.
- Frosting of windows. Caused by the abrasive action of the ash on the windshields.

8.3.8.6.3.3 Active Ash Cloud

- An active ash cloud will be visible from a considerable distance.
- Changing engine conditions. Surging, torching from the tailpipe and flameouts can occur. Engine temperatures can change unexpectedly, and a bright orange glow can appear at the engine inlet.
- Airspeed. If volcanic ash fouls the pitot tube, the indicated airspeed can decrease or fluctuate erratically.

- Pressurization. Cabin pressure can change, including possible loss of cabin pressurization.

8.3.8.6.3.4 Post Flight

Malta Air has implemented CFM56-7B S/B 72-0789, Revision 02, December 14, 2011/LEAP-1B-72-00-0004-01A-930A-D. Accordingly, Malta Air has identified selected engines to their CFM representative for inclusion in a volcanic ash ingestion leading indicator program. Aircraft fitted with these selected engines can be identified by means of a Notice to Crew in the Aircraft Technical Log.

For the selected engines, the Commander must ensure that the following visual inspections after each flight in concentrations of volcanic ash are completed. Visually inspect the fan blades, inlet cowl, Outlet Guide Vanes (OGVs), spinner cone, splitter fairing, acoustical panels, and exhaust cowl for unusual damage or evidence of volcanic ash associated with erosion on the powerplant. Report unusual findings to Maintrol.

Pilots are also required to pay particular attention to windscreens and landing lights when performing exterior inspections, to look for any new signs of abrasion and to seek maintenance advice if they suspected that VA was encountered on the inbound flight.

8.3.8.6.3.5 FOPS Control

FOPS Control will be informed of a VA event by any or all of the agencies specified in the VA Event Response Procedure which will be actioned immediately under the supervision of the Deputy Director Flight Operations.

8.3.8.6.4 Recovery

8.3.8.6.4.1 VA Encounter

Whenever an ash encounter is confirmed when airborne, the QRH Volcanic Ash checklist shall be completed in its appropriate sequence. The following procedures, compatible with the B737 QRH, have been communicated to the TM CAD:

8.3.8.6.4.2 Commence the Manoeuvre (PF)

1. MCP HDG: Select reciprocal heading
2. HDG SEL: Engage
3. MCP ALT: Select Lower Altitude
4. LVL CHG: Engage
5. A/T: Disengage
6. THR Levers: Close

Exit the ash cloud as quickly as possible. Executing a 180-deg turn out of the ash cloud using a descending turn is the quickest exit strategy. Many ash clouds extend for hundreds of miles, so assuming that the encounter will end shortly can be false. Climbing out of the ash could result in increased engine debris build up as a result of increased EGT. The increased engine build up can cause total thrust loss.

8.3.8.6.4.3 Communicate (PM)

- Make a Mayday call. Inform ATC of intended heading and level. (PM)
- Set 7700 and select TA on the transponder mode selector. (PM)

8.3.8.6.4.4 Pan or Mayday

- The least level of distress declared will be a Mayday call. The MAYDAY can be downgraded to a PAN call if the aircraft's condition or its impact on other traffic as a result of the above manoeuvre does not endanger it or other traffic.

8.3.8.6.4.5 Heading and Flight Level

- When the encounter is momentary in nature and there are sufficient visual indications and it can be confirmed that the aircraft has cleared the VA, the Commander may elect to continue flying a course which is deemed to be free of VA.
- The crew are directed by the QRH to consider flying 180 degrees from the heading that existed at the time of the VA encounter. Malta Air crew will state the reciprocal heading that has been selected and activated. A crew shall initially select a heading 180 degrees away and remain on the new heading (terrain permitting) until it can be confirmed that the aircraft is clear of ash or other manoeuvring is required.
- Malta Air crew will confirm with ATC when they are clear of the ash, their intentions and when they can accept vectors again.
- Malta Air crew will state the required flight level/altitude that has been selected and activated. ATC may treat this manoeuvre as an Emergency Descent. In this case controllers may have to initiate emergency separation between following aircraft on the same track in order to provide separation between these aircraft and the aircraft turning and descending away from the ash. This is a matter for ATC to resolve and Malta Air crew will proceed as described above.
- The rate of descent will be very much less than that associated with a pressurisation failure induced emergency descent. This is because retaining altitude to increase time available is crucial to the success of the VA QRH checklist. Therefore the aircraft will descend at idle thrust, and at the speed that obtained at the time of the VA encounter. In order to reduce rate of descent and retain altitude, this speed may be reduced. This will give a variable rate of descent and no standard rate of descent can be predicted.

8.3.8.6.4.6 QRH Volcanic Ash Checklist

- Do the Volcanic Ash emergency checklist in the Boeing 737 QRH. Below is background information to the steps in the NNC intended to give crews a better understanding of the required actions. The main objective is to exit the ash cloud as quickly as possible and restart engines if needed.
- QRH Item 1: If volcanic dust fills the flight deck, the crew may need to use oxygen. Use flight deck oxygen at the 100 per cent setting.
- QRH Item 3: Turn the autothrottle off. This prevents the engines from increasing thrust above idle. Ash debris in the engine can result in reduced surge margins, and limiting the number of thrust adjustments improves the chances of engine recovery. If conditions allow, reduce thrust to idle immediately. By reducing thrust, engines may suffer less build up of molten debris on turbine blades and hot - section components. This reduces possible engine damage or flameout, or both, by decreasing EGT. Idle thrust allows engines to continue producing electrical power, bleed air for pressurization, and hydraulic power for airplane control.
- QRH Item 5: Turn ignition to FLT. In the event that the engines flame out or stall, use appropriate procedures to restart the engines. During restart, the engines may take longer than normal to reach idle thrust due to the combined effects of high altitude and volcanic ash ingestion. If an engine fails to start, try restarting it again immediately.
- QRH Items 7 & 8: Turn on engine and wing anti-ice and set air-conditioning packs to HIGH flow. These actions improve the engine stall margins by increasing the flow of bleed air.
- QRH Items 9: If possible, start the Auxiliary Power Unit (APU). The APU can power systems in the event of a multiple-engine power loss. It can also be used to restart engines through the use of APU bleed air.
- QRH Item 10: Monitor engine Exhaust Gas Temperature (EGT). Engines may accelerate to idle very slowly, especially at high altitudes. Slow acceleration may be incorrectly interpreted as a hung start or an engine malfunction. If N2 is steadily increasing, and EGT stays within limits, the start is progressing normally.

8.3.8.6.4.7 TCAS

- Follow QRH procedure and select TA.

8.3.8.6.4.8 NATS Policy**Safety and Procedural Considerations**

- Within the various contamination areas it is for the aircraft operator and aircraft commander to determine that it is safe to operate the aircraft in such airspace. ATS providers are not required to question a pilot's approval to fly through a Volcanic Ash Temporary Danger Area (TDA).

- ATS providers are to ensure that ATCOs and FISOs are aware that, should an aircraft unexpectedly encounter a significant concentration of volcanic ash, the pilot may (possibly without prior warning):
 - Execute a 180° turn;
 - Descend;
 - Reduce engine power; and
 - Disconnect auto-throttle.
- Pilots operating within an area of contamination may decline a vector or climb/descent instruction if that would take the aircraft into an area of higher contamination and if that would be contrary to the aircraft type and engine pairing tolerance level. A pilot may also request to leave an area if the aircraft is unable to sustain flight within the area. ATCOs and FISOs should accommodate any such request as expeditiously as operational safety considerations allow.

8.3.8.6.4.9 If the windscreen is obscured, and auto land capability is not available, it may be necessary to fly what is effectively an IMC approach and open the left or right Number 2 window to assess the landing flare.

8.3.8.6.4.10 Non Normal or Emergency Procedures

In the event of a depressurization on a flight operating over a NFZ, an emergency descent may penetrate the NFZ. It is important to understand that flight in a NFZ may, or may not, result in a volcanic ash encounter. The only circumstances in which a Malta Air aircraft would intentionally be in a NFZ is as a result of a prescribed action to address a current emergency situation. In the case of depressurization an emergency descent in accordance with manufacturer's procedures must be performed. Statistical data would indicate that the probability of a depressurization is very low.

Equally, statistical data would indicate that the probability of a VA encounter is very low. Therefore the possibility of a VA encounter combined with a depressurization is statistically highly unlikely.

In summary, various threat and associated recovery techniques are explained in the table below:

Threat	Recovery
Uncontained Fire on Board	Apply Manufacturers Procedure, Emergency Descent, Land at nearest suitable Airport
Engine Fire Severe Damage, Separation	Apply Manufacturers procedures, Drift-down, standard method Land at nearest suitable Airport
Engine failure	Apply Manufacturers procedures, Drift-down standard method Land at nearest suitable airport
Depressurization (slow or explosive)	Apply the Manufacturers procedure, Emergency Descent to 10,000 ft or MSA if higher, land at nearest suitable airport
Severe Turbulence	Apply Manufacturers procedure
Bomb Scare	Apply Company Procedures
Pilot Incapacitation Medical Emergency	Apply Company Procedures Land at nearest Suitable Airport
Crew/Passenger Medical Emergency	Apply Company Procedures
Pack Failure	1 Pack operation certified to maintain cabin pressurisation at current Flight level up to FL 410, no descent should be required in this case
Volcanic ash encounter	Apply Manufacturers procedure
Pitot Static System, Other malfunctions	Apply Manufacturers procedures
Nav System Failure	Declare Pan and request Radar vectoring away from the NFZ

8.3.8.7 Heavy Precipitation

Heavy precipitation can quickly lead to high levels of runway contamination so runway clearance/drainage rate must be closely monitored in order to assess if a diversion is necessary.

When encountered whilst in flight, heavy precipitation can be associated with significant downdrafts and windshear. The specific procedures in the FCOM Vol.1 Limitations apply.

8.3.8.8 Operation in a Sandy or Dusty Environment**Sandstorms**

Dust or sand is raised from the ground by the wind and carried upwards to a height depending on the meteorological conditions and the size of the particles. Whenever the deterioration in visibility is serious the phenomenon is generally known as a dust storm or sand storm.

Large sand particles are too heavy to be lifted far and some sandstorms consist merely of this surface sand, no more than three or four feet deep. To an aircraft coming in to land the surface may be blanketed but hangars/buildings stand out.

Small sand/dust particles can be carried up through the whole turbulent layer to some 3,000 ft and remain in suspension as long as the wind blows with sufficient strength. This type of storm can be prolonged and widespread, or it may occur in patches in association with minor squalls of wind. The diurnal variation of windspeed is usually marked in the regions affected and such storms rarely occur at night.

Dust/haze exists when fine particles raised during dust storms are gradually dissipating in the atmosphere. This gives a general haziness but rarely results in visibility less than 1,000 m.

A HABOOB is a severe sand/dust storm. The term is now restricted to summer dust storms originating with CB clouds. These storms can be seen approaching, usually at considerable distances airborne, as a solid wall of dust. Its arrival is characterised by a sudden increase of wind and a rapid deterioration of visibility.

A more widespread type of dust storm occurs with strong winds over a long area of desert. Both the strength of the wind and instability of the lower layers contribute to the dust laden atmosphere. Such storms can continue for many hours and cover a considerable area, visibility being reduced to hundreds, sometimes tens of metres, while they may extend upwards several thousand feet. These may continue well into the night although they are then generally less severe.

Dust Devils and sand-pillars are small whirlwinds formed by convection. They are usually quite small and can exceed gale force. These shall be avoided.

Sandstorms/dust storms constitute serious hazards to aircraft, and shall be avoided.

Operation in dust/sand storm poses a high risk of aircraft damage and may not be obvious to operating crew. Damage to engines, electronic equipment, pitot tubes and air conditioning systems is not unusual following a dust or sand encounter.

Should inadvertent aircraft exposure to Dust/Sandstorms occur enroute or whilst on the ground, contact Maintrol and the Duty Pilot prior to operating, file an ASR at the end of the duty.

The following dispatch and enroute planning criteria apply:

Dispatch:

OM(A) 8.1.2 applies. If an airport is forecasting Sandstorms (SS) or Dust storms (DS) +/-1 hour of ETA, the aircraft may dispatch to the airport so long as two suitable alternates are available. Crew should select the furthest suitable ALTN for the OFP and uplift fuel accordingly.

If conducting ground operations at an airport experiencing moderate to heavy DS/SS:

- i. Confirm local disembarkation/embarkation restrictions with dispatcher DO NOT commence boarding if boarding will necessitate the use of APU for Air conditioning.
- ii. Avoid APU usage if possible. If required, operate APU as per FCOM SP 16 Operation in a Sandy or Dusty Environment.
- iii. Do not start engines or attempt dispatch during active +DS/+SS or when visibilities reduce below 600 m.
- iv. Do not taxi on taxiways which are contaminated with drifting sand.
- v. Contact MAINTROL and the Duty Pilot for guidance.

Enroute:

It is prohibited to commence an approach at an airport under the following conditions:

- i. Active Heavy Sandstorms (+SS) or Heavy Dust storms (+DS) reported, or
- ii. Reported visibilities below 600 m or as per approach visibility requirements, whichever is greater.

8.3.8.9 Mountain Waves**General**

Mountain wave turbulence is a vertical air mass disturbance, the predominant feature of which is a long wave like movement of the air, but within which higher frequency gusting may also occur. It is caused by strong winds blowing over and across high mountain ranges such as the Alps and characterised by complete upward and downward movement of the large mass of air covering a wide area. Where the mountains are very high the wave effects extend to very much higher altitudes, in certain cases to in excess of 40,000 ft. The blocking effect of the mountains causes a rising air motion and acceleration on the windward side and a descending motion and deceleration of the wind as the wave subsides to leeward.

The possibility of wave turbulence can be assessed by considering the upper wind strength and direction in relation to the orientation of the mountain range with the most likely geographical location to encounter turbulence being downwind of the mountain mass.

Handling Procedures

The correct procedure for flying the aircraft in turbulence/severe turbulence/mountain wave conditions is described in detail in the Airplane Flight manual and in FCOM Vol 1, SP 16. The operational procedures for Jet Upset Recovery are presented in the QRH: Upset recovery Manoeuvre.

8.3.8.10 Significant Temperature Inversions

The most serious manifestation of significant temperature inversions is in the take-off/climb phase. Crews will notice higher than normal (up to and including red line EGT indications) in the initial climb, occasionally requiring thrust reduction to EGT limits, until clear of the inversion layer. This may present performance issues.

The maximum cruising altitude capability of the aeroplane can be significantly reduced if a temperature inversion, even of small magnitude, exists in the upper levels. This may prevent an aeroplane reaching its preferred cruising altitude. Should an aeroplane encounter an area of inversion once in the cruise at limiting altitude its buffet margins may be so eroded that a descent is necessary.

The winds above and below the inversion are significantly different. The 2,000 ft wind in NYO on approach to RW 09 is frequently a strong tailwind, and below the inversion becomes a headwind. The unexpected and unforecast tailwind generally presents added difficulty in slowing down during the approach.

Temperature inversions at lower levels in the atmosphere are frequently associated with deteriorating visibility and can prevent the clearance of fog for prolonged periods.

8.3.9 Wake Turbulence

8.3.9.1 Causes of Wake Turbulence

Aircraft wake turbulence is caused by wing tip vortices generated by the production of aerodynamic lift and by jet efflux. Wing tip vortices behind a heavy transport aircraft can have sufficient intensity to cause control difficulties and the potential for a turbulence upset for a lighter aircraft which encounters them. They are consequently considered to be a safety hazard which shall be avoided or the effects minimised. Jet efflux does not create a significant hazard in the context of commercial aircraft operation. Care needs to be taken by larger aircraft to avoid jet blast damage while manoeuvring on the ramp, taxiing or during engine ground runs.

The intensity of vortex turbulence is directly proportional to aircraft weight and increases at high angles of attack (i.e. as airspeed is decreased). Thus the most severe vortex activity is generated and most likely to be encountered in the vicinity of the airfield as, in these areas, airspeeds will be low and aircraft taking off will be at maximum weight. Vortex generation starts at the lift-off point with rotation for take-off and stops on landing as the nosewheel contacts the runway.

Vortices are formed at the aircraft wing tip and rotate in the direction shown in Figure 1. They become fully established about 200 – 400 ft behind a large aircraft and have a downward sinking motion. When generated close to the ground, the vortices move downwards to ground level and then spread out laterally on either side of the runway in still air. In still air conditions, vortices will persist for several minutes while their energy slowly dissipates. In windy or turbulent conditions, this dissipation is more rapid. With a crosswind, the two vortices will move downwind. In a light crosswind, this may cause the upwind vortex to become more or less stationary along the runway centreline as shown in figure 2.

This could well be the most adverse case as dissipation will also be slower in the light wind. In slightly stronger crosswinds not only will both vortices be seen to move away off beyond the leeward edge of the runway, but they will also decay more rapidly.

The Airbus A380 (A380) has been assigned a Super-Heavy wake turbulence category. A Boeing 747-800 (B748) and the Boeing 787 family (B787) have been assigned a Heavy wake turbulence category and is therefore the same as a Boeing 747-400 (B744) for wake turbulence. The 737-800/8200 are Medium.

Wake turbulence separation minima are based on a grouping of aircraft types into four categories according to the maximum certificated take-off mass as follows:

- a. SUPER (J) – aircraft types listed as such in ICAO Doc 8643 ‘Aircraft Type Designators’, latest edition;

- b. HEAVY (H) – all aircraft types of 136,000 kg or more, with the exception of aircraft types covered in point (a);
- c. MEDIUM (M) – aircraft types of less than 136,000 kg but more than 7,000 kg; and
- d. LIGHT (L) – aircraft types of 7,000 kg or less.

8.3.9.1.1 Arrivals

Arriving B737-800/8200 (outside 4 DME) if following a:

- B747/B748/B787/A380 – 5 nm gap, 3 mins non radar;
- B757 – 4 nm gap;
- B737 – No min wake turbulence gap.

8.3.9.1.2 Departures

Departing Criteria:

- The same runway;
- Parallel runways separated by less than 760 m (2,500 ft);
- Crossing runways if the projected flight path of the second aircraft will cross the projected flight path of the first aircraft at the same altitude or less than 300 m (1,000 ft) below;
- Parallel runways separated by 760 m (2,500 ft) or more, if the projected flight path of the second aircraft will cross the projected flight path of the first aircraft at the same altitude or less than 300 m (1,000 ft) below.

Departing B737-800/8200 from same position on a runway if following a:

- A380 – 3 mins;
- B747/B748/B787 – 2 minutes;
- B757 – No min wake turbulence gap;
- B737 – No min wake turbulence gap.

Departing B737-800/8200 from an intermediate holding point/intersection take-off if following;

- B744/B748/B787 (from full length) – 3 mins;
- B757 (from full length) – No min wake turbulence gap;
- B737 – No min wake turbulence gap.

A separation minimum of 4 minutes shall be applied for a B737-800/8200 aircraft when taking off behind an A380 aircraft from:

- An intermediate part of the same runway; or
- An intermediate part of a parallel runway separated by less than 760 m (2,500 ft).

8.3.9.2 Types of Upset

Figure 3 illustrates the type of turbulence deviations experienced when penetrating a vortex pair. When travelling in the same direction as the axis of the vortex core i.e. in the same direction as the preceding aircraft causing the wake, the effect can be a sink and loss of climb performance in the downdraft between the two vortex cores, or if laterally displaced relative to the cores a vigorous rolling action. When penetrating across a wake, the effect will be rapid pitch changes – up then down and finally up again.

8.3.9.3 Avoidance of Wake Turbulence

Aircraft have been classified by weight into “heavy”, “medium” and “light” and ATC apply distance separation criteria between the different classified types under radar control on the approach. Commanders may apply discretion as to acceptable separation between their aircraft and the previous take-off aircraft depending on wind conditions and expected lift-off point relative to the previous aircraft.

The downward displacement of the wake allows the following general rules for avoidance of wake turbulence caused by wingtip vortices of heavy swept wing aircraft to be formulated. In the absence of sufficient dissipating wind forces apply the following rules:

1. In following a large aircraft on approach it is desirable to fly the same, or a slightly higher path; but never a lower path. The use of a common ILS or VASI/glide slope by all aircraft is a desirable practice for this reason;
2. Flight directly under, and parallel to, the wake of another aircraft shall be avoided, because of the inherent sinking characteristic of the wake (See Fig.2).

- Figure 1**
Wake-turbulence
formation

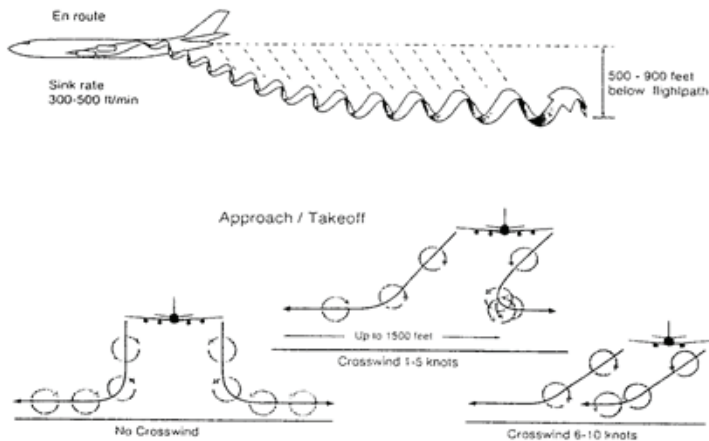
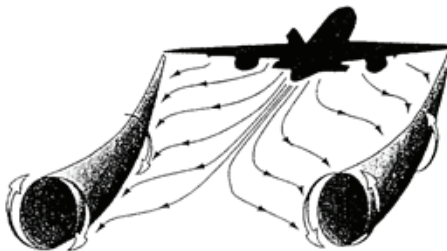


Figure 2

4. Take-off or landing shall be avoided immediately after a heavy aircraft has made a low pass (or missed approach) down the runway in use;

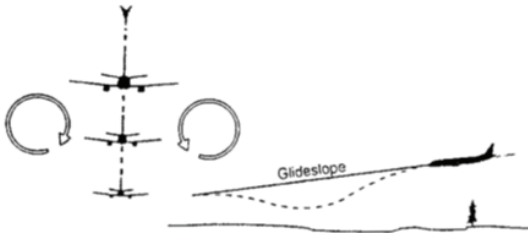


Figure 3a
Aircraft reaction to
wake-turbulence
encounter, approach
from shore-center
(rear view depiction)

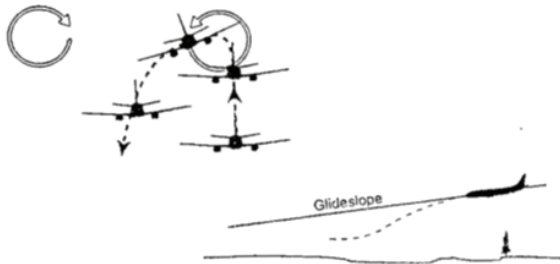
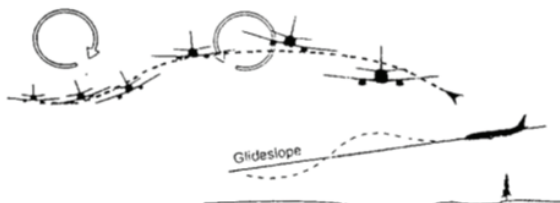


Figure 3b
Aircraft reaction to
wake-turbulence
encounter, approach
from below right
(rear view depiction)

Figure 3c
Aircraft reaction to
wake turbulence
encounter, rapid
approach from the
side (rear view
depiction)



5. Certain noise abatement and Emergency Turn procedures require a sharp turn immediately after take-off. When heavy aircraft are operating, the maximum practical separation time shall be allowed between take-off to permit dissipation of the high intensity vortices which develop in such

circumstances. It can be appreciated how hazardous it could be for an aircraft to penetrate such a vortex while banked at such a relatively low altitude;

6. Controllers and Commanders shall anticipate possible vortex encounters in light crosswind conditions. It should also be remembered that encounters can occur with vortices which have drifted at ground level, still intact and dangerously active from parallel and/or intersecting runways;
7. Except for the condition described in (vi) and (vii) above, it is usually safe for a lighter aircraft to follow the take-off of a heavy aircraft on the same runway, if the light aircraft lifts off after a shorter ground run and remains above the climbout path of the heavy aircraft while it is on a common course. If this is not possible, a take-off interval of at least two minutes is desirable. ATC, at busy airports in particular, should be advised what separation interval is required before accepting a lineup clearance;
8. Avoid flight within five miles behind a heavy jet when operating at the same altitude or within less than 1,000 below;
9. Use extreme caution when taxing behind a heavy jet. Static test data indicate that the area of concern is within 750 ft behind the tail of a heavy jet;
10. When operating in the same VMC environment as a heavy jet and being provided with radar vectoring, at least 5 miles spacing behind the heavy jet may be expected in order to ensure that wake turbulence problems are not encountered;
11. These precautions do not apply to “Heavy” jets following “Heavy” jets;
12. Test data indicates potential wake turbulence problems may exist when parallel runways, separated by less than 3,500 are being used in conditions of crosswind. The wake turbulence on one runway may drift across and effect operations on the other runway. Use at least a two minute interval behind a heavy jet departing the same runway, parallel runway – separated by less than 3,500', or crossing runways, if the departure courses converge after take-off.

8.3.10 Crew Members at Their Stations

CAT.OP.MPA.210, AMC1 CAT.OP.MPA.210(b)

General

Crew Members at Station – Flight Crew

During all critical phases of the flight, i.e. take-off, departure, approach and landing, all flight crew members required to be on flight deck duty, shall be at their station. (ORO.GEN.110(f))

During all other phases of flight, all flight crew members required to be on flight deck duty, shall remain at their station unless his absence is necessary for the performance of their duties in connection with the operation, or (before TOD) for physiological needs, provided at least one suitably qualified pilot remains at the controls of the aeroplane at all times. After TOD visits to the lavatory shall not occur.

Note: Flight crew shall not leave the flight deck in flight to intervene with an unruly passenger incident.

Checking CBs in Flight

It is best practise for crew to remain in their seats. Normally a crew can check CBs without vacating a seat. If it is necessary to vacate a crew seat for CB inspection the following guidance applies:

1. The Captain becomes the PF and remains in his seat.
2. Select “speaker” with volume turned up on ACP 1 & 2.
3. Select volume up on VHF 1 (operational ATC frequency) and VHF2 (121.5 usually).
4. The F/O will complete the CB inspection.

Crew Members at Station – Cabin Crew

On all decks of an aeroplane that are occupied by passengers, required cabin crew members shall be seated at their assigned stations during take-off and landing, and whenever deemed necessary by the Commander in the interest of safety.

The minimum altitude for releasing the CCMs from their seats shall be 3,000 ft AGL. This is achieved by pressing the Cabin Crew call bell once. When releasing the cabin, crews are reminded that the safety of the operation must take precedence over passenger service requirements. If the Commander considers that the departure routing could be affected by adverse weather he will delay releasing the Cabin Crew from their seats. Where weather forecasts indicate the possibility of adverse weather the Cabin Crew should be briefed prior to departure.

8.3.10.1 Controlled Rest for Fight Crew

CAT.OP.MPA.210

All crew members have a responsibility to be properly rested before flight and crews are expected to report fit for duty. However, in order to mitigate against unexpected fatigue, a controlled rest procedure for flight deck crew may be used on flights in compliance with the procedure below:

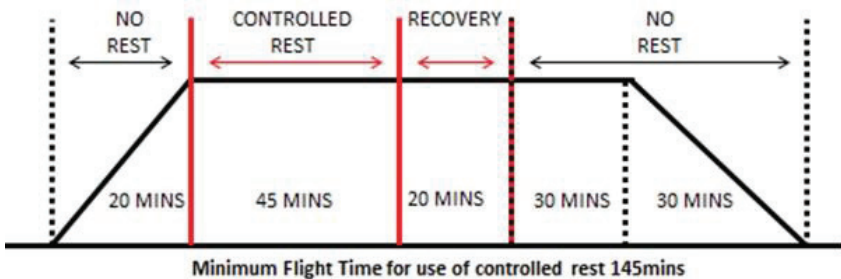
Controlled rest means a period of time ‘off task’ in the pilot’s seat and may include a period defined as ‘actual sleep’. Controlled rest may be used at the discretion of the Commander to manage unexpected fatigue. It cannot be planned prior to a flight. Unexpected fatigue is most likely to occur during extended periods of crew inactivity. Controlled rest can be used to mitigate against the effects of unexpected fatigue and increase alertness levels during the later phases of flight.

Controlled rest shall be used in conjunction with other on board fatigue management countermeasures such as, bright cockpit illumination at appropriate times, balanced eating and drinking, intellectual activity and physical exercise. A maximum rest time of 45 minutes has been chosen to limit deep sleep with consequent long recovery time (sleep inertia). The use of Controlled Rest procedure shall be in accordance with the following guidelines:

- Controlled rest shall only take place in the cruise during periods of low workload as determined by the commander.
- Controlled rest periods shall be agreed according to individual needs and the accepted principles of CRM.
- The Commander shall identify and brief the threats associated with the use of controlled rest on a particular flight.
- Only one crew member at a time shall take a controlled rest.
- The full harness shall be used/locked and the flight deck seat positioned to minimise unintentional interference with the controls.
- The commander shall ensure that the operational duties (including level change, frequency change, FIR crossing, etc.) of the non-resting crew member are clearly identified before the commencement of the rest period.
- The rest period shall be no longer than 45 minutes (in order to limit any actual sleep to approximately 30 minutes).
- After this 45-minute rest period, there should be a recovery period of 20 minutes during which sole control of the aeroplane will not be entrusted to the pilot who has availed of controlled rest.
- The non-resting flight crew member must remain alert. This may be achieved with the aid of crew alarm systems (clocks, etc.), but must include frequent cabin crew checks.

- The commander shall inform the SCCM of the intention of the flight crew member to take controlled rest, and the time of the end of that rest.
- The commander shall inform the SCCM that the F/D will call the cabin at 20 min intervals and if the call has not been initiated by the F/D the SCCM shall call the F/D in accordance with SOP.
- A minimum 20 minute period shall be allowed between rest & recovery periods to overcome the effects of sleep inertia and allow for adequate briefing.
- If necessary, a flight crew member may take more than one rest period, if time permits, on longer sectors, subject to the restrictions above.
- Controlled rest & recovery periods will terminate at least 30 minutes before top of descent.
- In the event of a non-normal situation arising or illumination of a Master Caution, the controlled rest period shall be immediately terminated.

Controlled Rest Flight Profile



8.3.11 Use of Restraint Devices for Crew and Passengers

CAT.OP.MPA.225

Seats, Safety Belts and Harnesses – Crew Members

During take-off and landing, and whenever deemed necessary by the Commander in the interest of safety, each crew member shall be properly secured by all safety belts and harnesses provided.

Flight Crew – Wearing of Safety Belt and Safety Harness

On boarding the aircraft, Commander and co-pilot will adjust and check the serviceability of the harness and its control and release mechanism.

Safety belt, and shoulder-harness will be worn by the occupants of all cockpit seats:

1. Prior to pushback, if practicable, but not later than commencement of taxi;

2. Below 10,000 ft;
3. Until vacating the runway after landing;
4. If a cockpit window cracks or de-laminates when the aircraft is pressurised;
5. At the first indication of circumstances which may effect the safety of the flight (indication of possible engine shutdown, etc.);
6. In conditions of turbulence.

Safety belts will be worn by Flight Crew members in the following circumstances:

1. By one Flight Crew member at the controls at all times;
2. By all Flight Crew members when occupying a pilot's seat.

Seats and Safety Belts – Passengers

Seat belts shall be used as long the seat belt sign is on. If allowed by the Commander, a limited service may continue if the seat belt sign is switched on during flight. When the seat belt sign is switched on before approach, the Cabin Crew(s) are expected to complete their work in order to prepare the cabin for landing, until the message “Cabin Crew seats for landing” is announced at which point cabin crew are expected to take their seats for landing.

Before take-off, all passengers shall be briefed on how to fasten and unfasten seat belts.

Before take-off and landing, and during taxiing, and whenever deemed necessary in the interest of safety, the cabin crew shall ensure that each passenger on board occupies a seat with his safety belt, or harness where provided, properly secured.

Multiple occupancy of aeroplane seats shall only occur by one adult and one infant who is properly secured by a supplementary loop belt or other restraint device.

The seat belts shall be fastened whenever the fasten seat belt sign is on.

Note: As a precaution, during fuelling and defueling, the seat belt sign shall be OFF, and the seat belts shall be unfastened.

8.3.12 Admission to Flight Deck

CAT.GEN.MPA.135, ORO.GEN.140

Malta Air ensures that no person, other than a flight crew member assigned to a flight is admitted to, or carried in the flight deck unless that person is:

- an operating crew member or

- a representative of the TM CAD or an inspecting authority if required to be there in the performance of his duties
- a person authorised by the NPFO

The commander shall ensure that:

1. admission to the flight crew compartment does not cause distraction or interference with the operation of the flight; and
2. all persons carried in the flight crew compartment are made familiar with the relevant safety procedures.

The commander shall make the final decision regarding the admission to the flight crew compartment.

8.3.12.1 Persons Entitled or Authorised to Use the Jump Seat

Inspectors of the TM CAD have rights of access and use of the jump seat in the exercise of their official duties. Members or employees of aviation organisations or aeronautical agencies may be specifically approved to travel in the jump seat for specific observation duties or familiarisation by the NPFO. In the exercise of discretion when giving approval to occupy the jump seat, Commanders will consider both the suitability of the person and the best interests of the Company.

The NPFO may authorise any person to travel in any seat in the cabin and in vacant flight deck and cabin jumpseats, including without a ticket.

8.3.13 Use of Vacant Crew Seats Supernumerary Seat/Jump Seat

The supernumerary seat/jump seat may only be occupied at the discretion and with the express approval of the Commander. This approval may not unreasonably be withheld for legitimate checking personnel, training and familiarisation, observers monitoring aircraft systems or performance, or positioning crew.

The Captain shall satisfy himself that a jumpseat passenger is not of any known or apparent mental instability, is not under the influence of alcohol or drugs or any prohibited or prescribed substances, is able bodied/not prevented through any infirmity from reacting correctly to instructions and reacting correctly in an emergency situation.

Fare Paying/Revenue Passengers

A revenue passenger shall not occupy any jumpseat under any circumstances.

8.3.13.1 Flight Deck Jump Seats

Valid company identification refers to Ryanair Group identification.

All flight deck jump seat passengers shall satisfy one of the following criteria:

- TM CAD flight operations personnel carrying out their official duties;
- Ryanair Group Airline Compliance Auditors;
- Ryanair Group Airline Management Pilots;
- Technical and other aviation industry personnel approved by the NPFO;
- Ryanair Group Airline personnel on duty travel carrying valid company identification;
- Ryanair Group Airline personnel travelling on concession travel carrying valid company identification;
- Ryanair Group Airline pilot in uniform on privilege travel, or;
- Be a Malta Air instructor travelling to SIM/Training locations to carry out their duties.

Qualifying personnel will only use a cockpit jumpseat in the following circumstances:

1. The Captain is given advance notification of the name of the person who intends to travel on the flight deck jumpseat.
2. This person is in possession of a valid concessionary/duty travel ticket.
3. Pilots in uniform on privilege travel with valid company identification, Management Pilots with valid company identification and instructors traveling to SIM/Training locations to carry out their duties in possession of valid company identification.
4. Operations has the list of States whose State Security Policy permits authorised persons to occupy a flight deck jumpseat only when there is no seat available in the cabin.

Note: Personnel on operational duties requiring occupation of a flight deck jumpseat are permitted to occupy a flight deck jumpseat regardless of seating availability in the cabin.

The Commander of the aircraft is ultimately responsible for the safety and security of the aircraft and must satisfy himself with the identity of all jump seat passengers prior to departure.

8.3.13.2 Cabin Jumpseats

A person travelling on Ryanair Group concession travel may occupy a jumpseat in the cabin of the aircraft subject to the following restrictions:

1. The Captain is given advance notice of the name of the person who is travelling on the concession/duty travel ticket who intends to occupy the jumpseat.
2. The passenger is in possession of a valid concession ticket issued by Malta Air/Ryanair Group
3. Any personnel entitled to occupy a flight deck jumpseat.

Note: Qualifying personnel will only use a jumpseat if there is no seat in the cabin.

Aeronautical Inspectors may occupy the jumpseats or passenger seats for the purposes of carrying out their duties as an inspector. Subject to his discretion, the Commander should accept the allocated person but may refuse to do so only for safety reasons. If such an allocation has been refused by the Commander, he may not then offer the seat to another staff passenger for personal travel.

8.3.13.3 Locked Flight Deck Door Concept

ORO.SEC.100

Normal Procedures

The FLIGHT DECK DOOR must be locked once the forward entry door is closed and should remain locked at all times until the completion of the flight, indicated by the opening of the forward entry door. Formal confirmation of the door being locked prior to departure will be made at the “FLIGHT DECK WINDOWS AND COCKPIT DOOR” challenge on the “BEFORE START CHECKLIST”.

Communications with the Cabin Crew while the cabin door is locked will be via the CABIN INTERPHONE.

At any time that the main aircraft doors are closed with passengers on board the Flight Crew shall not allow entry to the flight deck except in an emergency or in circumstances where one crew member can safely leave his/her seat to verify visually through the viewing lens that the vestibule area is clear.

A crew member shall not leave his/her seat during critical phases of flight for the purpose of opening the flight deck door. Critical phases of flight are defined as the phase of operation from the commencement of taxi to top of climb or from top of descent to the parking gate.

Ideally crews would not normally open the flight deck door or leave the flight deck until established in the cruise

Should either the Captain or First Officer find it necessary to leave the flight deck during flight the following additional procedures apply:

1. Fasten Seat Belt Sign...ON.
2. Establish communication with the CSS and confirm that all passengers are seated.
3. The CSS must remain in the forward galley area while the cabin door is unlocked and the Captain or First Officer has vacated their seat.
4. The door shall only be opened from inside by means of the door handle.
5. A cabin crew member shall lock the door and remain on the flight deck and verify that the immediate area is clear before allowing access to the returning crew member.

Note 1: The door shall only be unlocked with the agreement of either both pilots, or the pilot and cabin crew on the flight deck.

Note 2: The cabin crew member shall NOT operate any control on the flight deck or occupy any seat at the controls of the aircraft.

Note 3: It is a requirement that a crew member in the flight deck be standing directly facing the door in the flight deck whenever the door is open. It is not permitted to use the door stop to lock or hold the door in the open position.

Note 4: The cabin crew member shall not be 'NEW' crew member as defined by their certificate of competence.

The flight deck door must not be left open or unlocked at any time.

8.3.14 Incapacitation of Crew Members

Pilot incapacitation represents an operational risk and is defined operationally as “any physiological or psychological state or situation that adversely affects performance.” There are degrees of incapacitation and strategies to manage these are detailed in this section.

8.3.14.1 Crew Member Incapacitation

When a flight crew member is incapacitated, control of the aircraft reverts to the other pilot. After landing, the Commander may elect to taxi the aircraft to the parking location as the F/O has access to thrust and brakes to stop the aircraft at any time. The Commander (or next senior crewmember if the Commander was incapacitated) shall contact Ryanair Operations and the Duty Pilot as soon as practicable.

Operations/Duty Pilot will inform Crewing and Rostering if the crewmember is being stood down due to incapacitation, and the crewmember shall not be assigned flight duties until the medical requirements applicable to that licence/attestation holder are completed.

8.3.14.1.1 Guidance for Flight Crew After Use of Emergency Oxygen in Flight

If Emergency Oxygen is used as part of a QRH Checklist (Rapid decompression/Smoke Fire Fumes etc.) contact the Duty Pilot and, if the decision is to stand down the crew or crew member, the minimum time is one rest period.

8.3.14.2 General

Incapacitation is a real air safety hazard which occurs more frequently than many of the other emergencies which are the subject of routine training. Incapacitation can occur in many forms varying from obvious sudden death to subtle, partial loss of function. It occurs in all age groups and during all phases of flight.

Sudden and total incapacity are obvious but subtle incapacity is insidious and may be difficult to recognise and take appropriate action. This section sets out intervention criteria for different phases of flight which are particularly important as the basis for the other pilot making a corrective input or taking control of the flight.

8.3.14.3 Recognition

The keys to early recognition of incapacitation are deviations or variation from SOP, path control, and, standard calls and responses which may be identified by the application of:

1. Standard Operating Procedures.
2. Routine monitoring and cross-checking of flight instruments, particularly during critical phases of flight.
3. Use of crew concept method of operation.
4. Use of the “Two Communications Rule”.

Flight Crew shall suspect “subtle incapacitation” if a crew member does not respond appropriately to two verbal communications, **OR** a crew member does not respond appropriately to a verbal communication associated with a significant deviation from a standard flight profile.

“Subtle” incapacitation means what it says! Any lack of, or inappropriate response to normal stimuli should cause the remaining Flight Crew member to consider some level of incapacitation.

8.3.14.4 Action

If an incapacitation occurs: (time permitting refer SEP Manual 5.1.11.8)

1. The remaining Flight Crew member shall take over the controls and maximise use of “automatics”.
2. “No 1 to the Flight Deck”, admit CSS after receiving correct password using flight deck access system switch.
3. Declare PAN or MAYDAY as appropriate to the severity of the incapacitation. ATC will acknowledge your MAYDAY.
4. The incapacitated Flight Crew member will be appropriately restrained, and his seat will be moved back to ensure this crewmember has no access to controls.
5. Cabin Crew will be used (minimum number of two) to remove the incapacitated Flight Crew member from his seat if required, in accordance with OMA-SEP procedures, e.g., CPR/AED.
6. If there is a Ryanair Group qualified pilot on board, (subject to verification of ID), this pilot may assist and occupy a pilot seat. A non-Ryanair Group pilot may NOT occupy a pilot seat. Cabin crew may NOT occupy a pilot seat.
7. Conduct PIOSEE, NITS, PA, plan to land at nearest suitable airport and advise ATC.

8.3.14.5 Overview Guidance

If you don't feel well – SAY SO! Let the other Flight Crew member fly. The fully fit Flight Crew member takes over and keeps the aircraft under control maintaining a safe flight profile.

8.3.14.6 Operational Concept

Two-communication Rule

The “two-communication” rule was developed to meet the need for a method of detecting subtle (including cognitive) incapacitations before they become operationally critical. The rule states: “Flight crew members should have a high index of suspicion of a ‘subtle’ incapacitation any time a crew member does not

respond appropriately to two verbal communications, or any time a crew member does not respond appropriately to any verbal communication associated with a significant deviation from a standard operating procedure or a standard flight profile.”

Obvious and Subtle Incapacitation

Incapacitations can be divided into several operational classifications such as impairment, cognitive, obvious and subtle. Obvious incapacitations are those immediately apparent to the other crew members. The course of onset can be “sudden” or “insidious” and complete loss of function can occur. Subtle incapacitations are frequently partial in nature and can be insidious because the affected pilot may look well and continue to operate but at a less than optimum level of performance. The pilot may not be aware of their problem or capable of rationally evaluating it.

Cognitive Incapacitation

A particular category of incapacitations has been identified as “cognitive.” The problem created by these incapacitations is how to deal with a pilot who is “mentally disoriented, mentally incapacitated or obstinate, while physically able and vocally responsive.” In most cases of cognitive incapacitation, the pilot demonstrates manifestly inappropriate behaviour involving action or inaction, and the inappropriate behaviour is associated with failures of comprehension, perception, or judgement.

Crew Member Impairment

Gastroenteritis accounts for 75% of ‘incapacitation’ episodes. Recent UKCAA CMO guidance on gastroenteritis uses the term ‘impairment’ when the crewmember can continue to function as the Pilot Monitoring. This guidance advises crew members to seek AME advice if symptoms are not resolved after 48 hrs.

Control of the Incapacitation Risk

This is dependent upon effective operational monitoring. All flight crew members must be aware of the aircraft status. This is one of the most important reasons for following standard operating procedures (SOPs), flying standard flight profiles, and making standard calls and responses, particularly when physical control inputs are unnecessary, e.g., auto-coupled operations.

Duty Pilot (DP) Support

The DP shall be contacted for all incapacitation events, including impairment, referred to in this section. Upon notification, the DP shall liaise with OCC and notify the associated NPFO to obtain guidance on required process and facilitation.

8.3.15 Cabin Safety Requirements

8.3.15.0 Cabin Safety Requirements

All Malta Air aircraft are equipped with means for emergency evacuation including ropes in the flight deck and slides attached to L1, L2, R1 and R2 doors on the B737 800/8200 and MED L, MED R on the B737-8200 only. This is in compliance with CAT.IDE.A.265.

All Malta Air aircraft are equipped with Life Jackets in compliance with CAT.IDE.A.285(a).

Malta Air aircraft are equipped with floatation devices, in compliance with CAT.IDE.A.285(a)(1).

For take-off and landing all means of assistance for emergency evacuation that deploy automatically are armed. CAT.OP.MPA.220.

8.3.15.1 Cabin Preparation

CAT.OP.MPA.230

Stowage of Baggage and Cargo

Only hand baggage which can be adequately and securely stowed is allowed to be carried in the passenger cabin. All baggage on board which might cause injury or damage, or obstruct aisles and exits if displaced, must be adequately and securely stowed to take account of the following:

1. Each item must be stowed only in a location that is capable of restraining it;
2. Mass limitations placarded on or adjacent to stowages must not be exceeded;
3. Under-seat stowage's must not be used unless the seat is equipped with a restraint bar and the baggage is of such a size that it may adequately be restrained by this equipment;
4. Items must not be stowed in toilets or against bulkheads that are incapable of restraining articles against movement forwards, sideways or upwards and unless the bulkheads carry a placard specifying the greatest mass that may be placed there;
5. Baggage placed in lockers must not be of such a size that they prevent latched doors from being closed securely;
6. Baggage must not be placed where it can impede access to emergency equipment; and
7. Checks must be made before take-off, before landing, and whenever the commander illuminates the fasten seat belts signs to ensure that baggage is stowed where it cannot impede evacuation from the aircraft or cause injury by falling (or other movement) as may be appropriate to the phase of flight.

8.3.15.1.1 Cellos/Paintings/Bulk Items in the Cabin

In exceptional cases it may be necessary to carry bulky and/or fragile items of baggage, e.g. cellos, or fine art paintings, in the cabin. This can only be done provided that the Commander checks that the baggage item is adequately secured/lashed to passenger seats and confirms that aisles, emergency exits, and escape paths are not obstructed, and that an emergency evacuation will not be impeded. In these cases the passenger purchases a seat for the item. The item is included in the headcount and is noted on the LID SI Box as 'ITEMSEAT' or on the paper loadsheet. "TOB 160, including 1 cello". This TOB figure is documented on the Electronic Flight Log/Paper Voyage report/journey log.

8.3.15.1.2 Securing of the Passenger Cabin and Galleys

CAT.OP.MPA.230

The CSS will ensure that before taxiing, take-off and landing all exits and escape paths are unobstructed, and that all equipment and baggage is properly secured.

With regard to galley items a specific check must be made that containers and trolleys are secured with restraining levers/bars, locker doors are latched, all equipment secured with the appropriate devices provided, and all fluids in open containers such as hot cups emptied.

The CSS will report to the Commander that the cabin is "SECURE" for take-off and landing having complied with the actions and checks required, and that the other Cabin Crew have fastened their safety harnesses.

The Commander shall ensure that this report is received before take-off and landing, and whenever deemed necessary in the interests of safety, that all equipment and baggage is properly secured. (If the cabin has not been secured before landing a go-around is required.)

The CSS will then return to her/his assigned position for take-off and landing and fasten her/his safety harness.

8.3.15.2 Passenger Seating in the Context of an Emergency Evacuation

CAT.OP.MPA.165

Passengers must be seated where, in the event that an emergency evacuation is required, they may best assist and not hinder evacuation from the aircraft. In the preparation for an emergency evacuation (time available) (CAT.OP.MPA.220) able bodied persons (ABPs) will be briefed and re-seated to assist at emergency exits in accordance with the Malta Air Safety and Emergency Procedures Manual. Passengers are seated where they, in the event of an emergency evacuation, may best assist and not hinder the evacuation from the aircraft. Those passengers who are allocated direct access seats must appear to be reasonably fit, strong and able to assist the rapid evacuation of the aircraft in an emergency. (Please refer to SEP 5.4.5 for type specific reserved seating procedures.)

Cabin Crew must ensure the following are not seated in direct access seats:

- Passengers under 16 years of age.
- Passengers unwilling or unable to help in the event of an emergency.
- Passengers travelling with an infant.
- Persons who necessitate the use of a seat belt extender (which could get tangled in an emergency).
- Passengers with reduced mobility, blind/visual impairment or who require special assistance.
- Deportees or prisoners in custody.

8.3.15.2.1 Passenger Seating at Over-wing Exits

Unless in cases of low passenger number or mass and balance limitations, cabin crew shall ensure that there shall be at least one (1) briefed ABP at each of the overwing exits onboard the aircraft prior to departure and for taxi, take-off and landing.

8.3.15.3 Passenger Embarkation and Disembarkation Procedures

See Part A Section 8.2.2.

8.3.15.4 Procedures for Refuelling/Defuelling with Passengers on Board or Embarking/Disembarking

CAT.OP.MPA.195

See Part A Section 8.2.2.

These procedures also comply with various Airport Directives, Bylaws and industry standards and meet the Company's obligation to its passengers to operate safely.

The procedures require the liaison and co-ordination of the Ramp Agent, the Engineer or other person supervising the refuelling, the Flight Crew members and CCMs on board.

One qualified pilot shall be in the flight deck during fuelling operations with passengers on board. This pilot shall be capable of handling emergency procedures concerning fire protection and fire-fighting, handling communications and initiating and directing an evacuation.

The following are the minimum egress requirements:

1. Aircraft equipped with automatic inflatable slides;

2. The main door normally used for passenger disembarkation open, clear of obstructions and manned;
3. Loading bridge, gangway, or passenger or internal aircraft steps in position at the aircraft door;
4. Rear exit doors closed or, if open, passenger steps positioned at the open door;
5. The areas beneath all exit doors clear of obstructions; and
6. Aisle to be kept clear, except for passengers embarking or disembarking with their carry-on luggage.

8.3.15.5 Carriage of Special Categories of Passengers (SCPs)

CAT.OP.MPA.155, AMC1 CAT.OP.MPA.155(b)

1. Persons requiring special conditions, assistance and/or devices when carried on a flight shall be considered as SCPs including at least:
 - a. Persons with Reduced Mobility (PRMs) who, without prejudice to Regulation (EC) No 1107/2006, are understood to be any person whose mobility is reduced due to any physical disability, sensory or locomotory, permanent or temporary, intellectual disability or impairment, any other cause of disability, or age;
 - b. infants and unaccompanied children; and
 - c. deportees, inadmissible passengers or prisoners in custody.
2. SCPs shall be carried under conditions that ensure the safety of the aircraft and its occupants according to procedures established by Malta Air.
3. SCPs shall not be allocated, nor occupy, seats that permit direct access to emergency exits or where their presence could:
 - a. impede crew members in their duties;
 - b. obstruct access to emergency equipment; or
 - c. impede the emergency evacuation of the aircraft.
4. The commander shall be notified in advance when SCPs are to be carried on board.

Direct Access Seat

‘Direct access’ means a seat from which a passenger can proceed directly to the exit without entering an aisle or passing around an obstruction.

Direct access Seating on the Boeing 737-800 is Rows 1ABC, 16 and 17.

Direct access Seating on the Boeing 737-8200 is Rows 1BC, 2ABC, 17, 18, 28 and 29.

8.3.15.6 Smoking

CAT.OP.MPA.240

8.3.15.6.1 Policy

Malta Air operates a no-smoking policy on all its flights. There shall be no smoking in any part of the aircraft at any time by any person.

8.3.15.6.2 Implementation

The passenger pre-flight briefing covers the smoking regulations.

8.3.15.7 Infectious Disease Suspected on Board

In cases of suspected communicable/infectious disease on board the commander shall be notified, and he shall promptly advise ATC and request that port health is notified with the information listed below:

1. Aircraft identification;
2. Departure aerodrome;
3. Destination aerodrome;
4. Estimated time of arrival;
5. Number of persons on board;
6. Number of suspected cases on board; and
7. Nature of the public health risk, if known.

OCC shall be advised by the most expeditious means and shall provide the most up to date instructions to the crew as soon as practicable.

Port Health (Public Health)

Every international airport in the world has a Port Health Unit. The function of the unit is:

- Health control – The prevention of importation of communicable diseases.
- Medical surveillance of all immigrants.

As a matter of routine they will only meet flights if they have been requested to do so by an airline's Operations Department.

A request for Port Health will usually cause a short delay at the end of the flight. Doctors from Port Health will assess the nature of the illness, the number of people affected, the flight details, and where the aircraft has come from, before allowing any passengers or Aircrew to disembark. They may then refer the

suspected passengers to isolation and may take details of nearby passengers, e.g. four rows in front and four behind. The Public Health Inspector will give surveillance forms to those who require them.

8.3.16 Contents and Timing of Passenger Briefing

CAT.OP.MPA.170, AMC1 CAT.OP.MPA.170

8.3.16.1 Prior to Boarding

At check-in attention of passengers is drawn to the list of prohibited articles, including dangerous goods, displayed and which they are not allowed to carry on their person, or in their hand baggage. They are questioned on the packing of checked baggage and the inclusion of certain articles, see Chapter 10, Security for details.

Prior to boarding passengers will be instructed on the 'no smoking' requirements while boarding, and instructed or warned of any other safety matters relevant to their boarding such as excessive or bulky hand baggage, use of walkways and routing to avoid aircraft structure and servicing equipment, and ramp conditions.

8.3.16.2 Prior to Take-off

When the aircraft doors have been closed and the slides armed the passengers will be briefed on the regulations and procedures itemised below and concurrently the demonstrations specified above will be given:

- Instructions to fasten seat belts for take-off, including how to fasten and unfasten them, and the recommendation to keep seat belts fastened while in their seats during flight;
- Location and use of oxygen masks;
- The Malta Air policy and requirements of no smoking on all its flights;
- Location and use of life jackets;
- Back of the seat to be in the upright position and the tray table stowed;
- Location of emergency exits;
- The importance of leaving hand luggage behind in case of evacuation;
- Location and use of floor proximity escape path markings;
- Stowage of hand baggage;
- The use and stowage of portable electronic devices;
- The location and the contents of the safety briefing card and the need for passengers to review it prior take-off;
- Compliance with cabin signs and crewmember instructions.

8.3.16.3 After Take-off

- CSS will make an announcement reminding passengers of the “no smoking” regulations and the use of safety belts.
- CSS will include a Malta Air safety recommendation that seat belts are kept fastened at all times.

If a turbulence is encountered a further announcement instructing the passengers to ensure that their seat belt is tightly/securely fastened if turbulence is encountered and the “FASTEN SEAT BELT” sign illuminated. The CSS will make an announcement advising caution when opening overhead compartments.

8.3.16.4 Before Landing

CSS instructs or reminds the passengers:

- Of the smoking regulations, and no further smoking until inside the terminal building;
- To fasten seat belts/safety harnesses for landing;
- To put the seat back in the upright position and Stow the tray table;
- To re-stow hand baggage; and
- The use and stowage of portable electronic devices.
- The location and the contents of the safety briefing card and the need for passengers to review it prior landing;
- The importance of leaving hand baggage behind in case of evacuation.

8.3.16.5 After Landing

CSS instructs or reminds passengers:

- To observe the no smoking instruction until inside the terminal;
- The use and stowage of portable electronic devices;
- To keep their seat belts/safety harnesses fastened until the aircraft has stopped or reached the gate and the engines have been shut down; and
- Warn of any safety matters affecting their disembarkation such as the need for care when opening over head lockers, and conditions on the ramp.

Emergencies in Flight

If an emergency occurs during flight, the passengers will be instructed in such emergency action as may be appropriate to the circumstances.

The procedures to be used have been developed for various scenarios and are contained in the Safety and Emergency Procedures Manual.

The call to initiate a line of communication with the Cabin Crew in an emergency is *“No 1 to the flight deck”*. This will inform the Cabin Crew that an emergency situation exists. On hearing this call the CSS will proceed to the cabin interphone and await a communication from the Captain.

8.3.16.6 NITS and Security

Security procedures associated with locked cockpit door are still valid in an emergency situation. A pilot must view the cabin and forward galley area prior to opening the cockpit door. It is therefore preferable that the NITS drill is accomplished over the cabin interphone with the cockpit door locked. The final decision as to how this communication occurs rests with the Commander.

8.3.17 Cosmic or Solar Radiation

Malta Air shall take account of the in-flight exposure to cosmic radiation of all crew members while on duty (including positioning) and shall take the following measures for those crew liable to be subject to exposure of more than 1mSv per year:

1. Assess their exposure;
2. Take into account the assessed exposure when organising working schedules with a view to reduce the doses of highly exposed crew members;
3. Inform the crew members concerned of the health risks their work involves;
4. Ensure that the working schedules for female crew members, once they have notified Malta Air that they are pregnant, keep the equivalent dose to the foetus as low as can reasonably be achieved and in any case ensure that the dose does not exceed 1 mSv for the remainder of the pregnancy;
5. Ensure that individual records are kept for those crew members who are liable to high exposure. These exposures are to be notified to the individual on an annual basis, and also upon leaving Malta Air.

Malta Air has an approved cosmic radiation monitoring programme in place in accordance with Maltese Subsidiary Legislation 585.01.

For the purposes of this programme each sector is operated at maximum altitude, and the corresponding dosage for this altitude/time is assigned to this sector.

The current annual flight profile of any individual Malta Air pilot is significantly below the benchmark dosage requiring individual monitoring. Individual annual dosage reported is available from the Dir Flt Ops.

8.3.18 Policy on the Use of Automation, AFDS

The Flight Director system shall not be switched off when serviceable during flight operations. Crews are reminded that Flight Directors are to be used in all phases of flight unless it is directed to be turned off by procedure or checklist, and that flight director OFF flying for the maintaining of pilot proficiency practice shall NOT be practiced during line operations.

Procedures for re-engaging of AFDS automation is provided in FCOM Vol 1.

The Flight Director system shall not be switched off when serviceable during flight operations.

8.3.18.1 Practise Hand Flying

It is also recognised that “hand flying” the aircraft is a required skill and Commanders, having due consideration to workload and environmental conditions, are authorised to allow pilots to hand fly the aircraft in controlled airspace, only when the fasten seatbelt sign is switched on. Passenger safety and comfort should not be compromised.

Practice hand flying is restricted to below FL150, unless otherwise instructed by a non-normal checklist. The autopilot, when available, shall be engaged anytime the fasten seatbelt sign is OFF.

Practice hand flying is prohibited in the following circumstances:

- Thunderstorm activity
- Towering cumulus activity
- Areas of known or forecast windshear or turbulence
- Areas affected by gale warning
- Busy TMAs (high density traffic, numerous frequency changes, multiple procedural altitude restrictions, high pilot workload, etc...)
- While any Non-Normal procedure is in progress when the autopilot is available and it is appropriate to use it
- Outside controlled airspace (including Class E), except for take-off and landing
- VFR traffic announced by ATC or on ATIS or NOTAM

8.4 Low Visibility Operations

(SPA.LVO.100)

Malta Air shall only conduct the following Low Visibility Operations (LVO) when approved by the TM CAD:

- Low visibility take-off (LVTO) operation;
- Standard category II (CAT II) operation;
- Standard category III (CAT III) operation.

8.4.1 Classification of LVTO Operations

GM1 SPA.LVO.100 (a)

Low-visibility Operations

Take-off operations are classified as ‘normal take-off operations’ with an RVR at or above 550 m and ‘LVTO operations’ with an RVR below 550 m.

Only LVTO operations in an RVR of less than 400 m require a specific approval.

Low-Visibility Take-Off (LVTO) Operations in an RVR of Less than 400 m

AMC1 SPA.LVO.100(a)

Low-visibility Operations

1. Required RVR
 - a. for multi-engined aeroplanes which, in the event of a critical engine failure at any point during take-off, can either stop or continue the take-off to a height of 1500 ft above the aerodrome while clearing obstacles by the required margins, the criteria in Table 2 should apply:

Table 2 LVTO operations – RVR versus facilities

Minimum RVR (m)	Facilities
300 m (day)	Centre line markings; and Runway edge lights.
300 m (night)	Centre line markings; and Runway edge lights; and Runway end lights or centre line lights.
150 m	Centre line markings; and Runway end lights; and Runway edge lights; and Runway centre line lights.

Minimum RVR (m)	Facilities
125 m	Centre line markings; and Runway end lights; and Runway edge lights (spaced 60 m or less); and Runway centre line lights (spaced 15 m or less).

- b. the reported RVR value representative of the initial part of the take-off run can be replaced by pilot assessment.
- c. the minimum RVR value specified in Table 2 should be achieved for all reporting points representative of the parts of the runway from the point at which the aircraft commences the take-off until the calculated accelerate-stop distance from that point.

Visual Segment for Take-off

GM2 SPA.LVO.100 (a)

Low-visibility Operations

The value of 125 m RVR for take-off with 15 m centre line light spacing has been selected because flight deck geometry means that this will provide at least a 90 m visual segment for the large majority of aircraft types. In a 90 m visual segment the pilot is expected to be able to see six centre line light intervals (seven centre line lights) at 15 m spacing once lined up on the runway centre line.

8.4.2 Instrument Approach Operations in Low Visibility Conditions – Classification of Standard Approach Operations

GM1 SPA.LVO.100 (b)

The different types of approach and landing operations are classified according to the lowest DH (or MDH) and RVR applicable to the approach type. The classification of approach types does not depend on the technology used for the approach. The lowest minima specified do not take account of ‘operational credits’ that may allow for lower operating minima.

The classification does not subdivide CAT III operations into CAT IIIA, IIIB, and IIIC. The actual minima applicable to any operation depends on the aircraft equipment and the specific LVO approval held by Malta Air.

The AFM for aircraft certified for CAT III operations states the lowest usable DH:

- Category III: DH below 100 feet but not less than 50 feet.

8.4.2.1 Instrument Approach Operations in Low Visibility Conditions – Cat II Operations

AMC1 SPA.LVO.100 (b)

Low-visibility Operations

For CAT II operations in Malta Air, the following provisions apply:

1. The DH should be determined by the use of a radio altimeter and be not lower than the highest of:
 - a. the minimum DH specified in the AFM;
 - b. the applicable Obstacle Clearance Height (OCH) for the category of aircraft;
 - c. the DH to which the flight crew is qualified to operate; or
 - d. 100 ft.
2. The lowest RVR minima to be used for CAT II operations are specified in Table 3:

Table 3 CAT II Operation Minima: RVR (m) Versus DH (ft)

Aircraft Categories		Auto-coupled to Below DH*	
		A, B, C (RVR m)	D (RVR m)
DH (ft)	100 – 120	300	300/350*
	121 – 140	400	400
	141 – 199	450	450

*: An RVR of 300 m may be used for a Category D aeroplane conducting an autoland.

Note 1: HUD N/A MAY.

Note 2: Malta Air is approved for CAT II (100 ft DH, 300 m RVR).

Note 3: CAT II instrument approach operations should only be conducted using a CAT II IAP (AMC2 SPA.LVO.110 (a)).

Note 4: For CAT II instrument approach operations, a PA runway category II or category III should be used. The following visual aids should be available: Category II approach lights, standard runway markings and Category II runway lights. (AMC3 SPA.LVO.110).

8.4.2.2 Instrument Approach Operations in Low-visibility Conditions – Cat III Operations

AMC2 SPA.LVO.100 (b)

Low-visibility Operations

For CAT III operations in Malta Air, the following provisions apply:

- 1. The DH should be determined by the use of a radio altimeter and be not lower than:
 - a. the minimum DH specified in the AFM;
 - b. the DH to which the flight crew is qualified to operate.
- 2. The lowest RVR to be used for CAT III operations should be determined in accordance with Table 5:

Table 5 CAT III Operation Minima: RVR (m) Versus DH (ft)

DH (ft)	Roll-out Control/guidance System	RVR (m)*
50-99	Not required	175*

*: Malta Air is approved for CAT III (50 ft DH, 175 m RVR).

Note 1: CAT III instrument approach operations should only be conducted using a CAT III IAP AMC2 SPA.LVO.110 (b).

Note 2: For CAT III instrument approach operations, a PA runway category III should be used. The following visual aids should be available: Category III approach lights, standard runway markings and category III runway lights. (AMC3 SPA.LVO.110).

Instrument Approach Operations in Low-visibility Conditions – Equipment Certification for Low Visibility Approach Operations Other Than EFVS

GM2 SPA.LVO.100 (b)

Certification Requirements of CS-AWO:

Aircraft suitable for low visibility approach operations are certified according to the minimum usable DH which is stated in the AFM.

Legacy systems may be described as capable of ‘CAT 3A’ or ‘CAT IIIA’ operations. This implies a minimum DH of less than 100 ft but not less than 50 ft. Operations to a DH of less than 100 ft but not less than 50 ft will typically require a fail-passive automatic landing system.

CAT III Fail-passive Operations

GM3 SPA.LVO.100 (b)

During an automatic landing the pilot needs to monitor the performance of the aircraft system, not in order to detect a failure that is better done by the monitoring devices built into the system, but so as to know precisely the flight situation. In the final stages the pilot should establish visual contact and, by the time the pilot reaches the DH, the pilot should have checked the aircraft position relative to the approach or runway centre line lights.

For this the pilot will need sight of horizontal elements (for roll reference) and part of the touchdown area. The pilot should check for lateral position and cross-track velocity and, if not within the pre-stated lateral limits, the pilot should carry out a missed approach procedure. The pilot should also check longitudinal progress and sight of the landing threshold is useful for this purpose, as is sight of the TDZ lights.

8.4.2.3 Operations Utilising EVS

AMC6 SPA.LVO.100

Not applicable to Malta Air operations.

8.4.2.4 Instrument Approach Operations in Low-visibility Conditions – Effect on Landing Minima of Temporarily Failed or Downgraded Equipment for Approach Operations with a DH Below 200 ft

GM4 SPA.LVO.100 (b), AMC3 SPA.LVO.100 (b)

Low-visibility Operations

The instructions for the effect on landing minima of temporarily failed or downgraded equipment are intended for use both before flight and during flight. It is, however, not expected that the pilot-in-command/commander would consult such instructions after passing 1000 ft above the aerodrome.

If failures of ground aids are announced at such a late stage, the approach could be continued at the pilot-in-command/commander's discretion.

If failures are announced before such a late stage in the approach, their effect on the approach should be considered as described in Table 6, and the approach may have to be abandoned.

1. Only those facilities mentioned in Table 6 should be acceptable to be used to determine the effect of temporarily failed or downgraded equipment on the required RVR for CAT II/III approach operations.
2. The following conditions should be applied to Table 6 below:
 - a. multiple failures of runway/FATO lights other than those indicated in Table 6 are not acceptable;

- b. failures of approach and runway/FATO lights are acceptable at the same time, and the most demanding consequence should be applied;
- c. for approach operations with a DH below 200 ft, a combination of deficiencies in runway/FATO lights and RVR assessment equipment are not permitted; and
- d. failures other than ILS, GLS and MLS affect the RVR only and not the DH.

Table 6 Failed or Downgraded Equipment – Effect on Landing Minima CAT II/III Operations

Failed or downgraded equipment	Effect on landing minima			
	CAT III no DH	CAT III DH < 50 ft	CAT III DH ≥ 50 ft	CAT II
Navaid stand-by transmitter	Not allowed	RVR 200 m	No effect	
Outer marker (ILS)	No effect if the required height versus glide path can be checked using other means, e.g. DME fix			
Middle marker (ILS)	No effect			
DME	No effect if replaced by RNAV (GNSS) information or the outer marker.			
RVR assessment systems	At least one RVR value to be available on the aerodrome	On runways equipped with two or more RVR assessment units, one may be inoperative		
Approach lights	No effect	Not allowed for operations with DH > 50 ft		Not allowed
Approach lights except the last 210m	No effect			Not allowed
Approach lights except the last 420m	No effect			
Standby power for approach lights	No effect			
Standby power for runway lights with 1-second switchover time	No effect	Not allowed	Day: RVR 550m Night: RVR 550m	Day: RVR 550m Night: RVR 550m
Edge lights	No effect	Day: no effect Night: RVR 550 m	Day: no effect Night: RVR 550m	Day: no effect Night: not allowed
Threshold lights	No effect	No effect	Day: no effect Night: RVR 550m	Day: no effect Night: not allowed
Runway end lights	No effect if centre line lights are serviceable			
Centre line lights	Day: RVR 200m Night: not allowed	Not allowed	Day: RVR 300m Night: RVR 400m	Day: RVR 350m Night: RVR 550m (400m with HUD or Autoland)
Centre line lights spacing increased to 30m	RVR 150m		No effect	
TDZ lights	No effect	Day: RVR 200m Night: RVR 300m	Day: RVR 300m Night: RVR 550m, 350m with HUD or Autoland	
Taxiway light system	No effect			

Note: When the airfield anemometer is failed or downgraded, the Commander shall:

- Assess whether to land.
- Decide which pilot conducts the landing.
- PM monitors wind presentation on the ND.
- PF/PM assesses windsock information (other ground source).
- Captain only landing at night.

8.4.2.5 Flight Crew Actions in Case of Autopilot Failure at or Below DH in Fail-passive CAT III Operations

GM1 SPA.LVO.105 (c)

For operations to actual RVR values less than 300 m, in the event of an autopilot failure at or below DH a missed approach shall be executed.

8.4.3 LVO Approval

SPA.LVO.105

Specific Approval Criteria

Malta Air has obtained a specific approval for low visibility operations (as stated in the Operations Specifications), having demonstrated that:

1. For low visibility approach operations, the aircraft has been certified in the AFM for the intended operations;
2. The flight crew members are competent to conduct the intended operation and a training and checking programme for the flight crew members and relevant personnel involved in the flight preparation has been established, in accordance with (SPA.LVO.120) as described in OMD 2.1.9 and OMD 2.6.3;
3. Operating procedures for the intended operations have been established as described in FCOM;
4. Any relevant changes to the Minimum Equipment List (MEL) have been made;
5. Any relevant changes to the maintenance programme have been made;
6. Procedures have been established to ensure the suitability of aerodromes, including instrument flight procedures, for the intended operations, in accordance with SPA.LVO.110 as described in OMC 15.5; and
7. For the intended operations, a safety assessment has been carried out, and performance indicators have been established to monitor the level of safety.

8.4.3.1 Aircraft Certification for the Intended Operations

AMC1 SPA.LVO.105 (a)

Malta Air aircraft used for low-visibility approach operations are equipped in accordance with the applicable airworthiness requirements and certified in the AFM as follows:

- For CAT II operations, the aircraft should be certified for CAT II operations.
- For CAT III operations, the aircraft should be certified for CAT III operations.

8.4.3.2 Safety Assessment – Monitoring, Data Collection and Performance Indicators

AMC1 SPA.LVO.105 (g)

Malta Air monitors LVOs in order to validate the effectiveness of the applicable aircraft flight guidance systems, training, flight crew procedures, and aircraft maintenance programme, and to identify hazards.

1. Data is collected whenever an LVO is attempted regardless of whether the approach is abandoned, is unsatisfactory, or is concluded successfully. This data shall be recorded on the auto coupled approach report form. The data should include records of the following:
 - a. occasions when it was not possible to commence an approach due to deficiencies or unserviceabilities of related airborne equipment;
 - b. occasions when approaches were discontinued, including the reasons for discontinuing the approach and the height above the runway at which the approach was discontinued;
 - c. occasions when system abnormalities required pilot intervention to ensure a continued approach or safe landing;
 - d. landing performance, whether or not the aircraft landed satisfactorily within the desired touchdown area with acceptable lateral velocity or cross-track error. The approximate lateral and longitudinal position of the actual touchdown point in relation to the runway centre line and the runway threshold, respectively, should be recorded.
2. Data about LVOs should be collected by means of the Malta Air's flight data monitoring programme supplemented by other means including reports submitted by flight crew.
3. Performance indicators include the following:
 - a. the rate of unsuccessful low-visibility approaches, i.e. the number of attempted approaches terminating in discontinued approaches, approaches where pilot intervention was required to ensure a

- continued approach or safe landing or where landing performance was unsatisfactory, compared to the number of low-visibility approaches attempted;
 - b. measures of performance of the airborne equipment for low-visibility approaches;
 - c. safety performance indicators related to other specific risks associated with LVOs.
4. The following information are retained for at least 5 years:
- a. the total number of low-visibility approaches or operations with an operational approval attempted or completed, including practice approaches, by aircraft type; and
 - b. reports of unsatisfactory approaches and/or landings, by runway and aircraft registration, in the following categories:
 - i. airborne equipment faults;
 - ii. ground facility difficulties;
 - iii. missed approaches because of Air Traffic Control (ATC) instructions;
 - iv. other reasons.

8.4.3.3 Safety Assessment Prior to Obtaining an Approval

AMC2 SPA.LVO.105 (g)

Prior to commencing LVOs, Malta Air demonstrated to the TMCAD that such operations will achieve an acceptable level of safety. This required Malta Air to gather data from operations using the relevant systems and procedures and conduct safety assessments taking that data into account.

8.4.3.4 Successful Approach and Landing

GM1 SPA.LVO.105 (g)

Specific Approval Criteria

1. An approach may be considered to be successful if:
 - a. from 500 ft to start of the flare:
 - i. speed is maintained within ± 5 kt of the intended speed, disregarding rapid fluctuations due to turbulence;
 - ii. no relevant system failure occurs; and
 - b. from 300 ft to the DH:
 - i. no excess deviation occurs; and

- ii. no centralised warning (including aural) gives a missed approach procedure command.
- 2. A landing may be considered to be successful if:
 - a. no relevant system failure occurs;
 - b. no flare failure occurs;
 - c. longitudinal touchdown is beyond a point on the runway 150 m after the threshold and before the end of the Touchdown Zone (TDZ) (750 m from the threshold);
 - d. lateral touchdown with the outboard landing gear is not outside the TDZ edge;
 - e. sink rate is not excessive;
 - f. bank angle does not exceed a bank angle limit.

8.4.3.5 Suitable Aerodromes – Assessment

The Flight Operations Performance Engineering Department maintains a register of Aerodromes and Runway ends assessed and approved for Low Visibility Operations (LVO). Any approved aerodrome (OMC Appendix A, Approved Aerodromes List) with an LVO landing procedure that has not passed an assessment will be notified to Crew by Company NOTAM or Airfield Brief (AFB) as not authorised.

8.4.3.6 Verification Autoland

When instructed by Company NOTAM, the Captain shall complete a Verification Autoland in accordance with the Company NOTAM. The minimum RVR/VIS requirements and any Crew Qualification requirements for the Verification Autoland will be specified in the Company NOTAM.

Note: The Verification Autoland shall be recorded on the auto coupled approach report form.

8.4.4 Operating Procedures – Cat II/Cat III

SPA.LVO.110, AMC3 SPA.LVO.105 (c), AMC4 SPA.LVO.105 (c)

Specific Approval Criteria

Malta Air shall ensure that only aerodromes, including instrument flight procedures, suitable for the intended operations are used for LVOs.

Malta Air only conducts CAT II or CAT III operations when:

- Each aircraft concerned is certified for operations with a Decision Height (DH) below 200 ft, and signed off as CATII/III in the Technical Log and on the centre pedestal indicating compliance with applicable regulations

For CAT II operations, the following applies:

- The flight crew consists of at least two pilots.
- The approach is flown using a certified system (autopilot/autoland) as identified in the AFM.
- All height callouts below 200 ft above the runway threshold elevation are determined by the use of a radio altimeter.
- The DH is determined by the use of a radio altimeter.
- At DH, the following visual references are distinctly visible and identifiable to the pilot:
 - A segment of at least three consecutive lights, which are the centre line of the approach lights or TDZ lights or runway centre line lights or edge lights or a combination of these.
 - A visual reference that should include a lateral element of the ground pattern, such as an approach lighting crossbar, or the landing threshold, or a barrette of the TDZ lighting unless the operation is conducted using a HUD or an equivalent system to touchdown.

For CAT III operations, the following applies:

- The flight crew consists of at least two pilots.
- The approach is flown using a certified system (autopilot/autoland) as identified in the AFM.
- All height callouts below 200 ft above the runway threshold elevation are determined by the use of a radio altimeter.
- The DH is determined by the use of a radio altimeter.
- At DH, the following visual references are distinctly visible and identifiable to the pilot:
 - A segment of at least three consecutive lights, which are the centre line of the approach lights or TDZ lights or runway centre line lights or edge lights or a combination of these.

Note: The approach/autoland shall be recorded on the auto coupled approach report form.

8.4.5 Aerodrome Related Requirements

Malta Air shall not use an aerodrome for LVOs below a visibility of 800 m unless:

- The aerodrome has been approved for such operations by the State of the aerodrome; and
- Low Visibility Procedures (LVP) have been established.

If the term LVP is not used, the equivalent procedures that adhere to the requirements of LVP at the aerodrome must be stated in the Jeppesen charts in use thereby assuring an equivalent level of safety. If an LVP or equivalent procedure is not presented in the Jeppesen charts or Airfield Brief, autolands shall not be conducted in LVP.

Practice CAT II/III Approaches

Practice CAT II/III approaches are permitted to unprotected runways, with visibility 1,500 m or greater and ceiling 500 ft or higher, including at night.

Verify all Landing Distance Required for all practice autolands. During a practice approach, system malfunction in itself does not require a Go-Around. In a practise CAT II/III approach the captain may decide to take over manually and land provided the approach can be continued safely.

Autoland Upgrade

Ideally the autoland for the purpose of an autoland upgrade will be conducted to an operational and protected CATII/III runway. However a satisfactory autoland for upgrade is valid if conducted on any runway with a serviceable and operational ILS.

8.4.6 Competence of the Flight Crew for the Intended Operations – Experience in Type or Class, or as Pilot-in-command/commander

SPA.LVO.120, AMC1 SPA.LVO.120 (a)

1. Malta Air shall ensure that the flight crew is competent to conduct the intended operations.
2. Malta Air shall ensure that each flight crew member successfully completes training and checking for all types of LVOs for which an approval has been granted. Such training and checking shall:
 - a. include initial and recurrent training and checking;
 - b. include normal, abnormal and emergency procedures;
 - c. be tailored to the type of technologies used in the intended operations; and

- d. take into account the human factor risks associated with the intended operations.
3. Malta Air shall keep records of the training and qualifications of the flight crew members.
4. The training and checking shall be conducted by appropriately qualified personnel. In the case of flight and flight simulation training and checking, the personnel providing the training and conducting the checks shall be qualified in accordance with Annex I (Part-FCL) to Regulation (EU) No 1178/2011.

To ensure that the flight crew is competent to conduct the intended operations, Malta Air has assessed the risks associated with the conduct of low-visibility approach operations by pilots new to the aircraft type or class and take the necessary mitigations. Such mitigations include an increment to the visibility or RVR for LVOs in accordance with OMA 5.1.3.1.

Prior to conducting LVO Malta Air, ensures that each flight crew member:

- complies with the training and checking requirements prescribed in the operations manual Part D, including Flight Simulation Training Device (FSTD) training, in operating to the limiting values of RVR/VIS (visibility) and DH specific to the operation and the aircraft type;
- is qualified in accordance with the standards prescribed in the operations manual;
- training and checking is conducted in accordance with a detailed syllabus.

The syllabus for an initial Low Visibility Operations qualification includes three practice approaches and landings (CAT II or CAT III as appropriate) to be carried out during Line Flying under Supervision.

To clarify this, a First or Second Officer may conduct these practice approaches, in CAT I conditions or better, under the supervision of a Line Captain who is himself LVO qualified on the aircraft being flown. A Captain completing this part of his initial LVO training will require the supervision of either a Line Training Captain or TRE.

The Flight Crew qualification is specific to the operation (take-off, CAT II/III) and the aircraft type. The Malta Air approvals for Flight Crew operating on the B737-800/8200 are as follows:

Take-off: 125 m RVR

Landing CAT II 100 ft DH/300 m RVR

Landing CAT III 50 ft DH/175 m RVR

8.4.6.1 CAT II/III ILS

For any CAT II or CAT III approach the ILS must be promulgated as operating to the required standard, i.e. “ILS CAT II” or “ILS CAT III” as appropriate. The performance criteria are contained in ICAO document Annex 10. In order to prevent electromagnetic interference with the ILS ATC is required to establish an ILS CRITICAL AREA. The associated “sensitive area” must be protected from infringement by taxiing aircraft and other vehicles when any automatic approach/landing is in progress. The effect of such interference, should it occur, is excessive aileron and elevator activity during the approach. Anytime an Autoland is planned the cabin crew must be notified to make a PA to instruct passengers to switch OFF all PED’s (PA as per SEP manual). If a go-around is completed as a result of suspected spurious electromagnetic interference, the aircraft Commander shall instruct the Cabin Crew to ensure that all portable electronic devices are switched off before a subsequent approach is carried out. On landing a Technical Log entry shall be made and a ASR submitted.

8.4.6.2 Competence of the Flight Crew for the Intended Operations – Recent Experience for Cat II and Cat III Approach Operations

AMC3 SPA.LVO.120 (a)

To be considered competent:

1. Pilots authorised to conduct low-visibility approach operations should complete at least two approaches using Ryanair’s procedures for low-visibility approach operations during the validity period of each operator proficiency check or periodic demonstration of competence, unless credits related to recent experience when operating more than one type are defined in the operational suitability data established in accordance with Regulation (EU) No 748/2012.
2. If a flight crew member is authorised to operate as pilot flying and pilot monitoring, the flight crew member should complete the required number of approaches in each operating capacity.

8.4.7 Operating Procedures for LVOs

AMC1 SPA.LVO.105 (c)

Malta Air has established procedures and instructions to be used for LVOs in this manual and FCOM Vol 1, containing the duties of flight crew members during taxiing, take-off, approach, flare, landing, rollout and missed approach operations, as appropriate.

Prior to commencing an LVO, the pilot-in-command/commander shall be satisfied that:

- The status of the visual and non-visual facilities is as required;

- Appropriate LVPs are in force according to information received from Air Traffic Services (ATS);
- If LVPs are required for such operations, LVPs are in effect: and
- The Flight crew members are appropriately ‘qualified’;
- The aircraft is ‘qualified’ (technical log and CAT I/II/III placard status);
- ILS ‘qualified’;
- Runway ‘qualified’.

General

LVOs include the following when the applicable RVR is less than 550 m.

- manual take-off,
- auto-coupled approach followed by auto-flare, auto-landing and manual rollout.

Procedures

Malta Air procedures and instructions applicable to LVO on B737-800/8200 series aircraft are also stated in the following manuals: FCOM Vol 1, OMD and MEL.

The following parameters for LVO are approved by the TM CAD:

1. Malta Air has been approved for CAT II and CAT III (minimums as indicated on OPS SPEC Certificate).
2. Max. headwind component: 25 kts.
3. Max crosswind component: 20 kts.
4. Max tailwind component: 10 kts.
5. Dual Channel Approach.
6. Two engine approach, flaps 30/40.
7. No reported windshear.
8. Glideslope capture by 1500 ft AAL.
9. Mid-point RVR 125 m, and stop end RVR 75 m.
10. The Flight Crew consists of 2 qualified flight crew members (LHS/RHS).

Runway Markings and Lighting System

Runway lighting and markings must conform to the specifications of ICAO Annex 14.

The significant addition to CAT I systems are the supplementary approach lights on the inner 300 metres (red barrettes), and the touchdown zone lighting.

The inner 300 m (1,000 ft) of the HIALS is supplemented with red approach barrettes. Their colour indicates clearly the pre-threshold area. The lateral dimensions of the red barrettes is the same as the white touchdown zone lights.

Between the two there is a full width green bar marking the runway threshold.

Stop-end lighting: To indicate that the aircraft is coming towards the end of the runway during the stopping roll, at 910 m (3,000 ft) from the stop-end, the centreline lights are alternatively red and white, and for the last 300 m (1,000 ft) they become red only.

Approach lighting system patterns are shown in the Jeppesen General Airway Manual (Intro/Chart Legend/Charting Symbol Legend).

Further information is available in the All Weather Operations Guide which is a Flight Operations Controlled Document. This document must be revised in conjunction with the Malta Air Operations Manual Part A and Boeing FCOM.

8.4.8 Minimum Equipment

(SPA.LVO.130)

The minimum equipment that has to be serviceable at the commencement of an LVO in accordance with the aircraft flight manual (AFM) is stated in the MEL.

The commander shall be satisfied that the status of the aircraft and of the relevant airborne systems is appropriate for the specific operation to be conducted.

8.5 ETOPS – Extended Range Operations for Two-engined Aeroplanes

Malta Air does not have ETOPS Approval.

8.6 Minimum Equipment List

ORO.MLR.105, AMC1 ORO.MLR.105(c)

A minimum equipment list (MEL) is established as specified under 8.a.3 of Annex IV to Regulation (EU) 2018/1139, based on the relevant master minimum equipment list (MMEL) as defined in the data established in accordance with Regulation (EC) No 748/2012.

The MEL and any amendment thereto shall be approved by the TM CAD. Malta Air shall submit an amended MEL to TM CAD within 90 days from the effective date specified in the approval page of the MMEL.

In addition to the list of MEL items, the MEL shall contain:

- a preamble, including guidance and definitions for flight crews and maintenance personnel using the MEL;
- the revision status of the MMEL upon which the MEL is based and the revision status of the MEL;
- the scope, extent and purpose of the MEL.

Malta Air has;

- reviewed the MEL to ensure that rectification intervals for each inoperative instrument, item of equipment or function listed in the MEL are established, and ensures that the rectification interval in the MEL is not less restrictive than the corresponding rectification interval in the MMEL;
- establish an effective rectification programme;
- ensured that operation of the aircraft after expiry of the rectification interval specified in the MEL occurs only when the defect has been rectified or the rectification interval has been extended in accordance with approved maintenance procedures.

Malta Air has established a procedure controlled by the NPCA and NPFO and approved by the TM CAD for 'one time extension' of category B, C and D rectification intervals. NPCA and NPFO have approved the specific duties and responsibilities for controlling extensions. This procedure is subject to the rules below:

1. The extension of the rectification interval is within the scope of the MMEL for the aircraft type;
2. The extension of the rectification interval is, as a maximum, of the same duration as the rectification interval specified in the MEL;

3. The rectification interval extension is not used as a normal means of conducting MEL item rectification and is used only when events beyond the control of Malta Air have precluded rectification;
4. The TM CAD is notified of any extension of the applicable rectification interval; and
5. A plan to accomplish the rectification at the earliest opportunity is established.

The operational (O) and maintenance (M) procedures referenced in the MEL are compliant with the MMEL and are documented in the DDG and shall be completed prior to dispatch. The DDG and MEL are on board all Malta Air operating aircraft and are readily accessible to aircrew.

Subject to a specific case-by-case approval by the TM CAD, Malta Air may operate an aircraft with inoperative instruments, items of equipment or functions outside the constraints of the MEL but within the constraints of the MMEL, provided that:

- The concerned instruments, items of equipment or functions are within the scope of the MMEL as defined in the data established in accordance with Regulation (EC) No 1702/2003;
- The approval is not used as a normal means of conducting operations outside the constraints of the approved MEL and is used only when events beyond the control of Malta Air have precluded the MEL compliance;
- A description of specific duties and responsibilities for controlling the operation of the aircraft under such approval is established by the Malta Air; and
- A plan to rectify the inoperative instruments, items of equipment or functions or to return operating the aircraft under the MEL constraints at the earliest opportunity is established.

8.6.1 Minimum Equipment List MEL

CAT.IDE.A.105

GENERAL

The MEL is a document that lists the equipment that may be temporarily inoperative, subject to certain conditions, at the commencement of flight. This document is prepared by Malta Air for particular aircraft taking account of their aircraft configuration and all those individual variables that cannot be addressed

at MMEL level, such as operating environment, route structure, geographic location, aerodromes where spare parts and maintenance capabilities are available etc., in accordance with a procedure approved by the TM CAD.

- Where non-safety related equipment serves a second function, such as lockable cockpit door, Malta Air has developed and includes operational contingency procedures in the MEL in case of an equipment malfunction.
- Where non-safety related equipment is part of another aircraft system, for example the electrical system, procedures should be developed and included in the MEL for deactivating and securing in case of malfunction. In these cases, the item should be listed in the MEL, with compensating provisions and deactivation instructions if applicable. The rectification interval will be dependent on the secondary function of the item and the extent of its effect on other systems.

If an item is not listed in the MEL the commander can verify the serviceability of the part by the use of contacting MAINTROL.

The MEL provides the Commander with the necessary authority to operate a flight with inoperative equipment. The Commander retains the final discretion to accept and operate the aircraft in accordance with the MEL. In planning a flight, the Commander will take account of any operational constraints imposed on him as a consequence of having to operate with inoperative equipment.

Whereas the format of the MEL lists equipment which may be inoperative for dispatch, it does not consider items which are clearly necessary for airworthiness such as flight controls etc, nor does it list items which are obviously not required for safe operation such as passenger reading lights, etc. It is recognised that acceptance of multiple "MEL items" which individually might be safe but taken together may interact to compound the operational situation.

The NPFO controls the MEL/DDG content with the approval of the TM CAD and provides only current MEL/DDG which are carried on board.

EXTENT OF THE MEL Minimum equipment list

The MEL shall be used when required before the commencement of the flight (i.e before the aircraft first moves under its own power). If a failure occurs between the commencement of the flight and the start of the take-off, any decision to continue the flight is subject to pilot judgement and good airmanship. It is prudent for the commander to refer to the MEL before any decision to continue the flight is taken.

It is prudent to review the MEL as unservicabilities arise to ensure the most expeditious access to Maintenance. This frequently includes interaction with Maintrol during flight, relayed by ATC or handling agents enroute

8.6.2 Configuration Deviation List (CDL)

The CDL contains items of secondary airframe structure or engine parts which may be missing or otherwise non-functioning, but for which the aircraft may be dispatched. The CDL is promulgated as an appendix in the AFM and may be included in the MEL.

Operation with a “CDL item” may incur a performance penalty or an operational limit, cognisance of which may also be needed at the planning stage of a flight as well as during flight.

If an MEL/CDL item is entered into the LIDO flight planning system, the OFP is capable of reflecting certain performance penalties and operational limits. MEL and CDL items are also accounted for in the OPT take-off and landing performance solution.

8.7 Non Commercial Operations

ORO.AOC.125

Malta Air conducts a limited number of Non Commercial flights using aircraft listed in the Operations Specifications Certificate. A Non Commercial flight is a flight conducted entirely at the expense of Malta Air. If passengers are on board they are non-revenue passengers.

All non-commercial flights except Check flights and Display flights must be operated in accordance with the operating procedures specified in Annexes III, IV and V of Commission Regulation (EU) 965/2012 and the Malta Air Operations Manual, as if the flight were commercial air transport.

The type of flight shall be specified in flight related documents such as OFP and other equivalent documents. ORO.AOC.125(d). Non commercial flights are operated in accordance with the Malta Air Operations Manuals suite, as if the flight were commercial air transport. AMC2 ORO.AOC.125(a).

Check Flights are operated under the Malta Air OMA 8.7 or a Permit to Fly.

Display Flights are operated under an associated Flight Authorisation issued by the TMCAD (which would include any flight conditions specified by EASA under Part 21).

A company notam will indicate the status of the non-commercial flight.

On any flight without Cabin Crew, it is the responsibility of the Pilot in Command to complete a security check of the galley, cabin and toilets of the aircraft. The Pilot in Command is to ensure that all equipment is properly secured. This duty maybe delegated to the First Officer. Both forward doors shall be armed. Care must be taken to disarm the doors before they are opened. The carriage of non-

revenue passengers on non-commercial flights without cabin crew is permitted, provided that the Pilot in Command ensures compliance with the additional provisions and procedures described in Section 8.7.3.2.

8.7.1 Malta Air Conducts the Following Types of Non Commercial Operations

GM1 ORO.AOC.125(a)(2)

- Demonstration Flights;
- Check Flights;
- Relocation Flights;
- Training Flights (other than line training);
- Other Non-commercial flights.

Additional considerations or special approvals specified for Non-Commercial flights as listed in this section are provided to the operating crew, including:

- identification of the applicable requirements, CAT or Permit/Flight Authorisation
- a clear identification of any differences between operating procedures used when conducting commercial air transport and check flights, including such notification to TM CAD in advance;
- a means of ensuring that all personnel involved in the operation are fully familiar with the associated procedures.

8.7.1.1 Demonstration Flights

ARO.GEN.310(a), ORO.AOC.125(a)(2)

The purpose of a demonstration flight is to demonstrate the aircraft's handling, performance, and functionalities to the TM CAD, the media, or to actual or potential buyers or lessees. These flights are conducted under the normal requirements of the Malta Air Operations Manual, as if the flight were commercial air transport and contain no non-standard manoeuvres or profiles. The flights must be authorised by the NPFO.

1. 'Route proving flight' is a flight to demonstrate the aircraft's flying characteristics and systems to the TM CAD, for verification of compliance with the operational requirements. ARO.GEN.310(a).
2. A public relations flight or a commercial 'fly-around' is a flight targeting official or media representatives.

These flights may or may not have cabin crew as authorised by the NPFO.

8.7.1.2 Maintenance Check Flights (MCFs)**1. Levels of maintenance check flights**

SPO.SPEC.MCF.110

The Company carries out Level ‘A’ and Level ‘B’ Maintenance Check Flights (MCFs).

Level ‘A’ check flights are flights where the use of abnormal or emergency procedures is expected as defined in the Aircraft Flight Manual or where it is required to prove the functioning of a backup system or other safety device.

The Level ‘A’ MCFs conducted by The Company are

- a. Check Flight for Renewal of Certificate of Airworthiness/End of Lease/Full Check
- b. Check Flight for Elevator Hydraulic Power Off
- c. Check Flight to Correct Asymmetrical Flight Control Forces

Note: End of Lease Check flights are elective Check Flights for the purpose of demonstrating an aircraft and may not follow a maintenance procedure but will be treated as MCFs as the items checked are as per Level ‘A’ Flights.

Level ‘B’ check flights are any other maintenance check flight other than a Level ‘A’ check flight. No use of abnormal or emergency procedures is required on these flights.

2. Flight Programme for “Level A” flights

SPO.SPEC.MCF.105

A full flight programme for “Level A” flights is published in the Ryanair Maintenance Check Flight Manual

3. Maintenance Check Flight Manual for Level A Flights

SPO.SPEC.MCF.110

All operations and associated procedures to be used in “Level A” check flights are detailed in the Ryanair Maintenance Check Flight Manual.

This manual will be updated by a person nominated by NPFO and such updates will be promulgated in Docunet to all qualified Maintenance Check Pilots.

4. Flight Crew Requirements for “Level A” maintenance check flights

SPO.SPEC.MCF.115

“Level A” maintenance check flights will be conducted only by qualified and current Maintenance Check Flight Pilots. The entry requirements, course syllabus and recency requirements for Maintenance Check Pilots are detailed in Ops Part D.

5. Flight Crew Training Course for “Level A” maintenance check flights

SPO.SPEC.MCF.120

To qualify as a Maintenance Check Flight Pilot the candidate will meet the entry requirements and pass the course as detailed in OMD.

The training course followed on one aircraft category is considered valid for all aircraft types of that category. Any differences between the aircraft used in training and the aircraft to be flown on the check flight will be detailed in the Ryanair Maintenance Check Flight Manual.

6. Crew Composition and persons onboard

SPO.SPEC.MCF.125

- a. The crew composition and requirement for a task specialist is detailed for each type of “Level A” check flights in the Ryanair Maintenance Check Flight Manual.
- b. The PIC shall ensure that no unnecessary personnel are on board for the Maintenance Check Flight (MCF).
- c. Task Specialists are specially qualified Engineering staff. The requirement for and the duties of the task specialist are detailed for each type of “Level A” Check flight in the Ryanair Maintenance Check Flight Manual.
- d. Any MCP qualified pilot from any AOC in the Ryanair Group may operate a Maintenance Check Flight on an 9H registered aircraft in Malta Air subject to the following provisions;

ORO.AOC.125(a)

- i. The MCP qualification is current and in compliance with the Ryanair Maintenance Check Flight Manual.
- ii. The MCF crew on Malta Air (9H registered) aircraft shall have a current EASA Part-FCL Licence, or a Non-EASA Licence with an TMCAD validation issued in accordance with EASA Part-FCL.
- iii. The MCF crew may be over 65 years of age.
- iv. The Malta Air NPFO accepts the MCF crew prior to the commencement of the MCF.
- v. Rostering of MCF is done by the operating crew AOC and all legality checks must be valid.
- vi. The FTL records are maintained in the operating crew AOC crew management system and shared with the Malta Air NPFO ASAP on completion of the MCF.

- vii. Any reports required are filed on the operating crew AOC reporting system and shared ASAP with the Malta Air NPFO and Safety Manager (a non Malta Air pilot will not have access to the Malta Air reporting system).
 - viii. The Malta Air Operations Manuals suite applies for the conduct of the flight. Relevant differences shall be briefed to the operating crew by the Malta Air NPFO or his nominee.
 - ix. The aircraft registration shall be filed in the flight plan and is the callsign for the flight.
 - x. Operating crew AOC recency requirements apply.
7. Simulated abnormal or emergency procedures in flight
- SPO.SPEC.MCF.130
- By way of derogation from point SPO.OP.185 a task specialist may be onboard for a “Level A” Maintenance Check flight if the task specialist is required to meet the intention of the flight and has been identified in the flight programme section of the Ryanair Maintenance Check Flight Manual.
8. Flight Time Limitations and rest requirements
- SPO.SPEC.MCF.135
- Maintenance Check Flights are a normal flight duty and the duty time and rest requirements are as per a line flight as detailed in this manual, Chapter 7.
9. Systems and Equipment
- SPO.SPEC.MCF.140
- When a maintenance check flight is intended to check the proper functioning of a system or equipment that system shall be identified as potentially unreliable and appropriate mitigation measures shall be agreed prior to the flight in order to minimise risks to flight safety. This identification and discussion of mitigation will be done in the pre-flight briefing between flight crew and (if applicable task specialist) using the guidance available in the Ryanair Maintenance Check Flight Manual.
10. Cockpit Voice Recorder, Flight Data Recorder and Data Link Recording Equipt
- SPO.SPEC.MCF.145
- When the aircraft is engaged in a maintenance check flight the requirement for CVR and DFDR is unchanged from normal operations.

8.7.1.3 Relocation Flights

The following flights are operated in accordance with the normal requirements of the Malta Air Operations Manual, as if the flight were commercial air transport, except that Malta Air may designate a pilot for the flight as Pilot in Command or co-pilot, who has attained the age of 65 or greater.

AMC2 ORO.AOC.125(a)

Non-commercial operational differences are described below.

Non-Malta Air crew may operate designated non-commercial operations on a Malta Air (9H registered) aircraft. Both the Pilot in Command and the co-pilot must:

- Be from the same (Boeing) Ryanair Group AOC.
- Hold a current EASA Part-FCL License, or a Non-EASA License with an TMCAD validation issued in accordance with EASA Part-FCL.
- Hold an EASA Class 1 medical certificate in accordance with Part-MED.
- Be current and recent on the aircraft type/variant to be operated.
- Operate the flight in accordance with the Malta Air Operations Manual suite. Prior to the commencement of the flight, both crewmembers must be briefed of any differences by the Malta Air NPFO or his nominee.
- Either or both flight crew members may be over 65 years of age.

The Malta Air NPFO accepts the non Malta Air operating crew prior to the commencement of the flight. Rostering of non commercial flights under this section of the Operations Manual is done by the operating crew AOC and all legality checks must be valid. The FTL records are maintained in the operating crew AOC crew management system and shared with the Malta Air NPFO ASAP on completion of the flight(s). Any reports required are filed on the operating crew AOC reporting system and shared ASAP with the Malta Air NPFO and Safety Manager, (a non Malta Air pilot will not have access to the Malta Air reporting system).

1. **‘Ferry Flight *’:** On Occasion it may be necessary for Malta Air to operate a flight for maintenance purposes.

This flight will be operated with any additional briefing material required made available to the operating crew. The Operations Control Centre will coordinate with operating crew and with the Performance Department for any specialist evaluations required for flight planning. This flight is authorised by the Malta Air NPFO or his nominee in advance.

Examples of ferry flights conducted by Malta Air:

- a. Unpressurised ferry flight,

- b. Gear-down ferry flight,
 - c. AOG maintenance ferry flight.
2. **‘Delivery Flight’**: a flight to transfer the aircraft from the acceptance location (manufacturer, refurbishment location, previous owner, lessor/lessee, long-term storage) to a delivery location. Malta Air has approval from TM CAD to conduct delivery flights in accordance with the Operations Specifications Certificate.

OMD details a special crew qualification to be designated Pilot in Command of a delivery flight and minimum experience levels for the co-pilot. If a co-pilot is assigned to operate, this must be authorised by the NPFO.

If non-revenue passengers are carried, the number is limited to 19 and each will be approved by the NPFO. The requirements of carriage of passengers without cabin crew will be applied.

Report time for crew on Delivery flights is normally 60 mins prior to the planned departure time. This may be reduced by the commander to not less than 45 mins. Operational procedures are detailed in the Delivery Flight Brief (Boeing). The OFP is prepared by Jeppesen and issued by Boeing Dispatch. The format is the same as the standard LIDO OFP.

Malta Air applies ‘Dispatch Permissible’ minima to aerodromes identified as adequate enroute aerodromes for Delivery flights. This means that at the relevant time of potential use, the forecast weather must be such that will allow the conduct of the relevant Instrument Flight Procedure to make an approach to the aerodrome.

Diversion Distance for non-commercial operations Delivery flights, is 90 minutes flying time at an OEI cruising speed according to the AFM in still air standard conditions based on the actual take-off mass (624 nm for the B737-8200).

In order to apply this extended diversion distance, the following conditions must be met:

- a. The aircraft may not dispatch with any deferred defect of a critical system.
- b. No Dangerous Goods carried in the aircraft holds.

This flight is authorised by the Malta Air NPFO or his nominee.

3. **‘Positioning Flight’**: a flight to position the aircraft to a location from which a further flight will be performed. This flight is authorised by the Malta Air NPFO or his nominee in advance. Positioning flights carry no fare paying passengers.

4. **‘Recovery Flight *’:** a flight to position a serviceable aircraft from its current location to an adequately secure location for various reasons e.g. to remove it from a hazardous area. This flight is authorised by the Malta Air NPFO or his nominee in advance.

Note: * Ferry flights and Recovery flights which operate under a flight permit require a briefing from the duty pilot prior to dispatch.

8.7.1.4 Training Flights: Base Training

The conduct of Base Training is described in OMD. Base training flights are authorised by the NPFO.

Weather Minima and Flight Limitations for Base Training Flights

It is the Instructors responsibility to ensure that training is not carried out in conditions of ceiling, visibility and crosswind that are inappropriate to the exercise involved and the experience, competency or capacity to learn of the trainee.

The lowest minima, ceiling, visibility and crosswind associated with flying training shall be those laid down in the Part D, Training Manual.

During base training only the Flight Crew involved, including a safety pilot if required representatives of the TM CAD and those listed below may be carried on board.

1. Malta Air/Ryanair Group staff who have a particular function to carry out on the flight may be authorised by the Training Instructor, provided they have been listed as crew members; and
2. Non Malta Air/Ryanair Group personnel who have a specific reason to be on board may be authorised by the NPFO.

8.7.1.5 Other Non-Commercial Flights

1. Private flights or leisure flights,
2. Charity flights.

These flights are performed at the full cost of Malta Air. For these flights, the operational control remains with Malta Air. The normal requirements of the Malta Air Operations Manual, as if the flight were commercial air transport apply and these flights are authorised by the NPFO. For cabin crewing requirements see 8.7.3.2.

8.7.1.6 Display Flight

Display flights require a flight authorisation from the TM CAD. The conduct of a display flights is described in the Flight Authorisation documentation. The operating crew will comply with the provisions specified in the Flight Authorisation. Only essential crew may be on board.

8.7.2 Safety Management and Risk Assessment AMC1 ORO.AOC.125(a)(2)(ii)

Malta Air manages the risks associated with flights described in this section under the Malta Air Safety Management System as described in the Safety Management Manual. The NPFO may require completion of a Safety Risk Assessment for any non-commercial flight. Each safety risk assessment will follow established Malta Air protocols and will include at least the following:

- a. Flight profile (including manoeuvres to be performed, any simulated abnormal situations in flight, duties and responsibilities of the crew members);
- b. Levels of functional equipment and systems (MEL, CDL);
- c. Operating procedures, minima, and dispatch criteria;
- d. Flight and duty time limitations and rest requirements, cumulative and transient fatigue;
- e. Selection, composition, and training of flight and cabin crew; and
- f. Categories of passengers on board.
- g. Fatigue risks arising from any combination of non-commercial flights and commercial flights

8.7.3 Crew Requirements

A non-Malta Air operating crew under OMA 8.7 must be from the same AOC, i.e. either a Ryanair or Ryanair Sun crew. No interoperable crewing is permitted.

8.7.3.1 Non-Commercial Flights: Operating Crew

Non-commercial Flights are operated in accordance with the normal requirements of the Malta Air Operations Manual, as if the flight were commercial air transport, except that Malta Air may designate a pilot for the flight as Pilot in Command or co-pilot, who has attained the age of 65 or greater (*AMC2 ORO.AOC.125(a)*).

Maintenance Check Flights

Any pilot operating a Maintenance Check Flight on a Malta Air (9H registered) aircraft must have completed training to comply with Malta Air OMA and Malta Air OMD training and checking criteria.

Maintenance Check Pilots-Seattle will be Maintenance Check Pilots who have also completed at least 1 Boeing Customer Acceptance Flight or completed pre-Customer Acceptance Flight Simulator training. These pilots are qualified to accept new aircraft from the Boeing Delivery Centre, Seattle.

First Officers rostered for Maintenance Check flights will have a minimum of 1,500 hrs flight time on type.

The Maintenance Check Pilot shall be the Pilot in Command and the responsibility for the safe completion of the Maintenance Check flight in accordance with the guidance contained in this manual lies solely with him or her.

Responsibilities and Duties of the individual crew members are as detailed in the Malta Air Flight Crew Operations Manual. Where differences/additions to duties exist in a specific test sequence they are detailed in the relevant chapter of the MCF Manual and as appropriate in the Permit to Fly. These duties shall be thoroughly briefed by the Pilot in Command prior to the flight.

Options	MCF Pilot (Pilot in Command)	Co-Pilot	Total Crew
1. Engineer available	MCP or MCPS	F/O 1500 hrs minimum	3
2. Engineer unavailable	MCP or MCPS	MCP or MCPS	2

Delivery Flights

Pilots are qualified to act as Delivery Pilots once the training program as detailed in the Malta Air Operations Manual Part D is completed. This qualification allows crew to conduct delivery flights. The PIC shall have completed three delivery flights prior to being the PIC on a delivery flight.

8.7.3.2 Non Commercial Flights without Cabin Crew

ORO.CC.100, AMC1 ORO.CC.100(d)(2), AMC3 ORO.AOC.125(a)(2)(ii), GM1 ORO.CC.100(D)(2), CAT.OP.MPA.155

Non Commercial flights may be carried out without cabin crew if the passenger number is 19 or less. The following differences and restrictions apply:

- 1. Passengers are restricted as follows
 - a. The maximum number of passengers is 19.
 - b. Revenue (ticketed) passengers are prohibited.
 - c. Each passenger must be approved by the Chief Pilot and or Security Manager. Such approval will be communicated to the Captain before the flight.

- d. All passengers must be 'known' i.e. a staff member or directly vouched for by Chief Pilot/Security Manager.
 - e. The number of SCP/PRMs permitted depends on having an assigned crewmember or other passenger to each SCP/PRM who will assist the SCP/PRM in an emergency/evacuation. The number of SCP/PRMs permitted is determined by the NPFO.
 - f. Passengers and crew will be subject to security screening at the departure airfield.
2. Flight crew assigned to these duties are trained in first aid procedures and familiar with operational differences applicable to these flights.
- a. Communication and coordination between flight crew and passengers:
 - i. The flight deck door remains open during all non-critical phases of flight enabling cabin surveillance and communication with passengers,
 - ii. During critical phases of flight, the flight deck door is closed and locked – the passenger will use the cabin attendant call button, which the flight crew can hear, to get flight crew attention. (Both ACP PA selected),
 - iii. Direction from the flight deck will be made on the PA system initially,
 - iv. During non-critical phases of flight as the flight deck door is unlocked passengers may communicate directly with the flight crew.
 - b. Flight Crew member Incapacitation
 - i. The non-incapacitated flight crew member will direct passengers to secure and/or remove the incapacitated crew member from his/her station. First Aid should be rendered by the passenger(s) with direction from the operating pilot via PA if appropriate. The operating crew member will declare an Emergency and initiate a diversion.
 - c. Cabin surveillance will be achieved by securing the flight deck door in the OPEN position by use of the foot operated lock. During checklist operation pre-departure and pre-landing, a visual check will be made of the cabin by the PM. A PA instructing all passengers to be seated will be made before this check. During the cruise the non-flying pilot will conduct a check of the cabin approximately every 30 mins if safe to do so.

- d. Procedures for rapid egress from the aircraft in case of rapid disembarkation or evacuation are included in the passenger briefing. The flight crew will refer the passengers to the relevant section of the safety card as well as physically pointing out the exits.
- e. The operation and use of emergency exits is included in the passenger briefing and the flight crew will assist with evacuation if possible as per standard evacuation procedures.
- f. The location and use of Oxygen is included in the passenger briefing.
- g. The location and use of life jackets is included in the passenger briefing.
- h. Seating of passengers will be as far forward in the cabin as can be allowed by Weight and Balance. If passengers need to occupy the MID zone they should be seated in the front rows of the MID Zone. This will facilitate cabin surveillance from the flight deck as well as rapid egress using the L1 door. (If needed any emergency exit may be used).
- i. Passenger briefing will be completed by a member of the flight crew using the general safety brief on EFB and by reference to the 'Non-Commercial Ops without Cabin Crew Checklist' to provide information on communication methods and the location and use of equipment not included in the general safety briefing such as a fire extinguisher, first-aid equipment (e.g. first-aid kit, defibrillator if fitted), smoke hood, etc. In the event of a requirement to use this equipment inflight, the non-flying pilot will use the equipment and utilize the assistance of passenger(s) if appropriate.
- j. If ALL the passengers are qualified (crew members or engineering staff) the Captain may elect not to deliver the general safety brief and instead brief using the 'Non-Commercial Ops without Cabin Crew Checklist' to concentrate on specific differences pertaining to the actual flight.
- k. In the case of additional emergency/safety equipment being onboard, passengers will be shown its location and briefed on its operation by the briefing crew member at the briefing stage. Examples of such equipment would be portable ELT and life raft on a delivery flight.

8.7.3.3 Non-Commercial Ops Without Cabin Crew Checklist

- Passenger load Max 19
- Passenger approval received from Chief Pilot and/or Security Manager
- Passenger e-briefing Completed

- Passengers shown location and instructions for use on the equipment
- First Aid Kit
- Fire Extinguisher
- Smoke Hoods
- Torches
- Slides/Doors
- Life raft/portable ELT when on board
- Passengers briefed on requirement to be secured prior to Take-Off and Landing and when fasten seatbelt sign ON during flight
- Passengers briefed on use of the Cabin Crew Call button and Cabin Crew Interphone to gain Flight Crew attention
- Passenger seating assigned by the flight crew to facilitate surveillance from the flight-deck. Passengers will be seated as adjacent to the L1/R1 exits as compatible with W&B requirements
- Flight Crew Arm and Disarm all slides
- Flight Deck door CLOSED for take-off and landing and remain OPEN during all non-critical phases of flight
- PA selected on both ACPs to enable hearing cabin crew call bell when used

8.7.3.4 Flight and Duty Time Limitations and Rest Requirements for Commercial and Non-Commercial Operations

AMC2 ORO.AOC.125(a)(2)(ii), ORO.FTL.210

When conducting a mix of commercial and non-commercial flights in the same duty period the provisions of the Malta Air FTL scheme shall apply. Additionally, any potential increase in fatigue associated with the non-commercial flight, e.g. degree of difficulty or complexity, shall be considered.

8.7.3.4.1 Flight and Duty Time Limitations and Rest Requirements for Mixed Operations

AMC1 ORO.AOC.125(a)

When conducting a mix of commercial and non-commercial flights by non Malta Air crew members, the operating crew AOC ensures:

- The flight duty complies with the provisions of 'Flight times and duty periods' of the operating crew to ensure compliance with Subpart FTL. ORO.FTL.210.
- Mixed operations are included in its safety risk management process to ensure that the fatigue risks arising from any combination of commercial and non-commercial flights do not affect the operations safety margins.

8.8 Oxygen Requirements

8.8.1 Use of supplemental oxygen

(CAT.OP.MPA.285)

The commander shall ensure that flight crew members engaged in performing duties essential to the safe operation of an aircraft in flight use supplemental oxygen continuously whenever the cabin altitude exceeds 10,000 ft for a period of more than 30 minutes and whenever the cabin altitude exceeds 13,000 ft.

8.8.2 Supplemental oxygen — Pressurised Aeroplanes

(CAT.IDE.A.235)

1. Malta Air aircraft are pressurised, operate at pressure altitudes above 10,000 ft and are equipped with supplemental oxygen equipment that is capable of storing and dispensing the oxygen supplies in accordance with Table 1.
2. Malta Air aircraft are pressurised, operated at pressure altitudes above 25,000 ft and are equipped with:
 - a. Quick donning types of masks for flight crew members;
 - b. Sufficient spare outlets and masks or portable oxygen units with masks distributed evenly throughout the passenger compartment, to ensure immediate availability of oxygen for use by each required cabin crew member;
 - c. An oxygen dispensing unit connected to oxygen supply terminals immediately available to each cabin crew member, additional crew member and occupants of passenger seats, wherever seated; and
 - d. A device to provide a warning indication to the flight crew of any loss of pressurisation.
3. The individual oxygen dispensing units are automatically deployable.

4. The total number of dispensing units and outlets exceed the number of seats by at least 10%. The extra units are evenly distributed throughout the passenger compartment.

Oxygen Minimum Requirements for Pressurised Aeroplanes

Supply for	Duration and Cabin Pressure Altitude
1. Occupants of flight crew compartment seats on flight crew compartment duty.	a. The entire flying time when the cabin pressure altitude exceeds 13,000 ft. b. The remainder of the flying time when the cabin pressure altitude exceeds 10,000 ft but does not exceed 13,000 ft, after the initial 30 minutes at these altitudes, but in no case less than: 30 minutes' supply for aeroplanes certified to fly at altitudes not exceeding 25,000 ft; and 2 hours' supply for aeroplanes certified to fly at altitudes of more than 25,000 ft.
2. Required cabin crew members.	a. The entire flying time when the cabin pressure altitude exceeds 13,000 ft, but not less than 30 minutes' supply. b. The remainder of the flying time when the cabin pressure altitude exceeds 10,000 ft, but does not exceed 13,000 ft, after the initial 30 minutes at these altitudes.
Supply for	Duration and Cabin Pressure Altitude
3. 100% of passengers (*)	The entire flying time when the cabin pressure altitude exceeds 15,000 ft, but in no case less than 10 minutes' supply.
4. 30% of passengers (*)	The entire flying time when the cabin pressure altitude exceeds 14,000 ft, but does not exceed 15,000 ft.
5. 10% of passengers (*)	The remainder of the flying time when the cabin pressure altitude exceeds 10,000 ft, but does not exceed 14,000 ft, after the initial 30 minutes at these altitudes.
(*) Passenger numbers Table 1 refer to passengers actually carried on board, including persons younger than 24 months.	

8.8.2.1 Determination of Oxygen

AMC1 CAT.IDE.A.235

In the determination of the amount of supplemental oxygen required for the routes to be flown, it is assumed that the aeroplane will descend in accordance with the emergency procedures specified in the operations manual, without exceeding its operating limitations, to a flight altitude that will allow the flight to be completed safely (i.e. flight altitudes ensuring adequate terrain clearance, navigational accuracy, hazardous weather avoidance etc.).

The amount of supplemental oxygen shall be determined on the basis of cabin pressure altitude, flight duration and on the assumption that a cabin pressurisation failure will occur at the pressure altitude or point of flight that is most critical from the standpoint of oxygen need.

- The amount of oxygen required for Flight Crew, including all persons in the cockpit supplied from the Flight Crew oxygen supply source is as specified in Operations manual Part B, Vol 1, Performance Dispatch and MEL Sec 35.

Following a cabin pressurisation failure, the cabin pressure altitude shall be considered to be the same as the aeroplane pressure altitude.

If a flight deck Oxygen mask has been removed make a Technical Log entry. Do not attempt to clean a flight deck Oxygen mask visor without consulting engineering.

8.8.2.2 Supplemental Oxygen – Pressurised Aeroplanes

(AMC2 CAT.IDE.A.235)

OXYGEN REQUIREMENTS FOR FLIGHT CREW COMPARTMENT SEAT OCCUPANTS AND CABIN CREW IN ADDITION TO THE REQUIRED MINIMUM NUMBER OF CABIN CREW.

1. For the purpose of supplemental oxygen supply, flight crew compartment seat occupants who are:
 - supplied with oxygen from the flight crew source of oxygen will be considered as flight crew members; and
 - not supplied with oxygen by the flight crew source of oxygen will be considered as passengers.
2. Cabin crew members in addition to the minimum number of cabin crew and additional crew members will be considered as passengers for the purpose of supplemental oxygen supply.

The amount of supplemental oxygen required can be determined from data provided in the AOM.

8.8.2.3 Supplemental Oxygen — Non-pressurised Aeroplanes

(CAT.IDE.A.240)

Malta Air does not operate unpressurised revenue flights. Unpressurised ferry flights may be dispatched with NPFO approval and the requirements shall be specified in the ferry flight approval.

Oxygen Minimum Requirements for Non-pressurised Aeroplanes

Supply For	Duration and Cabin Pressure Altitude
1. Occupants of flight crew compartment seats on flight crew compartments duty and crew members assisting flight crew in their duties.	The entire flying time at pressure altitudes above 10,000 ft.
2. Required cabin crew members.	The entire flying time at pressure altitudes above 13,000 ft and for any period exceeding 30 minutes at pressure altitudes above 10,000 ft but not exceeding 13,000 ft.
3. Additional crew members and 100% of passengers (*).	The entire flying time at pressure altitudes above 13,000 ft.
4. 10% of passengers (*).	The entire flying time after 30 minutes at pressure altitudes above 10,000 ft but not exceeding 13,000 ft.
(*) Passenger numbers in Table 1 refer to passengers actually carried on board, including persons younger than 24 months.	

8.8.2.4 Crew Protective Breathing Equipment

CAT.IDE.A.245

All Malta Air aircraft are equipped with protective breathing equipment (PBE) to protect the eyes, nose and mouth and to provide oxygen for a period of at least 15 minutes and their locations are indicated in the SEP Manual.

(e) Quick donning masks on Malta Air aircraft facilitate flight crew interphone system and radio communications (CAT.IDE.A.170, 330).

8.8.2.4.1 Crew Protective Breathing Equipment

AMC1 CAT.IDE.A.245

PROTECTIVE BREATHING EQUIPMENT (PBE)

- Malta Air has a PBE installed in the Flight deck, procedures for use are presented in the Safety and Emergency Procedures Manual. (CAT.IDE.A.235, 240).

8.8.2.5 First-aid Oxygen

CAT.IDE.A.230

Malta Air aircraft are equipped with a supply of undiluted oxygen for passengers who, for physiological reasons, might require oxygen following a cabin depressurisation.

The oxygen supply is calculated in compliance with applicable regulations. (CAT.IDE.A.235)

There shall be a sufficient number of dispensing units, but in no case less than two, with a means for cabin crew to use the supply.

The first-aid oxygen equipment shall be capable of generating a mass flow to each user of at least 4 litres STPD per minute.

The location of the first-aid oxygen bottles is given in the Safety and Emergency Procedures Manual, and Part B, Vol 2, Chap 1. Passengers can pre order supplemental oxygen. To accommodate this requirement all aircraft are fitted with 2 extra Oxygen bottles, Malta Air aircraft have 6 bottles on board.

The amount of first-aid oxygen required is specified in the MEL. Also an additional amount of first-aid oxygen required as specified in the MEL is carried in accordance with CAT.IDE.A.230.

When Oxygen is used this is entered in the Cabin defects Log, the SCCM will advise the Commander to include this entry in the tech log. If no engineer is available at a destination the Commander shall contact Maintrol and raise a HIL on Maintrol instructions. Nothing in this procedure shall prevent any passenger receiving as much oxygen as is required for medicinal or emergency reasons.

8.9 Procedures Related to the Use of Type B EFB Applications

8.9.1 Electronic Flight Bag

Malta Air has approval for the use of Type B applications for portable Electronic Flight Bag (EFB) in accordance with SPA.EFB.100.

The EFB (C-PED) is a company issued and authorised device, approved for use in non-transmitting mode only during flight so as to not affect the performance of aircraft systems, equipment or the ability to operate the aircraft. Procedures in FCOM Vol 1 require the EFB to be switched to 'AIRPLANE MODE' prior to pushback to comply with this requirement.

As part of the approval process both the EFB hardware and software have undergone a comprehensive safety risk assessment. Human-Machine Interface (HMI) is also evaluated. Additionally, operational trials were conducted to

identify potential malfunctions and failure scenarios, analyse their operational effects, and, if necessary, propose mitigation means which are contained later in this section.

The internal EFB Policy and Procedures Manual contains procedures for the EFB Administrator to manage and maintain the EFB, and each EFB function. EFB Flight Crew training is conducted in accordance with OMD 2.1.38 and 3.1.36.

Crew members must ensure that the EFB is charged and in good condition for operational use. This includes battery, screen and overall condition. Crew are expected to maintain the unit in the protective casing provided.

Utilisation of device 'sleep mode' is recommended whenever possible. This is a battery conservation technique.

The Following EFB Hardware is Approved for Operational Use:

- iPad Air
- iPad Air 2
- iPad 9.7'
- iPad 10.2'

The Following Type B Applications are Approved for Operational Use:

- Electronic Flight Log (EFL)
- Boeing OPT (Performance)
- Boeing OPT (Mass and Balance)
- Jeppesen FD Pro (Navigational Charts)
- Docunet
- Flight Dispatch Folder (FDF)

New App versions are advised to crew via memo. Current App versions are listed in the EFB Admin update emails.

Electronic Signature

AMC1 CAT.POL.MAB.105(c)

Electronic Signature is required for submission of eLID, EFL and OPT Weight and Balance. The Ryanair Group airlines computerised electronic Loading Information Document, Electronic Flight Log and mass and balance system uses an electronic PIN Code on a tablet screen in accordance with AMC1 CAT.POL.MAB.105(c). The use of the PIN code is, from a legal and responsibility point of view, considered to be fully equivalent to signature by hand.

Management of Change

AMC1 ORO.GEN200(a)(3)(e)

Malta Air has a requirement to manage changes to existing systems, equipment, programmes, products and services; and introduction of new equipment or procedures.

Hazards may inadvertently be introduced into the operation whenever change occurs. Effective safety risk management requires that hazards that are a by-product of change are systematically and proactively identified and that strategies to manage the safety risks of the consequences of hazards are developed, implemented and subsequently evaluated.

Modifications to an EFB may have to be introduced, either by the EFB system suppliers, the EFB applications developers, or by the operator itself. The modifications which:

- a. do not bring any change to the calculation algorithm and/or to the HMI of a type B application,
- b. introduce a new Type A app or modify an existing one (provided its software classification remains Type A),
- c. do not introduce any additional functionality to an existing Type B application, or
- d. update an existing database necessary to use an existing Type B, may be introduced by the operator without the need to notify the competent authority. These changes should, nevertheless, be controlled and properly tested prior to use in flight.

The modifications in the following non-exhaustive list are considered to meet these criteria:

- a. Operating system updates;
- b. Chart or airport database update;
- c. Update to introduce fixes (patch); and
- d. Type A application installation and modification.

For all other types of modification, the operator should apply the change management procedure approved by the competent authority in accordance with rule ARO.GEN.310(c). Chapter 3 of the Malta Air Safety Management Manual documents the change management process.

The formal process for the management of change takes into account the criticality of the EFB and Type B applications. Criticality is closely related to safety risk. Criticality relates to the potential consequences of failing to properly manage the change. It may relate to the consequence of equipment being improperly operated or an activity being incorrectly executed.

Determination of which process and activities are safety critical is normally a consideration that should be made during the system design process; however it becomes relevant during a situation of change. Clearly, some activities are more essential for safe delivery of services than others.

Equipment and activities that have higher safety criticality should be reviewed as part of the change process to ensure that appropriate mitigation actions are taken to control potentially emerging safety risks.

Change Impact Assessment Procedure

1. Identify the nature and scope of the changes (e.g. Algorithm changes, HMI changes, New Type B applications, Type B functionality changes, Type B database changes, Operating system changes, Hardware changes);
2. Identify hazards related to implementing the proposed change and their possible consequences;

8.9.1.1 Use of Mounting Device

There is an EFB mounting device fitted to the sliding window in the flight deck. EFB may be used during all phases of flight provided it is retained in its mounting device during critical phases of flight (Taxi out/Takeoff/Climb/Descent/Approach/Taxi in). Crew may remove the device from its mount during pre-flight briefing on the ground and for approach briefings during cruise. The EFB will not normally be handled below 1000 ft AGL.

Note: If the mounting device is not available, then crew must place the EFB in their flight bag during critical phases of flight.

8.9.1.2 EFB Flight Crew Operating Procedures

Any EFB defects or degradation of any kind must be reported as soon as practical via Zendesk the dedicated EFB support line in order to ensure no delays to the operation occur.

8.9.1.3 EFB Pre-flight Procedures

Crew must satisfy themselves that the EFB device is suitable for operational use during pre-flight preparation.

Applications to be used during flight will be opened at check-in to confirm their availability. The database validity for all operational applications must be checked against the validity published in the Crew Alert section of the Lido briefing pack.

8.9.1.4 EFB Normal Cockpit Procedures

Normal cockpit procedures allow the use of a single EFB for Malta Air operations. The F/O's EFB is considered the primary device for operational reference. Both crew are permitted to reference individual EFBs for appropriate phases of flight. PF shall handover control of the aircraft when referencing the EFB to ensure that flight crew members do not become preoccupied with the EFB system at the same time.

Pre-flight performance calculations shall normally be performed using both EFB's. The Captain and co-pilot must independently perform the take-off performance calculation and inputs to confirm accuracy. If any differences are evident between the separate calculations then the reasons for the difference must be established. ‘

During taxi, the taxi charts shall always be active on the co-pilots EFB.

Operating policy is not to have two heads in the cockpit at any time. EFB may be referenced at any time once this policy is adhered to.

8.9.1.5 EFB Phase of Flight Procedures

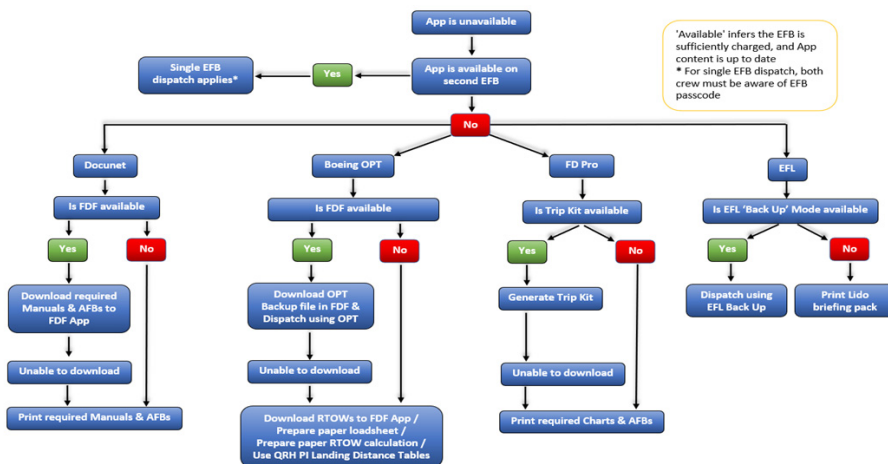
Briefings shall be conducted with the EFB ‘in hand’ in order to promote CRM. The mounting device provided shall be used to stow the EFB during all critical phases of flight. The PM may have the SID charts active during departure, and the Approach charts active during arrival if required.

8.9.1.6 Contingency or Abnormal Procedures

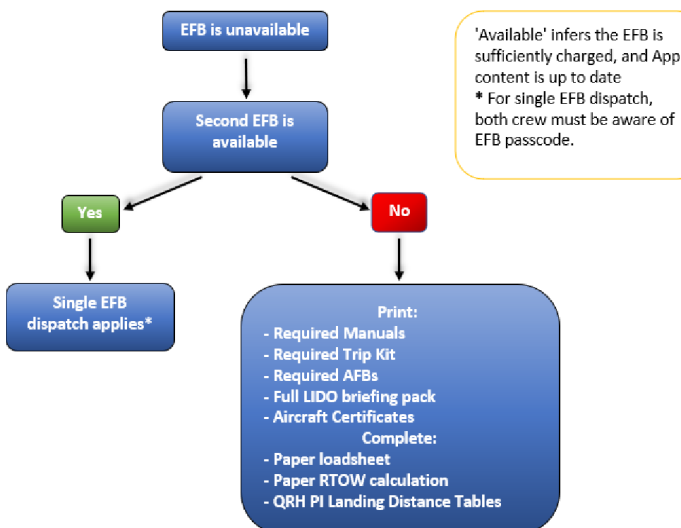
If any system malfunction (e.g. display unit interference) is encountered that could be related to the EFB device being active, immediately disconnect and switch off the device.

Devices can overheat when left in direct sunlight for extended periods of time, rendering the device unusable. EFB operating temperature (0° to 35° C) will be adhered to in order to ensure operability. The EFB will be mounted when not in use. It is not permissible to store the device on the flight deck glareshield.

8.9.1.6.1 EFB Contingency Flow Chart – App(s) Unavailable



8.9.1.6.2 EFB Contingency Flow Chart – EFB(s) Unavailable



Note: Crew will contact Operations to discuss options prior to commencing large printing tasks. Printing should be completed using the MFP printer in the crewroom where possible (2 pages to 1 sheet).

Downloading to FDF App:

Select the required document within the FDF App to download. A grey icon indicates that the document has not been downloaded yet. A green icon indicates that the document is up to date. An orange icon indicates that an updated version of the document is available for download. If you see an orange icon, you must download the required document prior to use.

8.9.1.6.3 Single EFB Failure Prior to Departure

Malta Air operations are authorised for single EFB use. However, if an EFB becomes inoperative prior to departure, a replacement will be sought in the base. If a replacement EFB is not available, then crew will follow the steps contained in the 'EFB Contingency Flow Chart' to determine an alternative means of compliance.

For single EFB operations, when the F/O has completed the calculation and noted the speeds, the OPT App will be reset and the serviceable EFB will be handed to the Captain. The Captain and co-pilot must independently perform the take-off performance calculation to confirm accuracy.

8.9.1.6.4 Single EFB Failure During Flight

Should one EFB become inoperative during flight, then the flight may be continued using the second EFB. Malta Air operations are authorised for single EFB dispatch.

8.9.1.6.5 Dual EFB Failure Prior to Departure

Should both EFBs become inoperative prior to departure, replacements will be sought in the base. If one replacement EFB is available, then crew will follow guidance contained in 'Single EFB failure prior to departure' above. If no replacement EFBs are available, then crew will follow the steps contained in the 'EFB Contingency Flow Chart' to determine an alternative means of compliance. Paper Trip Kits are authorised for Malta Air operations. Should the dual EFB failure occur out of base, contact Malta Air operations. The crew will be provided with a replacement EFB device or paper backup documentation. An ASR shall be filed in the event of dual EFB failure.

8.9.1.6.6 Dual EFB Failure During Flight

Should both EFBs become inoperative during flight, with no back up paper charts available on board, crew should use available FMC information and obtain necessary enroute or approach details from the ATC unit on VHF to allow safe approach and landing at the destination/diversion airport. Crew will also be aware that much of this information required for landing is readily available in the FMC. FMC info may be used in this case to limit the time spent transmitting on a potentially busy frequency. Landing performance data shall be extracted from the QRH Performance Inflight section. An ASR shall be filed in the event of dual EFB failure.

8.9.1.6.7 EFB Not Charged Prior to Departure (Both Aircraft USB Chargers Unavailable)

Portable EFB batteries shall normally be charged above 70% at the pre-flight stage.

Flight Duty Period	Minimum Battery Power %
< 6 hours	60%
< 5 hours	50%
< 4 hours	40%
< 3 hours	30%
< 2 hours	20%

Moderate use of the EFB will result in approx. 7% battery discharge per hour. Should either device have less than the minimum battery required for your duty, and with both aircraft USB chargers unavailable, crew will follow the steps contained in the 'EFB Contingency Flow Chart – EFB(s) unavailable' to determine an alternative means of compliance.

8.9.1.7 In-flight Charging

The internal battery is the primary power supply for the EFB. A USB power supply is located on each side of the flight deck if required. EFB may be connected to the USB power supply. Charging is allowed during all phases of flight. The EFB being charged must be secured in the mounting device.

The only approved source for inflight charging is the installed USB power supply.

Caution: Disconnect the EFB from the power supply when the flight deck is unattended.

8.9.1.8 Post Flight Procedures

Crew shall ensure the relevant flight documents have been forwarded to the appropriate recipients.

The crew will check to ensure that the following documentation which has been previously submitted electronically prior to dispatch has been acknowledged:

Note: Flight information must be submitted by the PM at the end of each sector. Flight information must be verified by the Captain at the end of each duty.

1. OPT take-off performance calculations (including W&B).
2. Electronic Flight Log (EFL).

Any EFB hardware or software defects shall be promptly reported via Zendesk or the dedicated EFB support line as appropriate.

Crew will confirm that they are in possession of their own EFB post duty.

8.9.1.9 Boeing OPT (On Board Performance Tool) Application

The Boeing OPT is a software application for computing aircraft mass and balance, all engine climb gradient, take-off and landing performance. OPT calculates “real-time” RTOWs for all actual conditions existing at time of departure, such as variations of aircraft configuration, ambient conditions and runway length. The airport, runway, take-off obstacle database and aircraft weights and W&B envelope is maintained and administrated by the Ryanair Performance Department. OPT allows crew to adjust performance for MEL & CDL items. Flight crew may also enter airport/runway & NOTAM information that is unavailable in the database. Dispatch landing distances, In-flight landing performance data and QRH non normal configuration landing distances are also available for use. Crew must ensure the application content is up to date prior to flight by crosschecking database validity against the Crew Alert section of the OFP.

8.9.1.10 Jeppesen (Electronic Charting) Application

The Jeppesen FD Pro App is an electronic charting application providing terminal and enroute aeronautical charts as well as the associated airways manual. The application also hosts Airfield Briefing (AFB) documents as published by the performance department. These are available via the *INFO/WX* tab on the Jeppesen FD Pro App. Crew must ensure the application content is up to date prior to flight by crosschecking database validity against the Crew Alert section of the OFP. Updates will occur with the 28-day AIRAC cycle and mid cycle revisions. Crew can expect to update their Jeppesen FD Pro App database weekly.

Note: The use of Jeppesen IFW (In-Flight Weather) should not be used to support tactical decisions and/or as a substitute for certified aircraft systems (e.g., weather radar).

8.9.1.11 Docunet (Electronic Manuals) Application

Flight manuals will be stored in an electronic format accessible on the EFB device. The Docunet App is the primary reference for all manuals except the QRH. Crews must use the paper QRH when operationally necessary. Should crew need to reference the QRH on the EFB, ensure the screen displays the full content, selecting the ‘fit page to screen’ icon within the App if necessary. Electronic manual revisions will be in the form of a file comprising the entire revised manual or section. Crew will be advised when a revision to a manual has been made and available to download.

8.9.1.12 Flight Dispatch Folder (FDF) Application

The FDF App is a secondary source of flight information for crew. FDF provides access to Performance (RTOWs and AFBs), Navigational (Terminal and Enroute Charts) and Flight Documentation (Flight Manuals) required for dispatch. Should any operational App become unavailable, then crew should follow the guidance in 8.9.1.6.1 (EFB Contingency Flow Chart – App(s) unavailable). If FDF is required for use as a contingency measure, then crew must select the required flight documentation within the FDF App and download prior to flight. The FDF App should only be used in the event of any failure that prevents flight crew from accessing or updating operational application data (Boeing OPT or Docunet).

8.9.1.13 Electronic Flight Log (EFL) Application

The EFL App provides required flight information in electronic format. The EFL must be completed correctly and verified by the Commander on completion of the series of sectors which constitute his assigned flight duty. The EFL contains details of flight times, delay reports, fuel data, FDT, DT, HIL and weather data and electronic briefing pack. The electronic flight plan (eFPL) is accessed within the EFL App and submitted to the storage database electronically at the end of flight. EFL contains all required flight documentation; OFP (eFPL), ATS flight plan, weather, NOTAMs, company NOTAMs, crew alerts, additional info, RAIM, significant weather charts, upper air data charts, and vertical profile information.

8.10 Emergency and Special Operating Procedures**8.10.1 General Criteria to Improve Safety and Handle Emergencies**

Malta Air operating philosophy and Malta Air operational policies are presented in FCOM Vol 1 and QRH introduction, and additional guidance is provided in the Boeing FCTM, Chapter 8, Non Normal Operations.

8.10.2 International Emergency Procedures, Distress Communications

The OMC, Jeppesen Publications (Pubs) > General Airway Manual > Emergency Data – ICAO > Distress and Urgency Radiotelephony Communication Procedures contains a section which gathers together extracts from the various ICAO publications on Emergency Procedures.

The section covers:

- Emergency Procedures,
- Unlawful Interference (Special Emergency),
- Emergency Descent,
- Distress and Urgency R.T. Communications Procedures,
- Communications Failure,

- Interception, and
- Search and Rescue.

The section also identifies variations to these procedures which are specific to individual countries.

Flight Crew members shall have a general understanding and appreciation of these procedures, and in addition some specific knowledge of certain items including distress frequencies, transponder “squawk” codes used in these procedures, and the Communication Failure procedures as well as the Distress and Urgency message format.

8.10.2.1 Distress Procedures

These procedures are presented in the OMC, Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data* – ICAO.

8.10.2.1.1 Intercepting a Distress Call

These procedures are presented in the OMC, Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data* – ICAO.

8.10.2.2 Urgency Procedures

These procedures are presented in the OMC, Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data* – ICAO.

8.10.2.3 Information on Emergency and Survival Equipment Carried

(AMC1 CAT.GEN.MPA.145)

Malta Air shall ensure that there are available for immediate communication to rescue coordination centres, lists containing information on the emergency and survival equipment carried on board all of our aeroplanes. The information shall include, as applicable, the number, colour and type of life-rafts and pyrotechnics, details of emergency medical supplies, water supplies and the type and frequencies of emergency portable radio equipment.

8.10.3 Declaration of Emergency, Alerting ATC to Aircraft Problems

“MAYDAY MAYDAY MAYDAY Malta Air call sign”

Squawk 7700.

8.10.3.1 Reporting Malfunctions or System Failures

Any aircraft malfunction or system failure which significantly or potentially degrades the normal levels of safety for the continuation of the flight shall be reported to ATC either on an advisory basis or because facilitation or assistance may be required.

Additionally, the degree of severity of other problems such as an impending fuel situation or environmental difficulty (turbulence, thunderstorms, icing etc.) warrant contact with ATC and possible facilitation or assistance in terms of a change of clearance or flight plan.

It is not practically possible to give guidance on how to deal with every abnormal or emergency situation. Once actions have been taken to bring the immediate problems under control and to stabilise the situation, then judgement will dictate whether the subsequent follow-up action should be precautionary or an emergency should be declared.

Certain situations may become more serious with the passing of time (classically, for example, an impending fuel shortage) and it is important that in such a deteriorating situation a timely emergency declaration shall be made.

An emergency situation is one in which there is a real potential for an accident and where the aircraft requires facilitation and assistance and priority over other traffic in order to minimise the potential risk.

8.10.3.2 Situations Warranting Declaration of an Emergency

A situation in which the aircraft may not be able to reach a suitable airfield for landing is clearly a full emergency. Other cases where the Commander has cause to doubt the aircraft's serviceability to land, or where a warning circuit gives evidence of hazard, shall also be treated as an emergency. Typical examples would be:

1. Unsafe landing gear indications or wheel or tyre damage;
2. Suspected complete brake failure;
3. Indications of fire or smoke in the passenger cabin or freight compartment;
4. Unexplainable adverse response from fly controls or degradation in aircraft control capability;
5. Complete electrical generator failure; and
6. Insufficient instrument/navigation capability for the approach.

Where it is considered to be an emergency situation, the Commander will declare an emergency with ATC and indicate his intentions and the assistance required.

Given the nature of the emergency ATC can be expected to have the airport emergency services fully deployed and take any further action considered appropriate.

For SSR procedures, refer to Jeppesen Publications (Pubs) > *General Airway Manual* > *ATC* > *Flight Procedures SSR Transponder Operating Procedures* > *Emergency Procedures*.

8.10.3.3 Actions to be Taken After Declaration of Emergency

In the case of an emergency, the Commander will take the following actions:

1. Direct the Cabin Crew to prepare for an emergency landing;
2. Advise passengers on the P.A. of the situation (time available);
3. Brief the co-pilot on his required function and elaborate on his intentions;
4. Determine whether a delay in landing is necessary to allow full deployment of ground emergency services;
5. Where possible, ensure that the full procedures detailed in the appropriate QRH checklist have been completed. (Don't rush the process if there is no benefit from an early landing. If time available, manage the situation;
6. Decide and brief the crew on emergency procedures to be followed after landing; and
7. Safety harnesses tight and locked for landing.

8.10.3.4 Precautionary Landing

8.10.3.4.1 A precautionary landing is one where the Commander considers the fire, rescue/medical services might be required after landing, but the aircraft is expected to be capable of manoeuvring under its own power. Typical examples would include:

1. Single engine landing;
2. Hydraulic failure, but leaving the aircraft with the standby brake system;
3. Landing gear unsafe light ON but mechanically visual indicators showing locked down;
4. Flapless landing;
5. A fire warning, but apart from the light no other indications of fire.

If it is considered that the situation requires only precautionary actions to be taken, then advise ATC of the nature of the malfunction or problem and to what extent the operation of the aircraft is constrained or limited.

If required, indicate what facilitation or assistance would be needed or helpful.

If it is considered desirable for fire and rescue services to stand-by, do not refer to them as the "emergency services". According to the procedures of the airport authority these services may be deployed to the landing runway.

8.10.3.4.2 Actions in the Event of a Precautionary Landing

In the case of a precautionary landing the Commander will take the following actions:

1. Inform the Cabin Staff of the situation and that action shall be limited to full observance of the normal landing checks;
2. Advise the Cabin Crew what if anything he intends to say to passengers and the element of discretion needed in dealing with passengers;
3. Advise the Cabin Crew of any procedures required after landing and that they may or may not expect to see the fire and rescue services in attendance;
4. Ensure that the abnormal or emergency procedure outlined in the Emergency Checklist have been completed. (Don't rush the process. Time is expected to be available so it shall be used to manage the situation – e.g. extend the approach/circuit as appropriate, see Section 8.10.1);
5. Based on the nature of the unserviceability determine with ATC, any facilitation required whether an approach priority is desirable or whether a delay would facilitate deployment of fire and rescue services;
6. If it is considered necessary to inform passengers, care will be taken not to over-dramatise the situation or use operational jargon. A lot of passengers get upset with the mention of malfunction and systems not working as they have no concept of the use of back-up or standby systems. Any announcement shall convey a sense of routine.

8.10.3.5 Actions after Emergency/Precautionary Landing

After an emergency or precautionary landing, the Commander will inform ATC whether he can manoeuvre under his own power and whether he wishes to dispense with the fire and rescue services.

In some cases, an "escort" of the fire services to the ramp may be necessary or desirable but in other cases this practice is neither necessary nor desirable.

In cases of fire or smoke indication, the airport procedures could call for the situation to be dealt with away from terminal buildings so that the aircraft may be required to hold its position.

The Commander must keep abreast of the situation which can be confused by a three-way relationship involving ATC and the fire services, and continually assess the possible need for an emergency evacuation.

8.10.3.6 Reporting

"All aircraft shall report to the appropriate aeronautical communications station, as soon as possible, any hazardous flight conditions encountered during a flight" – Article 44 Air Navigation (Operations) Order, 1986.

The report shall include such details as may be pertinent to the safety of other aircraft.

After landing all crew will consult with the Commander to ensure that all pertinent information is captured, and affording the Commander the opportunity to debrief the crew prior to any other intervention/distraction.

Then the Commander shall file a ASR giving all relevant details of the emergency, malfunction and the subsequent action taken including the emergency or precautionary landing.

8.10.3.7 PAN PAN PAN

Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data* – ICAO provides definitions and guidance for PAN PAN PAN call.

8.10.4 Diversion in Case of Serious Mechanical Failure

Malta Air policy and operating philosophy is as presented in the Boeing FCTM, Chapter 8, Non-Normal Operations.

8.10.4.0.1 Public Address System

8.10.4.0.1.1 Malta Air shall not operate an aeroplane with a maximum approved passenger seating configuration of more than 19 unless a public address system is installed.

8.10.4.0.1.2 The public address system required by this paragraph must:

1. Operate independently of the interphone systems except for handsets, headsets, microphones, selector switches and signalling devices;
2. Be readily accessible for immediate use from each required Flight Crew member station;
3. For each required floor level passenger emergency exit which has an adjacent cabin crew seat, have a microphone which is readily accessible to the seated cabin crew member, except that one microphone may serve more than one exit, provided the proximity of the exits allows unassisted verbal communication between seated cabin crew members;
4. Be capable of operation within 10 seconds by a cabin crew member at each of those stations in the compartment from which its use is accessible; and
5. Be audible and intelligible at all passenger seats, toilets and cabin crew seats and work stations.

8.10.5 Bird Strike on Final Approach

8.10.5.1 An Aircraft is Hit by Birds While on Final Approach to Land

Should the pilot continue the approach or initiate a go-around?

Having encountered birds, the question to be answered is “what is the damage to the aircraft and what effect will this have on the safe conduct of the flight?”.

The full extent of any damage, to the engines and/or the control surfaces and landing gear, may not be apparent until applying power, configuring, or manoeuvring the aircraft. It might therefore be the case that, if a go-around is initiated, the pilot rapidly finds himself in a situation where the runway is disappearing beneath him but the aircraft cannot safely fly the missed approach procedure.

Therefore, in the above scenario, it is advisable to continue the approach to land.

8.10.5.2 Flocks of Birds Seen on Final Approach

A pilot sees a flock of birds ahead of him on final approach. Should the pilot continue the approach or initiate the go-around? Having seen the birds, the question to be answered is “if a go-around is initiated, how likely is it that the aircraft will avoid a bird strike?” There are two matters to consider. Firstly, when a flock of birds perceives a threat, it typically moves upwards, potentially into the path of the aircraft initiating a go-around. Secondly, the greater the engine thrust, the greater the damage caused by ingesting birds – it is probable that less damage will be caused if the birds are hit while engines are at low speed or idle. Therefore, in the scenario described above, unless a go-around can be achieved with a responsible degree of confidence that the aircraft will not hit birds, it is less hazardous to continue that approach to land.

8.10.6 Flight Data Recorder

CAT.IDE.A.190

Malta Air aircraft are equipped with Digital Flight Data Recorders in accordance with Air Ops.

8.10.7 ATC Avoidance Manoeuvres at High ALT

Guidance for high altitude high speed flight is given in FCTM, chapter 3 and guidance for optimum altitude in FCOM Vol 1.

1. 10° bank shall be selected at FL 300 for operations above FL300, and 25° bank shall be selected below FL 300. If the airfield brief specifies bank angle for an approach/go-around, this shall be selected as appropriate.

2. ATC Avoidance manoeuvre at high altitude in response to phrases “Avoiding Action” and “immediately” shall result in the following actions by MAY crews:
 - Execute the required action immediately and acknowledge the instruction.
 - Do NOT disengage the AutoThrottle or AutoPilot in the execution of the ATC instruction.
 - HDG mode shall be immediately engaged and the required heading selected.
 - The Bank Angle selector shall not be changed to achieve increased bank angle.
 - The aircraft shall always be in a position to respond to a RA, i.e. not conducting a manoeuvre that may lead to an excursion from the flight envelope at high altitude.
 - Radius of turn required for a controlled turn is 15nm at or below FL 190 and 22.5 nm at or above FL 200. 10 deg bank at or above FL 300 satisfies the 22.5 nm radius of turn requirement.

8.10.8 Medical Emergency in Flight

The primary task is to conduct the flight safely and then to endeavour to present the patient in a suitable airport as safely and as expeditiously as possible. The following guidance applies:

1. Flight deck advised by Cabin Crew.

Give:

- Age.
- Sex.
- Chief complaint.
- Level of consciousness.

Note: This allows medics to take the correct equipment to the passenger/patient and minimise any delay in the treatment or management of their condition once on the ground.

2. PAN × 3, inform ATC that “MEDICAL SERVICES” are required. ATC may assign a discrete frequency. In this case PM deals with discrete frequency, PF has control of the aircraft and operational ATC.

3. ATC will usually seek a report on the patient, this information may be passed to medical professionals, e.g. MEDAIRE. A VHF patch on discrete frequency may be established to enable crew to receive professional medical advice.
4. Plan to land at the nearest suitable airport. In this context the Malta Air/Ryanair Group network in Europe is extensive and resourced to deal with these events. The most suitable airport is a Malta Air/Ryanair Group Base, followed by a Malta Air/Ryanair Group destination. If neither of these is in equivalent proximity to other suitable airports then consider the closest reasonably large airport. It is probably prudent to avoid very large airports especially those which the crews are fundamentally unfamiliar.
5. Consider the level of medical service likely to be available, estimated time to landing (including direct routings from ATC). Consider crew familiarity with the intended destination. It may be quicker to fly a little farther to a familiar destination than to take a short cut to an unfamiliar airport.
6. The flight shall continue to be conducted at all times in accordance with the terms of the OM(A) and FCOM Vol I. A Medical emergency shall not be considered a reason to rush an approach or deviate from Malta Air SOP.
7. When time and operational conditions permit, the Flight Crew shall, after landing, confirm with the Tower that the medical services have been alerted.
8. It is likely the patient has consumed oxygen and this may need to be replenished prior to next departure.
9. When the aircraft arrives at the stand CSS shall make a PA requiring all passengers to remain seated so that the paramedic can access the aircraft and commence treatment on the passenger/patient. Disembarkation can commence via exits not involving the medics/patient.

8.10.8.1 Deceased Passenger

Where a passenger is pronounced dead on board (by a Doctor/Nurse) the decision for the crew is to return to departure point, divert or continue to destination. Operational experience shows that returning to departure point or continuing to destination provide the best level of support for family members. Diverting to an alternate will require the deceased passenger to be disembarked along with traveling companion at a location that was not part of any travel arrangements and can lead to additional stress at what will be inevitably a difficult time for passengers and crew.

8.10.9 Laser or High-Powered Light Exposure

If the aircraft is illuminated (attacked) by a laser or high-powered lighting, the following actions and precautions are recommended:

- Maintain aircraft control. Consider re-engaging the autopilot (if disengaged) and/or transferring aircraft control to the pilot less affected or less likely to be affected by the laser attack.
- Do not look into the beam and shield your eyes to the maximum extent possible. Note that even a low power “eye-safe laser” can impair vision for up to 8 minutes.
- If the laser or high-powered light strikes the vision of the PF, turn your head away, look down, and transfer aircraft control.
- Increase cockpit lighting to mitigate the laser beam’s impact on vision.
- If on approach and landing is not assured, consider executing a missed approach.
- Inform ATC as soon as possible, especially if diverging from the cleared flight path.
- If vision is impacted, avoid rubbing eyes to reduce the potential for corneal abrasion.

Following the laser attack and when flight safety permits:

- Advise ATC and include as much detail as possible (e.g., description of the laser’s/light’s location, direction of beam, colour, and the length of exposure).
- Submit ASR, including above detail and the ATC Callsigns and frequencies notified.
- If you have any post-event physiological concerns, seek guidance from your AME.

Regardless of the severity or impact on the flight, report all events as per OMA 11.3.4.

8.10.10 Global Aeronautical Distress and Safety System (GADSS)

CAT.IDE.A.190, 280, 285, CAT.GEN.MPA.205, 210

Regulation (EU) 2015/2338, Global Aeronautical Distress and Safety System (GADSS) delivers aircraft tracking (AT), autonomous distress tracking (ADT), and post flight localization and recovery (PFLR).

The GADSS objectives are to ensure:

- timely detection of aircraft in distress

- tracking of aircraft in distress and timely, accurate determination of end-of-flight location
- timely initiation and accurate direction of SAR actions
- efficient and effective SAR operations
- timely retrieval of Flight Recorder Data

The Malta Air Operations Control Centre shall maintain oversight and ensure operational compliance with respect to GADSS regulation. Malta Air's Operations Control Centre shall identify and coordinate GADSS requirements when generating operational flight plans, and ensure AT, ADT, PFLR compliance when the flight planned route transits through airspace which does not provide required four-dimensional tracking services at regular time intervals (less than 15 minutes) which includes:

- aircraft's latitude and longitude,
- altitude,
- and time at that position.

8.10.10.1 General Description

Ryanair DAC maintains, as part of the system for exercising operational control over the flights, an aircraft tracking system via GlobalBeaconSM systems and service contracts.

Malta Air's flights shall be tracked from take-off to landing, except when the planned route and the planned diversion routes remain within airspace whereby:

1. ATS surveillance service is normally provided which is supported by ATC surveillance systems locating the aircraft at time intervals with adequate duration; **and**
2. the operator has provided to competent air navigation service providers necessary contact information.

Malta Air's aircraft, tracking systems, and associated procedures comply with Regulation (EU) 2015/2338 requirements to ensure four-dimensional tracking (latitude, longitude, altitude, and time) when operating within designated airspace.

8.10.10.2 Operational Control and Response (GADSS)

GADSS compliance is an Operations Control function and requires no additional crew procedures.

Malta Air Operations Control Centre is responsible for scheduling, equipage, notification, tracking, and response actions associated with aircraft operating in GADSS-affected airspace, as outlined in the Operations Control Centre Manual.

Aircraft Tracking

Aircraft tracking systems and services provide four-dimensional tracking (latitude, longitude, altitude, and time) via:

- Automated position reporting at least every 15 minutes
- Latitude, longitude, altitude and time stamp available (as a minimum)
- Latency from plane to operational control max 10 minutes
- Maintenance of operational control
- Recording and preservation aircraft tracking data

Autonomous Distress Tracking

In response to recognised events (e.g., significant divergence from flight plan/flight level, high rates of descent, or loss of thrust on both engines), ADT systems alert the Operations Control Centre and transmits AT information at reduced time intervals. Upon receiving an ADT alert, the Operations Control Shift Leader will implement appropriate response procedures and contact the Duty Pilot.

Post Flight Localization and Recovery

CAT.IDE.A.190, 280, 285

GADSS information management and PFLR procedures are outlined within the Malta Air Operations Control Manual and integrated into the Emergency Response Plan.

Malta Air aircraft are equipped with robust and automatic means via EASA specification components to accurately determine, following an accident where the aeroplane is severely damaged, the aircraft's end-of-flight location:

- Malta Air aircraft flight recorders and cockpit voice recorders are fitted with EASA-compliant, 90-day transmitting Underwater Locator Beacons.
- When scheduled on HLA routings after January 1st 2019, Malta Air Operations Control Centre will ensure the equipage of EASA-compliant, Low-Frequency Underwater Locating Devices. The device's installation, location (above aft galley ceiling panels, next to flight data recorder), and operation are independent of the crew. Aircraft equipped with Low-

Frequency Underwater Locating Devices are identifiable by Technical Log Notice to Crew: *'THIS AIRCRAFT IS FITTED WITH A FIXED LOW FREQUENCY UNDERWATER LOCATING DEVICE. INSTALLED IAW ERT-SB-1612'*.

8.10.11 Rapid Disembarkation

A non-normal situation may occur at any time while boarding/disembarking. Rapid disembarkation is a precautionary egress from the aircraft in situations assessed by the Commander or in his absence the FO or CSS (following the chain of command) as deviating from normal conditions but not being an immediate emergency, i.e. not posing an immediate threat to passengers and crew members on board, but which may escalate into an emergency.

A rapid disembarkation occurs with the aircraft parked on stand and an immediate disembarkation is required using airstairs/air bridge or attached stairs. Rapid disembarkation shall be applied when a non-normal situation arises that requires an expeditious disembarkation but is not immediately life-threatening.

Possible reasons for a rapid disembarkation include a security alert, smoke or fumes in the cabin, overheating PEDs, a large and/or uncontrollable fuel spillage, a fire outside the aircraft which could threaten the aircraft, or any situation that the Commander, FO or CSS decides that the passengers should leave the aircraft urgently.

The Command to commence a Rapid Disembarkation is similar to that of an evacuation.

“Disembark the passengers immediately using all available doors, disembark the passengers immediately using all available doors.”

The cabin crew actions will be to stop the boarding if it is in progress and disembark passengers using the aircraft doors, but **not** the overwing exits or escape slides.

- L1 door using Airbridge, mobile steps or Airstairs as appropriate,
- L2 door if mobile steps are available,
- R2/R1 door if an Ambulift is available and there are PRMs on board.

The cabin crew will instruct the passengers to leave their belongings behind. The aim will be to disembark the passengers as quickly as possible but in a controlled manner not using the slides.

At ALL times the Commander or CSS retains the authority to upgrade to a full evacuation if they decides that the situation has deteriorated and has now become immediately life threatening.

They will make the Evacuation PA and in response the Cabin Crew will deploy the available escape slide(s).

Communications with ATC and members of ground staff are critical during this event.

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9 DANGEROUS GOODS AND WEAPONS

9.1 Policy on the Transport of Dangerous Goods

9.1.1 Approval for the Transport of Dangerous Goods

CAT.GEN.MPA.200, SPA.DG.105

Dangerous goods can only be carried according to the International Civil Aviation Organization's Technical Instructions for the Safe Transport of Dangerous Goods by Air (Technical Instructions).

Malta Air (the company) (AOC No. MT-57) holds an EASA approval for the transport of dangerous goods by air.

Malta Air uses the 'IATA Dangerous Goods Regulations (DGR)' as the everyday reference. The IATA Dangerous Goods Regulations fully comply with the ICAO Technical Instructions.

It is company policy not to carry third party cargo or dangerous goods. The company DG approval includes carriage of company goods and spare parts that qualify as DG.

The Shipper seeks approvals, permissions and exemptions and will provide relevant documentation to Malta Air before the flight. Ryanair Engineering are the designated Shipper of Dangerous Goods. They will ensure all shipments will be in accordance with applicable State Variations and Provisions.

Malta Air's process of Acceptance of Dangerous Goods is described in Section 9.3 of this chapter in detail. At the acceptance point the handling agent will accept the Dangerous Goods shipment in accordance with the ICAO Technical Instructions by referring to the IATA Dangerous Goods Regulation (DGR) from the Shipper.

Ryanair Ground Operations have oversight of Acceptance and Training per service level agreement. The Ground Operations Manual is available in the EFB and describes details.

The provisions in the ICAO Technical Instructions Doc 9284 allow a limited number of items to be carried by passengers and crew. Exemptions for carriage of weapons and ammunitions of Armed Escorts/Diplomatic Protection Officers is regulated in 9.7.2 and requires prior approval by the TM CAD.

Captain Alice Ballestriero is assigned responsibility for the Dangerous Goods approval. Contact details: trsbasecaptain@maltaair.com

She reports to the NPFO.

9.1.2 Forbidden Dangerous Goods

CAT.GEN.MPA.200 (c), GM1 CAT.GEN.MPA.200

Any article or substance which, as presented for transport, is liable to explode, dangerously react, produce a flame or dangerous evolution of heat or dangerous emission of toxic, corrosive or flammable gases or vapours under conditions normally encountered in transport must not be carried on aircraft under any circumstance.

Malta Air does not hold any exemptions or approvals to carry forbidden dangerous goods.

9.1.3 General Exceptions

9.1.3.1 Airworthiness and Operational Items

(CAT.GEN.MPA.200(b)(1))

An approval is not required for dangerous goods which are required to be aboard the aircraft as:

1. Items for airworthiness or operating reasons or for the health of passengers or crew, such as batteries, fire extinguishers, first-aid kits, insecticides, air fresheners, life rafts, escape slides, life- saving appliances, portable oxygen supplies, tritium signs, smoke hoods, passenger service units;
2. Aerosols, alcoholic beverages, perfumes, colognes, liquefied gas lighters and portable electronic devices containing lithium metal or lithium ion cells or batteries carried aboard an aircraft by the operator for use or sale on the aircraft during the flight or series of flights, but excluding nonrefillable gas lighters and those lighters liable to leak when exposed to reduced pressure;
3. Dry ice intended for use in food and beverage service aboard the aircraft; and
4. Electronic devices such as electronic flight bags, personal entertainment devices, and credit card readers, containing lithium metal or lithium ion cells or batteries and spare lithium batteries for such devices carried aboard an aircraft by Malta Air Ltd. for use on the aircraft during the flight or series of flights, (provided that the batteries meet the provisions of Table 9.1.5.5, Item 1). Spare lithium batteries must be individually protected so as to prevent short circuits when not in use.

Conditions for the carriage and use of these electronic devices and for the carriage of spare batteries must be provided in the SEP Manual and enable flight crew, cabin crew and other employees to carry out the functions for which they are responsible.

5. Alcohol-based hand sanitizers and alcohol-based cleaning products carried aboard an aircraft by the operator for use on the aircraft during the flight or series of flights for the purposes of passenger and crew hygiene.

Crew members shall not carry on board spare batteries or cells for these devices.

Note: Dangerous goods intended as replacements for those referred to in 9.1.3.1 1, 2 and 3 above may not be carried without the approval referred to in 9.1.1 and unless consigned and accepted for transport in accordance with the ICAO Technical Instructions.

9.1.3.2 Medical Aid for a Patient

(CAT.GEN.MPA.200 (b)(1))

An approval is not required for dangerous goods which:

1. Are placed on board an aircraft with the approval of the operator; or
2. Form part of the permanent equipment of the aircraft when it has been adapted for specialised use, to provide, during flight, medical aid for a patient, such as gas cylinders, drugs, medicines, other medical material (e.g. sterilising wipes) and wet cell or lithium batteries, providing:
 - a. The gas cylinders have been manufactured specifically for the purpose of containing and transporting that particular gas;
 - b. The drugs and medicines and other medical matter are under the control of trained personnel during the time when they are in use;
 - c. The equipment containing wet cell batteries is kept, and when necessary secured, in an upright position to prevent spillage of the electrolyte; and
 - d. Proper provision is made to stow and secure all the equipment during take-off and landing and at all other times when deemed necessary by the commander in the interests of safety.
 - e. Lithium metal or lithium ion cells or batteries and spare lithium batteries are individually protected so as to prevent short circuits when not in use.

These dangerous goods may also be carried on a flight made by the same aircraft to collect a patient or after that patient has been delivered (e.g. training flights and positioning flights prior to or after maintenance), when it is impracticable to load or unload the goods at the time of the flight on which the patient is carried.

Note: The dangerous goods carried may differ from those identified above due to the needs of the patient. These provisions apply both to dedicated air ambulances and to temporarily modified aircraft.

9.1.3.3 Unaccompanied Baggage

An approval is not required for dangerous goods contained within items of unaccompanied baggage being sent provided that:

- 1. The unaccompanied baggage has been manifested by an approved company representative in compliance with security and dangerous goods regulations;
- 2. The dangerous goods may only be those that are permitted by and in accordance with 9.1.5 to be carried in checked baggage; and
- 3. The unaccompanied baggage is marked with the words “unaccompanied hold baggage certificate”;
- 4. The unaccompanied baggage is consigned as a piece of cargo.

TMP

Example Only

UNACCOMPANIED HOLD BAGGAGE CERTIFICATE OF SECURITY MEASURES
(I CONFIRM THAT THE BAG(S) DETAILED, WHICH TO BE ARE CARRIED ON FLIGHT:

FR2194

DATE

DD/MM/YY

HAVE BEEN SUBJECTED TO AT LEAST ONE OF THE FOLLOWING SECURITY CONTROLS:

METHOD 1

SEARCHED BY HAND

METHOD 2

X-RAY SCREENED FROM TWO SEPARATE BAG ORIENTATIONS BY THE SAME OPERATOR AT THE SAME LOCATION

METHOD 3

X-RAY SCREENED USING AN AUTHORISED EXPLOSIVES DETECTION SYSTEM WHERE THE IMAGE IS ASSESSED BY AN X-RAY OPERATOR

TIME

10:35

BAG TAG NUMBERS

RUSH BAG NUMBERS

XH419111

XH706745

COMPLETED BY

NAME OF SECURITY AGENTS

SIGNATURE OF SECURITY AGENT

COMPANY/ORGANISATION

DATE

9.1.4 Company Approval for Dangerous Goods Carried by Passenger or Crew

There is no restriction of the carriage of employees on an aircraft carrying dangerous goods which are permitted on a passenger aircraft, providing the requirements of the Technical Instructions are complied with. It is intended this be interpreted as meaning they have duties concerned with the preparation or undertaking of a flight or on the ground once the aircraft has landed, although not necessarily in connection with an aircraft.

9.1.5 Items That May Be Carried by Passengers and Crew

(CAT.GEN.MPA.200 (b)(2))

Certain items of dangerous goods may be carried on-board an aircraft by passengers and Aircrew, in limited quantities only, and are exempt from the provisions of the Technical Instructions.

The attached table refers to revised IATA Regulations.

Provisions for Dangerous Goods Carried by Passengers or Crew (MALTA AIR TABLE 2.3.A) 2025

Note: The calculation used to determine watt hours is Volts (V) x Ampere hour (Ah) – Watt hours (Wh).

Dangerous goods must not be carried in or as passengers or crew, checked or carry-on baggage, except as otherwise provided below. Dangerous goods permitted in carry-on baggage are also permitted “on one’s person”, except where otherwise specified.

The pilot-in-command must be informed of the location				
Permitted in or as carry-on baggage				
Permitted in or as checked baggage				
The approval of the operator is required				
Alcoholic beverages, when in retail packagings, containing more than 24% but not more than 70% alcohol by volume, in receptacles not exceeding 5L, with a total net quantity per person of 5L. Note: <i>Alcoholic beverages containing 24% or less alcohol by volume are not subject to any restrictions.</i>	NO	YES	YES	NO
Ammunition, securely packaged (in Div. 1.4S, UN 0012 or UN 0014 only), in quantities not exceeding 5 kg gross weight per person for that person’s own use. Allowances for more than one person must not be combined into one or more packages.	YES	YES	NO	NO
Avalanche rescue backpack, one (1) per person, containing cartridges of compressed gas in Div. 2.2. May also be equipped with a pyrotechnic trigger mechanism containing no more than 200 mg net of Div. 1.4S. The backpack must be packed in such a manner that it cannot be accidentally activated. The airbags within the backpacks must be fitted with pressure relief valves.	YES	YES	YES	NO
Baggage with installed lithium batteries non-removable batteries exceeding 0.3 g lithium metal or 2.7 Wh.	FORBIDDEN			
Baggage with installed lithium batteries: • non-removable batteries. Batteries must contain no more than 0.3 g lithium metal or for lithium ion must not exceed 2.7 Wh; • removable batteries. Batteries must be removed if baggage is to be checked in. Removed batteries must be carried in the cabin.	NO	YES	YES	NO
Batteries, spare/loose, including lithium batteries, non-spillable batteries, nickel-metal hydride batteries and dry batteries (see Note 1) for portable electronic devices must be carried in carry-on baggage only. Articles which have the primary purpose as a power source, e.g. power banks are considered as spare batteries. These batteries must be individually protected to prevent short circuits. Lithium metal batteries: the lithium metal content must not exceed 2 g (see Note 2) Lithium ion batteries: the Watt-hour rating must not exceed 100 Wh (see Note 2) Each person is limited to a maximum of 20 spare batteries. *The operator may approve the carriage of more than 20 batteries. Non-spillable batteries: must be 12 V or less and 100 Wh or less. Each person is limited to a maximum of 2 spare batteries (see Note 3)	NO*	NO	YES	NO
Camping stoves and fuel containers that have contained a flammable liquid fuel, with empty fuel tank and/or fuel container (see Note 13 for details)	YES	YES	NO	NO
Chemical Agent Monitoring Equipment, when carried by staff members of the Organization for the Prohibition of Chemical Weapons on official travel.	YES	YES	YES	NO
Disabling devices such as mace, pepper spray, etc. containing an irritant or incapacitating substance are forbidden on the person, in checked and carry-on baggage.	FORBIDDEN			

The pilot-in-command must be informed of the location				
Permitted in or as carry-on baggage				
Permitted in or as checked baggage				
The approval of the operator is required				
Dry ice (carbon dioxide, solid), in quantities not exceeding 2.5 kg per person when used to pack perishables not subject to these Regulations in checked or carry-on baggage, provided the baggage (package) permits the release of carbon dioxide gas. Checked baggage must be marked “dry ice” or “carbon dioxide, solid” and with the net weight of dry ice or an indication that there is 2.5 kg or less dry ice.	YES	YES	YES	NO
e-cigarettes (including e-cigars, e-pipes, other personal vaporizers) containing batteries must be individually protected to prevent accidental activation (see Note 1).	NO	NO	YES	NO
Electro shock weapons (e.g. Tasers) containing dangerous goods such as explosives, compressed gases, lithium batteries, etc. are forbidden in carry-on baggage or checked baggage or on the person.	FORBIDDEN			
Fuel cells containing fuel, powering portable electronic devices (e.g. cameras, cellular phones, laptop computers and camcorders), see Note 4 for details.	NO	NO	YES	NO
Fuel cell cartridges, spare for portable electronic devices, see Note 4 for details.	NO	YES	YES	NO
Gas cartridges, small, non-flammable containing carbon dioxide or other suitable gas in Division 2.2. Up to two (2) small cartridges fitted into a self-inflating personal safety device, intended to be worn by a person, such as a life jacket or vest. Not more than two (2) devices per passenger and up to two (2) spare small cartridges per device, not more than four (4) cartridges up to 50 mL water capacity for other devices.	YES	YES	YES	NO
Gas cylinders, non-flammable, non-toxic worn for the operation of mechanical limbs. Also, spare cylinders of a similar size if required to ensure an adequate supply for the duration of the journey.	NO	YES	YES	NO
Hair styling equipment containing a hydrocarbon gas cartridge, up to one (1) per passenger or crew-member, provided that the safety cover is securely fitted over the heating element. This hair styling equipment must not be used on board the aircraft. Spare gas cartridges for such hair styling equipment are not permitted in checked or carry-on baggage.	NO	YES	YES	NO
Insulated packagings containing refrigerated liquid nitrogen (dry shipper), fully absorbed in a porous material containing only non-dangerous goods.	NO	YES	YES	NO
Internal combustion or fuel cell engines, must meet A70 (see Note 5 for details).	NO	YES	NO	NO
Lithium Batteries: Portable electronic devices (PED) containing lithium metal or lithium ion cells or batteries, including medical devices such as portable oxygen concentrators (POC) and consumer electronics such as cameras, mobile phones, laptops and tablets (see Note 1). For lithium metal batteries the lithium metal content must not exceed 2 g and for lithium ion batteries the Watt-hour rating must not exceed 100 Wh. Devices in checked baggage must be completely switched off and must be protected from damage. Each person is limited to a maximum of 15 PED. *The operator may approve the carriage of more than 15 PED.	NO*	YES	YES	NO

The pilot-in-command must be informed of the location				
Permitted in or as carry-on baggage				
Permitted in or as checked baggage				
The approval of the operator is required				
Lithium batteries, spare/loose, including power banks, see Batteries, spare/loose				
Lithium battery-powered electronic devices. Lithium ion batteries for portable (including medical) electronic devices, a Wh rating exceeding 100 Wh but not exceeding 160 Wh. For portable medical electronic devices only, lithium metal batteries with a lithium metal content exceeding 2 g but not exceeding 8 g. Devices in checked baggage must be completely switched off and must be protected from damage.	YES	YES	YES	NO
Lithium batteries, spare/loose with a Watt-hour rating exceeding 100 Wh but not exceeding 160 Wh for consumer electronic devices and PMED or with a lithium metal content exceeding 2 g but not exceeding 8 g for PMED only. Maximum of two spare batteries in carry-on baggage only. These batteries must be individually protected to prevent short circuits.	YES	NO	YES	NO
Matches, safety (one small packet) or a small cigarette lighter that does not contain unabsorbed liquid fuel, other than liquefied gas, intended for use by an individual when carried on the person. Lighter fuel and lighter refills are not permitted on one's person or in checked or carry-on baggage. 1. "Strike anywhere" matches are forbidden for air transport. 2. "Blue Flame" or "Cigar" lighters are not permitted on one's person, carry-on or checked baggage. 3. Cigarette lighters should have two independent actions by the user to activate ignition. 4. Cigarette lighters, powered by a lithium ion or lithium metal battery without a safety cap or means of protection against unintentional activation are not permitted on one's person, carry-on or checked baggage (refer also to Note 2 – Point 5).	NO	ON ONE'S PERSON		NO
Mobility Aids: Battery-powered wheelchairs or other similar mobility devices with non-spillable wet batteries, nickel-metal hydride batteries or dry batteries (see Note 6).	YES	YES	NO	YES
Mobility Aids: Battery-powered wheelchairs or other similar mobility devices with spillable batteries or with lithium ion batteries (see Note 7 and Note 8 for details).	YES	YES	NO	YES
Mobility Aids: Battery-powered wheelchairs or other similar mobility devices with lithium ion batteries where the design of the mobility aid does not provide adequate protection for the battery(ies) (see Note 8 3(b)ii. for details).	YES	NO	YES	YES
Non-radioactive medicinal or toiletry articles (including aerosols) such as hair sprays, perfumes, colognes and medicines containing alcohol; and Non-flammable, non-toxic (Division 2.2) aerosols, with no subsidiary hazard, for sporting or home use. The total net quantity of non-radioactive medicinal or toiletry articles and non-flammable, non-toxic (Division 2.2) aerosols must not exceed 2 kg or 2 L and the net quantity of each single article must not exceed 0.5 kg or 0.5 L. Release valves on aerosols must be protected by a cap or other suitable means to prevent inadvertent release of the contents.	NO	YES	YES	NO

The pilot-in-command must be informed of the location					
Permitted in or as carry-on baggage					
Permitted in or as checked baggage					
The approval of the operator is required					
Oxygen or air, gaseous, cylinders required for medical use. The cylinder must not exceed 5 kg gross weight. Note: <i>Liquid oxygen systems are forbidden for transport.</i>	YES	YES	YES	YES	YES
Permeation devices, must meet A41 (see Note 9 for details).	NO	YES	NO	NO	
Radioisotopic cardiac pacemakers or other devices, including those powered by lithium batteries, implanted into a person or fitted externally.	NO	ON ONE'S PERSON		NO	
Security-type equipment (Note 10 for details).	YES	YES	NO	NO	
Security-type attaché cases, cash boxes, cash bags, etc. incorporating dangerous goods, such as lithium batteries and/or pyrotechnic material, except as provided in Note 10 are totally forbidden. See entry in 4.2 – List of Dangerous Goods.	FORBIDDEN				
Specimens, non-infectious packed with small quantities of flammable liquid, must meet A180 (see Note 11 for details).	NO	YES	YES	NO	
Thermometer, medical or clinical, which contains mercury, one (1) per person for personal use, when in its protective case.	NO	YES	NO	NO	
Thermometer or barometer, mercury filled carried by a representative of a government weather bureau or similar official agency (see Note 12.1 and Note 14).	YES	NO	YES	YES	

Remark: State Variations in IATA DGR section 2.3 Dangerous Good Carried by Passengers or Crew do not apply to the Malta Air network.

Notes 1–14 provide additional information to the table above. If you require support during operations, contact the Ramp Agent, DUBOPS and GOPS for assistance.

Note 1: Portable Electronic Devices (PED) (Including Medical Devices) Containing Batteries and Spare Batteries

For the purpose of these Regulations, battery-powered electronic device means the equipment or apparatus for which the batteries will provide electrical power for its operation. These devices (PED), which may include medical devices such as portable oxygen concentrators (POC) and consumer electronics such as cameras, mobile phones, laptops and tablets containing batteries when carried by passengers or crew for personal use, which should be carried in carry-on baggage. Batteries and heating elements must be isolated in portable electronic devices capable of generating extreme heat, by removal of the heating element, battery or other components. These provisions apply to dry batteries, nickel-metal hydride batteries, lithium batteries and wet, non-spillable batteries. Additional specific requirements for lithium batteries and wet, non-spillable batteries are set out (Note 2 and 3) respectively).

If devices are carried in checked baggage:

1. measures must be taken to protect the device from damage and to prevent unintentional activation;
2. the device must be completely switched off (not in sleep or hibernation mode).

Electronic cigarettes including e-cigars and other personal vaporisers containing batteries must be in carry-on baggage only. Recharging of these devices and/or batteries on board the aircraft is not permitted, and measures must be taken to prevent accidental activation.

Spare batteries must be individually protected to prevent short circuits by placement in the original retail packaging or by otherwise insulating terminals, e.g. by taping over exposed terminals or placing each battery in a separate plastic bag or protective pouch and carried in carry-on baggage only. Each person is limited to a maximum of 20 spare batteries; however, the operator may approve the carriage of more than 20 spare batteries.

Lithium Battery-Powered Electronic Devices

For the purpose of these Regulations (DG), lithium battery-powered electronic device means the equipment or apparatus for which the lithium cells or batteries will provide electrical power for its operation. These devices are permitted in checked and carry-on baggage with the approval of the operator as follows:

1. Portable medical electronic devices (PMED), such as automated external defibrillators (AED), portable oxygen concentrators (POC) and continuous positive airway pressure (CPAP), containing lithium metal or lithium ion cells or batteries may be carried by passengers for medical use as follows:
 - a. For lithium metal or lithium alloy batteries, a lithium content exceeding 2 g, but not exceeding 8 g; or
 - b. For lithium-ion batteries, a watt-hour rating exceeding 100 Wh, but not exceeding 160 Wh;
 - c. Batteries must be of a type that meets the requirements of the UN Manual of Tests and Criteria, Part III, subsection 38.3.
2. Portable electronic devices, such as power tools, video cameras and laptops containing lithium ion batteries as follows:
 - a. Lithium-ion batteries with a watt-hour rating exceeding 100 Wh, but not exceeding 160 Wh;

- b. Batteries must be of a type that meets the requirements of the UN Manual of Tests and Criteria, Part III, subsection 38.3.
3. If devices are carried in checked baggage, Points a and b of Note 1 apply.

(For spare lithium metal batteries with a lithium content exceeding 2 g and lithium ion batteries with a Watt-hour rating exceeding 100 Wh see Point b of “Goods Acceptable with Operator Approval as Carry-on Baggage Only” (Ch.9–Pag.23). For electronic devices with lithium metal batteries with a lithium content not exceeding 2 g and lithium ion batteries with a Watt-hour rating not exceeding 100 Wh see Note 1 above.)

Note 2: Additional requirements for lithium batteries:

1. each installed or spare battery must not exceed:
 - a. for lithium metal or lithium alloy batteries, a lithium content of not more than 2 g or
 - b. for lithium ion batteries, a watt-hour rating of not more than 100 Wh.
2. batteries must be of a type that meets the requirements of the UN Manual of Tests and Criteria, Part III, subsection 38.3;
3. each person is limited to a maximum of 15 PED; however, the operator may approve the carriage of more than 15 PED;
4. articles containing lithium metal or lithium ion cells or batteries, the primary purpose of which is to provide power to another device, e.g. power banks, are permitted in carry-on baggage only. These articles must be individually protected to prevent short circuits by placement in the original retail packaging or by otherwise insulating terminals, e.g. by taping over exposed terminals or placing each battery in a separate plastic bag or protective pouch;
5. electronic cigarette lighters (see also **Matches, safety (one small packet) or a small cigarette lighter**) powered by lithium batteries, the following conditions must also be met:
 - a. only lighters with a safety cap or means of protection against unintentional activation are permitted on one’s person;
 - b. recharging of these devices and/or batteries on board the aircraft is not permitted and measures must be taken to prevent accidental activation.

6. baggage with installed lithium batteries, with a lithium metal content exceeding 0.3 g or a Watt-hour rating exceeding 2.7 Wh:
 - a. if the baggage is to be checked in, the lithium battery must be removed from the baggage and the lithium battery must be carried in the cabin; or
 - b. the baggage must be carried in the cabin;
 - c. baggage where the lithium battery exceeds the limits in (6) and cannot be removed is forbidden for carriage.

Note 3: Additional requirements for non-spillable wet batteries:

Each person is limited to a maximum of two spare batteries in carry-on baggage only, subject to the following conditions:

1. batteries must meet the requirements of Special Provision A67 and must not contain any free or unabsorbed liquid;
2. the voltage of each battery must not exceed 12 V and the Watt-hour rating must not exceed 100 Wh;
3. each spare battery must be protected from short circuit by insulation of the battery terminal.

Note 4: Fuel Cells Contained in Portable Electronic Devices

Fuel cells used to power portable electronic devices (for example cameras, cellular phones, laptop computers and camcorders) and spare fuel cell cartridges, under the following conditions:

1. fuel cells and fuel cell cartridges may only contain flammable liquids, corrosive substances, liquefied flammable gas, water-reactive substances or hydrogen in metal hydride;
2. refuelling of fuel cells on board an aircraft is not permitted except that the installation of a spare cartridge is allowed;
3. the maximum quantity of fuel in any fuel cell or fuel cell cartridge must not exceed:
 - a. for liquids, 200 mL;
 - b. for solids 200 g;
 - c. for liquefied gases, 120 mL for non-metallic fuel cells or fuel cell cartridges or 200 mL for metal fuel cells or fuel cell cartridges;
 - d. for hydrogen in metal hydride the fuel cell cartridges must have a water capacity of 120 mL or less.

4. each fuel cell and each fuel cell cartridge must conform to IEC 62282-6-100 Ed. 1, including Amendment 1 and must be marked with a manufacturer's certification that it conforms to the specification. In addition, each fuel cell cartridge must be marked with the maximum quantity and type of fuel in the cartridge;
5. no more than two spare fuel cell cartridges may be carried in checked baggage, carry-on baggage, or on the person;
6. fuel cells containing fuel are permitted in carry-on baggage only;
7. interaction between fuel cells and integrated batteries in a device must conform to IEC 62282-6-100 Ed. 1, including Amendment 1. Fuel cells whose sole function is to charge a battery in the device are not permitted;
8. fuel cells must be of a type that will not charge batteries when the portable electronic device is not in use and must be durably marked by the manufacturer: "APPROVED FOR CARRIAGE IN AIRCRAFT CABIN ONLY" to so indicate; and
9. in addition to the languages which may be required by the State of origin for the marks specified above, English should be used.

Note 5: Internal Combustion or Fuel Cell Engines

Flammable liquid powered internal combustion or fuel cell engines being carried separately or incorporated into a machine or other apparatus, without batteries or other dangerous goods may be accepted in checked baggage only provided that the engine must comply with the following requirements of Special Provision A70:

1. the engine is powered by a fuel that does not meet the classification criteria for any class or division; or
2. the fuel tank of the vehicle, machine or other apparatus has never contained any fuel, or the fuel tank has been flushed and purged of vapours and adequate measures taken to nullify the hazard;
3. the passenger has provided the operator with written or electronic documentation stating that a flushing and purging procedure has been followed; and
4. the entire fuel system of the engine has no free liquid and all fuel lines are sealed or capped or securely connected to the machinery or apparatus.

Note 6: Wheelchairs/Mobility Aids with Non-spillable Wet Batteries, Nickel-Metal Hydride Batteries or Dry Batteries

Battery-powered wheelchairs or other similar mobility aids for use by passengers whose mobility is restricted by either a disability, their health or age, or a temporary mobility problem (e.g. broken leg), with non-spillable wet batteries which comply with Special Provision A67 or nickel-metal hydride batteries which comply with Special Provision A199 or dry batteries which comply with Special Provision A123 (for details on Special Provision A67, A123 and A199 refer to “Additional/information” text at the end of Note 6). These batteries must meet the following requirements:

Note: Additional guidance on the end-to-end processes associated with the carriage of mobility aids is available in the IATA Guidance on the Transport of Mobility Aids at the following link – <https://www.iata.org/en/programs/passenger/accessibility/>

1. The mobility aid must be prepared for transport to prevent:
 - a. unintentional activation; and
 - b. non-spillable batteries are not permitted to contain any free or unabsorbed liquid.
2. The operator must secure, by use of straps, tie-downs or other restraint devices, a battery powered mobility aid with installed battery(ies). The mobility aid, the battery(ies) electrical cabling and controls must be protected from damage including by the movement of baggage, mail or cargo.
3. The operator must verify that:
 - a. the passenger has confirmed that the battery(ies) is a non-spillable wet battery that complies with Special Provision A67*, or a nickel-metal hydride battery that complies with Special Provision A199** or dry battery that complies with Special Provision A123***;
 - b. the battery terminals are protected from short circuits, e.g. by being enclosed within a battery container;
 - c. the battery(ies) is either:
 - i. adequately protected against damage by the design of the mobility aid and securely attached to the wheelchair or mobility aid. The electrical circuits must be isolated following the manufacturer’s instructions; or
 - ii. removed from the mobility aid following the manufacturer’s instructions.

4. A passenger may carry a maximum of:
 - a. one spare wet, non-spillable battery meeting Special Provision A67; or
 - b. two spare nickel-metal hydride batteries meeting Special Provision A199 or dry batteries meeting Special Provision A123.
5. The operator must ensure that any battery(ies) removed from the wheelchair/mobility aid and any spare batteries are carried in strong, rigid packagings which must be carried in the cargo compartment.
6. The operator must inform the pilot-in-command of the location of mobility aids with installed battery(ies), removed battery(ies) and spare battery(ies).

Additional Information:

***A67** Wet cell batteries can be considered as non-spillable provided that they are capable of withstanding the vibration and pressure differential as per regulatory DG specs, without leakage of battery fluid.

Note: Non-spillable type batteries which are an integral part of, and necessary for the operation of, mechanical or electronic equipment must be securely fastened in the battery holder on the equipment and protected in such a manner so as to prevent damage and short circuits.

The words “not restricted” and the special provision number A67 must be included in the description of the substance on the Air Waybill (as required by DG Regulations) when an Air Waybill is issued; this is required to indicate that the special provision has been applied.

****A199** The UN number UN 3496 is only applicable in sea transport. Nickel-metal hydride batteries or nickel-metal hydride battery-powered devices, equipment or vehicles having the potential of a dangerous evolution of heat are not subject to these Regulations provided they are prepared for transport so as to prevent:

1. a short circuit (e.g. in the case of batteries, by the effective insulation of exposed terminals; or, in the case of equipment, by disconnection of the battery and protection of exposed terminals); and
2. unintentional activation.

The words “not restricted” and the special provision number A199 must be included in the description of the substance on the Air Waybill (as required by DG Regulations) when an Air Waybill is issued; this is required to indicate that the special provision has been applied.

*****A123** This entry applies to Batteries, electric storage, not otherwise listed in Subsection 4.2 – List of Dangerous Goods. Examples of such batteries are: alkali-manganese, zinc-carbon and nickel-cadmium batteries. Any electrical battery or battery powered device, equipment or vehicle having the potential of a dangerous evolution of heat must be prepared for transport so as to prevent:

1. a short-circuit (e.g. in the case of batteries, by the effective insulation of exposed terminals; or, in the case of equipment, by disconnection of the battery and protection of exposed terminals); and
2. unintentional activation.

The words “not restricted” and the special provision number A123 must be included in the description of the substance on the Air Waybill (as required by DG Regulations) when an Air Waybill is issued; this is required to indicate that the special provision has been applied.

Note 7: Wheelchairs/Mobility Aids with Spillable Batteries

Battery-powered wheelchairs or other similar mobility aids for use by passengers whose mobility is restricted by either a disability, their health or age, or a temporary mobility problem (e.g. broken leg), with spillable batteries. These batteries must meet the following requirements:

Note: Additional guidance on the end-to-end processes associated with the carriage of mobility aids is available in the IATA Guidance on the Transport of Mobility Aids at the following link – <https://www.iata.org/en/programs/passenger/accessibility/>

1. The operator must secure, by use of straps, tie-downs or other restraint devices, a battery powered mobility aid with installed battery(ies). The mobility aid, the battery(ies), electrical cabling and controls must be protected from damage including by the movement of baggage, mail or cargo.
2. The operator must verify that:
 - a. the battery terminals are protected from short circuits, e.g. by being enclosed within a battery container;

- b. the battery(ies) is fitted, where feasible, with spill-resistant vent caps;
- c. the battery(ies) is either:
 - i. adequately protected against damage by the design of the mobility aid and securely attached to the wheelchair or mobility aid. The electrical circuits must be isolated following the manufacturer's instructions; or
 - ii. removed from the mobility aid following the manufacturer's instructions when the mobility aid cannot be maintained in an upright position (refer to point 3 below).
- 3. The operator must load, stow, secure and unload a mobility aid with a spillable battery(ies) in an up-right position. If the wheelchair or mobility aid cannot be loaded, stowed, secured and unloaded always in an upright position or if the mobility aid does not adequately protect the battery(ies), the operator must remove the battery(ies). The removed battery(ies) must be carried in strong, rigid packagings as follows:
 - a. packagings must be leak-tight, impervious to battery fluid and be protected against upset by securing to pallets or by securing them in cargo compartments using appropriate means of securement (other than by bracing with freight or baggage) such as by use of restraining straps, brackets or holders;
 - b. battery(ies) must be protected against short circuits, secured upright in these packagings and surrounded by compatible absorbent material sufficient to absorb their total liquid contents; and
 - c. these packagings must be marked "BATTERY, WET, WITH WHEELCHAIR" or "BATTERY, WET, WITH MOBILITY AID" and be labelled with the "Corrosive" label and with the "Package Orientation" label.
- 4. The operator must inform the pilot-in-command of the location of mobility aids with installed batteries and removed batteries.

Note 8: Wheelchairs/Mobility Aids with Lithium Batteries

Lithium ion battery powered wheelchairs or other similar mobility aids for use by passengers whose mobility is restricted by either a disability, their health or age, or a temporary mobility problem (e.g. broken leg), subject to the following conditions:

Note: Additional guidance on the end-to-end processes associated with the carriage of mobility aids is available in the IATA Guidance on the Transport of Mobility Aids at the following link – <https://www.iata.org/en/programs/passenger/accessibility/>

1. The batteries must be of a type which meets the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3.
2. The operator must secure, by use of straps, tie-downs or other restraint devices, a battery powered mobility aid with installed battery(ies). The mobility aid, the battery(ies), electrical cabling and controls must be protected from damage including by the movement of baggage, mail or cargo.
3. The operator must verify:
 - a. the battery terminals are protected from short circuits, e.g. by being enclosed within a battery container;
 - b. the battery(ies) is either
 - i. adequately protected against damage by the design of the mobility aid and securely attached to the wheelchair or mobility aid. The electrical circuits must be isolated following the manufacturer's instructions; or
 - ii. removed from the mobility aid following the manufacturer's instructions. Each battery removed from the mobility aid must not exceed 300 Wh.
4. A passenger may carry a maximum of one spare lithium ion battery not exceeding 300 Wh or two spare batteries each not exceeding 160 Wh.
5. The operator must ensure that any battery(ies) removed from the mobility aid and any spare battery(ies) are carried in the passenger cabin. The removed or spare battery(ies) must be protected from damage (e.g. by placing each battery in a protective pouch).
6. The operator must inform the pilot-in-command of the location of the mobility aid with installed battery(ies), removed battery(ies) and spare battery(ies).
7. It is recommended that passengers make advance arrangements with each operator.
8. When a lithium-ion battery(ies) remains installed in the mobility aid, there is no watt-hour limit.

Note 9: Permeation Devices

In checked baggage only permeation devices for calibrating air quality monitoring equipment. These devices must comply with the following requirements of Special Provision A41:

1. each device must be constructed of a material compatible with the dangerous goods it contains;
2. the total quantity of dangerous goods in each device is limited to 2 mL and the device must not be liquid full at 55°C;
3. each permeation device must be placed in a sealed, high impact-resistant, tubular inner packaging of plastic or equivalent material. Sufficient absorbent material must be contained in the inner packaging to completely absorb the contents of the device. The closure of the inner packaging must be securely held in place with wire, tape or other positive means;
4. each inner packaging must be contained in a secondary packaging constructed of metal or plastic having a minimum thickness of 1.5 mm. The secondary packaging must be hermetically sealed;
5. the secondary packaging must be securely packed in strong outer packaging. The completed package must be capable of withstanding, without breakage or leakage of any inner packaging and without significant reduction in effectiveness:
 - a. the following free drops onto a rigid, non-resilient, flat and horizontal surface from a height of 1.8 m:
 - one drop flat on the bottom;
 - one drop flat on the top;
 - one drop flat on the long side;
 - one drop flat on the short side;
 - one drop on a corner at the junction of three intersecting edges; and
 - b. a force applied to the top surface for a duration of 24 hours, equivalent to the total weight of identical packages if stacked to a height of 3 m (including the test sample).

Note: Each of the above tests may be performed on different but identical packages.

6. the gross weight of the completed package must not exceed 30 kg.

Note 10: Security-Type Equipment

Security type equipment such as attaché cases, cash boxes, cash bags, etc. incorporating dangerous goods as part of this equipment, for example lithium batteries or pyrotechnic material, may be carried as checked baggage only if the equipment complies with the following:

1. the equipment must be equipped with an effective means of preventing accidental activation;
2. if the equipment contains an explosive or pyrotechnic substance or an explosive article, this article or substance must be excluded from Class 1 by the appropriate national authority of the State of Manufacture. The appropriate national authority may exclude an article or substance from Class 1 by virtue of test results and the Class 1 definition;
3. if the equipment contains lithium cells or batteries, these cells or batteries must comply with the following restrictions:
 - a. for a lithium metal cell, the lithium content is not more than 1 g;
 - b. for a lithium metal battery, the aggregate lithium content is not more than 2 g;
 - c. for lithium ion cells, the Watt-hour rating is not more than 20 Wh;
 - d. for lithium ion batteries, the Watt-hour rating is not more than 100 Wh;
 - e. each cell or battery is of the type proven to meet the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3.
4. if the equipment contains gases to expel dye or ink, only gas cartridges and receptacles, small, containing gas with a capacity not exceeding 50 mL, containing no constituents subject to these Regulations other than a Division 2.2 gas, are allowed. The release of gas must not cause extreme annoyance or discomfort to crew members so as to prevent the correct performance of assigned duties. In case of accidental activation all hazardous effects must be confined within the equipment and must not produce extreme noise;
5. security type equipment that is defective or that has been damaged is forbidden for transport.

Note 11: Non-Infectious Specimens Packed with Small Quantities of Flammable Liquids

In checked or carry-on baggage non-infectious specimens, such as specimens of mammals, birds, amphibians, reptiles, fish, insects and other invertebrates containing small quantities of flammable liquids provided that the following requirements of Special Provision A180 are complied with:

1. specimens are:
 - a. wrapped in paper towel and/or cheesecloth moistened with alcohol or an alcohol solution and then placed in a plastic bag that is heat-sealed. Any free liquid in the bag must not exceed 30 mL; or
 - b. placed in vials or other rigid containers with no more than 30 mL of alcohol or an alcohol solution;
2. the prepared specimens are then placed in a plastic bag that is then heat-sealed;
3. the bagged specimens are then placed inside another plastic bag with absorbent material then heat sealed;
4. the finished bag is then placed in a strong outer packaging with suitable cushioning material;
5. the total quantity of flammable liquid per outer packaging must not exceed 1 L; and
6. the completed package is marked “scientific research specimens, not restricted”.

The following dangerous goods, as listed are permitted on aircraft as carry-on baggage only and with the approval of the operator(s):

- a. A mercurial barometer or mercurial thermometer carried by a representative of a government weather bureau or similar official agency. The barometer or thermometer must be packed in a strong outer packaging, having a sealed inner liner or a bag of strong leak-proof and puncture-resistant material impervious to mercury, which will prevent the escape of mercury from the package irrespective of its position. The pilot-in-command must be informed of the location of the barometer or thermometer.

- b. Spare lithium batteries including articles containing lithium metal or lithium ion cells or batteries, the primary purpose of which is to provide power to another device, e.g. power banks, are permitted in carry-on baggage as follows:
 - i. no more than 2 lithium ion batteries with a watt-hour rating exceeding 100 Wh but not exceeding 160 Wh or 2 lithium metal batteries, with a lithium content exceeding 2 g but not exceeding 8 g. Lithium metal batteries are only permitted for portable medical electronic devices (PMED), such as automated external defibrillators (AED), portable oxygen concentrators (POC) and continuous positive airway pressure (CPAP);
 - ii. spare batteries must be individually protected so as to prevent short circuits (by placement in original retail packaging or by otherwise insulating terminals, e.g. by taping over exposed terminals or placing each battery in a separate plastic bag or protective pouch);
 - iii. batteries must be of a type that meet the requirements of the UN Manual of Tests and Criteria, Part III, subsection 38.3. No more than two individually protected spare batteries per person may be carried.

Note 12: Goods Acceptable with Operator Approval as Carry-on Baggage Only

The following dangerous goods, as listed in 1) and 2) below, are permitted on aircraft as carry-on baggage only and with the approval of Malta Air:

- 1. A mercurial barometer or mercurial thermometer carried by a representative of a government weather bureau or similar official agency. The barometer or thermometer must be packed in a strong outer packaging, having a sealed inner liner or a bag of strong leak-proof and puncture-resistant material impervious to mercury, which will prevent the escape of mercury from the package irrespective of its position. The pilot-in-command must be informed of the location of the barometer or thermometer.

2. Spare lithium batteries including articles containing lithium metal or lithium ion cells or batteries, the primary purpose of which is to provide power to another device, e.g. power banks, are permitted in carry-on baggage as follows:
 - a. No more than 2 lithium ion batteries with a watt-hour rating exceeding 100 Wh but not exceeding 160 Wh or 2 lithium metal batteries, with a lithium content exceeding 2 g but not exceeding 8 g. Lithium metal batteries are only permitted for portable medical electronic devices (PMED), such as automated external defibrillators (AED), portable oxygen concentrators (POC) and continuous positive airway pressure (CPAP);
 - b. Spare batteries must be individually protected so as to prevent short circuits (by placement in original retail packaging or by otherwise insulating terminals, e.g. by taping overexposed terminals or placing each battery in a separate plastic bag or protective pouch);
 - c. Batteries must be of a type that meet the requirements of the UN Manual of Tests and Criteria, Part III, subsection 38.3. No more than two individually protected spare batteries per person may be carried.

Note 13: Camping Stoves and Fuel Containers that have Contained a Flammable Liquid Fuel

Camping stoves and fuel containers for camping stoves that have contained a flammable liquid fuel may be carried provided the fuel tank of the camping stove and/or fuel container has been completely drained of all liquid fuel and action has been taken to nullify the danger. To nullify the danger, the empty fuel tank and/or container must be allowed to drain for at least 1 hour, the fuel tank and/or container must then be left uncapped for a minimum of 6 hours to allow any residual fuel to evaporate. Alternative methods, such as adding cooking oil to the fuel tank and/or container to elevate the flash point of any residual liquid above the flash point of flammable liquid and then emptying the fuel tank and/or container, are equally acceptable. The fuel tank and/or container must then have the cap securely fastened and be wrapped in an absorbent material such as paper towel and placed in a polyethylene or equivalent bag. The top of the bag must then be sealed or gathered and closed with an elastic band or twine.

Note: Provided the above cleaning method is followed in accordance with these Regulations, the fuel stove or container can be classified as non-hazardous. However, to control the carriage of these items, they are listed in Table 1 Provisions for Dangerous Goods Carried by Passengers or Crew.

Note 14: The barometer or thermometer must be packed in a strong outer packaging, having a sealed inner liner or a bag of strong leak-proof and puncture-resistant material impervious to mercury, which will prevent the escape of mercury from the package irrespective of its position.

9.1.5.1 Restrictions

AIR OPS Ref. CAT.GEN.MPA.AIR OPS Ref. CAT.GEN.MPA.200

Medical Oxygen

Malta Air as an operator does not allow passengers to carry their own medical oxygen on board. They must contact the company prior to travel. Malta Air will supply a bottle of portable oxygen for the passenger as Malta Air carry more than the minimum requirement of portable oxygen. There is a charge for this facility.

This extra bottle is not for the passenger to take with them at their destination.

Human Remains

Malta Air will carry human remains on selected flights. Malta Air will only accept human remains from a Regulated Agent.

The Commander will be furnished with the necessary paperwork and the human remains will be recorded on the LID/loadsheet as 'Human remains' or 'HUM'.

Organs

Organs that are NOT packed in dry ice are to be stowed in the flight deck. Organs that are packed in dry ice are to be stowed in the cabin either in an empty bar box, hatbin or behind trolleys (size permitting) and subject to the Captain's approval.

Ashes

The carriage of ashes is permitted. Passengers must ensure that they are packed securely in a suitable container. Ashes can only be carried as hand baggage (may be carried in addition to the usual baggage allowance). The urn/casket must have a secure screw top/lid.

Articles and substances which would otherwise be classed as dangerous goods are excluded from the provisions of AIR OPS Subpart R provided they are carried for use in flight for medical aid for a patient, (Gas cylinders, drugs, medicines, other medicinal material (such as sterilising wipes) and lithium batteries are the dangerous goods which are normally provided for use in flight as

medical aid for a patient. However, what is carried may depend on the needs of the patient. These dangerous goods are not those which are part of the normal equipment of the aeroplane) provided that:

1. Gas cylinders have been manufactured specifically for the purpose of containing and transporting that particular gas;
2. Drugs, medicines and other medical matter are under the control of trained personnel during the time when they are in use in the aeroplane;
3. Proper provision is made to stow and secure all the equipment during take-off and landing and at other times when deemed necessary by the Commander in the interests of safety; or
4. They are carried by passengers or crew members (as listed in the appendix and to the extent specified in the Technical Instructions).

Note: From time to time it may be necessary the CC to remove bags from the cabin and have them placed in the hold. In such circumstances the passenger must be asked if there are any spare batteries for the mobility aid and the passenger must be instructed to remove the battery before placing the bag in the hold. The spare batteries will then remain with the passenger in the cabin for the duration of the flight.

Portable Oxygen Concentrators (POCs)

Refer to OM(A) SEP 2.17.4.7.

An approval is not required for those dangerous goods which, according to the Technical Instructions, can be carried by passengers or crew members.

Passengers or crew are forbidden to carry dangerous goods either as or in carry-on baggage, checked baggage or on their person unless the dangerous goods are permitted in accordance with the table below and:

1. carried by passengers or crew for personal use only;
2. contained in baggage that has been separated from its owner during transit (e.g. lost baggage or improperly routed baggage); or
3. contained within items of excess baggage sent as cargo as permitted by 9.1.3.3.

The entry in the table that most appropriately describes the item or article must be selected. For instance, electronic cigarettes must meet the requirements of the entry for “Battery-powered portable electronic smoking devices” not the entry for lithium batteries or non-spillable batteries.

An item or article that contains multiple dangerous goods must meet all applicable entries. For instance, the restrictions and conditions for entries 1) and 14) apply to an avalanche backpack that contains lithium batteries and gas cartridges.

Active devices must meet defined standards for electromagnetic radiation to ensure that the operation of the devices does not interfere with aircraft systems.

Where an entry requires compliance with specific UN tests or Special Provisions, if considered necessary (e.g. to grant the operator's approval for carriage), passengers should be able to confirm that the applicable requirements have been met. For items such as batteries, the passenger should be able to obtain confirmation from the manufacturer or distributor of the item.

If there is any doubt prior to a flight as whether a Special Provision has been complied with by a passenger, the Captain shall consult with the ground operations team at the airport or the Ryanair DAC Operations Control Centre, who will ensure the query is directed to the appropriate personnel. Malta Air will not transport such goods if confirmation of compliance is not obtained.

Note: Should it be necessary to transfer carry-on baggage to the hold, cabin crew must verify that the baggage contains no dangerous goods that are permitted for carriage in carry-on baggage only (e.g. spare lithium batteries, large PED's etc).

Refer to ryanair.com for the complete list of prohibited articles on company aircraft.

Note 1: Wet/ Wet Cell Batteries are prohibited on company aircraft.

Note 2: Lithium ION batteries are prohibited in any baggage placed in any hold of the aircraft.

Note 3: The following dangerous goods may be commonly carried by passengers on other modes of transport, however, they are prohibited either as or in carry-on baggage or checked baggage:

- personal medical oxygen devices that utilize liquid oxygen;
- electroshock weapons (e.g. tasers) containing dangerous goods such as explosives, compressed gases, lithium batteries, etc.;
- "strike anywhere" matches;
- lighter fuel and lighter refills;
- premixing burner lighter without a means of protection against unintentional activation; and

- battery-powered lighters powered by a lithium ion or lithium metal battery (e.g. laser plasma lighters, tesla coil lighters, flux lighters, arc lighters and double arc lighters) without a safety cap or means of protection against unintentional activation.

Note 4: Exceptions found in the Technical Instructions from the restrictions on carriage by passengers and crew (e.g. by application of a Special Provision) are not reproduced in the tables below. The following dangerous goods are not subject to the Technical Instructions:

- Radio-pharmaceuticals contained within the body of a person as the result of medical treatment; and
- Energy efficient lamps when in retail packaging and intended for personal or home use.

Note 5: Air Cylinders for purposes such as scuba diving: if empty or at a pressure less than 200 kPa at 20° (2 Bar or 29 PSI) air cylinders are not classified as dangerous goods so are permitted for carriage by passenger or crew.

9.1.5.2 Loading of Battery Powered Mobility Aids – General Requirements

A battery powered mobility aid with installed batteries must be secured, by use of straps, tie-downs or other restraint devices.

The mobility aid, the batteries, electrical cabling and controls must be protected from damage including by the movement of baggage, mail or cargo.

The operator must verify that:

1. the battery terminals are protected from short circuits (e.g. by being enclosed within a battery container); and
2. the battery is either:
 - a. securely attached to the mobility aid and the electrical circuits are isolated following the manufacturer's instructions; or
 - b. removed by the user, if the mobility aid is specifically designed to allow it to be, following the manufacturer's instructions.

Note: To check that electrical circuits are isolated (inactive), place the device into drive mode (i.e. not freewheel mode), see if the mobility aid will power up and if so whether use of the joystick results in the mobility aid moving. It must also be verified that the circuits of supplemental motorised systems such as seating systems have been inhibited to

prevent inadvertent operation, e.g. by the separation of cable connectors. If an electric mobility aid has not been made safe for carriage, it must not be loaded.

9.1.5.3 Additional Requirements for Non-spillable Wet Battery Powered Mobility Aids Which Comply with Special Provision A123 or A199

1. The passenger has confirmed that the battery is:
 - a. a non-spillable wet battery that complies with Special Provision A67;
 - b. a dry battery that complies with Special Provision A123; or
 - c. a nickel-metal hydride battery that complies with Special Provision A199.
2. The battery terminals are protected from short circuits (e.g. by being enclosed within a battery container);
3. The battery is either:
 - a. securely attached to the mobility aid and the electrical circuits are isolated following the manufacturer's instructions; or
 - b. removed by the user, if the mobility aid is specifically designed to allow it to be, following the manufacturer's instructions.
4. A maximum of one non-spillable wet spare battery is carried per passenger.

The operator must inform the pilot-in-command of the location of any mobility aids with installed batteries, removed batteries and spare batteries.

9.1.5.4 Additional Requirements for Spillable Battery Powered Mobility Aids

Where feasible, the battery is fitted with spill resistant-vent caps.

If the mobility aid cannot be loaded, stowed, secured and unloaded always in an upright position or if the mobility aid does not adequately protect the battery, the operator must remove the batteries and carry them in strong, rigid packagings, as follows:

1. packagings must be leak-tight, impervious to battery fluid and be protected against being overturned by securing them to pallets or by securing them in cargo compartments using appropriate means of securement;
2. batteries must be protected against short circuits, secured upright in these packagings and surrounded by compatible absorbent material sufficient to absorb their total liquid contents; and

3. these packagings must be marked “Battery, wet, with wheelchair” or “Battery, wet, with mobility aid” and be labelled with a “Corrosive” label and with package orientation labels.

The operator must inform the pilot-in-command of the location of any mobility aids with installed spillable batteries and removed batteries.

9.1.5.5 Additional Requirements for Mobility Aids Powered by Lithium Ion Batteries

Any battery removed from the mobility aid and any spare batteries must be carried in the cabin and protected from damage (e.g., by placing each battery in a protective pouch) and the battery terminals protected from short circuit (by insulating the terminals, e.g. by taping over exposed terminals).

A removed battery must not exceed 300 Watt-hours (Wh). In addition, one spare not exceeding 300 Wh or two spares not exceeding 160 Wh are permitted.

The operator must inform the pilot-in-command of the location of any mobility aids with installed lithium ion batteries, removed batteries and spare batteries.

Note: The calculation used to determine watt hours is:

$$\text{Volts} \times \text{ampere hour (Ah)} = \text{watt hours.}$$

Table 1 Provisions for Dangerous Goods Carried by Passengers or Crew
(ICAO DOC 9284 – Ed. 2025–2026)

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
Batteries				
1) Lithium batteries (including portable electronic devices)	Yes ((except for g) and h))	Yes	(see c) and d))	a. each battery must be of a type which meets the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3; b. each battery must not exceed the following: • for lithium metal batteries, a lithium content of 2 grams; or • for lithium ion batteries, a Watt-hour rating of 100 Wh; c. each battery may exceed 100 Wh but not exceed 160 Wh Watt-hour rating for lithium ion with the approval of the operator; d. each battery may exceed 2 grams but not exceed 8 grams lithium content for lithium metal for portable medical electronic devices with the approval of the operator; e. batteries contained in portable electronic devices should be carried as carry-on baggage; however, if carried as checked baggage: • measures must be taken to prevent unintentional activation and to protect the devices from damage; and • the devices must be completely switched off (not in sleep or hibernation mode); f. batteries and heating elements must be isolated in portable electronic devices capable of generating extreme heat, which could cause a fire if activated, by removal of the heating element, battery or other components; g. spare batteries, including power banks: • must be carried as carry-on baggage; and • must be individually protected so as to prevent short circuits (by placement in original retail packaging or by otherwise insulating terminals, e.g. by taping over exposed terminals or placing each battery in a separate plastic bag or protective pouch); h. baggage equipped with a lithium battery(ies) exceeding: • for lithium metal batteries, a lithium content of 0.3 grams; or • for lithium ion batteries, a Watt-hour rating of 2.7 Wh must be carried as carry-on baggage unless the battery(ies) is removed from the baggage, in which case the battery(ies) must be carried in accordance with g); i. no more than two spare batteries meeting the requirements of c) or d) may be carried per person.

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
2) Non-spillable wet, nickelmetal hydride, and dry batteries	Yes	Yes	No	a. for a non-spillable battery: i. must meet the requirements of Special Provision A67; ii. each battery must not exceed a voltage of 12 volts and a Watt-hour rating of 100 Wh; iii. each battery must be protected from short circuit by the effective insulation of exposed terminals; iv. no more than two spare batteries per person may be carried; and v. if contained in equipment, the equipment must be either protected from unintentional activation, or each battery must be disconnected and its exposed terminals insulated. b. for a dry battery or nickel-metal hydride battery, each battery must comply with Special Provision A123 or A199, respectively; and c. batteries and heating elements must be isolated in battery powered equipment capable of generating extreme heat, by removal of the heating element, battery or other components.
3) Battery-powered portable electronic smoking devices (e.g. e-cigarettes, ecigs, ecigars, epipes, personal vaporizers, electronic nicotine delivery systems)	No	Yes	No	a. if powered by lithium batteries, each battery must comply with restrictions of 1) a), b) and g); b. the devices and/or batteries must not be recharged on board the aircraft; and c. measures must be taken to prevent unintentional activation of the heating element while on board the aircraft.

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
4) Mobility aids (e.g. wheelchairs) powered by: - spillable batteries; - non-spillable wet batteries; - dry batteries; - nickel-metal hydride batteries; or - lithium ion batteries.	Yes	(see e)	Yes	a. for use by passengers whose mobility is restricted by either a disability, their health or age, or a temporary mobility problem (e.g. broken leg); b. the passenger should make advance arrangements with each operator and provide information on the type of battery installed and on the handling of the mobility aid (including instructions on how to isolate the battery); c. in the case of a dry battery or nickel-metal hydride battery, each battery must comply with Special Provision A123 or A199, respectively; d. in the case of a non-spillable wet battery: i. each battery must comply with Special Provision A67; and ii. a maximum of one spare battery may be carried per passenger; e. in the case of a lithium ion battery: i. each battery must be of a type which meets the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3; ii. when the mobility aid does not provide adequate protection to the battery: • the battery must be removed in accordance with the manufacturer's instructions; • the battery must not exceed 300 Wh; • the battery terminals must be protected from short circuit (by insulating the terminals, e.g. by taping over exposed terminals); • the battery must be protected from damage (e.g. by placing each battery in a protective pouch); and the battery must be carried in the cabin. iii. a maximum of one spare battery not exceeding 300 Wh or two spare batteries not exceeding 160 Wh each may be carried. Spare batteries must be carried in the cabin.
Flames and Fuel Sources				
5) Cigarette lighter Small packet of safety matches	No	(see b))	No	a. no more than one per person; b. must be carried on the person; c. must not contain unabsorbed liquid fuel (other than liquefied gas); and d. if a cigarette lighter is powered by lithium batteries, each battery must comply with restrictions of 1) a), b) and g) and 3) b) and c).
6) Alcoholic beverages containing more than 24 per cent but not more than 70 per cent alcohol by volume	Yes	Yes	No	a. must be in retail packagings; and b. no more than 5 L total net quantity per person. Note: Alcoholic beverages containing not more than 24 per cent alcohol by volume are not subject to any restrictions.

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
7) Internal combustion engines or fuel cell engines	Yes	No	No	Measures must be taken to nullify the hazard. For more information, Special Provision A70 below refers: A70: Internal combustion or fuel cell engines or machinery, being shipped either separately or incorporated into a vehicle, machine or other apparatus, without batteries or other dangerous goods, are not subject to these Regulations when carried as cargo or baggage, provided that: a. for flammable liquid powered engines: - the engine is powered by a fuel that does not meet the classification criteria for any class or division; or - the fuel tank of the vehicle, machine or other apparatus has never contained any fuel or the fuel tank has been flushed and purged of vapours and adequate measures taken to nullify the hazard; - the shipper has provided the operator with written or electronic documentation stating that a flushing and purging procedure has been followed; and - the entire fuel system of the engine has no free liquid that is subject to these Regulations and all fuel lines are sealed or capped or securely connected to the engine and vehicle, machinery or apparatus. b. for flammable gas powered internal combustion or fuel cell engines: - the entire fuel system must have been flushed, purged and filled with a non-flammable gas or fluid to nullify the hazard; and - the final pressure of the non-flammable gas used to fill the system does not exceed 200 kPa at 20°C; - the shipper has made prior arrangements with the operator; and - the shipper has provided the operator with written or electronic documentation stating that the flushing, purging and filling procedure has been followed and that the final contents of the engine(s) have been tested and verified to be non-flammable.
8) Fuel cells containing fuel	No	Yes	No	a. fuel cell cartridges may only contain flammable liquids, corrosive substances, liquefied flammable gas, water reactive substances or hydrogen in metal hydride;
Spare fuel cell cartridges	Yes	Yes	No	b. refuelling of fuel cells on board an aircraft is not permitted except that the installation of a spare cartridge is allowed;

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
				c. the maximum quantity of fuel in any fuel cell or fuel cell cartridge must not exceed: • for liquids 200 mL; • for solids 200 grams; • for liquefied gases, 120 mL for non-metallic fuel cell cartridges or 200 mL for metal fuel cell or fuel cell cartridges; and • for hydrogen in metal hydride, the fuel cell or fuel cell cartridges must have a water capacity of 120 mL or less; d. each fuel cell and each fuel cell cartridge must conform to IEC 62282-6-100 Ed. 1, including Amendment 1, and must be marked with a manufacturer's certification that it conforms to the specification. In addition, each fuel cell cartridge must be marked with the maximum quantity and type of fuel in the cartridge; e. fuel cell cartridges containing hydrogen in metal hydride must comply with the requirements in Special Provision A162 (described at the end of this table); f. fuel cell cartridges containing hydrogen in metal hydride must comply with the requirements in Special Provision A162; g. no more than two spare fuel cell cartridges may be carried by a passenger;
				h. fuel cells containing fuel are permitted in carry-on baggage only; i. interaction between fuel cells and integrated batteries in a device must conform to IEC 62282-6-100 Ed. 1, including Amendment 1. Fuel cells whose sole function is to charge a battery in the device are not permitted; j. fuel cells must be of a type that will not charge batteries when the portable electronic device is not in use and must be durably marked by the manufacturer: "APPROVED FOR CARRIAGE IN AIRCRAFT CABIN ONLY" to so indicate; and k. in addition to the languages which may be required by the State of Origin for the markings specified above, English should be used.
Gases in Cylinders and Cartridges				
9) Cylinders of oxygen or air required for medical use	Yes	Yes	Yes	a. no more than 5 kg gross mass per cylinder; b. cylinders, valves and regulators, where fitted, must be protected from damage which could cause inadvertent release of the contents; c. advance arrangements recommended; and d. the pilot-in-command must be informed of the number of oxygen or air cylinders loaded on board the aircraft and their loading location(s).
10) Cartridges of Division 2.2 worn for the operation of mechanical limbs	Yes	Yes	No	Spare cartridges of a similar size are also allowed, if required, to ensure an adequate supply for the duration of the journey.

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Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
11) Cartridge of hydrocarbon gas contained in hair styling equipment	Yes	Yes	No	a. no more than one per person; b. the safety cover must be securely fitted over the heating element; and c. spare cartridges must not be carried.
12) Cartridges of Division 2.2 with no subsidiary hazard fitted into a self-inflating personal safety device, intended to be worn by a person, such as a lifejacket or vest	Yes	Yes	Yes	a. no more than two personal safety devices per person; b. the personal safety device(s) must be packed in such a manner that they cannot be accidentally activated; c. must be for inflation purposes; d. no more than two cartridges are fitted into each device; and e. no more than two spare cartridges per device.
13) Cartridges of Division 2.2 with no subsidiary hazard for other than a self-inflating personal safety device	Yes	Yes	Yes	a. no more than four cartridges per person; and b. the water capacity of each cartridge must not exceed 50 mL. Note: For carbon dioxide, a gas cartridge with a water capacity of 50 mL is equivalent to a 28 g cartridge.
14) Cartridges and cylinders of Division 2.2 with no subsidiary hazard contained in an avalanche rescue backpack	Yes	Yes	Yes	a. no more than one avalanche rescue backpack per person; b. the backpack must be packed in such a manner that it cannot be accidentally activated; c. may contain a pyrotechnic trigger mechanism which must not contain more than 200 mg net of Division 1.4S; and d. the airbags within the backpack must be fitted with pressure relief valves.
Radioactive Material				
15) Radioisotopic cardiac pacemakers or other medical devices	n/a (see restrictions)	n/a (see restrictions)	No	Must be implanted into a person or fitted externally as the result of medical treatment.
Mercury				
16) Small medical or clinical thermometer which contains mercury	Yes	No	No	a. no more than one per person; and b. must be in its protective case.
Other Dangerous Goods				

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
17) Non-radioactive medicinal articles (including aerosols), toiletry articles (including aerosols) and aerosols in Division 2.2 with no subsidiary hazard	Yes	Yes	No	a. no more than 0.5 kg or 0.5 L total net quantity per single article; b. no more than 2 kg or 2 L total net quantity of all articles (e.g. four aerosol cans of 0.5 L each) per person; c. release valves on aerosols must be protected by a cap or other suitable means to prevent inadvertent release of the contents; and d. the release of gas must not cause extreme annoyance or discomfort to crew members so as to prevent the correct performance of assigned duties.
18) Dry ice	Yes	Yes	Yes	a. no more than 2.5 kg per person; b. used to pack perishables that are not subject to the Technical Instructions; c. the package must permit the release of carbon dioxide gas; and d. when carried as checked baggage, each package must be marked: i. “DRY ICE” or “CARBON DIOXIDE, SOLID”; and ii. the net weight of dry ice or an indication that the net weight is 2.5 kg or less.
19) Cartridges in Division 1.4S (UN 0012 or UN 0014 only)	Yes	No	Yes	a. no more than 5 kg gross mass per person; b. must be securely packaged; c. must not include ammunition with explosive or incendiary projectiles; and d. allowances for more than one person must not be combined into one or more packages.
20) Permeation devices	Yes	No	No	Instructions on how to package permeation devices for calibrating air quality monitoring equipment are found in Special Provision A41 (refer to OMA 9.1.5 – Note 9).
21) Non-infectious specimens in flammable solutions	Yes	Yes	No	Instructions on how to package and mark specimens are found in Special Provision A180 (refer to OMA 9.1.5 – Note 11).
22) Refrigerated liquid nitrogen	Yes	Yes	No	Must be contained in insulated packagings (e.g. dry shippers) that would not allow the build-up of pressure and be fully absorbed in a porous material so that there is no free liquid that could be released from the packaging. Refer to Special Provision A152 for more information.
23) Dangerous goods incorporated in security-type equipment, such as attaché cases, cash boxes, cash bags, etc.	Yes	No	Yes	The security-type equipment must be equipped with an effective means of preventing accidental activation and the dangerous goods incorporated in the equipment must meet the conditions of Special Provision A178 (refer to OMA 9.1.5 – Note 10 for description).

Special Provisions Mentioned in Table 1

Some Special Provisions are already described in paragraphs above.

Additionally, see below:

A162: Fuel cell cartridges containing hydrogen in a metal hydride transported under this entry must have a water capacity less than or equal to 120 ml.

The fuel cell cartridges must be designed and constructed to prevent fuel leakage under normal conditions of transport. Each cartridge design type, including cartridges integral to a fuel cell, must be subjected to and must pass the required tests.

A152: Insulated packagings conforming to the requirements of Packing Instruction 202 containing refrigerated liquid nitrogen fully absorbed in a porous material are not subject to these Regulations provided the design of the insulated packaging would not allow the build-up of pressure within the container and would not permit the release of any refrigerated liquid nitrogen irrespective of the orientation of the insulated packaging and any outer packaging or overpack used is closed in a way that will not allow the build-up of pressure within that packaging or overpack.

The words “not restricted” and the special provision number A152 must be included in the description of the substance on the Air Waybill (as required by DG Regulation) when an Air Waybill is issued; this is required to indicate that the special provision has been applied.

A213: Lithium batteries in conformity with IATA DGR section 3.9.2.6.1 (f) containing both primary lithium metal cells and rechargeable lithium ion cells must be assigned to UN 3090 or UN 3091, as appropriate. When such batteries are transported in accordance with Section IB of Packing Instruction 968 or in accordance with Section II of 969 or 970 (i.a.w. IATA DG Manual), the total lithium content of all lithium metal cells contained in the battery must not exceed 1.5 g and the total capacity of all lithium ion cells contained in the battery must not exceed 10 Wh (the responsible of the DG acceptance process shall confirm this i.a.w. OM-A 9.3.1).

The Organization for the Prohibition of Chemical Weapons (OPCW) and government agencies listed in the table below may carry specified instruments containing dangerous goods when:

1. carried by staff members on official travel;
2. contained in baggage that has been separated from its owner during transit (e.g. lost baggage or improperly routed baggage); or

3. contained within items of excess baggage sent as cargo as permitted by 9.1.3.3.

Table 2 Provisions for Instruments Carried by OPCW and Government Agencies (ICAO DOC 9284 – Ed. 2025–2026)

Dangerous Goods	Location		Approval of the Operator(s) is Required	Restrictions
	Checked Baggage	Carry-on Baggage		
1) Instruments containing radioactive material (i.e. chemical agent monitor (CAM) and/or rapid alarm and identification device monitor (RAID-M))	Yes	Yes	Yes	a. the instruments must not exceed the activity limits for *excepted packages; b. must be securely packed; and c. must be carried by staff members of the Organization for the Prohibition of Chemical Weapons (OPCW) on official travel.
2) A mercurial barometer or mercurial thermometer	No	Yes	Yes	a. must be carried by a representative of a government weather bureau or similar official agency; b. must be packed in a strong outer packaging, having a sealed inner liner or a bag of strong leakproof and puncture-resistant material impervious to mercury, which will prevent the escape of mercury from the package irrespective of its position; and c. the pilot-in-command must be informed of the barometer or thermometer.

9.1.6 Provision of Information to Company Personnel & Passengers

(CAT.GEN.MPA.200 (f))

9.1.6.1 Information to Passengers

Ryanair Group Airlines publish information on the company website for the types and quantities of dangerous goods which may be carried on board an aircraft, as well as those that are forbidden.

Additionally, notices informing passengers of the types of dangerous goods which they are forbidden to transport aboard an aircraft are prominently displayed where tickets are issued, passengers are checked-in (including automated check-in) and at aircraft boarding areas.

Passengers cannot complete the check in process until they have indicated that they understand these restrictions as stated in the Terms and Conditions.

9.1.7 Marking and Labelling of Packages

Articles and substances meeting the dangerous goods classification criteria are assigned a 'UN Number' under the United Nations classification system. This consists a four-digit number preceded by the capital letters 'UN'. Packages of dangerous goods must be marked with the UN Number(s) applicable to their contents.

Dangerous goods bearing any UN Number, hazard label; the radioactive material, excepted package handling label; the lithium battery mark; the environmentally hazardous substances mark; or the excepted or limited quantities mark must not be loaded on an aircraft; or the excepted or limited quantities marking must not be loaded on an aircraft (except as identified in 9.1.3, and as company spare parts properly packaged, labelled and documented).

Packages containing dangerous goods can also be identified by labels indicating the hazard of the goods by their class or division or by the presence of certain handling labels/marks.

Note: When dangerous goods marks or labels are seen on items not declared as dangerous goods, it is often an indication that they do contain such goods. Undeclared dangerous goods must not be loaded on an aircraft and reporting procedures are implemented (see 9.6.4).

During the course of air transport, including storage, dangerous goods marks and labels must not be covered or obscured by any part of or attachment to the packaging or any other label or marking.

CLASS 1 – EXPLOSIVE

Class 1 (with exploding bomb symbol) – explosives generally not permitted on an aircraft.



* Division and compatibility group

Class 1 (without exploding bomb symbol) – Divisions 1.4B, 1.4F, 1.5 and 1.6 – explosives not permitted on an aircraft in normal circumstances.



** Compatibility group

CLASS 2 – GASES

Flammable gas (Division 2.1) Non-flammable, non-toxic gas (Division 2.2) Toxic gas (Division 2.3)



CLASS 3 – FLAMMABLE LIQUID



CLASS 4 – FLAMMABLE SOLIDS; SUBSTANCES LIABLE TO SPONTANEOUS COMBUSTION; SUBSTANCES WHICH, IN CONTACT WITH WATER, EMIT FLAMMABLE GASES

Flammable solid (Division 4.1) Substance liable to spontaneous combustion (Division 4.2) Substance which, in contact with water, emits flammable gas (Division 4.3)



CLASS 5 – OXIDISING SUBSTANCES AND ORGANIC PEROXIDES

Oxidising substance (Division 5.1) Organic peroxide (Division 5.2) (flame may be black or white)



CLASS 6– TOXIC AND INFECTIOUS SUBSTANCES

Toxic substance (Division 6.1) Infectious substance (Division 6.2)



The bottom part of the label should bear the inscription:
“INFECTIOUS SUBSTANCE
– In case of damage or leakage
immediately notify public
health Authority”.

CLASS 7 – RADIOACTIVE MATERIAL

Category I



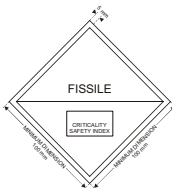
Category II



Category III



Criticality safety index label



CLASS 8 – CORROSIVE



CLASS 9 – MISCELLANEOUS

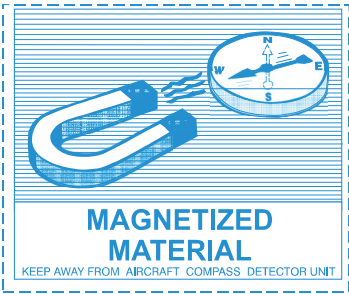
Class 9 label for Section IA and IB lithium battery shipments



HANDLING LABELS

Packages of dangerous goods may also bear labels providing handling information; these are:

Magnetized material



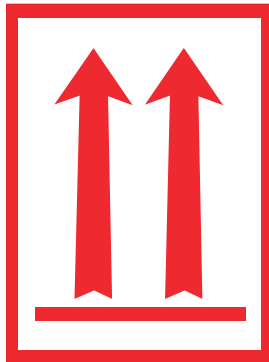
Cargo aircraft only



Cryogenic liquid label



Package orientation



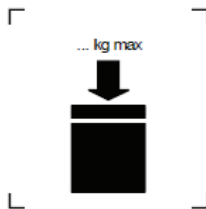
(red or black)

Keep away from heat

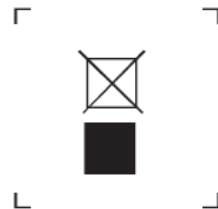


Intermediate Bulk Containers (IBCs) are only permitted for the transport of UN 3077 Environmentally hazardous substance, solid, n.o.s. The maximum permitted stacking load applicable when the IBC is in use must be displayed on a symbol as follows:

IBCs capable of being stacked



IBCs NOT capable of being stacked



BATTERY MARK



* Place for UN Numbers(s)

Application of the lithium battery mark to a consignment of lithium batteries (of any type) indicates that the Shipper has determined specific requirements have been met. Consignments borne this label without the Class 9 label do not need to be accompanied by a dangerous goods transport document (Shipper's Declaration) and no acceptance check is required.

EXCEPTED QUANTITIES MARK

Packages containing excepted quantities of dangerous goods can be identified from the following:



Hatching and symbol of the same colour, black or red, on white or suitable contrasting background.

* Place for class or, when assigned, the division number(s).

** Place for name of shipper or consignee, if not shown elsewhere on the package.

LIMITED QUANTITIES MARK

Packages containing limited quantities of dangerous goods can be identified from the following:



Many dangerous goods when in reasonably limited quantities present a reduced hazard during transport and can safely be carried in good quality packaging's that have not been tested and marked as is required for UN Specification packaging's required for larger quantities of dangerous goods. Packages containing limited quantities of dangerous goods must be marked with a diamond shaped mark. When presented for carriage by air, the mark must additionally include a "Y" which indicates compliance with the provisions of the ICAO Technical Instructions, some of which are more stringent than those of the UN Model Regulations and of other modes of transport.

Note: The mark depicted here but without the 'Y' indicates that the package contains dangerous goods in limited quantities as permitted by surface transport regulations (ADR/IMDG) which may not be acceptable for air transport. A package so marked and offered for transport in the absence of a dangerous goods transport document must be reported to the appropriate Authority where the goods are discovered as a discovery of undeclared dangerous goods.

ENVIRONMENTALLY HAZARDOUS SUBSTANCES MARK

Packages containing environmentally hazardous substances (UN Nos. 3077 and 3082) must be durably marked with the environmentally hazardous substance mark with the exception of packages containing a net quantity per single or inner packagings with contents of 5 L or less for liquids; or having a net mass per single or inner packaging contents of 5 kg or less for solids.

9.2 Duties of All Personnel Involved

9.2.1 Detailed Assignments of Responsibilities

(CAT.GEN.MPA.200 (d))

Alice Ballestriero is the Person Nominated Responsible for:

Operator's Dangerous goods Approval of Malta Air:

- Oversight of the carriage of dangerous goods.
- Liaison between Malta Air and Ryanair HQ in Dublin.
- The shipper seeks approvals, permissions and exemptions and will provide relevant documentation to Malta Air before the flight.
- GOPS have oversight over the Acceptance Procedures carried out by Handling Agents on behalf of the company.
- Responding to queries regarding the carriage of dangerous goods.

The Shipper of DG (Ryanair Engineering)	The shipper ensures all approvals, permissions and exemptions are applied for and received in time before shipping company spare parts or property containing DG.
Persons receiving or handling general cargo, mail and stores	- Recognition of undeclared dangerous goods. - Dealing with dangerous goods that are found damaged or leaking during processing for transport. If there is a dangerous goods incident or accident, or if undeclared dangerous goods are detected, a report is made to the appropriate Authority.
Acceptance of DG: Persons receiving or handling dangerous goods (Ground Handling Agents)	- Acceptance procedures for dangerous goods are carried out by Handling Agents as required by the Technical Instructions. - Inspection procedures during the processing of dangerous goods for transport are carried out as required by the Technical Instructions. - Dangerous goods are loaded, segregated, stowed and secured on an aircraft in accordance with the Technical Instructions. - Generation of written information to the commander (NOTOC). - Provision of written information about dangerous goods loaded on board to the commander for signature. - Retention of documentation on the ground. - Recognition of undeclared dangerous goods. - If there is a dangerous goods incident or accident, or if undeclared dangerous goods are detected, a report is made to the appropriate Authority (see 9.6.4).
Commander of the Flight	The Captain will then countersign the NOTOC confirming receipt of the information therein. The Captain will then tear off one portion of the signed counter copy and give it back to the ground handling agent for retention on the ground. All flight documentation to include the NOTOC must be retained at the station for a period of three months after shipping.
Reservations	- Ensuring that information is provided with the passenger ticket or in another manner such that prior to or during the check-in process the passenger receives the information. - Considering passenger requests for approval of the Company for items of dangerous goods requiring such approval.

Persons handling passengers	• Ensuring that the provisions concerning passengers and dangerous goods are complied with. • Ensuring that notices are displayed in sufficient number and prominence at each of the places at an airport where tickets are issued, passengers checked in and aircraft boarding areas maintained, and at any other location where passengers are checked in. • With the aim of preventing dangerous goods which passengers are not permitted to have from being taken on board an aircraft in their baggage, seeking confirmation from a passenger about the contents of any item where there are suspicions that it may contain dangerous goods. • When baggage intended as carry-on is taken by the operator and placed into the cargo compartment for carriage, seeking confirmation from the passenger that dangerous goods which are only permitted in carry-on baggage (e.g. lithium batteries, including power banks) have been removed. • Ensuring that the discovery of prohibited dangerous goods (after a passenger has checked in) is reported to the appropriate Authority via approved company reporting procedures (see 9.6.4).
Cabin Crew	• Ensuring that the provisions concerning passengers and dangerous goods are complied with. • When baggage intended as carry-on is taken by the operator and placed into the cargo compartment for carriage, seeking confirmation from the passenger that dangerous goods which are only permitted in carry-on baggage (e.g. lithium batteries, including power banks) have been removed. • Responding to a dangerous goods incident or accident in the cabin. Ensuring that a dangerous goods incident or accident in the cabin, or the discovery of prohibited dangerous goods (after a passenger has boarded), is reported via approved company reporting procedures and/to the appropriate Authority (see 9.6.4).

Operations Personnel	• If there is an aircraft incident or accident, information is passed to emergency services and state Authorities as required by the Technical Instructions (see 9.6.2). • If there is a dangerous goods incident or accident, or if undeclared dangerous goods are detected a report is made via approved company reporting procedures and/to the appropriate Authority (see 9.6.4).
Flight Crew	• Recognition of undeclared dangerous goods. • Signature of NOTOC to indicate receipt of information. • If an in-flight emergency occurs, as soon as the situation permits, passage of details of dangerous goods on board to the appropriate Air Traffic Services Unit. • Responding to a dangerous goods incident or accident in the cabin (if operation does not have cabin crew). • If there is a dangerous goods incident or accident, or if undeclared dangerous goods are detected a report is made to the appropriate Authority (see 9.6.4).
Trainers	• Provision of initial and recurrent dangerous goods training commensurate with the responsibilities of the personnel concerned or the functions for which they are responsible.
Compliance Monitoring Manager, Auditors and Safety Manager	• Ensuring that activities are monitored for compliance with dangerous goods requirements and that these activities are carried out properly under the supervision of the relevant head of functional area. • Ensuring the initiation and follow-up of internal occurrence/accident investigations.

9.3 **Guidance on the Requirements for Acceptance, Handling and Stowage**

(SPA.DG.105)

9.3.1 **Acceptance Check**

It is company policy not to carry third party cargo or dangerous goods. The company Dangerous Goods approval includes carriage of company goods and spare parts that qualify as DG.

Before a consignment consisting of a package or overpack containing dangerous goods is first accepted for carriage by air, the handling agent on behalf of Malta Air must, by use of the ICAO Dangerous Goods Checklist for Non-Radioactive Shipment, verify the following:

1. The documentation or, when provided, the electronic data is compliant with the applicable requirements;
2. The quantity of dangerous goods stated on the dangerous goods transport document is within the limits per package on a passenger or cargo aircraft as appropriate;

3. The package, overpack or freight container marks accord with the details stated on the accompanying dangerous goods transport document and is clearly visible;
4. Where required, the letter in the packaging specification marking designating the packing group for which the design type has been successfully tested is appropriate for the dangerous goods contained within. This does not apply to overpacks where the specification marking is not visible;
5. Proper shipping names, UN numbers, labels, and special handling instructions appearing on the interior package(s) are clearly visible or reproduced on the outside of an overpack;
6. The labelling of the package, overpack or freight container is as required for the consignment;
7. The outer packaging of a combination package or the single packaging is permitted by the applicable packing instruction, and when visible is of the type stated on the accompanying dangerous goods transport document;
8. The package or overpack does not contain different dangerous goods which require segregation from each other; and
9. The package, overpack, freight container or Unit Load Device (ULD) is not leaking and there is no indication that its integrity has been compromised.

Note 1: An acceptance check is not required for dangerous goods in excepted quantities, radioactive material in excepted packages and lithium batteries consigned in accordance with Section II of the applicable packing instruction.

Note 2: Persons conducting dangerous goods acceptance checks must have received dangerous goods training commensurate with this responsibility.

9.3.2 Inspections for Damage or Leakage

(SPA.DG.105)

A package or overpack containing dangerous goods must not be loaded onto an aircraft or into a ULD unless it has been inspected immediately prior to loading and found free from evidence of leakage or damage. A ULD must not be loaded aboard an aircraft unless the device has been inspected and found free from any evidence of leakage from or damage to any dangerous goods contained therein. Packages or overpacks containing dangerous goods must be inspected for signs of damage or leakage upon unloading from the aircraft or ULD.

9.3.3 Prohibition on the Carriage of Dangerous Goods Within a Cabin Occupied by Passengers

(SPA.DG.105)

Dangerous goods must not be carried in the cabin of an aircraft occupied by passengers or on the flight deck, except as provided for in the Technical Instructions.

9.3.4 Prohibition on the Carriage of Passengers with ‘Cargo Aircraft Only’ Dangerous Goods

(SPA.DG.105)

Dangerous goods identified as suitable for transport only on a cargo aircraft must not be carried on an aircraft on which passengers are being carried. In this context “passenger” excludes a crew member, an operator’s employee (see 9.1.4 above), an authorised representative of an Authority and a person with duties in respect of a particular shipment of dangerous goods or other cargo on board.

9.3.5 Segregation, Separation and CAO Accessibility

(SPA.DG.105)

Dangerous goods must be loaded, stowed and secured on an aircraft as required by the Technical Instructions. This includes segregating packages from each other when they contain incompatible dangerous goods, the separation of explosives of different division numbers and compatibility groups (when required), securing packages in a manner that will prevent any movement. Dangerous goods must also be protected so they cannot be damaged by the movement of baggage, mail, stores or other cargo.

Segregation of Incompatible Dangerous Goods

Hazard Label	Class or Division								
	1	2	3	4.2	4.3	5.1	5.2	6	8
1	Note 1	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2		Note 2
2	Note 2								
3	Note 2					X			
4.2	Note 2					X			
4.3	Note 2								X
5.1	Note 2		X	X					
5.2	Note 2								
6									
8	Note 2				X				

An “X” at the intersection of a row and column indicates that packages containing these classes of dangerous goods may not be stowed next to or in contact with each other, or in a position which would allow interaction in the event of leakage of the contents. Thus, a package containing Class 3 dangerous goods may not be stowed next to or in contact with a package containing Division 5.1 dangerous goods.

Note 1: ‘Only Division 1.4S is permitted for carriage on passenger aircraft’.

Note 2: This class or division must not be stowed together with explosives other than those in Division 1.4, Compatibility Group S.

Note 3: Dangerous goods checklist for a non-radioactive shipment, for information purpose only:

DANGEROUS GOODS CHECKLIST FOR A NON-RADIOACTIVE SHIPMENT

The recommended checklist appearing on the following pages is intended to verify shipments at origin. Copies of the checklist can be obtained from:

Website: <http://www.iata.org/whatwedo/cargo/dgr/Pages/download.aspx>

Never accept or refuse a shipment before all items have been checked.

Is the following information correct for each entry?

SHIPPER'S DECLARATION FOR DANGEROUS GOODS (DGD)

Air Waybill No.:	Origin:	Destination:
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	YES	NO*	N/A
1. Two copies in English and in the IATA format including the air certification statement. This question may be indicated as not applicable "N/A" only when the Shipper's Declaration data is submitted electronically [8.0.2.1, 8.1.1, 8.1.2, 8.1.6.12]	☐	☐	☐
2. Full name and address of Shipper and Consignee [8.1.6.1, 8.1.6.2]	☐	☐	☐
3. If the Air Waybill number is not shown, enter it. [8.1.6.3]	☐	☐	☐
4. The number of pages shown. This question may be indicated as not applicable "N/A" only when the Shipper's Declaration data is submitted electronically [8.1.6.4]	☐	☐	☐
5. The applicable Aircraft Type deleted or not shown [8.1.2.5.2, 8.1.6.5]	☐	☐	☐
6. If the Airport or City of Departure or Destination is not shown, enter it. [8.1.6.6 and 8.1.6.7]	☐	☐	☐
7. If the "Active" deleted or not shown [8.1.2.5.2, 8.1.6.8]	☐	☐	☐
Identified by prefix [8.1.6.9.1, Step 1]	☐	☐	☐
9. Technical name in brackets for entries with * [8.1.6.9.1, Step 2]	☐	☐	☐
10. Class and Compatibility Group, [8.1.6.9.1, Step 3]	☐	☐	☐
11. Subsidiary following Class or Division [8.1.6.9.1, Step 4]	☐	☐	☐
12. Packing Group	☐	☐	☐
Quantity and Type of Packages	☐	☐	☐
13. Number and Type of Packages (applicable) within per package	☐	☐	☐
14. Quantity and unit of measure (applicable) within per package limit [8.1.6.9.2, Step 6]	☐	☐	☐
15. For Class 1 (Explosives), the net quantity or explosive mass followed by unit of measurement [8.1.6.9.2, Step 6]	☐	☐	☐
16. When different dangerous goods are packed in one container, the following rules are complied with:	☐	☐	☐
16.1 - Compatible according to Table 9.3.A	☐	☐	☐
16.2 - Conditions met for UN packages containing Division 6.2 [5.0.2.11 (g) & (h)]	☐	☐	☐
16.3 - Wording "All packed in one (type of packaging)" [8.1.6.9.2, Step 7]	☐	☐	☐
16.4 - Calculation of "Q" value which must not exceed 1 [5.0.2.11 (g) & (h)]	☐	☐	☐
17. Overpack	☐	☐	☐
17.1 - Compatible according to Table 9.3.A	☐	☐	☐
17.2 - Wording "Overpack Used" [8.1.6.9.2, Step 7]	☐	☐	☐
17.3 - If more than one overpack is used, identification marks shown and total quantity of dangerous goods [8.1.6.9.2, Step 7]	☐	☐	☐
Packing Instructions	☐	☐	☐
18. Packing Instruction Number [8.1.6.9.3, Step 8]	☐	☐	☐
19. For lithium batteries in compliance with Section IB, "IB" follows the packing instruction [8.1.6.9.3, Step 8]	☐	☐	☐
Authorizations	☐	☐	☐
20. Check all verifiable special provisions. The Special Provision Number A1, A2, A4, A5, A51, A81, A88, A99, A130, A190, A191, A201, A202, A211, A212, A331 if used [8.1.6.9.4, Step 9]	☐	☐	☐
21. Indication that governmental authorization is attached, including a copy in English and additional approvals for other items under [8.1.6.9.4, Step 9]	☐	☐	☐
Additional Handling Information	☐	☐	☐
22. Additional handling information shown for self-reactive and related substances of Division 4.1 and organic peroxides of Division 5.2, or samples thereof, PBE, infectious and controlled substances, fireworks (UN0336 & UN0337) and viscous flammable liquids [8.1.6.11]	☐	☐	☐
23. Name of Signatory and Date indicated and Signature of Shipper [8.1.6.13, 8.1.6.14 and 8.1.6.15]	☐	☐	☐
24. Amendment or alteration signed by Shipper [8.1.2.6]	☐	☐	☐

*IF ANY BOX IS CHECKED "NO", DO NOT ACCEPT THE SHIPMENT AND GIVE A DUPLICATE COPY OF THIS COMPLETED FORM TO THE SHIPPER.

9.3.6 Loading of Dry Ice

(SPA.DG.105)

N/A – Malta Air will only carry dry ice under the provisions laid down in OMA CH09.

9.3.7 Loading of Magnetised Material

(SPA.DG.105)

N/A

9.3.8 Loading of Radioactive Material

(SPA.DG.105)

N/A

9.3.9 Loading of UN 2211, Polymeric beads, Expandable or UN 3314, Plastics Moulding Compound

N/A

9.3.10 Notification to Captain (NOTOC)

(AMC SPA.DG.110(a))

The pilot-in-command of a company aircraft in which dangerous goods are to be carried shall as early as practicable before departure of the aircraft be provided with accurate and legible written or printed information concerning dangerous goods that are to be carried as company freight. Personnel with responsibilities for operational control of the aircraft (e.g. the flight operations officer, flight dispatcher, or designated ground personnel responsible for flight operations) must be provided with the same information that is provided to the pilot-in-command (e.g. a copy of the written information provided to the pilot-in-command). This is to facilitate notifying emergency services and authorities of the dangerous goods on board in the event of an aircraft accident or incident.

Prior to dispatch on each sector, the PIC shall ensure that the Operations Control Centre receives a copy of the NOTOC. The process for this is as follows:

1. The PIC will take a picture of the NOTOC using the company EFB.
2. The picture of the NOTOC will be emailed to dgs@ryanair.com.

If the journey is multi sector then a new NOTOC will be produced prior to the commencement of each sector detailing all new and transit loads. One copy must be left at the departure Station.

Note: At non-scheduled stops a new NOTOC need not be issued if the Dangerous Goods on the aircraft has not been altered and the incoming Commander will command the outgoing flight.

This information must include the following:

- a. The date of the flight;
- b. The air waybill number (when issued);
- c. The proper shipping name (the technical name(s) shown on the dangerous goods transport document is not required) and UN Number or ID number;
- d. The class or division, and subsidiary hazard(s) corresponding to the subsidiary hazard label(s) applied, by numerals, and (in the case of Class 1) the compatibility group;
- e. The packing group shown on the dangerous goods transport document;
- f. The number of packages and their exact loading location. For radioactive material see (g) below;
- g. The net quantity, or gross mass if applicable, of each package, except that this does not apply to radioactive material or other dangerous goods where the net quantity or gross mass is not required on the dangerous goods transport document. For a consignment consisting of multiple packages containing dangerous goods bearing the same proper shipping name and UN number, only the total quantity and an indication of the quantity of the largest and smallest package at each loading location need to be provided;
- h. For radioactive material, the number of packages, overpacks or freight containers, their category, their Transport Index (if applicable) and their exact loading location;
- i. Whether the package must be carried on cargo aircraft only;
- j. The aerodrome at which the package(s) is to be unloaded;
- k. Where applicable, an indication that the dangerous goods are being carried under a State exemption; and
- l. Signed confirmation, or some other indication, from the person responsible for loading the aircraft that there was no evidence of any damage to or leakage from the packages or any leakage from the unit load devices loaded on the aircraft.

On the completion of the flight the commander will present the dangerous goods documentation to the dispatcher and confirm their location. Should the aircraft not be met on arrival by ground staff then the commander will place the dangerous goods paperwork with the dangerous goods. Where possible commanders should ensure that the dangerous goods are removed from the aircraft

- Note 1:** For UN 1845 Carbon dioxide, solid (dry ice), the information detailed above may be replaced by the UN number, proper shipping name, class, total quantity in each cargo compartment on the aircraft and the aerodrome at which the package(s) is to be unloaded.
- Note 2:** For UN 3480 (Lithium ion batteries) and UN 3090 (Lithium metal batteries), the information detailed above may be replaced by the UN number, proper shipping name, class, total quantity at each specific loading location, the aerodrome at which the package(s) is to be unloaded and whether the package must be carried on cargo aircraft only. A full NOTOC is required when such batteries are carried under a State exemption.
- Note 3:** For consumer commodities, the information provided may be either the gross mass of each package or the average gross mass of the packages as shown on the dangerous goods transport document.

The following dangerous goods need not appear on the NOTOC:

- Dangerous goods packed in excepted quantities.
- Biological substance, Category B.
- Genetically modified micro-organisms.
- Genetically modified organisms.
- Lithium ion/sodium ion batteries (including lithium ion polymer batteries); Lithium ion/sodium ion batteries contained in equipment; and Lithium ion/sodium batteries packed with equipment when meeting the Section II requirements of the applicable Packing Instruction.
- Lithium metal batteries (including lithium alloy batteries), Lithium metal batteries contained in equipment, and Lithium metal batteries packed with equipment when meeting the Section II requirements of the applicable Packing Instruction.
- Magnetized material with field strengths causing a compass deflection of not more than 2 degrees at a distance of 4.6 m.
- Radioactive material, excepted package (UN 2908, UN 2909, UN 2910 or UN 2911).

9.3.11 Availability of NOTOC on the Ground for the Duration of Flight

(SPA.DG.110)

A legible copy of the information to the pilot-in command must be retained on the ground. This copy must have an indication on it, or with it, that the pilot-in-command has received the information.

9.3.12 Retention of Documents

ORO.MLR.115 (b)(4),, SPA.DG.110(f)

At least one copy of the documents appropriate to the transport by air of a consignment of dangerous goods (including consignments that fail their acceptance check) must be retained for a minimum period of three months, or such other period as specified by the States concerned, after the flight on which the dangerous goods were transported. As a minimum, the documents which must be retained are the dangerous goods transport document (Shipper's Declaration), the acceptance checklist (when this is in a form which requires completion) including identification of the person who completed it, and the NOTOC (if the goods were carried).

Notification of Dangerous Goods will be stored for a period of 3 months in the Malta Air storage facility.

9.4 Recognition of Undeclared/Hidden Dangerous Goods

(CAT.GEN.MPA.200(e))

9.4.1 'Hidden' Dangerous Goods

Personnel must be alert to indications that undeclared dangerous goods are present within cargo, mail or stores. Personnel interfacing with passengers must be alert to indications that prohibited dangerous goods are carried by passengers or within their baggage.

Note: ***THE DISCOVERY OF UNDECLARED OR MIS-DECLARED DANGEROUS GOODS OR THE DISCOVERY OF DANGEROUS GOODS FORBIDDEN FOR CARRIAGE BY PASSENGERS (DISCOVERED AFTER THE CHECK-IN PROCESS) MUST BE REPORTED TO THE COMPANY AND THE COMPETENT AUTHORITY – SEE 9.6.4.***

The following is a list of general descriptions that are often used for items in company freight or in passengers' baggage and the types of dangerous goods that may be included in any item bearing that description.

Aircraft on ground (AOG) spares — may contain explosives (flares or other pyrotechnics), chemical oxygen generators, unserviceable tyre assemblies, cylinders of compressed gas (oxygen, carbon dioxide or fire extinguishers), paint, adhesives, aerosols, life-saving appliances, first aid kits, fuel in equipment, wet or lithium batteries, matches, etc.

Automobile parts/supplies (car, motor, motorcycle) — may contain ferro-magnetic material which may not meet the definition for magnetized material but which may be subject to special stowage requirements due to the possibility of affecting aircraft instruments (see "Magnets"). May also contain engines, including fuel cell engines, carburettors or fuel tanks which contain or

have contained fuel, wet or lithium batteries, compressed gases in tyre inflation devices, fire extinguishers, shocks/struts with nitrogen, air bag inflators/air bag modules, flammable adhesives, paints, sealants and solvents, etc.

Battery-powered devices/equipment — may contain wet or lithium batteries.

Breathing apparatus — may indicate cylinders of compressed air or oxygen, chemical oxygen generators or refrigerated liquefied oxygen.

Camping equipment — may contain flammable gases (butane, propane, etc.), flammable liquids (kerosene, gasoline, etc.) or flammable solids (hexamine, matches, etc.) or other dangerous goods.

Cars, car parts — see automobile parts, etc.

Chemicals — may contain items meeting any of the criteria for dangerous goods, particularly flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances.

Consolidated consignments (groupages) — may contain any of the defined classes of dangerous goods.

Cryogenic (liquid) — indicates refrigerated liquefied gases such as argon, helium, neon, nitrogen, etc.

Company Materials — such as aircraft parts, may contain dangerous goods as an integral part, e.g. chemical oxygen generators in a passenger service unit (PSU), various compressed gases such as oxygen, carbon dioxide and nitrogen, gas lighters, aerosols, fire extinguishers, flammable liquids such as fuels, paints and adhesives and corrosive material such as batteries. Other items such as flares, first aid kits, life-saving appliances, matches, magnetized material, etc.

Cylinders — may contain compressed or liquefied gas.

Dental apparatus — may contain flammable resins or solvents, compressed or liquefied gas, mercury and radioactive material.

Diagnostic specimens — may contain infectious substances.

Diving equipment — may contain cylinders (such as scuba tanks, vest bottles, etc.) of compressed gas (air, oxygen, etc.), high intensity diving lamps which can generate extremely high heat when operated in air. In order to be carried safely, the bulb or battery must be disconnected.

Drilling and mining equipment — may contain explosive(s) and/or other dangerous goods.

Dry shipper (vapour shipper) — may contain free liquid nitrogen. Dry shippers are subject to these Regulations (DG) when they permit the release of any free liquid nitrogen irrespective of the orientation of the packaging.

Electrical/electronic equipment — may contain magnetised materials, mercury in switch gear, electron tubes, wet or lithium batteries or fuel cells or fuel cell cartridges that contain or have contained fuel.

Electrically-powered apparatus (wheelchairs, lawn mowers, golf carts, etc.) — may contain wet or lithium batteries or fuel cells or fuel cell cartridges that contain or have contained fuel.

Expeditionary equipment — may contain explosives (flares), flammable liquids (gasoline), flammable gas (camping gas) or other dangerous goods.

Film crew and media equipment — may contain explosive pyrotechnic devices, generators incorporating internal combustion engines, wet or lithium batteries, fuel, heat-producing items, etc.

Frozen embryos — may be packed in refrigerated liquefied gas or dry ice (solid carbon dioxide).

Frozen fruit, vegetables, etc. — may be packed in dry ice.

Fuel control units — may contain flammable liquids.

Hot-air balloon — may contain cylinders with flammable gas, fire extinguishers, engines (internal combustion), batteries, etc.

Household goods — may contain items meeting any of the criteria for dangerous goods. Examples include flammable liquids such as solvent-based paint, adhesives, polishes, aerosols (for passengers, those not permitted under ICAO Technical Instructions 8.1.1.2), bleach, corrosive oven or drain cleaners, ammunition, matches, etc.

Instruments — may conceal barometers, manometers, mercury switches, rectifier tubes, thermometers, etc. containing mercury.

Laboratory/testing equipment — may contain items meeting any of the criteria for dangerous goods, particularly flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances, lithium batteries, cylinders of compressed gas, etc.

Machinery parts — may contain flammable adhesives, paints, sealants and solvents, wet and lithium batteries, mercury, cylinders of compressed or liquefied gas, etc.

Magnets and other items of similar material — may individually or cumulatively meet the definition of magnetised material (“Any material which, when packed for air transport, has a maximum magnetic field strength sufficient to cause a compass deflection of more than 2 degrees at a distance of 2.1 m from any point on the surface of the assembled package. The magnetic field strength at the compass producing a 2 degree deflection is taken to be 0.418 Alm (0.00525 Gauss)” IATA DG Regulation).

Medical supplies/equipment — may contain items meeting any of the criteria for dangerous goods, particularly flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances, lithium batteries.

Metal construction material — may contain ferro-magnetic material which may be subject to special stowage requirements due to the possibility of affecting aircraft instruments.

Metal fencing — may contain ferro-magnetic material which may be subject to special stowage requirements due to the possibility of affecting aircraft instruments.

Metal piping — may contain ferro-magnetic material which may be subject to special stowage requirements due to the possibility of affecting aircraft instruments.

Parts of Automobile (Car, Motor, Motorcycle) — may contain wet batteries, etc.

Passengers Baggage | Promotional Material — may contain items meeting any of the criteria for dangerous goods. Examples include fireworks, flammable household liquids, corrosive oven or drain cleaners, flammable gas or liquid lighter refills or camping stove cylinders, matches, ammunition, bleach, aerosols (those not permitted under “Dangerous Goods Carried by Passengers or Crew”), etc.

Pharmaceuticals — may contain items meeting any of the criteria for dangerous goods, particularly radioactive material, flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances.

Photographic supplies/equipment — may contain items meeting any of the criteria for dangerous goods, particularly heat-producing devices, flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances, lithium batteries.

Racing car or motorcycle team equipment — may contain engines, including fuel cell engines, carburettors or fuel tanks which contain fuel or residual fuel, flammable aerosols, cylinders of compressed gases, nitromethane, other fuel additives, wet batteries, lithium batteries, etc.

Refrigerators — may contain liquefied gases or an ammonia solution.

Repair kits — may contain organic peroxides and flammable adhesives, solvent-based paints, resins, etc.

Samples for testing — may contain items meeting any of the criteria for dangerous goods, particularly infectious substances, flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances.

Semen — may be packed with dry ice or refrigerated liquefied gas (see also dry shipper).

Sporting goods/sports team equipment — may contain cylinders of compressed or liquefied gas (air, carbon dioxide, etc.), lithium batteries, propane torches, first aid kits, flammable adhesives, aerosols, etc.

Swimming pool chemicals — may contain oxidising or corrosive substances.

Switches in electrical equipment or instruments — may contain mercury.

Tool boxes — may contain explosives (power rivets), compressed gases or aerosols, flammable gases (Butane cylinders or torches), flammable adhesives or paints, corrosive liquids, lithium batteries, etc.

Torches — micro torches and utility lighters may contain flammable gas and be equipped with an electronic starter. Larger torches may consist of a torch head (often with a self-igniting switch) attached to a container or cylinder of flammable gas.

Unaccompanied passengers' baggage/personal effects — may contain items meeting any of the criteria for dangerous goods such as fireworks, flammable household liquids, corrosive oven or drain cleaners, flammable gas or liquid lighter refills or camping stove cylinders, matches, bleach, aerosols, etc. (those not permitted under Dangerous Goods Carried by Passengers or Crew).

Note: Excess baggage carried as cargo may contain certain dangerous goods (see 1.4.3 (1)).

Vaccines — may be packed in dry ice.

Note: Articles and substances which do not fall within the definitions of dangerous goods as shown in these Regulations and which in the event of leakage may cause a serious clean-up problem or corrosion to aluminium on a long term basis must be checked by the shipper to at

least ensure that the packaging is adequate to prevent leakage during transportation. These may include brine, powdered or liquid dyes, pickled foodstuffs, etc.

9.4.1.1 Identification of Dangerous Goods Through X-Ray Screening

Persons conducting security screening of cargo should be alert to the presence of dangerous goods within packages that are not marked and labelled as dangerous goods and/or not accompanied by a Shipper's Declaration. In particular, items such as aerosols, ammunition, gas cylinders (camping gas, cylinders attached to life-jackets, etc.), cigarette lighters and wet acid batteries can be readily identified from x-ray images. Information provided on an air waybill or marked on a package often indicates that a consignment contains no dangerous goods. In the absence of such annotation by the shipper, should suspicions be raised by the size and shape of the contents of a package, consideration should be given to opening and hand-searching the consignment to verify that no undeclared dangerous goods are present.

Consignments of dangerous goods that have been properly marked, labelled and declared to the operator (where approved for carriage) are commonly processed separately from general freight. Should consignments bearing UN numbers, proper shipping names or hazard labels be discovered within general freight, when separate arrangements exist, this should be queried. It may be that no shipper's declaration accompanies the consignment; as such the consignment of dangerous goods would be considered 'undeclared'.

9.4.1.2 Safety Data Sheets

REACH (**R**egistration, **E**valuation, **A**uthorisation & **r**estriction of **C**hemicals) is a European Union regulation controlling chemicals in Europe. REACH requires for many substances and mixtures, a Safety Data Sheet (SDS) to be provided either before or at the time of first delivery. Section 14 of the EU format SDS provides basic classification information, i.e. UN number, proper shipping name, Class/Division and Packing Group.

9.4.1.3 GHS Consumer Labelling (Overview)

Some everyday household items bear consumer warning labels which may or may not indicate they are classified as dangerous goods in air transport. All over the world there are different laws on how to identify the hazardous properties of chemicals (called ‘classification’) and how information about these hazards is then passed to users (through consumer supply labels and safety data sheets for workers). This can be confusing because the same chemical can have different hazard descriptions in different countries. For example, a chemical could be labelled for supply as ‘toxic’ in one country, but not in another. For this reason, the UN brought together experts from different countries to create the Globally Harmonized System of Classification and Labelling of Chemicals (GHS).

The GHS has been implemented within Europe by the Regulation on Classification, Labelling and Packaging of Substances and Mixtures (known as the CLP Regulation).

GHS Labels

Products bearing the following GHS labels ARE classified as dangerous goods:



Note: A product bearing the GHS corrosive label (depicted far right above) is NOT classified as dangerous goods if the signal word ‘Danger’ and hazard statement ‘causes serious eye damage’ applies.

Products bearing the following GHS labels (and none of the above) are NOT classified as dangerous goods:



9.5 Emergency Situations

(AMC1 SPA.DG.105(b))

9.5.1 Provision of Information for Use in Responding to In-Flight Emergencies

(SPA.DG.110)

For those dangerous goods for which a dangerous goods NOTOC is required, the commander of an aircraft is provided with information which can be used on board to assist in planning the response to an emergency arising in-flight involving the dangerous goods., the 'Emergency Response Guidance for Aircraft Incidents Involving Dangerous Goods' (Doc 9481), which is accessible onboard as part of the Malta Air SEP Manual..

9.5.1.1 Flight Crew Dangerous Goods Incident Procedures

Please refer to Chapter 4 – Emergency Procedures – Malta Air Safety and Emergency Procedures Manual.

9.5.1.2 Cabin Crew Dangerous Goods Incident Procedures

Please refer to Chapter 4 – Emergency Procedures – Malta Air Safety and Emergency Procedures Manual.

9.6 Incidents and Accident Report

Special Notification Requirements in the Event of an Accident or Occurrence When Dangerous Goods are Being Carried or Have Been Offered for Air Transport Without Having Been Prepared and Declared in Accordance with the ICAO Technical Instructions.

9.6.1 Information to be Provided by the Pilot-In-Command in the Event of an In-flight Emergency

(AMC1 SPA.DG.105(b))

If an in-flight emergency occurs and the situation permits, the commander must inform the appropriate Air Traffic Services Unit of any dangerous goods on board. This information should include the proper shipping name, class/division, identified subsidiary hazard(s), compatibility group for explosives, quantity and location on board.

9.6.2 Information to be Provided by the Company in the Event of an Aircraft Accident or Serious Incident Where Dangerous Goods Carried as Company Freight may be Involved

If a company aircraft carrying dangerous goods as company freight is involved in an accident or serious incident where the dangerous goods may be involved, the company provides information, without delay, to emergency services responding to the accident or serious incident about the dangerous goods on board, as shown on the copy of the information to the pilot-in-command (NOTOC). The information must be sufficient to enable any hazards created by the dangerous goods to be minimised and include the proper shipping name, UN number, class/division, any identified subsidiary hazard(s), the compatibility group for explosives, the quantity and the location on board the aircraft. As soon as possible, the company also provides this information to the TM CAD and the appropriate authority of the State in which the accident or serious incident occurred. In the first instance, the TM CAD should be alerted to the incident or accident by phone using the following number:

Director General for Civil Aviation (DGCA): +356 7742 9549

TM CAD Duty Officer: +356 7924 5205

9.6.3 Information to be Provided by the Company in the Event of an Aircraft Incident

(AMC1 SPA.DG.105(b))

In the event of an aircraft incident, the company must if requested to do so, provide information without delay to the emergency services responding to the incident and to the appropriate Authority of the State in which the incident occurred, about the dangerous goods on board, as shown on the copy of the information to the pilot-in-command (NOTOC). For aircraft accidents and serious incidents, see 9.6.2.

9.6.4 Dangerous Goods Accident and Incident Reports

CAT.GEN.MPA.200(e)

Definitions:

Dangerous goods accident: An occurrence associated with and related to the transport of dangerous goods by air which results in fatal or serious injury to a person or major property or environmental damage.

Dangerous goods incident: An occurrence other than a dangerous goods accident associated with and related to the transport of dangerous goods by air, not necessarily occurring on board an aircraft, which results in injury to a person, property or environmental damage, fire, breakage, spillage, leakage of fluid or radiation or other evidence that the integrity of the packaging has not been

maintained. Any occurrence relating to the transport of dangerous goods which seriously jeopardises an aircraft or its occupants is also deemed to be a dangerous goods incident.

Note 1: The Commander has the authority to disembark any person, or any part of the hold baggage or Company Freight, that may represent a potential hazard to the safety of the aircraft or its occupants.

Note 2: A dangerous goods accident or incident may also constitute an aircraft accident or incident as specified in ICAO Annex 13 — Aircraft Accident and Incident Investigation.

Malta Air Ltd. must report dangerous goods accidents and incidents to the appropriate Authorities in Malta in accordance with the reporting requirements.

Malta Air must report to the Maltese Authorities any occasion when:

1. Dangerous goods are discovered to have been carried when not correctly loaded, segregated, separated or secured.
2. Dangerous goods are discovered to have been carried without information having been provided to the pilot-in-command (when required) or the information is inadequate.

Malta Air Ltd. must report any occasion where undeclared or incorrectly declared dangerous goods are discovered in cargo or mail. The report must be made to the appropriate authorities in Malta.

Malta Air must report any occasion where non-permitted dangerous goods are discovered by Malta Air Ltd. (or third party) either in the baggage or on passengers (after check-in) or crew members. Such a report must be made to the Maltese Authorities.

In addition to the requirements of the ICAO Technical Instructions for the reporting of dangerous goods occurrences (above), ORO.GEN.160 requires that any incident which endangers or which, if not corrected, would endanger an aircraft, its occupants or any other person is reported to Competent Authority Safety Data. Dangerous goods occurrences reportable under the Mandatory Occurrence Reporting Scheme include:

- Dangerous goods found not to have been secured to prevent movement
- Damage to packages of dangerous goods
- NOTOC errors where dangerous goods have not been stowed in accordance with loading instructions

- Failure to prepare electric wheelchairs in order to prevent accidental activation
- Electric wheelchairs found not to have been stowed and secured correctly
- Leakage of dangerous goods from passenger baggage

Note: Dangerous goods occurrences meeting the criteria of ORO.GEN.160 also meet the definition of a dangerous goods accident or incident (above), reportable in accordance with CAT.GEN.MPA.200(e). Accordingly, the report must be made to TM CAD within 72 hours (rather than 96), unless exceptional circumstances prevent this.

A dangerous goods accident or dangerous goods incident not meeting the criteria of ORO.GEN.160 must be reported to the Competent Authority within 72 hours, unless exceptional circumstances prevent this. If necessary, a subsequent report shall be made as soon as possible giving all the details that were not known at the time the first report was sent. If a report has been made verbally, written confirmation shall be sent as soon as possible. Any type of accident or incident must be reported irrespective of whether the dangerous goods are in cargo, mail, stores, passengers' baggage or crew baggage.

In compliance with Regulation (EU) No. 376/2014 and 1018/2015 on the reporting, analysis and follow-up of occurrences in civil aviation aircraft, the company stores occurrence reports on a database capable of producing an output that is ECCAIRS compatible. The company submits Mandatory Occurrence Reports to the Competent Authority in this format.

The first and any subsequent report shall be as precise as possible and contain such of the following data that are relevant:

- Date of the incident or accident or the finding of undeclared or misdeclared dangerous goods.
- Location, the flight number and flight date.
- Description of the goods and the reference number of the air waybill, pouch, baggage tag, ticket, etc.
- Proper shipping name (including the technical name, if appropriate) and UN/ID number, when known.
- Class or division and any subsidiary hazard.
- Type of packaging, and the packaging specification marking on it.
- Quantity of dangerous goods.
- Name and address of the shipper, passenger, etc.
- Any other relevant details.

- Suspected cause of the incident or accident.
- Action taken.
- Any other reporting action taken.
- Name, title, address and telephone number of the person making the report.

Copies of relevant documents and any photographs taken should be attached to a report.

Note: IF SAFE TO DO SO, THE DANGEROUS GOODS INVOLVED IN THE ACCIDENT OR INCIDENT SHOULD BE HELD PENDING INVESTIGATION.

9.6.5 Removal of Contamination

(SPA.DG.105)

In the event of a spillage or leakage of dangerous goods within an aircraft, the position where the dangerous goods or ULD was stowed on the aircraft must be inspected for damage or contamination and any hazardous contamination removed. The hazard of the dangerous goods within packages concerned may be established by checking the entry on the NOTOC for that loading position or from hazard labels applied to the packages. Persons responding in the event of damage to or leakage of dangerous goods from packages must:

- Identify the hazards and wear appropriate protective clothing;
- Avoid handling the package or keep handling to a minimum;
- Inspect adjacent packages for contamination and put aside any that may have been contaminated;
- Arrange for decontamination of the aircraft and equipment; and
- In the case of infectious material, inform the appropriate public health authority or veterinary authority, and provide information to any other countries of transit where persons may have been exposed to danger; and notify the shipper and/or the consignee.

If it is evident that a package containing radioactive material is damaged or leaking, or if it is suspected that the package may have leaked or been damaged, access to the package must be restricted and a qualified person must, as soon as possible, assess the extent of contamination and the resultant radiation level of the package. The scope of the assessment must include the package, the aircraft, the adjacent loading and unloading areas and, if necessary, all other material which has been carried in the aircraft. When necessary, additional steps for the protection of persons, property and the environment must be taken in accordance with provisions established by the relevant competent authority, to overcome and minimise the consequences of such leakage or damage. An aeroplane which has

been contaminated by radioactive materials must be immediately taken out of service and not returned until the radiation level at any accessible surface and the non-fixed contamination are not more than the values specified in the Technical Instructions. In the event of non-compliance with any limit in the Technical Instructions applicable to radiation level or contamination, the operator must ensure the shipper is informed if the non-compliance is identified during transport; take immediate steps to mitigate the consequences of the non-compliance; and communicate the non-compliance to the shipper and relevant competent Authority(ies), respectively, as soon as practicable and immediately whenever an emergency situation has developed or is developing.

9.7 Carriage of Weapons

9.7.1 Conditions Under Which Weapons, Munitions of War and Sporting Weapons May Be Carried

CAT.GEN.MPA.155, CAT.GEN.MPA.160

General

The carriage of firearms (including sporting weapons) and/or ammunition is not ordinarily permitted on Malta Air Aircraft.

9.7.2 Armed Escorts/Diplomatic Protection Officers

In accordance with the provisions of the Transport Malta Civil Aviation Directorate, Air Navigation Order, Article 42, and Operations Advisory Notice (OAN) 07/18, Malta Air may apply to the Director General Transport Malta Civil Aviation Directorate for a Personal Protection Officer may apply for an exemption from the prohibition on carrying weapons or munitions on Maltese Registered Aircraft.

The following conditions shall apply:

- Applications must be made on a flight by flight basis.
- Applications must be submitted to the Director General Transport Malta Civil Aviation Directorate at least 48 hours before the intended time of departure of the flight concerned.
- The person carrying the items must secure approval from the State of origin as well as any State that the aircraft overflies or lands in.
- The packing, labelling and marking of the items must be in accordance with the ICAO Technical Instructions.
- The person carrying the items must advise the airport operators involved.
- Firearms and/or ammunition must be carried in the hold of the aircraft and must not be accessible to passengers during the flight. Ammunition must be stored in a separate compartment.

The flight crew will be advised in advance of the scheduled flight that a personal protection officer/s will be travelling on board.

9.8 Portable Electronic Devices

CAT.GEN.MPA.140, AMC1 CAT.GEN.MPA.140

No person shall use any PED that may adversely affect the performance of the aircraft systems and equipment on board a Malta Air aircraft. Approved PEDs may be used throughout the flight, subject to the conditions in this section.

9.8.1 PEDs Allowed for all Phases of Flight

PEDs such as tablets, e-readers and mobile phones weighing one kilo or less may be used during all phases of the flight provided Airplane or Flight mode is selected after the main cabin door is closed. Airplane or Flight mode allows PEDs functions such as MP3 players, games and built in cameras, etc. to be used with the cellular signal transmission disabled. Bluetooth accessories including wireless keyboard, headphones may be used during the flight but must remain switched off for taxi, take-off approach and landing.

9.8.2 Prohibited PEDs

The following PEDs are forbidden on Malta Air aircraft:

- E-cigarettes.
- Remote control transmitters.
- VHF scanner transmitters and receivers and any device with cellular network service enabled.
- FM receivers.

9.8.3 Medical PEDs

Certain approved medical PEDs such as Portable Oxygen Concentrators maybe allowed to be used during flight.

9.8.4 Transmitting PED (T-PED)

Transmitting PEDs may be used only when the main aircraft door is open, except during extended delays or when the commander assesses that the use of T-PED's is required for safety and security purpose (which is a reportable event).

9.8.5 Low Vis Ops PEDs OFF

Attention is drawn to the fact that all PEDs must be switched-off for all low visibility operations. Flight Crew must brief Cabin Crew on LVO operations to ensure that the respective safety PA is made.

9.9 Portable Electronic Devices Affecting Performance of the Aircraft's Systems and Equipment

AMC2 CAT.GEN.MPA.140

The provisions in this section provide means to prevent that portable electronic devices (PEDs) on board aircraft adversely affect the performance of the aircraft's systems and equipment. This section addresses operation of PEDs in the passenger compartment and flight compartment. Furthermore, it addresses the specific case of PEDs qualified and under configuration control by Malta Air called controlled PEDs (C-PEDs).

9.9.1 General Principle

As the general principle all PEDs (including transmitting PEDs (T-PEDs)) are switched-off at the start of the flight when the passengers have boarded and all doors have been closed, until a passenger door has been opened at the end of the flight.

9.9.2 PEDs and Phases of Flight

SEP 2.24 indicates Malta Air policy on use of PEDs during all phases of flight.

9.9.3 PEDs Affecting Aircraft Systems

1. If at any time the Commander believes that the use of PEDs may be affecting an aircraft system, a cabin announcement shall be made instructing passengers to switch-off all portable electronic devices.
2. Passenger use of equipment during the flight is monitored;
3. Suspect equipment is switched off; and
4. Particular attention is given to passenger misuse of equipment that could include a built-in transmitting function.
5. The commander may for any reason and during any phase of flight require deactivation and stowage of PEDs.

9.9.4 Thermal Runaways of Batteries

Procedures for Thermal runaway of batteries, in particular lithium batteries, and potential resulting fire have been promulgated to aircrew.

9.9.5 Occurrences of Suspected or Confirmed Interference

Occurrences of suspected or confirmed interference that have potential safety implications shall be reported to the TM CAD by Malta Air from the ASR filed by the commander. Where possible, to assist follow-up and technical investigation, the ASR shall describe the offending device, identify the brand name and model number, its location in the aircraft at the time of the occurrence, interference symptoms and the results of actions taken by the crew. The cooperation of the device owner will be sought by obtaining contact details.

9.9.6 PEDs Not Accessible During the Flight

PEDs shall be switched off, when not accessible for deactivation during flight.

9.9.7 Portable Electronic Devices**DEFINITIONS****Categories of PEDs**

PEDs include the following two categories:

Non-intentional transmitters can non-intentionally radiate RF transmissions, sometimes referred to as spurious emissions. This category includes, but is not limited to, calculators, cameras, radio receivers, audio and video players, electronic games and toys; when these devices are not equipped with a transmitting function.

Intentional transmitters radiate RF transmissions on specific frequencies as part of their intended function. In addition, they may radiate non-intentional transmissions like any PED. The term ‘transmitting PED’ (T-PED) is used to identify the transmitting capability of the PED. Intentional transmitters are transmitting devices such as RF based remote control equipment, which may include some toys, two-way radios (sometimes referred to as “private mobile radio”), mobile phones of any type, satellite phones, computers with mobile phone data connection, wireless local area network (WLAN) or Bluetooth capability. After deactivation of the transmitting capability, e.g. by activating the so-called ‘flight mode’ or ‘flight safety mode’, the T-PED remains a PED having non-intentional emissions.

Definition of the Switched-off Status

Many PEDs are not completely disconnected from the internal power source when switched off. The switching function may leave some remaining functionality, e.g. data storage, timer, clock, etc. These devices can be considered switched off when in the deactivated status. The same applies to devices having no transmitting capability and are operated by coin cells without further deactivation capability, e.g. wrist watches.

Electromagnetic Interference (EMI)

The two classes of EMI to be addressed can be described as follows:

1. Front door coupling is the possible disturbance to an aircraft system as received by the antenna of the system and mainly in the frequency band used by the system. Any PED internal oscillation has the potential to radiate low level signals in the aviation frequency bands. Through this disturbance especially the instrument landing system (ILS) and the VHF omni range (VOR) navigation system may indicate erroneous information.

2. Back door coupling is the possible disturbance of aircraft systems by electromagnetic fields generated by transmitters at a level which could exceed on short distance (i.e. within the aircraft) the electromagnetic field level used for the aircraft system certification. This disturbance may then lead to system malfunction.

9.10 Annexes: List of Dangerous Goods with Drill Reference Numbers and Chart of Drills

Annex 9.A – Alphabetical List of Dangerous Goods with Drill Codes (ref. Table 3) followed by a numerical list by UN Number of Dangerous Goods with Drill Codes (ref. Table 4).

Annex 9.B – Aircraft Emergency Response Drills (Reproduced from ICAO Doc 9481 Edition 2025–2026) (ref. Table 9.B).

The following tables are reproduced from ICAO Emergency Response Guidance for Aircraft Incidents Involving Dangerous Goods (DOC 9481 AN/928) 2025–2026 Edition.

Refer also to OM(A) SEP Chapter 9 for additional information. “Subsidiary Risk” and “Subsidiary Hazard” have identical meaning.

The drill code assigned to the articles below is based on their primary hazard. The drill letter may need to be altered if there are also subsidiary hazard(s) to consider. Subsidiary hazards, when applicable, are assigned as Note 1:

UN 3537 – Articles containing flammable gas, n.o.s.*

UN 3538 – Articles containing non-flammable, non toxic gas, n.o.s.*

UN 3539 – Articles containing toxic gas, n.o.s.*

UN 3540 – Articles containing flammable liquid, n.o.s.*

UN 3541 – Articles containing flammable solid, n.o.s.*

UN 3542 – Articles containing a substance liable to spontaneous combustion, n.o.s.*

UN 3543 – Articles containing a substance which emits flammable gas in contact with water, n.o.s.*

UN 3544 – Articles containing oxidizing substance, n.o.s.*

UN 3545 – Articles containing organic peroxide, n.o.s.*

UN 3546 – Articles containing toxic substance, n.o.s.*

UN 3547 – Articles containing corrosive substance, n.o.s.*

UN 3548 – Articles containing miscellaneous dangerous goods, n.o.s.*

- Note 1:** (ICAO DOC 9284 – Part 2): Subsidiary Hazards must be representative of the primary hazard posed by the other dangerous goods contained within the article. When only one item of dangerous goods is present in the article, the subsidiary hazard(s), if any, is the subsidiary hazard(s) identified in column 4 (“Subsidiary Risk”) of the Dangerous Goods list. If the article contains more than one item of dangerous goods and these could react dangerously with one another during the transport, each of the dangerous goods must be enclosed separately (refer to Note 2).
- Note 2:** (ICAO DOC 9284 – Part 4; 1.1.8): Dangerous goods must not be packed together in the same outer packaging with dangerous or other goods if they react dangerously with each other and cause:
1. Combustion and/or evolution of considerable heat;
 2. Evolution of flammable, toxic or asphyxiant gases;
 3. The formation of corrosive substances; or
 4. The formation of unstable substances.

Annex 9.A: Alphabetical List of Dangerous Goods with Drill Codes

The list of proper shipping names used for dangerous goods, arranged in alphabetical order, is shown below.

For each entry, a drill code is shown and this should be used to find the appropriate drill on the chart of Aircraft Emergency Response Drills. The United Nations identification number is also listed to permit confirmation that the correct entry has been selected.

In a few cases, besides the proper shipping name, the UN number, packing group and/or the subsidiary hazard must also be used to identify the correct drill code. Where this is necessary, the possible UN numbers, packing groups and/or subsidiary hazards are given below the proper shipping name and the relevant drill code can be selected. These additional details are shown on the pilot-in-command’s notification form.

In determining the alphabetical order of the proper shipping names, numbers and the terms n.o.s., alpha-, beta-, meta-, omega-, sec-, tert-, a-, b-, m-, N-, n-, O-, o- and p- have been ignored. Where names comprise more than one word, they have been alphabetized as if they were a single word.

The following tables are provided to crews with the required information to enable them to carry out their responsibilities in respect of emergencies involving Dangerous Goods as required by ICAO Annex 18.

They contain:

- Alphabetical listing of dangerous goods with Drill Code for each entry.
- List of Dangerous Goods by UN Number with Drill Code for each entry.
- The drill table.

Refer to OM(A) SEP Section 9.2 for guidance on their use.

The following tables are reproduced from ICAO Emergency Response Guidance for Aircraft Incidents Involving Dangerous Goods (DOC 9481 AN/928) 2025–2026 Edition.

“Subsidiary Risk” and “Subsidiary Hazard” have identical meaning.

Table 3 Alphabetical List of Dangerous Goods with Drill Codes

			<i>Drill</i>		<i>Proper Shipping Name</i>
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	
1088	3H	Acetal	3518	2PX	Adsorbed gas, toxic, oxidizing, corrosive, n.o.s.*
1089	3H	Acetaldehyde	1950		Aerosols
1841	9L	Acetaldehyde ammonia	10L		Division 2.1 without subsidiary risk
2332	3L	Acetaldehyde oxime	10C		Division 2.1 with subsidiary risk 8
2789	8F	Acetic acid, glacial	10CP		Division 2.1 with subsidiary risks 6.1 and 8
2789	8F	Acetic acid solution	2L		Division 2.2 without subsidiary risk
2790	8L	Acetic acid solution	2X		Division 2.2 with subsidiary risk 5.1
1715	8F	Acetic anhydride	2C		Division 2.2 with subsidiary risk 8
1090	3H	Acetone	2CP		Division 2.2 with subsidiary risks 6.1 and 8
1541	6L	Acetone cyanohydrin, stabilized	2P		Division 2.3 without subsidiary risk or Division 2.2 with subsidiary risk 6.1 (including tear gas devices)
1091	3L	Acetone oils	10P		Division 2.3 with subsidiary risk 2.1 or Division 2.1 with subsidiary risk 6.1 (including tear gas devices)
1648	3L	Acetonitrile	0331	1L	Agent, blasting, type B
1716	8L	Acetyl bromide	0332	1L	Agent, blasting, type E
1717	3C	Acetyl chloride	1002	2L	Air, compressed
1001	10L	Acetylene, dissolved	3165	3CP	Aircraft hydraulic power unit fuel tank
3374	10L	Acetylene, solvent free	1003	2X	Air, refrigerated liquid
1898	8L	Acetyl iodide	3274	3C	Alcoholates solution, n.o.s.*
2621	3L	Acetyl methyl carbinol	3065	3L	Alcoholic beverages
2713	6L	Acridine	1987	3L	Alcohols, n.o.s.*
2607	3L	Acrolein dimer, stabilized	1986		Alcohols, flammable, toxic, n.o.s.*
1092	6H	Acrolein, stabilized	3HP		Packing Group I or II
2074	6L	Acrylamide, solid	3P		Packing Group III
3426	6L	Acrylamide solution	1989		Aldehydes, n.o.s.*
2218	8F	Acrylic acid, stabilized	3H		Packing Group I or II
1093	3P	Acrylonitrile, stabilized	3L		Packing Group III
1133	3L	Adhesives	1988		Aldehydes, flammable, toxic, n.o.s.*
2205	6L	Adiponitrile	3HP		Packing Group I or II
3511	2L	Adsorbed gas, n.o.s.*	3P		Packing Group III
3510	10L	Adsorbed gas, flammable, n.o.s.*	2839	6L	Aldol
3513	2X	Adsorbed gas, oxidizing, n.o.s.*	3206	4C	Alkali metal alcoholates, self- heating, corrosive, n.o.s.*
3512	2P	Adsorbed gas, toxic, n.o.s.*	1421	4W	Alkali metal alloy, liquid, n.o.s.
3516	2CP	Adsorbed gas, toxic, corrosive, n.o.s.*	1389	4W	Alkali metal amalgam, liquid
3514	10P	Adsorbed gas, toxic, flammable, n.o.s.*			
3517	10C	Adsorbed gas, toxic, flammable, corrosive, n.o.s.*			
3515	2PX	Adsorbed gas, toxic, oxidizing, n.o.s.*			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3401	4W	Alkali metal amalgam, solid	2580	8L	Aluminium bromide solution
1390	4W	Alkali metal amides	1394	4W	Aluminium carbide
1391	4W	Alkali metal dispersion	1726	8L	Aluminium chloride, anhydrous
3482	4W	Alkali metal dispersion, flammable	2581	8L	Aluminium chloride solution
3205	4L	Alkaline earth metal alcoholates, n.o.s.*	1395	4PW	Aluminium ferrosilicon powder
1393	4W	Alkaline earth metal alloy, n.o.s.	2463	4W	Aluminium hydride
1392	4W	Alkaline earth metal amalgam, liquid	1438	5L	Aluminium nitrate
3402	4W	Alkaline earth metal amalgam, solid	1397	4PW	Aluminium phosphide
1391	4W	Alkaline earth metal dispersion	3048	6W	Aluminium phosphide pesticide
3482	4W	Alkaline earth metal dispersion, flammable	1309	3L	Aluminium powder, coated
3140	6L	Alkaloid salts, liquid, n.o.s.*	1396	4W	Aluminium powder, uncoated
1544	6L	Alkaloid salts, solid, n.o.s.*	3170	4W	Aluminium remelting by-products
3140	6L	Alkaloids, liquid, n.o.s.*	2715	3L	Aluminium resinate
1544	6L	Alkaloids, solid, n.o.s.*	1398	4W	Aluminium silicon powder, uncoated
3145	8L	Alkylphenols, liquid, n.o.s.	3170	4W	Aluminium smelting by-products
2430	8L	Alkylphenols, solid, n.o.s.	2733	3C	Amines, flammable, corrosive, n.o.s.*
2584	8L	Alkylsulphonic acids, liquid	2735	8L	Amines, liquid, corrosive, n.o.s.*
2586	8L	Alkylsulphonic acids, liquid	2734	8F	Amines, liquid, corrosive, flammable, n.o.s.*
2583	8L	Alkylsulphonic acids, solid	3259	8L	Amines, solid, corrosive, n.o.s.*
2585	8L	Alkylsulphonic acids, solid	2673	6L	2-Amino-4-chlorophenol
2571	8L	Alkylsulphuric acids	2946	6L	2-Amino-5-diethylaminopentane
2333	3P	Allyl acetate	3317	3E	2-Amino-4,6-dinitrophenol, wetted
1098	6F	Allyl alcohol	3055	8L	2-(2-Aminoethoxy)ethanol
2334	6H	Allylamine	2815	8P	N-Aminoethylpiperazine
1099	3P	Allyl bromide	2512	6L	Aminophenols
1100	3P	Allyl chloride	2671	6L	Aminopyridines
1722	6CF	Allyl chloroformate	1005	2CP	Ammonia, anhydrous
2335	3P	Allyl ethyl ether	2073	2L	Ammonia solution
2336	3P	Allyl formate	2672	8L	Ammonia solution
2219	3L	Allyl glycidyl ether	3318	2CP	Ammonia solution
1723	3C	Allyl iodide	1546	6L	Ammonium arsenate
1545	6F	Allyl isothiocyanate, stabilized	1439	5L	Ammonium dichromate
1724	8F	Allyltrichlorosilane, stabilized	1843	6L	Ammonium dinitro-o-cresolate, solid
2870	4W	Aluminium borohydride	3424	6L	Ammonium dinitro-o-cresolate solution
2870	4W	Aluminium borohydride in devices	2505	6L	Ammonium fluoride
1725	8L	Aluminium bromide, anhydrous	2854	6L	Ammonium fluorosilicate

<i>Drill</i>			<i>Drill</i>		
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>
1727	8L	Ammonium hydrogendifluoride, solid	0245	1L	Ammunition, smoke, white phosphorus
2817	8P	Ammonium hydrogendifluoride solution	0246	1L	Ammunition, smoke, white phosphorus
2506	8L	Ammonium hydrogen sulphate	0018	1CP	Ammunition, tear-producing
2859	6L	Ammonium metavanadate	0019	1CP	Ammunition, tear-producing
0222	1L	Ammonium nitrate	0301	1CP	Ammunition, tear-producing
1942	5L	Ammonium nitrate	2017	6C	Ammunition, tear-producing, non-explosive
2067	5L	Ammonium nitrate based fertilizer	0020	1P	Ammunition, toxic*
2071	9L	Ammonium nitrate based fertilizer	0021	1P	Ammunition, toxic*
3375	5L	Ammonium nitrate emulsion	2016	6L	Ammunition, toxic, non-explosive
3375	5L	Ammonium nitrate gel	1104	3L	Amyl acetates
2426	5L	Ammonium nitrate, liquid	2819	8L	Amyl acid phosphate
3375	5L	Ammonium nitrate suspension	1106	3C	Amylamine
0402	1L	Ammonium perchlorate	2620	3L	Amyl butyrates
1442	5L	Ammonium perchlorate	1107	3L	Amyl chloride
1444	5L	Ammonium persulphate	1108	3H	n-Amylene
0004	1L	Ammonium picrate	1109	3L	Amyl formates
1310	3E	Ammonium picrate, wetted	1111	3L	Amyl mercaptan
2818	8P	Ammonium polysulphide solution	1110	3L	n-Amyl methyl ketone
2861	6L	Ammonium polyvanadate	1112	3L	Amyl nitrate
2683	8FP	Ammonium sulphide solution	1113	3H	Amyl nitrite
0171	1L	Ammunition, illuminating	1728	8L	Amyltrichlorosilane
0254	1L	Ammunition, illuminating	1547	6L	Aniline
0297	1L	Ammunition, illuminating	1548	6L	Aniline hydrochloride
0009	1L	Ammunition, incendiary	2431	6L	Anisidines
0010	1L	Ammunition, incendiary	2222	3L	Anisole
0247	1L	Ammunition, incendiary	1729	8L	Anisoyl chloride
0300	1L	Ammunition, incendiary	3141	6L	Antimony compound, inorganic, liquid, n.o.s.*
0243	1L	Ammunition, incendiary, white phosphorus	1549	6L	Antimony compound, inorganic, solid, n.o.s.*
0244	1L	Ammunition, incendiary, white phosphorus	1550	6L	Antimony lactate
0362	1L	Ammunition, practice	1730	8L	Antimony pentachloride, liquid
0488	1L	Ammunition, practice	1731	8L	Antimony pentachloride solution
0363	1L	Ammunition, proof	1732	8P	Antimony pentafluoride
0015	1L	Ammunition, smoke	1551	6L	Antimony potassium tartrate
0016	1L	Ammunition, smoke	2871	6L	Antimony powder
0303	1L	Ammunition, smoke	1733	8L	Antimony trichloride

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1006	2L	Argon, compressed	3546	6L	Articles containing toxic substance, n.o.s.*
1951	2L	Argon, refrigerated liquid	0486	1L	Articles, EEI
1558	6L	Arsenic	0349	3L	Articles, explosive, n.o.s.*
1553	6L	Arsenic acid, liquid	0350	1L	Articles, explosive, n.o.s.*
1554	6L	Arsenic acid, solid	0351	1L	Articles, explosive, n.o.s.*
1562	6L	Arsenical dust	0352	1L	Articles, explosive, n.o.s.*
2760	3P	Arsenical pesticide, liquid, flammable, toxic*	0353	1L	Articles, explosive, n.o.s.*
2994	6L	Arsenical pesticide, liquid, toxic*	0354	1L	Articles, explosive, n.o.s.*
2993	6F	Arsenical pesticide, liquid, toxic, flammable*	0355	1L	Articles, explosive, n.o.s.*
2759	6L	Arsenical pesticide, solid, toxic*	0356	1L	Articles, explosive, n.o.s.*
1555	6L	Arsenic bromide	0462	1L	Articles, explosive, n.o.s.*
1556	6L	Arsenic compound, liquid, n.o.s.*	0463	1L	Articles, explosive, n.o.s.*
1557	6L	Arsenic compound, solid, n.o.s.*	0464	1L	Articles, explosive, n.o.s.*
1559	6L	Arsenic pentoxide	0465	1L	Articles, explosive, n.o.s.*
1560	6L	Arsenic trichloride	0466	1L	Articles, explosive, n.o.s.*
1561	6L	Arsenic trioxide	0467	1L	Articles, explosive, n.o.s.*
2188	10P	Arsine	0468	1L	Articles, explosive, n.o.s.*
3522	10P	Arsine, adsorbed	0469	1L	Articles, explosive, n.o.s.*
3542	4L	Articles containing a substance liable to spontaneous combustion, n.o.s.*	0470	1L	Articles, explosive, n.o.s.*
3543	4W	Articles containing a substance which emits flammable gas in contact with water, n.o.s.*	0471	1L	Articles, explosive, n.o.s.*
3547	8L	Articles containing corrosive substance, n.o.s.*	0472	1L	Articles, explosive, n.o.s.*
3537	10L	Articles containing flammable gas, n.o.s.*	0486	1L	Articles, explosive, extremely insensitive
3540	3L	Articles containing flammable liquid, n.o.s.*	3164	2L	Articles, pressurized, hydraulic
3541	3L	Articles containing flammable solid, n.o.s.*	3164	2L	Articles, pressurized, pneumatic
3548	9L	Articles containing miscellaneous dangerous goods, n.o.s.*	0380	1S	Articles, pyrophoric
3538	2L	Articles containing non-flammable, non toxic gas, n.o.s.*	0428	1L	Articles, pyrotechnic
3545	5L	Articles containing organic peroxide, n.o.s.*	0429	1L	Articles, pyrotechnic
3544	5L	Articles containing oxidizing substance, n.o.s.*	0430	1L	Articles, pyrotechnic
3539	2P	Articles containing toxic gas, n.o.s.*	0431	1L	Articles, pyrotechnic
			0432	3L	Articles, pyrotechnic
			2584	8L	Arylsulphonic acids, liquid
			2586	8L	Arylsulphonic acids, liquid
			2583	8L	Arylsulphonic acids, solid
			2585	8L	Arylsulphonic acids, solid
			2212	9L	Asbestos, amphibole*
			2590	9L	Asbestos, chrysotile

<i>Drill</i>			<i>Drill</i>		
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>
3334	9A	Aviation regulated liquid, n.o.s.*	1736	8W	Benzoyl chloride
3335	9A	Aviation regulated solid, n.o.s.*	1737	6C	Benzyl bromide
3242	3L	Azodicarbonamide	1738	6C	Benzyl chloride
1400	4W	Barium	1739	8L	Benzyl chloroformate
1854	4W	Barium alloys, pyrophoric	2619	8F	Benzyl dimethylamine
0224	1P	Barium azide	1886	6L	Benzylidene chloride
1571	3EP	Barium azide, wetted	2653	6L	Benzyl iodide
2719	5P	Barium bromate	1566	6L	Beryllium compound, n.o.s.*
1445	5P	Barium chlorate, solid	2464	5P	Beryllium nitrate
3405	5P	Barium chlorate solution	1567	6F	Beryllium powder
1564	6L	Barium compound, n.o.s.*	1327	3L	Bhusa
1565	6L	Barium cyanide	2251	3L	Bicyclo [2.2.1] hepta-2-5-diene, stabilized
2741	5P	Barium hypochlorite	3373	11L	Biological substance, Category B
1446	5P	Barium nitrate	3291	11L	Biomedical waste, n.o.s.
1884	6L	Barium oxide	2782	3P	Bipyridilium pesticide, liquid, flammable, toxic*
1447	5P	Barium perchlorate, solid	3016	6L	Bipyridilium pesticide, liquid, toxic*
3406	5P	Barium perchlorate solution	3015	6F	Bipyridilium pesticide, liquid, toxic, flammable*
1448	5P	Barium permanganate	2781	6L	Bipyridilium pesticide, solid, toxic*
1449	5P	Barium peroxide	2837	8L	Bisulphates, aqueous solution
3292	4W	Batteries, containing sodium	2693	8L	Bisulphites, aqueous solution, n.o.s.*
3028	8L	Batteries, dry, containing potassium hydroxide solid	0027	1L	Black powder
3496	9L	Batteries, nickel-metal hydride	0028	1L	Black powder, compressed
2794	8L	Batteries, wet, filled with acid	0028	1L	Black powder in pellets
2795	8L	Batteries, wet, filled with alkali	0033	1L	Bombs
2800	8L	Batteries, wet, non-spillable	0034	1L	Bombs
2796	8L	Battery fluid, acid	0035	1L	Bombs
2797	8L	Battery fluid, alkali	0291	1L	Bombs
3171	9L	Battery-powered equipment	0037	1L	Bombs, photo-flash
3171	9L	Battery-powered vehicle	0038	1L	Bombs, photo-flash
1990	9N	Benzaldehyde	0039	1L	Bombs, photo-flash
1114	3H	Benzene	0299	1L	Bombs, photo-flash
2225	8L	Benzenesulphonyl chloride	2028	8L	Bombs, smoke, non-explosive
1885	6L	Benzidine	0399	1L	Bombs with flammable liquid
2224	6L	Benzonitrile	0400	1L	Bombs with flammable liquid
2587	6L	Benzoquinone	0042	1L	Boosters
2226	8L	Benzotrichloride			
2338	3L	Benzotrifluoride			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0283	1L	Boosters	2515	6L	Bromoform
0225	1L	Boosters with detonator	2341	3L	1-Bromo-3-methylbutane
0268	1L	Boosters with detonator	2342	3L	Bromomethylpropanes
1312	3L	Borneol	3241	3L	2-Bromo-2-nitropropane-1,3-diol
2692	8L	Boron tribromide	2343	3L	2-Bromopentane
1741	2CP	Boron trichloride	2344	3L	Bromopropanes
1008	2CP	Boron trifluoride	2345	3L	3-Bromopropyne
1742	8L	Boron trifluoride acetic acid complex, liquid	2419	10L	Bromotrifluoroethylene
3419	8L	Boron trifluoride acetic acid complex, solid	1009	2L	Bromotrifluoromethane
3519	2CP	Boron trifluoride, adsorbed	1570	6L	Brucine
2604	8F	Boron trifluoride diethyl etherate	0043	1L	Bursters
2851	8L	Boron trifluoride dihydrate	1010	10L	Butadienes and hydrocarbon mixture, stabilized
2965	4FW	Boron trifluoride dimethyl etherate	1010	10L	Butadienes, stabilized
1743	8L	Boron trifluoride propionic acid complex, liquid	1011	10L	Butane
3420	8L	Boron trifluoride propionic acid complex, solid	2346	3L	Butanedione
1450	5L	Bromates, inorganic, n.o.s.*	1120	3L	Butanols
3213	5L	Bromates, inorganic, aqueous solution, n.o.s.*	1123	3L	Butyl acetates
1744	8P	Bromine	1718	8L	Butyl acid phosphate
2901	2PX	Bromine chloride	2348	3L	Butyl acrylates, stabilized
1745	5CP	Bromine pentafluoride	1125	3C	n-Butylamine
1744	8P	Bromine solution	2738	6L	N-Butylaniline
1746	5CP	Bromine trifluoride	2709	3L	Butylbenzenes
3425	8L	Bromoacetic acid, solid	2743	6CF	n-Butyl chloroformate
1938	8L	Bromoacetic acid solution	2747	6L	tert-Butylcyclohexyl chloroformate
1569	6F	Bromoacetone	1012	10L	Butylene
2513	8L	Bromoacetyl bromide	3022	3H	1,2-Butylene oxide, stabilized
2514	3L	Bromobenzene	1128	3L	n-Butyl formate
1694	6i	Bromobenzyl cyanides, liquid	3255	4C	tert-Butyl hypochlorite
3449	6L	Bromobenzyl cyanides, solid	2690	6L	N,n-Butylimidazole
1126	3L	1-Bromobutane	2485	6F	n-Butyl isocyanate
2339	3L	2-Bromobutane	2484	6F	tert-Butyl isocyanate
1887	6L	Bromochloromethane	2347	3L	Butyl mercaptan
2688	6L	1-Bromo-3-chloropropane	2227	3L	n-Butyl methacrylate, stabilized
2340	3L	2-Bromoethyl ethyl ether	2350	3L	Butyl methyl ether
			2351	3L	Butyl nitrites
			1914	3L	Butyl propionates
			2667	6L	Butyltoluenes

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1747	8F	Butyltrichlorosilane	1748	5L	Calcium hypochlorite mixture, dry
2956	3E	5-tert-Butyl-2,4,6-trinitro-m-xylene	2208	5L	Calcium hypochlorite mixture, dry
2352	3L	Butyl vinyl ether, stabilized	3485	5C	Calcium hypochlorite mixture, dry, corrosive
2716	6L	1,4-Butynediol	3486	5C	Calcium hypochlorite mixture, dry, corrosive
1129	3L	Butyraldehyde	2844	4W	Calcium manganese silicon
2840	3L	Butyraldoxime	1454	5L	Calcium nitrate
2820	8L	Butyric acid	1910	8L	Calcium oxide
2739	8L	Butyric anhydride	1455	5L	Calcium perchlorate
2411	3P	Butyronitrile	1456	5L	Calcium permanganate
2353	3C	Butyryl chloride	1457	5L	Calcium peroxide
1572	6L	Cacodylic acid	1360	4PW	Calcium phosphide
2570	6L	Cadmium compound*	1855	4W	Calcium, pyrophoric
1407	4W	Caesium	1313	3L	Calcium resinate
2682	8L	Caesium hydroxide	1314	3L	Calcium resinate, fused
2681	8L	Caesium hydroxide solution	1405	4W	Calcium silicide
1451	5L	Caesium nitrate	2717	3L	Camphor
1401	4W	Calcium	1130	3L	Camphor oil
1855	4W	Calcium alloys, pyrophoric	3508	9L	Capacitor, asymmetric
1573	6L	Calcium arsenate	3499	9L	Capacitor, electric double layer
1574	6L	Calcium arsenate and calcium arsenite mixture, solid	2829	8L	Caproic acid
1402	4W	Calcium carbide	2758	3P	Carbamate pesticide, liquid, flammable, toxic*
1452	5L	Calcium chlorate	2992	6L	Carbamate pesticide, liquid, toxic*
2429	5L	Calcium chlorate, aqueous solution	2991	6F	Carbamate pesticide, liquid, toxic, flammable*
1453	5L	Calcium chlorite	2757	6L	Carbamate pesticide, solid, toxic*
1403	4W	Calcium cyanamide	1361	4L	Carbon
1575	6L	Calcium cyanide	1362	4L	Carbon, activated
1923	4L	Calcium dithionite	1013	2L	Carbon dioxide
1404	4W	Calcium hydride	2187	2L	Carbon dioxide, refrigerated liquid
1923	4L	Calcium hydrosulphite	1845	9L	Carbon dioxide, solid
1748	5L	Calcium hypochlorite, dry	1131	3HP	Carbon disulphide
3485	5C	Calcium hypochlorite, dry, corrosive	1016	10P	Carbon monoxide, compressed
2880	5L	Calcium hypochlorite, hydrated	2516	6L	Carbon tetrabromide
3487	5C	Calcium hypochlorite, hydrated, corrosive	1846	6L	Carbon tetrachloride
2880	5L	Calcium hypochlorite, hydrated mixture	2417	2CP	Carbonyl fluoride
3487	5C	Calcium hypochlorite, hydrated mixture, corrosive	2204	10P	Carbonyl sulphide

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0049	1L	Cartridges, flash	0447	1L	Cases, combustible, empty, without primer
0050	1L	Cartridges, flash	2969	9L	Castor beans
0014	3L	Cartridges for tools, blank	2969	9L	Castor flake
0005	1L	Cartridges for weapons	2969	9L	Castor meal
0006	1L	Cartridges for weapons	2969	9L	Castor pomace
0007	1L	Cartridges for weapons	1719	8L	Caustic alkali liquid, n.o.s.*
0321	1L	Cartridges for weapons	3292	4W	Cells, containing sodium
0348	1L	Cartridges for weapons	2000	3L	Celluloid
0412	1L	Cartridges for weapons	2002	4L	Celluloid, scrap
0014	3L	Cartridges for weapons, blank	1333	3L	Cerium
0326	1L	Cartridges for weapons, blank	3078	4W	Cerium
0327	1L	Cartridges for weapons, blank	0457	1L	Charges, bursting, plastics bonded
0338	1L	Cartridges for weapons, blank	0458	1L	Charges, bursting, plastics bonded
0413	1L	Cartridges for weapons, blank	0459	1L	Charges, bursting, plastics bonded
0012	3L	Cartridges for weapons, inert projectile	0460	3L	Charges, bursting, plastics bonded
0328	1L	Cartridges for weapons, inert projectile	0048	1L	Charges, demolition
0339	1L	Cartridges for weapons, inert projectile	0056	1L	Charges, depth
0417	1L	Cartridges for weapons, inert projectile	0442	1L	Charges, explosive, commercial
0277	1L	Cartridges, oil well	0443	1L	Charges, explosive, commercial
0278	1L	Cartridges, oil well	0444	1L	Charges, explosive, commercial
0275	1L	Cartridges, power device	0445	3L	Charges, explosive, commercial
0276	1L	Cartridges, power device	0271	1L	Charges, propelling
0323	3L	Cartridges, power device	0272	1L	Charges, propelling
0381	1L	Cartridges, power device	0415	1L	Charges, propelling
0054	1L	Cartridges, signal	0491	1L	Charges, propelling
0312	1L	Cartridges, signal	0242	1L	Charges, propelling, for cannon
0405	3L	Cartridges, signal	0279	1L	Charges, propelling, for cannon
0012	3L	Cartridges, small arms	0414	1L	Charges, propelling, for cannon
0339	1L	Cartridges, small arms	0059	1L	Charges, shaped
0417	1L	Cartridges, small arms	0439	1L	Charges, shaped
0014	3L	Cartridges, small arms, blank	0440	1L	Charges, shaped
0327	1L	Cartridges, small arms, blank	0441	3L	Charges, shaped
0338	1L	Cartridges, small arms, blank	0237	1L	Charges, shaped, flexible, linear
0055	3L	Cases, cartridge, empty, with primer	0288	1L	Charges, shaped, flexible, linear
0379	1L	Cases, cartridge, empty, with primer	0060	1L	Charges, supplementary, explosive
0446	1L	Cases, combustible, empty, without primer	3316	9L	Chemical kit
			3315	6L	Chemical sample, toxic

<i>Drill</i>			<i>Drill</i>		
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>
3500	2L	Chemical under pressure, n.o.s.*	3427	6L	Chlorobenzyl chlorides, solid
3503	2C	Chemical under pressure, corrosive, n.o.s.*	1127	3L	Chlorobutanes
3501	10L	Chemical under pressure, flammable, n.o.s.*	3437	6L	Chlorocresols, solid
3505	10C	Chemical under pressure, flammable, corrosive, n.o.s.*	2669	6L	Chlorocresols solution
3504	10P	Chemical under pressure, flammable, toxic, n.o.s.*	1974	2L	Chlorodifluorobromomethane
3502	2P	Chemical under pressure, toxic, n.o.s.*	2517	10L	1-Chloro-1,1-difluoroethane
2075	6L	Chloral, anhydrous, stabilized	1018	2L	Chlorodifluoromethane
1458	5L	Chlorate and borate mixture	1973	2L	Chlorodifluoromethane and chloropentafluoroethane mixture
1459	5L	Chlorate and magnesium chloride mixture, solid	1577	6L	Chlorodinitrobenzenes, liquid
3407	5L	Chlorate and magnesium chloride mixture solution	3441	6L	Chlorodinitrobenzenes, solid
1461	5L	Chlorates, inorganic, n.o.s.*	2232	6L	2-Chloroethanal
3210	5L	Chlorates, inorganic, aqueous solution, n.o.s.*	1888	6A	Chloroform
2626	5L	Chloric acid, aqueous solution	3277	6C	Chloroformates, toxic, corrosive, n.o.s.*
1017	2PX	Chlorine	2742	6CF	Chloroformates, toxic, corrosive, flammable, n.o.s.*
3520	2PX	Chlorine, adsorbed	2745	6C	Chloromethyl chloroformate
2548	2PX	Chlorine pentafluoride	2354	3P	Chloromethyl ethyl ether
1749	2PX	Chlorine trifluoride	2236	6L	3-Chloro-4-methylphenyl isocyanate, liquid
1462	5L	Chlorites, inorganic, n.o.s.*	3428	6L	3-Chloro-4-methylphenyl isocyanate, solid
1908	8L	Chlorite solution	2237	6L	Chloronitroanilines
3250	6C	Chloroacetic acid, molten	3409	6L	Chloronitrobenzenes, liquid
1751	6C	Chloroacetic acid, solid	1578	6L	Chloronitrobenzenes, solid
1750	6C	Chloroacetic acid solution	2433	6L	Chloronitrotoluenes, liquid
1695	6Fi	Chloroacetone, stabilized	3457	6L	Chloronitrotoluenes, solid
2668	6F	Chloroacetonitrile	1020	2L	Chloropentafluoroethane
3416	6i	Chloroacetophenone, liquid	2904	8L	Chlorophenolates, liquid
1697	6i	Chloroacetophenone, solid	2905	8L	Chlorophenolates, solid
1752	6Ci	Chloroacetyl chloride	2021	6L	Chlorophenols, liquid
2019	6L	Chloroanilines, liquid	2020	6L	Chlorophenols, solid
2018	6L	Chloroanilines, solid	1753	8L	Chlorophenyltrichlorosilane
2233	6L	Chloroanisidines	1580	6L	Chloropicrin
1134	3L	Chlorobenzene	1581	2P	Chloropicrin and methyl bromide mixture
2234	3L	Chlorobenzotrifluorides	1582	2P	Chloropicrin and methyl chloride mixture
2235	6L	Chlorobenzyl chlorides, liquid	1583	6L	Chloropicrin mixture, n.o.s.*

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2507	8L	Chloroplatinic acid, solid	1136	3L	Coal tar distillates, flammable
1991	3HP	Chloroprene, stabilized	1139	3L	Coating solution
1278	3H	1-Chloropropane	3550	6L	Cobalt dihydroxide powder
2356	3H	2-Chloropropane	2001	3L	Cobalt naphthenates, powder
2849	6L	3-Chloropropanol-1	1318	3L	Cobalt resinate, precipitated
2456	3H	2-Chloropropene	0382	1L	Components, explosive train, n.o.s.*
2511	8L	2-Chloropropionic acid	0383	1L	Components, explosive train, n.o.s.*
2822	6L	2-Chloropyridine	0384	3L	Components, explosive train, n.o.s.*
2987	8L	Chlorosilanes, corrosive, n.o.s.	0461	1L	Components, explosive train, n.o.s.*
2986	8F	Chlorosilanes, corrosive, flammable, n.o.s.	1956	2L	Compressed gas, n.o.s.*
2985	3C	Chlorosilanes, flammable, corrosive, n.o.s.	1954	10L	Compressed gas, flammable, n.o.s.*
3361	6C	Chlorosilanes, toxic, corrosive, n.o.s.*	3156	2X	Compressed gas, oxidizing, n.o.s.*
3362	6CF	Chlorosilanes, toxic, corrosive, flammable, n.o.s.*	1955	2P	Compressed gas, toxic, n.o.s.*
2988	4FW	Chlorosilanes, water-reactive, flammable, corrosive, n.o.s.	3304	2CP	Compressed gas, toxic, corrosive, n.o.s.*
1754	8W	Chlorosulphonic acid	1953	10P	Compressed gas, toxic, flammable, n.o.s.*
1021	2L	1-Chloro-1,2,2,2-tetrafluoroethane	3305	10C	Compressed gas, toxic, flammable, corrosive, n.o.s.*
2238	3L	Chlorotoluenes	3303	2X	Compressed gas, toxic, oxidizing, n.o.s.*
1579	6L	4-Chloro-o-toluidine hydrochloride, solid	3306	2CX	Compressed gas, toxic, oxidizing, corrosive, n.o.s.*
3410	6L	4-Chloro-o-toluidine hydrochloride solution	8000	9L	Consumer commodity
3429	6L	Chlorotoluidines, liquid	0248	1L	Contrivances, water-activated*
2239	6L	Chlorotoluidines, solid	0249	1L	Contrivances, water-activated*
1983	2L	1-Chloro-2,2,2-trifluoroethane	1585	6L	Copper acetoarsenite
1022	2L	Chlorotrifluoromethane	1586	6L	Copper arsenite
2599	2L	Chlorotrifluoromethane and trifluoromethane azeotropic mixture	2776	3P	Copper based pesticide, liquid, flammable, toxic*
1755	8L	Chromic acid solution	3010	6L	Copper based pesticide, liquid, toxic*
1756	8L	Chromic fluoride, solid	3009	6F	Copper based pesticide, liquid, toxic, flammable*
1757	8L	Chromic fluoride solution	2775	6L	Copper based pesticide, solid, toxic*
2720	5L	Chromium nitrate	2721	5L	Copper chlorate
1758	8W	Chromium oxychloride	2802	8L	Copper chloride
1463	5CP	Chromium trioxide, anhydrous	1587	6L	Copper cyanide
2240	8L	Chromosulphuric acid	1363	4L	Copra
3291	11L	Clinical waste, unspecified, n.o.s.	0065	1L	Cord, detonating
1023	10P	Coal gas, compressed	0102	1L	Cord, detonating

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0289	1L	Cord, detonating	2022	6C	Cresylic acid
0290	1L	Cord, detonating	1143	6Fi	Crotonaldehyde
0104	1L	Cord, detonating, mild effect	1143	6Fi	Crotonaldehyde, stabilized
0066	1L	Cord, igniter	3472	8L	Crotonic acid, liquid
1760	8L	Corrosive liquid, n.o.s.*	2823	8L	Crotonic acid, solid
3264	8L	Corrosive liquid, acidic, inorganic, n.o.s.*	1144	3H	Crotonylene
3265	8L	Corrosive liquid, acidic, organic, n.o.s.*	1761	8P	Cupriethylenediamine solution
3266	8L	Corrosive liquid, basic, inorganic, n.o.s.*	0070	3L	Cutters, cable, explosive
3267	8L	Corrosive liquid, basic, organic, n.o.s.*	1588	6L	Cyanides, inorganic, solid, n.o.s.*
2920	8F	Corrosive liquid, flammable, n.o.s.*	1935	6L	Cyanide solution, n.o.s.*
3093	8X	Corrosive liquid, oxidizing, n.o.s.*	1026	10P	Cyanogen
3301	8L	Corrosive liquid, self-heating, n.o.s.*	1889	6C	Cyanogen bromide
2922	8P	Corrosive liquid, toxic, n.o.s.*	1589	2CP	Cyanogen chloride, stabilized
3094	8W	Corrosive liquid, water-reactive, n.o.s.*	2670	8L	Cyanuric chloride
1759	8L	Corrosive solid, n.o.s.*	2601	10L	Cyclobutane
3260	8L	Corrosive solid, acidic, inorganic, n.o.s.*	2744	6CF	Cyclobutyl chloroformate
3261	8L	Corrosive solid, acidic, organic, n.o.s.*	2518	6L	1,5,9-Cyclododecatriene
3262	8L	Corrosive solid, basic, inorganic, n.o.s.*	2241	3L	Cycloheptane
3263	8L	Corrosive solid, basic, organic, n.o.s.*	2603	3P	Cycloheptatriene
2921	8S	Corrosive solid, flammable, n.o.s.*	2242	3L	Cycloheptene
3084	8X	Corrosive solid, oxidizing, n.o.s.*	1145	3H	Cyclohexane
3095	8S	Corrosive solid, self-heating, n.o.s.*	1915	3L	Cyclohexanone
2923	8P	Corrosive solid, toxic, n.o.s.*	2256	3H	Cyclohexene
3096	8W	Corrosive solid, water-reactive, n.o.s.*	1762	8L	Cyclohexenyltrichlorosilane
1364	4L	Cotton waste, oily	2243	3L	Cyclohexyl acetate
1365	4L	Cotton, wet	2357	8F	Cyclohexylamine
3024	3P	Coumarin derivative pesticide, liquid, flammable, toxic*	2488	6F	Cyclohexyl isocyanate
3026	6L	Coumarin derivative pesticide, liquid, toxic*	3054	3L	Cyclohexyl mercaptan
3025	6F	Coumarin derivative pesticide, liquid, toxic, flammable*	1763	8L	Cyclohexyltrichlorosilane
3027	6L	Coumarin derivative pesticide, solid, toxic*	0391	1L	Cyclonite and cyclotetramethylenetetranitramine mixture, desensitized
2076	6C	Cresols, liquid	0391	1L	Cyclonite and cyclotetramethylenetetranitramine mixture, wetted
3455	6C	Cresols, solid	0483	1L	Cyclonite, desensitized
			0072	1L	Cyclonite, wetted
			2940	4L	Cyclooctadiene phosphines
			2520	3L	Cyclooctadienes

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2358	3L	Cyclooctatetraene	0364	1L	Detonators for ammunition
1146	3H	Cyclopentane	0365	1L	Detonators for ammunition
2244	3L	Cyclopentanol	0366	3L	Detonators for ammunition
2245	3L	Cyclopentanone	0029	1L	Detonators, non-electric
2246	3H	Cyclopentene	0267	1L	Detonators, non-electric
1027	10A	Cyclopropane	0455	3L	Detonators, non-electric
0484	1L	Cyclotetramethylenetetranitramine, desensitized	1957	10L	Deuterium, compressed
0226	1L	Cyclotetramethylenetetranitramine, wetted	3150	10L	Devices, small, hydrocarbon gas powered
0391	1L	Cyclotrimethylenetrinitramine and cyclotetramethylenetetranitramine mixture, desensitized	1148	3L	Diacetone alcohol
0391	1L	Cyclotrimethylenetrinitramine and cyclotetramethylenetetranitramine mixture, wetted	2359	3CP	Diallylamine
0483	1L	Cyclotrimethylenetrinitramine, desensitized	2360	3P	Diallyl ether
0072	1L	Cyclotrimethylenetrinitramine, wetted	2651	6L	4,4'-Diaminodiphenylmethane
2046	3L	Cymenes	2841	3P	Di-n-amylamine
3363	9L	Dangerous goods in apparatus	0074	1L	Diazodinitrophenol, wetted
3363		Dangerous goods in articles	2434	8L	Dibenzylchlorosilane
3363	9L	Dangerous goods in machinery	1911	10P	Diborane
1868	3P	Decaborane	2648	6L	1,2-Dibromobutan-3-one
1147	3L	Decahydronaphthalene	2872	6L	Dibromochloropropanes
2247	3L	n-Decane	1941	9L	Dibromodifluoromethane
0132	1L	Deflagrating metal salts of aromatic nitro-derivatives, n.o.s.	2664	6L	Dibromomethane
3379	3E	Desensitized explosive, liquid, n.o.s.*	2248	8F	Di-n-butylamine
3380	3E	Desensitized explosive, solid, n.o.s.*	2873	6L	Dibutylaminoethanol
0360	1L	Detonator assemblies, non-electric	1149	3L	Dibutyl ethers
0361	1L	Detonator assemblies, non-electric	1764	8i	Dichloroacetic acid
0500	3L	Detonator assemblies, non-electric	2649	6i	1,3-Dichloroacetone
0030	1L	Detonators, electric	1765	8i	Dichloroacetyl chloride
0255	1L	Detonators, electric	1590	6L	Dichloroanilines, liquid
0456	3L	Detonators, electric	3442	6L	Dichloroanilines, solid
0511	1L	Detonators, electronic	1591	6L	o-Dichlorobenzene
0512	1L	Detonators, electronic	1916	6F	2,2'-Dichlorodiethyl ether
0513	1L	Detonators, electronic	1028	2L	Dichlorodifluoromethane
0073	1L	Detonators for ammunition	2602	2L	Dichlorodifluoromethane and difluoroethane azeotropic mixture
			2249	6F	Dichlorodimethyl ether, symmetrical
			2362	3L	1,1-Dichloroethane
			1150	3L	1,2-Dichloroethylene
			1029	2L	Dichlorofluoromethane

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2465	5L	Dichloroisocyanuric acid, dry	1959	10L	1,1-Difluoroethylene
2465	5L	Dichloroisocyanuric acid salts	3252	10L	Difluoromethane
2490	6L	Dichloroisopropyl ether	1768	8L	Difluorophosphoric acid, anhydrous
1593	6L	Dichloromethane	2376	3H	2,3-Dihydropyran
2650	6L	1,1-Dichloro-1-nitroethane	2361	3C	Diisobutylamine
1152	3L	Dichloropentanes	2050	3L	Diisobutylene, isomeric compounds
2250	6L	Dichlorophenyl isocyanates	1157	3L	Diisobutyl ketone
1766	8L	Dichlorophenyltrichlorosilane	1902	8L	Diisooctyl acid phosphate
1279	3L	1,2-Dichloropropane	1158	3CH	Diisopropylamine
2750	6L	1,3-Dichloropropanol-2	1159	3H	Diisopropyl ether
2047	3L	Dichloropropenes	2521	6F	Diketene, stabilized
2189	10P	Dichlorosilane	2252	3L	1,2-Dimethoxyethane
1958	2L	1,2-Dichloro-1,1,2,2- tetrafluoroethane	2377	3L	1,1-Dimethoxyethane
2565	8L	Dicyclohexylamine	1032	10L	Dimethylamine, anhydrous
2687	3L	Dicyclohexylammonium nitrite	1160	3C	Dimethylamine, aqueous solution
2048	3L	Dicyclopentadiene	2378	3P	2-Dimethylaminoacetone
2372	3L	1,2-Di-(dimethylamino) ethane	2051	8F	2-Dimethylaminoethanol
1465	5L	Didymium nitrate	3302	6L	2-Dimethylaminoethyl acrylate, stabilized
1202	3L	Diesel fuel	2522	6L	2-Dimethylaminoethyl methacrylate, stabilized
2373	3H	Diethoxymethane	2253	6L	N,N-Dimethylaniline
2374	3L	3,3-Diethoxypropene	2457	3H	2,3-Dimethylbutane
1154	3CH	Diethylamine	2379	3C	1,3-Dimethylbutylamine
2686	8F	2-Diethylaminoethanol	2262	8L	Dimethylcarbamoyl chloride
2684	3C	3-Diethylaminopropylamine	1161	3L	Dimethyl carbonate
2432	6L	N,N-Diethylaniline	2263	3L	Dimethylcyclohexanes
2049	3L	Diethylbenzene	2264	8F	N,N-Dimethylcyclohexylamine
2366	3L	Diethyl carbonate	1162	3C	Dimethyldichlorosilane
1767	8F	Diethyldichlorosilane	2380	3L	Dimethyldiethoxysilane
0075	1L	Diethyleneglycol dinitrate, desensitized	2707	3L	Dimethyldioxanes
2079	8L	Diethylenetriamine	2381	3P	Dimethyl disulphide
1155	3AH	Diethyl ether	1033	10L	Dimethyl ether
2685	8F	N,N-Diethylethylenediamine	2265	3L	N,N-Dimethylformamide
1156	3L	Diethyl ketone	2382	6F	Dimethylhydrazine, symmetrical
1594	6L	Diethyl sulphate	1163	6CH	Dimethylhydrazine, unsymmetrical
2375	3L	Diethyl sulphide	2044	10L	2,2-Dimethylpropane
2751	8L	Diethylthiophosphoryl chloride	2266	3C	Dimethyl-N-propylamine
1030	10L	1,1-Difluoroethane			

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1595	6C	Dimethyl sulphate	1601	6L	Disinfectant, solid, toxic, n.o.s.*
1164	3H	Dimethyl sulphide	3253	8L	Disodium trioxosilicate
2267	6C	Dimethyl thiophosphoryl chloride	1167	3AH	Divinyl ether, stabilized
0489	1L	DINGU	1771	8L	Dodecyltrichlorosilane
1596	6L	Dinitroanilines	1845	9L	Dry ice
1597	6L	Dinitrobenzenes, liquid	2801	8L	Dye intermediate, liquid, corrosive, n.o.s.*
3443	6L	Dinitrobenzenes, solid	1602	6L	Dye intermediate, liquid, toxic, n.o.s.*
1598	6L	Dinitro-o-cresol	3147	8L	Dye intermediate, solid, corrosive, n.o.s.*
1067	2PX	Dinitrogen tetroxide	3143	6L	Dye intermediate, solid, toxic, n.o.s.*
0489	1L	Dinitroglucuril	2801	8L	Dye, liquid, corrosive, n.o.s.*
0076	1P	Dinitrophenol	1602	6L	Dye, liquid, toxic, n.o.s.*
0077	1P	Dinitrophenolates	3147	8L	Dye, solid, corrosive, n.o.s.*
1321	3EP	Dinitrophenolates, wetted	3143	6L	Dye, solid, toxic, n.o.s.*
1599	6L	Dinitrophenol solution	3257	9L	Elevated temperature liquid, n.o.s.*
1320	3EP	Dinitrophenol, wetted	3256	3L	Elevated temperature liquid, flammable, n.o.s.*
0078	1L	Dinitroresorcinol	3258	9L	Elevated temperature solid, n.o.s.*
1322	3E	Dinitroresorcinol, wetted	3529	10L	Engine, fuel cell, flammable gas powered
0406	1L	Dinitrosobenzene	3528	3L	Engine, fuel cell, flammable liquid powered
2038	6L	Dinitrotoluenes, liquid	3530	9L	Engine, internal combustion
1600	6L	Dinitrotoluenes, molten	3529	10L	Engine, internal combustion, flammable gas powered
3454	6L	Dinitrotoluenes, solid	3528	3L	Engine, internal combustion, flammable liquid powered
1165	3L	Dioxane	3082	9L	Environmentally hazardous substance, liquid, n.o.s.*
1166	3L	Dioxolane	3077	9L	Environmentally hazardous substance, solid, n.o.s.*
2052	3L	Dipentene	2558	6F	Epichlorohydrin
1698	6i	Diphenylamine chloroarsine	2023	6F	Epibromohydrin
1699	6i	Diphenylchloroarsine, liquid	2752	3L	1,2-Epoxy-3-ethoxypropane
3450	6L	Diphenylchloroarsine, solid	3272	3L	Esters, n.o.s.*
1769	8L	Diphenyldichlorosilane	1035	10L	Ethane
1770	8L	Diphenylmethyl bromide	1961	10L	Ethane, refrigerated liquid
0079	1L	Dipicrylamine	1170	3L	Ethanol
0401	1L	Dipicryl sulphide	2491	8L	Ethanolamine
2852	3E	Dipicryl sulphide, wetted	2491	8L	Ethanolamine solution
2383	3C	Dipropylamine			
2384	3H	Di-n-propyl ether			
2710	3L	Dipropyl ketone			
1903	8L	Disinfectant, liquid, corrosive, n.o.s.*			
3142	6L	Disinfectant, liquid, toxic, n.o.s.*			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3475	3L	Ethanol and gasoline mixture
3475	3L	Ethanol and motor spirit mixture
3475	3L	Ethanol and petrol mixture
1170	3L	Ethanol solution
3271	3L	Ethers, n.o.s.*
1173	3L	Ethyl acetate
2452	10L	Ethylacetylene, stabilized
1917	3i	Ethyl acrylate, stabilized
1170	3L	Ethyl alcohol
1170	3L	Ethyl alcohol solution
1036	10L	Ethylamine
2270	3CH	Ethylamine, aqueous solution
2271	3L	Ethyl amyl ketone
2272	6L	N-Ethylaniline
2273	6L	2-Ethylaniline
1175	3L	Ethylbenzene
2274	6L	N-Ethyl-N-benzylaniline
2753	6L	N-Ethylbenzyltoluidines, liquid
3460	6L	N-Ethylbenzyltoluidines, solid
1176	3L	Ethyl borate
1891	3P	Ethyl bromide
1603	6F	Ethyl bromoacetate
2275	3L	2-Ethylbutanol
1177	3L	2-Ethylbutyl acetate
1179	3L	Ethyl butyl ether
1178	3L	2-Ethylbutyraldehyde
1180	3L	Ethyl butyrate
1037	10A	Ethyl chloride
1181	6F	Ethyl chloroacetate
1182	6CF	Ethyl chloroformate
2935	3L	Ethyl 2-chloropropionate
2826	8F	Ethyl chlorothioformate
1862	3L	Ethyl crotonate
1892	6i	Ethyl dichloroarsine
1183	4HW	Ethyl dichlorosilane
1962	10A	Ethylene
3138	10L	Ethylene, acetylene and propylene mixture refrigerated liquid

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1135	6F	Ethylene chlorohydrin
1604	8F	Ethylenediamine
1605	6L	Ethylene dibromide
1184	3P	Ethylene dichloride
1153	3L	Ethylene glycol diethyl ether
1171	3L	Ethylene glycol monoethyl ether
1172	3L	Ethylene glycol monoethyl ether acetate
1188	3L	Ethylene glycol monomethyl ether
1189	3L	Ethylene glycol monomethyl ether acetate
1185	6FH	Ethyleneimine, stabilized
1040	10P	Ethylene oxide
1041	10L	Ethylene oxide and carbon dioxide mixture
1952	2L	Ethylene oxide and carbon dioxide mixture
3300	10P	Ethylene oxide and carbon dioxide mixture
3297	2L	Ethylene oxide and chlorotetrafluoroethane mixture
3070	2L	Ethylene oxide and dichlorodifluoromethane mixture
3298	2L	Ethylene oxide and pentafluoroethane mixture
2983	3P	Ethylene oxide and propylene oxide mixture
3299	2L	Ethylene oxide and tetrafluoroethane mixture
1040	10P	Ethylene oxide with nitrogen
1038	10A	Ethylene, refrigerated liquid
1155	3AH	Ethyl ether
2453	10L	Ethyl fluoride
1190	3H	Ethyl formate
2276	3C	2-Ethylhexylamine
2748	6C	2-Ethylhexyl chloroformate
2385	3L	Ethyl isobutyrate
2481	6F	Ethyl isocyanate
1192	3L	Ethyl lactate
2363	3N	Ethyl mercaptan
2277	3L	Ethyl methacrylate, stabilized

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1039	10L	Ethyl methyl ether	1373	4L	Fibres, animal, n.o.s.
1193	3L	Ethyl methyl ketone	1353	3L	Fibres impregnated with weakly nitrated nitrocellulose, n.o.s.
1194	3P	Ethyl nitrite solution	1373	4L	Fibres, synthetic, n.o.s.
2524	3L	Ethyl orthoformate	1372	4L	Fibres, vegetable
2525	6L	Ethyl oxalate	1373	4L	Fibres, vegetable, n.o.s.
2435	8L	Ethylphenyldichlorosilane	3360	3L	Fibres, vegetable, dry
2386	3C	1-Ethylpiperidine	1324	3L	Films, nitrocellulose base
1195	3L	Ethyl propionate	1774	8L	Fire extinguisher charges
2615	3H	Ethyl propyl ether	1044	2L	Fire extinguishers
2754	6L	N-Ethyltoluidines	2623	3L	Firelighters, solid
1196	3C	Ethyltrichlorosilane	0333	1L	Fireworks
0081	1L	Explosive, blasting, type A	0334	1L	Fireworks
0082	1L	Explosive, blasting, type B	0335	1L	Fireworks
0331	1L	Explosive, blasting, type B	0336	1L	Fireworks
0083	1L	Explosive, blasting, type C	0337	3L	Fireworks
0084	1L	Explosive, blasting, type D	3316	9L	First aid kit
0241	1L	Explosive, blasting, type E	2216	9L	Fish meal, stabilized
0332	1L	Explosive, blasting, type E	1374	4L	Fish meal, unstabilized
1197	3L	Extracts, liquid	2216	9L	Fish scrap, stabilized
1373	4L	Fabrics, animal, n.o.s.	1374	4L	Fish scrap, unstabilized
1353	3L	Fabrics impregnated with weakly nitrated nitrocellulose, n.o.s.	1993		Flammable liquid, n.o.s.*
1373	4L	Fabrics, synthetic, n.o.s.	3H		Packing Group I or II
1373	4L	Fabrics, vegetable, n.o.s.	3L		Packing Group III
1606	6L	Ferric arsenate	2924		Flammable liquid, corrosive, n.o.s.*
1607	6L	Ferric arsenite	3CH		Packing Group I or II
1773	8L	Ferric chloride, anhydrous	3C		Packing Group III
2582	8L	Ferric chloride solution	1992		Flammable liquid, toxic, n.o.s.*
1466	5L	Ferric nitrate	3HP		Packing Group I or II
1323	3L	Ferrocium	3P		Packing Group III
1408	4PW	Ferrosilicon	3286	3CP	Flammable liquid, toxic, corrosive, n.o.s.*
1608	6L	Ferrous arsenate	3180	3C	Flammable solid, corrosive, inorganic, n.o.s.*
2793	4L	Ferrous metal borings	2925	3C	Flammable solid, corrosive, organic, n.o.s.*
2793	4L	Ferrous metal cuttings	3178	3L	Flammable solid, inorganic, n.o.s.*
2793	4L	Ferrous metal shavings	1325	3L	Flammable solid, organic, n.o.s.*
2793	4L	Ferrous metal turnings			
1043	2L	Fertilizer ammoniating solution			
1372	4L	Fibres, animal			

<i>Drill</i>			<i>Drill</i>		
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>
3176	3L	Flammable solid, organic, molten, n.o.s.*	3473	3L	Fuel cell cartridges contained in equipment
3097	3X	Flammable solid, oxidizing, n.o.s.*	3476	4W	Fuel cell cartridges contained in equipment
3179	3P	Flammable solid, toxic, inorganic, n.o.s.*	3477	8L	Fuel cell cartridges contained in equipment
2926	3P	Flammable solid, toxic, organic, n.o.s.*	3478	10L	Fuel cell cartridges contained in equipment
0093	1L	Flares, aerial	3479	10L	Fuel cell cartridges contained in equipment
0403	1L	Flares, aerial	3473	3L	Fuel cell cartridges packed with equipment
0404	3L	Flares, aerial	3476	4W	Fuel cell cartridges packed with equipment
0420	1L	Flares, aerial	3477	8L	Fuel cell cartridges packed with equipment
0421	1L	Flares, aerial	3478	10L	Fuel cell cartridges packed with equipment
0092	1L	Flares, surface	3479	10L	Fuel cell cartridges packed with equipment
0418	1L	Flares, surface	1780	8L	Fumaryl chloride
0419	1L	Flares, surface	3359	9L	Fumigated cargo transport unit
0094	1L	Flash powder	1199	6F	Furaldehydes
0305	1L	Flash powder	2389	3H	Furan
1045	2PX	Fluorine, compressed	2874	6L	Furfuryl alcohol
2642	6L	Fluoroacetic acid	2526	3C	Furfurylamine
2941	6L	Fluoroanilines	0102	1L	Fuse, detonating
2387	3L	Fluorobenzene	0290	1L	Fuse, detonating
1775	8L	Fluoroboric acid	0104	1L	Fuse, detonating, mild effect
1776	8L	Fluorophosphoric acid, anhydrous	0103	1L	Fuse, igniter
2856	6L	Fluorosilicates, n.o.s.*	1201	3L	Fusel oil
1778	8L	Fluorosilicic acid	0101	1L	Fuse, non-detonating
1777	8W	Fluorosulphonic acid	0105	3L	Fuse, safety
2388	3L	Fluorotoluenes	0106	1L	Fuzes, detonating
2209	8i	Formaldehyde solution	0107	1L	Fuzes, detonating
1198	3Ci	Formaldehyde solution, flammable	0257	1L	Fuzes, detonating
1779	8F	Formic acid	0367	3L	Fuzes, detonating
3412	8L	Formic acid	0408	1L	Fuzes, detonating
0099	1L	Fracturing devices, explosive	0409	1L	Fuzes, detonating
1863	3L	Fuel, aviation, turbine engine	0410	1L	Fuzes, detonating
3473	3L	Fuel cell cartridges	0316	1L	Fuzes, igniting
3476	4W	Fuel cell cartridges			
3477	8L	Fuel cell cartridges			
3478	10L	Fuel cell cartridges			
3479	10L	Fuel cell cartridges			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0317	1L	Fuzes, igniting	0372	1L	Grenades, practice
0368	3L	Fuzes, igniting	0452	1L	Grenades, practice
2803	8L	Gallium	1467	5L	Guanidine nitrate
2037		Gas cartridges	0113	1L	Guanyl nitrosaminoguanylidene hydrazine, wetted
	10L	Division 2.1 without subsidiary risk	0114	1L	Guanyl nitrosaminoguanyltetrazene, wetted
	2L	Division 2.2 without subsidiary risk	0027	1L	Gunpowder
	2X	Division 2.2 with subsidiary risk 5.1	0028	1L	Gunpowder, compressed
	2P	Division 2.3 without subsidiary risk	0028	1L	Gunpowder in pellets
	10P	Division 2.3 with subsidiary risk 2.1	2545	4L	Hafnium powder, dry
	10C	Division 2.3 with subsidiary risk 2.1 and 8	1326	3L	Hafnium powder, wetted
	2PX	Division 2.3 with subsidiary risk 5.1	3151	9L	Halogenated monomethyldiphenylmethanes, liquid
	2CX	Division 2.3 with subsidiary risk 5.1 and 8	3152	9L	Halogenated monomethyldiphenylmethanes, solid
	2CP	Division 2.3 with subsidiary risk 8	1327	3L	Hay
1202	3L	Gas oil	1202	3L	Heating oil, light
1203	3H	Gasoline	1046	2L	Helium, compressed
3158	2L	Gas, refrigerated liquid, n.o.s.*	1963	2L	Helium, refrigerated liquid
3312	10L	Gas, refrigerated liquid, flammable, n.o.s.*	3296	2L	Heptafluoropropane
3311	2X	Gas, refrigerated liquid, oxidizing, n.o.s.*	3056	3L	n-Heptaldehyde
3167	10L	Gas sample, non-pressurized, flammable, n.o.s.	1206	3H	Heptanes
3169	2P	Gas sample, non-pressurized, toxic, n.o.s.	2278	3L	n-Heptene
3168	10P	Gas sample, non-pressurized, toxic, flammable, n.o.s.	2661	6L	Hexachloroacetone
3245	9L	Genetically modified micro- organisms	2729	6L	Hexachlorobenzene
3245	9L	Genetically modified organisms	2279	6L	Hexachlorobutadiene
2192	10P	Germane	2646	6L	Hexachlorocyclopentadiene
3523	10P	Germane, adsorbed	2875	6L	Hexachlorophene
2689	6L	Glycerol alpha-monochlorohydrin	1781	8L	Hexadecyltrichlorosilane
2622	3P	Glycidaldehyde	2458	3H	Hexadiene
0284	1L	Grenades	1611	6L	Hexaethyl tetraphosphate
0285	1L	Grenades	1612	2P	Hexaethyl tetraphosphate and compressed gas mixture
0292	1L	Grenades	2420	2CP	Hexafluoroacetone
0293	1L	Grenades	2552	6L	Hexafluoroacetone hydrate, liquid
0110	3L	Grenades, practice	3436	6L	Hexafluoroacetone hydrate, solid
0318	1L	Grenades, practice	2193	2L	Hexafluoroethane
			1782	8L	Hexafluorophosphoric acid

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1858	2L	Hexafluoropropylene
1207	3L	Hexaldehyde
2280	8L	Hexamethylenediamine, solid
1783	8L	Hexamethylenediamine solution
2281	6L	Hexamethylene diisocyanate
2493	3C	Hexamethyleneimine
1328	3L	Hexamethylenetetramine
1208	3H	Hexanes
0079	1L	Hexanitrodiphenylamine
0392	1L	Hexanitrostilbene
2282	3L	Hexanols
2370	3H	1-Hexene
0391	1L	Hexogen and cyclotetramethylenetetranitramine mixture, desensitized
0391	1L	Hexogen and cyclotetramethylenetetranitramine mixture, wetted
0483	1L	Hexogen, desensitized
0072	1L	Hexogen, wetted
0118	1L	Hexolite
0118	1L	Hexotol
0393	1L	Hexotonal
0079	1L	Hexyl
1784	8L	Hexyltrichlorosilane
0484	1L	HMX, desensitized
0226	1L	HMX, wetted
2029	8FP	Hydrazine, anhydrous
2030	8P	Hydrazine, aqueous solution
3293	6L	Hydrazine, aqueous solution
3484	8FP	Hydrazine aqueous solution, flammable
1787	8L	Hydriodic acid
1788	8L	Hydrobromic acid
1964	10L	Hydrocarbon gas mixture, compressed, n.o.s.*
1965	10L	Hydrocarbon gas mixture, liquefied, n.o.s.*
3150	10L	Hydrocarbon gas refills for small devices

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3295		Hydrocarbons, liquid, n.o.s.
3H		Packing Group I or II
3L		Packing Group III
1789	8L	Hydrochloric acid
1613	6L	Hydrocyanic acid, aqueous solution
1790	8P	Hydrofluoric acid
1786	8P	Hydrofluoric acid and sulphuric acid mixture
2034	10L	Hydrogen and methane mixture, compressed
1048	2CP	Hydrogen bromide, anhydrous
1050	2CP	Hydrogen chloride, anhydrous
2186	2CP	Hydrogen chloride, refrigerated liquid
1049	10L	Hydrogen, compressed
1613	6L	Hydrogen cyanide, aqueous solution
3294	6F	Hydrogen cyanide, solution in alcohol
1051	6H	Hydrogen cyanide, stabilized
1614	6L	Hydrogen cyanide, stabilized
1740	8L	Hydrogendifluorides, solid, n.o.s.
3471	8L	Hydrogendifluorides, solution, n.o.s.
1052	8P	Hydrogen fluoride, anhydrous
3468	10L	Hydrogen in a metal hydride storage system
3468	10L	Hydrogen in a metal hydride storage system contained in equipment
3468	10L	Hydrogen in a metal hydride storage system packed with equipment
2197	2CP	Hydrogen iodide, anhydrous
3149	5C	Hydrogen peroxide and peroxyacetic acid mixture stabilized
2014	5C	Hydrogen peroxide, aqueous solution
2984	5L	Hydrogen peroxide, aqueous solution
2015	5C	Hydrogen peroxide, aqueous solution, stabilized
2015	5C	Hydrogen peroxide, stabilized
1966	10L	Hydrogen, refrigerated liquid
3526	10P	Hydrogen selenide, adsorbed
2202	10P	Hydrogen selenide, anhydrous
1053	10P	Hydrogen sulphide
0508	1L	1-Hydroxybenzotriazole, anhydrous

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3474	3E	1-Hydroxybenzotriazole monohydrate	2283	3L	Isobutyl methacrylate, stabilized
2865	8L	Hydroxylamine sulphate	2394	3L	Isobutyl propionate
3212	5L	Hypochlorites, inorganic, n.o.s.*	2045	3H	Isobutyraldehyde
1791	8L	Hypochlorite solution	2529	3C	Isobutyric acid
0121	1L	Igniters	2284	3P	Isobutyronitrile
0314	1L	Igniters	2395	3C	Isobutyryl chloride
0315	1L	Igniters	2478	3P	Isocyanates, flammable, toxic, n.o.s.*
0325	1L	Igniters	2478	3P	Isocyanate solution, flammable, toxic, n.o.s.*
0454	3L	Igniters	2206	6L	Isocyanate solution, toxic, n.o.s.*
2269	8L	3,3'-Iminodipropylamine	3080	6F	Isocyanate solution, toxic, flammable, n.o.s.*
2900	11Y	Infectious substance, affecting animals	2206	6L	Isocyanates, toxic, n.o.s.*
2814	11Y	Infectious substance, affecting humans	3080	6F	Isocyanates, toxic, flammable, n.o.s.*
1968	2L	Insecticide gas, n.o.s.*	2285	6F	Isocyanatobenzotrifluorides
3354	10L	Insecticide gas, flammable, n.o.s.*	2287	3H	Isoheptene
1967	2P	Insecticide gas, toxic, n.o.s.*	2288	3H	Isohexene
3355	10P	Insecticide gas, toxic, flammable, n.o.s.*	1216	3H	Isooctene
3495	8P	Iodine	2371	3H	Isopentenenes
3498	8L	Iodine monochloride, liquid	2289	8L	Isophoronediamine
1792	8L	Iodine monochloride, solid	2290	6L	Isophorone diisocyanate
2495	5CP	Iodine pentafluoride	1218	3H	Isoprene, stabilized
2390	3L	2-Iodobutane	1219	3L	Isopropanol
2391	3L	Iodomethylpropanes	2403	3L	Isopropenyl acetate
2392	3L	Iodopropanes	2303	3L	Isopropenylbenzene
1376	4L	Iron oxide, spent	1220	3L	Isopropyl acetate
1994	6H	Iron pentacarbonyl	1793	8L	Isopropyl acid phosphate
1376	4L	Iron sponge, spent	1219	3L	Isopropyl alcohol
1969	10L	Isobutane	1221	3CH	Isopropylamine
1212	3L	Isobutanol	1918	3L	Isopropylbenzene
1213	3L	Isobutyl acetate	2405	3L	Isopropyl butyrate
2527	3L	Isobutyl acrylate, stabilized	2947	3L	Isopropyl chloroacetate
1212	3L	Isobutyl alcohol	2407	6CF	Isopropyl chloroformate
2045	3H	Isobutyl aldehyde	2934	3L	Isopropyl 2-chloropropionate
1214	3CH	Isobutylamine	2406	3L	Isopropyl isobutyrate
1055	10L	Isobutylene	2483	6H	Isopropyl isocyanate
2393	3L	Isobutyl formate	1222	3L	Isopropyl nitrate
2528	3L	Isobutyl isobutyrate	2409	3L	Isopropyl propionate
2486	6F	Isobutyl isocyanate			

<i>Drill</i>			<i>Drill</i>		
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>
2907	3L	Isosorbide dinitrate mixture	3307	2X	Liquefied gas, toxic, oxidizing, n.o.s.*
3251	3L	Isosorbide-5-mononitrate	3310	2CX	Liquefied gas, toxic, oxidizing, corrosive, n.o.s.*
0124	1L	Jet perforating guns, charged	1415	4W	Lithium
0494	1L	Jet perforating guns, charged	1410	4W	Lithium aluminium hydride
1223	3L	Kerosene	1411	4HW	Lithium aluminium hydride, ethereal
1224	3L	Ketones, liquid, n.o.s.*	3536	12FZ	Lithium batteries installed in cargo transport unit
3497	4L	Krill meal	1413	4W	Lithium borohydride
1056	2L	Krypton, compressed	2830	4W	Lithium ferrosilicon
1970	2L	Krypton, refrigerated liquid	1414	4W	Lithium hydride
1616	6L	Lead acetate	2805	4W	Lithium hydride, fused solid
1617	6L	Lead arsenates	2680	8L	Lithium hydroxide
1618	6L	Lead arsenites	2679	8L	Lithium hydroxide solution
0129	1L	Lead azide, wetted	1471	5L	Lithium hypochlorite, dry
2291	6L	Lead compound, soluble, n.o.s.*	1471	5L	Lithium hypochlorite mixture
1620	6L	Lead cyanide	3480	12FZ	Lithium ion batteries
1872	5L	Lead dioxide	3481	12FZ	Lithium ion batteries contained in equipment
1469	5P	Lead nitrate	3481	12FZ	Lithium ion batteries packed with equipment
1470	5P	Lead perchlorate, solid	3090	12FZ	Lithium metal batteries
3408	5P	Lead perchlorate solution	3091	12FZ	Lithium metal batteries contained in equipment
2989	3L	Lead phosphite, dibasic	3091	12FZ	Lithium metal batteries packed with equipment
0130	1L	Lead styphnate, wetted	2722	5L	Lithium nitrate
1794	8L	Lead sulphate	2806	4W	Lithium nitride
0130	1L	Lead trinitroresorcinate, wetted	1472	5L	Lithium peroxide
3072	9L	Life-saving appliances, not self-inflating	1417	4W	Lithium silicon
2990	9L	Life-saving appliances, self-inflating	1621	6L	London Purple
1057	10L	Lighter refills	3529	10L	Machinery, fuel cell, flammable gas powered
1057	10L	Lighters	3528	3L	Machinery, fuel cell, flammable liquid powered
0131	3L	Lighters, fuse	3530	9L	Machinery, internal combustion
3163	2L	Liquefied gas, n.o.s.*	3529	10L	Machinery, internal combustion, flammable gas powered
1058	2L	Liquefied gases	3528	3L	Machinery, internal combustion, flammable liquid powered
3161	10L	Liquefied gas, flammable, n.o.s.*	1869	3L	Magnesium
3157	2X	Liquefied gas, oxidizing, n.o.s.*			
3162	2P	Liquefied gas, toxic, n.o.s.*			
3308	2CP	Liquefied gas, toxic, corrosive, n.o.s.*			
3160	10P	Liquefied gas, toxic, flammable, n.o.s.*			
3309	10C	Liquefied gas, toxic, flammable, corrosive, n.o.s.*			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1869	3L	Magnesium alloys	1851	6L	Medicine, liquid, toxic, n.o.s.
1418	4SW	Magnesium alloys powder	3249	6L	Medicine, solid, toxic, n.o.s.
1419	4PW	Magnesium aluminium phosphide	3336		Mercaptan mixture, liquid, flammable, n.o.s.*
1622	6L	Magnesium arsenate	3H		Packing Group I or II
1473	5L	Magnesium bromate	3L		Packing Group III
2723	5L	Magnesium chlorate	1228	3P	Mercaptan mixture, liquid, flammable, toxic, n.o.s.*
2004	4W	Magnesium diamide	3071	6F	Mercaptan mixture, liquid, toxic, flammable, n.o.s.*
2853	6L	Magnesium fluorosilicate	3336		Mercaptans, liquid, flammable, n.o.s.*
2950	4W	Magnesium granules, coated	3H		Packing Group I or II
2010	4W	Magnesium hydride	3L		Packing Group III
1474	5L	Magnesium nitrate	1228	3P	Mercaptans, liquid, flammable, toxic, n.o.s.*
1475	5L	Magnesium perchlorate	3071	6F	Mercaptans, liquid, toxic, flammable, n.o.s.*
1476	5L	Magnesium peroxide	0448	1L	5-Mercaptotetrazol-1-acetic acid
2011	4PW	Magnesium phosphide	1623	6L	Mercuric arsenate
1418	4SW	Magnesium powder	1624	6L	Mercuric chloride
2624	4W	Magnesium silicide	1625	6L	Mercuric nitrate
2807	9M	Magnetized material	1626	6L	Mercuric potassium cyanide
2215	8L	Maleic anhydride	1627	6L	Mercurous nitrate
2215	8L	Maleic anhydride, molten	2809	8P	Mercury
2647	6L	Malononitrile	1629	6L	Mercury acetate
2210	4SW	Maneb	1630	6L	Mercury ammonium chloride
2210	4SW	Maneb preparation	2778	3P	Mercury based pesticide, liquid, flammable, toxic*
2968	4W	Maneb preparation, stabilized	3012	6L	Mercury based pesticide, liquid, toxic*
2968	4W	Maneb stabilized	3011	6F	Mercury based pesticide, liquid, toxic, flammable*
2724	5L	Manganese nitrate	2777	6L	Mercury based pesticide, solid, toxic*
1330	3L	Manganese resinate	1631	6L	Mercury benzoate
0133	1L	Mannitol hexanitrate, wetted	1634	6L	Mercury bromides
2254	3L	Matches, fusee	2024	6L	Mercury compound, liquid, n.o.s.*
1944	3L	Matches, safety	2025	6L	Mercury compound, solid, n.o.s.*
1331	3L	Matches, 'strike anywhere'	3506	8L	Mercury contained in manufactured articles
1945	3L	Matches, wax 'vesta'	1636	6L	Mercury cyanide
3291	11L	Medical waste, n.o.s.	0135	1L	Mercury fulminate, wetted
3549	11Y	Medical waste, Category A affecting animals			
3549	11Y	Medical waste, Category A affecting humans			
3248	3P	Medicine, liquid, flammable, toxic, n.o.s.			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1637	6L	Mercury gluconate	1060	10L	Methylacetylene and propadiene mixture, stabilized
1638	6L	Mercury iodide	1919	3Hi	Methyl acrylate, stabilized
1639	6L	Mercury nucleate	1234	3H	Methylal
1640	6L	Mercury oleate	2554	3H	Methylallyl chloride
1641	6L	Mercury oxide	1061	10L	Methylamine, anhydrous
1642	6L	Mercury oxycyanide, desensitized	1235	3CH	Methylamine, aqueous solution
1643	6L	Mercury potassium iodide	1233	3L	Methylamyl acetate
1644	6L	Mercury salicylate	2294	6L	N-Methylaniline
1645	6L	Mercury sulphate	2937	6L	alpha-Methylbenzyl alcohol, liquid
1646	6L	Mercury thiocyanate	3438	6L	alpha-Methylbenzyl alcohol, solid
1229	3L	Mesityl oxide	1062	2P	Methyl bromide
3281	6L	Metal carbonyls, liquid, n.o.s.*	1647	6L	Methyl bromide and ethylene dibromide mixture, liquid
3466	6L	Metal carbonyls, solid, n.o.s.*	2643	6i	Methyl bromoacetate
2881	4L	Metal catalyst, dry*	3371	3L	2-Methylbutanal
1378	4L	Metal catalyst, wetted*	2397	3L	3-Methylbutan-2-one
1332	3L	Metaldehyde	2459	3H	2-Methyl-1-butene
3182	3L	Metal hydrides, flammable, n.o.s.*	2460	3H	2-Methyl-2-butene
1409	4W	Metal hydrides, water-reactive, n.o.s.*	2561	3H	3-Methyl-1-butene
3208	4W	Metallic substance, water-reactive, n.o.s.*	2945	3C	N-Methylbutylamine
3209	4SW	Metallic substance, water-reactive, self-heating, n.o.s.*	2398	3L	Methyl tert-butyl ether
3089	3L	Metal powder, flammable, n.o.s.	1237	3L	Methyl butyrate
3189	4L	Metal powder, self-heating, n.o.s.*	1063	10L	Methyl chloride
3181	3L	Metal salts of organic compounds, flammable, n.o.s.*	1912	10L	Methyl chloride and methylene chloride mixture
2396	3P	Methacrylaldehyde, stabilized	2295	6F	Methyl chloroacetate
2531	8L	Methacrylic acid, stabilized	1238	6F	Methyl chloroformate
3079	6F	Methacrylonitrile, stabilized	1239	6F	Methyl chloromethyl ether
2614	3L	Methallyl alcohol	2933	3L	Methyl 2-chloropropionate
1971	10L	Methane, compressed	2534	10P	Methylchlorosilane
1972	10L	Methane, refrigerated liquid	2296	3H	Methylcyclohexane
3246	6C	Methanesulphonyl chloride	2617	3L	Methylcyclohexanols
1230	3L	Methanol	2297	3L	Methylcyclohexanone
2605	6F	Methoxymethyl isocyanate	2298	3H	Methylcyclopentane
2293	3L	4-Methoxy-4-methylpentan-2-one	2299	6L	Methyl dichloroacetate
3092	3L	1-Methoxy-2-propanol	1242	4HW	Methyldichlorosilane
1231	3H	Methyl acetate	1193	3L	Methyl ethyl ketone

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2300	6L	2-Methyl-5-ethylpyridine	2508	8L	Molybdenum pentachloride
2454	10L	Methyl fluoride	2054	8F	Morpholine
1243	3H	Methyl formate	1649	6L	Motor fuel anti-knock mixture
2301	3H	2-Methylfuran	3483	6F	Motor fuel anti-knock mixture, flammable
3023	6F	2-Methyl-2-heptanethiol	1203	3H	Motor spirit
2302	3L	5-Methylhexan-2-one	2956	3E	Musk xylene
1244	6F	Methylhydrazine	1334	3L	Naphthalene, crude
2644	6L	Methyl iodide	2304	3L	Naphthalene, molten
2053	3L	Methyl isobutyl carbinol	1334	3L	Naphthalene, refined
1245	3L	Methyl isobutyl ketone	2077	6L	alpha-Naphthylamine
2480	6H	Methyl isocyanate	1650	6L	beta-Naphthylamine, solid
1246	3L	Methyl isopropenyl ketone, stabilized	3411	6L	beta-Naphthylamine solution
2477	6F	Methyl isothiocyanate	1651	6L	Naphthylthiourea
2400	3L	Methyl isovalerate	1652	6L	Naphthylurea
1928	4HW	Methyl magnesium bromide in ethyl ether	1971	10L	Natural gas, compressed
1064	10P	Methyl mercaptan	1972	10L	Natural gas, refrigerated liquid
1247	3L	Methyl methacrylate monomer, stabilized	1065	2L	Neon, compressed
2535	3C	4-Methylmorpholine	1913	2L	Neon, refrigerated liquid
2535	3C	N-Methylmorpholine	1259	6H	Nickel carbonyl
2606	6F	Methyl orthosilicate	1653	6L	Nickel cyanide
2461	3H	Methylpentadiene	2725	5L	Nickel nitrate
2560	3L	2-Methylpentan-2-ol	2726	5L	Nickel nitrite
2437	8L	Methylphenyldichlorosilane	1654	6L	Nicotine
2399	3C	1-Methylpiperidine	3144	6L	Nicotine compound, liquid, n.o.s.*
1248	3H	Methyl propionate	1655	6L	Nicotine compound, solid, n.o.s.*
2612	3AH	Methyl propyl ether	1656	6L	Nicotine hydrochloride, liquid
1249	3L	Methyl propyl ketone	3444	6L	Nicotine hydrochloride, solid
2536	3H	Methyltetrahydrofuran	1656	6L	Nicotine hydrochloride solution
2533	6L	Methyl trichloroacetate	3144	6L	Nicotine preparation, liquid, n.o.s.*
1250	3C	Methyltrichlorosilane	1655	6L	Nicotine preparation, solid, n.o.s.*
2367	3L	alpha-Methylvaleraldehyde	1657	6L	Nicotine salicylate
1251	6CH	Methyl vinyl ketone, stabilized	3445	6L	Nicotine sulphate, solid
0136	1L	Mines	1658	6L	Nicotine sulphate solution
0137	1L	Mines	1659	6L	Nicotine tartrate
0138	1L	Mines	1477	5L	Nitrates, inorganic, n.o.s.
0294	1L	Mines	3218	5L	Nitrates, inorganic, aqueous solution, n.o.s.

<i>Drill</i>			<i>Drill</i>		
<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Code</i>	<i>Proper Shipping Name</i>
1796		Nitrating acid mixture	3270	3L	Nitrocellulose membrane filters
	8X	Packing Group I	2557	3L	Nitrocellulose mixture without plasticizer, without pigment
	8L	Packing Group II	2557	3L	Nitrocellulose mixture without plasticizer, with pigment
1826		Nitrating acid mixture, spent	2557	3L	Nitrocellulose mixture with plasticizer, without pigment
	8X	Packing Group I	2557	3L	Nitrocellulose mixture with plasticizer, with pigment
	8L	Packing Group II	0343	1L	Nitrocellulose, plasticized
2031		Nitric acid	2059		Nitrocellulose solution, flammable
	8L	Other than red fuming, with more than 20% and less than 65% nitric acid	3H		Packing Group I or II
	8X	Other than red fuming, with more than 70% nitric acid	3L		Packing Group III
	8L	Other than red fuming, with not more than 20% nitric acid	0342	1L	Nitrocellulose, wetted
	8X	Other than red fuming, with at least 65% but not more than 70% nitric acid	2556	3L	Nitrocellulose with alcohol
2032	8PX	Nitric acid, red fuming	2555	3E	Nitrocellulose with water
1975	2PX	Nitric oxide and dinitrogen tetroxide mixture	2307	6L	3-Nitro-4-chlorobenzotrifluoride
1975	2PX	Nitric oxide and nitrogen dioxide mixture	3434	6L	Nitrocresols, liquid
1660	2PX	Nitric oxide, compressed	2446	6L	Nitrocresols, solid
3273	3HP	Nitriles, flammable, toxic, n.o.s.*	2842	3L	Nitroethane
3276	6L	Nitriles, liquid, toxic, n.o.s.*	1066	2L	Nitrogen, compressed
3439	6L	Nitriles, solid, toxic, n.o.s.*	1067	2PX	Nitrogen dioxide
3275	6F	Nitriles, toxic, flammable, n.o.s.*	1977	2L	Nitrogen, refrigerated liquid
2627	5L	Nitrites, inorganic, n.o.s.*	2451	2X	Nitrogen trifluoride
3219	5L	Nitrites, inorganic, aqueous solution, n.o.s.*	2421	2PX	Nitrogen trioxide
1661	6L	Nitroanilines	0143	1P	Nitroglycerin, desensitized
2730	6L	Nitroanisoles, liquid	3357	3L	Nitroglycerin mixture, desensitized, liquid, n.o.s.*
3458	6L	Nitroanisoles, solid	3343	3E	Nitroglycerin mixture, desensitized, liquid flammable, n.o.s.*
1662	6L	Nitrobenzene	3319	3L	Nitroglycerin mixture, desensitized, solid, n.o.s.*
2305	8L	Nitrobenzenesulphonic acid	0144	1L	Nitroglycerin solution in alcohol
0385	1L	5-Nitrobenzotriazol	1204	3L	Nitroglycerin solution in alcohol
2306	6L	Nitrobenzotrifluorides, liquid	3064	3L	Nitroglycerin solution in alcohol
3431	6L	Nitrobenzotrifluorides, solid	0282	1L	Nitroguanidine
2732	6L	Nitrobromobenzene, liquid	1336	3E	Nitroguanidine, wetted
3459	6L	Nitrobromobenzene, solid	1798	8L	Nitrohydrochloric acid
0340	1L	Nitrocellulose	0133	1L	Nitromannite, wetted
0341	1L	Nitrocellulose	1261	3L	Nitromethane

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2538	3L	Nitronaphthalene	3103	5L	Organic peroxide type C, liquid*
1663	6L	Nitrophenols	3113	5S	Organic peroxide type C, liquid, temperature controlled*
3376	3E	4-Nitrophenylhydrazine	3104	5L	Organic peroxide type C, solid*
2608	3L	Nitropropanes	3114	5S	Organic peroxide type C, solid, temperature controlled*
1369	4L	p-Nitrosodimethylaniline	3105	5L	Organic peroxide type D, liquid*
0146	1L	Nitrostarch	3115	5S	Organic peroxide type D, liquid, temperature controlled*
1337	3E	Nitrostarch, wetted	3106	5L	Organic peroxide type D, solid*
1069	2CP	Nitrosyl chloride	3116	5S	Organic peroxide type D, solid, temperature controlled*
2308	8L	Nitrosylsulphuric acid, liquid	3107	5L	Organic peroxide type E, liquid*
3456	8L	Nitrosylsulphuric acid, solid	3117	5S	Organic peroxide type E, liquid, temperature controlled*
1664	6L	Nitrotoluenes, liquid	3108	5L	Organic peroxide type E, solid*
3446	6L	Nitrotoluenes, solid	3118	5S	Organic peroxide type E, solid, temperature controlled*
2660	6L	Nitrotoluidines (mono)	3109	5L	Organic peroxide type F, liquid*
0490	1L	Nitrotriazolone	3119	5S	Organic peroxide type F, liquid, temperature controlled*
0147	1L	Nitro urea	3110	5L	Organic peroxide type F, solid*
1070	2AX	Nitrous oxide	3120	5S	Organic peroxide type F, solid, temperature controlled*
2201	2AX	Nitrous oxide, refrigerated liquid	3313	4L	Organic pigments, self-heating
1665	6L	Nitroxylenes, liquid	3280	6L	Organoarsenic compound, liquid, n.o.s.*
3447	6L	Nitroxylenes, solid	3465	6L	Organoarsenic compound, solid, n.o.s.*
1920	3L	Nonanes	2762	3P	Organochlorine pesticide, liquid, flammable, toxic*
1799	8L	Nonyltrichlorosilane	2996	6L	Organochlorine pesticide, liquid, toxic*
2251	3L	2,5-Norbormadiene, stabilized	2995	6F	Organochlorine pesticide, liquid, toxic, flammable*
0490	1L	NTO	2761	6L	Organochlorine pesticide, solid, toxic*
1800	8L	Octadecyltrichlorosilane	3282	6L	Organometallic compound, liquid, toxic, n.o.s.*
2309	3L	Octadiene	3467	6L	Organometallic compound, solid, toxic, n.o.s.*
2422	2L	Octafluorobut-2-ene	3392	4L	Organometallic substance, liquid, pyrophoric*
1976	2L	Octafluorocyclobutane	3394	4W	Organometallic substance, liquid, pyrophoric, water reactive*
2424	2L	Octafluoropropane			
1262	3H	Octanes			
0484	1L	Octogen, desensitized			
0226	1L	Octogen, wetted			
0266	1L	Octol			
0266	1L	Octolite			
0496	1L	Octonal			
1191	3L	Octyl aldehydes			
1801	8L	Octyltrichlorosilane			
1071	10P	Oil gas, compressed			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3398	4W	Organometallic substance, liquid, water reactive*
3399	4FW	Organometallic substance, liquid, water reactive, flammable*
3391	4L	Organometallic substance, solid, pyrophoric*
3393	4W	Organometallic substance, solid, pyrophoric, water reactive*
3400	4L	Organometallic substance, solid, self-heating*
3395	4W	Organometallic substance, solid, water reactive*
3396	4W	Organometallic substance, solid, water reactive, flammable*
3397	4W	Organometallic substance, solid, water reactive, self-heating*
3278	6L	Organophosphorus compound, liquid, toxic, n.o.s.*
3464	6L	Organophosphorus compound, solid, toxic, n.o.s.*
3279	6F	Organophosphorus compound, toxic, flammable, n.o.s.*
2784	3P	Organophosphorus pesticide, liquid, flammable, toxic*
3018	6L	Organophosphorus pesticide, liquid, toxic*
3017	6F	Organophosphorus pesticide, liquid, toxic, flammable*
2783	6L	Organophosphorus pesticide, solid, toxic*
2788	6L	Organotin compound, liquid, n.o.s.*
3146	6L	Organotin compound, solid, n.o.s.*
2787	3P	Organotin pesticide, liquid, flammable, toxic*
3020	6L	Organotin pesticide, liquid, toxic*
3019	6F	Organotin pesticide, toxic, flammable*
2786	6L	Organotin pesticide, solid, toxic*
2471	6L	Osmium tetroxide
3139	5L	Oxidizing liquid, n.o.s.*
3098	5C	Oxidizing liquid, corrosive, n.o.s.*
3099	5P	Oxidizing liquid, toxic, n.o.s.*
1479	5L	Oxidizing solid, n.o.s.*
3085	5C	Oxidizing solid, corrosive, n.o.s.*

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3137	5F	Oxidizing solid, flammable, n.o.s.*
3100	5S	Oxidizing solid, self-heating, n.o.s.*
3087	5P	Oxidizing solid, toxic, n.o.s.*
3121	5W	Oxidizing solid, water-reactive, n.o.s.*
1072	2X	Oxygen, compressed
2190	2PX	Oxygen difluoride, compressed
3356	5L	Oxygen generator, chemical
1073	2X	Oxygen, refrigerated liquid
3509	9L	Packagings, discarded, empty, uncleaned
1263	3L	Paint
3066	8L	Paint
3470	8F	Paint, corrosive, flammable
3469		Paint, flammable, corrosive
	3CH	Packing Group I or II
	3CL	Packing Group III
1263	3L	Paint related material
3066	8L	Paint related material
3470	8F	Paint related material corrosive, flammable
3469		Paint related material, flammable, corrosive
	3CH	Packing Group I or II
	3CL	Packing Group III
1379	4L	Paper, unsaturated oil treated
2213	3L	Paraformaldehyde
1264	3L	Paraldehyde
1380	4P	Pentaborane
1669	6L	Pentachloroethane
3155	6L	Pentachlorophenol
0411	1L	Pentaerythrite tetranitrate
0150	1L	Pentaerythrite tetranitrate, desensitized
3344	3E	Pentaerythrite tetranitrate mixture desensitized, solid, n.o.s.*
0150	1L	Pentaerythrite tetranitrate, wetted
0411	1L	Pentaerythritol tetranitrate
0150	1L	Pentaerythritol tetranitrate, desensitized

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3344	3E	Pentaerythritol tetranitrate mixture desensitized, solid, n.o.s.*	1267	3L	Petroleum crude oil
0150	1L	Pentaerythritol tetranitrate, wetted	1268		Petroleum distillates, n.o.s.
3220	2L	Pentafluoroethane	3H		Packing Group I or II
2286	3L	Pentamethylheptane	3L		Packing Group III
2310	3P	Pentane-2,4-dione	1075	10L	Petroleum gases, liquefied
1265	3H	Pentanes	1268		Petroleum products, n.o.s.
1105	3L	Pentanols	3H		Packing Group I or II
1108	3H	1-Pentene	3L		Packing Group III
2705	8L	1-Pentol	3494	3P	Petroleum sour crude oil, flammable, toxic
0151	1L	Pentolite	2645	6i	Phenacyl bromide
1481	5L	Perchlorates, inorganic, n.o.s.	2311	6L	Phenetidines
3211	5L	Perchlorates, inorganic, aqueous solution, n.o.s.	2904	8L	Phenolates, liquid
1802	8X	Perchloric acid	2905	8L	Phenolates, solid
1873	5C	Perchloric acid	2312	6L	Phenol, molten
1670	6L	Perchloromethyl mercaptan	1671	6L	Phenol, solid
3083	2PX	Perchloryl fluoride	2821	6L	Phenol solution
3154	10L	Perfluoro (ethyl vinyl ether)	1803	8L	Phenolsulphonic acid, liquid
3153	10L	Perfluoro (methyl vinyl ether)	3346	3P	Phenoxyacetic acid derivative pesticide, liquid, flammable, toxic*
1266	3L	Perfumery products	3348	6L	Phenoxyacetic acid derivative pesticide, liquid, toxic*
1482	5L	Permanganates, inorganic, n.o.s.*	3347	6F	Phenoxyacetic acid derivative pesticide, liquid, toxic, flammable*
3214	5L	Permanganates, inorganic, aqueous solution, n.o.s.*	3345	6L	Phenoxyacetic acid derivative pesticide, solid, toxic*
1483	5L	Peroxides, inorganic, n.o.s.	2470	6L	Phenylacetone, liquid
3215	5L	Persulphates, inorganic, n.o.s.	2577	8i	Phenylacetyl chloride
3216	5L	Persulphates, inorganic, aqueous solution, n.o.s.	1672	6i	Phenylcarbamylamine chloride
3021	3P	Pesticide, liquid, flammable, toxic, n.o.s.*	2746	6C	Phenyl chloroformate
2902	6L	Pesticide, liquid, toxic, n.o.s.*	1673	6L	Phenylenediamines
2903	6F	Pesticide, liquid, toxic, flammable, n.o.s.*	2572	6L	Phenylhydrazine
2588	6L	Pesticide, solid, toxic, n.o.s.*	2487	6Fi	Phenyl isocyanate
0411	1L	PETN	2337	6F	Phenyl mercaptan
0150	1L	PETN, desensitized	1674	6L	Phenylmercuric acetate
3344	3E	PETN mixture desensitized, solid, n.o.s.*	2026	6L	Phenylmercuric compound, n.o.s.*
0150	1L	PETN, wetted	1894	6L	Phenylmercuric hydroxide
1203	3H	Petrol	1895	6L	Phenylmercuric nitrate
			2798	8L	Phenylphosphorus dichloride

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2799	8L	Phenylphosphorus thiodichloride
1804	8L	Phenyltrichlorosilane
1076	2CP	Phosgene
2940	4L	9-Phosphabicyclononanes
2199	10P	Phosphine
3525	10P	Phosphine, adsorbed
3453	8L	Phosphoric acid, solid
1805	8L	Phosphoric acid, solution
2834	8L	Phosphorous acid
1338	3L	Phosphorus, amorphous
1339	3L	Phosphorus heptasulphide
1939	8W	Phosphorus oxybromide
2576	8W	Phosphorus oxybromide, molten
1810	6C	Phosphorus oxychloride
2691	8W	Phosphorus pentabromide
1806	8W	Phosphorus pentachloride
2198	2CP	Phosphorus pentafluoride
3524	2CP	Phosphorus pentafluoride, adsorbed
1340	4FW	Phosphorus pentasulphide
1807	8W	Phosphorus pentoxide
1341	3W	Phosphorus sesquisulphide
1808	8W	Phosphorus tribromide
1809	6CW	Phosphorus trichloride
2578	8L	Phosphorus trioxide
1343	3W	Phosphorus trisulphide
1381	4P	Phosphorus, white, dry
1381	4P	Phosphorus, white, in solution
2447	4P	Phosphorus, white, molten
1381	4P	Phosphorus, white, under water
1381	4P	Phosphorus, yellow, dry
1381	4P	Phosphorus, yellow, in solution
1381	4P	Phosphorus, yellow, under water
2214	8L	Phthalic anhydride
2313	3L	Picolines
0153	1L	Picramide
0154	1L	Picric acid
1344	3E	Picric acid, wetted

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3364	3E	Picric acid, wetted
0282	1L	Picrite
1336	3E	Picrite, wetted
0155	1L	Picryl chloride
3365	3E	Picryl chloride, wetted
2368	3L	alpha-Pinene
1272	3L	Pine oil
2579	8L	Piperazine
2401	8F	Piperidine
3314	9L	Plastics moulding compound
2006	4L	Plastics, nitrocellulose-based, self-heating, n.o.s.*
2733	3C	Polyamines, flammable, corrosive, n.o.s.*
2735	8L	Polyamines, liquid, corrosive, n.o.s.*
2734	8F	Polyamines, liquid, corrosive, flammable, n.o.s.*
3259	8L	Polyamines, solid, corrosive, n.o.s.*
2315	9L	Polychlorinated biphenyls, liquid
3432	9L	Polychlorinated biphenyls, solid
3269	3L	Polyester resin kit
3527	3S	Polyester resin kit
3151	9L	Polyhalogenated biphenyls, liquid
3152	9L	Polyhalogenated biphenyls, solid
3151	9L	Polyhalogenated terphenyls, liquid
3152	9L	Polyhalogenated terphenyls, solid
2211	9L	Polymeric beads, expandable
3532	3L	Polymerizing substance, liquid, stabilized, n.o.s.*
3534	3L	Polymerizing substance, liquid, temperature controlled, n.o.s.*
3531	3L	Polymerizing substance, solid, stabilized, n.o.s.*
3533	3L	Polymerizing substance, solid, temperature controlled, n.o.s.*
2257	4W	Potassium
1677	6L	Potassium arsenate
1678	6L	Potassium arsenite
1870	4W	Potassium borohydride
1484	5L	Potassium bromate

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1485	5L	Potassium chlorate	0433	1L	Powder paste, wetted
2427	5L	Potassium chlorate, aqueous solution	0160	1L	Powder, smokeless
1679	6L	Potassium cuprocyanide	0161	1L	Powder, smokeless
1680	6L	Potassium cyanide, solid	0509	1L	Powder, smokeless
3413	6L	Potassium cyanide solution	0044	3L	Primers, cap type
1929	4L	Potassium dithionite	0377	1L	Primers, cap type
1812	6L	Potassium fluoride, solid	0378	1L	Primers, cap type
3422	6L	Potassium fluoride solution	0319	1L	Primers, tubular
2628	6L	Potassium fluoroacetate	0320	1L	Primers, tubular
2655	6L	Potassium fluorosilicate	0376	3L	Primers, tubular
1811	8P	Potassium hydrogendifluoride, solid	1210	3L	Printing ink
3421	8P	Potassium hydrogendifluoride solution	1210	3L	Printing ink related material
2509	8L	Potassium hydrogen sulphate	0167	1L	Projectiles
1929	4L	Potassium hydrosulphite	0168	1L	Projectiles
1813	8L	Potassium hydroxide, solid	0169	1L	Projectiles
1814	8L	Potassium hydroxide solution	0324	1L	Projectiles
1420	4W	Potassium metal alloys, liquid	0344	1L	Projectiles
3403	4W	Potassium metal alloys, solid	0345	3L	Projectiles
2864	6L	Potassium metavanadate	0346	1L	Projectiles
2033	8L	Potassium monoxide	0347	1L	Projectiles
1486	5L	Potassium nitrate	0424	1L	Projectiles
1487	5L	Potassium nitrate and sodium nitrite mixture	0425	1L	Projectiles
1488	5L	Potassium nitrite	0426	1L	Projectiles
1489	5L	Potassium perchlorate	0427	1L	Projectiles
1490	5L	Potassium permanganate	0434	1L	Projectiles
1491	5L	Potassium peroxide	0435	1L	Projectiles
1492	5L	Potassium persulphate	2200	10L	Propadiene, stabilized
2012	4PW	Potassium phosphide	1978	10L	Propane
1422	4W	Potassium sodium alloys, liquid	2402	3H	Propanethiols
3404	4W	Potassium sodium alloys, solid	1274	3L	n-Propanol
1382	4L	Potassium sulphide	0495	1L	Propellant, liquid
1382	4L	Potassium sulphide, anhydrous	0497	1L	Propellant, liquid
1847	8L	Potassium sulphide, hydrated	0498	1L	Propellant, solid
2466	5L	Potassium superoxide	0499	1L	Propellant, solid
0159	1L	Powder cake, wetted	0501	1L	Propellant, solid
0433	1L	Powder cake, wetted	1275	3H	Propionaldehyde
0159	1L	Powder paste, wetted	1848	8L	Propionic acid

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3463	8F	Propionic acid	2908	7L	Radioactive material, excepted package — empty packaging
2496	8L	Propionic anhydride	2911	7L	Radioactive material, excepted package — instruments or articles
2404	3P	Propionitrile	2910	7L	Radioactive material, excepted package — limited quantity of material
1815	3C	Propionyl chloride	2912	7L	Radioactive material, low specific activity (LSA-I)
1276	3L	n-Propyl acetate	3321	7L	Radioactive material, low specific activity (LSA-II)
1274	3L	Propyl alcohol, normal	3324	7L	Radioactive material, low specific activity (LSA-II), fissile
1277	3CH	Propylamine	3322	7L	Radioactive material, low specific activity (LSA-III)
2364	3L	n-Propylbenzene	3325	7L	Radioactive material, low specific activity (LSA-III), fissile
2740	6CF	n-Propyl chloroformate	3326	7L	Radioactive material, surface contaminated objects (SCO-I or SCO-II), fissile
1077	10L	Propylene	2913	7L	Radioactive material, surface contaminated objects (SCO-I, SCO-II or SCO-III)
2611	6F	Propylene chlorohydrin	2919	7L	Radioactive material, transported under special arrangement
2258	8F	1,2-Propylenediamine	3331	7L	Radioactive material, transported under special arrangement, fissile
1921	3HP	Propyleneimine, stabilized	2915	7L	Radioactive material, Type A package
1280	3H	Propylene oxide	3327	7L	Radioactive material, Type A package, fissile
2850	3L	Propylene tetramer	3332	7L	Radioactive material, Type A package, special form
1281	3H	Propyl formates	3333	7L	Radioactive material, Type A package, special form, fissile
2482	6F	n-Propyl isocyanate	2917	7L	Radioactive material, Type B(M) package
1865	3L	n-Propyl nitrate	3329	7L	Radioactive material, Type B(M) package, fissile
1816	8F	Propyltrichlorosilane	2916	7L	Radioactive material, Type B(U) package
3350	3P	Pyrethroid pesticide, liquid flammable, toxic*	3328	7L	Radioactive material, Type B(U) package, fissile
3352	6L	Pyrethroid pesticide, liquid, toxic*	3323	7L	Radioactive material, Type C package
3351	6F	Pyrethroid pesticide, liquid, toxic, flammable*	3330	7L	Radioactive material, Type C package, fissile
3349	6L	Pyrethroid pesticide, solid, toxic*	2978	7CP	Radioactive material, uranium hexafluoride
1282	3L	Pyridine			
1383	4L	Pyrophoric alloy, n.o.s.*			
3194	4W	Pyrophoric liquid, inorganic, n.o.s.*			
2845	4W	Pyrophoric liquid, organic, n.o.s.*			
1383	4L	Pyrophoric metal, n.o.s.*			
3200	4L	Pyrophoric solid, inorganic, n.o.s.*			
2846	4L	Pyrophoric solid, organic, n.o.s.*			
1817	8W	Pyrosulphuryl chloride			
1922	3C	Pyrrolidine			
2656	6L	Quinoline			
2909	7L	Radioactive material, excepted package — articles manufactured from natural uranium or depleted uranium or natural thorium			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2977	7CP	Radioactive material, uranium hexafluoride, fissile	1983	2L	Refrigerant gas R 133a
1856	4L	Rags, oily	3159	2L	Refrigerant gas R 134a
0391	1L	RDX and cyclotetramethylenetetranitramine mixture, desensitized	2517	10L	Refrigerant gas R 142b
0391	1L	RDX and cyclotetramethylenetetranitramine mixture, wetted	2035	10L	Refrigerant gas R 143a
0483	1L	RDX, desensitized	1030	10L	Refrigerant gas R 152a
0072	1L	RDX, wetted	2453	10L	Refrigerant gas R 161
2037		Receptacles, small, containing gas	2424	2L	Refrigerant gas R 218
	10L	Division 2.1 without subsidiary risk	3296	2L	Refrigerant gas R 227
	2L	Division 2.2 without subsidiary risk	3337	2L	Refrigerant gas R 404A
	2X	Division 2.2 with subsidiary risk 5.1	3338	2L	Refrigerant gas R 407A
	2P	Division 2.3 without subsidiary risk	3339	2L	Refrigerant gas R 407B
	10P	Division 2.3 with subsidiary risk 2.1	3340	2L	Refrigerant gas R 407C
	10C	Division 2.3 with subsidiary risk 2.1 and 8	2602	2L	Refrigerant gas R 500
	2PX	Division 2.3 with subsidiary risk 5.1	1973	2L	Refrigerant gas R 502
	2CX	Division 2.3 with subsidiary risk 5.1 and 8	2599	2L	Refrigerant gas R 503
	2CP	Division 2.3 with subsidiary risk 8	1082	10P	Refrigerant gas R 1113
1078	2L	Refrigerant gas, n.o.s.*	1959	10L	Refrigerant gas R 1132a
1028	2L	Refrigerant gas R 12	1858	2L	Refrigerant gas R 1216
1974	2L	Refrigerant gas R 12B1	2422	2L	Refrigerant gas R 1318
1022	2L	Refrigerant gas R 13	1976	2L	Refrigerant gas R C318
1009	2L	Refrigerant gas R 13B1	2857	2L	Refrigerating machines
1982	2A	Refrigerant gas R 14	3358	10L	Refrigerating machines
1029	2L	Refrigerant gas R 21	3291	11L	Regulated medical waste, n.o.s.
1018	2L	Refrigerant gas R 22	0173	3L	Release devices, explosive
1984	2A	Refrigerant gas R 23	1866	3L	Resin solution
3252	10L	Refrigerant gas R 32	2876	6L	Resorcinol
1063	10L	Refrigerant gas R 40	0174	3L	Rivets, explosive
2454	10L	Refrigerant gas R 41	0186	1L	Rocket motors
1958	2L	Refrigerant gas R 114	0280	1L	Rocket motors
1020	2L	Refrigerant gas R 115	0281	1L	Rocket motors
2193	2L	Refrigerant gas R 116	0510	1L	Rocket motors
1021	2L	Refrigerant gas R 124	0395	1L	Rocket motors, liquid fuelled
3220	2L	Refrigerant gas R 125	0396	1L	Rocket motors, liquid fuelled
			0250	1L	Rocket motors with hypergolic liquids
			0322	1L	Rocket motors with hypergolic liquids
			0180	1L	Rockets
			0181	1L	Rockets

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0182	1L	Rockets
0183	1L	Rockets
0295	1L	Rockets
0436	1L	Rockets
0437	1L	Rockets
0438	1L	Rockets
0502	1L	Rockets
0238	1L	Rockets, line-throwing
0240	1L	Rockets, line-throwing
0453	1L	Rockets, line-throwing
0397	1L	Rockets, liquid fuelled
0398	1L	Rockets, liquid fuelled
1286	3L	Rosin oil
1345	3L	Rubber scrap
1345	3L	Rubber shoddy
1287	3L	Rubber solution
1423	4W	Rubidium
2678	8L	Rubidium hydroxide
2677	8L	Rubidium hydroxide solution
3268	9L	Safety devices
0503	1L	Safety devices, pyrotechnic
0190	1L	Samples, explosive*
1386	4L	Seed cake
2217	4L	Seed cake
2630	6L	Selenates*
1905	8L	Selenic acid
2630	6L	Selenites*
3440	6L	Selenium compound, liquid, n.o.s.*
3283	6L	Selenium compound, solid, n.o.s.*
2657	6L	Selenium disulphide
2194	2CP	Selenium hexafluoride
2879	8P	Selenium oxychloride
3188	4C	Self-heating liquid, corrosive, inorganic, n.o.s.*
3185	4C	Self-heating liquid, corrosive, organic, n.o.s.*
3186	4L	Self-heating liquid, inorganic, n.o.s.*
3183	4L	Self-heating liquid, organic, n.o.s.*

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3187	4P	Self-heating liquid, toxic, inorganic, n.o.s.*
3184	4P	Self-heating liquid, toxic, organic, n.o.s.*
3192	4C	Self-heating solid, corrosive, inorganic, n.o.s.*
3126	4C	Self-heating solid, corrosive, organic, n.o.s.*
3190	4L	Self-heating solid, inorganic, n.o.s.*
3088	4L	Self-heating solid, organic, n.o.s.*
3127	4X	Self-heating solid, oxidizing, n.o.s.*
3191	4P	Self-heating solid, toxic, inorganic, n.o.s.*
3128	4P	Self-heating solid, toxic, organic n.o.s.*
3223	3L	Self-reactive liquid type C*
3233	3S	Self-reactive liquid type C, temperature controlled*
3225	3L	Self-reactive liquid type D*
3235	3S	Self-reactive liquid type D, temperature controlled*
3227	3L	Self-reactive liquid type E*
3237	3S	Self-reactive liquid type E, temperature controlled*
3229	3L	Self-reactive liquid type F*
3239	3S	Self-reactive liquid type F, temperature controlled*
3224	3L	Self-reactive solid type C*
3234	3S	Self-reactive solid type C, temperature controlled*
3226	3L	Self-reactive solid type D*
3236	3S	Self-reactive solid type D, temperature controlled*
3228	3L	Self-reactive solid type E*
3238	3S	Self-reactive solid type E, temperature controlled*
3230	3L	Self-reactive solid type F*
3240	3S	Self-reactive solid type F, temperature controlled*
1288	3L	Shale oil
0191	1L	Signal devices, hand
0373	3L	Signal devices, hand
0194	1L	Signals, distress

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0195	1L	Signals, distress	3378	5L	Sodium carbonate peroxyhydrate
0505	1L	Signals, distress	1495	5L	Sodium chlorate
0506	3L	Signals, distress	2428	5L	Sodium chlorate, aqueous solution
0192	1L	Signals, railway track, explosive	1496	5L	Sodium chloride
0193	3L	Signals, railway track, explosive	2659	6L	Sodium chloroacetate
0492	1L	Signals, railway track, explosive	2316	6L	Sodium cuprocyanide, solid
0493	1L	Signals, railway track, explosive	2317	6L	Sodium cuprocyanide solution
0196	1L	Signals, smoke	1689	6L	Sodium cyanide, solid
0197	1L	Signals, smoke	3414	6L	Sodium cyanide solution
0313	1L	Signals, smoke	0234	1L	Sodium dinitro-o-cresolate
0487	1L	Signals, smoke	1348	3EP	Sodium dinitro-o-cresolate, wetted
0507	3L	Signals, smoke	3369	3E	Sodium dinitro-o-cresolate, wetted
2203	10L	Silane	1384	4L	Sodium dithionite
1346	3L	Silicon powder, amorphous	1690	6L	Sodium fluoride, solid
1818	8L	Silicon tetrachloride	3415	6L	Sodium fluoride solution
1859	2CP	Silicon tetrafluoride	2629	6L	Sodium fluoroacetate
3521	2CP	Silicon tetrafluoride, adsorbed	2674	6L	Sodium fluorosilicate
1683	6L	Silver arsenite	1427	4W	Sodium hydride
1684	6L	Silver cyanide	2439	8L	Sodium hydrogendifluoride
1493	5L	Silver nitrate	2318	4L	Sodium hydrosulphide
1347	3E	Silver picrate, wetted	2949	8L	Sodium hydrosulphide, hydrated
1906	8L	Sludge acid	1384	4L	Sodium hydrosulphite
1907	8L	Soda lime	1823	8L	Sodium hydroxide, solid
1428	4W	Sodium	1824	8L	Sodium hydroxide solution
2812	8L	Sodium aluminate, solid	3551	12FZ	Sodium ion batteries
1819	8L	Sodium aluminate solution	3552	12FZ	Sodium ion batteries contained in equipment
2835	4W	Sodium aluminium hydride	1431	4C	Sodium methylate
2863	6L	Sodium ammonium vanadate	1289	3C	Sodium methylate solution
2473	6L	Sodium arsanilate	1825	8L	Sodium monoxide
1685	6L	Sodium arsenate	1498	5L	Sodium nitrate
1686	6L	Sodium arsenite, aqueous solution	1499	5L	Sodium nitrate and potassium nitrate mixture
2027	6L	Sodium arsenite, solid	1500	5P	Sodium nitrite
1687	6L	Sodium azide	2567	6L	Sodium pentachlorophenate
1426	4W	Sodium borohydride	3377	5L	Sodium perborate monohydrate
3320	8L	Sodium borohydride and sodium hydroxide solution	1502	5L	Sodium perchlorate
1494	5L	Sodium bromate	1503	5L	Sodium permanganate
1688	6L	Sodium cacodylate			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
1504	5L	Sodium peroxide
3247	5L	Sodium peroxoborate, anhydrous
1505	5L	Sodium persulphate
1432	4PW	Sodium phosphide
0235	1L	Sodium picramate
1349	3E	Sodium picramate, wetted
1385	4L	Sodium sulphide
1385	4L	Sodium sulphide, anhydrous
1849	8L	Sodium sulphide, hydrated
2547	5L	Sodium superoxide
3244	8L	Solids containing corrosive liquid, n.o.s.*
3175	3L	Solids containing flammable liquid, n.o.s.*
3243	6L	Solids containing toxic liquid, n.o.s.*
0204	1L	Sounding devices, explosive
0296	1L	Sounding devices, explosive
0374	1L	Sounding devices, explosive
0375	1L	Sounding devices, explosive
1827	8W	Stannic chloride, anhydrous
2440	8L	Stannic chloride pentahydrate
1433	4PW	Stannic phosphides
2676	10P	Stibine
1327	3L	Straw
1691	6L	Strontium arsenite
1506	5L	Strontium chlorate
1507	5L	Strontium nitrate
1508	5L	Strontium perchlorate
1509	5L	Strontium peroxide
2013	4PW	Strontium phosphide
1692	6L	Strychnine
1692	6L	Strychnine salts
0219	1L	Styphnic acid
0394	1L	Styphnic acid, wetted
2055	3L	Styrene monomer, stabilized
0482	1L	Substances, EVI, n.o.s.*
0357	1L	Substances, explosive, n.o.s.*
0358	1L	Substances, explosive, n.o.s.*

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0359	1L	Substances, explosive, n.o.s.*
0473	1L	Substances, explosive, n.o.s.*
0474	1L	Substances, explosive, n.o.s.*
0475	1L	Substances, explosive, n.o.s.*
0476	1L	Substances, explosive, n.o.s.*
0477	1L	Substances, explosive, n.o.s.*
0478	1L	Substances, explosive, n.o.s.*
0479	1L	Substances, explosive, n.o.s.*
0480	1L	Substances, explosive, n.o.s.*
0481	3L	Substances, explosive, n.o.s.*
0485	1L	Substances, explosive, n.o.s.*
0482	1L	Substances, explosive, very insensitive, n.o.s.*
2780	3P	Substituted nitrophenol pesticide, liquid, flammable, toxic*
3014	6L	Substituted nitrophenol pesticide, liquid, toxic*
3013	6F	Substituted nitrophenol pesticide, liquid, toxic, flammable*
2779	6L	Substituted nitrophenol pesticide, solid, toxic*
2967	8L	Sulphamic acid
1350	3L	Sulphur
1828	8W	Sulphur chlorides
1079	2CP	Sulphur dioxide
1080	2L	Sulphur hexafluoride
1830	8L	Sulphuric acid
2796	8L	Sulphuric acid
1831	8P	Sulphuric acid, fuming
1832	8L	Sulphuric acid, spent
2448	3L	Sulphur, molten
1833	8L	Sulphurous acid
2418	2CP	Sulphur tetrafluoride
1829	8L	Sulphur trioxide, stabilized
1834	6C	Sulphuryl chloride
2191	2P	Sulphuryl fluoride
1999	3L	Tars, liquid
1700	6F	Tear gas candles
1693	6i	Tear gas substance, liquid, n.o.s.*

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3448	6L	Tear gas substance, solid, n.o.s.*	2772	3P	Thiocarbamate pesticide, liquid, flammable, toxic*
3284	6L	Tellurium compound, n.o.s.*	3006	6L	Thiocarbamate pesticide, liquid, toxic*
2195	2CP	Tellurium hexafluoride	3005	6F	Thiocarbamate pesticide, liquid, toxic, flammable*
2319	3L	Terpene hydrocarbons, n.o.s.	2771	6L	Thiocarbamate pesticide, solid, toxic*
2541	3L	Terpinolene	2966	6L	Thioglycol
2504	6L	Tetrabromoethane	1940	8L	Thioglycolic acid
1702	6L	1,1,2,2-Tetrachloroethane	2936	6L	Thiolactic acid
1897	6L	Tetrachloroethylene	1836	8W	Thionyl chloride
1704	6L	Tetraethyl dithiopyrophosphate	2414	3H	Thiophene
2320	8L	Tetraethylenepentamine	2474	6L	Thiophosgene
1292	3L	Tetraethyl silicate	1837	8W	Thiophosphoryl chloride
3159	2L	1,1,1,2-Tetrafluoroethane	3341	4L	Thiourea dioxide
1081	10L	Tetrafluoroethylene, stabilized	1293	3L	Tinctures, medicinal
1982	2A	Tetrafluoromethane	—	2L	Tire assemblies inflated, unserviceable, damaged or above maximum rated pressure
2498	3L	1,2,3,6-Tetrahydrobenzaldehyde	3174	4L	Titanium disulphide
2056	3H	Tetrahydrofuran	1871	3W	Titanium hydride
2943	3L	Tetrahydrofurfurylamine	2546	4L	Titanium powder, dry
2698	8L	Tetrahydrophthalic anhydrides	1352	3L	Titanium powder, wetted
2410	3L	1,2,3,6-Tetrahydropyridine	2878	3L	Titanium sponge granules
2412	3L	Tetrahydrothiophene	2878	3L	Titanium sponge powders
3423	8L	Tetramethylammonium hydroxide, solid	1838	6C	Titanium tetrachloride
1835	8L	Tetramethylammonium hydroxide solution	2869	8L	Titanium trichloride mixture
2749	3H	Tetramethylsilane	2441	4C	Titanium trichloride mixture, pyrophoric
0207	1L	Tetranitroaniline	2441	4C	Titanium trichloride, pyrophoric
1510	6X	Tetranitromethane	0209	1L	TNT
2413	3L	Tetrapropyl orthotitanate	0388	1L	TNT and hexanitrostilbene mixture
0114	1L	Tetrazene, wetted	0388	1L	TNT and trinitrobenzene mixture
0407	1L	Tetrazol-1-acetic acid	0389	1L	TNT mixture containing trinitrobenzene and hexanitrostilbene
0504	1L	1H-Tetrazole	1356	3E	TNT, wetted
0208	1L	Tetryl	3366	3E	TNT, wetted
1857	4L	Textile waste, wet	1294	3L	Toluene
2573	5P	Thallium chlorate	2078	6L	Toluene diisocyanate
1707	6L	Thallium compound, n.o.s.*	1708	6L	Toluidines, liquid
2727	6X	Thallium nitrate			
2785	6L	4-Thiapentanal			
2436	3i	Thioacetic acid			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3451	6L	Toluidines, solid
1709	6L	2,4-Toluylenediamine, solid
3418	6L	2,4-Toluylenediamine solution
0329	1L	Torpedoes
0330	1L	Torpedoes
0451	1L	Torpedoes
0449	1L	Torpedoes, liquid fuelled
0450	1L	Torpedoes, liquid fuelled
3381	6L	Toxic by inhalation liquid, n.o.s.*
3382	6L	Toxic by inhalation liquid, n.o.s.*
3389	6C	Toxic by inhalation liquid, corrosive, n.o.s.*
3390	6C	Toxic by inhalation liquid, corrosive, n.o.s.*
3383	6F	Toxic by inhalation liquid, flammable, n.o.s.*
3384	6F	Toxic by inhalation liquid, flammable, n.o.s.*
3488	6CF	Toxic by inhalation liquid, flammable, corrosive, n.o.s.*
3489	6CF	Toxic by inhalation liquid, flammable, corrosive, n.o.s.*
3387	6X	Toxic by inhalation liquid, oxidizing, n.o.s.*
3388	6X	Toxic by inhalation liquid, oxidizing, n.o.s.*
3385	6W	Toxic by inhalation liquid, water-reactive, n.o.s.*
3386	6W	Toxic by inhalation liquid, water-reactive, n.o.s.*
3490	6FW	Toxic by inhalation liquid, water-reactive, flammable, n.o.s.*
3491	6FW	Toxic by inhalation liquid, water-reactive, flammable, n.o.s.*
3289	6C	Toxic liquid, corrosive, inorganic, n.o.s.*
2927	6C	Toxic liquid, corrosive, organic, n.o.s.*
2929	6F	Toxic liquid, flammable, organic, n.o.s.*
3287	6L	Toxic liquid, inorganic, n.o.s.*
2810	6L	Toxic liquid, organic, n.o.s.*
3122	6X	Toxic liquid, oxidizing, n.o.s.*
3123	6W	Toxic liquid, water-reactive, n.o.s.*

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3290	6C	Toxic solid, corrosive, inorganic, n.o.s.*
2928	6C	Toxic solid, corrosive, organic, n.o.s.*
3535	6F	Toxic solid, flammable, inorganic, n.o.s.*
2930	6F	Toxic solid, flammable, organic, n.o.s.*
3288	6L	Toxic solid, inorganic, n.o.s.*
2811	6L	Toxic solid, organic, n.o.s.*
3086	6X	Toxic solid, oxidizing, n.o.s.*
3124	6S	Toxic solid, self-heating, n.o.s.*
3125	6W	Toxic solid, water-reactive, n.o.s.*
3172	6L	Toxins, extracted from living sources, liquid, n.o.s.*
3462	6L	Toxins, extracted from living sources, solid, n.o.s.*
0212	1L	Tracers for ammunition
0306	1L	Tracers for ammunition
2610	3C	Triallylamine
2609	6L	Triallyl borate
2764	3P	Triazine pesticide, liquid, flammable, toxic*
2998	6L	Triazine pesticide, liquid, toxic*
2997	6F	Triazine pesticide, liquid, toxic, flammable*
2763	6L	Triazine pesticide, solid, toxic*
2542	6L	Tributylamine
3254	4L	Tributylphosphane
1839	8L	Trichloroacetic acid
2564	8L	Trichloroacetic acid solution
2442	8W	Trichloroacetyl chloride
2321	6L	Trichlorobenzenes, liquid
2322	6L	Trichlorobutene
2831	6L	1,1,1-Trichloroethane
1710	6A	Trichloroethylene
2468	5L	Trichloroisocyanuric acid, dry
1295	4HW	Trichlorosilane
2574	6L	Tricresyl phosphate
1296	3CH	Triethylamine
2259	8L	Triethylenetetramine

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2323	3L	Triethyl phosphite	0218	1L	Trinitrophenetole
2699	8N	Trifluoroacetic acid	0154	1L	Trinitrophenol
3057	2CP	Trifluoroacetyl chloride	1344	3E	Trinitrophenol, wetted
1082	10P	Trifluorochloroethylene, stabilized	3364	3E	Trinitrophenol, wetted
2035	10L	1,1,1-Trifluoroethane	0208	1L	Trinitrophenylmethylnitramine
1984	2A	Trifluoromethane	0219	1L	Trinitroresorcinol
3136	2A	Trifluoromethane, refrigerated liquid	0394	1L	Trinitroresorcinol, wetted
2942	6L	2-Trifluoromethylaniline	0209	1L	Trinitrotoluene
2948	6L	3-Trifluoromethylaniline	0388	1L	Trinitrotoluene and hexanitrostilbene mixture
2324	3L	Triisobutylene	0388	1L	Trinitrotoluene and trinitrobenzene mixture
2616	3L	Triisopropyl borate	0389	1L	Trinitrotoluene mixture containing trinitrobenzene and hexanitrostilbene
2438	6FW	Trimethylacetyl chloride	1356	3E	Trinitrotoluene, wetted
1083	10L	Trimethylamine, anhydrous	3366	3E	Trinitrotoluene, wetted
1297		Trimethylamine, aqueous solution	2260	3C	Tripropylamine
	3CH	Packing Group I or II	2057	3L	Tripropylene
	3C	Packing Group III	2501	6L	Tris-(1-aziridinyl) phosphine oxide solution
2325	3L	1,3,5-Trimethylbenzene	0390	1L	Tritonal
2416	3L	Trimethyl borate	2196	2CP	Tungsten hexafluoride
1298	3CH	Trimethylchlorosilane	1299	3L	Turpentine
2326	8L	Trimethylcyclohexylamine	1300	3L	Turpentine substitute
2327	8L	Trimethylhexamethylenediamines	—	2L	Tyre assemblies inflated, unserviceable, damaged or above maximum rated pressure
2328	6L	Trimethylhexamethylene diisocyanate	2330	3L	Undecane
2329	3L	Trimethyl phosphite	3507	6C	Uranium hexafluoride, radioactive material, excepted package
0153	1L	Trinitroaniline	1511	5C	Urea hydrogen peroxide
0213	1L	Trinitroanisole	0220	1L	Urea nitrate
0214	1L	Trinitrobenzene	1357	3E	Urea nitrate, wetted
0386	1L	Trinitrobenzenesulphonic acid	3370	3E	Urea nitrate, wetted
1354	3E	Trinitrobenzene, wetted	2058	3L	Valeraldehyde
3367	3E	Trinitrobenzene, wetted	2502	8FW	Valeryl chloride
0215	1L	Trinitrobenzoic acid	3285	6L	Vanadium compound, n.o.s.*
1355	3E	Trinitrobenzoic acid, wetted	2443	8W	Vanadium oxytrichloride
3368	3E	Trinitrobenzoic acid, wetted	2862	6L	Vanadium pentoxide
0155	1L	Trinitrochlorobenzene	2444	8W	Vanadium tetrachloride
3365	3E	Trinitrochlorobenzene, wetted			
0216	1L	Trinitro-m-cresol			
0387	1L	Trinitrofluorenone			
0217	1L	Trinitronaphthalene			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2475	8W	Vanadium trichloride	1387	4L	Wool waste, wet
2931	6L	Vanadyl sulphate	3342	4L	Xanthates
3166	9L	Vehicle, flammable gas powered	2036	2L	Xenon
3166	9L	Vehicle, flammable liquid powered	2591	2L	Xenon, refrigerated liquid
3166	9L	Vehicle, fuel cell, flammable gas powered	1307	3L	Xylenes
3166	9L	Vehicle, fuel cell, flammable liquid powered	3430	6L	Xylenols, liquid
1301	3H	Vinyl acetate, stabilized	2261	6L	Xylenols, solid
1085	10L	Vinyl bromide, stabilized	1711	6L	Xylidines, liquid
2838	3L	Vinyl butyrate, stabilized	3452	6L	Xylidines, solid
1086	10L	Vinyl chloride, stabilized	1701	6L	Xylyl bromide, liquid
2589	6F	Vinyl chloroacetate	3417	6L	Xylyl bromide, solid
1302	3H	Vinyl ethyl ether, stabilized	1512	5L	Zinc ammonium nitrite
1860	10L	Vinyl fluoride, stabilized	1712	6L	Zinc arsenate
1303	3H	Vinylidene chloride, stabilized	1712	6L	Zinc arsenate and zinc arsenite mixture
1304	3H	Vinyl isobutyl ether, stabilized	1712	6L	Zinc arsenite
1087	10L	Vinyl methyl ether, stabilized	1435	4W	Zinc ashes
3073	6CF	Vinylpyridines, stabilized	2469	5L	Zinc bromate
2618	3L	Vinyltoluenes, stabilized	1513	5L	Zinc chlorate
1305	3CH	Vinyltrichlorosilane	2331	8L	Zinc chloride, anhydrous
0286	1L	Warheads, rocket	1840	8L	Zinc chloride solution
0287	1L	Warheads, rocket	1713	6L	Zinc cyanide
0369	1L	Warheads, rocket	1931	9L	Zinc dithionite
0370	1L	Warheads, rocket	1436	4SW	Zinc dust
0371	1L	Warheads, rocket	2855	6L	Zinc fluorosilicate
0221	1L	Warheads, torpedo	1931	9L	Zinc hydrosulphate
3148	4W	Water-reactive liquid, n.o.s.*	1514	5L	Zinc nitrate
3129	4CW	Water-reactive liquid, corrosive, n.o.s.*	1515	5L	Zinc permanganate
3130	4PW	Water-reactive liquid, toxic, n.o.s.*	1516	5L	Zinc peroxide
2813	4W	Water-reactive solid, n.o.s.*	1714	4PW	Zinc phosphide
3131	4CW	Water-reactive solid, corrosive, n.o.s.*	1436	4SW	Zinc powder
3132	4FW	Water-reactive solid, flammable, n.o.s.*	2714	3L	Zinc resinate
3133	4WX	Water-reactive solid, oxidizing, n.o.s.*	2009	4L	Zirconium, dry
3135	4SW	Water-reactive solid, self-heating, n.o.s.*	2858	3L	Zirconium, dry
3134	4PW	Water-reactive solid, toxic, n.o.s.*	1437	3L	Zirconium hydride
1306	3L	Wood preservatives, liquid	2728	5L	Zirconium nitrate
			0236	1L	Zirconium picramate
			1517	3E	Zirconium picramate, wetted

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
2008	4L	Zirconium powder, dry
1358	3L	Zirconium powder, wetted
1932	4L	Zirconium scrap
1308		Zirconium suspended in a flammable liquid
	3H	Packing Group I or II
	3L	Packing Group III
2503	8L	Zirconium tetrachloride

Table 4 Numerical List of Dangerous Goods with Drill Codes

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
—	2L	Tire assemblies inflated, unserviceable, damaged or above maximum rated pressure	0028	1L	Black powder in pellets
—	2L	Tyre assemblies inflated, unserviceable, damaged or above maximum rated pressure	0028	1L	Gunpowder, compressed
0004	1L	Ammonium picrate	0028	1L	Gunpowder in pellets
0005	1L	Cartridges for weapons	0029	1L	Detonators, non-electric
0006	1L	Cartridges for weapons	0030	1L	Detonators, electric
0007	1L	Cartridges for weapons	0033	1L	Bombs
0009	1L	Ammunition, incendiary	0034	1L	Bombs
0010	1L	Ammunition, incendiary	0035	1L	Bombs
0012	3L	Cartridges for weapons, inert projectile	0037	1L	Bombs, photo-flash
0012	3L	Cartridges, small arms	0038	1L	Bombs, photo-flash
0014	3L	Cartridges for tools, blank	0039	1L	Bombs, photo-flash
0014	3L	Cartridges for weapons, blank	0042	1L	Boosters
0014	3L	Cartridges, small arms, blank	0043	1L	Bursters
0015	1L	Ammunition, smoke	0044	3L	Primers, cap type
0016	1L	Ammunition, smoke	0048	1L	Charges, demolition
0018	1CP	Ammunition, tear-producing	0049	1L	Cartridges, flash
0019	1CP	Ammunition, tear-producing	0050	1L	Cartridges, flash
0020	1P	Ammunition, toxic*	0054	1L	Cartridges, signal
0021	1P	Ammunition, toxic*	0055	3L	Cases, cartridge, empty, with primer
0027	1L	Black powder	0056	1L	Charges, depth
0027	1L	Gunpowder	0059	1L	Charges, shaped
0028	1L	Black powder, compressed	0060	1L	Charges, supplementary, explosive
			0065	1L	Cord, detonating
			0066	1L	Cord, igniter
			0070	3L	Cutters, cable, explosive
			0072	1L	Cyclonite, wetted
			0072	1L	Cyclotrimethylenetrinitramine, wetted

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0072	1L	Hexogen, wetted	0124	1L	Jet perforating guns, charged
0072	1L	RDX, wetted	0129	1L	Lead azide, wetted
0073	1L	Detonators for ammunition	0130	1L	Lead styphnate, wetted
0074	1L	Diazodinitrophenol, wetted	0130	1L	Lead trinitroresorcinate, wetted
0075	1L	Diethyleneglycol dinitrate, desensitized	0131	3L	Lighters, fuse
0076	1P	Dinitrophenol	0132	1L	Deflagrating metal salts of aromatic nitro-derivatives, n.o.s.
0077	1P	Dinitrophenolates	0133	1L	Mannitol hexanitrate, wetted
0078	1L	Dinitroresorcinol	0133	1L	Nitromannite, wetted
0079	1L	Dipicrylamine	0135	1L	Mercury fulminate, wetted
0079	1L	Hexanitrodiphenylamine	0136	1L	Mines
0079	1L	Hexyl	0137	1L	Mines
0081	1L	Explosive, blasting, type A	0138	1L	Mines
0082	1L	Explosive, blasting, type B	0143	1P	Nitroglycerin, desensitized
0083	1L	Explosive, blasting, type C	0144	1L	Nitroglycerin solution in alcohol
0084	1L	Explosive, blasting, type D	0146	1L	Nitrostarch
0092	1L	Flares, surface	0147	1L	Nitro urea
0093	1L	Flares, aerial	0150	1L	Pentaerythrite tetranitrate, desensitized
0094	1L	Flash powder	0150	1L	Pentaerythrite tetranitrate, wetted
0099	1L	Fracturing devices, explosive	0150	1L	Pentaerythritol tetranitrate, desensitized
0101	1L	Fuse, non-detonating	0150	1L	Pentaerythritol tetranitrate, wetted
0102	1L	Cord, detonating	0150	1L	PETN, desensitized
0102	1L	Fuse, detonating	0150	1L	PETN, wetted
0103	1L	Fuse, igniter	0151	1L	Pentolite
0104	1L	Cord, detonating, mild effect	0153	1L	Picramide
0104	1L	Fuse, detonating, mild effect	0153	1L	Trinitroaniline
0105	3L	Fuse, safety	0154	1L	Picric acid
0106	1L	Fuzes, detonating	0154	1L	Trinitrophenol
0107	1L	Fuzes, detonating	0155	1L	Picryl chloride
0110	3L	Grenades, practice	0155	1L	Trinitrochlorobenzene
0113	1L	Guanyl nitrosaminoguanylidene hydrazine, wetted	0159	1L	Powder cake, wetted
0114	1L	Guanyl nitrosaminoguanyltetrazene, wetted	0159	1L	Powder paste, wetted
0114	1L	Tetrazene, wetted	0160	1L	Powder, smokeless
0118	1L	Hexolite	0161	1L	Powder, smokeless
0118	1L	Hexotol	0167	1L	Projectiles
0121	1L	Igniters	0168	1L	Projectiles

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0169	1L	Projectiles	0226	1L	Cyclotetramethylenetetranitramine, wetted
0171	1L	Ammunition, illuminating	0226	1L	HMX, wetted
0173	3L	Release devices, explosive	0226	1L	Octogen, wetted
0174	3L	Rivets, explosive	0234	1L	Sodium dinitro-o-cresolate
0180	1L	Rockets	0235	1L	Sodium picramate
0181	1L	Rockets	0236	1L	Zirconium picramate
0182	1L	Rockets	0237	1L	Charges, shaped, flexible, linear
0183	1L	Rockets	0238	1L	Rockets, line-throwing
0186	1L	Rocket motors	0241	1L	Explosive, blasting, type E
0190	1L	Samples, explosive*	0242	1L	Charges, propelling, for cannon
0191	1L	Signal devices, hand	0243	1L	Ammunition, incendiary, white phosphorus
0192	1L	Signals, railway track, explosive	0245	1L	Ammunition, smoke, white phosphorus
0193	3L	Signals, railway track, explosive	0246	1L	Ammunition, smoke, white phosphorus
0194	1L	Signals, distress	0247	1L	Ammunition, incendiary
0195	1L	Signals, distress	0248	1L	Contrivances, water-activated*
0196	1L	Signals, smoke	0250	1L	Rocket motors with hypergolic liquids
0197	1L	Signals, smoke	0254	1L	Ammunition, illuminating
0204	1L	Sounding devices, explosive	0255	1L	Detonators, electric
0207	1L	Tetranitroaniline	0257	1L	Fuzes, detonating
0208	1L	Tetryl	0266	1L	Octol
0208	1L	Trinitrophenylmethylnitramine	0266	1L	Octolite
0209	1L	TNT	0267	1L	Detonators, non-electric
0209	1L	Trinitrotoluene	0268	1L	Boosters with detonator
0212	1L	Tracers for ammunition	0271	1L	Charges, propelling
0213	1L	Trinitroanisole	0275	1L	Cartridges, power device
0214	1L	Trinitrobenzene	0276	1L	Cartridges, power device
0215	1L	Trinitrobenzoic acid	0277	1L	Cartridges, oil well
0216	1L	Trinitro-m-cresol	0278	1L	Cartridges, oil well
0217	1L	Trinitronaphthalene	0279	1L	Charges, propelling, for cannon
0218	1L	Trinitrophenetole	0280	1L	Rocket motors
0219	1L	Styphnic acid	0281	1L	Rocket motors
0219	1L	Trinitroresorcinol	0282	1L	Nitroguanidine
0220	1L	Urea nitrate	0282	1L	Picrite
0221	1L	Warheads, torpedo	0283	1L	Boosters
0222	1L	Ammonium nitrate			
0224	1P	Barium azide			
0225	1L	Boosters with detonator			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0284	1L	Grenades	0327	1L	Cartridges, small arms, blank
0285	1L	Grenades	0328	1L	Cartridges for weapons, inert projectile
0286	1L	Warheads, rocket	0329	1L	Torpedoes
0287	1L	Warheads, rocket	0330	1L	Torpedoes
0288	1L	Charges, shaped, flexible, linear	0331	1L	Agent, blasting, type B
0289	1L	Cord, detonating	0331	1L	Explosive, blasting, type B
0290	1L	Cord, detonating	0332	1L	Agent, blasting, type E
0290	1L	Fuse, detonating	0332	1L	Explosive, blasting, type E
0291	1L	Bombs	0333	1L	Fireworks
0292	1L	Grenades	0334	1L	Fireworks
0293	1L	Grenades	0335	1L	Fireworks
0294	1L	Mines	0336	1L	Fireworks
0295	1L	Rockets	0337	3L	Fireworks
0296	1L	Sounding devices, explosive	0338	1L	Cartridges for weapons, blank
0297	1L	Ammunition, illuminating	0338	1L	Cartridges, small arms, blank
0299	1L	Bombs, photo-flash	0339	1L	Cartridges for weapons, inert projectile
0300	1L	Ammunition, incendiary	0339	1L	Cartridges, small arms
0301	1CP	Ammunition, tear-producing	0340	1L	Nitrocellulose
0303	1L	Ammunition, smoke	0341	1L	Nitrocellulose
0305	1L	Flash powder	0342	1L	Nitrocellulose, wetted
0306	1L	Tracers for ammunition	0343	1L	Nitrocellulose, plasticized
0312	1L	Cartridges, signal	0344	1L	Projectiles
0313	1L	Signals, smoke	0345	3L	Projectiles
0314	1L	Igniters	0346	1L	Projectiles
0315	1L	Igniters	0347	1L	Projectiles
0316	1L	Fuzes, igniting	0348	1L	Cartridges for weapons
0317	1L	Fuzes, igniting	0349	3L	Articles, explosive, n.o.s.*
0318	1L	Grenades, practice	0350	1L	Articles, explosive, n.o.s.*
0319	1L	Primers, tubular	0351	1L	Articles, explosive, n.o.s.*
0320	1L	Primers, tubular	0352	1L	Articles, explosive, n.o.s.*
0321	1L	Cartridges for weapons	0353	1L	Articles, explosive, n.o.s.*
0322	1L	Rocket motors with hypergolic liquids	0354	1L	Articles, explosive, n.o.s.*
0323	3L	Cartridges, power device	0355	1L	Articles, explosive, n.o.s.*
0324	1L	Projectiles	0356	1L	Articles, explosive, n.o.s.*
0325	1L	Igniters	0357	1L	Substances, explosive, n.o.s.*
0326	1L	Cartridges for weapons, blank	0358	1L	Substances, explosive, n.o.s.*
0327	1L	Cartridges for weapons, blank			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0359	1L	Substances, explosive, n.o.s.*	0390	1L	Tritonal
0360	1L	Detonator assemblies, non-electric	0391	1L	Cyclonite and cyclotetramethylenetetranitramine mixture, desensitized
0361	1L	Detonator assemblies, non-electric	0391	1L	Cyclonite and cyclotetramethylenetetranitramine mixture, wetted
0362	1L	Ammunition, practice	0391	1L	Cyclotrimethylenetrinitramine and cyclotetramethylenetetranitramine mixture, desensitized
0363	1L	Ammunition, proof	0391	1L	Cyclotrimethylenetrinitramine and cyclotetramethylenetetranitramine mixture, wetted
0364	1L	Detonators for ammunition	0391	1L	Hexogen and cyclotetramethylenetetranitramine mixture, desensitized
0365	1L	Detonators for ammunition	0391	1L	Hexogen and cyclotetramethylenetetranitramine mixture, wetted
0366	3L	Detonators for ammunition	0391	1L	RDX and cyclotetramethylenetetranitramine mixture, desensitized
0367	3L	Fuzes, detonating	0391	1L	RDX and cyclotetramethylenetetranitramine mixture, wetted
0368	3L	Fuzes, igniting	0392	1L	Hexanitrostilbene
0369	1L	Warheads, rocket	0393	1L	Hexotonal
0370	1L	Warheads, rocket	0394	1L	Styphnic acid, wetted
0371	1L	Warheads, rocket	0394	1L	Trinitroresorcinol, wetted
0372	1L	Grenades, practice	0395	1L	Rocket motors, liquid fuelled
0373	3L	Signal devices, hand	0396	1L	Rocket motors, liquid fuelled
0374	1L	Sounding devices, explosive	0397	1L	Rockets, liquid fuelled
0375	1L	Sounding devices, explosive	0398	1L	Rockets, liquid fuelled
0376	3L	Primers, tubular	0399	1L	Bombs with flammable liquid
0377	1L	Primers, cap type	0400	1L	Bombs with flammable liquid
0378	1L	Primers, cap type	0401	1L	Dipicryl sulphide
0379	1L	Cases, cartridge, empty, with primer	0402	1L	Ammonium perchlorate
0380	1S	Articles, pyrophoric	0403	1L	Flares, aerial
0381	1L	Cartridges, power device	0404	3L	Flares, aerial
0382	1L	Components, explosive train, n.o.s.*	0405	3L	Cartridges, signal
0383	1L	Components, explosive train, n.o.s.*	0406	1L	Dinitrosobenzene
0384	3L	Components, explosive train, n.o.s.*	0407	1L	Tetrazol-1-acetic acid
0385	1L	5-Nitrobenzotriazol	0408	1L	Fuzes, detonating
0386	1L	Trinitrobenzenesulphonic acid			
0387	1L	Trinitrofluorenone			
0388	1L	TNT and hexanitrostilbene mixture			
0388	1L	TNT and trinitrobenzene mixture			
0388	1L	Trinitrotoluene and hexanitrostilbene mixture			
0388	1L	Trinitrotoluene and trinitrobenzene mixture			
0389	1L	TNT mixture containing trinitrobenzene and hexanitrostilbene			
0389	1L	Trinitrotoluene mixture containing trinitrobenzene and hexanitrostilbene			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
0409	1L	Fuzes, detonating	0445	3L	Charges, explosive, commercial
0410	1L	Fuzes, detonating	0446	1L	Cases, combustible, empty, without primer
0411	1L	Pentaerythrite tetranitrate	0447	1L	Cases, combustible, empty, without primer
0411	1L	Pentaerythritol tetranitrate	0448	1L	5-Mercaptotetrazol-1-acetic acid
0411	1L	PETN	0449	1L	Torpedoes, liquid fuelled
0412	1L	Cartridges for weapons	0450	1L	Torpedoes, liquid fuelled
0413	1L	Cartridges for weapons, blank	0451	1L	Torpedoes
0414	1L	Charges, propelling, for cannon	0452	1L	Grenades, practice
0415	1L	Charges, propelling	0453	1L	Rockets, line-throwing
0417	1L	Cartridges for weapons, inert projectile	0454	3L	Igniters
0417	1L	Cartridges, small arms	0455	3L	Detonators, non-electric
0418	1L	Flares, surface	0456	3L	Detonators, electric
0419	1L	Flares, surface	0457	1L	Charges, bursting, plastics bonded
0420	1L	Flares, aerial	0458	1L	Charges, bursting, plastics bonded
0421	1L	Flares, aerial	0459	1L	Charges, bursting, plastics bonded
0424	1L	Projectiles	0460	3L	Charges, bursting, plastics bonded
0425	1L	Projectiles	0461	1L	Components, explosive train, n.o.s.*
0426	1L	Projectiles	0462	1L	Articles, explosive, n.o.s.*
0427	1L	Projectiles	0463	1L	Articles, explosive, n.o.s.*
0428	1L	Articles, pyrotechnic	0464	1L	Articles, explosive, n.o.s.*
0429	1L	Articles, pyrotechnic	0465	1L	Articles, explosive, n.o.s.*
0430	1L	Articles, pyrotechnic	0466	1L	Articles, explosive, n.o.s.*
0431	1L	Articles, pyrotechnic	0467	1L	Articles, explosive, n.o.s.*
0432	3L	Articles, pyrotechnic	0468	1L	Articles, explosive, n.o.s.*
0433	1L	Powder cake, wetted	0469	1L	Articles, explosive, n.o.s.*
0433	1L	Powder paste, wetted	0470	1L	Articles, explosive, n.o.s.*
0434	1L	Projectiles	0471	1L	Articles, explosive, n.o.s.*
0435	1L	Projectiles	0472	1L	Articles, explosive, n.o.s.*
0436	1L	Rockets	0473	1L	Substances, explosive, n.o.s.*
0437	1L	Rockets	0474	1L	Substances, explosive, n.o.s.*
0438	1L	Rockets	0475	1L	Substances, explosive, n.o.s.*
0439	1L	Charges, shaped	0476	1L	Substances, explosive, n.o.s.*
0440	1L	Charges, shaped	0477	1L	Substances, explosive, n.o.s.*
0441	3L	Charges, shaped	0478	1L	Substances, explosive, n.o.s.*
0442	1L	Charges, explosive, commercial	0479	1L	Substances, explosive, n.o.s.*
0443	1L	Charges, explosive, commercial	0480	1L	Substances, explosive, n.o.s.*
0444	1L	Charges, explosive, commercial			

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0481	3L	Substances, explosive, n.o.s.*	0507	3L	Signals, smoke
0482	1L	Substances, EVI, n.o.s.*	0508	1L	1-Hydroxybenzotriazole, anhydrous
0482	1L	Substances, explosive, very insensitive, n.o.s.*	0509	1L	Powder, smokeless
0483	1L	Cyclonite, desensitized	0510	1L	Rocket motors
0483	1L	Cyclotrimethylenetrinitramine, desensitized	0511	1L	Detonators, electronic
0483	1L	Hexogen, desensitized	0512	1L	Detonators, electronic
0483	1L	RDX, desensitized	0513	1L	Detonators, electronic
0484	1L	Cyclotetramethylenetetranitramine, desensitized	1001	10L	Acetylene, dissolved
0484	1L	HMXX, desensitized	1002	2L	Air, compressed
0484	1L	Octogen, desensitized	1003	2X	Air, refrigerated liquid
0485	1L	Substances, explosive, n.o.s.*	1005	2CP	Ammonia, anhydrous
0486	1L	Articles, EEI	1006	2L	Argon, compressed
0486	1L	Articles, explosive, extremely insensitive	1008	2CP	Boron trifluoride
0487	1L	Signals, smoke	1009	2L	Bromotrifluoromethane
0488	1L	Ammunition, practice	1009	2L	Refrigerant gas R 13B1
0489	1L	DINGU	1010	10L	Butadienes and hydrocarbon mixture, stabilized
0489	1L	Dinitroglycoluril	1010	10L	Butadienes, stabilized
0490	1L	Nitrotriazolone	1011	10L	Butane
0490	1L	NTO	1012	10L	Butylene
0491	1L	Charges, propelling	1013	2L	Carbon dioxide
0492	1L	Signals, railway track, explosive	1016	10P	Carbon monoxide, compressed
0493	1L	Signals, railway track, explosive	1017	2PX	Chlorine
0494	1L	Jet perforating guns, charged	1018	2L	Chlorodifluoromethane
0495	1L	Propellant, liquid	1018	2L	Refrigerant gas R 22
0496	1L	Octonal	1020	2L	Chloropentafluoroethane
0497	1L	Propellant, liquid	1020	2L	Refrigerant gas R 115
0498	1L	Propellant, solid	1021	2L	1-Chloro-1,2,2,2-tetrafluoroethane
0499	1L	Propellant, solid	1021	2L	Refrigerant gas R 124
0500	3L	Detonator assemblies, non-electric	1022	2L	Chlorotrifluoromethane
0501	1L	Propellant, solid	1022	2L	Refrigerant gas R 13
0502	1L	Rockets	1023	10P	Coal gas, compressed
0503	1L	Safety devices, pyrotechnic	1026	10P	Cyanogen
0504	1L	1H-Tetrazole	1027	10A	Cyclopropane
0505	1L	Signals, distress	1028	2L	Dichlorodifluoromethane
0506	3L	Signals, distress	1028	2L	Refrigerant gas R 12
			1029	2L	Dichlorofluoromethane
			1029	2L	Refrigerant gas R 21

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1030	10L	1,1-Difluoroethane	1067	2PX	Nitrogen dioxide
1030	10L	Refrigerant gas R 152a	1069	2CP	Nitrosyl chloride
1032	10L	Dimethylamine, anhydrous	1070	2AX	Nitrous oxide
1033	10L	Dimethyl ether	1071	10P	Oil gas, compressed
1035	10L	Ethane	1072	2X	Oxygen, compressed
1036	10L	Ethylamine	1073	2X	Oxygen, refrigerated liquid
1037	10A	Ethyl chloride	1075	10L	Petroleum gases, liquefied
1038	10A	Ethylene, refrigerated liquid	1076	2CP	Phosgene
1039	10L	Ethyl methyl ether	1077	10L	Propylene
1040	10P	Ethylene oxide	1078	2L	Refrigerant gas, n.o.s.*
1040	10P	Ethylene oxide with nitrogen	1079	2CP	Sulphur dioxide
1041	10L	Ethylene oxide and carbon dioxide mixture	1080	2L	Sulphur hexafluoride
1043	2L	Fertilizer ammoniating solution	1081	10L	Tetrafluoroethylene, stabilized
1044	2L	Fire extinguishers	1082	10P	Refrigerant gas R 1113
1045	2PX	Fluorine, compressed	1082	10P	Trifluorochloroethylene, stabilized
1046	2L	Helium, compressed	1083	10L	Trimethylamine, anhydrous
1048	2CP	Hydrogen bromide, anhydrous	1085	10L	Vinyl bromide, stabilized
1049	10L	Hydrogen, compressed	1086	10L	Vinyl chloride, stabilized
1050	2CP	Hydrogen chloride, anhydrous	1087	10L	Vinyl methyl ether, stabilized
1051	6H	Hydrogen cyanide, stabilized	1088	3H	Acetal
1052	8P	Hydrogen fluoride, anhydrous	1089	3H	Acetaldehyde
1053	10P	Hydrogen sulphide	1090	3H	Acetone
1055	10L	Isobutylene	1091	3L	Acetone oils
1056	2L	Krypton, compressed	1092	6H	Acrolein, stabilized
1057	10L	Lighter refills	1093	3P	Acrylonitrile, stabilized
1057	10L	Lighters	1098	6F	Allyl alcohol
1058	2L	Liquefied gases	1099	3P	Allyl bromide
1060	10L	Methylacetylene and propadiene mixture, stabilized	1100	3P	Allyl chloride
1061	10L	Methylamine, anhydrous	1104	3L	Amyl acetates
1062	2P	Methyl bromide	1105	3L	Pentanol
1063	10L	Methyl chloride	1106	3C	Amylamine
1063	10L	Refrigerant gas R 40	1107	3L	Amyl chloride
1064	10P	Methyl mercaptan	1108	3H	n-Amylene
1065	2L	Neon, compressed	1108	3H	1-Pentene
1066	2L	Nitrogen, compressed	1109	3L	Amyl formates
1067	2PX	Dinitrogen tetroxide	1110	3L	n-Amyl methyl ketone
			1111	3L	Amyl mercaptan

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1112	3L	Amyl nitrate	1162	3C	Dimethyldichlorosilane
1113	3H	Amyl nitrite	1163	6CH	Dimethylhydrazine, unsymmetrical
1114	3H	Benzene	1164	3H	Dimethyl sulphide
1120	3L	Butanols	1165	3L	Dioxane
1123	3L	Butyl acetates	1166	3L	Dioxolane
1125	3C	n-Butylamine	1167	3AH	Divinyl ether, stabilized
1126	3L	1-Bromobutane	1170	3L	Ethanol
1127	3L	Chlorobutanes	1170	3L	Ethanol solution
1128	3L	n-Butyl formate	1170	3L	Ethyl alcohol
1129	3L	Butyraldehyde	1170	3L	Ethyl alcohol solution
1130	3L	Camphor oil	1171	3L	Ethylene glycol monoethyl ether
1131	3HP	Carbon disulphide	1172	3L	Ethylene glycol monoethyl ether acetate
1133	3L	Adhesives	1173	3L	Ethyl acetate
1134	3L	Chlorobenzene	1175	3L	Ethylbenzene
1135	6F	Ethylene chlorohydrin	1176	3L	Ethyl borate
1136	3L	Coal tar distillates, flammable	1177	3L	2-Ethylbutyl acetate
1139	3L	Coating solution	1178	3L	2-Ethylbutyraldehyde
1143	6Fi	Crotonaldehyde	1179	3L	Ethyl butyl ether
1143	6Fi	Crotonaldehyde, stabilized	1180	3L	Ethyl butyrate
1144	3H	Crotonylene	1181	6F	Ethyl chloroacetate
1145	3H	Cyclohexane	1182	6CF	Ethyl chloroformate
1146	3H	Cyclopentane	1183	4HW	Ethyldichlorosilane
1147	3L	Decahydronaphthalene	1184	3P	Ethylene dichloride
1148	3L	Diacetone alcohol	1185	6FH	Ethyleneimine, stabilized
1149	3L	Dibutyl ethers	1188	3L	Ethylene glycol monomethyl ether
1150	3L	1,2-Dichloroethylene	1189	3L	Ethylene glycol monomethyl ether acetate
1152	3L	Dichloropentanes	1190	3H	Ethyl formate
1153	3L	Ethylene glycol diethyl ether	1191	3L	Octyl aldehydes
1154	3CH	Diethylamine	1192	3L	Ethyl lactate
1155	3AH	Diethyl ether	1193	3L	Ethyl methyl ketone
1155	3AH	Ethyl ether	1193	3L	Methyl ethyl ketone
1156	3L	Diethyl ketone	1194	3P	Ethyl nitrite solution
1157	3L	Diisobutyl ketone	1195	3L	Ethyl propionate
1158	3CH	Diisopropylamine	1196	3C	Ethyltrichlorosilane
1159	3H	Diisopropyl ether	1197	3L	Extracts, liquid
1160	3C	Dimethylamine, aqueous solution	1198	3Ci	Formaldehyde solution, flammable
1161	3L	Dimethyl carbonate			

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1199	6F	Furaldehydes	1238	6F	Methyl chloroformate
1201	3L	Fusel oil	1239	6F	Methyl chloromethyl ether
1202	3L	Diesel fuel	1242	4HW	Methyldichlorosilane
1202	3L	Gas oil	1243	3H	Methyl formate
1202	3L	Heating oil, light	1244	6F	Methylhydrazine
1203	3H	Gasoline	1245	3L	Methyl isobutyl ketone
1203	3H	Motor spirit	1246	3L	Methyl isopropenyl ketone, stabilized
1203	3H	Petrol	1247	3L	Methyl methacrylate monomer, stabilized
1204	3L	Nitroglycerin solution in alcohol	1248	3H	Methyl propionate
1206	3H	Heptanes	1249	3L	Methyl propyl ketone
1207	3L	Hexaldehyde	1250	3C	Methyltrichlorosilane
1208	3H	Hexanes	1251	6CH	Methyl vinyl ketone, stabilized
1210	3L	Printing ink	1259	6H	Nickel carbonyl
1210	3L	Printing ink related material	1261	3L	Nitromethane
1212	3L	Isobutanol	1262	3H	Octanes
1212	3L	Isobutyl alcohol	1263	3L	Paint
1213	3L	Isobutyl acetate	1263	3L	Paint related material
1214	3CH	Isobutylamine	1264	3L	Paraldehyde
1216	3H	Isooctene	1265	3H	Pentanes
1218	3H	Isoprene, stabilized	1266	3L	Perfumery products
1219	3L	Isopropanol	1267	3L	Petroleum crude oil
1219	3L	Isopropyl alcohol	1268		Petroleum distillates, n.o.s.
1220	3L	Isopropyl acetate		3H	Packing Group I or II
1221	3CH	Isopropylamine		3L	Packing Group III
1222	3L	Isopropyl nitrate	1268		Petroleum products, n.o.s.
1223	3L	Kerosene		3H	Packing Group I or II
1224	3L	Ketones, liquid, n.o.s.*		3L	Packing Group III
1228	3P	Mercaptan mixture, liquid, flammable, toxic, n.o.s.*	1272	3L	Pine oil
1228	3P	Mercaptans, liquid, flammable, toxic, n.o.s.*	1274	3L	n-Propanol
1229	3L	Mesityl oxide	1274	3L	Propyl alcohol, normal
1230	3L	Methanol	1275	3H	Propionaldehyde
1231	3H	Methyl acetate	1276	3L	n-Propyl acetate
1233	3L	Methylamyl acetate	1277	3CH	Propylamine
1234	3H	Methylal	1278	3H	1-Chloropropane
1235	3CH	Methylamine, aqueous solution	1279	3L	1,2-Dichloropropane
1237	3L	Methyl butyrate	1280	3H	Propylene oxide

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1281	3H	Propyl formates	1324	3L	Films, nitrocellulose base
1282	3L	Pyridine	1325	3L	Flammable solid, organic, n.o.s.*
1286	3L	Rosin oil	1326	3L	Hafnium powder, wetted
1287	3L	Rubber solution	1327	3L	Bhusa
1288	3L	Shale oil	1327	3L	Hay
1289	3C	Sodium methylate solution	1327	3L	Straw
1292	3L	Tetraethyl silicate	1328	3L	Hexamethylenetetramine
1293	3L	Tinctures, medicinal	1330	3L	Manganese resinate
1294	3L	Toluene	1331	3L	Matches, 'strike anywhere'
1295	4HW	Trichlorosilane	1332	3L	Metalddehyde
1296	3CH	Triethylamine	1333	3L	Cerium
1297		Trimethylamine, aqueous solution	1334	3L	Naphthalene, crude
	3CH	Packing Group I or II	1334	3L	Naphthalene, refined
	3C	Packing Group III	1336	3E	Nitroguanidine, wetted
1298	3CH	Trimethylchlorosilane	1336	3E	Picrite, wetted
1299	3L	Turpentine	1337	3E	Nitrostarch, wetted
1300	3L	Turpentine substitute	1338	3L	Phosphorus, amorphous
1301	3H	Vinyl acetate, stabilized	1339	3L	Phosphorus heptasulphide
1302	3H	Vinyl ethyl ether, stabilized	1340	4FW	Phosphorus pentasulphide
1303	3H	Vinylidene chloride, stabilized	1341	3W	Phosphorus sesquisulphide
1304	3H	Vinyl isobutyl ether, stabilized	1343	3W	Phosphorus trisulphide
1305	3CH	Vinyltrichlorosilane	1344	3E	Picric acid, wetted
1306	3L	Wood preservatives, liquid	1344	3E	Trinitrophenol, wetted
1307	3L	Xylenes	1345	3L	Rubber scrap
1308		Zirconium suspended in a flammable liquid	1345	3L	Rubber shoddy
	3H	Packing Group I or II	1346	3L	Silicon powder, amorphous
	3L	Packing Group III	1347	3E	Silver picrate, wetted
1309	3L	Aluminium powder, coated	1348	3EP	Sodium dinitro-o-cresolate, wetted
1310	3E	Ammonium picrate, wetted	1349	3E	Sodium picramate, wetted
1312	3L	Borneol	1350	3L	Sulphur
1313	3L	Calcium resinate	1352	3L	Titanium powder, wetted
1314	3L	Calcium resinate, fused	1353	3L	Fabrics impregnated with weakly nitrated nitrocellulose, n.o.s.
1318	3L	Cobalt resinate, precipitated	1353	3L	Fibres impregnated with weakly nitrated nitrocellulose, n.o.s.
1320	3EP	Dinitrophenol, wetted	1354	3E	Trinitrobenzene, wetted
1321	3EP	Dinitrophenolates, wetted	1355	3E	Trinitrobenzoic acid, wetted
1322	3E	Dinitroresorcinol, wetted	1356	3E	TNT, wetted
1323	3L	Ferrocium			

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1356	3E	Trinitrotoluene, wetted	1385	4L	Sodium sulphide
1357	3E	Urea nitrate, wetted	1385	4L	Sodium sulphide, anhydrous
1358	3L	Zirconium powder, wetted	1386	4L	Seed cake
1360	4PW	Calcium phosphide	1387	4L	Wool waste, wet
1361	4L	Carbon	1389	4W	Alkali metal amalgam, liquid
1362	4L	Carbon, activated	1390	4W	Alkali metal amides
1363	4L	Copra	1391	4W	Alkali metal dispersion
1364	4L	Cotton waste, oily	1391	4W	Alkaline earth metal dispersion
1365	4L	Cotton, wet	1392	4W	Alkaline earth metal amalgam, liquid
1369	4L	p-Nitrosodimethylaniline	1393	4W	Alkaline earth metal alloy, n.o.s.
1372	4L	Fibres, animal	1394	4W	Aluminium carbide
1372	4L	Fibres, vegetable	1395	4PW	Aluminium ferrosilicon powder
1373	4L	Fabrics, animal, n.o.s.	1396	4W	Aluminium powder, uncoated
1373	4L	Fabrics, synthetic, n.o.s.	1397	4PW	Aluminium phosphide
1373	4L	Fabrics, vegetable, n.o.s.	1398	4W	Aluminium silicon powder, uncoated
1373	4L	Fibres, animal, n.o.s.	1400	4W	Barium
1373	4L	Fibres, synthetic, n.o.s.	1401	4W	Calcium
1373	4L	Fibres, vegetable, n.o.s.	1402	4W	Calcium carbide
1374	4L	Fish meal, unstabilized	1403	4W	Calcium cyanamide
1374	4L	Fish scrap, unstabilized	1404	4W	Calcium hydride
1376	4L	Iron oxide, spent	1405	4W	Calcium silicide
1376	4L	Iron sponge, spent	1407	4W	Caesium
1378	4L	Metal catalyst, wetted*	1408	4PW	Ferrosilicon
1379	4L	Paper, unsaturated oil treated	1409	4W	Metal hydrides, water-reactive, n.o.s.*
1380	4P	Pentaborane	1410	4W	Lithium aluminium hydride
1381	4P	Phosphorus, white, dry	1411	4HW	Lithium aluminium hydride, ethereal
1381	4P	Phosphorus, white, in solution	1413	4W	Lithium borohydride
1381	4P	Phosphorus, white, under water	1414	4W	Lithium hydride
1381	4P	Phosphorus, yellow, dry	1415	4W	Lithium
1381	4P	Phosphorus, yellow, in solution	1417	4W	Lithium silicon
1381	4P	Phosphorus, yellow, under water	1418	4SW	Magnesium alloys powder
1382	4L	Potassium sulphide	1418	4SW	Magnesium powder
1382	4L	Potassium sulphide, anhydrous	1419	4PW	Magnesium aluminium phosphide
1383	4L	Pyrophoric alloy, n.o.s.*	1420	4W	Potassium metal alloys, liquid
1383	4L	Pyrophoric metal, n.o.s.*	1421	4W	Alkali metal alloy, liquid, n.o.s.
1384	4L	Sodium dithionite	1422	4W	Potassium sodium alloys, liquid
1384	4L	Sodium hydrosulphite	1423	4W	Rubidium

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1426	4W	Sodium borohydride	1471	5L	Lithium hypochlorite, dry
1427	4W	Sodium hydride	1471	5L	Lithium hypochlorite mixture
1428	4W	Sodium	1472	5L	Lithium peroxide
1431	4C	Sodium methylate	1473	5L	Magnesium bromate
1432	4PW	Sodium phosphide	1474	5L	Magnesium nitrate
1433	4PW	Stannic phosphides	1475	5L	Magnesium perchlorate
1435	4W	Zinc ashes	1476	5L	Magnesium peroxide
1436	4SW	Zinc dust	1477	5L	Nitrates, inorganic, n.o.s.
1436	4SW	Zinc powder	1479	5L	Oxidizing solid, n.o.s.*
1437	3L	Zirconium hydride	1481	5L	Perchlorates, inorganic, n.o.s.
1438	5L	Aluminium nitrate	1482	5L	Permanganates, inorganic, n.o.s.*
1439	5L	Ammonium dichromate	1483	5L	Peroxides, inorganic, n.o.s.
1442	5L	Ammonium perchlorate	1484	5L	Potassium bromate
1444	5L	Ammonium persulphate	1485	5L	Potassium chlorate
1445	5P	Barium chlorate, solid	1486	5L	Potassium nitrate
1446	5P	Barium nitrate	1487	5L	Potassium nitrate and sodium nitrite mixture
1447	5P	Barium perchlorate, solid	1488	5L	Potassium nitrite
1448	5P	Barium permanganate	1489	5L	Potassium perchlorate
1449	5P	Barium peroxide	1490	5L	Potassium permanganate
1450	5L	Bromates, inorganic, n.o.s.*	1491	5L	Potassium peroxide
1451	5L	Caesium nitrate	1492	5L	Potassium persulphate
1452	5L	Calcium chlorate	1493	5L	Silver nitrate
1453	5L	Calcium chlorite	1494	5L	Sodium bromate
1454	5L	Calcium nitrate	1495	5L	Sodium chlorate
1455	5L	Calcium perchlorate	1496	5L	Sodium chlorite
1456	5L	Calcium permanganate	1498	5L	Sodium nitrate
1457	5L	Calcium peroxide	1499	5L	Sodium nitrate and potassium nitrate mixture
1458	5L	Chlorate and borate mixture	1500	5P	Sodium nitrite
1459	5L	Chlorate and magnesium chloride mixture, solid	1502	5L	Sodium perchlorate
1461	5L	Chlorates, inorganic, n.o.s.*	1503	5L	Sodium permanganate
1462	5L	Chlorites, inorganic, n.o.s.*	1504	5L	Sodium peroxide
1463	5CP	Chromium trioxide, anhydrous	1505	5L	Sodium persulphate
1465	5L	Didymium nitrate	1506	5L	Strontium chlorate
1466	5L	Ferric nitrate	1507	5L	Strontium nitrate
1467	5L	Guanidine nitrate	1508	5L	Strontium perchlorate
1469	5P	Lead nitrate	1509	5L	Strontium peroxide
1470	5P	Lead perchlorate, solid			

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1510	6X	Tetranitromethane	1574	6L	Calcium arsenate and calcium arsenite mixture, solid
1511	5C	Urea hydrogen peroxide	1575	6L	Calcium cyanide
1512	5L	Zinc ammonium nitrite	1577	6L	Chlorodinitrobenzenes, liquid
1513	5L	Zinc chlorate	1578	6L	Chloronitrobenzenes, solid
1514	5L	Zinc nitrate	1579	6L	4-Chloro-o-toluidine hydrochloride, solid
1515	5L	Zinc permanganate	1580	6L	Chloropicrin
1516	5L	Zinc peroxide	1581	2P	Chloropicrin and methyl bromide mixture
1517	3E	Zirconium picramate, wetted	1582	2P	Chloropicrin and methyl chloride mixture
1541	6L	Acetone cyanohydrin, stabilized	1583	6L	Chloropicrin mixture, n.o.s.*
1544	6L	Alkaloid salts, solid, n.o.s.*	1585	6L	Copper acetoarsenite
1544	6L	Alkaloids, solid, n.o.s.*	1586	6L	Copper arsenite
1545	6F	Allyl isothiocyanate, stabilized	1587	6L	Copper cyanide
1546	6L	Ammonium arsenate	1588	6L	Cyanides, inorganic, solid, n.o.s.*
1547	6L	Aniline	1589	2CP	Cyanogen chloride, stabilized
1548	6L	Aniline hydrochloride	1590	6L	Dichloroanilines, liquid
1549	6L	Antimony compound, inorganic, solid, n.o.s.*	1591	6L	o-Dichlorobenzene
1550	6L	Antimony lactate	1593	6L	Dichloromethane
1551	6L	Antimony potassium tartrate	1594	6L	Diethyl sulphate
1553	6L	Arsenic acid, liquid	1595	6C	Dimethyl sulphate
1554	6L	Arsenic acid, solid	1596	6L	Dinitroanilines
1555	6L	Arsenic bromide	1597	6L	Dinitrobenzenes, liquid
1556	6L	Arsenic compound, liquid, n.o.s.*	1598	6L	Dinitro-o-cresol
1557	6L	Arsenic compound, solid, n.o.s.*	1599	6L	Dinitrophenol solution
1558	6L	Arsenic	1600	6L	Dinitrotoluenes, molten
1559	6L	Arsenic pentoxide	1601	6L	Disinfectant, solid, toxic, n.o.s.*
1560	6L	Arsenic trichloride	1602	6L	Dye intermediate, liquid, toxic, n.o.s.*
1561	6L	Arsenic trioxide	1602	6L	Dye, liquid, toxic, n.o.s.*
1562	6L	Arsenical dust	1603	6F	Ethyl bromoacetate
1564	6L	Barium compound, n.o.s.*	1604	8F	Ethylenediamine
1565	6L	Barium cyanide	1605	6L	Ethylene dibromide
1566	6L	Beryllium compound, n.o.s.*	1606	6L	Ferric arsenate
1567	6F	Beryllium powder	1607	6L	Ferric arsenite
1569	6F	Bromoacetone	1608	6L	Ferrous arsenate
1570	6L	Brucine	1611	6L	Hexaethyl tetraphosphate
1571	3EP	Barium azide, wetted			
1572	6L	Cacodylic acid			
1573	6L	Calcium arsenate			

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1612	2P	Hexaethyl tetraphosphate and compressed gas mixture	1653	6L	Nickel cyanide
1613	6L	Hydrocyanic acid, aqueous solution	1654	6L	Nicotine
1613	6L	Hydrogen cyanide, aqueous solution	1655	6L	Nicotine compound, solid, n.o.s.*
1614	6L	Hydrogen cyanide, stabilized	1655	6L	Nicotine preparation, solid, n.o.s.*
1616	6L	Lead acetate	1656	6L	Nicotine hydrochloride, liquid
1617	6L	Lead arsenates	1656	6L	Nicotine hydrochloride solution
1618	6L	Lead arsenites	1657	6L	Nicotine salicylate
1620	6L	Lead cyanide	1658	6L	Nicotine sulphate solution
1621	6L	London Purple	1659	6L	Nicotine tartrate
1622	6L	Magnesium arsenate	1660	2PX	Nitric oxide, compressed
1623	6L	Mercuric arsenate	1661	6L	Nitroanilines
1624	6L	Mercuric chloride	1662	6L	Nitrobenzene
1625	6L	Mercuric nitrate	1663	6L	Nitrophenols
1626	6L	Mercuric potassium cyanide	1664	6L	Nitrotoluenes, liquid
1627	6L	Mercurous nitrate	1665	6L	Nitroxyls, liquid
1629	6L	Mercury acetate	1669	6L	Pentachloroethane
1630	6L	Mercury ammonium chloride	1670	6L	Perchloromethyl mercaptan
1631	6L	Mercury benzoate	1671	6L	Phenol, solid
1634	6L	Mercury bromides	1672	6i	Phenylcarbamylamine chloride
1636	6L	Mercury cyanide	1673	6L	Phenylenediamines
1637	6L	Mercury gluconate	1674	6L	Phenylmercuric acetate
1638	6L	Mercury iodide	1677	6L	Potassium arsenate
1639	6L	Mercury nucleate	1678	6L	Potassium arsenite
1640	6L	Mercury oleate	1679	6L	Potassium cuprocyanide
1641	6L	Mercury oxide	1680	6L	Potassium cyanide, solid
1642	6L	Mercury oxycyanide, desensitized	1683	6L	Silver arsenite
1643	6L	Mercury potassium iodide	1684	6L	Silver cyanide
1644	6L	Mercury salicylate	1685	6L	Sodium arsenate
1645	6L	Mercury sulphate	1686	6L	Sodium arsenite, aqueous solution
1646	6L	Mercury thiocyanate	1687	6L	Sodium azide
1647	6L	Methyl bromide and ethylene dibromide mixture, liquid	1688	6L	Sodium cacodylate
1648	3L	Acetonitrile	1689	6L	Sodium cyanide, solid
1649	6L	Motor fuel anti-knock mixture	1690	6L	Sodium fluoride, solid
1650	6L	beta-Naphthylamine, solid	1691	6L	Strontium arsenite
1651	6L	Naphthylthiourea	1692	6L	Strychnine
1652	6L	Naphthylurea	1692	6L	Strychnine salts
			1693	6i	Tear gas substance, liquid, n.o.s.*

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1694	6i	Bromobenzyl cyanides, liquid	1736	8W	Benzoyl chloride
1695	6Fi	Chloroacetone, stabilized	1737	6C	Benzyl bromide
1697	6i	Chloroacetophenone, solid	1738	6C	Benzyl chloride
1698	6i	Diphenylamine chloroarsine	1739	8L	Benzyl chloroformate
1699	6i	Diphenylchloroarsine, liquid	1740	8L	Hydrogendifluorides, solid, n.o.s.
1700	6F	Tear gas candles	1741	2CP	Boron trichloride
1701	6L	Xylyl bromide, liquid	1742	8L	Boron trifluoride acetic acid complex, liquid
1702	6L	1,1,2,2-Tetrachloroethane	1743	8L	Boron trifluoride propionic acid complex, liquid
1704	6L	Tetraethyl dithiopyrophosphate	1744	8P	Bromine
1707	6L	Thallium compound, n.o.s.*	1744	8P	Bromine solution
1708	6L	Toluidines, liquid	1745	5CP	Bromine pentafluoride
1709	6L	2,4-Toluylenediamine, solid	1746	5CP	Bromine trifluoride
1710	6A	Trichloroethylene	1747	8F	Butyltrichlorosilane
1711	6L	Xylidines, liquid	1748	5L	Calcium hypochlorite, dry
1712	6L	Zinc arsenate	1748	5L	Calcium hypochlorite mixture, dry
1712	6L	Zinc arsenate and zinc arsenite mixture	1749	2PX	Chlorine trifluoride
1712	6L	Zinc arsenite	1750	6C	Chloroacetic acid solution
1713	6L	Zinc cyanide	1751	6C	Chloroacetic acid, solid
1714	4PW	Zinc phosphide	1752	6Ci	Chloroacetyl chloride
1715	8F	Acetic anhydride	1753	8L	Chlorophenyltrichlorosilane
1716	8L	Acetyl bromide	1754	8W	Chlorosulphonic acid
1717	3C	Acetyl chloride	1755	8L	Chromic acid solution
1718	8L	Butyl acid phosphate	1756	8L	Chromic fluoride, solid
1719	8L	Caustic alkali liquid, n.o.s.*	1757	8L	Chromic fluoride solution
1722	6CF	Allyl chloroformate	1758	8W	Chromium oxychloride
1723	3C	Allyl iodide	1759	8L	Corrosive solid, n.o.s.*
1724	8F	Allyltrichlorosilane, stabilized	1760	8L	Corrosive liquid, n.o.s.*
1725	8L	Aluminium bromide, anhydrous	1761	8P	Cupriethylenediamine solution
1726	8L	Aluminium chloride, anhydrous	1762	8L	Cyclohexenyltrichlorosilane
1727	8L	Ammonium hydrogendifluoride, solid	1763	8L	Cyclohexyltrichlorosilane
1728	8L	Amyltrichlorosilane	1764	8i	Dichloroacetic acid
1729	8L	Anisoyl chloride	1765	8i	Dichloroacetyl chloride
1730	8L	Antimony pentachloride, liquid	1766	8L	Dichlorophenyltrichlorosilane
1731	8L	Antimony pentachloride solution	1767	8F	Diethyldichlorosilane
1732	8P	Antimony pentafluoride	1768	8L	Difluorophosphoric acid, anhydrous
1733	8L	Antimony trichloride	1769	8L	Diphenyldichlorosilane

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1770	8L	Diphenylmethyl bromide	1809	6CW	Phosphorus trichloride
1771	8L	Dodecyltrichlorosilane	1810	6C	Phosphorus oxychloride
1773	8L	Ferric chloride, anhydrous	1811	8P	Potassium hydrogendifluoride, solid
1774	8L	Fire extinguisher charges	1812	6L	Potassium fluoride, solid
1775	8L	Fluoroboric acid	1813	8L	Potassium hydroxide, solid
1776	8L	Fluorophosphoric acid, anhydrous	1814	8L	Potassium hydroxide solution
1777	8W	Fluorosulphonic acid	1815	3C	Propionyl chloride
1778	8L	Fluorosilicic acid	1816	8F	Propyltrichlorosilane
1779	8F	Formic acid	1817	8W	Pyrosulphuryl chloride
1780	8L	Fumaryl chloride	1818	8L	Silicon tetrachloride
1781	8L	Hexadecyltrichlorosilane	1819	8L	Sodium aluminate solution
1782	8L	Hexafluorophosphoric acid	1823	8L	Sodium hydroxide, solid
1783	8L	Hexamethylenediamine solution	1824	8L	Sodium hydroxide solution
1784	8L	Hexyltrichlorosilane	1825	8L	Sodium monoxide
1786	8P	Hydrofluoric acid and sulphuric acid mixture	1826		Nitrating acid mixture, spent
1787	8L	Hydriodic acid		8X	Packing Group I
1788	8L	Hydrobromic acid		8L	Packing Group II
1789	8L	Hydrochloric acid	1827	8W	Stannic chloride, anhydrous
1790	8P	Hydrofluoric acid	1828	8W	Sulphur chlorides
1791	8L	Hypochlorite solution	1829	8L	Sulphur trioxide, stabilized
1792	8L	Iodine monochloride, solid	1830	8L	Sulphuric acid
1793	8L	Isopropyl acid phosphate	1831	8P	Sulphuric acid, fuming
1794	8L	Lead sulphate	1832	8L	Sulphuric acid, spent
1796		Nitrating acid mixture	1833	8L	Sulphurous acid
	8X	Packing Group I	1834	6C	Sulphuryl chloride
	8L	Packing Group II	1835	8L	Tetramethylammonium hydroxide solution
1798	8L	Nitrohydrochloric acid	1836	8W	Thionyl chloride
1799	8L	Nonyltrichlorosilane	1837	8W	Thiophosphoryl chloride
1800	8L	Octadecyltrichlorosilane	1838	6C	Titanium tetrachloride
1801	8L	Octyltrichlorosilane	1839	8L	Trichloroacetic acid
1802	8X	Perchloric acid	1840	8L	Zinc chloride solution
1803	8L	Phenolsulphonic acid, liquid	1841	9L	Acetaldehyde ammonia
1804	8L	Phenyltrichlorosilane	1843	6L	Ammonium dinitro-o-cresolate, solid
1805	8L	Phosphoric acid, solution	1845	9L	Carbon dioxide, solid
1806	8W	Phosphorus pentachloride	1845	9L	Dry ice
1807	8W	Phosphorus pentoxide	1846	6L	Carbon tetrachloride
1808	8W	Phosphorus tribromide			

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1847	8L	Potassium sulphide, hydrated	1903	8L	Disinfectant, liquid, corrosive, n.o.s.*
1848	8L	Propionic acid	1905	8L	Selenic acid
1849	8L	Sodium sulphide, hydrated	1906	8L	Sludge acid
1851	6L	Medicine, liquid, toxic, n.o.s.	1907	8L	Soda lime
1854	4W	Barium alloys, pyrophoric	1908	8L	Chlorite solution
1855	4W	Calcium alloys, pyrophoric	1910	8L	Calcium oxide
1855	4W	Calcium, pyrophoric	1911	10P	Diborane
1856	4L	Rags, oily	1912	10L	Methyl chloride and methylene chloride mixture
1857	4L	Textile waste, wet	1913	2L	Neon, refrigerated liquid
1858	2L	Hexafluoropropylene	1914	3L	Butyl propionates
1858	2L	Refrigerant gas R 1216	1915	3L	Cyclohexanone
1859	2CP	Silicon tetrafluoride	1916	6F	2,2'-Dichlorodiethyl ether
1860	10L	Vinyl fluoride, stabilized	1917	3i	Ethyl acrylate, stabilized
1862	3L	Ethyl crotonate	1918	3L	Isopropylbenzene
1863	3L	Fuel, aviation, turbine engine	1919	3Hi	Methyl acrylate, stabilized
1865	3L	n-Propyl nitrate	1920	3L	Nonanes
1866	3L	Resin solution	1921	3HP	Propyleneimine, stabilized
1868	3P	Decaborane	1922	3C	Pyrrolidine
1869	3L	Magnesium	1923	4L	Calcium dithionite
1869	3L	Magnesium alloys	1923	4L	Calcium hydrosulphite
1870	4W	Potassium borohydride	1928	4HW	Methyl magnesium bromide in ethyl ether
1871	3W	Titanium hydride	1929	4L	Potassium dithionite
1872	5L	Lead dioxide	1929	4L	Potassium hydrosulphite
1873	5C	Perchloric acid	1931	9L	Zinc dithionite
1884	6L	Barium oxide	1931	9L	Zinc hydrosulphite
1885	6L	Benzidine	1932	4L	Zirconium scrap
1886	6L	Benzylidene chloride	1935	6L	Cyanide solution, n.o.s.*
1887	6L	Bromochloromethane	1938	8L	Bromoacetic acid solution
1888	6A	Chloroform	1939	8W	Phosphorus oxybromide
1889	6C	Cyanogen bromide	1940	8L	Thioglycolic acid
1891	3P	Ethyl bromide	1941	9L	Dibromodifluoromethane
1892	6i	Ethylchloroarsine	1942	5L	Ammonium nitrate
1894	6L	Phenylmercuric hydroxide	1944	3L	Matches, safety
1895	6L	Phenylmercuric nitrate	1945	3L	Matches, wax 'vesta'
1897	6L	Tetrachloroethylene	1950		Aerosols
1898	8L	Acetyl iodide			
1902	8L	Diisooctyl acid phosphate			

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	10L	Division 2.1 without subsidiary risk	1971	10L	Methane, compressed
	10C	Division 2.1 with subsidiary risk 8	1971	10L	Natural gas, compressed
	10CP	Division 2.1 with subsidiary risks 6.1 and 8	1972	10L	Methane, refrigerated liquid
	2L	Division 2.2 without subsidiary risk	1972	10L	Natural gas, refrigerated liquid
	2X	Division 2.2 with subsidiary risk 5.1	1973	2L	Chlorodifluoromethane and chloropentafluoroethane mixture
	2C	Division 2.2 with subsidiary risk 8	1973	2L	Refrigerant gas R 502
	2CP	Division 2.2 with subsidiary risks 6.1 and 8	1974	2L	Chlorodifluorobromomethane
	2P	Division 2.3 without subsidiary risk or Division 2.2 with subsidiary risk 6.1 (including tear gas devices)	1974	2L	Refrigerant gas R 12B1
	10P	Division 2.3 with subsidiary risk 2.1 or Division 2.1 with subsidiary risk 6.1 (including tear gas devices)	1975	2PX	Nitric oxide and dinitrogen tetroxide mixture
			1975	2PX	Nitric oxide and nitrogen dioxide mixture
1951	2L	Argon, refrigerated liquid	1976	2L	Octafluorocyclobutane
1952	2L	Ethylene oxide and carbon dioxide mixture	1976	2L	Refrigerant gas R C318
1953	10P	Compressed gas, toxic, flammable, n.o.s.*	1977	2L	Nitrogen, refrigerated liquid
1954	10L	Compressed gas, flammable, n.o.s.*	1978	10L	Propane
1955	2P	Compressed gas, toxic, n.o.s.*	1982	2A	Refrigerant gas R 14
1956	2L	Compressed gas, n.o.s.*	1982	2A	Tetrafluoromethane
1957	10L	Deuterium, compressed	1983	2L	1-Chloro-2,2,2-trifluoroethane
1958	2L	1,2-Dichloro-1,1,2,2-tetrafluoroethane	1983	2L	Refrigerant gas R 133a
1958	2L	Refrigerant gas R 114	1984	2A	Refrigerant gas R 23
1959	10L	1,1-Difluoroethylene	1984	2A	Trifluoromethane
1959	10L	Refrigerant gas R 1132a	1986		Alcohols, flammable, toxic, n.o.s.*
1961	10L	Ethane, refrigerated liquid	3HP		Packing Group I or II
1962	10A	Ethylene	3P		Packing Group III
1963	2L	Helium, refrigerated liquid	1987	3L	Alcohols, n.o.s.*
1964	10L	Hydrocarbon gas mixture, compressed, n.o.s.*	1988		Aldehydes, flammable, toxic, n.o.s.*
1965	10L	Hydrocarbon gas mixture, liquefied, n.o.s.*	3HP		Packing Group I or II
1966	10L	Hydrogen, refrigerated liquid	3P		Packing Group III
1967	2P	Insecticide gas, toxic, n.o.s.*	1989		Aldehydes, n.o.s.*
1968	2L	Insecticide gas, n.o.s.*	3H		Packing Group I or II
1969	10L	Isobutane	3L		Packing Group III
1970	2L	Krypton, refrigerated liquid	1990	9N	Benzaldehyde
			1991	3HP	Chloroprene, stabilized
			1992		Flammable liquid, toxic, n.o.s.*
			3HP		Packing Group I or II
			3P		Packing Group III

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1993		Flammable liquid, n.o.s.*	8L		Other than red fuming, with more than 20% and less than 65% nitric acid
	3H	Packing Group I or II			
	3L	Packing Group III	8X		Other than red fuming, with more than 70% nitric acid
1994	6H	Iron pentacarbonyl	8L		Other than red fuming, with not more than 20% nitric acid
1999	3L	Tars, liquid			
2000	3L	Celluloid	8X		Other than red fuming, with at least 65% but not more than 70% nitric acid
2001	3L	Cobalt naphthenates, powder	2032	8PX	Nitric acid, red fuming
2002	4L	Celluloid, scrap	2033	8L	Potassium monoxide
2004	4W	Magnesium diamide	2034	10L	Hydrogen and methane mixture, compressed
2006	4L	Plastics, nitrocellulose-based, self-heating, n.o.s.*	2035	10L	Refrigerant gas R 143a
2008	4L	Zirconium powder, dry	2035	10L	1,1,1-Trifluoroethane
2009	4L	Zirconium, dry	2036	2L	Xenon
2010	4W	Magnesium hydride	2037		Gas cartridges
2011	4PW	Magnesium phosphide			
2012	4PW	Potassium phosphide	10L		Division 2.1 without subsidiary risk
2013	4PW	Strontium phosphide	2L		Division 2.2 without subsidiary risk
2014	5C	Hydrogen peroxide, aqueous solution	2X		Division 2.2 with subsidiary risk 5.1
2015	5C	Hydrogen peroxide, aqueous solution, stabilized	2P		Division 2.3 without subsidiary risk
			10P		Division 2.3 with subsidiary risk 2.1
2015	5C	Hydrogen peroxide, stabilized	10C		Division 2.3 with subsidiary risk 2.1 and 8
2016	6L	Ammunition, toxic, non-explosive	2PX		Division 2.3 with subsidiary risk 5.1
2017	6C	Ammunition, tear-producing, non-explosive	2CX		Division 2.3 with subsidiary risk 5.1 and 8
2018	6L	Chloroanilines, solid	2CP		Division 2.3 with subsidiary risk 8
2019	6L	Chloroanilines, liquid	2037		Receptacles, small, containing gas
2020	6L	Chlorophenols, solid	10L		Division 2.1 without subsidiary risk
2021	6L	Chlorophenols, liquid	2L		Division 2.2 without subsidiary risk
2022	6C	Cresylic acid	2X		Division 2.2 with subsidiary risk 5.1
2023	6F	Epichlorohydrin	2P		Division 2.3 without subsidiary risk
2024	6L	Mercury compound, liquid, n.o.s.*	10P		Division 2.3 with subsidiary risk 2.1
2025	6L	Mercury compound, solid, n.o.s.*	10C		Division 2.3 with subsidiary risk 2.1 and 8
2026	6L	Phenylmercuric compound, n.o.s.*	2PX		Division 2.3 with subsidiary risk 5.1
2027	6L	Sodium arsenite, solid	2CX		Division 2.3 with subsidiary risk 5.1 and 8
2028	8L	Bombs, smoke, non-explosive	2CP		Division 2.3 with subsidiary risk 8
2029	8FP	Hydrazine, anhydrous			
2030	8P	Hydrazine, aqueous solution			
2031		Nitric acid			

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2038	6L	Dinitrotoluenes, liquid	2193	2L	Refrigerant gas R 116
2044	10L	2,2-Dimethylpropane	2194	2CP	Selenium hexafluoride
2045	3H	Isobutyl aldehyde	2195	2CP	Tellurium hexafluoride
2045	3H	Isobutyraldehyde	2196	2CP	Tungsten hexafluoride
2046	3L	Cymenes	2197	2CP	Hydrogen iodide, anhydrous
2047	3L	Dichloropropenes	2198	2CP	Phosphorus pentafluoride
2048	3L	Dicyclopentadiene	2199	10P	Phosphine
2049	3L	Diethylbenzene	2200	10L	Propadiene, stabilized
2050	3L	Diisobutylene, isomeric compounds	2201	2AX	Nitrous oxide, refrigerated liquid
2051	8F	2-Dimethylaminoethanol	2202	10P	Hydrogen selenide, anhydrous
2052	3L	Dipentene	2203	10L	Silane
2053	3L	Methyl isobutyl carbinol	2204	10P	Carbonyl sulphide
2054	8F	Morpholine	2205	6L	Adiponitrile
2055	3L	Styrene monomer, stabilized	2206	6L	Isocyanate solution, toxic, n.o.s.*
2056	3H	Tetrahydrofuran	2206	6L	Isocyanates, toxic, n.o.s.*
2057	3L	Tripropylene	2208	5L	Calcium hypochlorite mixture, dry
2058	3L	Valeraldehyde	2209	8i	Formaldehyde solution
2059		Nitrocellulose solution, flammable	2210	4SW	Maneb
	3H	Packing Group I or II	2210	4SW	Maneb preparation
	3L	Packing Group III	2211	9L	Polymeric beads, expandable
2067	5L	Ammonium nitrate based fertilizer	2212	9L	Asbestos, amphibole*
2071	9L	Ammonium nitrate based fertilizer	2213	3L	Paraformaldehyde
2073	2L	Ammonia solution	2214	8L	Phthalic anhydride
2074	6L	Acrylamide, solid	2215	8L	Maleic anhydride
2075	6L	Chloral, anhydrous, stabilized	2215	8L	Maleic anhydride, molten
2076	6C	Cresols, liquid	2216	9L	Fish meal, stabilized
2077	6L	alpha-Naphthylamine	2216	9L	Fish scrap, stabilized
2078	6L	Toluene diisocyanate	2217	4L	Seed cake
2079	8L	Diethylenetriamine	2218	8F	Acrylic acid, stabilized
2186	2CP	Hydrogen chloride, refrigerated liquid	2219	3L	Allyl glycidyl ether
2187	2L	Carbon dioxide, refrigerated liquid	2222	3L	Anisole
2188	10P	Arsine	2224	6L	Benzonitrile
2189	10P	Dichlorosilane	2225	8L	Benzenesulphonyl chloride
2190	2PX	Oxygen difluoride, compressed	2226	8L	Benzotrichloride
2191	2P	Sulphuryl fluoride	2227	3L	n-Butyl methacrylate, stabilized
2192	10P	Germane	2232	6L	2-Chloroethanal
2193	2L	Hexafluoroethane	2233	6L	Chloroanisidines

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2234	3L	Chlorobenzotrifluorides	2271	3L	Ethyl amyl ketone
2235	6L	Chlorobenzyl chlorides, liquid	2272	6L	N-Ethylaniline
2236	6L	3-Chloro-4-methylphenyl isocyanate, liquid	2273	6L	2-Ethylaniline
2237	6L	Chloronitroanilines	2274	6L	N-Ethyl-N-benzylaniline
2238	3L	Chlorotoluenes	2275	3L	2-Ethylbutanol
2239	6L	Chlorotoluidines, solid	2276	3C	2-Ethylhexylamine
2240	8L	Chromosulphuric acid	2277	3L	Ethyl methacrylate, stabilized
2241	3L	Cycloheptane	2278	3L	n-Heptene
2242	3L	Cycloheptene	2279	6L	Hexachlorobutadiene
2243	3L	Cyclohexyl acetate	2280	8L	Hexamethylenediamine, solid
2244	3L	Cyclopentanol	2281	6L	Hexamethylene diisocyanate
2245	3L	Cyclopentanone	2282	3L	Hexanols
2246	3H	Cyclopentene	2283	3L	Isobutyl methacrylate, stabilized
2247	3L	n-Decane	2284	3P	Isobutyronitrile
2248	8F	Di-n-butylamine	2285	6F	Isocyanatobenzotrifluorides
2249	6F	Dichlorodimethyl ether, symmetrical	2286	3L	Pentamethylheptane
2250	6L	Dichlorophenyl isocyanates	2287	3H	Isoheptene
2251	3L	Bicyclo [2.2.1] hepta-2-5-diene, stabilized	2288	3H	Isohexene
2251	3L	2,5-Norbornadiene, stabilized	2289	8L	Isophoronediamine
2252	3L	1,2-Dimethoxyethane	2290	6L	Isophorone diisocyanate
2253	6L	N,N-Dimethylaniline	2291	6L	Lead compound, soluble, n.o.s.*
2254	3L	Matches, fusee	2293	3L	4-Methoxy-4-methylpentan-2-one
2256	3H	Cyclohexene	2294	6L	N-Methylaniline
2257	4W	Potassium	2295	6F	Methyl chloroacetate
2258	8F	1,2-Propylenediamine	2296	3H	Methylcyclohexane
2259	8L	Triethylenetetramine	2297	3L	Methylcyclohexanone
2260	3C	Tripropylamine	2298	3H	Methylcyclopentane
2261	6L	Xylenols, solid	2299	6L	Methyl dichloroacetate
2262	8L	Dimethylcarbamoyl chloride	2300	6L	2-Methyl-5-ethylpyridine
2263	3L	Dimethylcyclohexanes	2301	3H	2-Methylfuran
2264	8F	N,N-Dimethylcyclohexylamine	2302	3L	5-Methylhexan-2-one
2265	3L	N,N-Dimethylformamide	2303	3L	Isopropenylbenzene
2266	3C	Dimethyl-N-propylamine	2304	3L	Naphthalene, molten
2267	6C	Dimethyl thiophosphoryl chloride	2305	8L	Nitrobenzenesulphonic acid
2269	8L	3,3'-Iminodipropylamine	2306	6L	Nitrobenzotrifluorides, liquid
2270	3CH	Ethylamine, aqueous solution	2307	6L	3-Nitro-4-chlorobenzotrifluoride
			2308	8L	Nitrosylsulphuric acid, liquid

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2309	3L	Octadiene	2347	3L	Butyl mercaptan
2310	3P	Pentane-2,4-dione	2348	3L	Butyl acrylates, stabilized
2311	6L	Phenetidines	2350	3L	Butyl methyl ether
2312	6L	Phenol, molten	2351	3L	Butyl nitrites
2313	3L	Picolines	2352	3L	Butyl vinyl ether, stabilized
2315	9L	Polychlorinated biphenyls, liquid	2353	3C	Butyryl chloride
2316	6L	Sodium cuprocyanide, solid	2354	3P	Chloromethyl ethyl ether
2317	6L	Sodium cuprocyanide solution	2356	3H	2-Chloropropane
2318	4L	Sodium hydrosulphide	2357	8F	Cyclohexylamine
2319	3L	Terpene hydrocarbons, n.o.s.	2358	3L	Cyclooctatetraene
2320	8L	Tetraethylenepentamine	2359	3CP	Diallylamine
2321	6L	Trichlorobenzenes, liquid	2360	3P	Diallyl ether
2322	6L	Trichlorobutene	2361	3C	Diisobutylamine
2323	3L	Triethyl phosphite	2362	3L	1,1-Dichloroethane
2324	3L	Triisobutylene	2363	3N	Ethyl mercaptan
2325	3L	1,3,5-Trimethylbenzene	2364	3L	n-Propylbenzene
2326	8L	Trimethylcyclohexylamine	2366	3L	Diethyl carbonate
2327	8L	Trimethylhexamethylenediamines	2367	3L	alpha-Methylvaleraldehyde
2328	6L	Trimethylhexamethylene diisocyanate	2368	3L	alpha-Pinene
2329	3L	Trimethyl phosphite	2370	3H	1-Hexene
2330	3L	Undecane	2371	3H	Isopentenenes
2331	8L	Zinc chloride, anhydrous	2372	3L	1,2-Di-(dimethylamino) ethane
2332	3L	Acetaldehyde oxime	2373	3H	Diethoxymethane
2333	3P	Allyl acetate	2374	3L	3,3-Diethoxypropene
2334	6H	Allylamine	2375	3L	Diethyl sulphide
2335	3P	Allyl ethyl ether	2376	3H	2,3-Dihydropyran
2336	3P	Allyl formate	2377	3L	1,1-Dimethoxyethane
2337	6F	Phenyl mercaptan	2378	3P	2-Dimethylaminoacetonitrile
2338	3L	Benzotrifluoride	2379	3C	1,3-Dimethylbutylamine
2339	3L	2-Bromobutane	2380	3L	Dimethyldiethoxysilane
2340	3L	2-Bromoethyl ethyl ether	2381	3P	Dimethyl disulphide
2341	3L	1-Bromo-3-methylbutane	2382	6F	Dimethylhydrazine, symmetrical
2342	3L	Bromomethylpropanes	2383	3C	Dipropylamine
2343	3L	2-Bromopentane	2384	3H	Di-n-propyl ether
2344	3L	Bromopropanes	2385	3L	Ethyl isobutyrate
2345	3L	3-Bromopropyne	2386	3C	1-Ethylpiperidine
2346	3L	Butanedione	2387	3L	Fluorobenzene

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2388	3L	Fluorotoluenes	2427	5L	Potassium chlorate, aqueous solution
2389	3H	Furan	2428	5L	Sodium chlorate, aqueous solution
2390	3L	2-Iodobutane	2429	5L	Calcium chlorate, aqueous solution
2391	3L	Iodomethylpropanes	2430	8L	Alkylphenols, solid, n.o.s.
2392	3L	Iodopropanes	2431	6L	Anisidines
2393	3L	Isobutyl formate	2432	6L	N,N-Diethylaniline
2394	3L	Isobutyl propionate	2433	6L	Chloronitrotoluenes, liquid
2395	3C	Isobutyryl chloride	2434	8L	Dibenzylchlorosilane
2396	3P	Methacrylaldehyde, stabilized	2435	8L	Ethylphenyldichlorosilane
2397	3L	3-Methylbutan-2-one	2436	3i	Thioacetic acid
2398	3L	Methyl tert-butyl ether	2437	8L	Methylphenyldichlorosilane
2399	3C	1-Methylpiperidine	2438	6FW	Trimethylacetyl chloride
2400	3L	Methyl isovalerate	2439	8L	Sodium hydrogendifluoride
2401	8F	Piperidine	2440	8L	Stannic chloride pentahydrate
2402	3H	Propanethiols	2441	4C	Titanium trichloride mixture, pyrophoric
2403	3L	Isopropenyl acetate	2441	4C	Titanium trichloride, pyrophoric
2404	3P	Propionitrile	2442	8W	Trichloroacetyl chloride
2405	3L	Isopropyl butyrate	2443	8W	Vanadium oxytrichloride
2406	3L	Isopropyl isobutyrate	2444	8W	Vanadium tetrachloride
2407	6CF	Isopropyl chloroformate	2446	6L	Nitrocresols, solid
2409	3L	Isopropyl propionate	2447	4P	Phosphorus, white, molten
2410	3L	1,2,3,6-Tetrahydropyridine	2448	3L	Sulphur, molten
2411	3P	Butyronitrile	2451	2X	Nitrogen trifluoride
2412	3L	Tetrahydrothiophene	2452	10L	Ethylacetylene, stabilized
2413	3L	Tetrapropyl orthotitanate	2453	10L	Ethyl fluoride
2414	3H	Thiophene	2453	10L	Refrigerant gas R 161
2416	3L	Trimethyl borate	2454	10L	Methyl fluoride
2417	2CP	Carbonyl fluoride	2454	10L	Refrigerant gas R 41
2418	2CP	Sulphur tetrafluoride	2456	3H	2-Chloropropene
2419	10L	Bromotrifluoroethylene	2457	3H	2,3-Dimethylbutane
2420	2CP	Hexafluoroacetone	2458	3H	Hexadiene
2421	2PX	Nitrogen trioxide	2459	3H	2-Methyl-1-butene
2422	2L	Octafluorobut-2-ene	2460	3H	2-Methyl-2-butene
2422	2L	Refrigerant gas R 1318	2461	3H	Methylpentadiene
2424	2L	Octafluoropropane	2463	4W	Aluminium hydride
2424	2L	Refrigerant gas R 218	2464	5P	Beryllium nitrate
2426	5L	Ammonium nitrate, liquid	2465	5L	Dichloroisocyanuric acid, dry

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2465	5L	Dichloroisocyanuric acid salts	2508	8L	Molybdenum pentachloride
2466	5L	Potassium superoxide	2509	8L	Potassium hydrogen sulphate
2468	5L	Trichloroisocyanuric acid, dry	2511	8L	2-Chloropropionic acid
2469	5L	Zinc bromate	2512	6L	Aminophenols
2470	6L	Phenylacetoneitrile, liquid	2513	8L	Bromoacetyl bromide
2471	6L	Osmium tetroxide	2514	3L	Bromobenzene
2473	6L	Sodium arsanilate	2515	6L	Bromoform
2474	6L	Thiophosgene	2516	6L	Carbon tetrabromide
2475	8W	Vanadium trichloride	2517	10L	1-Chloro-1,1-difluoroethane
2477	6F	Methyl isothiocyanate	2517	10L	Refrigerant gas R 142b
2478	3P	Isocyanates, flammable, toxic, n.o.s.*	2518	6L	1,5,9-Cyclododecatiene
2478	3P	Isocyanate solution, flammable, toxic, n.o.s.*	2520	3L	Cyclooctadienes
2480	6H	Methyl isocyanate	2521	6F	Diketene, stabilized
2481	6F	Ethyl isocyanate	2522	6L	2-Dimethylaminoethyl methacrylate, stabilized
2482	6F	n-Propyl isocyanate	2524	3L	Ethyl orthoformate
2483	6H	Isopropyl isocyanate	2525	6L	Ethyl oxalate
2484	6F	tert-Butyl isocyanate	2526	3C	Furfurylamine
2485	6F	n-Butyl isocyanate	2527	3L	Isobutyl acrylate, stabilized
2486	6F	Isobutyl isocyanate	2528	3L	Isobutyl isobutyrate
2487	6Fi	Phenyl isocyanate	2529	3C	Isobutyric acid
2488	6F	Cyclohexyl isocyanate	2531	8L	Methacrylic acid, stabilized
2490	6L	Dichloroisopropyl ether	2533	6L	Methyl trichloroacetate
2491	8L	Ethanolamine	2534	10P	Methylchlorosilane
2491	8L	Ethanolamine solution	2535	3C	4-Methylmorpholine
2493	3C	Hexamethyleneimine	2535	3C	N-Methylmorpholine
2495	5CP	Iodine pentafluoride	2536	3H	Methyltetrahydrofuran
2496	8L	Propionic anhydride	2538	3L	Nitronaphthalene
2498	3L	1,2,3,6-Tetrahydrobenzaldehyde	2541	3L	Terpinolene
2501	6L	Tris-(1-aziridiny) phosphine oxide solution	2542	6L	Tributylamine
2502	8FW	Valeryl chloride	2545	4L	Hafnium powder, dry
2503	8L	Zirconium tetrachloride	2546	4L	Titanium powder, dry
2504	6L	Tetrabromoethane	2547	5L	Sodium superoxide
2505	6L	Ammonium fluoride	2548	2PX	Chlorine pentafluoride
2506	8L	Ammonium hydrogen sulphate	2552	6L	Hexafluoroacetone hydrate, liquid
2507	8L	Chloroplatinic acid, solid	2554	3H	Methylallyl chloride
			2555	3E	Nitrocellulose with water
			2556	3L	Nitrocellulose with alcohol

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2557	3L	Nitrocellulose mixture without plasticizer, without pigment	2599	2L	Chlorotrifluoromethane and trifluoromethane azeotropic mixture
2557	3L	Nitrocellulose mixture without plasticizer, with pigment	2599	2L	Refrigerant gas R 503
2557	3L	Nitrocellulose mixture with plasticizer, without pigment	2601	10L	Cyclobutane
2557	3L	Nitrocellulose mixture with plasticizer, with pigment	2602	2L	Dichlorodifluoromethane and difluoroethane azeotropic mixture
2558	6F	Epibromohydrin	2602	2L	Refrigerant gas R 500
2560	3L	2-Methylpentan-2-ol	2603	3P	Cycloheptatriene
2561	3H	3-Methyl-1-butene	2604	8F	Boron trifluoride diethyl etherate
2564	8L	Trichloroacetic acid solution	2605	6F	Methoxymethyl isocyanate
2565	8L	Dicyclohexylamine	2606	6F	Methyl orthosilicate
2567	6L	Sodium pentachlorophenate	2607	3L	Acrolein dimer, stabilized
2570	6L	Cadmium compound*	2608	3L	Nitropropanes
2571	8L	Alkylsulphuric acids	2609	6L	Triallyl borate
2572	6L	Phenylhydrazine	2610	3C	Triallylamine
2573	5P	Thallium chlorate	2611	6F	Propylene chlorohydrin
2574	6L	Tricresyl phosphate	2612	3AH	Methyl propyl ether
2576	8W	Phosphorus oxybromide, molten	2614	3L	Methallyl alcohol
2577	8i	Phenylacetyl chloride	2615	3H	Ethyl propyl ether
2578	8L	Phosphorus trioxide	2616	3L	Triisopropyl borate
2579	8L	Piperazine	2617	3L	Methylcyclohexanols
2580	8L	Aluminium bromide solution	2618	3L	Vinyltoluenes, stabilized
2581	8L	Aluminium chloride solution	2619	8F	Benzyltrimethylamine
2582	8L	Ferric chloride solution	2620	3L	Amyl butyrates
2583	8L	Alkylsulphonic acids, solid	2621	3L	Acetyl methyl carbinol
2583	8L	Arylsulphonic acids, solid	2622	3P	Glycidaldehyde
2584	8L	Alkylsulphonic acids, liquid	2623	3L	Firelighters, solid
2584	8L	Arylsulphonic acids, liquid	2624	4W	Magnesium silicide
2585	8L	Alkylsulphonic acids, solid	2626	5L	Chloric acid, aqueous solution
2585	8L	Arylsulphonic acids, solid	2627	5L	Nitrites, inorganic, n.o.s.*
2586	8L	Alkylsulphonic acids, liquid	2628	6L	Potassium fluoroacetate
2586	8L	Arylsulphonic acids, liquid	2629	6L	Sodium fluoroacetate
2587	6L	Benzoquinone	2630	6L	Selenates*
2588	6L	Pesticide, solid, toxic, n.o.s.*	2630	6L	Selenites*
2589	6F	Vinyl chloroacetate	2642	6L	Fluoroacetic acid
2590	9L	Asbestos, chrysotile	2643	6i	Methyl bromoacetate
2591	2L	Xenon, refrigerated liquid	2644	6L	Methyl iodide
			2645	6i	Phenacyl bromide

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2646	6L	Hexachlorocyclopentadiene	2691	8W	Phosphorus pentabromide
2647	6L	Malononitrile	2692	8L	Boron tribromide
2648	6L	1,2-Dibromobutan-3-one	2693	8L	Bisulphites, aqueous solution, n.o.s.*
2649	6i	1,3-Dichloroacetone	2698	8L	Tetrahydrophthalic anhydrides
2650	6L	1,1-Dichloro-1-nitroethane	2699	8N	Trifluoroacetic acid
2651	6L	4,4'-Diaminodiphenylmethane	2705	8L	1-Pentol
2653	6L	Benzyl iodide	2707	3L	Dimethyldioxanes
2655	6L	Potassium fluorosilicate	2709	3L	Butylbenzenes
2656	6L	Quinoline	2710	3L	Dipropyl ketone
2657	6L	Selenium disulphide	2713	6L	Acridine
2659	6L	Sodium chloroacetate	2714	3L	Zinc resinate
2660	6L	Nitrotoluidines (mono)	2715	3L	Aluminium resinate
2661	6L	Hexachloroacetone	2716	6L	1,4-Butynediol
2664	6L	Dibromomethane	2717	3L	Camphor
2667	6L	Butyltoluenes	2719	5P	Barium bromate
2668	6F	Chloroacetonitrile	2720	5L	Chromium nitrate
2669	6L	Chlorocresols solution	2721	5L	Copper chlorate
2670	8L	Cyanuric chloride	2722	5L	Lithium nitrate
2671	6L	Aminopyridines	2723	5L	Magnesium chlorate
2672	8L	Ammonia solution	2724	5L	Manganese nitrate
2673	6L	2-Amino-4-chlorophenol	2725	5L	Nickel nitrate
2674	6L	Sodium fluorosilicate	2726	5L	Nickel nitrite
2676	10P	Stibine	2727	6X	Thallium nitrate
2677	8L	Rubidium hydroxide solution	2728	5L	Zirconium nitrate
2678	8L	Rubidium hydroxide	2729	6L	Hexachlorobenzene
2679	8L	Lithium hydroxide solution	2730	6L	Nitroanisoles, liquid
2680	8L	Lithium hydroxide	2732	6L	Nitrobromobenzene, liquid
2681	8L	Caesium hydroxide solution	2733	3C	Amines, flammable, corrosive, n.o.s.*
2682	8L	Caesium hydroxide	2733	3C	Polyamines, flammable, corrosive, n.o.s.*
2683	8FP	Ammonium sulphide solution	2734	8F	Amines, liquid, corrosive, flammable, n.o.s.*
2684	3C	3-Diethylaminopropylamine	2734	8F	Polyamines, liquid, corrosive, flammable, n.o.s.*
2685	8F	N,N-Diethylethylenediamine	2735	8L	Amines, liquid, corrosive, n.o.s.*
2686	8F	2-Diethylaminoethanol	2735	8L	Polyamines, liquid, corrosive, n.o.s.*
2687	3L	Dicyclohexylammonium nitrite	2738	6L	N-Butylaniline
2688	6L	1-Bromo-3-chloropropane	2739	8L	Butyric anhydride
2689	6L	Glycerol alpha-monochlorohydrin			
2690	6L	N,n-Butylimidazole			

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2740	6CF	n-Propyl chloroformate	2780	3P	Substituted nitrophenol pesticide, liquid, flammable, toxic*
2741	5P	Barium hypochlorite	2781	6L	Bipyridilium pesticide, solid, toxic*
2742	6CF	Chloroformates, toxic, corrosive, flammable, n.o.s.*	2782	3P	Bipyridilium pesticide, liquid, flammable, toxic*
2743	6CF	n-Butyl chloroformate	2783	6L	Organophosphorus pesticide, solid, toxic*
2744	6CF	Cyclobutyl chloroformate	2784	3P	Organophosphorus pesticide, liquid, flammable, toxic*
2745	6C	Chloromethyl chloroformate	2785	6L	4-Thiapentanal
2746	6C	Phenyl chloroformate	2786	6L	Organotin pesticide, solid, toxic*
2747	6L	tert-Butylcyclohexyl chloroformate	2787	3P	Organotin pesticide, liquid, flammable, toxic*
2748	6C	2-Ethylhexyl chloroformate	2788	6L	Organotin compound, liquid, n.o.s.*
2749	3H	Tetramethylsilane	2789	8F	Acetic acid, glacial
2750	6L	1,3-Dichloropropanol-2	2789	8F	Acetic acid solution
2751	8L	Diethylthiophosphoryl chloride	2790	8L	Acetic acid solution
2752	3L	1,2-Epoxy-3-ethoxypropane	2793	4L	Ferrous metal borings
2753	6L	N-Ethylbenzyltoluidines, liquid	2793	4L	Ferrous metal cuttings
2754	6L	N-Ethyltoluidines	2793	4L	Ferrous metal shavings
2757	6L	Carbamate pesticide, solid, toxic*	2793	4L	Ferrous metal turnings
2758	3P	Carbamate pesticide, liquid, flammable, toxic*	2794	8L	Batteries, wet, filled with acid
2759	6L	Arsenical pesticide, solid, toxic*	2795	8L	Batteries, wet, filled with alkali
2760	3P	Arsenical pesticide, liquid, flammable, toxic*	2796	8L	Battery fluid, acid
2761	6L	Organochlorine pesticide, solid, toxic*	2796	8L	Sulphuric acid
2762	3P	Organochlorine pesticide, liquid, flammable, toxic*	2797	8L	Battery fluid, alkali
2763	6L	Triazine pesticide, solid, toxic*	2798	8L	Phenylphosphorus dichloride
2764	3P	Triazine pesticide, liquid, flammable, toxic*	2799	8L	Phenylphosphorus thiodichloride
2771	6L	Thiocarbamate pesticide, solid, toxic*	2800	8L	Batteries, wet, non-spillable
2772	3P	Thiocarbamate pesticide, liquid, flammable, toxic*	2801	8L	Dye intermediate, liquid, corrosive, n.o.s.*
2775	6L	Copper based pesticide, solid, toxic*	2801	8L	Dye, liquid, corrosive, n.o.s.*
2776	3P	Copper based pesticide, liquid, flammable, toxic*	2802	8L	Copper chloride
2777	6L	Mercury based pesticide, solid, toxic*	2803	8L	Gallium
2778	3P	Mercury based pesticide, liquid, flammable, toxic*	2805	4W	Lithium hydride, fused solid
2779	6L	Substituted nitrophenol pesticide, solid, toxic*	2806	4W	Lithium nitride
			2807	9M	Magnetized material
			2809	8P	Mercury
			2810	6L	Toxic liquid, organic, n.o.s.*

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2811	6L	Toxic solid, organic, n.o.s.*	2858	3L	Zirconium, dry
2812	8L	Sodium aluminate, solid	2859	6L	Ammonium metavanadate
2813	4W	Water-reactive solid, n.o.s.*	2861	6L	Ammonium polyvanadate
2814	11Y	Infectious substance, affecting humans	2862	6L	Vanadium pentoxide
2815	8P	N-Aminoethylpiperazine	2863	6L	Sodium ammonium vanadate
2817	8P	Ammonium hydrogendifluoride solution	2864	6L	Potassium metavanadate
2818	8P	Ammonium polysulphide solution	2865	8L	Hydroxylamine sulphate
2819	8L	Amyl acid phosphate	2869	8L	Titanium trichloride mixture
2820	8L	Butyric acid	2870	4W	Aluminium borohydride
2821	6L	Phenol solution	2870	4W	Aluminium borohydride in devices
2822	6L	2-Chloropyridine	2871	6L	Antimony powder
2823	8L	Crotonic acid, solid	2872	6L	Dibromochloropropanes
2826	8F	Ethyl chlorothioformate	2873	6L	Dibutylaminoethanol
2829	8L	Caproic acid	2874	6L	Furfuryl alcohol
2830	4W	Lithium ferrosilicon	2875	6L	Hexachlorophene
2831	6L	1,1,1-Trichloroethane	2876	6L	Resorcinol
2834	8L	Phosphorous acid	2878	3L	Titanium sponge granules
2835	4W	Sodium aluminium hydride	2878	3L	Titanium sponge powders
2837	8L	Bisulphates, aqueous solution	2879	8P	Selenium oxychloride
2838	3L	Vinyl butyrate, stabilized	2880	5L	Calcium hypochlorite, hydrated
2839	6L	Aldol	2880	5L	Calcium hypochlorite, hydrated mixture
2840	3L	Butyraldohime	2881	4L	Metal catalyst, dry*
2841	3P	Di-n-amylamine	2900	11Y	Infectious substance, affecting animals
2842	3L	Nitroethane	2901	2PX	Bromine chloride
2844	4W	Calcium manganese silicon	2902	6L	Pesticide, liquid, toxic, n.o.s.*
2845	4W	Pyrophoric liquid, organic, n.o.s.*	2903	6F	Pesticide, liquid, toxic, flammable, n.o.s.*
2846	4L	Pyrophoric solid, organic, n.o.s.*	2904	8L	Chlorophenolates, liquid
2849	6L	3-Chloropropanol-1	2904	8L	Phenolates, liquid
2850	3L	Propylene tetramer	2905	8L	Chlorophenolates, solid
2851	8L	Boron trifluoride dihydrate	2905	8L	Phenolates, solid
2852	3E	Dipicryl sulphide, wetted	2907	3L	Isosorbide dinitrate mixture
2853	6L	Magnesium fluorosilicate	2908	7L	Radioactive material, excepted package — empty packaging
2854	6L	Ammonium fluorosilicate	2909	7L	Radioactive material, excepted package — articles manufactured from natural uranium or depleted uranium or natural thorium
2855	6L	Zinc fluorosilicate			
2856	6L	Fluorosilicates, n.o.s.*			
2857	2L	Refrigerating machines			

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2910	7L	Radioactive material, excepted package — limited quantity of material	2940	4L	Cyclooctadiene phosphines
			2940	4L	9-Phosphabicyclononanes
2911	7L	Radioactive material, excepted package — instruments or articles	2941	6L	Fluoroanilines
2912	7L	Radioactive material, low specific activity (LSA-I)	2942	6L	2-Trifluoromethylaniline
2913	7L	Radioactive material, surface contaminated objects (SCO-I, SCO-II or SCO-III)	2943	3L	Tetrahydrofurfurylamine
2915	7L	Radioactive material, Type A package	2945	3C	N-Methylbutylamine
2916	7L	Radioactive material, Type B(U) package	2946	6L	2-Amino-5-diethylaminopentane
2917	7L	Radioactive material, Type B(M) package	2947	3L	Isopropyl chloroacetate
2919	7L	Radioactive material, transported under special arrangement	2948	6L	3-Trifluoromethylaniline
2920	8F	Corrosive liquid, flammable, n.o.s.*	2949	8L	Sodium hydrosulphide, hydrated
2921	8S	Corrosive solid, flammable, n.o.s.*	2950	4W	Magnesium granules, coated
2922	8P	Corrosive liquid, toxic, n.o.s.*	2956	3E	5-tert-Butyl-2,4,6-trinitro-mxylene
2923	8P	Corrosive solid, toxic, n.o.s.*	2956	3E	Musk xylene
2924		Flammable liquid, corrosive, n.o.s.*	2965	4FW	Boron trifluoride dimethyl etherate
	3CH	Packing Group I or II	2966	6L	Thioglycol
	3C	Packing Group III	2967	8L	Sulphamic acid
2925	3C	Flammable solid, corrosive, organic, n.o.s.*	2968	4W	Maneb preparation, stabilized
2926	3P	Flammable solid, toxic, organic, n.o.s.*	2968	4W	Maneb stabilized
2927	6C	Toxic liquid, corrosive, organic, n.o.s.*	2969	9L	Castor beans
2928	6C	Toxic solid, corrosive, organic, n.o.s.*	2969	9L	Castor flake
2929	6F	Toxic liquid, flammable, organic, n.o.s.*	2969	9L	Castor meal
2930	6F	Toxic solid, flammable, organic, n.o.s.*	2969	9L	Castor pomace
2931	6L	Vanadyl sulphate	2977	7CP	Radioactive material, uranium hexafluoride, fissile
2933	3L	Methyl 2-chloropropionate	2978	7CP	Radioactive material, uranium hexafluoride
2934	3L	Isopropyl 2-chloropropionate	2983	3P	Ethylene oxide and propylene oxide mixture
2935	3L	Ethyl 2-chloropropionate	2984	5L	Hydrogen peroxide, aqueous solution
2936	6L	Thiolactic acid	2985	3C	Chlorosilanes, flammable, corrosive, n.o.s.
2937	6L	alpha-Methylbenzyl alcohol, liquid	2986	8F	Chlorosilanes, corrosive, flammable, n.o.s.
			2987	8L	Chlorosilanes, corrosive, n.o.s.
			2988	4FW	Chlorosilanes, water-reactive, flammable, corrosive, n.o.s.
			2989	3L	Lead phosphite, dibasic
			2990	9L	Life-saving appliances, self-inflating

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2991	6F	Carbamate pesticide, liquid, toxic, flammable*	3024	3P	Coumarin derivative pesticide, liquid, flammable, toxic*
2992	6L	Carbamate pesticide, liquid, toxic*	3025	6F	Coumarin derivative pesticide, liquid, toxic, flammable*
2993	6F	Arsenical pesticide, liquid, toxic, flammable*	3026	6L	Coumarin derivative pesticide, liquid, toxic*
2994	6L	Arsenical pesticide, liquid, toxic*	3027	6L	Coumarin derivative pesticide, solid, toxic*
2995	6F	Organochlorine pesticide, liquid, toxic, flammable*	3028	8L	Batteries, dry, containing potassium hydroxide solid
2996	6L	Organochlorine pesticide, liquid, toxic*	3048	6W	Aluminium phosphide pesticide
2997	6F	Triazine pesticide, liquid, toxic, flammable*	3054	3L	Cyclohexyl mercaptan
2998	6L	Triazine pesticide, liquid, toxic*	3055	8L	2-(2-Aminoethoxy)ethanol
3005	6F	Thiocarbamate pesticide, liquid, toxic, flammable*	3056	3L	n-Heptaldehyde
3006	6L	Thiocarbamate pesticide, liquid, toxic*	3057	2CP	Trifluoroacetyl chloride
3009	6F	Copper based pesticide, liquid, toxic, flammable*	3064	3L	Nitroglycerin solution in alcohol
3010	6L	Copper based pesticide, liquid, toxic*	3065	3L	Alcoholic beverages
3011	6F	Mercury based pesticide, liquid, toxic, flammable*	3066	8L	Paint
3012	6L	Mercury based pesticide, liquid, toxic*	3066	8L	Paint related material
3013	6F	Substituted nitrophenol pesticide, liquid, toxic, flammable*	3070	2L	Ethylene oxide and dichlorodifluoromethane mixture
3014	6L	Substituted nitrophenol pesticide, liquid, toxic*	3071	6F	Mercaptan mixture, liquid, toxic, flammable, n.o.s.*
3015	6F	Bipyridilium pesticide, liquid toxic, flammable*	3071	6F	Mercaptans, liquid, toxic, flammable, n.o.s.*
3016	6L	Bipyridilium pesticide, liquid toxic*	3072	9L	Life-saving appliances, not self-inflating
3017	6F	Organophosphorus pesticide, liquid, toxic, flammable*	3073	6CF	Vinylpyridines, stabilized
3018	6L	Organophosphorus pesticide, liquid, toxic*	3077	9L	Environmentally hazardous substance, solid, n.o.s.*
3019	6F	Organotin pesticide, liquid, toxic, flammable*	3078	4W	Cerium
3020	6L	Organotin pesticide, liquid, toxic*	3079	6F	Methacrylonitrile, stabilized
3021	3P	Pesticide, liquid, flammable, toxic, n.o.s.*	3080	6F	Isocyanate solution, toxic, flammable, n.o.s.*
3022	3H	1,2-Butylene oxide, stabilized	3080	6F	Isocyanates, toxic, flammable, n.o.s.*
3023	6F	2-Methyl-2-heptanethiol	3082	9L	Environmentally hazardous substance, liquid, n.o.s.*
			3083	2PX	Perchloryl fluoride
			3084	8X	Corrosive solid, oxidizing, n.o.s.*
			3085	5C	Oxidizing solid, corrosive, n.o.s.*
			3086	6X	Toxic solid, oxidizing, n.o.s.*

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3087	5P	Oxidizing solid, toxic, n.o.s.*	3120	5S	Organic peroxide type F, solid, temperature controlled*
3088	4L	Self-heating solid, organic, n.o.s.*	3121	5W	Oxidizing solid, water-reactive, n.o.s.*
3089	3L	Metal powder, flammable, n.o.s.	3122	6X	Toxic liquid, oxidizing, n.o.s.*
3090	12FZ	Lithium metal batteries	3123	6W	Toxic liquid, water-reactive, n.o.s.*
3091	12FZ	Lithium metal batteries contained in equipment	3124	6S	Toxic solid, self-heating, n.o.s.*
3091	12FZ	Lithium metal batteries packed with equipment	3125	6W	Toxic solid, water-reactive, n.o.s.*
3092	3L	1-Methoxy-2-propanol	3126	4C	Self-heating solid, corrosive, organic, n.o.s.*
3093	8X	Corrosive liquid, oxidizing, n.o.s.*	3127	4X	Self-heating solid, oxidizing, n.o.s.*
3094	8W	Corrosive liquid, water-reactive, n.o.s.*	3128	4P	Self-heating solid, toxic, organic, n.o.s.*
3095	8S	Corrosive solid, self-heating, n.o.s.*	3129	4CW	Water-reactive liquid, corrosive, n.o.s.*
3096	8W	Corrosive solid, water-reactive, n.o.s.*	3130	4PW	Water-reactive liquid, toxic, n.o.s.*
3097	3X	Flammable solid, oxidizing, n.o.s.*	3131	4CW	Water-reactive solid, corrosive, n.o.s.*
3098	5C	Oxidizing liquid, corrosive, n.o.s.*	3132	4FW	Water-reactive solid, flammable, n.o.s.*
3099	5P	Oxidizing liquid, toxic, n.o.s.*	3133	4WX	Water-reactive solid, oxidizing, n.o.s.*
3100	5S	Oxidizing solid, self-heating, n.o.s.*	3134	4PW	Water-reactive solid, toxic, n.o.s.*
3103	5L	Organic peroxide type C, liquid*	3135	4SW	Water-reactive solid, self-heating, n.o.s.*
3104	5L	Organic peroxide type C, solid*	3136	2A	Trifluoromethane, refrigerated liquid
3105	5L	Organic peroxide type D, liquid*	3137	5F	Oxidizing solid, flammable, n.o.s.*
3106	5L	Organic peroxide type D, solid*	3138	10L	Ethylene, acetylene and propylene mixture, refrigerated liquid
3107	5L	Organic peroxide type E, liquid*	3139	5L	Oxidizing liquid, n.o.s.*
3108	5L	Organic peroxide type E, solid*	3140	6L	Alkaloid salts, liquid, n.o.s.*
3109	5L	Organic peroxide type F, liquid*	3140	6L	Alkaloids, liquid, n.o.s.*
3110	5L	Organic peroxide type F, solid*	3141	6L	Antimony compound, inorganic, liquid, n.o.s.*
3113	5S	Organic peroxide type C, liquid, temperature controlled*	3142	6L	Disinfectant, liquid, toxic, n.o.s.*
3114	5S	Organic peroxide type C, solid, temperature controlled*	3143	6L	Dye intermediate, solid, toxic, n.o.s.*
3115	5S	Organic peroxide type D, liquid, temperature controlled*	3143	6L	Dye, solid, toxic, n.o.s.*
3116	5S	Organic peroxide type D, solid, temperature controlled*	3144	6L	Nicotine compound, liquid, n.o.s.*
3117	5S	Organic peroxide type E, liquid, temperature controlled*	3144	6L	Nicotine preparation, liquid, n.o.s.*
3118	5S	Organic peroxide type E, solid, temperature controlled*	3145	8L	Alkylphenols, liquid, n.o.s.
3119	5S	Organic peroxide type F, liquid, temperature controlled*	3146	6L	Organotin compound, solid, n.o.s.*

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3147	8L	Dye intermediate, solid, corrosive, n.o.s.*	3166	9L	Vehicle, fuel cell, flammable liquid powered
3147	8L	Dye, solid, corrosive, n.o.s.*	3167	10L	Gas sample, non-pressurized, flammable, n.o.s.
3148	4W	Water-reactive liquid, n.o.s.*	3168	10P	Gas sample, non-pressurized, toxic, flammable, n.o.s.
3149	5C	Hydrogen peroxide and peroxyacetic acid mixture stabilized	3169	2P	Gas sample, non-pressurized, toxic, n.o.s.
3150	10L	Devices, small, hydrocarbon gas powered	3170	4W	Aluminium remelting by-products
3150	10L	Hydrocarbon gas refills for small devices	3170	4W	Aluminium smelting by-products
3151	9L	Halogenated monomethyldiphenylmethanes, liquid	3171	9L	Battery-powered equipment
3151	9L	Polyhalogenated biphenyls, liquid	3171	9L	Battery-powered vehicle
3151	9L	Polyhalogenated terphenyls, liquid	3172	6L	Toxins, extracted from living sources, liquid, n.o.s.*
3152	9L	Halogenated monomethyldiphenylmethanes, solid	3174	4L	Titanium disulphide
3152	9L	Polyhalogenated biphenyls, solid	3175	3L	Solids containing flammable liquid, n.o.s.*
3152	9L	Polyhalogenated terphenyls, solid	3176	3L	Flammable solid, organic, molten, n.o.s.*
3153	10L	Perfluoro (methyl vinyl ether)	3178	3L	Flammable solid, inorganic, n.o.s.*
3154	10L	Perfluoro (ethyl vinyl ether)	3179	3P	Flammable solid, toxic, inorganic, n.o.s.*
3155	6L	Pentachlorophenol	3180	3C	Flammable solid, corrosive, inorganic, n.o.s.*
3156	2X	Compressed gas, oxidizing, n.o.s.*	3181	3L	Metal salts of organic compounds, flammable, n.o.s.*
3157	2X	Liquefied gas, oxidizing, n.o.s.*	3182	3L	Metal hydrides, flammable, n.o.s.*
3158	2L	Gas, refrigerated liquid, n.o.s.*	3183	4L	Self-heating liquid, organic, n.o.s.*
3159	2L	Refrigerant gas R 134a	3184	4P	Self-heating liquid, toxic, organic, n.o.s.*
3159	2L	1,1,1,2-Tetrafluoroethane	3185	4C	Self-heating liquid, corrosive, organic, n.o.s.*
3160	10P	Liquefied gas, toxic, flammable, n.o.s.*	3186	4L	Self-heating liquid, inorganic, n.o.s.*
3161	10L	Liquefied gas, flammable, n.o.s.*	3187	4P	Self-heating liquid, toxic, inorganic, n.o.s.*
3162	2P	Liquefied gas, toxic, n.o.s.*	3188	4C	Self-heating liquid, corrosive, inorganic, n.o.s.*
3163	2L	Liquefied gas, n.o.s.*	3189	4L	Metal powder, self-heating, n.o.s.*
3164	2L	Articles, pressurized, hydraulic	3190	4L	Self-heating solid, inorganic, n.o.s.*
3164	2L	Articles, pressurized, pneumatic	3191	4P	Self-heating solid, toxic, inorganic, n.o.s.*
3165	3CP	Aircraft hydraulic power unit fuel tank	3192	4C	Self-heating solid, corrosive, inorganic, n.o.s.*
3166	9L	Vehicle, flammable gas powered			
3166	9L	Vehicle, flammable liquid powered			
3166	9L	Vehicle, fuel cell, flammable gas powered			

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3194	4W	Pyrophoric liquid, inorganic, n.o.s.*	3236	3S	Self-reactive solid type D, temperature controlled*
3200	4L	Pyrophoric solid, inorganic, n.o.s.*	3237	3S	Self-reactive liquid type E, temperature controlled*
3205	4L	Alkaline earth metal alcoholates, n.o.s.*	3238	3S	Self-reactive solid type E, temperature controlled*
3206	4C	Alkali metal alcoholates, self-heating, corrosive, n.o.s.*	3239	3S	Self-reactive liquid type F, temperature controlled*
3208	4W	Metallic substance, water-reactive, n.o.s.*	3240	3S	Self-reactive solid type F, temperature controlled*
3209	4SW	Metallic substance, water-reactive, self-heating, n.o.s.*	3241	3L	2-Bromo-2-nitropropane-1,3-diol
3210	5L	Chlorates, inorganic, aqueous solution, n.o.s.*	3242	3L	Azodicarbonamide
3211	5L	Perchlorates, inorganic, aqueous solution, n.o.s.	3243	6L	Solids containing toxic liquid, n.o.s.*
3212	5L	Hypochlorites, inorganic, n.o.s.*	3244	8L	Solids containing corrosive liquid, n.o.s.*
3213	5L	Bromates, inorganic, aqueous solution, n.o.s.*	3245	9L	Genetically modified micro-organisms
3214	5L	Permanganates, inorganic, aqueous solution, n.o.s.*	3245	9L	Genetically modified organisms
3215	5L	Persulphates, inorganic, n.o.s.	3246	6C	Methanesulphonyl chloride
3216	5L	Persulphates, inorganic, aqueous solution, n.o.s.	3247	5L	Sodium peroxoborate, anhydrous
3218	5L	Nitrates, inorganic, aqueous solution, n.o.s.	3248	3P	Medicine, liquid, flammable, toxic, n.o.s.
3219	5L	Nitrites, inorganic, aqueous solution, n.o.s.*	3249	6L	Medicine, solid, toxic, n.o.s.
3220	2L	Pentafluoroethane	3250	6C	Chloroacetic acid, molten
3220	2L	Refrigerant gas R 125	3251	3L	Isosorbide-5-mononitrate
3223	3L	Self-reactive liquid type C*	3252	10L	Difluoromethane
3224	3L	Self-reactive solid type C*	3252	10L	Refrigerant gas R 32
3225	3L	Self-reactive liquid type D*	3253	8L	Disodium trioxosilicate
3226	3L	Self-reactive solid type D*	3254	4L	Tributylphosphane
3227	3L	Self-reactive liquid type E*	3255	4C	tert-Butyl hypochlorite
3228	3L	Self-reactive solid type E*	3256	3L	Elevated temperature liquid, flammable, n.o.s.*
3229	3L	Self-reactive liquid type F*	3257	9L	Elevated temperature liquid, n.o.s.*
3230	3L	Self-reactive solid type F*	3258	9L	Elevated temperature solid, n.o.s.*
3233	3S	Self-reactive liquid type C, temperature controlled*	3259	8L	Amines, solid, corrosive, n.o.s.*
3234	3S	Self-reactive solid type C, temperature controlled*	3259	8L	Polyamines, solid, corrosive, n.o.s.*
3235	3S	Self-reactive liquid type D, temperature controlled*	3260	8L	Corrosive solid, acidic, inorganic, n.o.s.*
			3261	8L	Corrosive solid, acidic, organic, n.o.s.*
			3262	8L	Corrosive solid, basic, inorganic, n.o.s.*

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3263	8L	Corrosive solid, basic, organic, n.o.s.*	3291	11L	Clinical waste, unspecified, n.o.s.
3264	8L	Corrosive liquid, acidic, inorganic, n.o.s.*	3291	11L	Medical waste, n.o.s.
3265	8L	Corrosive liquid, acidic, organic, n.o.s.*	3291	11L	Regulated medical waste, n.o.s.
3266	8L	Corrosive liquid, basic, inorganic, n.o.s.*	3292	4W	Batteries, containing sodium
3267	8L	Corrosive liquid, basic, organic, n.o.s.*	3292	4W	Cells, containing sodium
3268	9L	Safety devices	3293	6L	Hydrazine, aqueous solution
3269	3L	Polyester resin kit	3294	6F	Hydrogen cyanide, solution in alcohol
3270	3L	Nitrocellulose membrane filters	3295		Hydrocarbons, liquid, n.o.s.
3271	3L	Ethers, n.o.s.*	3H		Packing Group I or II
3272	3L	Esters, n.o.s.*	3L		Packing Group III
3273	3HP	Nitriles, flammable, toxic, n.o.s.*	3296	2L	Heptafluoropropane
3274	3C	Alcoholates solution, n.o.s.*	3296	2L	Refrigerant gas R 227
3275	6F	Nitriles, toxic, flammable, n.o.s.*	3297	2L	Ethylene oxide and chlorotetrafluoroethane mixture
3276	6L	Nitriles, liquid, toxic, n.o.s.*	3298	2L	Ethylene oxide and pentafluoroethane mixture
3277	6C	Chloroformates, toxic, corrosive, n.o.s.*	3299	2L	Ethylene oxide and tetrafluoroethane mixture
3278	6L	Organophosphorus compound, liquid, toxic, n.o.s.*	3300	10P	Ethylene oxide and carbon dioxide mixture
3279	6F	Organophosphorus compound, toxic, flammable, n.o.s.*	3301	8L	Corrosive liquid, self-heating, n.o.s.*
3280	6L	Organoarsenic compound, liquid, n.o.s.*	3302	6L	2-Dimethylaminoethyl acrylate, stabilized
3281	6L	Metal carbonyls, liquid, n.o.s.*	3303	2X	Compressed gas, toxic, oxidizing, n.o.s.*
3282	6L	Organometallic compound, liquid, toxic, n.o.s.*	3304	2CP	Compressed gas, toxic, corrosive, n.o.s.*
3283	6L	Selenium compound, solid, n.o.s.*	3305	10C	Compressed gas, toxic, flammable, corrosive, n.o.s.*
3284	6L	Tellurium compound, n.o.s.*	3306	2CX	Compressed gas, toxic, oxidizing, corrosive, n.o.s.*
3285	6L	Vanadium compound, n.o.s.*	3307	2X	Liquefied gas, toxic, oxidizing, n.o.s.*
3286	3CP	Flammable liquid, toxic, corrosive, n.o.s.*	3308	2CP	Liquefied gas, toxic, corrosive, n.o.s.*
3287	6L	Toxic liquid, inorganic, n.o.s.*	3309	10C	Liquefied gas, toxic, flammable, corrosive, n.o.s.*
3288	6L	Toxic solid, inorganic, n.o.s.*	3310	2CX	Liquefied gas, toxic, oxidizing, corrosive, n.o.s.*
3289	6C	Toxic liquid, corrosive, inorganic, n.o.s.*	3311	2X	Gas, refrigerated liquid, oxidizing, n.o.s.*
3290	6C	Toxic solid, corrosive, inorganic, n.o.s.*			
3291	11L	Biomedical waste, n.o.s.			

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3312	10L	Gas, refrigerated liquid, flammable, n.o.s.*	3H		Packing Group I or II
3313	4L	Organic pigments, self-heating	3L		Packing Group III
3314	9L	Plastics moulding compound	3336		Mercaptans, liquid, flammable, n.o.s.*
3315	6L	Chemical sample, toxic	3H		Packing Group I or II
3316	9L	Chemical kit	3L		Packing Group III
3316	9L	First aid kit	3337	2L	Refrigerant gas R 404A
3317	3E	2-Amino-4,6-dinitrophenol, wetted	3338	2L	Refrigerant gas R 407A
3318	2CP	Ammonia solution	3339	2L	Refrigerant gas R 407B
3319	3L	Nitroglycerin mixture, desensitized, solid, n.o.s.*	3340	2L	Refrigerant gas R 407C
3320	8L	Sodium borohydride and sodium hydroxide solution	3341	4L	Thiourea dioxide
3321	7L	Radioactive material, low specific activity (LSA-II)	3342	4L	Xanthates
3322	7L	Radioactive material, low specific activity (LSA-III)	3343	3E	Nitroglycerin mixture, desensitized, liquid flammable, n.o.s.*
3323	7L	Radioactive material, Type C package	3344	3E	Pentaerythrite tetranitrate mixture desensitized, solid, n.o.s.*
3324	7L	Radioactive material, low specific activity (LSA-II), fissile	3344	3E	Pentaerythritol tetranitrate mixture desensitized, solid, n.o.s.*
3325	7L	Radioactive material, low specific activity (LSA-III), fissile	3344	3E	PETN mixture desensitized, solid, n.o.s.*
3326	7L	Radioactive material, surface contaminated objects (SCO-I or SCO-II), fissile	3345	6L	Phenoxyacetic acid derivative pesticide, solid, toxic*
3327	7L	Radioactive material, Type A package, fissile	3346	3P	Phenoxyacetic acid derivative pesticide, liquid, flammable, toxic*
3328	7L	Radioactive material, Type B(U) package, fissile	3347	6F	Phenoxyacetic acid derivative pesticide, liquid, toxic, flammable*
3329	7L	Radioactive material, Type B(M) package, fissile	3348	6L	Phenoxyacetic acid derivative pesticide, liquid, toxic*
3330	7L	Radioactive material, Type C package, fissile	3349	6L	Pyrethroid pesticide, solid, toxic*
3331	7L	Radioactive material, transported under special arrangement, fissile	3350	3P	Pyrethroid pesticide, liquid flammable, toxic*
3332	7L	Radioactive material, Type A package, special form	3351	6F	Pyrethroid pesticide, liquid, toxic, flammable*
3333	7L	Radioactive material, Type A package, special form, fissile	3352	6L	Pyrethroid pesticide, liquid, toxic*
3334	9A	Aviation regulated liquid, n.o.s.*	3354	10L	Insecticide gas, flammable, n.o.s.*
3335	9A	Aviation regulated solid, n.o.s.*	3355	10P	Insecticide gas, toxic, flammable, n.o.s.*
3336		Mercaptan mixture, liquid, flammable, n.o.s.*	3356	5L	Oxygen generator, chemical
			3357	3L	Nitroglycerin mixture, desensitized, liquid, n.o.s.*
			3358	10L	Refrigerating machines
			3359	9L	Fumigated cargo transport unit

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3360	3L	Fibres, vegetable, dry	3387	6X	Toxic by inhalation liquid, oxidizing, n.o.s.*
3361	6C	Chlorosilanes, toxic, corrosive, n.o.s.*	3388	6X	Toxic by inhalation liquid, oxidizing, n.o.s.*
3362	6CF	Chlorosilanes, toxic, corrosive, flammable, n.o.s.*	3389	6C	Toxic by inhalation liquid, corrosive, n.o.s.*
3363	9L	Dangerous goods in apparatus	3390	6C	Toxic by inhalation liquid, corrosive, n.o.s.*
3363		Dangerous goods in articles	3391	4L	Organometallic substance, solid, pyrophoric*
3363	9L	Dangerous goods in machinery	3392	4L	Organometallic substance, liquid, pyrophoric*
3364	3E	Picric acid, wetted	3393	4W	Organometallic substance, solid, pyrophoric, water reactive*
3364	3E	Trinitrophenol, wetted	3394	4W	Organometallic substance, liquid, pyrophoric, water reactive*
3365	3E	Picryl chloride, wetted	3395	4W	Organometallic substance, solid, water reactive*
3365	3E	Trinitrochlorobenzene, wetted	3396	4W	Organometallic substance, solid, water reactive, flammable*
3366	3E	TNT, wetted	3397	4W	Organometallic substance, solid, water reactive, self-heating*
3366	3E	Trinitrotoluene, wetted	3398	4W	Organometallic substance, liquid, water reactive*
3367	3E	Trinitrobenzene, wetted	3399	4FW	Organometallic substance, liquid, water reactive, flammable*
3368	3E	Trinitrobenzoic acid, wetted	3400	4L	Organometallic substance, solid, self-heating*
3369	3E	Sodium dinitro-o-cresolate, wetted	3401	4W	Alkali metal amalgam, solid
3370	3E	Urea nitrate, wetted	3402	4W	Alkaline earth metal amalgam, solid
3371	3L	2-Methylbutanal	3403	4W	Potassium metal alloys, solid
3373	11L	Biological substance, Category B	3404	4W	Potassium sodium alloys, solid
3374	10L	Acetylene, solvent free	3405	5P	Barium chlorate solution
3375	5L	Ammonium nitrate emulsion	3406	5P	Barium perchlorate solution
3375	5L	Ammonium nitrate gel	3407	5L	Chlorate and magnesium chloride mixture solution
3375	5L	Ammonium nitrate suspension	3408	5P	Lead perchlorate solution
3376	3E	4-Nitrophenylhydrazine	3409	6L	Chloronitrobenzenes, liquid
3377	5L	Sodium perborate monohydrate	3410	6L	4-Chloro-o-toluidine hydrochloride solution
3378	5L	Sodium carbonate peroxyhydrate	3411	6L	beta-Naphthylamine solution
3379	3E	Desensitized explosive, liquid, n.o.s.*	3412	8L	Formic acid
3380	3E	Desensitized explosive, solid, n.o.s.*	3413	6L	Potassium cyanide solution
3381	6L	Toxic by inhalation liquid, n.o.s.*			
3382	6L	Toxic by inhalation liquid, n.o.s.*			
3383	6F	Toxic by inhalation liquid, flammable, n.o.s.*			
3384	6F	Toxic by inhalation liquid, flammable, n.o.s.*			
3385	6W	Toxic by inhalation liquid, water-reactive, n.o.s.*			
3386	6W	Toxic by inhalation liquid, water-reactive, n.o.s.*			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3414	6L	Sodium cyanide solution	3449	6L	Bromobenzyl cyanides, solid
3415	6L	Sodium fluoride solution	3450	6L	Diphenylchloroarsine, solid
3416	6i	Chloroacetophenone, liquid	3451	6L	Toluidines, solid
3417	6L	Xylyl bromide, solid	3452	6L	Xylidines, solid
3418	6L	2,4-Toluylenediamine solution	3453	8L	Phosphoric acid, solid
3419	8L	Boron trifluoride acetic acid complex, solid	3454	6L	Dinitrotoluenes, solid
3420	8L	Boron trifluoride propionic acid complex, solid	3455	6C	Cresols, solid
3421	8P	Potassium hydrogendifluoride solution	3456	8L	Nitrosylsulphuric acid, solid
3422	6L	Potassium fluoride solution	3457	6L	Chloronitrotoluenes, solid
3423	8L	Tetramethylammonium hydroxide, solid	3458	6L	Nitroanisoles, solid
3424	6L	Ammonium dinitro-o-cresolate solution	3459	6L	Nitrobromobenzene, solid
3425	8L	Bromoacetic acid, solid	3460	6L	N-Ethylbenzyltoluidines, solid
3426	6L	Acrylamide solution	3462	6L	Toxins, extracted from living sources, solid, n.o.s.*
3427	6L	Chlorobenzyl chlorides, solid	3463	8F	Propionic acid
3428	6L	3-Chloro-4-methylphenyl isocyanate, solid	3464	6L	Organophosphorus compound, solid, toxic, n.o.s.*
3429	6L	Chlorotoluidines, liquid	3465	6L	Organoarsenic compound, solid, n.o.s.*
3430	6L	Xylenols, liquid	3466	6L	Metal carbonyls, solid, n.o.s.*
3431	6L	Nitrobenzotrifluorides, solid	3467	6L	Organometallic compound, solid, toxic, n.o.s.*
3432	9L	Polychlorinated biphenyls, solid	3468	10L	Hydrogen in a metal hydride storage system
3434	6L	Nitrocresols, liquid	3468	10L	Hydrogen in a metal hydride storage system contained in equipment
3436	6L	Hexafluoroacetone hydrate, solid	3468	10L	Hydrogen in a metal hydride storage system packed with equipment
3437	6L	Chlorocresols, solid	3469		Paint, flammable, corrosive
3438	6L	alpha-Methylbenzyl alcohol, solid		3CH	Packing Group I or II
3439	6L	Nitriles, solid, toxic, n.o.s.*		3CL	Packing Group III
3440	6L	Selenium compound, liquid, n.o.s.*	3469		Paint related material, flammable, corrosive
3441	6L	Chlorodinitrobenzenes, solid		3CH	Packing Group I or II
3442	6L	Dichloroanilines, solid		3CL	Packing Group III
3443	6L	Dinitrobenzenes, solid	3470	8F	Paint, corrosive, flammable
3444	6L	Nicotine hydrochloride, solid	3470	8F	Paint related material corrosive, flammable
3445	6L	Nicotine sulphate, solid	3471	8L	Hydrogendifluorides, solution, n.o.s.
3446	6L	Nitrotoluenes, solid	3472	8L	Crotonic acid, liquid
3447	6L	Nitroxylenes, solid	3473	3L	Fuel cell cartridges
3448	6L	Tear gas substance, solid, n.o.s.*			

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3473	3L	Fuel cell cartridges contained in equipment	3486	5C	Calcium hypochlorite mixture, dry, corrosive
3473	3L	Fuel cell cartridges packed with equipment	3487	5C	Calcium hypochlorite, hydrated, corrosive
3474	3E	1-Hydroxybenzotriazole monohydrate	3487	5C	Calcium hypochlorite, hydrated mixture, corrosive
3475	3L	Ethanol and gasoline mixture	3488	6CF	Toxic by inhalation liquid, flammable, corrosive, n.o.s.*
3475	3L	Ethanol and motor spirit mixture	3489	6CF	Toxic by inhalation liquid, flammable, corrosive, n.o.s.*
3475	3L	Ethanol and petrol mixture	3490	6FW	Toxic by inhalation liquid, water-reactive, flammable, n.o.s.*
3476	4W	Fuel cell cartridges	3491	6FW	Toxic by inhalation liquid, water-reactive, flammable, n.o.s.*
3476	4W	Fuel cell cartridges contained in equipment	3494	3P	Petroleum sour crude oil, flammable, toxic
3476	4W	Fuel cell cartridges packed with equipment	3495	8P	Iodine
3477	8L	Fuel cell cartridges	3496	9L	Batteries, nickel-metal hydride
3477	8L	Fuel cell cartridges contained in equipment	3497	4L	Krill meal
3477	8L	Fuel cell cartridges packed with equipment	3498	8L	Iodine monochloride, liquid
3478	10L	Fuel cell cartridges	3499	9L	Capacitor, electric double layer
3478	10L	Fuel cell cartridges contained in equipment	3500	2L	Chemical under pressure, n.o.s.*
3478	10L	Fuel cell cartridges packed with equipment	3501	10L	Chemical under pressure, flammable, n.o.s.*
3479	10L	Fuel cell cartridges	3502	2P	Chemical under pressure, toxic, n.o.s.*
3479	10L	Fuel cell cartridges contained in equipment	3503	2C	Chemical under pressure, corrosive, n.o.s.*
3479	10L	Fuel cell cartridges packed with equipment	3504	10P	Chemical under pressure, flammable, toxic, n.o.s.*
3480	12FZ	Lithium ion batteries	3505	10C	Chemical under pressure, flammable, corrosive, n.o.s.*
3481	12FZ	Lithium ion batteries contained in equipment	3506	8L	Mercury contained in manufactured articles
3481	12FZ	Lithium ion batteries packed with equipment	3507	6C	Uranium hexafluoride, radioactive material, excepted package
3482	4W	Alkali metal dispersion, flammable	3508	9L	Capacitor, asymmetric
3482	4W	Alkaline earth metal dispersion, flammable	3509	9L	Packagings, discarded, empty, uncleaned
3483	6F	Motor fuel anti-knock mixture, flammable	3510	10L	Adsorbed gas, flammable, n.o.s.*
3484	8FP	Hydrazine aqueous solution, flammable	3511	2L	Adsorbed gas, n.o.s.*
3485	5C	Calcium hypochlorite, dry, corrosive	3512	2P	Adsorbed gas, toxic, n.o.s.*
3485	5C	Calcium hypochlorite mixture, dry, corrosive	3513	2X	Adsorbed gas, oxidizing, n.o.s.*

<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>	<i>UN No.</i>	<i>Drill Code</i>	<i>Proper Shipping Name</i>
3514	10P	Adsorbed gas, toxic, flammable, n.o.s.*	3534	3L	Polymerizing substance, liquid, temperature controlled, n.o.s.*
3515	2PX	Adsorbed gas, toxic, oxidizing, n.o.s.*	3535	6F	Toxic solid, flammable, inorganic, n.o.s.*
3516	2CP	Adsorbed gas, toxic, corrosive, n.o.s.*	3536	12FZ	Lithium batteries installed in cargo transport unit
3517	10C	Adsorbed gas, toxic, flammable, corrosive, n.o.s.*	3537	10L	Articles containing flammable gas, n.o.s.*
3518	2PX	Adsorbed gas, toxic, oxidizing, corrosive, n.o.s.*	3538	2L	Articles containing non- flammable, non toxic gas, n.o.s.*
3519	2CP	Boron trifluoride, adsorbed	3539	2P	Articles containing toxic gas, n.o.s.*
3520	2PX	Chlorine, adsorbed	3540	3L	Articles containing flammable liquid, n.o.s.*
3521	2CP	Silicon tetrafluoride, adsorbed	3541	3L	Articles containing flammable solid, n.o.s.*
3522	10P	Arsine, adsorbed	3542	4L	Articles containing a substance liable to spontaneous combustion, n.o.s.*
3523	10P	Germane, adsorbed	3543	4W	Articles containing a substance which emits flammable gas in contact with water, n.o.s.*
3524	2CP	Phosphorus pentafluoride, adsorbed	3544	5L	Articles containing oxidizing substance, n.o.s.*
3525	10P	Phosphine, adsorbed	3545	5L	Articles containing organic peroxide, n.o.s.*
3526	10P	Hydrogen selenide, adsorbed	3546	6L	Articles containing toxic substance, n.o.s.*
3527	3S	Polyester resin kit	3547	8L	Articles containing corrosive substance, n.o.s.*
3528	3L	Engine, fuel cell, flammable liquid powered	3548	9L	Articles containing miscellaneous dangerous goods, n.o.s.*
3528	3L	Engine, internal combustion, flammable liquid powered	3549	11Y	Medical waste, Category A affecting animals
3528	3L	Machinery, fuel cell, flammable liquid powered	3549	11Y	Medical waste, Category A affecting humans
3528	3L	Machinery, internal combustion, flammable liquid powered	3550	6L	Cobalt dihydroxide powder
3529	10L	Engine, fuel cell, flammable gas powered	8000	9L	Consumer commodity
3529	10L	Engine, internal combustion, flammable gas powered			
3529	10L	Machinery, fuel cell, flammable gas powered			
3529	10L	Machinery, internal combustion, flammable gas powered			
3530	9L	Engine, internal combustion			
3530	9L	Machinery, internal combustion			
3531	3L	Polymerizing substance, solid, stabilized, n.o.s.*			
3532	3L	Polymerizing substance, liquid, stabilized, n.o.s.*			
3533	3L	Polymerizing substance, solid, temperature controlled, n.o.s.*			

Annex 9.B: Aircraft Emergency Response Drills (Reproduced from ICAO Doc 9481 Edition 2025–2026)

The Aircraft Emergency Response Drills as shown in Table 9.B below are for crew guidance in case of a Dangerous Goods incident in flight occurs which is, or might be, related to a particular package or packages containing Dangerous Goods.

When such a package has been identified, the corresponding entry on the pilot-in-command's dangerous goods notification form should be found. The applicable drill code may be given on the notification form, or if not given it can be found by noting the proper shipping name or the UN number on the notification form and by using the alphabetical or numerical list of dangerous goods given in Tables 3 and 4, respectively.

The drill code assigned to an item of dangerous goods consists of a number plus one or two letters. Referring to the chart of emergency response drills, each drill number corresponds to a line of information concerning the hazard posed by that substance and guidance on the preferable action that should be taken. The drill letter is shown separately on the drill chart; it indicates other possible hazards of the substance. In some cases, the guidance given by the drill number may be further refined by the information given by the drill letter(s).

Table: Aircraft Emergency Response Drills						
1. COMPLETE APPROPRIATE AIRCRAFT EMERGENCY PROCEDURES.						
2. CONSIDER LANDING AS SOON AS PRACTICABLE.						
3. USE DRILL FROM THE CHART BELOW.						
Drill No.	Inherent Hazard	Hazard to Aircraft	Hazard to Occupants	Spill or Leak Procedure	Firefighting Procedure	Additional Considerations
1	Explosion may cause structural failure	Fire and/or explosion	As indicated by the drill letter(s)	Use 100% oxygen; no smoking	All agents according to availability; use standard fire procedure	Possible abrupt loss of pressurization
2	Gas, non-flammable, pressure may create hazard in fire	Minimal	As indicated by the drill letter(s)	Use 100% oxygen; establish and maintain maximum ventilation for “A”, “i” or “P” drill letter	All agents according to availability; use standard fire procedure	Possible abrupt loss of pressurization
3	Flammable liquid or solid	Fire and/or explosion	Smoke, fumes and heat, and as indicated by the drill letter(s)	Use 100% oxygen; establish and maintain maximum ventilation; no smoking; minimum electrics	All agents according to availability; no water on “W” drill letter	Possible abrupt loss of pressurization
4	Spontaneously combustible or pyrophoric when exposed to air	Fire and/or explosion	Smoke, fumes and heat, and as indicated by the drill letter(s)	Use 100% oxygen; establish and maintain maximum ventilation	All agents according to availability; no water on “W” drill letter	Possible abrupt loss of pressurization; minimum electrics if “F” or “H” drill letter
5	Oxidizer, may ignite other materials, may explode in heat of a fire	Fire and/or explosion, possible corrosion damage	Eye, nose and throat irritation; skin damage on contact	Use 100% oxygen; establish and maintain maximum ventilation	All agents according to availability; no water on “W” drill letter	Possible abrupt loss of pressurization
6	Toxic*, may be fatal if inhaled, ingested, or absorbed by skin	Contamination with toxic* liquid or solid	Acute toxicity, effects may be delayed	Use 100% oxygen; establish and maintain maximum ventilation; do not touch without gloves	All agents according to availability; no water on “W” drill letter	Possible abrupt loss of pressurization; minimum electrics if “F” or “H” drill letter

Table: Aircraft Emergency Response Drills**1. COMPLETE APPROPRIATE AIRCRAFT EMERGENCY PROCEDURES.****2. CONSIDER LANDING AS SOON AS PRACTICABLE.****3. USE DRILL FROM THE CHART BELOW.**

Drill No.	Inherent Hazard	Hazard to Aircraft	Hazard to Occupants	Spill or Leak Procedure	Firefighting Procedure	Additional Considerations
7	Radiation from broken/unshielded packages	Contamination with spilled radioactive material	Exposure to radiation, and personnel contamination	Do not move packages; avoid contact	All agents according to availability	Call for a qualified person to meet the aircraft
8	Corrosive, fumes disabling if inhaled or in contact with skin	Possible corrosion damage	Eye, nose and throat irritation; skin damage on contact	Use 100% oxygen; establish and maintain maximum ventilation; do not touch without gloves	All agents according to availability; no water on "W" drill letter	Possible abrupt loss of pressurization; minimum electrics if "F" or "H" drill letter
9	No general inherent risk	As indicated by the drill letter	As indicated by the drill letter	Use 100% oxygen; establish and maintain maximum ventilation if "A" drill letter	All agents according to availability	None
10	Gas, flammable, high fire risk if any ignition source present	Fire and/or explosion	Smoke, fumes and heat, and as indicated by the drill letter	Use 100% oxygen; establish and maintain maximum ventilation; no smoking; minimum electrics	All agents according to availability	Possible abrupt loss of pressurization
11	Infectious substances may affect humans or animals if inhaled, ingested or absorbed through the mucous membrane or an open wound	Contamination with Infectious substances	Delayed infection to humans or animals	Do not touch. Minimum re-circulation and ventilation in affected area	All agents according to availability. No water on "Y" drill letter	Call for a qualified person to meet the aircraft

Table: Aircraft Emergency Response Drills						
1. COMPLETE APPROPRIATE AIRCRAFT EMERGENCY PROCEDURES.						
2. CONSIDER LANDING AS SOON AS PRACTICABLE.						
3. USE DRILL FROM THE CHART BELOW.						
Drill No.	Inherent Hazard	Hazard to Aircraft	Hazard to Occupants	Spill or Leak Procedure	Firefighting Procedure	Additional Considerations
12	Fire, heat, smoke, toxic and flammable vapour	Fire and/or explosion	Smoke, fumes, heat	Use 100% oxygen; establish and maintain maximum ventilation	All agents according to availability. Use water if available	Possible abrupt loss of pressurization; consider landing immediately
Drill Letter	Additional Hazard		Drill Letter	Additional Hazard		
A	ANAESTHETIC		S	SPONTANEOUSLY COMBUSTIBLE OR PYROPHORIC		
C	CORROSIVE					
E	EXPLOSIVE		W	IF WET GIVES OFF TOXIC* OR FLAMMABLE GAS		
F	FLAMMABLE					
H	HIGHLY IGNITABLE		X	OXIDIZER		
i	IRRITANT/TEAR PRODUCING		Y	DEPENDING ON THE TYPE OF INFECTIOUS SUBSTANCE, THE APPROPRIATE NATIONAL AUTHORITY MAY BE REQUIRED TO QUARANTINE INDIVIDUALS, ANIMALS, CARGO AND THE AIRCRAFT		
L	OTHER HAZARD LOW OR NONE					
M	MAGNETIC					
N	NOXIOUS					
P	TOXIC* (POISON)		Z	AIRCRAFT CARGO FIRE SUPPRESSION SYSTEM MAY NOT EXTINGUISH OR CONTAIN THE FIRE; CONSIDER LANDING IMMEDIATELY		
*Toxic has the same meaning as poison.						

The above table is reproduced from ICAO Dangerous Goods Regulations Emergency Response Guide DOC 9481 2025–2026 EDITION.

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10 SECURITY

Malta Air has established and maintains an approved Security Programme. The primary objective of the Malta Air Security programme is to safeguard the safety and security of Malta Air employees, passengers and other airport users.

The Security Programme addresses all prescribed security areas including but not limited to the following:

- Security of Passengers and Cabin Baggage,
- Security of hold baggage,
- Firearms and weapons including Personal Protection Officers,
- Screening procedures,
- Rushed Bag security policy,
- Aircraft Security,
- Company Mail,
- Catering,
- Staff recruitment and training.

The responsibility for security standards rests primarily with the Safety and Security Manager and with individual Departmental Managers. This section contains Security instructions, guidance, procedures, training and responsibilities, taking into account

- ORO.SEC.100;
- CAT.GEN.MPA.135; AMC1 CAT.GEN.MPA.135(a)(3)
- Regulation (EC) No. 300/2008 Common rules in the field of aviation security.
- Regulation (EU) 2018/1139 Annex IV, 8.d
- Annex 17 of the Chicago Convention-International Standards and Recommended practices (as amended)
- ECAC Doc 30 Part II

Some parts of the security instructions and guidance may be kept confidential

10.1 Malta Air Security Programme

The Malta Air Security Programme complies with the Requirements of the Maltese National Civil Aviation Security Programme and current EU Regulations. The Malta Air Security Programme is approved by the Aviation Security Malta (AVSEC) and is therefore recognised by all other member states as having fulfilled the requirements of Reg (EC) 300/2008.

The Malta Air Security Programme specifies the Malta Air Security Policy and Procedures and is a RESTRICTED Document. Malta Air Base Captains have been issued with a controlled copy of this Programme and will make it available to crews for reference. This programme shall NOT be removed from the Base Captain's custody.

States may conduct security audits, inspections and tests to verify compliance with applicable security requirements. Malta Air therefore provides EC and National Inspectors with access to aircraft for the purpose of audit/inspection.

Inspectors will carry proof of identity at all times which shall be inspected and verified by a crewmember.

The Malta Air form 'record of external inspection of security procedures' shall be completed and one copy given to the Inspector and the other returned to Malta Air. The CSS will have this form on each duty.

10.1.1 Enroute Security Threats

Security threats are made for a variety of reasons, normally aimed at causing distress and disruption to the schedule. Often, these sorts of threats are made as a written or verbal bomb threat/warning. Whilst the threats have been spurious, the level of disruption and inconvenience to our passengers has often been significant and Malta Air will seek to prosecute any person committing such an act.

Each threat/warning shall be taken seriously, recorded, reported, and relevant departments shall run their approved procedures with the aim to facilitate a safe and secure outcome. The actions to be taken by the Commander if the aircraft is in flight are detailed below. Actions to be taken if the aircraft is on the ground are detailed in the Security Programme and in the Malta Air OMA. When receiving a threat against a dispatched flight, the Operations Control Centre shall complete a full Threat Assessment according to established procedures to determine the risk associated and actions to be taken.

- **GREEN** – No credible threat exists. Diversion not required. Continue to destination as planned.
- **AMBER** – Credibility uncertain. The flight will continue to the planned destination or divert to a suitable alternate, as designated by the Pilot in Command, ATC, or the National Authorities.

- **RED** – There is a credible threat and the Commander should land at the nearest suitable airport as by the Commander or directed by ATC or the National Authorities.

ACTION TO BE TAKEN BY THE COMMANDER IN THE EVENT OF ENROUTE SECURITY THREAT

The Commander may be alerted to a security threat (e.g., explosive device, poison gas, etc.) through three sources:

- Written threat(s) found on board.
- Verbal threat(s) from passenger(s).
- Notification from Air Traffic Control.

The following procedures shall be used by flight crew when communicating with Air Traffic Control.

Note: Anytime a credible, clear, and immediate danger threatens the safety/security of the flight, divert to the nearest suitable alternate.

10.1.1.1 Written Threat(s) Found Onboard

1. Continue as per current ATC Clearance and Operational Flight Plan.
2. Contact the CSS and confirm:
 - a. If anything was found during Security Check/Search prior to boarding,
 - b. If anything was found whilst conducting on-board duties during the flight,
 - c. How the written threat was found onboard,
 - d. Where was the written threat found onboard.
3. Take a photograph of the written threat with vPOS and/or EFB.
4. Place the message in a sick bag/oven bag in order to protect it, handle the message as little as possible.

The Commander shall conduct a PIOSEE review of the situation using the following questions:

- Does it specify your specific flight number, route, aircraft, and/or departure/destination?
- Has the crew been advised of any high-profile passengers on board?
- Does it give a specific reason or make demands?

If the answers to the questions above are NO, and there is no other evident threat to the safety and security of the flight, the situation may be assessed as GREEN:

1. Continue to the planned destination.
2. There is no need to notify ATC.
3. Do not make any passenger announcements regarding the potential security threat.
4. Continue to monitor the onboard status according to SOP.
5. The Commander shall request Police to meet the aircraft upon arrival.
6. Prior to disembarking, advise the Police accordingly, producing the evidence.
7. After facilitating Police requirements, contact the Duty Pilot and OCC.
8. File ASR/CSR at the completion of the FDP.

If the situation cannot be assessed as GREEN, and the credibility is UNCERTAIN, the situation may be assessed as AMBER:

1. The flight will continue to the planned destination or divert to a suitable airport, as designated by the Pilot in Command, ATC, and/or the National Authorities.
2. Advise ATC. Commanders must be careful to accurately describe the message e.g. “vague threat scribbled on the back of a sick bag on the floor” or “typed note found in galley specifying our destination but not flight number”.
3. Follow ATC instructions if directed to divert.
4. Do not make any passenger announcements regarding the potential security threat.
5. Continue to monitor the onboard status.
6. The Commander shall request Police to meet the aircraft upon arrival.
7. Prior to disembarking, advise the Police accordingly and provide evidence of threat.
8. After facilitating Police requirements, contact the Duty Pilot and OCC.
9. File ASR/CSR at the completion of the FDP.

If the situation cannot be assessed as GREEN, and the credibility is CERTAIN, the situation should be assessed as RED:

1. The flight will divert to a suitable airport (or continue to the planned destination if this is determined to be the safest course of action), as designated by the Pilot in Command, ATC, and/or the National Authorities.
2. Advise ATC, specifically providing information in answer to the questions above and follow the associated procedure as specified in the Malta Air Operations Manual, Part A.
3. Follow ATC instructions if directed to divert.
4. Continue to monitor the onboard status.
5. Consider declaring PAN/MAYDAY. OMA 10.3.3, OMA 8.10.2.1–8.10.2.2 refers.
6. The Commander shall request Police to meet the aircraft upon arrival.
7. Pull the CVR CB in the Shutdown Checklist as per OMA 11.7.5.
8. Upon arrival, the Commander and crew shall follow instructions from Authorities to facilitate a safe and secure outcome.
9. After facilitating Police requirements, contact the Duty Pilot and OCC.
10. File ASR/CSR at the completion of the FDP.

10.1.1.2 Verbal Threat(s) From Passenger(s)

1. Is the passenger threatening the safety and/or security of the Aircraft?
2. Does the passenger claim to have some sort of weapon or device?
3. Does passenger appear credible, give plausible information or make specific demands/threats?
4. Has passenger previously been otherwise non-disruptive during the flight?
5. Does the passenger appear to be under the influence of any substance?

If the answer to questions 1, 2, 3, or 4 is NO, the passenger may be treated as merely disruptive and follow normal disruptive passenger procedures.

If the answer to either question 1, 2, or 3 above is **YES**, then:

1. Continue as per current ATC Clearance and Operational Flight Plan.
2. Gather as much information as possible.

3. Ensure the safety and security of the passengers, crew, aircraft using established procedures as specified in the Malta Air Operations Manual, Part A.
4. Advise ATC, if appropriate.
5. Continue to the planned destination if this is determined to be the safest course of action or divert to a suitable airport as designated by the Pilot in Command, ATC, and/or the National Authorities.
6. Continue to monitor the onboard status.
7. The Commander shall request Police to meet the aircraft upon arrival.
8. Prior to disembarking, advise the Police accordingly.
9. After facilitating Police requirements, contact the Duty Pilot and OCC.
10. File ASR/CSR at the completion of the FDP.

10.1.1.3 Notificaton From Air Traffic Control

1. Continue as per current ATC Clearance and Operational Flight Plan.
2. Confirm with ATC that the Company Operations Control Centre has been notified.
3. Follow OCC/ATC instructions.
4. Continue to monitor the onboard status.
5. If applicable, initiate associated procedure as specified in the Malta Air OMA.

In all cases of an alleged or actual security threat(s), the Commander must file an ASR, and the Cabin Crew file CSRs, upon completion of the flight duty period.

10.1.2 Discovery and Handling of a Suspect Object/Package While in Flight

Discovery and handling of a Suspect Object/Package while in flight is covered in the Security Programme and in the OMA-SEP.

10.1.3 Security Checks and Searches

Malta Air procedures for Security Checks and Searches are described in the Malta Air Security Programme.

OMA-SEP and Restricted Memo contain Aircrew Security Check and Search procedures.

Malta Air shall ensure that a passenger is not in any part of an aircraft in-flight that is not designed for the accommodation of passengers (CAT.GEN.MPA.165).

10.1.4 Sabotage

Sabotage is a deliberate act or omission intended to damage or destroy aircraft, facilities or property which directly or indirectly create an air safety hazard.

This is dealt with in the context of security precautions/measures and vigilance in general by Flight Crew members and employees of the airline/handling agency/state agencies/airport authority. Implementation of security standards based on the perceived threat are identified in the Malta Air Security Programme and the Malta Air OMA-SEP.

10.1.5 Hi-jacking – Unlawful Interference

A Hijack may occur at any time. The handling of a Hijack Incident and the procedures to be followed are contained in the Security Programme, the Malta Air OMA and the OMA-SEP. The term “unlawful interference” and “special security event” also refer to a Hijack situation. Refer also to Section 10.3, Airborne Security Events

10.1.5.1 Other Malta Air Security Policies

In a hijack/unlawful interference situation the immediate priority is to land the aircraft safely as soon as possible. Once on the ground maintain the flight deck secure, keep ATC advised of status and follow State Authorities’ instructions. On no account are the pilots to open the flight deck door until the unlawful interference/hijack situation is resolved. Use the deadbolt “Locked key inoperative” position to ensure the flight deck door is locked.

10.1.5.2 Hijack Procedures Airborne

A hijack situation in flight will invariably occur in the cabin and the flight deck will be made aware of this from the cabin. On no account are the pilots to open the flight deck door until the unlawful interference/hijack situation is resolved. Use the deadbolt “Locked key inoperative” position to ensure the flight deck door is locked.

- Set Transponder to 7500.

- As soon as possible inform ATC using correct phraseology and include “Flight deck secure” “MAYDAY-MAYDAY-MAYDAY; UNLAWFUL INTERFERENCE/HIJACK; FLIGHT DECK SECURE; MALTA AIR 123”.
- Plan to land as soon as possible and inform ATC. ATC will accept your landing location unless a State Authority prevents this. ATC will then suggest suitable airports where you can land safely.
- Follow instructions as directed by State agency or ATC.
- Avoid violent aircraft manoeuvres.

10.1.5.3 Hijack Procedures on Ground

Advise ATC. Follow their instructions. It is highly unlikely that the flight will be allowed to take off. You may be required to position the aircraft to another part of the airport.

After landing in a hijack situation complete all after landing checks. Ensure the flight deck door is locked using the dead bolt “Locked key inoperative” position. The flight deck must remain secure. The Commander shall keep the Authorities advised. The Commander should recognize that his assessment of the situation is limited to the confines of the aircraft. He must expect the Police or other Authorities to take control. It then becomes his duty to comply with their instructions to the best of his ability.

10.1.6 Crime on Board

The Commander has the authority to give commands for the purpose of securing the safety of the aircraft and of persons and property carried on board. Any person who does not obey such a lawful command shall be reported to the Garda/Police/Civil Authority when the aircraft lands.

While the ability of the Commander and the crew to deal with crime on board is limited by the resources immediately available, the Commander shall request through ATC or the Company for police to meet the aircraft on arrival.

Any other crime, such as stealing, personal abuse or assault, not affecting safety shall be dealt within a similar way. The Cabin Crew will take names and addresses of those affected and possible witnesses and preserve evidence as appropriate.

The Commander shall make a Captain’s Special Report on the matter detailing the incident and the information acquired. This may be used in any subsequent legal action.

10.1.7 Uncontained Disruptive Passengers

Following the escalation of a disruptive passenger incident while on the ground where all measures, including intervention by airport security personnel/local police has failed to resolve the situation the Commander must contact Dublin Operations for guidance on the security procedures to be followed.

For disruptive passenger events, the flight crew shall communicate the appropriate disruptive passenger threat level to ATC in accordance with ICAO Doc 9811 Annex 6.

Level 1 — Disruptive behaviour (verbal);

Level 2 — Physically abusive behaviour;

Level 3 — Life-threatening behaviour (or display of a weapon);

Level 4 — Attempted or actual breach of the flight crew compartment;

ATC shall decode the threat level and communicate as appropriate to local authorities.

10.1.8 Disruptive Passengers Report

Malta Air Commanders shall support their colleagues on board our aircraft when there are disruptive passenger incidents. It is Malta Air policy to prosecute disruptive passengers and Commanders shall ensure due process on the ground is completed including when such process will delay a subsequent departure. Commanders shall ensure disruptive passenger reports are filed and file an ASR for each disruptive passenger event.

10.2 Preventative Security Measures and Training**10.2.1 Preventative Security Measures**

These are in accordance with the Security Standard Code currently in force.

The Security Standard Codes are specified in the Malta Air Security Programme.

Standards 1, 2 and 3 are applied to all Malta Air flights as the basic level of security. Higher standards will be applied, when necessary, in accordance with the procedures laid down.

10.2.2 Training

Training in security procedures for Flight Crew and Cabin Crew will be conducted annually as part of the recurrent emergency and safety equipment training.

Training syllabi and checking programmes are specified in the Malta Air OMD – Training Manual and in the ASTO Manual.

10.3 Airborne Security Events

This section summarises the procedures for dealing with airborne security events.

10.3.1 Security Event

An airborne security event is one where an aircraft is either the subject of an actual or perceived threat, which may take the form of a hi-jack (unlawful interference), bomb alert or other onboard disruption, or the aircraft is perceived to pose a threat to a Country.

10.3.2 Other Security Alert Indicators

There are a range of indicators of suspicious aircraft/pilot activity which could highlight a potential security event to ATC and AD agencies (Air Defence), and lead to the initiation of intervention procedures. These include but are not limited to:

- No cleared flight profile/airspace infringement.
- Unauthorised deviation from cleared flight profile in the horizontal or vertical plane.
- Refusal or inability to comply with ATC instructions, including vectoring without good reason.
- Loss of RTF contact, particularly associated with flight profile deviation.
- Unauthorized SSR code changes, extended use of IDENT.
- Use of non standard phraseology by the crew or other covert attempts to highlight the situation (marked change in voice characteristics).
- Notification of a threat or incident from official or non official sources.
- Open RTF transmission from the cockpit.
- Non ATC related RTF transmission (political broadcast).
- Specific/non specific threat passed via a third party, e.g. Police/Public.

10.3.3 Aircraft Commander Responsibilities

Appropriate use of transponder code and RTF procedure is essential. The use of MAYDAY or PAN shall not be used in isolation to communicate the threat to ATC. Details of the security threat must be passed in the same transmission.

10.3.4 Integrity of the Flight Deck

Integrity of the flight deck shall be provided to ATC by including the phrase “FLIGHTDECK SECURE” in the transmission.

10.3.5 Aircraft Phone System (If Installed)

If the aircraft has a phone system on board, it shall not be switched off.

10.3.6 Commander and Malta Air Responsibility to Report

Following an act of unlawful interference on board an aircraft the Commander or in his absence Malta Air, shall submit without delay a report of such act to the designated local authority, the TM CAD.

10.3.7 False Alarms

False alarms are most frequently caused by Aircraft not maintaining continuous listening watch on the ATC frequency.

10.3.8 NAAs and False Alarms

The National Security Authorities will ask for an explanation of false alarms, and are entitled to interview a crew that triggers an alert, and follow up with the TM CAD, and are entitled to a copy of the Malta Air Investigation.

10.3.9 Quick Response Aircraft

A loss of communications combined with the aircraft deviating from its flight plan is likely to trigger a launch of the QRA (Quick Response Aircraft) which may result in an intercept.

10.3.10 Intercept

During an intercept the crew may be asked to complete certain manoeuvres to establish the security integrity of the crew. You **MUST** complete these manoeuvres whether passed by intercepting aircraft or ATC.

10.4 Key Points For Pilots

To avoid an incident ensure that:

- Communications with ATC maintained at all times.

- Distress frequency selected on Standby, with volume.
- BE AWARE of potential situations such as loss of comms, inadvertent A7500 selection which may indicate a potential security alert.
- COMMUNICATE clearly to all parties when you perceive actual or perceived security threat to your aircraft.
- THREAT ASSESSMENT-make contact with Operations ASAP to ensure all pertinent information is shared. (MAY has trained THREAT ASSESSORS available in OPS.)
- VOLUNTEER INFORMATION regarding the integrity of the flight deck and be clear and concise about the exact nature of the threat. If the flight deck is secure, include “FLIGHTDECK SECURE” in your transmission.
- Use appropriate RT phraseology, and special purpose transponder codes.
- Follow Government instructions whether given by radio or visual signals.

10.5 EC 300/2008

The wording of the security certification has been changed to take account of new European Security Requirement as specified in Regulation (EC) 300/2008.

- 10.5.1 There is no longer a requirement to conduct a security check however certain European Member States have chosen to retain the check requirement as a National variation. In the interest of standardisation, and to maintain our enhanced security procedure, cabin crew and ground crew (aircraft holds) must continue to conduct a security check prior to the departure of Malta Air flight. These procedures are specified in the OMA-SEP and GOPS manuals.

10.5.2 The Critical Part

The Critical Part is an area within the security restricted area of the airport. All EU airports and Norwegian airports into which we operate have confirmed that MAY flights will arrive into and depart from ‘Critical Part’. The airport/handling agent/Malta Air Operations is required to inform the Commander should the security status of the airport change. In such circumstances the Commander will ensure that a Security Search is completed and details recorded on the LID/paper Loadsheet as above.

10.6 Securing Aircraft

Malta Air aircraft are secured to comply with EU/National Security regulations. A Malta Air Commander must not leave an aircraft unattended and unprotected. He/She shall ensure that either:

1. The aircraft is under the care of an authorised person, or

2. In EU Member State Airport within a Critical Part, the security check is completed, external doors are closed, holds are closed and the aircraft is chocked or
3. In Non EU Member State Airport, or EU Member State airport without a Critical Part, the security search is completed, external doors are closed, stairs are withdrawn/removed/retracted, holds are closed and the aircraft is chocked or
4. If operating outside of EU Member State Airport or have diverted to an airport which is not a Malta Air destination or are parking at a stand which cannot be assured as being within Critical Part, the security search is completed, external doors are closed, stairs are withdrawn/removed/retracted, holds are closed and the aircraft is chocked.
5. Where the Commander is present on the aircraft, any person who is not “authorised” to be on board a Malta Air aircraft shall be referred to him/her for action as appropriate. Such persons must not be allowed unsupervised access to the aircraft, i.e. he/she must be accompanied at all times. If this is not practicable, he/she shall be denied access to the aircraft and shall be referred to Malta Air Operations Control.

This securing aircraft procedure does not apply to aircraft parked in a hangar that is locked or otherwise protected from unauthorised access.

10.6.1 Security Search

Crews will be aware of the difference between a Security Check and a Security Search. In addition to the MAY Security Check requirements, the following additional areas must be examined as part of a **Security Search**:

- Flight Deck – (if left unattended).
- Cabin – In case the visual check is not sufficient, physically verify that there is nothing between the seats or between the seats and cabin wall.
- Wheel wells – (part of the normal pre-flight inspection).
- External panels and hatches – (if there is evidence of unauthorised tampering).

10.6.2 In accordance with the new European Regulation a **Security Search** must be conducted:

- If there is a reason to believe that unauthorised persons have had access to the aircraft.
- Whenever an aircraft arrives into a critical part from a third country unless that third country is recognised as applying security standards equivalent to Regulations (EC) 300/2008*.

- If the aircraft was accessible outside the Critical Part and is moved into the Critical Part.
- If the aircraft arrives into or departs from outside a Critical Part.

A search shall be conducted as soon as possible after passenger disembarkation from the area to be searched and prior to departure subject to the following conditions:

- If the aircraft is in a critical part service providers are permitted to be onboard.
- If the aircraft is in a part other than a critical part service providers are permitted to be onboard provided that they remain under supervision.

Note: *A list of 3rd countries applying (EC) 300/2008 is provided on the [Operational] tab on the pilot EFB.

10.6.3 Security Search Procedure

- All elements of the security check procedure must be completed. The following areas must also be checked if they are accessible without the use of tools, keys, stairs/other aids and without breaking seals:

Engineering personnel will:

- Check the air conditioning access panel.

Flight crew will check:

- Flight deck if left unattended.
- Wheel wells
- The air conditioning access panel

Cabin crew will check:

- Areas between the seats and between the seat and wall.

Ground crew will check

- Aircraft hold
- Items contained in the aircraft hold.

The above areas will be visually checked initially. If the person conducting the search cannot verify that the area is empty/has not been tampered with a hand search shall be conducted. Items, such as passenger belongings, found during the security check/search procedure must be given to ground staff.

10.7 Recording Security Checks or Security Searches

Completion of the eLID/paper LID/paper Loadsheets, and the signature of the Captain is used to comply with the Security Check and Security Search administration requirements. The process is described below.

10.7.1 Recording Security Checks or Security Searches

The Commander shall clearly indicate on the eLID/paper LID/paper loadsheet to indicate on the eLID/paper LID/paper loadsheet whether a Security Check or Search has been completed.

eLID: The origin of the previous flight is automatically displayed on the eLID in EFL. The Commander must record the time (Z) the search or check was completed in the eLID.

Paper LID/Loadsheets: The origin of the previous flight must be recorded in the designated space on all paper LIDs/loadsheets regardless of whether a **Security Check** or a **Security Search** is conducted. The Commander must record the time (Z) the search or Check was completed in the space designated in the paper LID/loadsheets for this purpose.

10.8 Flight Crew Compartment Security (Flight Deck Door)

ORO.SEC.100, CAT.GEN.MPA.135

Malta Air aircraft are equipped with an approved flight crew compartment door which is capable of being locked and unlocked from either pilot's station and designed to meet the applicable airworthiness requirements.

This door is capable of being locked from the flight deck using deadbolts in which case it cannot be unlocked other than by using the deadbolts to unlock it.

It has means for flight crew to accept or deny access when requested from the cabin when electrically powered.

It may be locked or unlocked from the cabin or flight deck using a key including when no electrical power is available.

Procedures are established whereby which the cabin crew can notify the flight crew in the event of suspicious activity or security breaches in the cabin.

The flight deck door is closed prior to engine start for take-off and will be locked when required by security procedures or by the pilot-in-command until engine shut down after landing, except when deemed necessary for authorised persons to access or egress in compliance with national civil aviation security programmes.

Pending equipping of all Malta Air aircraft with means for monitoring from either pilot's station the entire door area outside the flight crew compartment to identify persons requesting entry and to detect suspicious behaviour or potential threat the following procedure applies:

Flight Deck Door procedures are contained in Chapter 8.

10.9 Emergency Response Plan

Detailed procedures are given in Air Carrier Security Program. Development of those procedures is described in Emergency Response Plan.

Information about an emergency situation shall be passed to the Director General for Civil Aviation (DGCA) and the BAAI Chief Investigator of Air Accident.

The following complete and accurate information should be given in the most expeditious manner:

1. Nature of the emergency
2. Type of aircraft
3. Aircraft registration
4. Number of persons on board
5. Estimated time of landing (if applicable)
6. The location and accessibility of site of serious incident/accident (if applicable)
7. Hazardous material on board (if applicable)
8. List of names of people on board

Immediately upon notification, a list of dangerous goods on board must be made available and within two hours of notification, a list of persons on board.

An occurrence report in writing containing the above-mentioned information shall be sent to Capt Charles Pace (DGCA) and the TM CAD Safety Management Unit (SMU).

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11 HANDLING NOTIFYING AND REPORTING ACCIDENTS INCIDENTS AND OCCURRENCES

Just Culture

Malta Air espouses a ‘Just Culture’ in the interests of the ongoing development of flight safety. This means that Malta Air supports the development, within all areas of the organisation, of a culture in which:

- Individuals are not punished for actions, omissions or decisions taken by them that are commensurate with their experience and training but which result in a reportable event; but
- Where gross negligence, wilful violations and destructive acts are not tolerated.

This position is in line with Regulation (EU) No 376/2014 and (EU) No 2015/1018.

Confidentiality of Reports

It is fundamental to the purpose of the reporting of incidents and accidents, that the knowledge gained from the investigation of these occurrences is disseminated so that we may all learn from them.

Without prejudice to the proper discharge of its responsibilities, Malta Air will not disclose the name of the person submitting the report or of a person to whom it relates unless required to do so by law, or the person concerned authorises disclosure.

In accordance with EU Regulation Malta Air will ensure that relevant safety information deriving from the analysis of reports, which have been subjected to disidentification, are made available to all parties so that they can be used for improving safety.

Should any safety follow-up action arising from a report be necessary, Malta Air will take all reasonable steps to avoid disclosing the identity of the reporter or of those individuals involved in any reportable occurrence.

‘Making a Disclosure in the Public Interest’, (Whistleblowing)

Where a reporter wishes to make a disclosure in the public interest, the preferred method of reporting is to use the Malta Air Confidential Reporting System by contacting the Malta Air Safety and Security Manager at the following email address confidential@maltaair.com.

Assurance Regarding Prosecution

Malta Air assures all staff and service providers that its primary concern in relation to the Occurrence Reporting Scheme (equivalent to MOR Scheme) is to secure free and uninhibited reporting and that it will not be Malta Air policy to institute proceedings in respect of unpremeditated or inadvertent breaches of the law which come to its attention only because they have been reported under the Scheme, except in cases of gross negligence.

11.1 Definitions and Responsibilities

11.1.1 Accident

Article 2 Definitions

An occurrence associated with the operation of an aircraft which takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked in which:

1. A person fatally or seriously injured as a result of:
 - Being in the aircraft; or
 - Direct contact with any part of the aircraft including parts which have become detached from the aircraft; or
 - Direct exposure to jet blast.

Except when the injuries are from natural causes, self-inflicted or inflicted by other persons, or when the injuries are to stowaways hiding outside the areas available to the passengers and crew; or

2. The aircraft sustains damage or structural failure which:
 - Adversely affects the structural strength, performance or flight characteristics of the aircraft; and
 - Would normally require major repair or replacement of the affected component.

Except for engine failure or damage when the damage is limited to the engine, its cowlings or accessories, or for damage limited to propellers, wing tips, antennas, tires, brakes, fairings, small dents or punctures, holes in the aircraft skin; or

3. The aircraft is missing or is completely inaccessible.

Note 1: For statistical uniformity only, an injury resulting in death within thirty days of the date of the accident is classified as a fatal injury.

Note 2: An aircraft is considered to be missing when the official search has been terminated and the wreckage has not been located.

11.1.2 Serious Injury**Article 2 Definitions**

An injury which is sustained by a person in an accident and which:

1. Requires hospitalisation for more than 48 hours, commencing within seven days from the date the injury was received; or
2. Results in a fracture of any bone (except simple fractures of fingers, toes, or nose); or
3. Involves lacerations which cause severe haemorrhage, nerve, muscle or tendon damage; or
4. Involves injury to any internal organ; or
5. Involves second or third degree burns, or any burns affecting more than 5 percent of the body surface; or
6. Involves verified exposure to infectious substances or injurious radiation.

11.1.3 Incident**Article 2 Definitions**

An occurrence, other than an accident associated with the operation of an aircraft or could affect the safety of operation.

11.1.4 Serious Incident**Article 2 Definitions**

An accident involving circumstances indicating that there was a high probability of an accident, and associated with the operation of an aircraft which, in the case of a manned aircraft, takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked.

Note: The difference between an accident and a serious accident lies in the result.

11.1.5 ‘Occurrence’

An Occurrence means any safety-related event which endangers or which, if not corrected or addressed, could endanger an aircraft, its occupants or any other person and includes in particular an accident or serious incident. Malta Air/ Ryanair Group personnel are required to report any event (Occurrence) that happens while they are on duty.

11.1.6 Responsibilities

Responsibility of the Commander/Crew

CAT.GEN.MPA.105

Whenever an accident, serious incident or occurrence or any other irregularity, which may or has resulted in an emergency occurs during flight/duty, the Commander/crew shall, if practicable, notify the appropriate ATS unit (including the declaration of emergency) and the nearest Malta Air facility by the fastest means available.

11.1.7 Response by the Operations Control

Ops Control conveys, by any available means, safety related information to the Commander that may be necessary for the safe conduct of the flight, including information related to any amendments to the flight plan that become necessary during or for the flight.

The Malta Air Emergency Response Plan (ERP) directs what tasks shall be completed and by whom in the organization when a crisis is declared.

11.2 Forms to be Used for Reporting

All Safety related reporting is performed through the Safety reporting system.

Should there be a prolonged failure of the Safety Reporting System the Safety Services Office shall advise all crew in the following manner:

- a. All crew will receive communication directly to their EFB/VPOS advising of the failure with instructions on alternative means of Safety Report submission.

Should individual crew members experience an issue with use of the Safety reporting system then Zendesk is available to individual crewmembers with issues with the app which is addressed within 24 hours.

By reporting in this manner through the Safety reporting system, company personnel meet their obligation to report to the Authority as described in this section.

Air Safety Reports (ASR)

Accidents and Incidents are defined in Regulation (EU) 996/2010 (as amended by Regulation (EU) No 376/2014) and 1018/2015. Should there be a prolonged failure of the Safety reporting system, the Safety Services Office will advise crews on the alternative means of reporting.

For any of the events referred to above, operating crews shall file a ASR, normally on completion of duty or as soon as possible thereafter, but in any case to enable Malta Air to have filed the required reports to the Authorities within 72 hrs of the time of the event

All incidents shall be reported as specified in these manuals. If there is a doubt about whether an event fits the definitions of an accident or incident a report shall be filled anyway.

An incident that involves a system failure, damage to the aircraft, bird-strike, lighting strike, severe turbulence and/or airframe exceedences a corresponding tech.log entry shall be made and included in the ASR.

Airprox (Airmiss) Report

An airprox report will be made where a pilot considers that his aircraft may/or has been endangered by the proximity of another aircraft during flight to the extent that a definite risk of collision existed, the incident will be reported to the appropriate Air Traffic Control unit in a standardised form. Reports will only be made when a genuine risk has occurred.

Malta Air Confidential Reporting Service (CRS)

The CRS is intended to enable you to bring your personal safety concerns to the Safety and Security Manager in complete confidence. This system is fully supported by Flight Operations Management. It is intended to give Aircrew an opportunity to report any event, error, discrepancy which they do not wish to report through normal channels. It is controlled at arm's length from management to guarantee confidentiality and the Company also guarantees immunity from disciplinary action under the Malta Air Just Culture process, unless the occurrence is excluded in legislation or independently pursued by a regulatory authority.

The CRS is not intended to be used for anonymous reporting. The Safety and Security Manager remains an independent member of the Safety Management System, which ensures that any matters brought to his attention remain confidential.

11.3 In the Event of an Accident/Incident/Occurrence

11.3.1 What Should Be Reported?

A reportable occurrence in relation to an aircraft means any incident which endangers or which, if not corrected, would endanger an aircraft, its occupants or any other person.

11.3.2 Who Should Report?

Any person subject to this Manual shall report any reportable occurrence/incident/accident of which they have positive knowledge, even though this may not be first hand, unless they have good reason to believe that appropriate details of the occurrence have already been, or will be, reported by someone else.

A report will also be submitted on any occurrence that involves an unsatisfactory condition, behaviour or procedure, which did not immediately endanger the aircraft but if allowed to continue uncorrected, or if repeated in other foreseeable circumstances, would create a hazard.

It is of great importance to the success of the Scheme that the reporters keep firmly in mind the concept of ‘endangering’ or ‘potentially endangering’, as used in the above definition, when deciding whether or not to submit a report. The primary objective of occurrence reporting is to monitor, disseminate and record for analysis, critical or potentially critical safety occurrences. It is not intended to collect and monitor the normal flow of day-to-day defects/incidents etc. The latter is an important part of the overall flight safety task but other procedures and systems exist to carry out this function.

11.3.3 Occurrence Reporting

ORO.GEN.160, AMC 1 ORO.GEN.160(a), 160(b), AMC 20-8

The Malta Air Safety and Security Manager shall report to the TM CAD, in Malta, and to the appropriate accident investigation agency in the state of occurrence, any accident, serious incident and occurrence as defined this chapter.

Without prejudice to the above Malta Air shall report to the TM CAD and to Boeing:

- Any incident, malfunction, technical defect, exceeding of technical limitations, occurrence that would highlight inaccurate, incomplete or ambiguous information contained in Boeing data.
- Other irregular circumstance that has or may have endangered the safe operation of the aircraft and that has not resulted in an accident or serious incident.

For any of the events referred to above, operating crews shall file a ASR, normally on completion of duty or as soon as possible thereafter, but in any case to enable Malta Air to have filed the required reports to the Authorities within 72 hrs of the time of the event.

Reports to the Authorities shall contain all pertinent information about the event known to Malta Air.

Where relevant, Malta Air Safety and Security Manager shall produce a follow-up report to provide details of actions Malta Air intends to take to prevent similar occurrences in the future, as soon as these actions have been identified.

Volcanic Ash clouds encountered in flight shall be reported to ATC and after completion of flight duty to Malta Air for transmission to the Authorities.

Occurrences to be reported are presented below.

11.3.4 List of Aircraft Occurrences to Be Reported

Note 1: Although this Part lists the majority of reportable occurrences, it is not completely comprehensive. Any other occurrences, which are judged by those involved to meet the criteria, shall also be reported.

Note 2: This Part does not include accidents.

Note 3: Occurrences to be reported are those where the safety of operation was or could have been endangered or which could have led to an unsafe condition. If in the view of the reporter an occurrence did not endanger the safety of the operation but if repeated in different but likely circumstances would create a hazard, then a report shall be made. What is judged to be reportable on one class of product, part or appliance may not be so on another and the absence or presence of a single factor, human or technical, can transform an occurrence into an accident or serious incident.

Note 4: Specific operational approvals, e.g. “RVSM” (reduced vertical separation minima), “RNAV” (area navigation), or a design or maintenance programme, may have specific reporting requirements for failures or malfunctions associated with that approval or programme.

11.3.4.1 Occurrences Related to the Operation of the Aircraft

Commission Implementing Regulation (EU) 2015/1018 – ANNEX I.

11.3.4.1.1 Flight Preparation

1. Use of incorrect data or erroneous entries into equipment used for navigation or performance calculations which has or could have endangered the aircraft, its occupants or any other person.
2. Carriage or attempted carriage of dangerous goods in contravention of applicable legislations including incorrect labelling, packaging, and handling of dangerous goods.

11.3.4.1.2 Aircraft Preparation

1. Incorrect fuel type or contaminated fuel.
2. Missing, incorrect or inadequate De-icing/Anti-icing treatment.

11.3.4.1.3 Take-off and Landing

1. Taxiway or runway excursion.
2. Actual or potential taxiway or runway incursion.
3. Final Approach and Take-off Area (FATO) incursion.
4. Any rejected take-off.
5. Inability to achieve required or expected performance during take-off, go around or landing.
6. Actual or attempted take-off, approach or landing with incorrect configuration setting.
7. Tail, blade/wingtip or nacelle strike during take-off or landing.
8. Approach continued against air operator stabilised approach criteria.
9. Continuation of an instrument approach below published minimums with inadequate visual references.
10. Precautionary or forced landing.
11. Short and long landing.
12. Hard landing.
13. Runway Awareness and Advisory System (RAAS) and Overrun Warning (ORW) System.
 - a. RAAS On Ground Caution.
 - i. Insufficient Runway Length (“ON RUNWAY XX, ZZ REMAINING”).
 - A. Report when not anticipated or RAAS callout distance less than OPT performance requirements.

- B. Report not required if RAAS caution occurs during backtracking.
- b. RAAS In Flight Caution.
 - i. Approaching Short Runway (“APPROACHING RUNWAY XX, ZZ AVAILABLE”).
 - A. Report when not anticipated or RAAS callout distance less than OPT performance requirements.
 - B. Report all Approaching Short Runway RAAS Cautions which result in a go-around.
 - ii. Short Runway (“CAUTION SHORT RUNWAY, SHORT RUNWAY”).
- c. RAAS Stabilised Approach Monitor Cautions (all).
- d. RAAS Non-Routine Callouts (all).
- e. ORW Overrun Warnings (all).
- f. RUNWAY INOP events not previously identified by AFB and/or OFP Co-NOTAM.

11.3.4.1.4 Any Phase of Flight

1. Loss of control.
2. Aircraft upset, exceeding normal pitch attitude, bank angle or airspeed inappropriate for the conditions.
3. Level bust.
4. Activation of any flight envelope protection, including stall warning, stick shaker, stick pusher and automatic protections.
5. Unintentional deviation from intended or assigned track of the lowest of twice the required navigation performance or 10 nautical miles.
6. Exceedance of aircraft flight manual limitation.
7. Operation with incorrect altimeter setting.
8. Jet blast or rotor and prop wash occurrences which have or could have endangered the aircraft, its occupants or any other person.
9. Misinterpretation of automation mode or of any flight deck information provided to the flight crew which has or could have endangered the aircraft, its occupants or any other person.

11.3.4.1.5 Other Types of Occurrences

1. Unintentional release of cargo or other externally carried equipment.

2. Loss of situational awareness (including environmental, mode and system awareness, spatial disorientation, and time horizon).
3. Any occurrence where the human performance has directly contributed to or could have contributed to an accident or a serious incident.

11.3.4.2 Technical Occurrences

11.3.4.2.1 Structure and Systems

1. Loss of any part of the aircraft structure in flight.
2. Loss of a system.
3. Loss of redundancy of a system.
4. Leakage of any fluid which resulted in a fire hazard or possible hazardous contamination of aircraft structure, systems, or equipment, or which has or could have endangered the aircraft, its occupants or any other person.
5. Fuel system malfunctions or defects, which had an effect on fuel supply and/or distribution.
6. Malfunction or defect of any indication system when this results in misleading indications to the crew.
7. Abnormal functioning of flight controls such as asymmetric or stuck/jammed flight controls (for example: lift (flaps/slats), drag (spoilers), attitude control (ailerons, elevators, rudder) devices).

11.3.4.2.2 Propulsion (Including Engines, Propellers and Rotor Systems) and Auxiliary Power Units (APUs)

1. Failure or significant malfunction of any part or controlling of a propeller, rotor or power plant.
2. Damage to or failure of main/tail rotor or transmission and/or equivalent systems.
3. Flameout, in-flight shutdown of any engine or APU when required (for example: ETOPS (Extended range Twin engine aircraft Operations), MEL (Minimum Equipment List)).
4. Engine operating limitation exceedance, including overspeed or inability to control the speed of any high-speed rotating component (for example: APU, air starter, air cycle machine, air turbine motor, propeller or rotor).
5. Failure or malfunction of any part of an engine, powerplant, APU or transmission resulting in any one or more of the following:
 - a. Thrust-reversing system failing to operate as commanded;
 - b. Inability to control power, thrust or rpm (revolutions per minute);

- c. Non-containment of components/debris.

11.3.4.3 Interaction with Air Navigation Services (ANS) and Air Traffic Management (ATM)

1. Unsafe ATC (Air Traffic Control) clearance.
2. Prolonged loss of communication with ATS (Air Traffic Service) or ATM Unit.
3. Conflicting instructions from different ATS Units potentially leading to a loss of separation.
4. Misinterpretation of radio-communication which has or could have endangered the aircraft, its occupants or any other person.
5. Intentional deviation from ATC instruction which has or could have endangered the aircraft, its occupants or any other person.

11.3.4.4 Emergencies and Other Critical Situations

1. Any event leading to the declaration of an emergency ('Mayday' or 'PAN call').
2. Any burning, melting, smoke, fumes, arcing, overheating, fire or explosion.
3. Contaminated air in the cockpit or in the passenger compartment which has or could have endangered the aircraft, its occupants or any other person.
4. Failure to apply the correct abnormal or emergency procedure by the flight or cabin crew to deal with an emergency.
5. Use of any emergency equipment or abnormal procedure affecting in-flight or landing performance.
6. Failure of any emergency or rescue system or equipment which has or could have endangered the aircraft, its occupants or any other person.
7. Uncontrollable cabin pressure.
8. Critically low fuel quantity or fuel quantity at destination below required final reserve fuel.
9. Any use of crew oxygen system by the crew.
10. Incapacitation of any member of the flight or cabin crew that results in the reduction below the minimum certified crew complement.
11. Crew fatigue impacting or potentially impacting their ability to perform safely their flight duties.
12. The use of any emergency equipment or prescribed emergency procedures in order to deal with a situation.

11.3.4.5 External Environment and Meteorology

1. A collision or a near collision on the ground or in the air, with another aircraft, terrain or obstacle (obstacle include vehicles).
2. ACAS RA (Airborne Collision Avoidance System, Resolution Advisory) (TCAS).
3. Activation of genuine ground collision system such as GPWS (Ground Proximity Warning System)/TAWS (Terrain Awareness and Warning System) 'warning'.
4. Wildlife strike including bird strike (see 11.4.3).
5. Foreign object damage/debris (FOD).
6. Unexpected encounter of poor runway surface conditions.
7. Wake-turbulence encounters.
8. Interference with the aircraft by firearms, fireworks, flying kites, laser illumination, high powered lights, lasers, Remotely Piloted Aircraft Systems, model aircraft or by similar means.
9. A lightning strike which resulted in damage to the aircraft or loss or malfunction of any aircraft system.
10. A hail encounter which resulted in damage to the aircraft or loss or malfunction of any aircraft system.
11. Severe turbulence encounter or any encounter resulting in injury to occupants or deemed to require a 'turbulence check' of the aircraft.
12. A significant wind shear or thunderstorm encounter which has or could have endangered the aircraft, its occupants or any other person.
13. Icing encounter resulting in handling difficulties, damage to the aircraft or loss or malfunction of any aircraft system.
14. Volcanic ash encounter.

11.3.4.6 Security

1. Bomb threat or hijack.
2. Difficulty in controlling intoxicated, violent or unruly passengers.
3. Discovery of a stowaway.

Note: SAFA/SACA/SANA/Ramp Inspection requires an ASR to be filed in accordance with OMA 8.3.0.4.

11.3.4.7 Reportable Events of PBN Operations

AMC2 ORO.GEN.160

1. A reportable event should be an event that adversely affects the safety of the operation and may be caused by actions or events external to the functioning of the aircraft navigation system.
2. Technical defects and the exceedance of technical limitations, including:
 - a. Significant navigation errors attributed to incorrect data or a database coding error;
 - b. Unexpected deviations in lateral/vertical flight path not caused by flight crew input or erroneous operation of equipment;
 - c. Significant misleading information without a failure warning;
 - d. Total loss or multiple navigation equipment failure; and
 - e. Loss of integrity, e.g., RAIM function, whereas integrity was predicted to be available during preflight planning, should be considered a reportable even and ASR shall be filed.
3. Malta Air will investigate any reportable event with the Aeronautical Service Provider to determine if it is due to an improperly coded procedure or a navigation database error and corrective actions for such an event will be issued to crews accordingly (e.g., issue of Co-Notam and/or note on Airfield Brief) for the time required to update the database.

11.3.4.8 Reportable Events in LVOs

AMC3 ORO.GEN.160

1. A reportable event should include:
 - a. Significant deviations from the flight path not caused by flight crew input;
 - b. Misleading information without flight deck effects;
 - c. Loss of airborne navigation equipment functions necessary for the operation;
 - d. Loss of functions or facilities at the aerodrome necessary for the operation, including aerodrome operating procedures, ATC operation, navigation facilities, visual aids and electrical supply;
 - e. Loss of other functions related to external infrastructure necessary for the operation; and
 - f. Any other event causing the approach or landing to be abandoned if occurring repeatedly.
2. The reports should be submitted to the aerodrome involved when relevant and in addition to the recipients prescribed by OMA 11.3.3.

11.3.5 Operational Flight Data Monitoring (OFDM) ORO.AOC.130

ORO.AOC.130

Flight data monitoring (FDM) is the pro-active use of digital flight data from routine operations to improve aviation safety. The flight data monitoring programme shall be non-punitive and contain adequate safeguards to protect the source(s) of the data. (Annex 1 definitions.) The OFDM functions applicable to the Safety and Security Manager are described in OMA 1.2.7.

Objectives of the Malta Air OFDM Programme AMC1 ORO.AOC.130:

1. Identify areas of operational risk and quantify current safety margins;
2. Identify and quantify operational risks by highlighting occurrences of non-standard, unusual or unsafe circumstances;
3. Use the FDM information on the frequency of such occurrences, combined with an estimation of the level of severity, to assess the safety risks and to determine which may become unacceptable if the discovered trend continues;
4. Put in place appropriate procedures for remedial action once an unacceptable risk, either actually present or predicted by trending, has been identified; and
5. Confirm the effectiveness of any remedial action by continued monitoring.

Full detail of the OFDM process can be found in the OFDM procedures document. Access to the document is limited to the OFDM trustees and other personnel involved in the OFDM programme.

Every crew member is responsible for reporting events. Significant risk-bearing incidents detected by FDM should therefore normally be the subject of mandatory occurrence reporting by the crew. If this is not the case, then they should submit a retrospective report that should be included under the normal process for reporting and analysing hazards, incidents and accidents.

If an air accident or serious incident occurs the commander, or if the commander is unable to report, other flight crew shall as expeditiously as possible, and latest within **72 hours** after the accident or serious incident occurred, send a report to the Authority. The examples of incidents listed above are always to be reported. If the person making the report is in doubt whether the incident is to be regarded as serious or not, the report shall be made regardless. Crew shall report to the Commander who makes the reporting as a professional courtesy.

11.3.6 Securing Documentation

The investigation of an accident/incident serves to uncover the chain of events leading to the event, its causal factors and consequently, to prevent re-occurrence. For that purpose, all Nominated Persons shall have all documents and papers at headquarters, at maintenance base and at aerodromes of last departure secured which may be of relevance.

Flight recorders comprise of two systems, a flight data recorder and a cockpit voice recorder. Flight Recorders must not be switched off during flight time.

Flight recorders must be de-activated upon completion of flight time following an accident or incident. The flight recorder must not be re-activated until clearance to do so has been issued by the NPFO.

For accidents and serious incidents the need for removal of the flight recorder records from the aircraft will be determined by the investigating authority in the state conducting the investigation with due regard to the seriousness of an occurrence and the circumstances, including the impact on the operation.

Malta Air and the commander must ensure to the greatest extent possible, in the event the aeroplane becomes involved in an accident or incident, that the preservation of all related flight recorder records, and if necessary the associated flight recorders, are secured and retained in safe custody.

11.4 Notification to ATC

CAT.GEN.MPA.105

The Commander shall notify air traffic control, as soon as practicable, in the event of an in-flight incident in any of the following categories:

- ACAS (TCAS) Resolution Advisory.
- Dangerous goods.

Any of the following SIGMET conditions must be reported to ATC immediately by R/T if the Commander considers that they are likely to affect the safety of other aircraft:

- All heavy, severe or widespread cases of hail.
- Icing.
- Line squalls.
- Mountain waves.
- Sandstorms or dust storms.
- Thunderstorms.

- Tropical revolving storms.
- Volcanic ash.
- Turbulence.

In the event of a lightning strike an ASR shall be filed, and the effect of the strike on the aircraft's instrumentation or systems will be noted. If flight instrumentation was not affected this shall be stated.

11.4.1 Turbulence

Levels of turbulence are defined as:

Light: Slight changes in attitude or altitude or changes in IAS of 5–15 kts.

- 'Light Chop': Slight bumpiness without changes in attitude, altitude or IAS. Occupants may feel a slight strain against seat belts though no difficulty is encountered in walking.

Moderate: Turbulence that is similar to light turbulence but of greater intensity, changes in attitude and or altitude occur, IAS fluctuates 16–25 kts but the aircraft remains in positive control at all times.

- 'Moderate Chop': Turbulence that is similar to light chop but of greater intensity causing rapid bumps or jolts without appreciable changes in attitude or altitude. Occupants feel definite strain against seat belts or shoulder harness. Unsecured objects are dislodged. Food service and walking are difficult.

Severe: Turbulence that causes large, abrupt changes in attitude or altitude. Aircraft may be momentarily out of control. IAS fluctuates by more than 25 kts. Occupants are forced against seat belts or shoulder harness. Unsecured objects are tossed about. Food service and walking are impossible.

Severe turbulence must be reported to ATC giving position, time (GMT), FL/Altitude, A/C type, intensity, in or near cloud and duration. A ASR must be filed and a Technical Log entry made. The Technical Log page reference will be included in the ASR to facilitate Engr admin.

11.4.2 Windshear

Windshear encountered on take-off or approach must be reported to ATC immediately by R/T if considered a hazard giving details of height and severity.

Severe windshear is considered to be uncontrollable changes from normal steady flight conditions below 1000 AGL, in excess of the following:

- 15 kts indicated airspeed.
- 500 fpm vertical speed.
- 5° pitch attitude.
- 1 dot displacement from the glideslope/glidepath.
- Unusual thrust levels for a significant period of time.

Occasional – less than 1/3 of the time.

Intermittent – 1/3 to 2/3 of the time.

Continuous – more than 2/3 of the time.

11.4.3 Bird Strikes and Hazards

The Commander shall immediately inform ATC whenever a potential bird hazard is observed. Bird strikes must be reported by completing a ASR 'Bird Strike' and there is no need to fill in any NAA or other national reporting forms.

11.4.4 Dangerous Goods Accident/Occurrence – Notification

If an in-flight emergency occurs and the situation permits, the Commander shall inform ATC of any Dangerous Goods on board.

Malta Air is required to report dangerous goods incidents and accidents to the TM CAD. An initial report shall be despatched within 72 hours of the event unless exceptional circumstances prevent this.

11.5 Submission of Written Reports on ATS, TCAS, Birdstrikes, DG Incidents, Unlawful Interference

In Malta Air the ASR is the required report to be filed. The Commander shall file a ASR on any accident or occurrence and on each safety related incident that has occurred while he was responsible for a specific flight or in charge of a crew. If members of his crew are involved, the Commander may require them to make an additional report.

International rules (ICAO Annex 13) specify that serious incidents will be investigated in the same way as accidents. In such cases, the company shall notify the Air Accident Investigation Agency, the TM CAD and the appropriate NAA.

11.5.1 Mandatory Reporting

Filing a ASR, ideally at the end of duty, enables the Malta Air SMS to review and classify reports as appropriate. This process also enables Malta Air to notify external agencies within the required timeframes.

11.5.2 Air Safety Reports (ASR)

Accidents and Incidents are defined in Regulation (EU) 996/2010 (as amended by Regulation (EU) No 376/2014). If electronic filing is unavailable, the SSO will publish a safety bulletin to crews describing the alternative method of filing a report.

All incidents shall be reported as specified in these manuals. If there is a doubt about whether an event fits the definitions of an accident or incident a report will be filled anyway. There are various sub categories in the electronic ASR, some are referred to below.

11.5.3 Airprox (Airmiss) Report

An airprox report will be made where a pilot considers that his aircraft may/or has been endangered by the proximity of another aircraft during flight to the extent that a definite risk of collision existed. The incident will be reported to the appropriate Air Traffic Control unit in a standardised form. Reports shall only be made when a genuine risk has occurred.

11.5.4 ASR Technical Reports

11.5.4.1 Go Arounds

Malta Air operates a “NO BLAME” policy with regard to go-arounds. When a go-around has commenced above 1000 ft AGL a report is not needed unless deemed necessary by the commander. Where a go around commenced below 1000 ft AGL an ASR shall be completed by the commander and subsequently the First Officer will complete a report as well.

11.5.4.2 Diversion

Malta Air requires operational reports to assess the nature of diversions, and the effectiveness of Malta Air SOPs and operations control in handling them.

An ASR shall be filed by the commander following any diversion, providing comprehensive details on the circumstances and the decision making process. A specific report can be accessed by typing “diversion” into the event description box and then choosing the diversion category from the drop down menu. The information contained in your report is often required by our customer standards department when answering customer inquiry’s in relation to same.

In the report, the cause of the diversion whether technical, operational, medical or weather related and the reasoning behind the PIC decision/action shall be documented in detail.

The following should be included:

- For weather related events, full details of the weather conditions (AIRMET, SIGMET, METAR, TAF, NOTAMs, etc.) including approach minima at the time of diversion.
- For operational event the reason for the diversion should include any ops control inputs at the time that affected the decision to divert.
- For medical events include whether the diversion was on the advice of a medical practitioner on board the aircraft.
- For technical events include any QRH actions taken, or MEL reference if required, if Maintrol was contacted and if they instructed a diversion, if a Technical Log entry is made ensure you include the Technical Log page number in the ASR.

11.5.5 Malta Air Confidential Reporting Service (CRS)

The CRS is intended to enable you to bring your personal safety concerns to the Safety and Security Manager in complete confidence. This system is fully supported by Flight Operations Management. It is intended to give Aircrew an opportunity to report any event, error, discrepancy which they do not wish to report through normal channels. It is controlled at arms length from management to guarantee confidentiality and the Company also guarantees immunity from disciplinary action under the Malta Air Just Culture process, unless the occurrence is excluded in legislation or independently pursued by a regulatory authority.

The Confidential Reporting System is intended to increase Malta Air's awareness and improve its of issues encountered in Flight operations by Aircrew. CRS is not intended to be used for anonymous reporting.

The Safety and Security Manager remains an independent member of the Safety Management System, which ensures that any matters brought to his attention remain confidential.

11.5.6 Investigations

AMC1 ORO.GEN.200(a)(3), (c)

The Safety and Security Manager is responsible for conducting a Company inquiry/investigation and preparing a report for the NPFO.

11.5.6.1 Base Investigations (BI)

A BI is conducted by the Base Captain or in his absence by a Management Pilot or other BC. Each BC and Management Pilot has received training on the conduct of Base Investigations. Each BI is reviewed at the weekly ASR review meeting where it is risk assessed and recommendations agreed for presentation to the NPFO. Any removal of crews from line operations is to facilitate the completion of the BI.

11.5.6.2 Formal Review

A Formal Review is conducted by a BC and is for less significant events than warrants a BI. Formal Reviews are submitted by the BC and all are reviewed at the weekly ASR review meeting. Recommendations when appropriate are furnished to the NPFO. Any removal of crews from line operations is to facilitate the completion of the Formal Review.

11.5.6.3 ASR Classification

All ASRs are reviewed using ARMS methodology and are Risk Assessed in accordance with the SMS.

11.5.6.4 Captain's Special Reports (CPTSR)

Captain's Special Reports shall be made when the Commander wishes to draw the attention of Flight Operations Management to any matter of significance which does not expose the operation to regulatory non-compliance and/or safety, security, or operational risks. Normally, any event which reflects actual or potential exposure to hazards, risks or regulatory non-compliance should be filed as an ASR.

EFL entries and remarks suffice for recording operational events (e.g., delays, track deviations), however, the PIC shall submit a CPTSR for events when prescribed by AFB or OFP Co-NOTAM (e.g., STN CDA non-compliances).

Significant breakdowns in pre-flight or turnaround processes (e.g., ground handling, de-icing, catering, PRM services) require an ASR from the flight crew and CSRs from appropriate cabin crewmembers.

Co-pilots may be asked to submit CPTSRs as appropriate by the Nominated Person Flight Operations or Safety and Security Manager, FSO or any Investigator.

Captain's Special Reports are normally handled by the DCP on the Nominated Person Flight Operations behalf and passed to the appropriate Department for their attention.

All submitted CPSRs are reviewed daily by the duty pilot and a weekly summary is provided to the NPFO.

11.6 Operations Personnel Reports to Commander

AMC1 CAT.GEN.MPA.100(b)

All personnel shall ensure that the Commander is advised immediately of any incident that has endangered or may have endangered safety during flight operations.

Personnel making a report to the commander shall provide the commander with all relevant information and be in a position to confirm such reports in writing later.

The Commander shall notify Malta Air Operations by the most expeditious means of any reportable occurrence or incident and provide OPS with as much relevant information as possible.

Malta Air provides a Duty Executive and Duty Pilot at weekends and National holidays both of whom will be advised of any reportable occurrence or incident. During business hours NPFO continuity provided by NPFO nominated management pilots, and these personnel are contactable through OPS.

11.7 Procedures for Preservation of Recordings Following a Reportable Event

CAT.GEN.MPA.195, CAT.IDE.A.185

11.7.1 Preservation of Recordings

1. Following an accident involving a Malta Air aircraft the commander and Malta Air shall, to the extent possible, preserve the original recorded data pertaining to that accident, as retained by the recorder for a period of 60 days unless otherwise directed by the investigating authority.
2. Malta Air will conduct operational checks and evaluations of the recordings to ensure the continued serviceability of the flight recorders which are required to be carried under Regulation (EU) 2019/1387.
3. Following an accident, a serious incident or an occurrence identified by the investigating authority, Malta Air shall preserve the original recorded data of the flight recorders for a period of 60 days or until otherwise directed by the investigating authority.
4. Additionally, when the Authority so directs, Malta Air shall preserve the original recorded data for a period of 60 days unless otherwise directed by the investigating Authority.
5. Malta Air shall save the recordings for the period of operating time as required by CAT.IDE.190 except that, for the purpose of testing and maintaining flight data recorders, up to one hour of the oldest recorded

material at the time of testing may be erased; and shall keep a document which presents the information necessary to retrieve and convert the stored data into engineering units.

6. Malta Air will comply with CAT.GEN.MPA.195(f) by applying Engineering CVR Protocol (EASA GDPR, Regulation (EU) 996/2010, Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016).

11.7.1.1 Commander's Responsibilities

The Commander's responsibilities with respect to Flight Recorders/CVRs are detailed in OM-A 1.4.3 (10).

Note: Should the Commander pull the CVR CB then he shall contact the duty pilot at the first available opportunity.

11.7.2 Production of Recordings

Malta Air shall, within a reasonable time after being requested to do so by the Authority, produce any recording made by a flight recorder or data link communications messages required to be recorded which are available or has been preserved.

11.7.3 Use of Recordings

1. The cockpit voice recorder recordings or data link communications message recordings may not be used for purposes other than for the investigation of an accident or incident subject to mandatory reporting except with the consent of all crew members concerned.
2. The flight data recorder recordings may not be used for purposes other than for the investigation of an accident or incident subject to mandatory reporting except when such records are:
 - a. Used by Malta Air for airworthiness or maintenance purposes only; or
 - b. De-identified; or
 - c. Disclosed under secure procedures.

11.7.4 Cockpit Voice Recorder

CAT.IDE.A.185, AMC1 CAT.IDE.A.185

Malta Air shall not operate an aeroplane unless it is equipped with a cockpit voice recorder which, with reference to a time scale, records:

1. Voice communications transmitted from or received on the flight deck by radio;

2. The aural environment of the flight deck, including without interruption, the audio signals received from each boom and mask microphone in use;
3. Voice communications of Flight Crew members on the flight deck using the aeroplane's interphone system;
4. Voice or audio signals identifying navigation or approach aids introduced into a headset or speaker; and
5. Voice communications of Flight Crew members on the flight deck using the public address system.

The cockpit voice recorder shall be capable of retaining information recorded during at least the last two hours for aeroplanes first issued with an individual C of A before the 1st January 2022. For aircraft issued with a C of A on or after this date the recording shall last for 25 hours.

In addition, depending on the availability of electrical power, the cockpit voice recorder must start to record as early as possible during the cockpit checks prior to engine start at the beginning of the flight until the cockpit checks immediately following engine shutdown at the end of the flight. The cockpit voice recorder must have a device to assist in locating that recorder in water.

The CVR is selected ON in accordance with FCOM procedures for first flight and when the anti-collision light is switched OFF after engine shut down on stand for the turnaround and subsequent sector.

Note: Power interruptions, e.g. GPU to APU frequently causes the CVR to switch off.

PF shall ensure the CVR is operating after power transfer prior to engine start.

11.7.5 List of MOR Requiring CVR Retention

It is mandatory to preserve CVR data after an incident/accident and it is the responsibility of the aircraft Commander to ensure that the yellow-collared CVR circuit breaker is pulled at the earliest opportunity on the ground following any of the following serious incidents:

1. Any flight where a MAYDAY is declared.
2. Emergency Descent declared.
3. Runway Incursion.
4. Aural EGPWS Warning after landing gate.
5. Altitude Bust (Pull CVR CB asap after next landing).

6. Take off Configuration Warning (TOC) for Flaps or STAB TRIM not in take-off position.
7. Engine failure or shutdown in flight.
8. Prolonged Loss of Communications (PLOC).
9. Controlled flight into terrain only marginally avoided including any aural EGPWS PULL UP Warning.
10. Aborted take-off on a closed runway or occupied runway.
11. Take-off on a closed or occupied runway (vehicle, aircraft or person).
12. Landing or attempted landing on a closed or occupied runway.
13. Gross failures to achieve predicted performance during take-off or initial climb.
14. Fire and smoke in passenger compartment, in cargo compartments or engine fires, even though such fires were extinguished by use of extinguishing agents.
15. Flight crew incapacitation in flight.
16. Events requiring the emergency use of oxygen by the Flight Crew (refer to OMA 8.3.14.1.1).
17. Multiple malfunctions of one or more aircraft systems seriously affecting the operation of the aircraft.
18. Fuel quantity requiring the declaration of an emergency by the pilot.
19. Take-off or landing incidents. Incidents such as undershooting, overrunning or running of the side of the runways.
20. Aircraft structural failures or engine disintegrations not classified as an accident.
21. Failures of more than one system in a redundancy system mandatory for flight guidance and navigation.
22. System failures, weather phenomena, operations outside the approved flight envelope or other occurrences which could have caused difficulties controlling the aircraft.
23. Any other incident where a Commander believes that CVR data would be useful in any subsequent investigation.
24. Near collisions requiring an avoidance manoeuvre to avoid a collision or an unsafe situation or when an avoidance action would have been appropriate (it is not intended to pull the CVR CB for TCAS or RA in the normal use of these systems).
25. Stick shaker event.

26. Pitch-up upset events/false glideslope capture.
27. EGPWS aural cautions TERRAIN, CAUTION TERRAIN and CAUTION OBSTACLE.

Note 1: Additionally, where take-off has commenced and a TOC warning is activated because of incorrectly configured flaps or stabiliser trim the aircraft will return and the CVR CB must be pulled.

Note 2: Maintrol must be contacted in the first instance and following their Instructions the CVR CB shall be pulled and an entry made in the Technical Log.

Note 3: Following any of the incidents above, the Commander shall pull the CVR circuit breaker and advise Operations Dublin of the incident and CVR action.

11.7.6 CVR Data Retention

On the ground, the Commander shall pull the CVR CB following any of the incidents listed above when the following actions are completed:

- The aircraft has come to a complete stop;
- The parking brake is set;
- If appropriate, all Non-Normal Checklist(s) complete;
- Aircraft recovery is being effected.

The decision to retain or reset the CVR data following such incident, can only be made following consultation with one of the authorised personnel listed below.

- NPFO or his nominated management pilots
- Deputy Chief Pilots
- Regional Base Captain
- Safety and Security Manager
- Flight Safety Officer

In the event that the authorised personnel deems it necessary to retain flight recorder data, he/she must communicate this decision to Engineering who shall record the communication on the RA164. The flight recorders are then securely placed in stores, preventing their inadvertent reactivation, repair or reinstallation in accordance with CAME Appendix A.

Procedures for the handling of flight recorders are further described in Chapter 5 of the SMM.

11.7.7 CVR CB Pulled

The aircraft shall NOT be dispatched with a pulled CVR CB.

11.7.8 Unserviceable CVR

The aircraft may be dispatched with an unserviceable CVR in accordance with the MEL. A Technical Log entry shall be made when the CVR CB is pulled.

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12 RULES OF THE AIR AND AIR TRAFFIC SERVICES**12.0 Introduction**

Various sections of the OMC Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2*, particularly Rules GEN and OPS a contain the required excerpts of ICAO documentation to address the ORO.MLR.100 requirements for the structure of OMA 12, Rules of the Air. They will reflect the Standardised European Rules of the Air (SERA) when these apply.

Compliance with the Rules of the Air

Rules of the Air apply to all Malta Air aircraft. Malta Air/Ryanair Group crews shall be familiar with and comply with the Rules of the Air.

Crew members shall also comply with the laws, regulations and procedures of those States in which operations are conducted and which are pertinent to the performance of their duties.

The Commander shall, in an emergency situation that requires immediate decision and action, take any action he considers necessary under the circumstances in the interests of safety. In such cases he may deviate from rules, operational procedures and methods in the interest of safety.

Simulated Abnormal Situations in Flight

CAT.OP.MPA.275

Malta Air prohibits simulation of abnormal or emergency situations in flight, or flight in IMC by artificial means, on any Malta Air flight.

12.1 Visual and Instrument Flight Rules

Visual Flight Rules is explained in Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2*.

Instrument Flight Rules are explained in Chapter 5 of the Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2*.

12.2 Territorial Application of the Rules of the Air

The Rules of the Air shall apply to aircraft bearing the nationality and registration marks of a Contracting State, wherever they may be, to the extent that they do not conflict with the rules published by the State having jurisdiction over the territory overflown. Therefore the rules of the Air apply to all Malta Air flight operations. The rules of the territories being overflown apply to Malta Air flight operations. Filed differences by States with ICAO whereby different procedures apply are explained in each country *Airway Manual* > *State Rules and Procedures* section, as applicable.

12.3 **Communications (Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2 – Chapter 3.6.5*)**

An aircraft operated as a controlled flight shall maintain continuous air-ground voice communication watch on the appropriate communication channel of, and establish two-way communication as necessary with, the appropriate air traffic control unit, except as may be prescribed by the appropriate ATS authority in respect of aircraft forming part of aerodrome traffic at a controlled aerodrome.

Note 1: SELCAL or similar automatic signalling devices satisfy the requirement to maintain an air-ground voice communication watch.

Note 2: The requirement for an aircraft to maintain an air-ground voice communication watch remains in effect after CPDLC has been established.

12.3.1 **Failure of Radio Communications Equipment**

These procedures are presented in the Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2 – Chapter 3.6.5.2 and Emergency Section for Related Communication Failure Information*.

12.4 **Information and Instructions Relating to the Interception of Civil Aircraft**

AMC1 CAT.GEN.MPA.180(a)(13)

These procedures are presented in the Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2 – Chapter 3.8*.

12.5 **Radio Listening Watch**

Radio listening watch shall be maintained on the ground and in the air: Comms 1: on the controlling VHF frequency.

Approaching the limits and outside the range of VHF coverage on the next expected frequency or international distress frequency, 121.5 MHz.

Comms 2: On 121.5 MHz when not being used for other purposes.

When approaching FIR boundaries, reconfirm that 121.5 is selected on COM 2.

In airspace controlled using HF by SELCAL, if a SELCAL watch for the flight has been established, otherwise by listening out on the primary HF frequency.

If intercepting a Distress Call but not involved in the distress, listen out, log the communications, and be prepared to assist if necessary.

12.6 Signals

Aerodrome Traffic signals and Air-Ground signals are presented in the Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2/Appendix 1 SIGNALS section*.

Search and Rescue signals are presented in the Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data – ICAO > section 8.4 Search and Rescue Signals*.

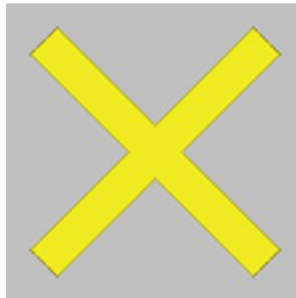
12.6.1 Visual Ground Signals

The full list of visual ground signals is contained in the OMC, Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2 > Appendix 1 SIGNALS section*. The following selection of ground signals are presented as they have had operational and safety significance in previous serious incidents. The full selection of ground signals is in the Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2 > Appendix 1 SIGNALS section*.

Closed Runways or Taxiways

Crosses of a single contrasting color, yellow or white, displayed horizontally on runways and taxiways or parts thereof indicate an area unfit for movement of aircraft.

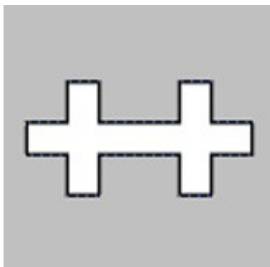
Figure 12.1



Glider Flights in Operation

A double white cross displayed horizontally in the signal area indicates that the aerodrome is being used by gliders and that glider flights are being performed.

Figure 12.2



12.6.2

Marshalling Signals

From a Signalman to an Aircraft

Note 1: These signals are designed for use by the signalman, with hands illuminated as necessary to facilitate observation by the pilot, and facing the aircraft in a position, for fixed-wing aircraft, on left side of aircraft, where best seen by the pilot.

Note 2: The meaning of the relevant signals remains the same if bats, illuminated wands or torchlights are held.

Note 3: The aircraft engines are numbered, for the signalman facing the aircraft, from right to left (i.e. Number one engine being the port outer engine).

Note 4: (Applicable to helicopters).

Note 5: References to wands may also be read to refer to daylight-fluorescent table-tennis bats or gloves (daytime only).

Note 6: References to the signalmen may also be read to refer to marshaller.

Additional signals from signalmen to pilot, and from pilot to signalmen, respectively, are described below, referring to:

- Brakes.
- Chocks.
- Ground power.
- Establishment of communications via interphone.

Prior to using the following signals, the signalman shall ascertain that the area within which an aircraft is to be guided is clear of objects which the aircraft might otherwise strike.

Note: The design of many aircraft is such that the path of the wing tips, engines and other extremities cannot always be monitored visually from the flight deck while the aircraft is being manoeuvred on the ground.

Proceed Under Further Guidance by Signalman



Indication to pilot that signalman takes over further guidance:

- Wave right hand above head level with wand pointing up,
- Left arm remains at side by knee.

Note: This signal has been deleted by ICAO for unknown reasons, but major airports incl. FRA continue to train their signalmen on this sign and make it public through their local airport regulations.

Identify Gate/Parking Position



Raise fully extended arms straight above head with wands pointing up.

Proceed to Next Signalman or as Directed by Tower/Ground Control



- Point both arms upward,
- Move and extend arms outward to sides of body, and
- Point with wands to direction of next signalman or taxi area.

Straight Ahead



- Bend extended arms at elbows, and
- Move wands up and down from chest height to head.

Turn



Turn left (from pilot's point of view):

- With right arm and wand extended at a 90 degree angle to body, and
- Make “come ahead” signal with left hand.

The rate of signal motion indicates to pilot the rate of aircraft turn: the faster the motion the sharper the turn.



Turn right (from pilot's point of view):

- With left arm and wand extended at a 90 degree angle to body, and
- Make “come ahead” signal with right hand.

The rate of signal motion indicates to pilot the rate of aircraft turn: the faster the motion the sharper the turn.

Normal Stop



- Fully extend arms and wands at a 90 degree angle to sides, and
- Slowly move to above head until wands cross.

The ICAO standard signals are suitable only during daylight or in brightly illuminated apron areas:

Set brakes:



- Raise Hand just above shoulder height with open palm,
- Ensuring eye contact with Flight Crew, close hand into a fist,

- Do not move until receipt of THUMBS UP acknowledgement from Flight Crew.

Release brakes:



- Raise hand just above shoulder height with hand closed in a fist,
- Ensuring eye contact with Flight Crew, open palm,
- Do not move until receipt of THUMBS UP acknowledgement from Flight Crew.

Alternative signals commonly used when it is dark:



Set brakes (when it is dark):

- Raise arms above head with wands pointing up,
- Then cross wands,
- Do not move until receipt of THUMBS UP or BRAKES ENGAGED acknowledgement from Flight Crew.

Release brakes (when it is dark):



- Raise arms above head with wands crossed,
- Then move wands into vertical position,
- Do not move until receipt of THUMBS UP or BRAKES RELEASED acknowledgement from Flight Crew.

Chocks



Chocks inserted:

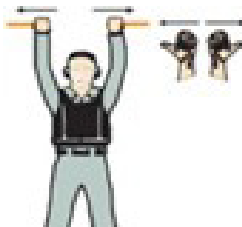
- With arms and wands fully extended above head,
- Move wands inward in a ‘jabbing’ motion until wands touch,
- Ensure acknowledgement is received from Flight Crew.

Alternative signal:

- Move hands inward until thumbs touch.

It is the Company standard procedure to communicate insertion of chocks via interphone whenever possible.

Chocks removed:



- With arms and wands fully extended above head,
- Move wands outward in a ›jabbing‹ motion,
- Do not remove chocks until authorized by Flight Crew.

Alternative signal:

- Move hands outward with thumbs pointing outward, too.

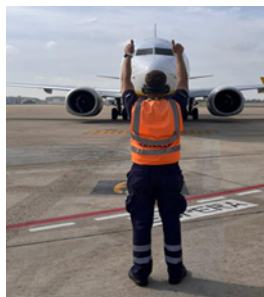
Start Engine(s)

It is the Company standard procedure to coordinate start of engines via interphone whenever possible.

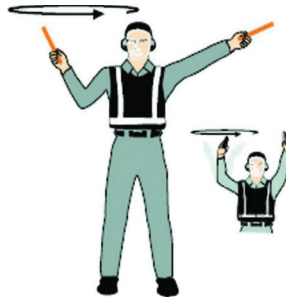
When interphone capability is not available for the pushback or start procedure: the use of standard ICAO hand signals will apply.

Self-Manoeuvring Stands (Headsets Not in Use)

- To indicate the ground crew's readiness for engine start, the marshaller stands on the designated marshalling position (i.e., "X-Spot", Ground Operations Manual refers), approximately 12 metres in front of the aircraft, or as per local apron design and indicates his readiness by raising both arms with thumbs up.



- Do not turn on the anti-collision light until the marshaller is on the designated marshalling position indicating that both handling pre-departure checks are complete and the ground crew is ready for start.
- In the absence of a marshaller on the X-spot indicating that the ground crew is ready for the engine start, the flight crew shall not select the anti-collision light ON unless to signal to the ground crew that a hazard exists.



- Raise right arm to head level with wand pointing up, and
- Start a circular motion with hand,
- At the same time, with left arm raised above head level, point to engine to be started,

Or:

- (Suitable during daylight or in brightly illuminated apron areas only) raise left arm above head level, with the number of fingers extended to indicate the number of the engine to be started

Cut Engines



- Extend arm with wand forward of body at shoulder level,
- Move hand and wand to top of left shoulder, and
- Draw wand to top of right shoulder in a slicing motion across throat.

Slow Down



- Move extended arms downwards in a ‘patting’ gesture.
- Moving wands up and down from waist to knees.

Slow Down Engine(s) on Indicated Side



- With arms down and wands towards ground,
- Wave either right (or left) wand up and down indicating that engine(s) on left (or right) side respectively should be slowed down.

Negative



- Hold right arm straight out at 90 degrees from shoulder, and

- Point wand down to ground,

Or:

- Display hand with ›thumbs down‹,
- Left hand remains at side by knee.

Fire



Shut down engine: left arm raised

Brakes: left arm lowered

- Move right hand in a ‘fanning’ motion from shoulder to knee,
- While at the same time pointing with left-hand wand to area of fire.

Night: same with wands.

Hold Position/Stand By



- Fully extend arms and wands downwards at a 45 degree angle to sides,
- Hold position until aircraft is clear for next manoeuvre.

Note: This sign is also commonly used to indicate “**Not clear to pushback/start engines/taxi!**” When observing the “Hold Position/Stand By” signal before pushback or engine start, the flight crew should not turn on the anti-collision light until the marshaller indicates the ground crew is ready to proceed.

Recommended Evacuation



Evacuation recommended based on ARFF and incident Commander’s assessment of external situation.

- Arm extended from body and held horizontal with hand upraised at eye level.
- Execute beckoning arm motion angled backward. Non-beckoning arm held against body.

Night: same with wands.

Recommended Stop



Recommend evacuation in progress be halted. Stop aircraft movement or other activity in progress.

- Arms in front of head, crossed at wrists. Night: same with wands.

Emergency Contained

No outside evidence of dangerous conditions or “all-clear”.

- Arms extended outward and down at a 45-degree angle. Arms moved inward below waistline simultaneously until wrists crossed, then extended outward to starting position (umpire’s “safe” signal). Night: same with wands.

Dispatch Aircraft

- Perform a standard salute with right hand or wand to dispatch the aircraft,
- Maintain eye contact with Flight Crew until aircraft has begun to taxi.

WARNING: Although signal shall be used only for dispatch of aircraft, it frequently happens that more than one person show this signal just to express ‘farewell’, while the walk-out assistant is still under or close to the aircraft, out of sight of the Flight Crew!

Note: The Company and handling agent walk-out assistants may have been instructed to use the signal affirmative/all clear instead of this signal to avoid confusion with other ‘greeting’ persons on the apron.

Do Not Touch Controls (Technical/Service Communication Signal)



- Extend right arm fully above head, and
- Close fist, or:
- Hold wand in horizontal position,
- Left arm remains at side by knee.

Note: This signal indicates to the Flight Crew that persons are working at flight controls or persons or equipment are standing in flight control travel areas.

Establish Communication Via Interphone



Request from cockpit to ground ICAO standard (suitable only during daylight or in brightly illuminated apron areas):

- Hold position,
- Move hands to cup both ears.

Commonly used alternative signal:

- Flash an external white light.

Acknowledgement from ground staff:

- Wave with headset.

When it is dark, illuminate the headset by a flashlight or repeatedly flash with illuminated wands.



Request from ground to cockpit ICAO standard (suitable only during daylight or in brightly illuminated apron areas):

- Extend both arms at 90 degrees from body, and
- Move hands to cup both ears.

When aircraft was moving:

- Do not approach aircraft until having clearly received acknowledgement from Flight Crew (see below) and aircraft holds position!

Commonly used alternative signals:

- Wave with headset.

When it is dark, illuminate the headset by a flashlight or repeatedly flash with illuminated wands.

Acknowledgement from Flight Crew:

- Hold position,
- Acknowledge the request by flashing an external white light.

12.6.3 From the Pilot of an Aircraft to a Signalman

Note: These signals are designed for use by a pilot in the cockpit with hands plainly visible to the signalman, and illuminated as necessary to facilitate observation by the signalman.



Brakes engaged:

- Raise arm and hand, with fingers extended, horizontally in front of face,
- Then clench fist.

Note: The moment the fist is clenched indicates the moment of brake engagement.



Brakes released:

- Raise arm, with fist clenched, horizontally in front of face,
- Then extend fingers.

Note: The moment the fingers are extended indicates the moment of brake release.

Chocks



Insert chocks:

- Arms extended, palms outwards.
- Move hands inwards to cross in front of face.

It is the Company's standard procedure to communicate insertion of chocks via interphone whenever possible.

Remove chocks:



- Hands crossed in front of face, palms outwards.
- Move arms outwards.

Ready to Start Engine(s)

It is the Company's standard procedure to coordinate start of engines via interphone whenever possible.



Raise the appropriate number of fingers on one hand indicating the number of the engine to be started.

Note: From the Flight Crew's view, the aircraft engines are numbered in flight direction, from left to right (i.e. No. 1 engine being the port outer engine).

Ground Power (Technical/Service Communication Signal)

Connect ground power:

- Hold arms fully extended above head,
- Open left hand horizontally, and
- Move finger tips of right hand into and touch open palm of left hand (forming a T).

At Night, illuminated wands can also be used to form the T above head.



Disconnect ground power:

Ground staff: Do not disconnect power until authorized by Flight Crew.

- Hold arms fully extended above head, with finger tips of right hand touching open horizontal palm of left hand (forming a T),
- Then move right hand away from the left.

At Night, illuminated wands can also be used to form the T above head.

12.7 Time

Coordinated Universal Time (UTC) shall be used and shall be expressed in hours and minutes and, when required, seconds of the 24-hour day beginning at midnight.

Wherever time is utilized in the application of data link communications, it shall be accurate to within 1 second of UTC.

12.8 ATC Clearances, Adherence to Flight Plan and Position Reports

SERA.8015(a), 8015(b)(1), 8015(b)(2), 8015(d)

A flight plan is submitted for every Malta Air Flight, Jeppesen Publications (Pubs) > General Airway Manual > ICAO Rules of the Air – Annex 2 > section 3.3 Flight Plans, 3.6.2 Adherence to Current Flight Plans.

The flight rules applicable at aerodromes, in controlled airspace and outside controlled airspace are presented in Jeppesen Publications (Pubs) > General Airway Manual > ICAO Rules of the Air – Annex 2 > section 5 IFR Flight Rules.

Position reports are presented in Jeppesen Publications (Pubs) > General Airway Manual > ICAO Rules of the Air – Annex 2 > section 3.6.3 Position Reports.

12.8.1 ATC Clearances

Air traffic control clearances shall be based solely on the requirements for providing air traffic control service.

An air traffic control clearance shall be obtained prior to operating a controlled flight. Such clearance shall be requested through the submission of a flight plan to an air traffic control unit.

The Commander (or the Co-Pilot, if instructed to do so by the Commander) shall inform ATC if an air traffic control clearance is not satisfactory. In such cases, ATC will issue an amended clearance, if practicable.

Any Malta Air aircraft operated on a controlled aerodrome shall not taxi on the manoeuvring area without clearance from the aerodrome control tower and shall comply with any instructions given by that unit.

Contents of clearances:

1. Aircraft identification as shown in the flight plan;
2. Clearance limit;
3. Route of flight,
4. The route of flight shall be detailed in each clearance when deemed necessary; and
5. The phrase 'cleared via flight planned route' shall not be used when granting a re-clearance;
6. Level(s) of flight for the entire route or part thereof and changes of levels if required;
7. Any necessary instructions or information on other matters such as approach or departure manoeuvres, communications and the time of expiry of the clearance.

The flight crew must read back to the air traffic controller safety-related parts of ATC clearances and instructions which are transmitted by voice.

The following items shall always be read back:

1. ATC route clearances;
2. Clearances and instructions to enter, land on, take off from, hold short of, cross, taxi and backtrack on any runway; and

3. Runway-in-use, altimeter settings, SSR codes, newly assigned communication channels, level instructions, heading and speed instructions; and
4. Transition levels, whether issued by the controller or contained in ATIS broadcasts.

Other clearances or instructions, including conditional clearances and taxi instructions, shall be read back or acknowledged by Flight Crew in a manner to clearly indicate that they have been understood and will be complied with.

12.8.2 Adherence to Flight Plan

SERA.8020

A flight plan is submitted for every Malta Air Flight.

1. Except as provided for in (b) all Malta Air aircrafts shall adhere to the current flight plan submitted for a controlled flight unless a request for a change has been made and clearance obtained from the appropriate air traffic control unit, or unless an emergency situation arises which necessitates immediate action by the aircraft, in which event as soon as circumstances permit, after such emergency authority is exercised, the appropriate air traffic services unit shall be notified of the action taken and that this action has been taken under emergency authority.
 - a. Unless otherwise authorised by the competent authority, or directed by the appropriate air traffic control unit, controlled Malta Air flights shall, in so far as practicable:
 - i. When on an established ATS route, operate along the defined centre line of that route; or
 - ii. When on any other route, operate directly between the navigation facilities and/or points defining that route.
 - b. Unless otherwise authorised by the competent authority, or directed by the appropriate air traffic control unit, all Malta Air aircrafts operating along an ATS route segment defined by reference to very high frequency omnidirectional radio ranges shall change over for its primary navigation guidance from the facility behind the aircraft to that ahead of it at, or as close as operationally feasible to, the changeover point, where established.
 - c. Deviation from the requirements in point (a) shall be notified to the appropriate ATS unit.

2. *Deviations from the current flight plan.* In the event that a controlled Malta Air flight inadvertently deviates from its current flight plan, the following action shall be taken:
 - a. Deviation from track: if the aircraft is off track, action shall be taken forthwith to adjust the heading of the aircraft to regain track as soon as practicable.
 - b. Deviation from the air traffic control assigned Mach number/indicated airspeed: the appropriate air traffic services unit shall be informed immediately.
 - c. Deviation from Mach number/true airspeed: if the sustained Mach number/true airspeed at cruising level varies by plus or minus Mach 0,02 or more, or plus or minus 19 km/h (10 kt) true airspeed or more from the current flight plan, the appropriate air traffic services unit shall be so informed.
 - d. Change in time estimate: except where ADS-C is activated and serviceable in airspace where ADSC services are provided, if the time estimate for the next applicable reporting point, flight information region boundary or destination aerodrome, whichever comes first, changes in excess of 2 minutes from that previously notified to air traffic services, or such other period of time as is prescribed by the competent authority, the flight crew shall notify the appropriate air traffic services unit as soon as possible.
 - e. Additionally, when an ADS-C agreement is in place, the air traffic services unit shall be informed automatically via data link whenever changes occur beyond the threshold values stipulated by the ADS-C event contract.
3. *Intended changes.* Requests for flight plan changes shall include information as indicated hereunder:
 - a. Change of cruising level: aircraft identification; requested new cruising level and cruising speed at this level, revised time estimates (when applicable) at subsequent flight information region boundaries.
 - b. Change of route:
 - i. Destination unchanged: aircraft identification; flight rules; description of new route of flight including related flight plan data beginning with the position from which requested change of route is to commence; revised time estimates; any other pertinent information.
 - ii. Destination changed: aircraft identification; flight rules; description of revised route of flight to revised destination aerodrome including related flight plan data, beginning with

the position from which requested change of route is to commence; revised time estimates; alternate aerodrome(s); any other pertinent information.

- c. Change of Mach number/true airspeed: aircraft identification; requested Mach number/true airspeed.

The flight rules applicable at aerodromes, in controlled airspace and outside controlled airspace are presented in Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2* > *section 5 IFR Flight Rules*.

12.8.3 Position Reports

Position reporting is described in Jeppesen Publications (Pubs) > *General Airway Manual* > *ICAO Rules of the Air – Annex 2* > *section 3.6.3 Position Reports*.

12.9 Visual Signals Used to Warn an Unauthorized Aircraft Flying in, or About to Enter a Restricted, Prohibited or Danger Area

By day and by Night, a series of projectiles discharged from the ground at intervals of 10 seconds, each showing, on bursting, red and green lights or stars will indicate to an unauthorized aircraft that it is flying in or about to enter a restricted, prohibited or danger area, and that the aircraft is to take such remedial action as may be necessary.

12.10 Procedures for Flight Crew Observing an Accident or Receiving a Distress Transmission (Publications (Pubs) > *General Airway Manual* > *Emergency Data – ICAO/Section 8.4*)

Flight Crew Observing an Accident

When a Commander observes that either another aircraft or a surface craft is in distress, the pilot shall, if possible and unless considered unreasonable or unnecessary:

1. Keep the craft in distress in sight until compelled to leave the scene or advised by the rescue coordination centre that it is no longer necessary;
2. Determine the position of the craft in distress;
3. As appropriate, report to the rescue coordination centre or air traffic services unit as much of the following information as possible:
 - Type of craft in distress, its identification and condition;
 - Its position, expressed in geographical or grid coordinates or in distance and true bearing from a distinctive landmark or from a radio navigation aid;
 - Time of observation expressed in hours and minutes UTC;

- Number of persons observed;
 - Whether persons have been seen to abandon the craft in distress;
 - On-scene weather conditions;
 - Apparent physical condition of survivors;
 - Apparent best ground access route to the distress site; and
4. Act as instructed by the rescue coordination centre or the traffic service unit.

If the first aircraft to reach the scene of an accident is not a search and rescue aircraft, it shall take charge of on-scene activities of all other aircraft subsequently arriving until the first search and rescue aircraft reaches the scene of the accident. If, in the meantime, such aircraft is unable to establish communication with the appropriate rescue co-ordination centre or air traffic services unit, it shall, by mutual agreement, hand over to an aircraft capable of establishing and maintaining such communications until the arrival of the first search and rescue aircraft.

When a ground signal has been displayed, the aircraft shall indicate whether the signal has been understood or not as indicated in this section, or if this is not practicable, by use of the appropriate visual signal as shown below.

12.11 Ground/Air Visual Codes for Use by Survivors, Signal Aids

These signals are presented in the Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data – ICAO/section 8.4 Search and Rescue Signals*.

12.12 Distress and Urgency Signals

This information is contained in the Jeppesen Publications (Pubs) > *General Airway Manual* > *Emergency Data – ICAO > section 2.5 Distress and Urgency Signals*, including general notes on emergency communication and cancelling of distress or urgency.

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13 AIRCRAFT LEASING

ORO.AOC.110

Any lease-in

1. Without prejudice to Regulation (EC) No 1008/2008, any lease agreement concerning Ryanair shall be subject to prior approval by the TM-CAD.
2. Malta Air shall not lease-in aircraft included in the list of operators subject to operational restrictions, registered in a State of which all operators under its oversight are subject to an operating ban or from an operator that is subject to an operating ban pursuant to Regulation (EC) No 2111/2005.

Lines of Responsibilities for Leasing

The following is a list of those responsible in Malta Air with regard to leasing of aircraft. The NPFO is responsible for keeping this Chapter up-to-date and notified to the TM-CAD. In addition, the following key post are responsible for:

1. NPFO is responsible for Lease Agreements;
2. Operations Control Manager is responsible for the day-to-day lease operations;
3. Compliance Manager (CMM) is responsible for auditing of the leasing policy and procedures;
4. NPFO is responsible for compliance of the leasing policy and procedures;
5. NPCA is responsible to ensure compliance with maintenance requirements.

Leasing Agreement

AMC1 ORO.AOC.110

General

Malta Air shall provide the TM-CAD with the following information for any lease agreement:

1. The aircraft type, registration markings and serial number, as soon as available;
2. The name and address of the registered owner;
3. A copy of the valid certificate of airworthiness;
4. A copy of the lease agreement or description of the lease provisions, except financial arrangements; and
5. Duration of the lease.

In case of wet lease-in a copy of the AOC of the third country operator and the areas of operation.

The information mentioned above shall be accompanied by a statement signed by the lessee that the parties to the lease agreement fully understand their respective responsibilities under the applicable regulations.

Terminology

Terms used in this paragraph have the following meaning:

Dry lease – is when the aeroplane is operated under the AOC of the lessee.

Wet lease – is when the aeroplane is operated under the AOC of the lessor.

13.1 Wet Leasing In

ORO.AOC.110(c)

Malta Air shall in advance of any Lease In activity seek approval from the TM-CAD.

13.1.1 Wet Leasing In from Any Operator

AMC1 ORO.AOC.110(c)

When seeking to wet lease in an aircraft from an any operator Malta Air shall provide the TM-CAD with the following information:

1. The aircraft type, registration markings and serial number, as soon as available;
2. The name and address of the registered owner;
3. A copy of the valid certificate of airworthiness;
4. A copy of the lease agreement or description of the lease provisions, except financial arrangements; and
5. Duration of the lease;
6. A copy of the Lessors AOC;
7. A copy of the Lessors Op Spec;
8. Certificate of Insurance;
9. Noise Certificate;
10. Airworthiness Review Certificate;
11. Aircraft weight Certificate.

The information mentioned above shall be accompanied by a statement signed by the lessee that the parties to the lease agreement fully understand their respective responsibilities under the applicable regulations.

13.1.2 Wet Lease In from 3rd Country Operator

AMC1 ORO.AOC.110(c)

If Malta Air is not intending to apply EU safety requirements for air operations and continuing airworthiness when wet leasing-in an aircraft registered in a third country, it shall demonstrate to the TM-CAD that the standards complied with are equivalent to the following requirements:

1. Annex IV (Part-CAT);
2. Part-ORO:
 - a. ORO.GEN.110 and Section 2 of Subpart GEN;
 - b. ORO.MLR, excluding ORO.MLR.105;
 - c. ORO.FC;
 - d. ORO.CC, excluding ORO.CC.200 and ORO.CC.210(a);
 - e. ORO.TC;
 - f. ORO.FTL, including related CS-FTL; and
 - g. ORO.SEC;
3. Annex V (Part-SPA), if applicable;
4. for continuing airworthiness management of the third-country operator, Part-M1 Subpart-B, Subpart-C and Part-CAMO, excluding M.A.707, and M.A.710;
5. for the maintenance organisation used by the third-country operator during the lease period: Part-I45;
6. retroactive airworthiness requirements in accordance with Part-26; and
7. Malta Air shall provide the TM-CAD with a full description of the flight time limitation scheme(s), operating procedures and safety assessment demonstrating compliance with the safety objectives set out in points (2)(a)–(f).

13.1.3 Records to be Kept

AMC2 ORO.AOC.110(c)

Malta Air shall maintain a record of occasions when lessors are used, for inspection by the TM-CAD. Malta Air uses an Crew Management System as its record keeping system for this purpose.

13.2 Wet Lease Out

AMC1 ORO.AOC.110(f)

Prior to the wet lease-out of an aircraft, Malta Air shall notify the competent authority.

When notifying the TM-CAD of its intention to wet lease-out an aircraft Malta Air shall provide the competent authority with the following information in the previously agreed manner.

1. the aircraft type, registration markings and serial number;
2. the name and address of the lessee;
3. a copy of the lease agreement or description of the lease provisions, except financial arrangements; and
4. the duration of the lease agreement.

13.3 Dry Lease In

ORO.AOC.110(d)

Malta Air when applying for an approval of the dry lease-in of an aircraft registered in a third country shall demonstrate to the TM-CAD that:

1. an operational need has been identified that cannot be satisfied through leasing an aircraft registered in the EU;
2. the duration of the dry lease-in does not exceed seven months in any 12 consecutive month period;
3. compliance with the applicable requirements of Regulation (EU) No 1321/2014 is ensured; and
4. the aircraft is equipped in accordance with the EU regulations for Air Operations.

13.4 Dry Lease Out

ORO.AOC.110(e)

Malta Air shall when intending to dry lease-out one of its aircraft apply for prior approval by the competent authority. The application shall be accompanied by copies of the intended lease agreement or description of the lease provisions, except financial arrangements, and all other relevant documentation.

13.5 Code Share

ORO.AOC.115

Malta Air does not currently use code share in its operation, as such no procedures are detailed for its use.

13.6 Ryanair Group

In order to maximise flexibility within the Ryanair Group, all members of the Ryanair Group will seek approval from their individual regulators to Wet lease IN from other members of the group.

All members of the Ryanair Group are required to comply with all conditions and stipulations by their individual regulators as detailed in their authority letters of approval as issued.

All leasing activity is subject to review annually, this activity takes place in advance of the commencement of the IATA Summer season.

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